

*Krishna's*

# Remedial Mathematics

*(For B.Pharm. First Semester Students of All Pharmacy Colleges Affiliated to  
U.P. Technical University, Lucknow and other Indian Universities.)*

*By*

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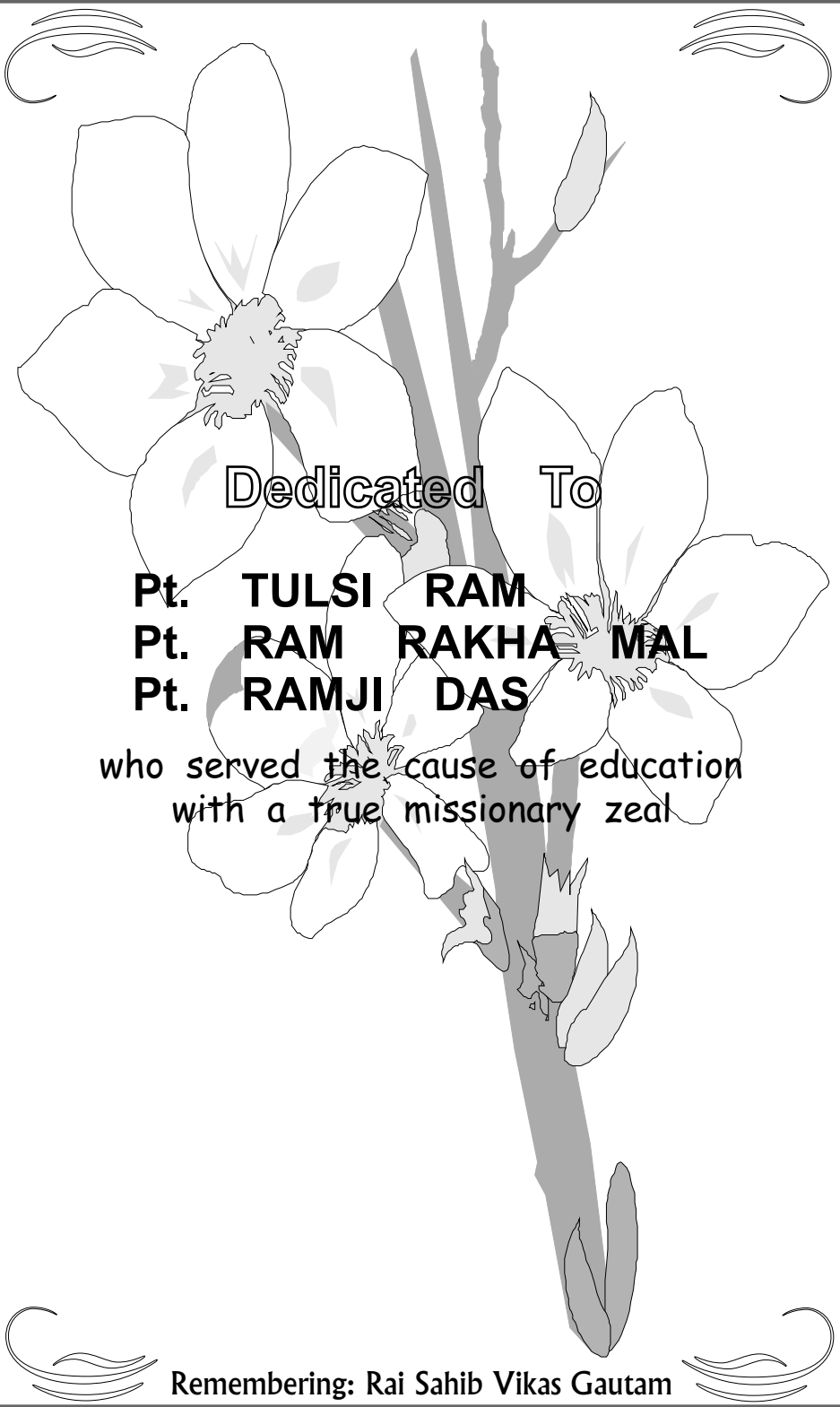
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Jai Shri Radhey Shyam

*Dedicated*  
to  
Lord  
Krishna

*Publishers*



Dedicated To

**Pt. TULSI RAM**

**Pt. RAM RAKHA MAL**

**Pt. RAMJI DAS**

who served the cause of education  
with a true missionary zeal

Remembering: Rai Sahib Vikas Gautam

# Preface to the Revised Edition

The authors feel great pleasure in presenting the **Thoroughly Revised Edition** of the book **REMEDIAL MATHEMATICS** and wish to record thanks to the teachers and students for their warm reception to the previous edition.

The present edition has been specially designed, made up-to-date and well organised in a systematic order according to the latest syllabus prescribed by **Uttar Pradesh Technical University, Lucknow** for **B.Pharm.** students of first semester.

The entire text has been thoroughly revised, improved and modified wherever necessary keeping in view the current trends and latest thoughts on the subject. A number of new problems selected from various university examination papers have been added in this new edition, thus making it more useful.

The authors have always endeavoured to keep the text update in the best interests of the students community — a gesture which the authors hope would be appreciated by the students and teachers alike.

In the last the authors wish to express their thanks to the Publishers for their painstaking in bringing out this book in the present nice form.

Suggestions for the improvement of the book will be thankfully received.

September, 2012

—*Authors*



# Preface to the First Edition

The authors feel great pleasure in presenting the **First Edition** of the book **REMEDIAL MATHEMATICS**. This book on mathematics has been primarily written according to the unified syllabus of “Remedial Mathematics” of first semester of all pharmacy colleges affiliated to U.P. Technical University, Lucknow and other Indian Universities.

The subject matter has been discussed in such a simple way that the students will find no difficulty to understand it. Each chapter of this book contains complete self-explanatory theory and a fairly large number of solved examples. Sufficient problems have also been selected from various university examination papers. We have tried to solve each problem in an elegant and more interesting way. Every effort has been made to explain the subject matter in such a simple way that the students can easily understand and feel encouraged to solve themselves the questions given in unsolved exercises. In each chapter, a number of unsolved questions with answers have been given so that students may have a lot of practice on that topic.

We have tried our best to keep the book free from misprints. The authors shall be grateful to the readers who point out errors and omissions which, inspite of all care, might have been there.

The authors in general hope that the present book will be warmly received by the students and teachers. We shall indeed be very thankful to our colleagues for their recommending this book to their students.

The authors wish to express their thanks to **Mr. S.K. Rastogi (M.D.)** and **Mr. Sugam Rastogi (E.D.)** and **entire team** of **KRISHNA Prakashan Media (P) Ltd., Meerut** for bringing out this book in the present nice form.

The authors will feel amply rewarded if the book serves the purpose for which it is meant. The suggestions for the improvement of the book are always welcome and will be highly appreciated.

July, 2007

—Authors

# Syllabus

## Remedial Mathematics

Pharmacy- I<sup>st</sup> Semester

Uttar Pradesh Technical University, Lucknow

PHARM-IIIM

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**Unit-I: Algebra:** Equations reducible to quadratics, Cramer's Rule Algebra of matrices.

1. Addition
2. Subtraction
3. Multiplication
4. Inverse of matrices

Simultaneous equation by matrices.

**Unit-II:** Certain co-ordinates, distance between two parts, area of triangle, locus of points, straight line, intercept form in normal.

**Unit-III:** Differential Calculus

1. Limit and function
2. Definition and formulation of differential calculus.
3. Rules of standard form of differential calculus.
4. Chain Rule, Parametric rule.

**Unit-IV:**

1. Standard form of Integral calculus
2. Partial fraction of Integral
3. Trigonometric function of Integral calculus.

**Unit-V:** Linear differential equation of order greater than one with constant coefficient complimentary function and particular Integral ( $x^n$ ,  $e^{ax}$ ) Trigonometric ( $\sin/\cos ex$ ).

## CONVERSION TABLES

| LENGTH (LINEAR MEASURE)       | MEASURE OF CAPACITY                            |
|-------------------------------|------------------------------------------------|
| 1 centimetre = 10 millimetres | 1 centilitre = 10 millilitres                  |
| 1 decimetre = 10 centimetres  | 1 decilitre = 10 centilitres                   |
| 1 metre = 10 decimetres       | 1 litre = 10 decilitres                        |
| 1 decametre = 10 metres       | 1 decalitre = 10 litres                        |
| 1 hectometre = 10 decametres  | 1 hectolitre = 10 decalitres                   |
| 1 kilometre = 10 hectometres  | 1 kilolitre = 10 hectolitres                   |
| WEIGHT MEASURE                | AREA (SQUARE MEASURE)                          |
| 1 centigram = 10 milligrams   | 1 square decimetre = 100 square centimetres    |
| 1 decigram = 10 centigrams    | 1 square metre = 100 square decimetres         |
| 1 gramme = 10 decigrams       | 1 are = 100 square metres                      |
| 1 decagram = 10 grams         | 1 hectare = 100 ares                           |
| 1 hectogram = 10 decagrams    | 1 square kilometre = 100 hectares              |
| 1 kilogram = 10 hectograms    | VOLUME (CUBIC MEASURE)                         |
| 1 myriagram = 10 kilograms    | 1 cubic decimetre = 1000 cubic centimetres     |
| 1 quintal = 100 kilograms     | 1 cubic metre or stere = 1000 cubic decimetres |
| 1 tonne = 1000 kilograms      |                                                |

## ABBREVIATIONS

|               |               |             |
|---------------|---------------|-------------|
| mm millimetre | ml millilitre | g gramme    |
| cm centimetre | l litre       | kg kilogram |
| m metre       | kl kilolitre  | a are       |
| km kilometre  | mg milligram  | ha hectare  |

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## **Appendix: Integration.....(01-32)**

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# Chapter

# 1



## Linear and Quadratic Equations

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### 1.1 Linear Equation

An equation which contains variables of degree one only is called a linear equation.

**For Example :** (i)  $3(x - 1) = 8$       (ii)  $\frac{z - 5}{2} - \frac{z - 3}{5} = \frac{1}{2}$

(iii)  $\frac{2 + n}{3 + n} = 5$       (iv)  $\frac{6y - 5}{2y} = \frac{7}{9}$

#### Operations used for Solving a Linear Equation.

- (i) We can add or subtract the same number on both sides of the linear equation.
  - (ii) We can multiply both sides of the linear equation by the same number.
  - (iii) We can divide both sides of the equation by the same non-zero number.
-



(iv) **Cross-multiplication** : If  $\frac{ax+b}{cx+d} = \frac{l}{m}$  then  $m(ax+b) = l(cx+d)$ . This

operation is called cross-multiplication.

(v) **Transposition** : Any term of a linear equation can be taken to other side by changing its sign. This operation is called transposition.

## Illustrative Examples

**Illustration 1 :** Solve :  $\frac{x+6}{4} + \frac{x-3}{5} = \frac{5x-4}{8}$ .

**Solution :** We have  $\frac{x+6}{4} + \frac{x-3}{5} = \frac{5x-4}{8}$

$$\text{or} \quad \frac{5(x+6) + 4(x-3)}{20} = \frac{5x-4}{8}$$

$$\text{or} \quad \frac{5x+30+4x-12}{5} = \frac{5x-4}{2}$$

$$\text{or} \quad \frac{9x+18}{5} = \frac{5x-4}{2}$$

$$\text{or} \quad 2(9x+18) = 5(5x-4)$$

[By cross-multiplication]

$$\text{or} \quad 18x+36 = 25x-20$$

$$\text{or} \quad 25x-18x = 36+20$$

[By transposition]

$$\text{or} \quad 7x = 56$$

$$\text{or} \quad x = \frac{56}{7} = 8.$$

$$\therefore x = 8.$$

**Illustration 2 :** Solve :  $\frac{3x+5}{4x+2} = \frac{3x+4}{4x+7}$ .

**Solution :** We have  $\frac{3x+5}{4x+2} = \frac{3x+4}{4x+7}$

$$\text{or} \quad (3x+5)(4x+7) = (4x+2)(3x+4) \quad [\text{By cross-multiplication}]$$

$$\text{or} \quad 12x^2 + 21x + 20x + 35 = 12x^2 + 16x + 6x + 8$$

$$\text{or} \quad 41x + 35 = 22x + 8$$

$$\text{or} \quad 41x - 22x = 8 - 35$$

[By transposition]

$$\text{or} \quad 19x = -27$$

$$\text{or} \quad x = -\frac{27}{19}.$$

$$\therefore x = -\frac{27}{19}.$$

## 1.2 Linear Equations in Two Variables

An equation which contains two variables of degree one only is called a linear equation in two variables.

An equation of the form,  $ax + by + c = 0$ , where  $a, b, c$  are real numbers ( $a \neq 0, b \neq 0$ ) is called a linear equation in two variables  $x$  and  $y$ .

**For example :** (i)  $5x + 4y - 9 = 0$       (ii)  $8x + 2y = 1$

(iii)  $0.3m + 0.2n = 27$       (iv)  $\sqrt{3}p - \sqrt{2}q = 0$

## 1.3 Simultaneous Linear Equations in Two Variables

When two or more than two linear equations in two unknowns  $x$  and  $y$  are satisfied by the same pair of values of  $x$  and  $y$  then these linear equations are called simultaneous linear equations.

**For example :** The system of linear equations

$$x + y = 2$$

$$x - y = 0$$

is satisfied by the values  $x = 1$  and  $y = 1$ .

## 1.4 Consistent and Inconsistent Systems of Linear Equations

Any two simultaneous linear equations in two variables  $x$  and  $y$  are said to be **consistent** if both equations can be satisfied by the same set of values of  $x$  and  $y$ .

If there does not exist any pair of values of  $x$  and  $y$  which satisfy both the equations then the two equations are said to be **inconsistent**.

## 1.5 Graphical Method of Solving Simultaneous Linear Equations

We know that every linear equation represents a straight line. On solving two simultaneous linear equations graphically the following three cases are possible :

**Case I. When the lines intersect** then the given equations are **consistent** and there exists a **unique solution** of two linear equations. The co-ordinates of the point of intersection give the unique solution.

**Case II. When the lines are parallel** then the given equations are **inconsistent** and have **no solution**.

**Case III. When the lines are coincident** then the given equations are said to be **dependent** equations and there are **infinitely many solutions**.

# Illustrative Examples

**Illustration 1 :** Solve graphically the following system of linear equations :

$$2x - y = 2, 4x - y = 8.$$

Also find the coordinates of the points where the lines meet the axis of  $x$ .

**Solution :** The equation  $2x - y = 2 \Rightarrow y = 2x - 2$  ... (I)

Here  $x = 0 \Rightarrow y = 2 \times 0 - 2 = -2$

and  $x = 2 \Rightarrow y = 2 \times 2 - 2 = 4 - 2 = 2.$

For (1), the table is

|     |    |   |
|-----|----|---|
| $x$ | 0  | 2 |
| $y$ | -2 | 2 |

Plot the points  $A(0, -2)$  and  $B(2, 2)$ . Join  $AB$  and produce the line on both sides to get the graph of the equation  $2x - y = 2$ .

Further, equation  $4x - y = 8$

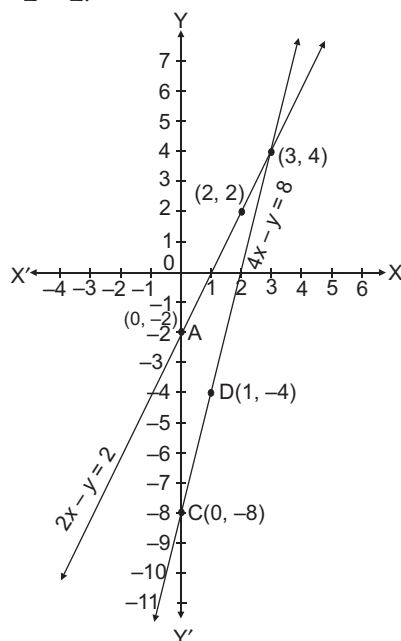
$\Rightarrow y = 4x - 8$  ... (2)

Here  $x = 0 \Rightarrow y = 4 \times 0 - 8 = -8$

and  $x = 1 \Rightarrow y = 4 \times 1 - 8 = 4 - 8 = -4.$

For (2), the table is

|     |    |    |
|-----|----|----|
| $x$ | 0  | 1  |
| $y$ | -8 | -4 |



Plot the points  $C(0, -8)$  and  $D(1, -4)$ . Join  $CD$  and produce the line on both sides to get the graph of  $4x - y = 8$ .

From the graph we see that the two lines intersect at the point  $(3, 4)$ . Hence  $x = 3, y = 4$  is the solution of the equations  $2x - y = 2$  and  $4x - y = 8$ .

Also, from the graph of  $2x - y = 2$  we see that it intersects the  $x$ -axis at the point  $(1, 0)$ .

Similarly the graph of  $4x - y = 8$  shows that this line intersects the  $x$ -axis at the point  $(2, 0)$ .

**Illustration 2 :** Solve graphically the following system of equations

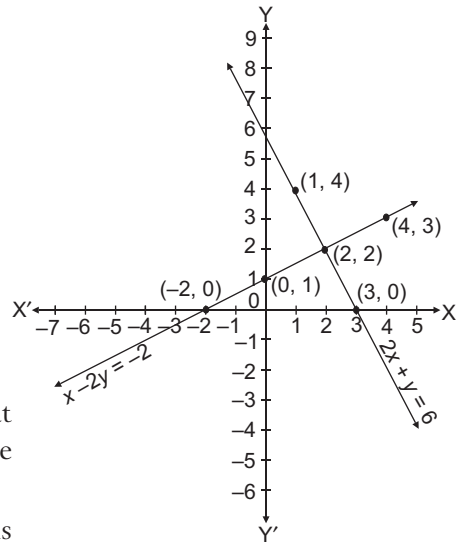
$$2x + y = 6, x - 2y = -2.$$

Also, find the coordinates of the points where the lines meet the axis of  $x$ .

**Solution :**

| Table for $2x + y = 6$ |   |   |
|------------------------|---|---|
| $x$                    | 1 | 3 |
| $y$                    | 4 | 0 |

| Table for $x - 2y = -2$ |   |   |
|-------------------------|---|---|
| $x$                     | 0 | 4 |
| $y$                     | 1 | 3 |



The graph shows that the lines intersect at the point  $(2, 2)$ . Hence  $x = 2$ ,  $y = 2$  is the required solution.

Further, the line  $2x + y = 6$  meets the  $x$ -axis at  $(3, 0)$ . And, the line  $x - 2y = -2$  meets the  $x$ -axis at  $(-2, 0)$ .

**Illustration 3 :** Solve the following system of equations graphically :

$$x - 2y = 6, 3x - 6y = 0.$$

**Solution :** We have  $x - 2y = 6 \Rightarrow y = \frac{x - 6}{2}$

Here  $x = 0 \Rightarrow y = -3$  and

$x = 6 \Rightarrow y = 0$

Thus, the table is

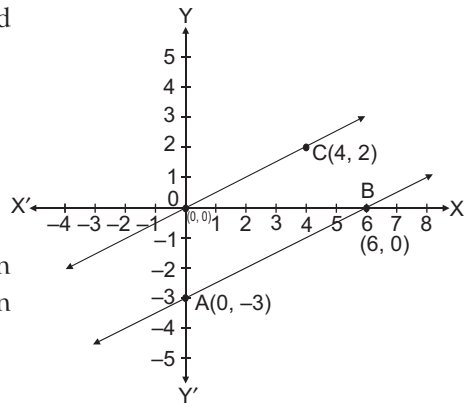
|     |    |   |
|-----|----|---|
| $x$ | 0  | 6 |
| $y$ | -3 | 0 |

Plot the points  $A(0, -3)$  and  $B(6, 0)$ . Join the points  $A$  and  $B$  and extend the line on both sides to get the graph of  $x - 2y = 6$ .

Again,  $3x - 6y = 0 \Rightarrow y = \frac{x}{2}$

Here  $x = 0 \Rightarrow y = 0$  and  $x = 4 \Rightarrow y = 2$ .

|     |   |   |
|-----|---|---|
| $x$ | 0 | 4 |
| $y$ | 0 | 2 |



Plot the points  $O(0, 0)$  and  $C(4, 2)$ . Join the points  $O$  and  $C$  and extend the line on both sides to get the graph of the line  $3x - 6y = 0$ .

From the graph we see that the two lines are parallel to each other and do not intersect. Hence the given system of equations is inconsistent and there is no solution.

## 1.6 Algebraic Methods of Solving Simultaneous Linear Equations

### (1) Substitution Method

**Step 1.** Find the value of any variable in terms of the other variable from any one equation.

**Step 2.** Substitute this value in the other equation and solve the simple equation obtained and find the value of one variable.

**Step 3.** Substitute the value of the variable obtained in any one equation to get the value of the other variable.

### Illustrative Examples

**Illustration 1 :** Solve for  $x$  and  $y$

$$x + y = 3$$

and

$$2x + 5y = 12.$$

**Solution :** We have  $x + y = 3$  ... (1)

$$2x + 5y = 12 \quad \dots (2)$$

From equation (1), we have

$$x = 3 - y \quad \dots (3)$$

Putting this value of  $x$  in equation (2), we get

$$2(3 - y) + 5y = 12$$

$$\text{or} \quad 6 - 2y + 5y = 12$$

$$\text{or} \quad 6 + 3y = 12 \quad \text{or} \quad 3y = 12 - 6 = 6$$

$$\text{or} \quad y = \frac{6}{3} = 2.$$

Putting  $y = 2$  in equation (3), we get

$$x = 3 - 2 = 1.$$

Hence  $x = 1, y = 2.$

### (2) Elimination Method.

**Step 1.** Multiply the equations by suitable non-zero constants so as to make equal the coefficients of the variable to be eliminated.

**Step 2.** Add the equations obtained if the terms with equal coefficients are of opposite sign and subtract if they are of the same sign.

**Step 3.** Solve the equation obtained after eliminating one variable to get the value of the other variable .

**Step 4.** Substitute the value of the variable obtained above in any one of the given equations and find the value of the other variable.

## Illustrative Examples

**Illustration 1 :** Solve for  $x$  and  $y$

$$3x + 4y = 17$$

and

$$2x + 9y = 5.$$

**Solution :** We have

$$3x + 4y = 17 \quad \dots(1)$$

$$2x + 9y = 5 \quad \dots(2)$$

Multiplying equation (1) by 2 and equation (2) by 3, we get

$$6x + 8y = 34 \quad \dots(3)$$

$$6x + 27y = 15 \quad \dots(4)$$

Subtracting equation (4) from equation (3), we get

$$-19y = 19 \quad \text{or} \quad y = -1$$

Putting  $y = -1$  in equation (1), we get

$$3x - 4 = 17 \quad \text{or} \quad 3x = 17 + 4 = 21$$

or

$$x = 7.$$

Hence

$$x = 7, y = -1.$$

### (3) Special Case Method

When the two given equations are such that the coefficients of the two variables in one equation are interchanged in the other equation then we add and subtract the given equations to obtain two new equations of the type  $x + y = a$  and  $x - y = b$ .

We can now solve  $x + y = a$  and  $x - y = b$  easily.

## Illustrative Examples

**Illustration 1 :** Solve for  $x$  and  $y$

$$47x + 31y = 63$$

and

$$31x + 47y = 15.$$

**Solution :** We have

$$47x + 31y = 63 \quad \dots(1)$$

and

$$31x + 47y = 15. \quad \dots(2)$$

Adding equations (1) and (2), we get

$$78x + 78y = 78 \quad \text{or} \quad 78(x + y) = 78$$

or

$$x + y = 1 \quad \dots(3)$$

Subtracting equation (2) from equation (1), we get

$$16x - 16y = 48 \quad \text{or} \quad 16(x - y) = 48$$

or  $x - y = 3$  ... (4)

Adding equations (3) and (4), we get

$$2x = 4 \quad \text{or} \quad x = 2.$$

Putting  $x = 2$  in equation (3), we get

$$2 + y = 1 \quad \text{or} \quad y = -1.$$

Hence  $x = 2, y = -1$ .

**Illustration 2 :** Solve for  $x$  and  $y$

$$\frac{1}{2x} - \frac{1}{y} = -1$$

and  $\frac{1}{x} + \frac{1}{2y} = 8.$

**Solution :** We have

$$\frac{1}{2x} - \frac{1}{y} = -1 \quad \dots (1)$$

$$\frac{1}{x} + \frac{1}{2y} = 8 \quad \dots (2)$$

Multiplying equation (2) by 2, we get

$$\frac{2}{x} + \frac{1}{y} = 16 \quad \dots (3)$$

Adding equations (1) and (3), we get

$$\frac{1}{2x} + \frac{2}{x} = 15 \quad \text{or} \quad \frac{5}{2x} = 15 \quad \text{or} \quad x = \frac{5}{2 \times 15} = \frac{1}{6}.$$

Putting  $x = \frac{1}{6}$  in equation (1), we get

$$\frac{1}{2 \times \frac{1}{6}} - \frac{1}{y} = -1 \quad \text{or} \quad \frac{1}{y} = \frac{1}{2 \times \frac{1}{6}} + 1$$

or  $\frac{1}{y} = 3 + 1 = 4 \quad \text{or} \quad y = \frac{1}{4}.$

Hence  $x = \frac{1}{6}, y = \frac{1}{4}.$

## 1.7 General Solution and Conditions for Solvability

Let  $a_1x + b_1y = c_1$  and  $a_2x + b_2y = c_2$  be the simultaneous equations.

**Case I.** If  $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ , then the given system has a unique solution.

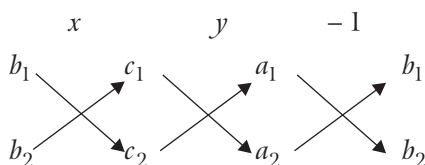
$$x = \frac{b_2 c_1 - b_1 c_2}{a_1 b_2 - a_2 b_1} \quad \text{and} \quad y = \frac{c_2 a_1 - c_1 a_2}{a_1 b_2 - a_2 b_1}$$

Thus solution is given by

$$\frac{x}{b_1 c_2 - b_2 c_1} = \frac{y}{c_1 a_2 - c_2 a_1} = \frac{-1}{a_1 b_2 - a_2 b_1}$$

In this case the two equations are said to be consistent.

**Cross Multiplication Method :** The above solution can be easily obtained with the help of the following diagram :



The arrows between two numbers mean that the numbers are to be multiplied. First multiply the numbers with downward arrow and from their product subtract the product of the numbers with upward arrow.

**Case II.** If  $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$ , then the given equations have infinite number of solutions and the two equations are said to be depended equations.

**Case III.** If  $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$ , then the system has no solution and the two equations are said to be inconsistent.

## 1.8 Homogeneous Linear Equations

If  $c_1 = c_2 = 0$ , then the equations become  $a_1x + b_1y = 0$  and  $a_2x + b_2y = 0$ .

- (i)  $\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \Rightarrow x = 0, y = 0$  is only solution, and
- (ii)  $\frac{a_1}{a_2} = \frac{b_1}{b_2} \Rightarrow$  the system has infinitely many solutions.

## Illustrative Examples

**Illustration 1 :** In each of the following system of equations, determine whether the system has a unique solution, no solution or infinitely many solutions. In case there is a unique solution, find it.

- (i)  $3x - 7y = 10, 2x - y = 3$
- (ii)  $-3x + 4y = 5, \frac{9}{2}x - 6y = \frac{15}{2}$
- (iii)  $4x + 7y = 10, 10x + \frac{35}{2}y = 25$



**Solution :** (i) We have

$$3x - 7y = 10 \quad \text{or} \quad 3x - 7y - 10 = 0 \quad \dots(1)$$

$$2x - y = 3 \quad \text{or} \quad 2x - y - 3 = 0 \quad \dots(2)$$

These equations are of the form

$$a_1 x + b_1 y + c_1 = 0, a_2 x + b_2 y + c_2 = 0$$

Here  $a_1 = 3, b_1 = -7, c_1 = -10, a_2 = 2, b_2 = -1, c_2 = -3$

$$\therefore \frac{a_1}{a_2} = \frac{3}{2}, \frac{b_1}{b_2} = \frac{-7}{-1} = \frac{7}{1}, \frac{c_1}{c_2} = \frac{-10}{-3} = \frac{10}{3}.$$

Since  $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ , therefore, a **unique solution** exists. Now by cross multiplication, we have

$$\begin{array}{ccccc} x & & y & & -1 \\ -7 & \nearrow & 10 & \nwarrow & 3 \\ -1 & \nwarrow & 3 & \nearrow & 2 \end{array}$$

$$\therefore \frac{x}{(-7) \times 3 - (-1) \times 10} = \frac{y}{10 \times 2 - 3 \times 3} = \frac{-1}{3 \times (-1) - 2 \times (-7)}$$

$$\text{or} \quad \frac{x}{-21 + 10} = \frac{y}{20 - 9} = \frac{-1}{-3 + 14}$$

$$\text{or} \quad \frac{x}{-11} = \frac{y}{11} = \frac{-1}{11}$$

$$\text{or} \quad \frac{x}{-11} = \frac{-1}{11} \quad \text{or} \quad x = \frac{11}{11} = 1,$$

$$\frac{y}{11} = \frac{-1}{11} \quad \text{or} \quad y = \frac{-11}{11} = -1$$

Hence,  $x = 1, y = -1$ .

(ii) We have  $-3x + 4y = 5 \quad \text{or} \quad -3x + 4y - 5 = 0,$

$$\frac{9x}{2} - 6y = \frac{15}{2} \quad \text{or} \quad 9x - 12y = 15$$

$$\text{or} \quad 9x - 12y - 15 = 0$$

These equations are of the form

$$a_1 x + b_1 y + c_1 = 0 \quad \text{and} \quad a_2 x + b_2 y + c_2 = 0$$

Here  $a_1 = -3, b_1 = 4, c_1 = -5, a_2 = 9, b_2 = -12, c_2 = -15$

$$\therefore \frac{a_1}{a_2} = \frac{-3}{9} = -\frac{1}{3}, \frac{b_1}{b_2} = \frac{4}{-12} = -\frac{1}{3}, \frac{c_1}{c_2} = \frac{-5}{-15} = \frac{1}{3}$$

$$\Rightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

Hence, the given equations are inconsistent having no solution.

(iii) We have  $4x + 7y = 10$  or  $4x + 7y - 10 = 0$ ,

$$10x + \frac{35}{2}y = 25 \quad \text{or} \quad 20x + 35y = 50$$

or  $20x + 35y - 50 = 0$

Here  $a_1 = 4, b_1 = 7, c_1 = -10, a_2 = 20, b_2 = 35, c_2 = -50$

$$\therefore \frac{a_1}{a_2} = \frac{4}{20} = \frac{1}{5}, \frac{b_1}{b_2} = \frac{7}{35} = \frac{1}{5}, \frac{c_1}{c_2} = \frac{-10}{-50} = \frac{1}{5}$$

$$\Rightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}.$$

Hence the given system of equations has **infinitely many solutions** and thus has no **unique solution**.

## Comprehensive Exercise 1

1. Solve the following system of simultaneous linear equations graphically :

(i)  $x + y = 3, 2x + 5y = 12$       (ii)  $3x - 5y = -1, 2x - y = -3$

(iii)  $x - 2y = 5, 2x - 4y = 10$

(iv)  $2x + 7y = 11, 5x + \frac{35}{2}y = 25$

(v)  $3x + 2y = 14, -x + 4y = 7$  (vi)  $4x + 6y = 9, 4x - 2y = -11$

2. Solve the following system of equations :

(i)  $3x - 5y = -1, x - y = -1$

(ii)  $11x + 15y + 23 = 0, 7x - 2y - 20 = 0$

(iii)  $\frac{x}{2} + \frac{y}{4} = 3, 2x - y = 4$       (iv)  $\frac{x}{3} + \frac{y}{4} = 3, \frac{5x}{6} - \frac{y}{8} = 4$

(v)  $\frac{8}{x} + \frac{3}{y} = 1, \frac{6}{x} - \frac{5}{y} = 8$       (vi)  $\frac{x+y}{4} - \frac{x-y}{3} = 1, \frac{x+y}{2} + \frac{x-y}{6} = 12$

3. In each of the following system of equations, find whether it has a unique, infinite or no solution.

(i)  $2x - y = 3, 6x - 3y = 2$

(ii)  $7x + 6y = 13, 2x + 3y = 4$

(iii)  $x - 5y + 3 = 0, 3x - 15y + 9 = 0$

(iv)  $5x + 2y = 12, 4x + \frac{8}{5}y = 3$

(v)  $3x + \frac{5}{8}y = 13, 2x - 3y = 5$

(vi)  $2x - \frac{10}{3}y = 4, 3x - 5y = 9$

## Answers

- |                              |                                           |
|------------------------------|-------------------------------------------|
| 1. (i) $1, y = 2$            | (ii) $x = -2, y = -1$                     |
| (iii) infinite solutions     | (iv) no solution                          |
| (v) $x = 3, y = \frac{5}{2}$ | (vi) $x = \frac{-3}{2}, y = \frac{5}{2}$  |
| 2. (i) $x = -2, y = -1$      | (ii) $x = 2, y = -3$                      |
| (iii) $x = 4, y = 4$         | (iv) $x = \frac{11}{2}, y = \frac{14}{3}$ |
| (v) $x = 2, y = -1$          | (vi) $x = 16, y = 4$                      |
| 3. (i) no solution           | (ii) unique solution                      |
| (iii) infinite solutions     | (iv) no solution                          |
| (v) unique solution          | (vi) no solution                          |

## 1.9 Quadratic Equation

An equation of the form  $ax^2 + bx + c = 0$ , where  $a, b, c$  are real numbers and  $a \neq 0$ , is called a **quadratic equation** with real coefficients.

Here,  $x$  is a variable and  $a, b, c$  are given real numbers which are called coefficients of the equation. The coefficient of  $x^2$  is  $a$ , the coefficient of  $x$  is  $b$  and  $c$  is called the **constant term** of the equation.

**Roots of an Equation.** The values of the variable satisfying a given equation are called its **roots**.

Thus,  $x = \alpha$  is a root of the equation  $f(x) = 0$  if  $f(\alpha) = 0$ .

**Solution Set of a Given Equation.** The set of all roots of an equation, in a given domain, is called the **solution set** of the equation.

## 1.10 Solving a Quadratic Equation

**(I) Solving a Pure Quadratic Equation.** The equation of the form  $ax^2 + c = 0$ , where  $a \neq 0$ , is known as a **pure quadratic equation**.

$$\text{Obviously, } ax^2 + c = 0 \Rightarrow ax^2 = -c \Rightarrow x^2 = -\frac{c}{a} \Rightarrow x = \pm \sqrt{-\frac{c}{a}}.$$

The equation  $ax^2 + bx + c = 0$ , where  $a \neq 0$ , and  $b \neq 0$ , is called a **complete quadratic equation**.

## Illustrative Examples

**Illustration 1 :** Solve :  $16x^2 - 25 = 0$ .

**Solution :** We have  $16x^2 - 25 = 0$

$$\Rightarrow 16x^2 = 25 \quad \Rightarrow x^2 = \frac{25}{16}$$

$$\Rightarrow x = \pm \sqrt{\frac{25}{16}} = \pm \frac{5}{4}.$$

Hence, the solution set is  $\left\{\frac{5}{4}, -\frac{5}{4}\right\}$ .

**(II) Method of Factorization for the Solution of a Quadratic Equation.**

**Illustration 2 :** Solve : (i)  $x^2 - 5x + 4 = 0$ , (ii)  $abx^2 - (a^2 + b^2)x + ab = 0$ ,

(iii)  $a(x^2 + 1) = x(a^2 + 1)$ .

**Solution :** (i) We have  $x^2 - 5x + 4 = 0$

$$\Rightarrow x^2 - 4x - x + 4 = 0$$

$$\Rightarrow x(x - 4) - 1(x - 4) = 0$$

$$\Rightarrow (x - 4)(x - 1) = 0$$

$$\Rightarrow x - 4 = 0 \quad \text{or} \quad x - 1 = 0$$

$$\Rightarrow x = 4 \quad \text{or} \quad x = 1.$$

Hence, the roots of the given equation are 4, 1.

(ii) We have  $abx^2 - (a^2 + b^2)x + ab = 0$

$$\Rightarrow abx^2 - a^2x + b^2x + ab = 0$$

$$\Rightarrow ax(bx - a) - b(bx - a) = 0$$

$$\Rightarrow (bx - a)(ax - b) = 0$$

$$\Rightarrow bx - a = 0 \quad \text{or} \quad ax - b = 0$$

$$\Rightarrow x = \frac{a}{b} \quad \text{or} \quad \frac{b}{a}.$$

Hence, the roots of the given equation are  $\frac{a}{b}, \frac{b}{a}$ .

(iii) We have  $ax^2 - (a^2 + 1)x + a = 0$

$$\Rightarrow ax^2 - a^2x - x + a = 0$$

$$\Rightarrow ax(x - a) - (x - a) = 0$$

$$\Rightarrow (x - a)(ax - 1) = 0$$

$$\Rightarrow (x - a) = 0 \quad \text{or} \quad (ax - 1) = 0$$

$$\Rightarrow x = a \quad \text{or} \quad \frac{1}{a}.$$

Hence, the roots of the given equation are  $a, \frac{1}{a}$ .

### (III) Method of Completing the Square for the Solution of a Quadratic Equation (Formula for the Solution of a Quadratic Equation).

(Shreedharacharyya's Method).

Let the given quadratic equation be  $ax^2 + bx + c = 0$ , where  $a, b, c \in R$  and  $a \neq 0$ .

Then roots of the equation  $ax^2 + bx + c = 0$ , where  $a \neq 0$ , are

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \text{ i.e., } x = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } x = \frac{-b - \sqrt{b^2 - 4ac}}{2a}.$$

## Illustrative Examples

**Illustration 1 :** (i) Solve :  $2x^2 - 4x + 3 = 0$ , (ii)  $25x^2 + 30x + 7 = 0$ ,

(iii)  $4x^2 - 4ax + (a^2 - b^2) = 0$ .

**Solution :** (i) Comparing the given equation  $2x^2 - 4x + 3 = 0$  with  $ax^2 + bx + c = 0$ , we have  $a = 2, b = -4$  and  $c = 3$ .

Using the formula for the solution of a quadratic equation, we have

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-(-4) \pm \sqrt{(-4)^2 - 4 \times 2 \times 3}}{2 \times 2} \\ &= \frac{4 \pm \sqrt{16 - 24}}{4} = \frac{4 \pm \sqrt{-8}}{4} = \frac{2 \pm \sqrt{-2}}{2} = \frac{2 \pm i\sqrt{2}}{2} \\ &= 1 \pm i \frac{\sqrt{2}}{2} = 1 \pm \frac{1}{\sqrt{2}} i. \end{aligned}$$

Hence, the roots of the given equation are  $1 + \frac{1}{\sqrt{2}} i$  and  $1 - \frac{1}{\sqrt{2}} i$ .

(ii) Comparing the given equation  $25x^2 + 30x + 7 = 0$  with  $ax^2 + bx + c = 0$ , we have  $a = 25, b = 30$  and  $c = 7$ .

Using the formula for the solution of a quadratic equation, we have

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-30 \pm \sqrt{(30)^2 - 4 \times 25 \times 7}}{2 \times 25} \\ &= \frac{-30 \pm \sqrt{900 - 700}}{50} = \frac{-30 \pm \sqrt{200}}{50} \\ &= \frac{-30 \pm 10\sqrt{2}}{50} = \frac{-3 \pm \sqrt{2}}{5}. \end{aligned}$$

Hence, the solution set of the given equation is  $\left\{ \frac{-3 + \sqrt{2}}{5}, \frac{-3 - \sqrt{2}}{5} \right\}$ .

(iii) Comparing the given equation  $4x^2 - 4ax + (a^2 - b^2) = 0$  with  $Ax^2 + Bx + C = 0$ , we have  $A = 4$ ,  $B = -4a$  and  $C = (a^2 - b^2)$ .

Using the formula for the solution of a quadratic equation, we have

$$\begin{aligned} x &= \frac{-B \pm \sqrt{B^2 - 4AC}}{2A} = \frac{-(-4a) \pm \sqrt{(-4a)^2 - 4 \times 4 \times (a^2 - b^2)}}{2 \times 4} \\ &= \frac{4a \pm \sqrt{16a^2 - 16(a^2 - b^2)}}{8} = \frac{4a \pm \sqrt{16b^2}}{8} = \frac{4a \pm 4b}{8}. \end{aligned}$$

Hence, the solution set of the given equation is  $\left\{ \frac{a+b}{2}, \frac{a-b}{2} \right\}$ .

## Comprehensive Exercise 2

Solve the following equations :

1.  $x^2 + 25 = 0$ .
2.  $16x^2 - 49 = 0$ .
3.  $x^2 - 8x - 48 = 0$ .
4.  $8x^2 - 22x - 21 = 0$ .
5.  $70x - 63 = 7x^2$ .
6.  $2x^2 - 11x = -5$ .
7.  $6x^2 + 7x - 10 = 0$ .
8.  $2x^2 - 10x = 3x - 15$ .
9.  $x^2 + 5x + 4 = 0$
10.  $\frac{1}{p+q+x} = \frac{1}{p} + \frac{1}{q} + \frac{1}{x}$

## Answers

1.  $\pm 5i$
2.  $\pm \frac{7}{4}$
3.  $12, -4$
4.  $\frac{7}{2}, -\frac{3}{4}$
5.  $1, 9$
6.  $\frac{1}{2}, 5$
7.  $-2, \frac{5}{6}$
8.  $\frac{3}{2}, 5$
9.  $-4, -1$
10.  $-p, -q$

## 1.11 Nature of the Roots of a Quadratic Equation

Let the quadratic equation be

$$ax^2 + bx + c = 0, \quad \dots(1)$$

where  $a, b, c \in R$  and  $a \neq 0$ .

We know that if  $\alpha, \beta$  are the roots of (1), then

$$\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \quad \text{and} \quad \beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}.$$

### Discriminant of a Quadratic Equation.

The expression ' $b^2 - 4ac$ ' upon which the nature of the roots of the quadratic equation  $ax^2 + bx + c = 0$  depends is called the **discriminant** of this equation and is denoted by the Greek letter  $\Delta$ .

Thus,  $\Delta = b^2 - 4ac = \text{discriminant of } ax^2 + bx + c = 0$ .

The roots of the equation  $ax^2 + bx + c = 0$  are

(i) real if and only if  $\Delta \geq 0$ ,

(ii) real and equal if and only if  $\Delta = 0$ ,

and (iii) imaginary and unequal if and only if  $\Delta < 0$ .

## Illustrative Examples

**Illustration 1:** Find the nature of the roots of the following equations without actually solving them :

(i)  $3x^2 - 8x - 3 = 0$       (ii)  $9x^2 - 6x + 1 = 0$       (iii)  $2x^2 + x + 5 = 0$ .

**Solution :** (i) The given equation is  $3x^2 - 8x - 3 = 0$ .

Its discriminant  $\Delta = (-8)^2 - 4 \times 3 \times (-3) = 64 + 36 = 100 > 0$ .

$\therefore$  The roots of the given equation are real and unequal.

Since  $\Delta = 100 = (10)^2$  is a perfect square and the coefficients of the equation are all rational, therefore the roots are rational and unequal.

(ii) The given equation is  $9x^2 - 6x + 1 = 0$ . Its discriminant

$$\Delta = (-6)^2 - 4 \times 9 \times 1 = 36 - 36 = 0.$$

$\therefore$  The roots of the equation are real and equal.

(iii) The given equation is  $2x^2 + x + 5 = 0$ .

Its discriminant  $\Delta = 1^2 - 4 \times 2 \times 5 = -39 < 0$ .

$\therefore$  The roots of the given equation are imaginary and unequal.

**Illustration 2:** For what value of  $m$  will the equation

$$x^2 - 2(1 + 3m)x + 7(3 + 2m) = 0 \text{ have equal roots?}$$

**Solution :** The given equation will have equal roots if its discriminant  $\Delta = 0$ .

Now,  $\Delta = 0 \Rightarrow 4(1 + 3m)^2 - 4 \times 7(3 + 2m) = 0$

$$\Rightarrow (1 + 3m)^2 - 7(3 + 2m) = 0$$

$$\begin{aligned}
 \Rightarrow & 9m^2 - 8m - 20 = 0 \\
 \Rightarrow & 9m^2 - 18m + 10m - 20 = 0 \\
 \Rightarrow & 9m(m - 2) + 10(m - 2) = 0 \\
 \Rightarrow & (m - 2)(9m + 10) = 0 \\
 \Rightarrow & m = 2 \quad \text{or} \quad m = -\frac{10}{9}.
 \end{aligned}$$

Hence, the roots will be equal for  $m = 2$  or  $m = -\frac{10}{9}$ .

**Illustration 3 :** The roots of the quadratic equation  $5x^2 - px + 1 = 0$  are real and distinct. Find the possible values of  $p$ .

**Solution :** The roots of the quadratic equation  $5x^2 - px + 1 = 0$  are real and distinct

$$\begin{aligned}
 \Rightarrow & \text{its discriminant } \Delta = (-p)^2 - 4 \times 5 \times 1 > 0 \\
 \Rightarrow & p^2 - 20 > 0 \quad \Rightarrow \quad p^2 > 20 \quad \Rightarrow \quad |p|^2 > 20 \\
 \Rightarrow & |p| > \sqrt{20} \quad \Rightarrow \quad p > \sqrt{20} \quad \text{or} \quad p < -\sqrt{20}.
 \end{aligned}$$

Hence,  $p$  may have any value less than  $-\sqrt{20}$  or any value greater than  $\sqrt{20}$ .

**Illustration 4 :** Determine the positive values of  $k$  for which the equations  $x^2 + kx + 64 = 0$  and  $x^2 - 8x + k = 0$  will both have real roots.

**Solution :** The equation  $x^2 + kx + 64 = 0$  will have real roots if its discriminant  $\geq 0$  i.e., if  $k^2 - 4 \times 1 \times 64 \geq 0$  i.e., if  $k^2 \geq 256$  i.e.,  $k \geq 16$ . [ $\because k$  is +ve]

The equation  $x^2 - 8x + k = 0$  will have real roots if its discriminant  $\geq 0$

i.e., if  $64 - 4k \geq 0$  i.e., if  $4k \leq 64$  i.e., if  $k \leq 16$ .

Thus, both the equations will have real roots if  $k \geq 16$  and  $k \leq 16$  i.e., if  $k = 16$ .

Hence, the only +ve value of  $k$  for which both the given equations has real roots is 16.

## 112 Relations between Roots and Coefficients

Let  $\alpha$  and  $\beta$  be the roots of the equation

$$ax^2 + bx + c = 0, \quad \dots(1)$$

where  $a, b, c \in R$  and  $a \neq 0$ .

Then, 
$$\alpha + \beta = -\frac{b}{a} = -\left(\frac{\text{coefficient of } x}{\text{coefficient of } x^2}\right)$$

and 
$$\alpha\beta = \frac{c}{a} = \left(\frac{\text{constant term}}{\text{coefficient of } x^2}\right).$$



## 113 Formation of a Quadratic Equation with Given Roots

The quadratic equation whose roots are  $\alpha$  and  $\beta$  is given by  $(x - \alpha)(x - \beta) = 0$

i.e.,  $x^2 - (\alpha + \beta)x + \alpha\beta = 0$

i.e.,  $x^2 - (\text{sum of the roots})x + \text{product of the roots} = 0.$

### Illustrative Examples

**Illustration 1 :** Without solving the equation  $3x^2 - 5x - 8 = 0$ , find the sum and the product of its roots.

**Solution :** Let  $\alpha, \beta$  be the roots of the equation  $3x^2 - 5x - 8 = 0$ .

Then,  $\alpha + \beta = -\left(\frac{\text{coefficient of } x}{\text{coefficient of } x^2}\right) = -\frac{(-5)}{3} = \frac{5}{3}$

and  $\alpha\beta = \frac{\text{constant term}}{\text{coefficient of } x^2} = \frac{-8}{3} = -\frac{8}{3}.$

**Illustration 2 :** If  $c, d$  are the roots of the equation  $(x - a)(x - b) - k = 0$ , prove that  $a, b$  are the roots of the equation  $(x - c)(x - d) + k = 0$ .

**Solution :** We have  $c, d$  are the roots of  $(x - a)(x - b) - k = 0$

$$\Rightarrow (x - a)(x - b) - k = (x - c)(x - d)$$

$$\Rightarrow (x - c)(x - d) + k = (x - a)(x - b)$$

$$\Rightarrow a, b \text{ are the roots of } (x - c)(x - d) + k = 0.$$

$$[\because a \text{ and } b \text{ are the roots of } (x - a)(x - b) = 0]$$

**Illustration 3 :** The roots  $\alpha, \beta$  of the equation  $x^2 + kx + 12 = 0$  are such that  $\alpha - \beta = 1$ , find  $k$ .

**Solution :** We have  $\alpha + \beta = -k$  and  $\alpha\beta = 12$ .

Also, it is given that  $\alpha - \beta = 1$ .

Now,  $(\alpha + \beta)^2 - (\alpha - \beta)^2 = 4\alpha\beta$

$$\Rightarrow (-k)^2 - 1 = 4 \times 12 \Rightarrow k^2 - 1 = 48$$

$$\Rightarrow k^2 = 49 \Rightarrow k = \pm 7.$$

**Illustration 4 :** If  $\alpha, \beta$  are the roots of the equation  $x^2 - 3x + k = 0$ , find  $k$  such that  $\alpha = 2\beta$ .

**Solution :** We have  $\alpha + \beta = -\frac{(-3)}{1} = 3$

and  $\alpha\beta = \frac{k}{1} = k.$

$$\begin{aligned}
 \text{Now,} \quad & \alpha + \beta = 3 \Rightarrow 2\beta + \beta = 3 \\
 \Rightarrow & 3\beta = 3 \Rightarrow \beta = 1. \quad [\because \alpha = 2\beta] \\
 \text{Also,} \quad & \alpha + \beta = 3 \text{ and } \beta = 1 \Rightarrow \alpha + 1 = 3 \Rightarrow \alpha = 2. \\
 \therefore & k = \alpha\beta = 2 \times 1 = 2.
 \end{aligned}$$

**Illustration 5 :** If 4 is a root of the equation  $x^2 - 5x + k = 0$ , find the value of  $k$  and the other root.

**Solution :** Since 4 is a root of  $x^2 - 5x + k = 0$ , therefore

$$4^2 - 5 \times 4 + k = 0 \Rightarrow k = 20 - 16 = 4.$$

Now, let  $\beta$  be the other root of  $x^2 - 5x + k = 0$ . ...(1)

Since 4 and  $\beta$  are the roots of (1), therefore

$$4 \cdot \beta = \frac{k}{1} = \frac{4}{1} = 4 \Rightarrow \beta = 1.$$

Hence,  $k = 4$  and the other root is 1.

**Illustration 6 :** If  $\alpha$  be a root of the equation  $4x^2 + 2x - 1 = 0$ , prove that  $4\alpha^3 - 3\alpha$  is the other root. [UPTU 2002]

**Solution :** Given  $\alpha$  be a root of the equation  $4x^2 + 2x - 1 = 0$ .

$$\text{Then} \quad 4\alpha^2 + 2\alpha - 1 = 0. \quad \dots(1)$$

Let  $\beta$  be the other root of the given equation.

$$\text{Then} \quad \alpha + \beta = -\frac{1}{2} \Rightarrow \beta = -\frac{1}{2} - \alpha. \quad \dots(2)$$

We have to show that  $\beta = 4\alpha^3 - 3\alpha$ .

$$\begin{aligned}
 \text{Now} \quad 4\alpha^3 - 3\alpha &= \alpha (4\alpha^2 + 2\alpha - 1) - \frac{1}{2} (4\alpha^2 + 2\alpha - 1) - \alpha - \frac{1}{2} \\
 &= \alpha \cdot 0 - \frac{1}{2} \cdot 0 - \alpha - \frac{1}{2}, \quad [\text{From (1)}] \\
 &= -\alpha - \frac{1}{2} = \beta. \quad [\text{From (2)}]
 \end{aligned}$$

Hence,  $4\alpha^3 - 3\alpha$  is the other root of the given equation.

**Illustration 7 :** Find the value of  $k$  for which the roots  $\alpha, \beta$  of the equation  $x^2 - 6x + k = 0$  satisfy the relation  $3\alpha + 2\beta = 20$ .

**Solution :** Since  $\alpha, \beta$  are the roots of  $x^2 - 6x + k = 0$ , therefore,

$$\alpha + \beta = 6 \quad \dots(1)$$

$$\text{and} \quad \alpha\beta = k. \quad \dots(2)$$

$$\text{Now,} \quad 3\alpha + 2\beta = 20 \Rightarrow \alpha + 2(\alpha + \beta) = 20$$

$$\Rightarrow \alpha + 2 \times 6 = 20 \quad [\text{Using (1)}]$$

$$\Rightarrow \alpha = 20 - 12 = 8.$$

Putting  $\alpha = 8$  in (1), we get  $\beta = 6 - 8 = -2$ .

Hence, from (2) we have  $k = \alpha\beta = 8 \times (-2) = -16$ .

**Illustration 8 :** Find the quadratic equation whose roots are 8 and  $-6$ .

**Solution :** We have, sum of the roots  $= 8 + (-6) = 8 - 6 = 2$

and product of the roots  $= 8 \times (-6) = -48$ .

Hence, the required equation is

$$x^2 - (\text{sum of the roots})x + \text{product of the roots} = 0$$

i.e.,  $x^2 - 2x - 48 = 0$ .

**Illustration 9 :** Form a quadratic equation with rational coefficients whose one root is  $2 + \sqrt{5}$ .

**Solution :** Let the roots of the required equation be  $\alpha$  and  $\beta$ .

Let  $\alpha = 2 + \sqrt{5}$ .

Then  $\beta = 2 - \sqrt{5}$  [ $\because$  In an equation with rational coefficients, irrational roots occur in conjugate pairs]

$\therefore \alpha + \beta = (2 + \sqrt{5}) + (2 - \sqrt{5}) = 4$

and  $\alpha\beta = (2 + \sqrt{5})(2 - \sqrt{5}) = 2^2 - (\sqrt{5})^2 = 4 - 5 = -1$ .

Hence, the required equation is  $x^2 - 4x - 1 = 0$ .

**Illustration 10 :** Form a quadratic equation with real coefficients whose one root is  $2 - 3i$ .

**Solution :** Let the roots of the required equation be  $\alpha$  and  $\beta$ .

Let  $\alpha = 2 - 3i$ .

Then  $\beta = 2 + 3i$  [ $\because$  In an equation with real coefficients, complex roots occur in conjugate pairs]

$\therefore \alpha + \beta = (2 - 3i) + (2 + 3i) = 4$

and  $\alpha\beta = (2 - 3i)(2 + 3i) = 2^2 + 3^2 = 13$ .

Hence, the required equation is  $x^2 - 4x + 13 = 0$ .

## Comprehensive Exercise 3

1. Find the nature of the roots of the following equations without actually solving them :

(i)  $2x^2 - 9x + 7 = 0$

(ii)  $4x^2 - 20x + 25 = 0$

(iii)  $2x^2 - x + 2 = 0$

(iv)  $x^2 + 2\sqrt{3}x - 1 = 0$

(v)  $3x^2 + 7x + 2 = 0$

(vi)  $3x^2 - 2x + 5 = 0$ .

2. For what values of  $m$  will the following equations have equal roots ?

(i)  $x^2 - 2(5 + 2m)x + 3(7 + 10m) = 0$

(ii)  $(m + 1)x^2 + 2(m + 3)x + (m + 8) = 0$

- (iii)  $(2m + 1)x^2 + 2(m + 3)x + (m + 5) = 0$   
 (iv)  $mx^2 + mx + 1 = -4x^2 - x$ .
3. Without solving the equation  $7x^2 - 8x + 49 = 0$ , find the sum and the product of its roots.
4. Find a quadratic equation whose roots are :  
 (i) 3, 8 (ii) 9, -11  
 (iii)  $5 + \sqrt{7}$ ,  $5 - \sqrt{7}$  (iv)  $-3 + i\sqrt{5}$ ,  $-3 - i\sqrt{5}$ .
5. Find a quadratic equation with rational coefficients whose one root is  $-4 + \sqrt{7}$ .
6. Find a quadratic equation with real coefficients whose one root is  $\frac{1}{2 - 3i}$ .
7. If  $\alpha, \beta$  are the roots of the equation  $x^2 - 2x + 3 = 0$ , find the equation whose roots are  $\alpha + 2$  and  $\beta + 2$ .

## Answers

1. (i) Rational and unequal (ii) Real and equal  
 (iii) Imaginary and unequal (iv) Real and unequal  
 (v) Rational and unequal (vi) Imaginary and unequal.
2. (i)  $2, \frac{1}{2}$  (ii)  $\frac{1}{3}$   
 (iii)  $\frac{-5 \pm \sqrt{41}}{2}$  (iv) 5, -3.
3. Sum =  $\frac{8}{7}$ , product = 7.
4. (i)  $x^2 - 11x + 24 = 0$  (ii)  $x^2 + 2x - 99 = 0$   
 (iii)  $x^2 - 10x + 18 = 0$  (iv)  $x^2 + 6x + 14 = 0$ .
5.  $x^2 + 8x + 9 = 0$ . 6.  $13x^2 - 4x + 1 = 0$ .
7.  $x^2 - 6x + 11 = 0$ .

## 114 Solution of Equations Reducible to Quadratic Forms

### Type I. Algebraic Equations Reducible to Quadratic Forms.

The general form of such an equation is  $px^{2n} + qx^n + r = 0$ , where  $p, q, r$  are constant coefficients.

Putting  $x^n = y$ , so that  $x^{2n} = (x^n)^2 = y^2$ , the given equation reduces to  $py^2 + qy + r = 0$ , which is a quadratic in  $y$ .

Now, we solve this equation for  $y$  and then we derive from it the values of  $x$ .

## Illustrative Examples

**Illustration 1 :** Solve :  $x^4 - 8x^2 - 9 = 0$ .

**Solution :** We have  $x^4 - 8x^2 - 9 = 0$

$$\Rightarrow y^2 - 8y - 9 = 0 \quad [\text{Putting } x^2 = y]$$

$$\Rightarrow y^2 - 9y + y - 9 = 0$$

$$\Rightarrow y(y - 9) + 1(y - 9) = 0$$

$$\Rightarrow (y - 9)(y + 1) = 0$$

$$\Rightarrow y - 9 = 0 \quad \text{or} \quad y + 1 = 0$$

$$\Rightarrow y = 9 \quad \text{or} \quad y = -1.$$

Now,  $y = 9 \Rightarrow x^2 = 9 \Rightarrow x = \pm 3$

and  $y = -1 \Rightarrow x^2 = -1$

$$\Rightarrow x = \pm \sqrt{-1} = \pm i.$$

Hence, the roots of the given equation are  $-3, 3, i$  and  $-i$ .

**Illustration 2 :** Solve :  $6x^{3/4} = 7x^{1/4} - 2x^{-1/4}$ .

**Solution :** We have  $6x^{3/4} = 7x^{1/4} - 2x^{-1/4}$

$$\Rightarrow 6x^{3/4} = 7x^{1/4} - \frac{2}{x^{1/4}}$$

$$\Rightarrow 6x = 7x^{1/2} - 2 \quad [\text{Multiplying both sides by } x^{1/4}]$$

$$\Rightarrow 6y^2 = 7y - 2 \quad [\text{Putting } x^{1/2} = y \text{ so that } x = (x^{1/2})^2 = y^2]$$

$$\Rightarrow 6y^2 - 7y + 2 = 0$$

$$\Rightarrow 6y^2 - 4y - 3y + 2 = 0$$

$$\Rightarrow 2y(3y - 2) - (3y - 2) = 0$$

$$\Rightarrow (3y - 2)(2y - 1) = 0$$

$$\Rightarrow y = \frac{2}{3} \quad \text{or} \quad y = \frac{1}{2}.$$

Now,  $y = \frac{2}{3} \Rightarrow x^{1/2} = \frac{2}{3} \Rightarrow x = \left(\frac{2}{3}\right)^2 = \frac{4}{9}$

and  $y = \frac{1}{2} \Rightarrow x^{1/2} = \frac{1}{2} \Rightarrow x = \left(\frac{1}{2}\right)^2 = \frac{1}{4}.$

Thus,  $x = \frac{1}{4}$  or  $x = \frac{4}{9}.$

Hence, the roots of the given equation are  $\frac{1}{4}$  and  $\frac{4}{9}.$

**Illustration 3 :** Solve :  $x^{2/3} + x^{1/3} - 2 = 0$ .

**Solution :** We have  $x^{2/3} + x^{1/3} - 2 = 0$

$$\Rightarrow y^2 + y - 2 = 0 \quad [\text{Putting } x^{1/3} = y \text{ so that } x^{2/3} = (x^{1/3})^2 = y^2]$$

$$\Rightarrow (y + 2)(y - 1) = 0 \Rightarrow y = -2 \text{ or } y = 1.$$

$$\text{Now, } y = -2 \Rightarrow x^{1/3} = -2 \Rightarrow x = (-2)^3 = -8$$

$$\text{and } y = 1 \Rightarrow x^{1/3} = 1 \Rightarrow x = 1^3 = 1.$$

$$\text{Thus, } x = -8 \text{ or } x = 1.$$

Hence, the solution set of the given equation is  $\{-8, 1\}$ .

**Illustration 4 :** Solve :  $(x^2 - 5x + 7)^2 - (x - 2)(x - 3) = 1$ .

**Solution :** We have  $(x^2 - 5x + 7)^2 - (x - 2)(x - 3) = 1$

$$\Rightarrow (x^2 - 5x + 7)^2 - (x^2 - 5x + 6) - 1 = 0$$

$$\Rightarrow (x^2 - 5x + 7)^2 - (x^2 - 5x + 7) = 0$$

$$\Rightarrow y^2 - y = 0 \quad [\text{Putting } x^2 - 5x + 7 = y]$$

$$\Rightarrow y(y - 1) = 0 \Rightarrow y = 0 \text{ or } y = 1.$$

$$\text{Now, } y = 0 \Rightarrow x^2 - 5x + 7 = 0$$

$$\Rightarrow x = \frac{5 \pm \sqrt{25 - 28}}{2} = \frac{5 \pm \sqrt{-3}}{2} = \frac{5 \pm i\sqrt{3}}{2}.$$

$$\text{Also, } y = 1 \Rightarrow x^2 - 5x + 7 = 1$$

$$\Rightarrow x^2 - 5x + 6 = 0$$

$$\Rightarrow (x - 3)(x - 2) = 0$$

$$\Rightarrow x = 3 \text{ or } x = 2.$$

Hence, the roots of the given equation are  $2, 3, \frac{5 + i\sqrt{3}}{2}$  and  $\frac{5 - i\sqrt{3}}{2}$ .

**Illustration 5 :** Solve :  $\left(x + \frac{1}{x}\right)^2 - \frac{3}{2}\left(x - \frac{1}{x}\right) = 4, x \neq 0$ .

[UPTU 2004]

**Solution :** We have  $\left(x + \frac{1}{x}\right)^2 - \frac{3}{2}\left(x - \frac{1}{x}\right) = 4$

$$\Rightarrow \left[\left(x - \frac{1}{x}\right)^2 + 4\right] - \frac{3}{2}\left(x - \frac{1}{x}\right) = 4$$

$$\Rightarrow y^2 + 4 - \frac{3}{2}y = 4 \quad \left[\text{Putting } x - \frac{1}{x} = y\right]$$

$$\Rightarrow y^2 - \frac{3}{2}y = 0 \Rightarrow y\left(y - \frac{3}{2}\right) = 0$$

$$\Rightarrow y = 0 \text{ or } y - \frac{3}{2} = 0$$

$$\Rightarrow y = 0 \quad \text{or} \quad y = \frac{3}{2}.$$

$$\text{Now, } y = 0 \Rightarrow x - \frac{1}{x} = 0$$

$$\Rightarrow x^2 - 1 = 0 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1.$$

$$\text{Again, } y = \frac{3}{2} \Rightarrow x - \frac{1}{x} = \frac{3}{2} \Rightarrow \frac{x^2 - 1}{x} = \frac{3}{2}$$

$$\Rightarrow 2x^2 - 2 = 3x \Rightarrow 2x^2 - 3x - 2 = 0$$

$$\Rightarrow 2x^2 - 4x + x - 2 = 0$$

$$\Rightarrow 2x(x - 2) + (x - 2) = 0$$

$$\Rightarrow (x - 2)(2x + 1) = 0$$

$$\Rightarrow x - 2 = 0 \quad \text{or} \quad 2x + 1 = 0$$

$$\Rightarrow x = 2 \quad \text{or} \quad x = -\frac{1}{2}.$$

Hence, the solution set of the given equation is  $\left\{1, -1, 2, -\frac{1}{2}\right\}$ .

**Illustration 6 :** Solve  $\left(\frac{2x+1}{x-1}\right)^4 - 10\left(\frac{2x+1}{x-1}\right)^2 + 9 = 0$ .

[UPTU 2005]

**Solution :** We have  $\left(\frac{2x+1}{x-1}\right)^4 - 10\left(\frac{2x+1}{x-1}\right)^2 + 9 = 0$

$$\Rightarrow y^2 - 10y + 9 = 0 \quad \left[ \text{Putting } \left(\frac{2x+1}{x-1}\right)^2 = y \right]$$

$$\Rightarrow y^2 - 9y - y + 9 = 0$$

$$\Rightarrow y(y - 9) - 1(y - 9) = 0$$

$$\Rightarrow (y - 1)(y - 9) = 0$$

$$\Rightarrow y = 1 \quad \text{or} \quad y = 9.$$

$$\text{Now, } y = 1 \Rightarrow \left(\frac{2x+1}{x-1}\right)^2 = 1 \Rightarrow \frac{2x+1}{x-1} = \pm 1$$

$$\Rightarrow \frac{2x+1}{x-1} = 1 \quad \text{or} \quad \frac{2x+1}{x-1} = -1$$

$$\Rightarrow 2x+1 = x-1 \quad \text{or} \quad 2x+1 = -x+1$$

$$\Rightarrow x = -2 \quad \text{or} \quad x = 0.$$

$$\text{Again } y = 9 \Rightarrow \left(\frac{2x+1}{x-1}\right)^2 = 9 \Rightarrow \frac{2x+1}{x-1} = \pm 3$$

$$\Rightarrow \frac{2x+1}{x-1} = 3 \quad \text{or} \quad \frac{2x+1}{x-1} = -3$$

$$\Rightarrow 2x + 1 = 3x - 3 \quad \text{or} \quad 2x + 1 = -3x + 3$$

$$\Rightarrow x = 4 \quad \text{or} \quad x = \frac{2}{5}.$$

Hence, the solution of the given equation is  $\left\{-2, 0, 4, \frac{2}{5}\right\}$ .

## Type II. Exponential Equations Reducible to Quadratic Forms.

Consider the equation  $ap^{2x} + bp^x + c = 0$ .

Putting  $p^x = y$  so that  $p^{2x} = (p^x)^2 = y^2$ , the given equation reduces to  $ay^2 + by + c = 0$ , which is a quadratic in  $y$ .

Now, we solve this equation for  $y$  and then we derive from it the values of  $x$ .

## Illustrative Examples

**Illustration 1 :** Solve :  $4^x - 3 \cdot 2^{x+3} = -128$ .

**Solution :** We have  $4^x - 3 \cdot 2^{x+3} = -128$

$$\Rightarrow (2^2)^x - 3 \cdot 2^3 \cdot 2^x = -128$$

$$\Rightarrow (2^x)^2 - 24 \cdot 2^x + 128 = 0$$

$$\Rightarrow y^2 - 24y + 128 = 0 \quad [\text{Putting } 2^x = y]$$

$$\Rightarrow (y - 16)(y - 8) = 0$$

$$\Rightarrow y = 16 \quad \text{or} \quad y = 8.$$

$$\text{Now, } y = 16 \Rightarrow 2^x = 16 \Rightarrow 2^x = 2^4 \Rightarrow x = 4$$

$$\text{and } y = 8 \Rightarrow 2^x = 8 \Rightarrow 2^x = 2^3 \Rightarrow x = 3.$$

Thus,  $x = 4$  or  $x = 3$ .

Hence, the roots of the given equation are 3 and 4.

**Illustration 2 :** Solve  $7^{x+1} + 7^{1-x} = 50$ .

**Solution :** We have  $7^{x+1} + 7^{1-x} = 50$

[UPTU 2007, 08]

$$\Rightarrow 7^x \cdot 7 + 7 \cdot 7^{-x} = 50$$

$$\Rightarrow 7^x \cdot 7 + \frac{7}{7^x} = 50$$

$$\Rightarrow 7y + \frac{7}{y} = 50 \quad [\text{Putting } 7^x = y]$$

$$\Rightarrow 7y^2 + 7 = 50y$$

$$\Rightarrow 7y^2 - 50y + 7 = 0$$

$$\Rightarrow 7y^2 - 49y - y + 7 = 0$$

$$\Rightarrow 7y(y - 7) - (y - 7) = 0$$

$$\Rightarrow (7y - 1)(y - 7) = 0$$



$$\Rightarrow y = \frac{1}{7} \quad \text{or} \quad y = 7.$$

$$\text{Now, } y = \frac{1}{7} \Rightarrow 7^x = \frac{1}{7}$$

$$\Rightarrow 7^x = 7^{-1} \Rightarrow x = -1$$

$$\text{and } y = 7 \Rightarrow 7^x = 7 \Rightarrow x = 1.$$

Thus,  $x = -1$  or  $x = 1$ .

Hence, the solution set of the given equation is  $\{-1, 1\}$ .

**Illustration 3 :** Solve  $[\sqrt{5+2\sqrt{6}}]^x + [\sqrt{5-2\sqrt{6}}]^x = 10$ .

[UPTU 2003]

**Solution :** We have  $(5+2\sqrt{6})(5-2\sqrt{6}) = 5^2 - (2\sqrt{6})^2 = 1$

$$\Rightarrow 5-2\sqrt{6} = \frac{1}{5+2\sqrt{6}} = (5+2\sqrt{6})^{-1}$$

$$\therefore [\sqrt{5+2\sqrt{6}}]^x + [\sqrt{5-2\sqrt{6}}]^x = 10$$

$$\Rightarrow (5+2\sqrt{6})^{x/2} + \frac{1}{(5+2\sqrt{6})^{x/2}} = 10$$

$$\Rightarrow y + \frac{1}{y} = 10 \quad [\text{Putting } (5+2\sqrt{6})^{x/2} = y]$$

$$\Rightarrow y^2 - 10y + 1 = 0 \Rightarrow y = 5 \pm 2\sqrt{6}.$$

$$\text{Now, } y = 5+2\sqrt{6} \Rightarrow (5+2\sqrt{6})^{x/2} = (5+2\sqrt{6})^1$$

$$\Rightarrow x/2 = 1 \Rightarrow x = 2.$$

$$\text{and } y = 5-2\sqrt{6} \Rightarrow (5+2\sqrt{6})^{x/2} = (5+2\sqrt{6})^{-1}$$

$$\Rightarrow x/2 = -1 \Rightarrow x = -2.$$

Hence, the solution set of the given equation is  $\{2, -2\}$ .

**Type III. Equations of the Form  $(x+a)(x+b)(x+c)(x+d)+k=0$ .**

Such type of equations are easily solvable if the sum of any two of the numbers  $a, b, c, d$  is equal to the sum of the other two. We take such pairs together and solve as illustrated below :

## Illustrative Examples

**Illustration 1 :** Solve :  $(x+9)(x-3)(x-7)(x+5) = 385$ .

**Solution :** We have  $(x+9)(x-3)(x-7)(x+5) = 385$

$$\Rightarrow [(x+9)(x-7)][(x+5)(x-3)] = 385$$

[Note that  $-7+9 = -3+5$ ]

$$\Rightarrow (x^2+2x-63)(x^2+2x-15) = 385$$

$$\Rightarrow (y-63)(y-15) = 385 \quad [\text{Putting } x^2+2x = y]$$

$$\Rightarrow y^2 - 78y + 945 = 385$$

$$\Rightarrow y^2 - 78y + 560 = 0$$

$$\Rightarrow y^2 - 70y - 8y + 560 = 0$$

$$\Rightarrow y(y - 70) - 8(y - 70) = 0$$

$$\Rightarrow (y - 70)(y - 8) = 0$$

$$\Rightarrow y = 70 \quad \text{or} \quad y = 8.$$

$$\text{Now, } y = 70 \Rightarrow x^2 + 2x = 70 \Rightarrow x^2 + 2x - 70 = 0$$

$$\begin{aligned} \Rightarrow x &= \frac{-2 \pm \sqrt{4 + 280}}{2} = \frac{-2 \pm \sqrt{284}}{2} \\ &= \frac{-2 \pm 2\sqrt{71}}{2} = -1 \pm \sqrt{71}. \end{aligned}$$

$$\text{Again, } y = 8 \Rightarrow x^2 + 2x = 8 \Rightarrow x^2 + 2x - 8 = 0$$

$$\Rightarrow (x + 4)(x - 2) = 0$$

$$\Rightarrow x = -4 \quad \text{or} \quad x = 2.$$

Hence, the roots of the given equation are  $2, -4, -1 \pm \sqrt{71}$ .

**Illustration 2 :** Solve :  $(2x + 7)(x^2 - 9)(2x - 5) = 91$ .

**Solution :** We have  $(2x + 7)(x^2 - 9)(2x - 5) = 91$

$$\Rightarrow (2x + 7)(x - 3)(x + 3)(2x - 5) = 91$$

$$\Rightarrow [(2x + 7)(x - 3)][(x + 3)(2x - 5)] = 91$$

$$\Rightarrow (2x^2 + x - 21)(2x^2 + x - 15) = 91$$

$$\Rightarrow (y - 21)(y - 15) = 91 \quad [\text{Putting } 2x^2 + x = y]$$

$$\Rightarrow y^2 - 36y + 315 = 91 \Rightarrow y^2 - 36y + 224 = 0$$

$$\Rightarrow y^2 - 28y - 8y + 224 = 0$$

$$\Rightarrow y(y - 28) - 8(y - 28) = 0$$

$$\Rightarrow (y - 28)(y - 8) = 0$$

$$\Rightarrow y = 28 \quad \text{or} \quad y = 8.$$

$$\text{Now, } y = 28 \Rightarrow 2x^2 + x = 28 \Rightarrow 2x^2 + x - 28 = 0$$

$$\Rightarrow 2x^2 + 8x - 7x - 28 = 0$$

$$\Rightarrow 2x(x + 4) - 7(x + 4) = 0$$

$$\Rightarrow (x + 4)(2x - 7) = 0$$

$$\Rightarrow x = -4 \quad \text{or} \quad x = \frac{7}{2}.$$

$$\text{Again, } y = 8 \Rightarrow 2x^2 + x = 8 \Rightarrow 2x^2 + x - 8 = 0$$

$$\Rightarrow x = \frac{-1 \pm \sqrt{1 + 64}}{4} = \frac{-1 \pm \sqrt{65}}{4}.$$

Hence, the roots of the given equation are  $-4$ ,  $\frac{7}{2}$ ,  $\frac{-1 + \sqrt{65}}{4}$  and  $\frac{-1 - \sqrt{65}}{4}$ .

**Illustration 3 :** Evaluate  $\sqrt{6 + \sqrt{6 + \sqrt{6 + \dots \infty}}}$ .

[UPTU 2011]

**Solution :** Let  $x = \sqrt{6 + \sqrt{6 + \sqrt{6 + \dots \infty}}}$ .

Then  $x = \sqrt{6 + x}$

$\Rightarrow x^2 = 6 + x$  [Squaring both sides]

$\Rightarrow x^2 - x - 6 = 0 \Rightarrow (x - 3)(x + 2) = 0$

$\Rightarrow x = 3$  or  $x = -2$ .

But, the given expression is +ve. So, we reject the value  $x = -2$ .

Hence, the value of the given expression is 3.

**Illustration 4 :** Find the value of:  $2 + \frac{1}{2 + \frac{1}{2 + \dots \infty}}$

**Solution :** Let the given expression be equal to  $x$ . Then,

$$x = 2 + \frac{1}{x} \Rightarrow x^2 = 2x + 1 \Rightarrow x^2 - 2x - 1 = 0$$

$$\Rightarrow x = \frac{-(-2) \pm \sqrt{(-2)^2 - 4 \times 1 \times (-1)}}{2} = \frac{2 \pm \sqrt{8}}{2} = \frac{2 \pm 2\sqrt{2}}{2}$$

$$= 1 \pm \sqrt{2}.$$

But, the given expression is +ve. So, we reject the value  $x = 1 - \sqrt{2}$ .

Hence, the value of the given expression is  $1 + \sqrt{2}$ .

**Type IV. Equations of the Form  $aX + \frac{b}{X} = c$ , where  $X$  is a function of  $x$ .**

Putting  $X = y$ , the given equation becomes  $ay + \frac{b}{y} = c$  or  $ay^2 - cy + b = 0$ , which

is a quadratic in  $y$ .

Now, we solve this equation for  $y$  and then we derive from it the values of  $x$ .

## Illustrative Examples

**Illustration 1:** Solve:  $\sqrt{\frac{x}{1-x}} + \sqrt{\frac{1-x}{x}} = \frac{13}{6}$ .

**Solution :** We have  $\sqrt{\frac{x}{1-x}} + \sqrt{\frac{1-x}{x}} = \frac{13}{6}$

$$\Rightarrow y + \frac{1}{y} = \frac{13}{6} \quad \left[ \text{Putting } \sqrt{\frac{x}{1-x}} = y \right]$$

$$\Rightarrow \frac{y^2 + 1}{y} = \frac{13}{6} \Rightarrow 6y^2 + 6 = 13y$$

$$\Rightarrow 6y^2 - 13y + 6 = 0 \Rightarrow 6y^2 - 9y - 4y + 6 = 0$$

$$\Rightarrow 3y(2y - 3) - 2(2y - 3) = 0 \Rightarrow (2y - 3)(3y - 2) = 0$$

$$\Rightarrow y = \frac{3}{2} \quad \text{or} \quad y = \frac{2}{3}.$$

$$\text{Now, } y = \frac{3}{2} \Rightarrow \sqrt{\frac{x}{1-x}} = \frac{3}{2} \Rightarrow \frac{x}{1-x} = \frac{9}{4}$$

$$\Rightarrow 4x = 9 - 9x \Rightarrow 13x = 9 \Rightarrow x = \frac{9}{13}.$$

$$\text{Again, } y = \frac{2}{3} \Rightarrow \sqrt{\frac{x}{1-x}} = \frac{2}{3} \Rightarrow \frac{x}{1-x} = \frac{4}{9}$$

$$\Rightarrow 9x = 4 - 4x \Rightarrow 13x = 4 \Rightarrow x = \frac{4}{13}.$$

Hence, the solution set of the given equation is  $\left\{ \frac{9}{13}, \frac{4}{13} \right\}$ .

**Illustration 2 :** Solve :  $\left( \frac{4x+1}{x+1} \right) + \left( \frac{x+1}{4x+1} \right) = \frac{5}{2}$ .

[UPTU 2009]

**Solution :** We have  $\left( \frac{4x+1}{x+1} \right) + \left( \frac{x+1}{4x+1} \right) = \frac{5}{2}$

$$\Rightarrow y + \frac{1}{y} = \frac{5}{2} \quad \left[ \text{Putting } \frac{4x+1}{x+1} = y \text{ so that } \frac{x+1}{4x+1} = \frac{1}{y} \right]$$

$$\Rightarrow \frac{y^2 + 1}{y} = \frac{5}{2} \Rightarrow 2y^2 + 2 = 5y \Rightarrow 2y^2 - 5y + 2 = 0$$

$$\Rightarrow 2y^2 - 4y - y + 2 = 0 \Rightarrow 2y(y - 2) - (y - 2) = 0$$

$$\Rightarrow (y - 2)(2y - 1) = 0 \Rightarrow y = 2 \quad \text{or} \quad y = \frac{1}{2}.$$

$$\text{Now, } y = 2 \Rightarrow \frac{4x+1}{x+1} = 2 \Rightarrow 2x + 2 = 4x + 1 \Rightarrow x = \frac{1}{2}.$$

$$\text{Again, } y = \frac{1}{2} \Rightarrow \frac{4x+1}{x+1} = \frac{1}{2} \Rightarrow 8x + 2 = x + 1$$

$$\Rightarrow 7x = -1 \Rightarrow x = -\frac{1}{7}.$$

Hence, the roots of the given equation are 1 and  $-\frac{1}{7}$ .

**Type V. Equations of the form**

$$(i) a \left( x^2 + \frac{1}{x^2} \right) + b \left( x + \frac{1}{x} \right) + c = 0 \quad (ii) a \left( x^2 + \frac{1}{x^2} \right) + b \left( x - \frac{1}{x} \right) + c = 0.$$

We have  $\left( x + \frac{1}{x} \right)^2 = x^2 + \frac{1}{x^2} + 2 \Rightarrow x^2 + \frac{1}{x^2} = \left( x + \frac{1}{x} \right)^2 - 2$

and  $\left( x - \frac{1}{x} \right)^2 = x^2 + \frac{1}{x^2} - 2 \Rightarrow x^2 + \frac{1}{x^2} = \left( x - \frac{1}{x} \right)^2 + 2.$

Thus, the given equations (i) and (ii) can be easily solved by putting  $x + \frac{1}{x} = y$  in case (i) and  $x - \frac{1}{x} = y$  in case (ii).

## Illustrative Examples

**Illustration 1 :** Solve by reducing to the quadratic equation

$$3 \left( x^2 + \frac{1}{x^2} \right) - 16 \left( x + \frac{1}{x} \right) + 26 = 0.$$

**Solution :** The given equation is  $3 \left( x^2 + \frac{1}{x^2} \right) - 16 \left( x + \frac{1}{x} \right) + 26 = 0.$

Put  $x + \frac{1}{x} = y$  i.e.,  $\left( x + \frac{1}{x} \right)^2 = y^2$  i.e.,  $x^2 + \frac{1}{x^2} = y^2 - 2.$

$$\therefore 3(y^2 - 2) - 16y + 26 = 0$$

$$\Rightarrow 3y^2 - 16y + 20 = 0$$

$$\Rightarrow (y - 2)(3y - 10) = 0$$

$$\Rightarrow y = 2 \text{ or } \frac{10}{3}.$$

Now,  $y = 2 \Rightarrow x + \frac{1}{x} = 2$

$$\Rightarrow x^2 - 2x + 1 = 0 \Rightarrow (x - 1)^2 = 0 \Rightarrow x = 1$$

and  $y = \frac{10}{3} \Rightarrow x + \frac{1}{x} = \frac{10}{3}$

$$\Rightarrow \frac{x^2 + 1}{x} = \frac{10}{3} \Rightarrow 3x^2 - 10x + 3 = 0$$

$$\Rightarrow (x - 3)(3x - 1) = 0 \Rightarrow x = 3 \text{ or } \frac{1}{3}.$$

Hence, the required solutions of the given equation are  $x = 3, \frac{1}{3}, 1.$

**Illustration 2 :** Solve by reducing to the quadratic equation

$$4 \left( x^2 + \frac{1}{x^2} \right) + 8 \left( x + \frac{1}{x} \right) + 3 = 0, x \neq 0, x \in R.$$

**Solution :** The given equation is  $4\left(x^2 + \frac{1}{x^2}\right) + 8\left(x + \frac{1}{x}\right) + 3 = 0$ .

Put  $x + \frac{1}{x} = y$  i.e.,  $\left(x + \frac{1}{x}\right)^2 = y^2$  i.e.,  $x^2 + \frac{1}{x^2} = y^2 - 2$ .

$\therefore 4(y^2 - 2) + 8y + 3 = 0 \Rightarrow 4y^2 + 8y + 3 - 8 = 0$

$\Rightarrow 4y^2 + 8y - 5 = 0$

$\Rightarrow (2y - 1)(2y + 5) = 0 \Rightarrow y = \frac{1}{2} \text{ or } -\frac{5}{2}$ .

Now,  $y = \frac{1}{2} \Rightarrow x + \frac{1}{x} = \frac{1}{2}$

$\Rightarrow \frac{x^2 + 1}{x} = \frac{1}{2} \Rightarrow 2x^2 - x + 2 = 0$ .

Here, discriminant  $= (-1)^2 - 4(2)(2) = 1 - 16 = -15$ .

Since the discriminant is negative, therefore real roots are not possible in this case.

Further  $y = -\frac{5}{2} \Rightarrow x + \frac{1}{x} = -\frac{5}{2}$

$\Rightarrow \frac{x^2 + 1}{x} = -\frac{5}{2} \Rightarrow 2x^2 + 5x + 2 = 0$

$\Rightarrow (x + 2)(2x + 1) = 0 \Rightarrow x = -2 \text{ or } -\frac{1}{2}$ .

Hence, the required solutions are  $x = -2, -\frac{1}{2}$ .

**Illustration 3 :** Solve by reducing to the quadratic equation

$$6\left(x^2 + \frac{1}{x^2}\right) - 25\left(x - \frac{1}{x}\right) + 12 = 0.$$

**Solution :** The given equation is  $6\left(x^2 + \frac{1}{x^2}\right) - 25\left(x - \frac{1}{x}\right) + 12 = 0$ .

Put  $x - \frac{1}{x} = y$  i.e.,  $\left(x - \frac{1}{x}\right)^2 = y^2$  i.e.,  $x^2 + \frac{1}{x^2} = y^2 + 2$ .

$\therefore 6(y^2 + 2) - 25y + 12 = 0$

$\Rightarrow 6y^2 - 25y + 24 = 0$

$\Rightarrow (2y - 3)(3y - 8) = 0$

$\Rightarrow y = \frac{3}{2} \text{ or } \frac{8}{3}$ .

Now,  $y = \frac{3}{2} \Rightarrow x - \frac{1}{x} = \frac{3}{2}$

$\Rightarrow \frac{x^2 - 1}{x} = \frac{3}{2} \Rightarrow 2x^2 - 3x - 2 = 0$

$\Rightarrow (x - 2)(2x + 1) = 0 \Rightarrow x = 2 \text{ or } -1/2$

$$\begin{aligned} \text{and} \quad y = \frac{8}{3} &\Rightarrow x - \frac{1}{x} = \frac{8}{3} \\ \Rightarrow \frac{x^2 - 1}{x} = \frac{8}{3} &\Rightarrow 3x^2 - 8x - 3 = 0 \\ \Rightarrow (x - 3)(3x + 1) = 0 &\Rightarrow x = 3 \text{ or } -1/3. \end{aligned}$$

Hence, the solutions of the given equation are  $x = 2, -\frac{1}{2}, 3$  and  $-\frac{1}{3}$ .

**Illustration 4 :** Solve by reducing to the quadratic equation

$$2\left(x^2 + \frac{1}{x^2}\right) - 3\left(x - \frac{1}{x}\right) - 4 = 0.$$

**Solution :** The given equation is  $2\left(x^2 + \frac{1}{x^2}\right) - 3\left(x - \frac{1}{x}\right) - 4 = 0$ .

$$\text{Put } x - \frac{1}{x} = y \text{ i.e., } \left(x - \frac{1}{x}\right)^2 = y^2 \text{ i.e., } x^2 + \frac{1}{x^2} = y^2 + 2.$$

$$\begin{aligned} \therefore 2(y^2 + 2) - 3y - 4 &= 0 \\ \Rightarrow 2y^2 - 3y &= 0 \Rightarrow y(2y - 3) = 0 \\ \Rightarrow y = 0 \text{ or } y &= \frac{3}{2}. \end{aligned}$$

$$\begin{aligned} \text{Now, } y = 0 &\Rightarrow x - \frac{1}{x} = 0 \\ \Rightarrow x^2 - 1 &= 0 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1 \end{aligned}$$

$$\begin{aligned} \text{and } y = \frac{3}{2} &\Rightarrow x - \frac{1}{x} = \frac{3}{2} \\ \Rightarrow \frac{x^2 - 1}{x} = \frac{3}{2} &\Rightarrow 2x^2 - 3x - 2 = 0 \\ \Rightarrow (x - 2)(2x + 1) &= 0 \Rightarrow x = 2 \text{ or } -\frac{1}{2}. \end{aligned}$$

Hence, the solutions of the given equation are  $x = 1, -1, 2$  or  $-\frac{1}{2}$ .

### Type VI. Irrational Equations Reducible to Quadratics:

Methods of solving such equations are illustrated by means of the following :

## Illustrative Examples

**Solution by Introducing a New Variable :**

**Illustration 1 :** Solve :  $\frac{3x-2}{2} + \sqrt{2x^2 - 5x + 3} = \frac{(x+1)^2}{3}$ .

**Solution :** We have  $\frac{3x-2}{2} + \sqrt{2x^2 - 5x + 3} = \frac{(x+1)^2}{3}$

$$\Rightarrow 3(3x - 2) + 6\sqrt{2x^2 - 5x + 3} = 2(x + 1)^2 = 2(x^2 + 2x + 1)$$

$$\Rightarrow 2x^2 - 5x + 8 - 6\sqrt{2x^2 - 5x + 3} = 0$$

$$\Rightarrow (2x^2 - 5x + 3) + 5 - 6\sqrt{2x^2 - 5x + 3} = 0$$

$$\Rightarrow y^2 + 5 - 6y = 0$$

$$[\text{Putting } \sqrt{2x^2 - 5x + 3} = y \text{ so that } 2x^2 - 5x + 3 = y^2]$$

$$\Rightarrow y^2 - 6y + 5 = 0$$

$$\Rightarrow (y - 5)(y - 1) = 0 \Rightarrow y = 5 \quad \text{or} \quad y = 1.$$

$$\text{Now, } y = 5 \Rightarrow \sqrt{2x^2 - 5x + 3} = 5$$

$$\Rightarrow 2x^2 - 5x + 3 = 25$$

$$\Rightarrow 2x^2 - 5x - 22 = 0$$

$$\Rightarrow x = \frac{5 \pm \sqrt{25 + 176}}{4} = \frac{5 \pm \sqrt{201}}{4}.$$

$$\text{Again, } y = 1 \Rightarrow \sqrt{2x^2 - 5x + 3} = 1 \Rightarrow 2x^2 - 5x + 3 = 1$$

$$\Rightarrow 2x^2 - 5x + 2 = 0 \Rightarrow (2x - 1)(x - 2) = 0$$

$$\Rightarrow x = \frac{1}{2} \quad \text{or} \quad x = 2.$$

Hence, the roots of the given equation are  $\frac{1}{2}, 2$  and  $\frac{5 \pm \sqrt{201}}{4}$ .

**Solution by Applying the Method of Identity :**

**Illustration 1 :** Solve :  $\sqrt{x^2 - 3x + 27} + \sqrt{x^2 - 3x + 11} = 8$ . [UPTU 2008]

**Solution :** The given equation is

$$\sqrt{x^2 - 3x + 27} + \sqrt{x^2 - 3x + 11} = 8.$$

$$\text{Let } L = \sqrt{x^2 - 3x + 27} \text{ and } M = \sqrt{x^2 - 3x + 11}$$

$$\text{Then } L + M = 8 \quad \dots(1)$$

$$\text{Also } L^2 - M^2 = (x^2 - 3x + 27) - (x^2 - 3x + 11) = 16$$

$$\text{or } (L + M)(L - M) = 16$$

$$\text{or } 8(L - M) = 16 \quad \text{or} \quad L - M = 2 \quad \dots(2)$$

Now from (1) and (2), we get

$$2L = 10 \quad \text{or} \quad L = 5 \quad \text{or} \quad L^2 = 25$$

$$\text{or } x^2 - 3x + 27 = 25 \quad \text{or} \quad x^2 - 3x + 2 = 0$$

$$\text{or } (x - 1)(x - 2) = 0 \quad \text{or} \quad x = 1, 2.$$

Hence, the roots of the given equation are 1 and 2.



**Illustration 2 :** Solve :  $\sqrt{3x^2 - 7x - 30} + \sqrt{2x^2 - 7x - 5} = x + 5$ .

**Solution :** The given equation is

$$\sqrt{3x^2 - 7x - 30} + \sqrt{2x^2 - 7x - 5} = x + 5.$$

Let  $L = \sqrt{3x^2 - 7x - 30}$  ,  $M = \sqrt{2x^2 - 7x - 5}$

Then  $L + M = x + 5$  ... (1)

Also  $L^2 - M^2 = (3x^2 - 7x - 30) - (2x^2 - 7x - 5) = x^2 - 25$

or  $(L + M)(L - M) = (x + 5)(x - 5)$

or  $(x + 5)(L - M) = (x + 5)(x - 5)$

or  $L - M = x - 5$  ... (2)

Now from (1) and (2), we get

$$2L = 2x \quad \text{or} \quad L = x \quad \text{or} \quad L^2 = x^2$$

or  $3x^2 - 7x - 30 = x^2$

or  $2x^2 - 7x - 30 = 0$

or  $(x - 6)(2x + 5) = 0$

or  $x = 6 \quad \text{or} \quad x = -\frac{5}{2}.$

Hence, the roots of the given equation are 6 and  $-\frac{5}{2}$ .

**Solution by the Method of Rationalisation :**

Suppose we have to solve equations of the form

$$\sqrt{ax + b} = k \quad \text{or} \quad \sqrt{ax + b} + \sqrt{cx + d} = k$$

or  $\sqrt{ax + b} + \sqrt{cx + d} = \sqrt{ex + f}.$

In such equations, we square both sides and then transpose so that the expression under radical sign alone is on one side. To remove the radical sign, we again square both sides. Thus, we get a quadratic equation in  $x$  and we solve it. However, in the process of squaring, sometimes **extraneous roots** are obtained. So, before writing the solution set, we check whether the roots obtained actually satisfy the given equation or not.

**Illustration 3 :** Solve :  $x + 2 = \sqrt{2x + 7}.$

**Solution :** We have  $x + 2 = \sqrt{2x + 7}$

$$\Rightarrow (x + 2)^2 = 2x + 7 \quad [\text{Squaring both sides}]$$

$$\Rightarrow x^2 + 2x - 3 = 0 \Rightarrow (x + 3)(x - 1) = 0$$

$$\Rightarrow x = -3 \quad \text{or} \quad x = 1.$$

Rejecting the extraneous root  $x = -3$  because it does not satisfy the given equation, we get  $x = 1$ .

Hence, the only root of the given equation is  $x = 1$ .

**Illustration 4 :** Solve :  $\sqrt{1-5x} + \sqrt{1-3x} = 2$ .

**Solution :** We have  $\sqrt{1-5x} + \sqrt{1-3x} = 2$

$$\Rightarrow \sqrt{1-5x} = 2 - \sqrt{1-3x}$$

$$\Rightarrow 1-5x = 4 - 4\sqrt{1-3x} + (1-3x) \quad [\text{Squaring both sides}]$$

$$\Rightarrow 4\sqrt{1-3x} = 4 + 2x \Rightarrow 2\sqrt{1-3x} = 2 + x$$

$$\Rightarrow 4(1-3x) = (2+x)^2 = 4 + 4x + x^2 \quad [\text{Squaring both sides}]$$

$$\Rightarrow x^2 + 16x = 0 \Rightarrow x(x+16) = 0$$

$$\Rightarrow x = 0 \quad \text{or} \quad x = -16.$$

We see that  $x = 0$  satisfies the given equation. But,  $x = -16$  does not satisfy the given equation and so we reject this extraneous root.

Hence, the only root of the given equation is  $x = 0$ .

**Illustration 5 :** Solve :  $\sqrt{x+5} + \sqrt{x+21} = \sqrt{6x+40}$ .

**Solution :** We have  $\sqrt{x+5} + \sqrt{x+21} = \sqrt{6x+40}$

$$\Rightarrow (x+5) + (x+21) + 2\sqrt{x+5}\sqrt{x+21} = 6x+40$$

[Squaring both sides]

$$\Rightarrow 2\sqrt{x+5}\sqrt{x+21} = 4x+14$$

$$\Rightarrow \sqrt{x+5}\sqrt{x+21} = 2x+7$$

$$\Rightarrow (x+5)(x+21) = (2x+7)^2 \quad [\text{Squaring both sides}]$$

$$\Rightarrow 3x^2 + 2x - 56 = 0 \Rightarrow (3x+14)(x-4) = 0$$

$$\Rightarrow x = -\frac{14}{3} \quad \text{or} \quad x = 4.$$

**Check :** For  $x = -\frac{14}{3}$ , we have

$$\sqrt{-\frac{14}{3}+5} + \sqrt{-\frac{14}{3}+21} = \sqrt{6 \times \left(-\frac{14}{3}\right) + 40}$$

$$\Rightarrow \sqrt{\frac{1}{3}} + \sqrt{\frac{49}{3}} = \sqrt{\frac{36}{3}} \Rightarrow \frac{1}{\sqrt{3}} + \frac{7}{\sqrt{3}} = \frac{6}{\sqrt{3}}$$

$$\Rightarrow 1 + 7 = 6, \text{ which is false.}$$

$\therefore x = -\frac{14}{3}$  is an extraneous root because it does not satisfy the given equation

and so we reject it.

For  $x = 4$ , we have

$$\sqrt{4+5} + \sqrt{4+21} = \sqrt{(6 \times 4) + 40}$$

$$\Rightarrow \sqrt{9} + \sqrt{25} = \sqrt{64}$$

$$\Rightarrow 3 + 5 = 8, \text{ which is true.}$$

$\therefore x = 4$  satisfies the given equation.

Hence, the only root of the given equation is  $x = 4$ .

## 1.15 Simultaneous Equations in Two Variables

Method of solving such equations are illustrated by means of the following :

### Illustrative Examples

**Illustration 1 :** Solve :

$$(i) \sqrt{\frac{x}{y}} + \sqrt{\frac{y}{x}} = \frac{5}{2}, x + y = 10$$

$$(ii) x^2 + y^2 = 185, x + y = 19$$

$$(iii) x^3 + y^3 = 91, x + y = 7$$

**Solution :** (i) We have

$$\sqrt{\frac{x}{y}} + \sqrt{\frac{y}{x}} = \frac{5}{2} \quad \dots(1)$$

$$\text{and } x + y = 10 \quad \dots(2)$$

From (1), we have

$$\frac{\sqrt{x}}{\sqrt{y}} + \frac{\sqrt{y}}{\sqrt{x}} = \frac{5}{2}$$

$$\text{or } \frac{x + y}{\sqrt{xy}} = \frac{5}{2}$$

$$\text{or } \frac{10}{\sqrt{xy}} = \frac{5}{2} \quad \text{[using (2)]}$$

$$\text{or } \sqrt{xy} = 4 \text{ or } xy = 16 \text{ or } x(10 - x) = 16 \quad \text{[using (2)]}$$

$$\text{or } 10x - x^2 - 16 = 0 \text{ or } x^2 - 10x + 16 = 0$$

$$\text{or } x = 2, 8$$

Now if  $x = 2$ , from (2),  $y = 10 - x = 10 - 2 = 8$

or if  $x = 8$ , from (2),  $y = 10 - x = 10 - 8 = 2$

Hence solution is  $x = 2, y = 8$ ;  $x = 8, y = 2$ .

$$(ii) \text{ We have } x^2 + y^2 = 185 \quad \dots(1)$$

$$\text{and } x + y = 19 \quad \dots(2)$$

From (2), we get

$$y = 19 - x$$

∴ from (1), we get

$$x^2 + (19 - x)^2 = 185$$

$$\text{or } x^2 + 361 + x^2 - 38x = 185$$

$$\text{or } 2x^2 - 38x + 176 = 0$$

$$\text{or } x - 19x + 88 = 0$$

$$\text{or } (x - 8)(x - 11) = 0$$

$$\text{or } x = 8, 11$$

$$\text{Now if } x = 8 \text{ then } y = 19 - x = 19 - 8 = 11$$

$$\text{or if } x = 11 \text{ then } y = 19 - x = 19 - 11 = 8$$

Hence the solution is  $x = 8, y = 11; x = 11, y = 8$

$$(iii) \text{ We have } x^3 + y^3 = 91 \quad \dots(1)$$

$$\text{and } x + y = 7 \quad \dots(2)$$

From (2), we get  $x = 7 - y$

$$\therefore \text{ from (1), we get } (7 - y)^3 + y^3 = 91$$

$$\{343 - y^3 - 21y(7 - y)\} + y^3 = 91$$

$$\text{or } 343 - 147y + 21y^2 = 91$$

$$\text{or } 21y^2 - 147y + 252 = 0$$

$$\text{or } y^2 - 7y + 12 = 0 \text{ or } y = 3, 4$$

$$\text{Now if } y = 3 \text{ then } x = 7 - y = 7 - 3 = 4$$

$$\text{if } y = 4 \text{ then } x = 7 - y = 7 - 4 = 3$$

Hence the solution is

$$x = 4, y = 3; x = 3, y = 4$$

## Comprehensive Exercise 4

Solve the following equations :

$$1. \quad (i) x^{2/3} + 2x^{1/3} - 8 = 0.$$

$$(ii) 3x^{1/2} + 3x^{-1/2} = 10.$$

$$(iii) 8x^{3/2} - 8x^{-3/2} = 63.$$

$$(iv) x^{-4} - 144 = 25x^{-2}.$$

$$(v) x + \sqrt{x} = \frac{6}{25}$$

$$2. \quad (i) 3^{2x} + 9 = 10 \cdot 3^x.$$

$$(ii) 5^{x+1} + 5^{2-x} = 5^3 + 1.$$

$$(iii) 4^x - 5 \cdot 2^x + 4 = 0.$$

$$(iv) 2^{2x+8} + 1 = 32 \cdot 2^x.$$

$$(v) 2^{2x+3} - 57 = 65 (2^x - 1).$$

$$(vi) 3^{x+2} + 3^{-x} = 3^2 + 1.$$

$$(vii) 2^{x^2} + 4^x = 8 + 1$$

$$(viii) 5^{1+x} + 5^{1-x} = 26$$

$$3. (i) \left( \frac{2x+3}{x+1} \right) + 6 \left( \frac{x+1}{2x+3} \right) - 7 = 0.$$

$$(ii) \left( \frac{x+2}{x-2} \right) - \left( \frac{x-2}{x+2} \right) = \frac{5}{6}.$$

$$(iii) 10 \sqrt{\frac{x}{x+3}} - \sqrt{\frac{x+3}{x}} = 3.$$

$$(iv) \sqrt{\frac{4x-1}{4x+1}} + \sqrt{\frac{4x+1}{4x-1}} = \frac{8}{3}.$$

$$4. (i) (x+1)(x+2)(x+3)(x+4) = 120.$$

[UPTU 2006]

$$(ii) x(x+1)^2(x+2) = 72.$$

$$(iii) 16x(x+1)(x+2)(x+3) = 9.$$

$$(iv) (x^2 + x - 6)(x^2 - 3x - 4) = 24.$$

$$(v) (x+1)(x+2)(x+3)(x+4) + 1 = 0.$$

$$5. (i) \sqrt{12 + \sqrt{12 + \sqrt{12 + \dots \infty}}}.$$

$$(ii) 4 + \frac{1}{4 + \frac{1}{4 + \frac{1}{4 + \dots \infty}}}.$$

$$(iii) \sqrt{20 + \sqrt{20 + \sqrt{20 + \dots \infty}}}$$

$$(iv) \sqrt{8 - \sqrt{8 - \sqrt{8 - \dots \infty}}}$$

$$6. (i) (5 + 2\sqrt{6})^{x^2-3} + (5 - 2\sqrt{6})^{x^2-3} = 10.$$

$$(ii) 2 \left( x^2 + \frac{1}{x^2} \right) - 3 \left( x + \frac{1}{x} \right) - 1 = 0.$$

$$(iii) 8 \left( x^2 + \frac{1}{x^2} \right) - 42 \left( x - \frac{1}{x} \right) + 29 = 0.$$

$$(iv) 3x^2 - 4x + \sqrt{3x^2 - 4x - 6} = 18.$$

$$7. (i) \sqrt{2x^2 + 5x - 2} - \sqrt{2x^2 + 5x - 9} = 1.$$

$$(ii) \sqrt{x} = (x - 2)$$

$$(iii) \sqrt{5x^2 + 7x + 2} + \sqrt{4x^2 + 7x + 18} = x + 4.$$

$$(iv) \sqrt{x+1} + \sqrt{2x} = 7.$$

$$(v) \sqrt{3x+1} - \sqrt{x-1} = 2.$$

$$(vi) \sqrt{2x+1} + \sqrt{3x+2} = \sqrt{5x+3}.$$

8. Solve the following simultaneous equations :

$$(i) \sqrt{\frac{x}{y}} + \sqrt{\frac{y}{x}} = \frac{10}{3}, x + y = 10$$

$$(ii) x + y = a + b, \frac{a}{x} + \frac{b}{y} = 2$$

$$(iii) \frac{1}{x} + \frac{1}{y} = \frac{5}{6}, 3x + 4y = 18$$

$$(iv) x + y = 20, xy = 64$$

$$(v) x + 2y = 1, x^2 + y^2 = 10$$

$$(vi) \frac{1}{x} + \frac{1}{y} = \frac{3}{4}, \frac{1}{x^2} + \frac{1}{y^2} = \frac{5}{16}$$

## Answers

$$1. (i) 8, -64. \quad (ii) 9, \frac{1}{9}. \quad (iii) 4, \frac{1}{4}. \quad (iv) \pm \frac{1}{3}, \pm \frac{1}{4}.$$

$$(v) \frac{1}{5}.$$

$$2. (i) 0, 2. \quad (ii) -1, 2. \quad (iii) 0, 2. \quad (iv) -4. \\ (v) 3, -3. \quad (vi) 0, -2. \quad (vii) -1, 3 \quad (viii) -1, 1$$

$$3. (i) -2, -\frac{3}{4}. \quad (ii) 10, -\frac{2}{5}. \quad (iii) 1. \quad (iv) -\frac{5}{16}.$$

$$4. (i) -6, 1, \frac{-5 \pm i\sqrt{39}}{2}. \quad (ii) 2, -4, -1 \pm i\sqrt{8}.$$

$$(iii) -\frac{3}{2}, \frac{-3 \pm \sqrt{10}}{2}. \quad (iv) 0, 1, \frac{1 \pm \sqrt{57}}{2}.$$

$$(v) \frac{-5 \pm \sqrt{5}}{2}.$$

$$5. (i) 4. \quad (ii) 2 + \sqrt{5}.$$

$$(iii) 5 \quad (iv) \frac{\sqrt{33} - 1}{2}$$

$$6. (i) \pm 2, \pm \sqrt{2}. \quad (ii) \frac{1}{2}, 2, \frac{-1 \pm i\sqrt{3}}{2}.$$

$$(iii) 2, -\frac{1}{2}, 4, -\frac{1}{4}. \quad (iv) 3, -\frac{5}{3}, \frac{1}{3}(2 \pm \sqrt{70}).$$

$$7. (i) 2, -\frac{9}{2}. \quad (ii) 4. \quad (iii) \frac{-7 \pm \sqrt{17}}{8}.$$

$$(iv) 8. \quad (v) 1, 5. \quad (vi) -\frac{1}{2}, -\frac{2}{3}.$$

$$8. (i) x = 1, y = 9; x = 9, y = 1$$

$$(ii) x = a, y = b; x = \frac{a+b}{2}, y = \frac{a+b}{2}$$

$$(iii) x = 2, y = 3; x = \frac{18}{5}, y = \frac{9}{5}$$

$$(iv) x = 16, y = 4; x = 4, y = 16$$

$$(v) \ x = 3, y = -1; x = -\frac{13}{5}, y = \frac{9}{5}$$

$$(vi) \ x = 2, y = 4; x = 4, y = 2.$$

## 1.16 Applied Problems on Quadratic Equations

In this section we shall discuss applications of quadratic equations in solving applied problems.

### Illustrative Examples

**Illustration 1 :** Divide 33 into two parts whose product is 242 . [UPTU 2001]

**Solution :** Let the two parts be  $x$  and  $33 - x$ . Then, according to the question

$$x(33 - x) = 242 \Rightarrow x^2 - 33x + 242 = 0$$

$$\Rightarrow x^2 - 22x - 11x + 242 = 0$$

$$\Rightarrow x(x - 22) - 11(x - 22) = 0$$

$$\Rightarrow (x - 22)(x - 11) = 0$$

$$\Rightarrow x = 22 \quad \text{or} \quad x = 11.$$

Hence, the required parts are 22 and 11.

**Illustration 2 :** Find two consecutive numbers whose squares have the sum 85.

**Solution :** Let the two consecutive numbers be  $x$  and  $x + 1$ .

$$\text{Then, } x^2 + (x + 1)^2 = 85$$

$$\Rightarrow x^2 + x^2 + 2x + 1 = 85$$

$$\Rightarrow 2x^2 + 2x - 84 = 0$$

$$\Rightarrow x^2 + x - 42 = 0$$

$$\Rightarrow (x + 7)(x - 6) = 0$$

$$\Rightarrow x = -7 \quad \text{or} \quad 6.$$

When  $x = -7$ , we have  $x + 1 = -7 + 1 = -6$

and when  $x = 6$ , we have  $x + 1 = 6 + 1 = 7$ .

Hence, the two required numbers are 6 and 7 or  $-7$  and  $-6$ .

**Illustration 3 :** The product of two successive positive multiples of 3 is 810. Find the numbers.

**Solution :** Let the required multiples of 3 be  $x$  and  $x + 3$ .

$$\text{Then, } x(x + 3) = 810$$

$$\Rightarrow x^2 + 3x - 810 = 0$$

$$\Rightarrow (x + 30)(x - 27) = 0$$

$$\Rightarrow x = 27 \quad \text{or} \quad -30.$$

Since only +ive multiple is required therefore  $-30$  is not acceptable.

$$\therefore x = 27 \quad \text{and} \quad x + 3 = 30.$$

Hence, the required numbers are 27 and 30.

**Illustration 4 :** A number exceeds its positive square root by 12. Find the number.

**Solution :** Let the number be  $x$ . Then, according to the question

$$x - \sqrt{x} = 12 \quad \Rightarrow \quad x - 12 = \sqrt{x} \quad \Rightarrow \quad (x - 12)^2 = x$$

[Squaring both sides]

$$\Rightarrow x^2 - 25x + 144 = 0 \quad \Rightarrow \quad (x - 16)(x - 9) = 0$$

$$\Rightarrow x = 16 \quad \text{or} \quad x = 9.$$

Since 9 cannot exceed its positive square root by 12, therefore the required number is 16.

**Illustration 5 :** A two digit number is such that the product of its digits is 8. When 63 is subtracted from the number the digits interchange their places. Find the number.

**Solution :** Let the number be  $10x + y$ .

$$\text{Then,} \quad xy = 8. \quad \dots(1)$$

$$\text{and} \quad 10x + y - 63 = 10y + x$$

$$\Rightarrow 9x - 9y = 63 \quad \Rightarrow \quad x - y = 7. \quad \dots(2)$$

Putting  $y = x - 7$  from (2) in (1), we get

$$x(x - 7) = 8 \quad \Rightarrow \quad x^2 - 7x - 8 = 0.$$

$$\Rightarrow (x - 8)(x + 1) = 0$$

$$\Rightarrow x = 8, \text{ neglecting the negative value.}$$

Putting  $x = 8$  in (1), we get  $y = 1$ .

Hence, the required number is  $10 \times 8 + 1 = 81$ .

**Illustration 6:** The sum of the ages of a man and his son is 35 years and the product of their ages is 150. Find their ages.

**Solution :** Let the man's age be  $x$  years. Then the son's age is  $35 - x$  years.

$$\text{Given that} \quad x(35 - x) = 150 \quad \Rightarrow \quad 35x - x^2 = 150$$

$$\Rightarrow x^2 - 35x + 150 = 0$$

$$\Rightarrow (x - 30)(x - 5) = 0$$

$$\Rightarrow x = 30 \text{ or } 5.$$

Since the father's age cannot be 5 years, therefore we have

father's age = 30 years and son's age = 5 years.

**Illustration 7 :** B is a point on the line segment AC such that it lies between A and C. If  $AC = 9$  cm and  $AB \times AC = 12(BC)^2$ , find BC.

**Solution :** Let  $BC = x$  cm, then  $AB = 9 - x$  cm.

$$\text{Given that} \quad AB \times AC = 12(BC)^2$$



$$\begin{aligned}
 \Rightarrow & 9(9 - x) = 12(x)^2 \\
 \Rightarrow & 81 - 9x = 12x^2 \quad \Rightarrow \quad 27 - 3x = 4x^2 \\
 \Rightarrow & 4x^2 + 3x - 27 = 0 \quad \Rightarrow \quad (x + 3)(4x - 9) = 0 \\
 \Rightarrow & x = \frac{9}{4} \quad \text{or} \quad -3.
 \end{aligned}$$

Since length cannot be -ive, therefore  $x = \frac{9}{4} = 2.25$  cm. Hence  $BC = 2.25$  cm.

**Illustration 8 :** The area of a right angled triangle is  $40 \text{ cm}^2$ . If its base is 2 cm less than its altitude, find its base and altitude.

**Solution :** Let the base of the triangle =  $x$  cm .

Then, its altitude =  $(x + 2)$  cm.

$$\therefore \text{Area of the triangle} = \frac{1}{2} x (x + 2) \text{ cm}^2.$$

According to the question,  $\frac{1}{2} x (x + 2) = 40$

$$\Rightarrow x(x + 2) = 80 \Rightarrow x^2 + 2x - 80 = 0$$

$$\Rightarrow (x + 10)(x - 8) = 0$$

$$\Rightarrow x = 8. \quad [\text{Neglecting } x = -10]$$

Hence, the base of the triangle = 8 cm. and its altitude = 10 cm.

**Illustration 9 :** A passenger train running at uniform speed, takes 2 hours less for a journey of 300 km, if its speed is increased by 5 km per hour from its usual speed. Find the usual speed.

**Solution :** Let the usual speed of the train be  $x$  km/hr.

Then, time taken to describe a distance of 300 km at  $x$  km/hr

$$= \left( \frac{300}{x} \right) \text{ hours.}$$

Also, time taken to describe a distance of 300 km at  $(x + 5)$  km/hr

$$= \left( \frac{300}{x + 5} \right) \text{ hours.}$$

According to question,  $\frac{300}{x + 5} = \left( \frac{300}{x} \right) - 2$

$$\Rightarrow 300 \left[ \frac{1}{x} - \frac{1}{x + 5} \right] = 2$$

$$\Rightarrow 150 \left[ \frac{(x + 5) - x}{x(x + 5)} \right] = 1$$

$$\Rightarrow x(x + 5) = 750$$

$$\Rightarrow x^2 + 5x - 750 = 0$$

$$\begin{aligned} \Rightarrow & x^2 + 30x - 25x - 750 = 0 \\ \Rightarrow & x(x + 30) - 25(x + 30) = 0 \\ \Rightarrow & (x + 30)(x - 25) = 0 \\ \Rightarrow & x = 25 \text{ or } x = -30. \end{aligned}$$

Rejecting  $x = -30$  because  $x$  cannot be -ve, we have  $x = 25$ .

Hence, the usual speed of the train = 25 km/hr.

**Illustration 10 :** A piece of cloth costs Rs. 35. If the piece were 4 metres longer and each metre costs Re. 1 less, the total cost would remain unchanged. How long is the piece?

**Solution :** Let the length of the cloth piece be  $x$  metres.

$$\text{Then, rate per metre} = \text{Rs.} \left( \frac{35}{x} \right).$$

If the piece were 4 metres longer, then the new length =  $(x + 4)$  metres.

Since the total cost remains unchanged, therefore, new rate per metre

$$= \text{Rs.} \left( \frac{35}{x + 4} \right).$$

$$\text{According to the question, } \frac{35}{x} - \frac{35}{x + 4} = 1$$

[ $\because$  Difference in rates = Re 1 per metre]

$$\Rightarrow 35 \left[ \frac{1}{x} - \frac{1}{x + 4} \right] = 1$$

$$\Rightarrow 35 \left[ \frac{x + 4 - x}{x(x + 4)} \right] = 1$$

$$\Rightarrow x(x + 4) = 140$$

$$\Rightarrow x^2 + 4x - 140 = 0$$

$$\Rightarrow (x + 14)(x - 10) = 0 \quad \Rightarrow x = 10.$$

Neglecting  $x = -14$  because the length of the cloth cannot be -ve]

Hence, the required length of the piece of cloth = 10 metres.

**Illustration 11 :** Two trains leave a railway station at the same time. The first train travels due west and the second train due north. The first train travels 5 km/hr faster than the second train. If after 2 hrs they are 50 km apart, find the average speed of each.

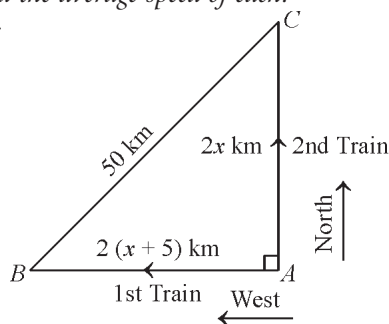
**Solution :** Let the speed of the second train be  $x$  km/hr, then the speed of the first train is  $(x + 5)$  km/hr.

Let the two trains start from the station  $A$ , then distance travelled by the first train in 2 hrs

$$= 2(x + 5) \text{ km} = AB, \text{ say}$$

and distance travelled by the second train in 2 hrs

$$= 2x \text{ km} = AC, \text{ say.}$$



Given that  $BC = 50$  km.

By Pythagoras Theorem, we have  $AB^2 + AC^2 = BC^2$

$$\Rightarrow [2(x+5)]^2 + [2x]^2 = (50)^2$$

$$\Rightarrow 4(x+5)^2 + 4x^2 = 2500$$

$$\Rightarrow x^2 + 10x + 25 + x^2 = 625$$

$$\Rightarrow 2x^2 + 10x - 600 = 0$$

$$\Rightarrow x^2 + 5x - 300 = 0$$

$$\Rightarrow (x+20)(x-15) = 0$$

$$\Rightarrow x = 15 \text{ or } -20.$$

Since speed cannot be negative, therefore  $x = 15$  and  $x + 5 = 20$ .

Hence, the speed of the first train is 20 km/hr and the speed of the second train is 15 km/hr.

**Illustration 12 :** The current of a stream runs at the rate of 2 km/hr. A motor boat goes 10 km upstream and back again to the starting point in 55 minutes. Find the speed of the motor boat in still water.

**Solution :** Let the speed of the motor boat in still water be  $x$  km/hr.

Then, its speed down stream =  $(x + 2)$  km/hr

and its speed upstream =  $(x - 2)$  km / hr.

$$\therefore \text{Time taken to cover 10 km upstream} = \left( \frac{10}{x-2} \right) \text{ hrs.}$$

$$\text{and the time taken to cover 10 km down stream} = \left( \frac{10}{x+2} \right) \text{ hrs.}$$

$$\text{According to the question, } \left( \frac{10}{x-2} \right) + \left( \frac{10}{x+2} \right) = \frac{55}{60}$$

$$\Rightarrow 10 \left[ \frac{1}{x-2} + \frac{1}{x+2} \right] = \frac{55}{60}$$

$$\Rightarrow 10 \left[ \frac{x+2+x-2}{(x-2)(x+2)} \right] = \frac{55}{60} = \frac{11}{12}$$

$$\Rightarrow 11(x^2 - 4) = 240x$$

$$\Rightarrow 11x^2 - 240x - 44 = 0$$

$$\Rightarrow 11x^2 - 242x + 2x - 44 = 0$$

$$\Rightarrow 11x(x-22) + 2(x-22) = 0$$

$$\Rightarrow (x-22)(11x+2) = 0$$

$$\Rightarrow x = 22. \quad \left[ \text{Neglecting } x = -\frac{2}{11} \text{ because the speed cannot be -ve} \right]$$

Hence, the speed of the motor boat in still water = 22 km/hr.

### Comprehensive Exercise 5

1. Divide 55 into two parts whose product is 516.
2. The sum of two numbers is 8 and the sum of their squares is 34. Find the numbers.
3. The sum of the squares of two consecutive even natural numbers is 340. Find the numbers.
4. The sum of a number and its reciprocal is  $2\frac{1}{20}$ . Find the number.
5. The sum of a number and its positive square root is  $\frac{6}{25}$ . Find the number.
6. The product of two consecutive even integers is 528. Find the integers.
7. The sum of the squares of three consecutive positive integers is 77. Find the numbers.
8. Two numbers are such that their sum is 54 and the product is 629. Find the numbers.
9. The sum of the squares of two consecutive odd positive integers is 290. Find the integers.
10. The sum of the squares of two numbers is 233 and one of the numbers is 3 less than twice the other number. Find the numbers.
11. The sum of the squares of two numbers is 130. The sum of the smaller number and twice the larger number is 25. Find the numbers.
12. A two digit number is four times the sum and three times the product of its digits. Find the number.
13. The area of a right angled triangle is  $54\text{ cm}^2$ . If its base is three times its altitude, find its base.
14. The perimeter of a rectangular plot is 42 m and its area is  $108\text{ m}^2$ . Find the dimensions of the plot.
15. A man sold a table for Rs. 75. If the percentage of his profit in this transaction is equal to the cost price of the table, find the cost price of the table.
16. A person covers a distance of 300 km by car in a certain time with a certain fixed uniform speed. If he increases his speed by 5 km/hr, he takes 2 hours less to cover the same distance. Find his original speed.

## Answers

1. 43 and 12.
2. 3 and 5.
3. 12 and 14.
4.  $\frac{5}{4}$  or  $\frac{4}{5}$ .
5.  $\frac{1}{25}$ .
6.  $(-24 \text{ and } -22)$  or  $(22 \text{ and } 24)$ .
7. 4, 5 and 6.
8. 37 and 17.
9. 11 and 13.
10.  $\left(-\frac{71}{5} \text{ and } -\frac{28}{5}\right)$  or  $(13 \text{ and } 8)$ .
11.  $(3 \text{ and } 11)$  or  $(7 \text{ and } 9)$ .
12. 24.
13. 18 cm.
14. Length 12 m and breadth 9 m.
15. Rs. 50.
16. 25 km/hr.

## Comprehensive Exercise 6

### Fill in the Blanks

Fill in the blanks “.....”, so that the following statements are complete and correct.

1. The roots of the equation  $x^2 - 8 = 0$  are .....
2. If the discriminant  $D = b^2 - 4ac = 0$  then the two roots are ..... and .....
3. The sum and product of the roots of the quadratic equation  $3x^2 - 7x - 5 = 0$  are ..... and ..... respectively.
4. The quadratic polynomial, whose zeros are 3 and  $-2$  is .....
5. A quadratic equation whose roots have the sum 5 and the product 6 is .....
6. If the product of the two roots of the equation  $4x^2 + 3x + k = 0$  is 3, the value of  $k$  is .....
7. If one root of the equation  $6x^2 - 12x + k = 0$  be the reciprocal of the other, then the value of  $k$  is .....

**Multiple Choice Questions**

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

8. The solution of the equation  $x^2 - 10x + 9 = 0$  is  
(a) 9 or 1 (b) -9 or -1  
(c) -9 or 1 (d) 9 or -1
9. The roots of the quadratic equation  $15x^2 - 11x + 3 = 0$  are  
(a) real and equal (b) real and unequal  
(c) imaginary (d) none of these
10. The value of  $k$  for which  $kx^2 - 8x + 4 = 0$  has two equal roots, is  
(a) -8 (b) 25 / 16  
(c) 4 (d) 16 / 25
11. If the equation  $x^2 + 4x + k = 0$  has real roots, the value of  $k$  is  
(a)  $k \leq 4$  (b)  $k \geq 4$   
(c)  $k \leq 0$  (d)  $k \geq 0$ .
12. The value of  $k$  for which  $x^2 - 8x + k = 0$  has coincident roots is  
(a) 16 (b) 4  
(c) -4 (d) 60
13. If sum and product of the roots of a quadratic equation is 9, the equation is  
(a)  $x^2 + 9x - 18 = 0$  (b)  $x^2 - 18x + 9 = 0$   
(c)  $x^2 + 9x - 9 = 0$  (d)  $x^2 - 9x + 9 = 0$
14. The sum of a positive number and its reciprocal is  $10/3$ . Then the number is  
(a) 3 or  $1/3$  (b) 4 or  $1/4$   
(c) -3 or  $-1/3$  (d) none of these

**True or False**

Write 'T' for true and 'F' for false statement.

15. The roots of equation  $x - \frac{2}{x} = 1$  are 1 and 3. [UPTU 2009]
16. If the discriminant  $D^2 = b^2 - 4ac$  is perfect square then the roots are rational and equal.
17. The nature of the roots of the equation  $3x^2 + 13x + 5 = 0$  are imaginary.
18. A quadratic equation whose one root is  $(5 + \sqrt{3})$  is  $x^2 - 10x + 22 = 0$ .
19. The sum of the roots of the equation  $2x^2 - 12x + 5 = 0$  is 7.
20. If one of the roots of the equation  $4x^2 + ax + 20 = 0$  is 4, then the value of  $a$  is -21.

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## Answers

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- |                    |                       |                |
|--------------------|-----------------------|----------------|
| 1. $\pm 2\sqrt{2}$ | 2. real, equal        | 3. $7/3, -5/3$ |
| 4. $(x-3)(x+2)$    | 5. $x^2 - 5x + 6 = 0$ | 6. 12          |
| 7. 6               | 8. (a)                | 9. (c)         |
| 10. (c)            | 11. (a)               | 12. (a)        |
| 13. (d)            | 14. (a)               | 15. $F$        |
| 16. $T$            | 17. $F$               | 18. $T$        |
| 19. $F$            | 20. $T$               |                |



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## Chapter

# 2



## Determinants

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### 2.1 Determinants of Order 2

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**D**efinition. Let  $a_{11}$ ,  $a_{12}$ ,  $a_{21}$ ,  $a_{22}$  be any four numbers (real or complex). The symbol

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$$

represents the number  $a_{11} a_{22} - a_{12} a_{21}$  and is called a **determinant of order 2**.

The numbers  $a_{11}$ ,  $a_{12}$ ,  $a_{21}$ ,  $a_{22}$  are called the **elements** of the determinant and the number  $a_{11} a_{22} - a_{12} a_{21}$  is called the **value** of the determinant.

Thus the value of a determinant of order two is equal to the product of the elements along the principal diagonal minus the product of the off-diagonal elements.

A determinant of order two has two rows and two columns.

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**Note.** In a determinant the number of rows is equal to the number of columns.

## 2.2 Determinants of Order 3

The symbol  $\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$  is called a **determinant of order 3** and its value is the number

$$a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}.$$

This is called the **expansion of the determinant** along its first row. In this expansion each element in the first row is multiplied by that determinant of the second order which is obtained by deleting the row and the column containing that element. Starting from the first the signs of the products are alternately positive and negative.

A determinant of order 3 has three rows and three columns. It has  $3 \times 3$  i.e., 9 elements.

We can find the value of a determinant of order 3 by expanding it along any of its rows or along any of its columns.

For example, we have

$$\begin{aligned} \Delta &= a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\ &= a_{11} (a_{22} a_{33} - a_{23} a_{32}) - a_{12} (a_{21} a_{33} - a_{31} a_{23}) \\ &\quad + a_{13} (a_{21} a_{32} - a_{31} a_{22}) \\ &= -a_{12} (a_{21} a_{33} - a_{31} a_{23}) + a_{22} (a_{11} a_{33} - a_{31} a_{13}) \\ &\quad - a_{32} (a_{11} a_{23} - a_{21} a_{13}) \\ &= -a_{12} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{33} \end{vmatrix} + a_{22} \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} - a_{32} \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix} \\ &= (-1)^{1+2} a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + (-1)^{2+2} a_{22} \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} \\ &\quad + (-1)^{3+2} a_{32} \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix} \end{aligned}$$

and this is the expansion of  $\Delta$  along its second column.

In the expansion of  $\Delta$  if  $a_{ij}$  is the element occurring in the  $i^{\text{th}}$  row and the  $j^{\text{th}}$  column, then to fix the positive or negative sign before it we should multiply it by  $(-1)^{i+j}$ .

## 2.3 Determinants of Order 4

The symbol  $\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{vmatrix}$  is called a **determinant of order 4** and

its value is the number

$$\begin{aligned} a_{11} \begin{vmatrix} a_{22} & a_{23} & a_{24} \\ a_{32} & a_{33} & a_{34} \\ a_{42} & a_{43} & a_{44} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} & a_{24} \\ a_{31} & a_{33} & a_{34} \\ a_{41} & a_{43} & a_{44} \end{vmatrix} \\ + a_{13} \begin{vmatrix} a_{21} & a_{22} & a_{24} \\ a_{31} & a_{32} & a_{34} \\ a_{41} & a_{42} & a_{44} \end{vmatrix} - a_{14} \begin{vmatrix} a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \\ a_{41} & a_{42} & a_{43} \end{vmatrix}. \end{aligned}$$

This is the expansion of  $\Delta$  along its first row.

The determinant of order 4 has 4 rows and 4 columns. We can expand it along any of its rows or columns as is the case of a determinant of order 3.

Proceeding in the same fashion we can evaluate the determinant of order greater than 4.

## 2.4 Minors and Cofactors

**Minors.** Consider the determinant

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}.$$

If we leave the row and the column passing through the element  $a_{ij}$ , then the second order determinant thus obtained is called the minor of the element  $a_{ij}$ . We shall denote the minor of the element  $a_{ij}$  by  $M_{ij}$ .

For example,

$$\text{the minor of the element } a_{21} = \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} = M_{21},$$

$$\text{the minor of the element } a_{32} = \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix} = M_{32}, \text{ and so on.}$$

In terms of the notation of minors, if we expand  $\Delta$  along the first row then

$$\begin{aligned} \Delta &= (-1)^{1+1} a_{11} M_{11} + (-1)^{1+2} a_{12} M_{12} + (-1)^{1+3} a_{13} M_{13} \\ &= a_{11} M_{11} - a_{12} M_{12} + a_{13} M_{13}. \end{aligned}$$

**Cofactors.** The minor  $M_{ij}$  multiplied by  $(-1)^{i+j}$  is called the cofactor of the

element  $a_{ij}$ . We shall denote the cofactor of an element by the corresponding capital letter. Then cofactor of  $a_{ij} = A_{ij} = (-1)^{i+j} M_{ij}$ .

For example,

the cofactor of the element  $a_{21} = A_{21} = (-1)^{2+1} M_{21} = - \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}$ ,

the cofactor of the element  $a_{32} = A_{32} = (-1)^{3+2} M_{32} = - \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix}$ .

Thus the cofactor of any element  $a_{ij} = (-1)^{i+j} \times$  the determinant obtained by leaving the row and the column passing through that element.

In terms of notation of cofactors, we have

$$\Delta = a_{11} A_{11} + a_{12} A_{12} + a_{13} A_{13}$$

or  $\Delta = a_{31} A_{31} + a_{32} A_{32} + a_{33} A_{33}$  and so on.

Therefore, in a determinant the sum of the products of the elements of any row or column with the corresponding cofactors is equal to the value of the determinant.

## Illustrative Examples

**Illustration 1 :** Evaluate the determinant  $\Delta = \begin{vmatrix} 2 & -1 & 5 \\ 1 & 2 & 1 \\ 3 & 1 & 3 \end{vmatrix}$  by expanding it along

the third row.

**Solution :** We have

$$\begin{aligned} \Delta &= (-1)^{3+1} 3 \cdot \begin{vmatrix} -1 & 5 \\ 2 & 1 \end{vmatrix} + (-1)^{3+2} 1 \cdot \begin{vmatrix} 2 & 5 \\ 1 & 1 \end{vmatrix} \\ &\quad + (-1)^{3+3} 3 \cdot \begin{vmatrix} 2 & -1 \\ 1 & 2 \end{vmatrix} \end{aligned}$$

( By definition )

$$= 3(-1 - 10) - (2 - 5) + 3(4 + 1) = -33 + 3 + 15 = -15.$$

**Illustration 2 :** Find the order and value of the following determinants:

(i)  $\begin{vmatrix} 4 & -3 \\ 2 & -1 \end{vmatrix}$

[UPTU 2008]

(ii)  $\begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & -3 \\ 2 & -2 & 1 \end{vmatrix}$ .

(iii)  $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$ .

**Solution :** (i) The determinant  $\begin{vmatrix} 4 & -3 \\ 2 & -1 \end{vmatrix}$  has 2 rows and 2 columns, so the order of this determinant is 2.

Let 
$$\Delta = \begin{vmatrix} 4 & -3 \\ 2 & -1 \end{vmatrix} = (4)(-1) - (-3)(2) = -4 + 6 = 2.$$

(ii) The determinant  $\begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & -3 \\ 2 & -2 & 1 \end{vmatrix}$  has 3 rows and 3 columns, so the order of this determinant is 3.

Let 
$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & -3 \\ 2 & -2 & 1 \end{vmatrix} = 1 \begin{vmatrix} 1 & -3 \\ -2 & 1 \end{vmatrix} - 1 \begin{vmatrix} 2 & -3 \\ 2 & 1 \end{vmatrix} + 1 \begin{vmatrix} 2 & 1 \\ 2 & -1 \end{vmatrix},$$

expanding the determinant along the first row

$$= 1[(1) \cdot (1) - (3) \cdot (-2)] - 1[(2) \cdot (1) - (-3) \cdot (2)] + 1[(2) \cdot (-1) - (1) \cdot (2)]$$

$$= (1 - 6) - (2 + 6) - (-2 - 2) = -5 - 8 + 4 = -9.$$

(iii) Let 
$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$$

$$= a \begin{vmatrix} b & f \\ f & c \end{vmatrix} - h \begin{vmatrix} h & f \\ g & c \end{vmatrix} + g \begin{vmatrix} h & b \\ g & f \end{vmatrix},$$

on expanding the determinant along the first row

$$= a(bc - f^2) - h(hc - gf) + g(hf - bg)$$

$$= abc + 2fgh - af^2 - bg^2 - ch^2.$$

**Illustration 3 :** Find the minors of the elements of the determinant

$$\Delta = \begin{vmatrix} 3 & 2 \\ -1 & 0 \end{vmatrix}.$$

**Solution :**  $M_{11}$  = minor of the element  $a_{11}$  ( $= 3$ ) is 0.

$M_{12}$  = minor of the element  $a_{12}$  ( $= 2$ ) is  $-1$ .

$M_{21}$  = minor of the element  $a_{21}$  ( $= -1$ ) is 2.

$M_{22}$  = minor of the element  $a_{22}$  ( $= 0$ ) is 3.

**Illustration 4 :** Write the cofactors of the elements of the determinant

$$\Delta = \begin{vmatrix} 3 & -1 & 1 \\ 1 & 4 & 7 \\ 4 & 2 & 3 \end{vmatrix}.$$

**Solution :**  $A_{11}$  = cofactor of the element  $a_{11}$  ( $= 3$ )

$$= (-1)^{1+1} \begin{vmatrix} 4 & 7 \\ 2 & 3 \end{vmatrix} = 12 - 14 = -2.$$

$A_{12}$  = cofactor of the element  $a_{12}$  ( $= -1$ )

$$= (-1)^{1+2} \begin{vmatrix} 1 & 7 \\ 4 & 3 \end{vmatrix} = -(3 - 28) = 25.$$

$A_{13}$  = cofactor of the element  $a_{13}$  ( $= 1$ )  $= (-1)^{1+3} \begin{vmatrix} 1 & 4 \\ 4 & 2 \end{vmatrix}$

$$= 2 - 16 = -14.$$

$A_{21}$  = cofactor of the element  $a_{21}$  ( $= 1$ )  $= (-1)^{2+1} \begin{vmatrix} -1 & 1 \\ 2 & 3 \end{vmatrix}$

$$= -(-3 - 2) = 5.$$

$A_{22}$  = cofactor of the element  $a_{22}$  ( $= 4$ )  $= (-1)^{2+2} \begin{vmatrix} 3 & 1 \\ 4 & 3 \end{vmatrix}$

$$= 9 - 4 = 5.$$

$A_{23}$  = cofactor of the element  $a_{23}$  ( $= 7$ )  $= (-1)^{2+3} \begin{vmatrix} 3 & -1 \\ 4 & 2 \end{vmatrix}$

$$= -(6 + 4) = -10.$$

$A_{31}$  = cofactor of the element  $a_{31}$  ( $= 4$ )  $= (-1)^{3+1} \begin{vmatrix} -1 & 1 \\ 4 & 7 \end{vmatrix}$

$$= -7 - 4 = -11.$$

$A_{32}$  = cofactor of the element  $a_{32}$  ( $= 2$ )  $= (-1)^{3+2} \begin{vmatrix} 3 & 1 \\ 1 & 7 \end{vmatrix}$

$$= -(21 - 1) = -20.$$

$A_{33}$  = cofactor of the element  $a_{33}$  ( $= 3$ )  $= (-1)^{3+3} \begin{vmatrix} 3 & -1 \\ 1 & 4 \end{vmatrix}$

$$= -(12 + 1) = -13.$$

## 2.5 Properties of Determinants

**Property 1.** *The value of a determinant does not change when rows and columns are interchanged,*

$$\text{i.e., } \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$

**Property 2.** *If any two rows (or two columns) of a determinant are interchanged the value of the determinant is multiplied by  $-1$ ,*

$$\text{i.e., } \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = - \begin{vmatrix} a_3 & b_3 & c_3 \\ a_2 & b_2 & c_2 \\ a_1 & b_1 & c_1 \end{vmatrix}.$$

**Corollary.** If any row (or column) of a determinant  $\Delta$  be passed over  $n$  rows (or columns), the resulting determinant will be  $(-1)^n \Delta$ .

**Property 3.** If all the elements of any row (or any column) of a determinant are multiplied by the same number  $k$ , the value of the new determinant is  $k$  times the value of the given determinant,

$$\text{i.e., } \begin{vmatrix} a_1 & b_1 & c_1 \\ ka_2 & kb_2 & kc_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = k \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}.$$

**Corollary 1.** If  $A$  be an  $n$ -rowed square matrix and  $k$  be any scalar then  
 $|k A| = k^n |A|.$

**Corollary 2.** If all the elements of any row (or any column) of a determinant are zero, the value of the determinant is zero.

$$\text{i.e., } \begin{vmatrix} 0 & 0 & 0 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0 \quad \text{or} \quad \begin{vmatrix} 0 & a_2 & a_3 \\ 0 & b_2 & b_3 \\ 0 & c_2 & c_3 \end{vmatrix} = 0.$$

**Property 4.** If two rows (or two columns) of a determinant are identical, the value of the determinant is zero,

$$\text{i.e., } \begin{vmatrix} a_1 & a_1 & c_1 \\ a_2 & a_2 & c_2 \\ a_2 & a_2 & c_2 \end{vmatrix} = 0 \quad \text{or} \quad \begin{vmatrix} a_1 & b_1 & c_1 \\ a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

**Property 5.** In a determinant the sum of the products of the elements of any row (or any column) with the cofactors of the corresponding elements of any other row (column) is zero.

Let  $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$  be a determinant of order 3.

Expanding  $\Delta$  along the first row, we get

$$\begin{aligned} \Delta &= a_1 (b_2 c_3 - b_3 c_2) - b_1 (a_2 c_3 - a_3 c_2) + c_1 (a_2 b_3 - a_3 b_2) \\ &= a_1 (-1)^{1+1} \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} + b_1 (-1)^{1+2} \begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1 (-1)^{1+3} \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix} \\ &= a_1 A_1 + b_1 B_1 + c_1 C_1. \end{aligned}$$

Similarly, we have

$$\begin{aligned} a_2 A_2 + b_2 B_2 + c_2 C_2 &= \Delta; \quad a_3 A_3 + b_3 B_3 + c_3 C_3 = \Delta; \\ a_1 A_1 + a_2 A_2 + a_3 A_3 &= \Delta \text{ and so on.} \end{aligned}$$

Again take the elements of the first row and cofactors of the corresponding elements of the third row then we have

$$\begin{aligned}
 & a_1 A_3 + b_1 B_3 + c_1 C_3 \\
 &= a_1 (-1)^{3+1} \begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix} + b_1 (-1)^{3+2} \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix} \\
 &\quad + c_1 (-1)^{3+3} \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \\
 &= a_1 (b_1 c_2 - b_2 c_1) - b_1 (a_1 c_2 - c_1 a_2) + c_1 (a_1 b_2 - a_2 b_1) = 0.
 \end{aligned}$$

Thus the sum of the products of the elements of any row (or column) with the cofactors of the corresponding elements of any other row (or column) is zero.

Similarly we have

$$a_2 A_1 + b_2 B_1 + c_2 C_1 = 0, a_1 B_1 + a_2 B_2 + a_3 B_3 = 0 \text{ etc.}$$

**Property 6.** If in a determinant each element in any row (or column) consists of the sum of two terms then the determinant can be expressed as the sum of two determinants of the same order

$$\begin{aligned}
 \text{i.e.,} \quad & \begin{vmatrix} a_1 + \alpha_1 & b_1 & c_1 \\ a_2 + \alpha_2 & b_2 & c_2 \\ a_3 + \alpha_3 & b_3 & c_3 \end{vmatrix} \\
 &= \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} \alpha_1 & b_1 & c_1 \\ \alpha_2 & b_2 & c_2 \\ \alpha_3 & b_3 & c_3 \end{vmatrix}.
 \end{aligned}$$

**Property 7.** If to the elements of a row (or column) of a determinant are added  $m$  times the corresponding elements of another row (or column), the value of the determinant thus obtained is equal to the value of the original determinant.

$$\text{Let } \Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \quad \text{and} \quad \Delta' = \begin{vmatrix} a_1 + mb_1 & b_1 & c_1 \\ a_2 + mb_2 & b_2 & c_2 \\ a_3 + mb_3 & b_3 & c_3 \end{vmatrix}$$

be the determinant obtained by adding to the elements of first column of  $\Delta$   $m$  times the elements of the second column.

$$\begin{aligned}
 \text{Now } \Delta' &= \begin{vmatrix} a_1 + mb_1 & b_1 & c_1 \\ a_2 + mb_2 & b_2 & c_2 \\ a_3 + mb_3 & b_3 & c_3 \end{vmatrix} \\
 &= \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} mb_1 & b_1 & c_1 \\ mb_2 & b_2 & c_2 \\ mb_3 & b_3 & c_3 \end{vmatrix}, \text{ by prop. 6} \\
 &= \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + m \begin{vmatrix} b_1 & b_1 & c_1 \\ b_2 & b_2 & c_2 \\ b_3 & b_3 & c_3 \end{vmatrix}, \text{ by prop. 3}
 \end{aligned}$$

$$= \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + 0,$$

since two columns of the second det. are identical

$$= \Delta.$$

**Property 8.** *If a determinant  $\Delta$  becomes zero when we put  $x = \alpha$  in it, then  $(x - \alpha)$  is a factor of  $\Delta$ .*

## 2.6 Product of Two Determinants of The Same Order

**Rule For The Multiplication Of Two Determinants Of The Third Order.**

$$\text{Let } \Delta_1 = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \quad \text{and} \quad \Delta_2 = \begin{vmatrix} \alpha_1 & \beta_1 & \gamma_1 \\ \alpha_2 & \beta_2 & \gamma_2 \\ \alpha_3 & \beta_3 & \gamma_3 \end{vmatrix}$$

be two determinants of the third order.

$$\text{Then } \Delta_1 \cdot \Delta_2 = \begin{vmatrix} a_1 \alpha_1 + b_1 \alpha_2 + c_1 \alpha_3 & a_1 \beta_1 + b_1 \beta_2 + c_1 \beta_3 & a_1 \gamma_1 + b_1 \gamma_2 + c_1 \gamma_3 \\ a_2 \alpha_1 + b_2 \alpha_2 + c_2 \alpha_3 & a_2 \beta_1 + b_2 \beta_2 + c_2 \beta_3 & a_2 \gamma_1 + b_2 \gamma_2 + c_2 \gamma_3 \\ a_3 \alpha_1 + b_3 \alpha_2 + c_3 \alpha_3 & a_3 \beta_1 + b_3 \beta_2 + c_3 \beta_3 & a_3 \gamma_1 + b_3 \gamma_2 + c_3 \gamma_3 \end{vmatrix}.$$

This is the *row-by-column multiplication rule* for writing down the product of two determinants of the third order in the form of a determinant of the third order. This rule is applicable for writing down the product of two determinants of the  $n^{\text{th}}$  order, where  $n$  is arbitrary. It should be noted that this multiplication rule is the same as the rule for multiplication of two matrices.

**Other ways of multiplying the determinants.** The value of a determinant does not change by interchanging the rows and columns. Therefore while writing down the product of two determinants of the same order, we can also follow the row-by-row multiplication rule, or the column-by-row multiplication rule or column-by-column multiplication rule.

## 2.7 Working Rule for Finding the Value of a Determinant

If the determinant is of order 2, we can at once find its value. But to find the value of a determinant of order  $\geq 3$ , we should always try to make zeros at maximum number of places in any particular row (or column) and then to expand the determinant along that row (or column). The property 7 helps us to make zeros.



## 2.8 Symbols and Notations to be Employed for finding the Values of a Determinant

For convenience we shall denote first, second, third rows of a determinant by  $R_1, R_2, R_3$  and columns by  $C_1, C_2, C_3$  etc.

If we change the  $i^{\text{th}}$  row of a determinant by interchanging it with  $j^{\text{th}}$  row, then we shall denote this operation by writing  $R_i \leftrightarrow R_j$ .

If we change the  $i^{\text{th}}$  column (or row) by multiplying its elements by some number  $k$  then we shall denote this operation by writing  $C_i \rightarrow kC_i$ .

If we change the  $i^{\text{th}}$  row by adding to it  $m$  times the corresponding elements of the  $j^{\text{th}}$  row then we shall denote this operation by writing  $R_i \rightarrow R_i + mR_j$ . It should be noted that in this operation only  $R_i$  will change while  $R_j$  will remain as it is.

### Illustrative Examples

**Illustration 1 :** Show that  $\begin{vmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{vmatrix} = 0$ .

**Solution :** Let  $\Delta = \begin{vmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{vmatrix}$ .

Applying  $C_3 \rightarrow C_3 + C_2$ , we get

$$\begin{aligned} \Delta &= \begin{vmatrix} 1 & a & a+b+c \\ 1 & b & b+c+a \\ 1 & c & a+b+c \end{vmatrix} \\ &= (a+b+c) \begin{vmatrix} 1 & a & 1 \\ 1 & b & 1 \\ 1 & c & 1 \end{vmatrix}, \text{ taking } a+b+c \text{ common from } C_3 \\ &= 0, \text{ since } C_1 \text{ and } C_3 \text{ are identical.} \end{aligned}$$

**Illustration 2 :** Without expanding the determinant, show that:

$$\begin{vmatrix} 0 & b & -c \\ -b & 0 & a \\ c & -a & 0 \end{vmatrix} = 0.$$

[UPTU 2002]

**Solution :** Let  $\Delta = \begin{vmatrix} 0 & b & -c \\ -b & 0 & a \\ c & -a & 0 \end{vmatrix}$ .

By property 1, changing columns into rows

$$\Delta = \begin{vmatrix} 0 & -b & c \\ b & 0 & -a \\ -c & a & 0 \end{vmatrix}$$

$$= (-1)^3 \begin{vmatrix} 0 & b & -c \\ -b & 0 & a \\ c & -a & 0 \end{vmatrix},$$

taking  $(-1)$  common from each column

$$= (-1)^3 \Delta = -\Delta$$

or  $2\Delta = 0$  or  $\Delta = 0.$

**Illustration 3 :** Without expanding show that  $\begin{vmatrix} b^2 & c^2 & bc & b+c \\ c^2 & a^2 & ca & c+a \\ a^2 & b^2 & ab & a+b \end{vmatrix} = 0.$

**Solution :** Let  $\Delta = \begin{vmatrix} b^2 & c^2 & bc & b+c \\ c^2 & a^2 & ca & c+a \\ a^2 & b^2 & ab & a+b \end{vmatrix}.$

Multiplying the 1st, 2nd and 3rd rows by  $a, b, c$  respectively, we get

$$\Delta = \frac{1}{abc} \begin{vmatrix} ab^2 & c^2 & abc & ab+ac \\ bc^2 & a^2 & bca & bc+ba \\ ca^2 & b^2 & cab & ca+cb \end{vmatrix}$$

$$= \frac{(abc)^2}{abc} \begin{vmatrix} bc & 1 & ab+ac \\ ca & 1 & bc+ba \\ ab & 1 & ca+cb \end{vmatrix},$$

taking  $abc$  common from 1st and 2nd columns

$$= abc \begin{vmatrix} bc & 1 & ab+bc+ca \\ ca & 1 & ab+bc+ca \\ ab & 1 & ab+bc+ca \end{vmatrix}, \text{ applying } C_3 \rightarrow C_3 + C_1$$

$$= abc (ab+bc+ca) \begin{vmatrix} bc & 1 & 1 \\ ca & 1 & 1 \\ ab & 1 & 1 \end{vmatrix},$$

taking  $ab+bc+ca$  common from third column

$$= 0, \text{ since } C_2 \text{ and } C_3 \text{ are identical.}$$

**Illustration 4 :** Show that  $\begin{vmatrix} b-c & c-a & a-b \\ c-a & a-b & b-c \\ a-b & b-c & c-a \end{vmatrix} = 0$ .

**Solution :** Let  $\Delta = \begin{vmatrix} b-c & c-a & a-b \\ c-a & a-b & b-c \\ a-b & b-c & c-a \end{vmatrix}$ .

Applying  $R_1 \rightarrow R_1 + R_2 + R_3$ , we get

$$\Delta = \begin{vmatrix} 0 & 0 & 0 \\ c-a & a-b & b-c \\ a-b & b-c & c-a \end{vmatrix} = 0,$$

since all elements of  $R_1$  are zero.

**Illustration 5 :** Show that  $\begin{vmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \\ \omega^2 & 1 & \omega \end{vmatrix} = 0$ , where  $\omega$  is a complex cube root of unity.

**Solution :** Let  $\Delta = \begin{vmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \\ \omega^2 & 1 & \omega \end{vmatrix}$ .

Applying  $C_1 \rightarrow C_1 + C_2 + C_3$ , we get

$$\Delta = \begin{vmatrix} 1+\omega+\omega^2 & \omega & \omega^2 \\ 1+\omega+\omega^2 & \omega^2 & 1 \\ 1+\omega+\omega^2 & 1 & \omega \end{vmatrix} = \begin{vmatrix} 0 & \omega & \omega^2 \\ 0 & \omega^2 & 1 \\ 0 & 1 & \omega \end{vmatrix}$$

$$[\because 1+\omega+\omega^2 = 0]$$

$$= 0, \quad \text{since all entries in } C_1 \text{ are 0.}$$

**Illustration 6 :** Without expanding the determinant, show that

$$\begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix} = \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} \text{ and evaluate it.}$$

[UPTU 2001]

**Solution :** Let  $\Delta = \begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix}$ .

Multiplying the 1st, 2nd and 3rd rows by  $a, b, c$  respectively, we get

$$\begin{aligned}
 \Delta &= \frac{1}{abc} \begin{vmatrix} a & a^2 & abc \\ b & b^2 & bca \\ c & c^2 & cab \end{vmatrix} \\
 &= \frac{abc}{abc} \begin{vmatrix} a & a^2 & 1 \\ b & b^2 & 1 \\ c & c^2 & 1 \end{vmatrix}, \text{ taking } abc \text{ common from 3rd column} \\
 &= - \begin{vmatrix} a & 1 & a^2 \\ b & 1 & b^2 \\ c & 1 & c^2 \end{vmatrix}, \text{ applying } C_2 \leftrightarrow C_3 \\
 &= \begin{vmatrix} 1 & 1 & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}, \text{ applying } C_1 | C_2.
 \end{aligned}$$

applying  $R_2 \rightarrow R_2 - R_1$  and  $R_3 \rightarrow R_3 - R_1$ , we get

$$\begin{aligned}
 \Delta &= \begin{vmatrix} 1 & a & a^2 \\ 0 & b-a & b^2-a^2 \\ 0 & c-a & c^2-a^2 \end{vmatrix} \\
 &= 1 \begin{vmatrix} b-a & b^2-a^2 \\ c-a & c^2-a^2 \end{vmatrix}, \text{ on expanding the determinant along } C_1 \\
 &= (b-a)(c-a) \begin{vmatrix} 1 & b+a \\ 1 & c+a \end{vmatrix},
 \end{aligned}$$

$$\begin{aligned}
 &\text{taking } (b-a) \text{ common from } R_1 \text{ and } (c-a) \text{ common from } R_2 \\
 &= (b-a)(c-a)\{c+a-(b+a)\} \\
 &= (b-a)(c-a)(c-b) = (a-b)(b-c)(c-a).
 \end{aligned}$$

**Illustration 7 :** Prove that

$$\begin{aligned}
 \Delta &= \begin{vmatrix} a & a^2 & bc \\ b & b^2 & ca \\ c & c^2 & ab \end{vmatrix} = \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} \\
 &= (a-b)(b-c)(c-a)(ab+bc+ca).
 \end{aligned}$$

**Solution :** Multiplying the first, second and third rows of  $\Delta$  on the left hand side by  $a, b$  and  $c$  respectively, we get

$$\Delta = \frac{1}{abc} \begin{vmatrix} a^2 & a^3 & abc \\ b^2 & b^3 & abc \\ c^2 & c^3 & abc \end{vmatrix} = \frac{abc}{abc} \begin{vmatrix} a^2 & a^3 & 1 \\ b^2 & b^3 & 1 \\ c^2 & c^3 & 1 \end{vmatrix},$$

taking  $abc$  common from  $C_3$

$$\begin{aligned}
 &= - \begin{vmatrix} a^2 & 1 & a^3 \\ b^2 & 1 & b^3 \\ c^2 & 1 & c^3 \end{vmatrix}, \text{ applying } C_2 \leftrightarrow C_3 \\
 &= \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}, \text{ applying } C_1 \leftrightarrow C_2 . \\
 \Delta &= \begin{vmatrix} 1 & a^2 & a^3 \\ 0 & b^2 - a^2 & b^3 - a^3 \\ 0 & c^2 - a^2 & c^3 - a^3 \end{vmatrix}, \\
 &\text{applying } R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1 \\
 &= 1 \begin{vmatrix} b^2 - a^2 & b^3 - a^3 \\ c^2 - a^2 & c^3 - a^3 \end{vmatrix}, \text{ expanding along first column} \\
 &= (b-a)(c-a) \begin{vmatrix} b+a & b^2 + a^2 + ab \\ c+a & c^2 + a^2 + ca \end{vmatrix} \\
 &= (b-a)(c-a) \begin{vmatrix} b+a & b^2 + a^2 + ab \\ c-b & c^2 - b^2 + ca - ab \end{vmatrix}, \\
 &\text{applying } R_2 \rightarrow R_2 - R_1 \\
 &= (b-a)(c-a)(c-b) \begin{vmatrix} b+a & b^2 + a^2 + ab \\ 1 & c+b+a \end{vmatrix}, \\
 &\text{taking } (c-b) \text{ common from } R_2 \\
 &= (b-a)(c-a)(c-b) \{ (a+b)(a+b+c) - (b^2 + a^2 + ab) \} \\
 &= (a-b)(b-c)(c-a)(ab+bc+ca).
 \end{aligned}$$

**Illustration 8 :** Prove that

$$\begin{aligned}
 &\equiv \begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = abc \left( 1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \\
 &= abc + ab + bc + ca.
 \end{aligned}$$

[UPTU 2004, 06]

**Solution :** Taking  $a, b, c$  common from first, second and third columns respectively, we get

$$\begin{aligned}
 \Delta &= abc \begin{vmatrix} \frac{1}{a} + 1 & \frac{1}{b} & \frac{1}{c} \\ \frac{1}{a} & \frac{1}{b} + 1 & \frac{1}{c} \\ \frac{1}{a} & \frac{1}{b} & \frac{1}{c} + 1 \end{vmatrix} \\
 &= abc \begin{vmatrix} 1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} & \frac{1}{b} & \frac{1}{c} \\ 1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} & \frac{1}{b} + 1 & \frac{1}{c} \\ 1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} & \frac{1}{b} & \frac{1}{c} + 1 \end{vmatrix}, \\
 &\quad \text{applying } C_1 \rightarrow C_1 + C_2 + C_3 \\
 &= abc \left( 1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \begin{vmatrix} 1 & \frac{1}{b} & \frac{1}{c} \\ 1 & \frac{1}{b} + 1 & \frac{1}{c} \\ 1 & \frac{1}{b} & \frac{1}{c} + 1 \end{vmatrix}, \\
 &\quad \text{taking } \left( 1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \text{ common from } C_1 \\
 &= abc \left( 1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \begin{vmatrix} 1 & \frac{1}{b} & \frac{1}{c} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}, \\
 &\quad \text{applying } R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1 \\
 &= abc \left( 1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) = abc + ab + bc + ca.
 \end{aligned}$$

**Illustration 9 :** Prove that

$$\Delta \equiv \begin{vmatrix} x - y - z & 2x & 2x \\ 2y & y - x - z & 2y \\ 2z & 2z & z - x - y \end{vmatrix} = (x + y + z)^3.$$

**Solution :** Applying  $R_1 \rightarrow R_1 + R_2 + R_3$ , we get

$$\begin{aligned}
 \Delta &= \begin{vmatrix} x + y + z & x + y + z & x + y + z \\ 2y & y - x - z & 2y \\ 2z & 2z & z - x - y \end{vmatrix} \\
 &= (x + y + z) \begin{vmatrix} 1 & 1 & 1 \\ 2y & y - x - z & 2y \\ 2z & 2z & z - x - y \end{vmatrix}
 \end{aligned}$$

$$= (x + y + z) \begin{vmatrix} 1 & 0 & 0 \\ 2y & -(x + y + z) & 0 \\ 2z & 0 & -(x + y + z) \end{vmatrix},$$

applying  $C_2 \rightarrow C_2 - C_1, C_3 \rightarrow C_3 - C_1$

$$= (x + y + z)^3.$$

**Illustration 10 :** Prove that

$$\Delta \equiv \begin{vmatrix} 1 + a^2 - b^2 & 2ab & -2b \\ 2ab & 1 - a^2 + b^2 & 2a \\ 2b & -2a & 1 - a^2 - b^2 \end{vmatrix} = (1 + a^2 + b^2)^3.$$

**Solution :** Applying  $C_1 \rightarrow C_1 - bC_3$  and  $C_2 \rightarrow C_2 + aC_3$ , we get

$$\Delta = \begin{vmatrix} 1 + a^2 + b^2 & 0 & -2b \\ 0 & 1 + a^2 + b^2 & 2a \\ b(1 + a^2 + b^2) & -a(1 + a^2 + b^2) & 1 - a^2 - b^2 \end{vmatrix}$$

$$= (1 + a^2 + b^2)^2 \begin{vmatrix} 1 & 0 & -2b \\ 0 & 1 & 2a \\ b & -a & 1 - a^2 - b^2 \end{vmatrix},$$

taking  $(1 + a^2 + b^2)$  common from  $C_1$  and  $C_2$ .

$$= (1 + a^2 + b^2)^2 \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 2a \\ b & -a & 1 - a^2 + b^2 \end{vmatrix},$$

applying  $C_3 \rightarrow C_3 + 2bC_1$

$$= (1 + a^2 + b^2)^2 (1 - a^2 + b^2 + 2a^2), \text{ expanding along first row}$$

$$= (1 + a^2 + b^2)^3.$$

**Illustration 11 :** Without expanding the determinant show that  $(a + b + c)$  is a factor of the following determinant :

$$\Delta \equiv \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}.$$

[UPTU 2003]

If  $a, b, c$  are positive and unequal, show that the value of  $\Delta$  is always negative.

**Solution :** Applying  $C_1 \rightarrow C_1 + C_2 + C_3$ , we get

$$\Delta = \begin{vmatrix} a + b + c & b & c \\ a + b + c & c & a \\ a + b + c & a & b \end{vmatrix} = (a + b + c) \begin{vmatrix} 1 & b & c \\ 1 & c & a \\ 1 & a & b \end{vmatrix}$$

$$= (a + b + c) \begin{vmatrix} 1 & b & c \\ 0 & c - b & a - c \\ 0 & a - b & b - c \end{vmatrix},$$

$$\begin{aligned} & \text{applying } R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1 \\ &= (a + b + c) \begin{vmatrix} c - b & a - c \\ a - b & b - c \end{vmatrix} \\ &= (a + b + c) \{-(b - c)^2 - (a - b)(a - c)\} \\ &= (a + b + c) (-a^2 - b^2 - c^2 + ab + bc + ca). \end{aligned}$$

Thus  $(a + b + c)$  is a factor of  $\Delta$ .

Now we shall prove the next part.

$$\begin{aligned} \text{We have } \Delta &= (a + b + c) (-a^2 - b^2 - c^2 + ab + bc + ca) \\ &= -\frac{1}{2} (a + b + c) (2a^2 + 2b^2 + 2c^2 - 2ab - 2bc - 2ca) \\ &= -\frac{1}{2} (a + b + c) \{(a - b)^2 + (b - c)^2 + (c - a)^2\} \\ &< 0, \text{ since } a, b, c \text{ are positive and unequal.} \end{aligned}$$

Therefore the given determinant is always negative if  $a, b, c$  are positive and unequal.

**Illustration 12 :** Prove that

$$\Delta \equiv \begin{vmatrix} (b + c)^2 & a^2 & a^2 \\ b^2 & (c + a)^2 & b^2 \\ c^2 & c^2 & (a + b)^2 \end{vmatrix} = 2abc (a + b + c)^3.$$

**Solution :** Applying  $C_1 \rightarrow C_1 - C_2$ ,  $C_2 \rightarrow C_2 - C_3$ , we get

$$\begin{aligned} \Delta &= \begin{vmatrix} (b + c + a)(b + c - a) & 0 & a^2 \\ 0 & (c + a + b)(c + a - b) & b^2 \\ (c + a + b)(c - a - b) & (c + a + b)(c - a - b) & (a + b)^2 \end{vmatrix} \\ &= (a + b + c)^2 \begin{vmatrix} b + c - a & 0 & a^2 \\ 0 & c + a - b & b^2 \\ c - a - b & c - a - b & (a + b)^2 \end{vmatrix}, \end{aligned}$$

taking  $(a + b + c)$  common from each of  $C_1$  and  $C_2$

$$= (a + b + c)^2 \begin{vmatrix} b + c - a & 0 & a^2 \\ 0 & c + a - b & b^2 \\ -2b & -2a & 2ab \end{vmatrix};$$

by  $R_3 \rightarrow R_3 - R_2 - R_1$



$$\begin{aligned}
 &= (a+b+c)^2 \begin{vmatrix} b+c & a^2/b & a^2 \\ b^2/a & c+a & b^2 \\ 0 & 0 & 2ab \end{vmatrix}, \text{ by } C_1 \rightarrow C_1 + \frac{1}{a} C_3 \\
 &\quad C_2 \rightarrow C_2 + \frac{1}{b} C_3 \\
 &= (a+b+c)^2 \cdot 2ab \begin{vmatrix} b+c & a^2/b \\ b^2/a & c+a \end{vmatrix}, \text{ expanding along } R_3 \\
 &= (a+b+c)^2 \cdot 2ab \{ (b+c)(c+a) - ab \} = 2abc (a+b+c)^3.
 \end{aligned}$$

**Illustration 13 :** Solve the equation  $\begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ x-4 & 2x-9 & 3x-16 \\ x-8 & 2x-27 & 3x-64 \end{vmatrix} = 0$ .

[UPTU 2007]

**Solution :** Applying  $R_2 \rightarrow R_2 - R_1$ ,  $R_3 \rightarrow R_3 - R_1$ , we get

$$\begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ -2 & -6 & -12 \\ -6 & -24 & -60 \end{vmatrix} = 0$$

or

$$\begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ 1 & 3 & 6 \\ 1 & 4 & 10 \end{vmatrix} = 0.$$

Expanding the determinant along the first row, we get

$$(x-2) \cdot 6 - (2x-3) \cdot 4 + (3x-4) \cdot 1 = 0$$

or  $6x - 12 - 8x + 12 + 3x - 4 = 0$

or  $x - 4 = 0$  or  $x = 4$ .

**Illustration 14 :** If  $a + b + c = 0$ , solve the equation

$$\begin{vmatrix} a-x & c & b \\ c & b-x & a \\ b & a & c-x \end{vmatrix} = 0.$$

**Solution :** Applying  $C_1 \rightarrow C_1 + C_2 + C_3$ , we get

$$\begin{vmatrix} a+b+c-x & c & b \\ a+b+c-x & b-x & a \\ a+b+c-x & a & c-x \end{vmatrix} = 0$$

or

$$(a+b+c-x) \begin{vmatrix} 1 & c & b \\ 1 & b-x & a \\ 1 & a & c-x \end{vmatrix} = 0$$

or

$$-x \begin{vmatrix} 1 & c & b \\ 0 & b-c-x & a-b \\ 0 & a-c & c-b-x \end{vmatrix} = 0,$$

$$\text{by } R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1$$

$$\begin{aligned}
 \text{or} \quad & -x \{(b-c-x)(c-b-x) - (a-c)(a-b)\} = 0 \\
 \text{or} \quad & x(x^2 - b^2 - c^2 + 2bc - a^2 + ab + ca - bc) = 0 \\
 \text{or} \quad & x(x^2 - a^2 - b^2 - c^2 + ab + bc + ca) = 0 \\
 \text{or} \quad & x = 0 \\
 \text{or} \quad & x^2 = a^2 + b^2 + c^2 - (ab + bc + ca) \\
 & = a^2 + b^2 + c^2 - \frac{1}{2} \{(a+b+c)^2 - (a^2 + b^2 + c^2)\} \\
 & = \frac{3}{2} (a^2 + b^2 + c^2). \\
 \therefore \quad & x = 0 \\
 \text{or} \quad & x = \pm \sqrt{\left\{ \frac{3}{2} (a^2 + b^2 + c^2) \right\}}.
 \end{aligned}$$

**Illustration 15 :** Without expanding evaluate

$$\begin{vmatrix} 1 & a & a^2 - bc \\ 1 & b & b^2 - ca \\ 1 & c & c^2 - ab \end{vmatrix}.$$

[UPTU 2008]

**Solution :** Let

$$\begin{aligned}
 \Delta &= \begin{vmatrix} 1 & a & a^2 - bc \\ 1 & b & b^2 - ca \\ 1 & c & c^2 - ab \end{vmatrix} = \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} + \begin{vmatrix} 1 & a & -bc \\ 1 & b & -ca \\ 1 & c & -ab \end{vmatrix} \\
 &= \begin{vmatrix} 1 & a & a^2 \\ 0 & b-a & b^2 - a^2 \\ 0 & c-a & c^2 - a^2 \end{vmatrix} + \begin{vmatrix} 1 & a & -bc \\ 0 & b-a & c(b-a) \\ 0 & c-a & b(c-a) \end{vmatrix},
 \end{aligned}$$

applying  $R_2 \rightarrow R_2 - R_1$  and  $R_3 \rightarrow R_3 - R_1$  in each determinant

$$\begin{aligned}
 &= (b-a)(c-a) \begin{vmatrix} 1 & a & a^2 \\ 0 & 1 & b+c \\ 0 & 1 & c+a \end{vmatrix} + (b-c)(c-a) \begin{vmatrix} 1 & a & -bc \\ 0 & 1 & c \\ 0 & 1 & b \end{vmatrix} \\
 &= (b-a)(c-a) \cdot \{ (c+a) - (b+a) \} + (b-a)(c-a) \cdot (b-c) \\
 &= (b-a)(c-a)(c-b) + (b-a)(c-a)(b-c) \\
 &= (b-a)(c-a)(c-b) - (b-a)(c-a)(c-b) \\
 &= 0.
 \end{aligned}$$

**Illustration 16 :** If  $x, y, z$  are all different and if

$$\begin{vmatrix} x & x^2 & 1+x^3 \\ y & y^2 & 1+y^3 \\ z & z^2 & 1+z^3 \end{vmatrix} = 0, \text{ prove that } xyz = -1.$$

**Solution :** We have

$$\begin{vmatrix} x & x^2 & 1+x^3 \\ y & y^2 & 1+y^3 \\ z & z^2 & 1+z^3 \end{vmatrix} = \begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} + \begin{vmatrix} x & x^2 & x^3 \\ y & y^2 & y^3 \\ z & z^2 & z^3 \end{vmatrix} \\ = - \begin{vmatrix} x & 1 & x^2 \\ y & 1 & y^2 \\ z & 1 & z^2 \end{vmatrix} + xyz \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix},$$

applying  $C_2 \leftrightarrow C_3$  in first det. and taking  $x, y, z$  common from  $R_1, R_2, R_3$  respectively of the second determinant

$$= \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} + xyz \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix},$$

applying  $C_1 \leftrightarrow C_2$  in first determinant

$$= (1 + x y z) \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} \\ = (1 + x y z) \begin{vmatrix} 1 & x & x^2 \\ 0 & y - x & y^2 - x^2 \\ 0 & z - x & z^2 - x^2 \end{vmatrix},$$

applying  $R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1$

$$= (1 + x y z) \begin{vmatrix} y - x & y^2 - x^2 \\ z - x & z^2 - x^2 \end{vmatrix} \\ = (1 + x y z) (y - x)(z - x) \begin{vmatrix} 1 & y + x \\ 1 & z + x \end{vmatrix} \\ = (1 + xyz) (y - x)(z - x) \{z + x - (y + x)\} \\ = (1 + xyz) (x - y)(y - z)(z - x).$$

Since  $x, y, z$  are all different therefore  $x - y \neq 0, y - z \neq 0, z - x \neq 0$ .

Hence  $(1 + xyz)(x - y)(y - z)(z - x) = 0$

$$\Rightarrow 1 + xyz = 0 \rightarrow xyz = -1.$$

## Comprehensive Exercise 1

1. Evaluate the following determinants :

$$(i) \begin{vmatrix} a & b \\ -b & a \end{vmatrix}$$

$$(ii) \begin{vmatrix} x+y & 1 \\ x^3 & x^2 - xy + y^2 \end{vmatrix}.$$

2. Evaluate each of the following determinants :

$$(i) \begin{vmatrix} 1 & 3 & -1 \\ -2 & 1 & 2 \\ 4 & 5 & 1 \end{vmatrix}$$

$$(ii) \begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 6 \\ x & y & z \end{vmatrix}.$$

3. Show that  $\begin{vmatrix} \sin 10^\circ & -\cos 10^\circ \\ \sin 80^\circ & \cos 80^\circ \end{vmatrix} = 1$ .

4. Find the cofactors of each element of the following determinants :

$$(i) \begin{vmatrix} 5 & 7 \\ 3 & -1 \end{vmatrix}$$

$$(ii) \begin{vmatrix} 0 & 1 & 2 \\ 1 & 4 & 1 \\ 5 & 3 & 7 \end{vmatrix}$$

$$(iii) \begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix}.$$

5. Using the method of cofactors find the value of the determinant

$$\begin{vmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{vmatrix}.$$

**Evaluate the following determinants :**

$$6. \begin{vmatrix} a-b & b-c & c-a \\ b-c & c-a & a-b \\ c-a & a-b & b-c \end{vmatrix}.$$

$$7. \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}.$$

$$8. \begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix}.$$

$$9. \begin{vmatrix} 1 & 1 & 1 \\ \alpha & \beta & \gamma \\ \beta+\gamma & \gamma+\alpha & \alpha+\beta \end{vmatrix}.$$

10. Without expanding the determinant, show that  $\left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$  is a

factor of the determinant  $\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix}$ .

11. Using properties of determinants show that

$$\begin{vmatrix} a+b+2c & a & b \\ c & 2a+b+c & b \\ c & a & a+2b+c \end{vmatrix} = 2(a+b+c)^3.$$

12. Using properties of determinants, prove the following results :

(i)  $\begin{vmatrix} x+9 & x & x \\ x & x+9 & x \\ x & x & x+9 \end{vmatrix} = 243(x+3).$

(ii)  $\begin{vmatrix} x+4 & x & x \\ x & x+4 & x \\ x & x & x+4 \end{vmatrix} = 16(3x+4).$

(iii)  $\begin{vmatrix} x+4 & 2x & 2x \\ 2x & x+4 & 2x \\ 2x & 2x & x+4 \end{vmatrix} = (5x+4)(4-x)^2.$

(iv)  $\begin{vmatrix} b+c & c+a & a+b \\ c+a & a+b & b+c \\ a+b & b+c & c+a \end{vmatrix} = 2 \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}.$

**Prove the following identities :**

13.  $\begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ x^3 & y^3 & z^3 \end{vmatrix} = xyz(x-y)(y-z)(z-x).$

14.  $\begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c).$

[UPTU 2005]

15.  $\begin{vmatrix} a+b+c & -c & -b \\ -c & a+b+c & -a \\ -b & -a & a+b+c \end{vmatrix} = 2(a+b)(b+c)(c+a).$

16.  $\begin{vmatrix} 1 & b+c & b^2+c^2 \\ 1 & c+a & c^2+a^2 \\ 1 & a+b & a^2+b^2 \end{vmatrix} = (a-b)(b-c)(c-a).$

$$17. \begin{vmatrix} -a^2 & ab & ac \\ ba & -b^2 & bc \\ ac & bc & -c^2 \end{vmatrix} = 4a^2b^2c^2.$$

Solve the following equations :

$$18. \begin{vmatrix} x+1 & 3 & 5 \\ 2 & x+2 & 5 \\ 2 & 3 & x+4 \end{vmatrix} = 0.$$

$$19. \begin{vmatrix} x+a & b & c \\ a & x+b & c \\ a & b & x+c \end{vmatrix} = 0.$$

$$20. \begin{vmatrix} 3x-8 & 3 & 3 \\ 3 & 3x-8 & 3 \\ 3 & 3 & 3x-8 \end{vmatrix} = 0.$$

## Answers

1. (i)  $a^2 + b^2$  (ii)  $y^3$ . 2. (i) 34 (ii)  $3x - z$ .
4. (i)  $A_{11} = -1, A_{12} = -3, A_{21} = -7, A_{22} = 5$ .  
 (ii)  $A_{11} = 25, A_{12} = -2, A_{13} = -17, A_{21} = -1, A_{22} = -10,$   
 $A_{23} = 5, A_{31} = -7, A_{32} = 2, A_{33} = -1$ .  
 (iii)  $A_{11} = a(b^2 - c^2), A_{12} = a(c - b), A_{13} = c - b,$   
 $A_{21} = b(c^2 - a^2), A_{22} = b(a - c), A_{23} = a - c,$   
 $A_{31} = c(a^2 - b^2), A_{32} = c(b - a), A_{33} = b - a$ .
5. 0. 6. 0.
7.  $abc + 2fgh - af^2 - bg^2 - ch^2$ .
8.  $(a - b)(b - c)(c - a)$ . 9. 0. 18. 1, 1, -9.
19.  $x = 0, -(a + b + c)$ . 20.  $x = \frac{2}{3}, \frac{11}{3}, \frac{11}{3}$ .

## 2.9 Applications of Determinants to Coordinate Geometry

**Area of a triangle.** We know in coordinate geometry the area of the triangle, whose vertices are  $(x_1, y_1), (x_2, y_2)$  and  $(x_3, y_3)$ , is given by

$$\Delta = \frac{1}{2} [x_1 (y_2 - y_3) + x_2 (y_3 - y_1) + x_3 (y_1 - y_2)].$$

Since we can express  $x_1 (y_2 - y_3) + x_2 (y_3 - y_1) + x_3 (y_1 - y_2)$  as a third order determinant

$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

therefore the area of the triangle with vertices  $(x_1, y_1)$ ,  $(x_2, y_2)$  and  $(x_3, y_3)$  is

$$\Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}.$$

Since the area has to be a positive quantity, we always take the absolute value of the above determinant for the area.

**Condition for three points to be collinear.** The three points  $A(x_1, y_1)$ ,  $B(x_2, y_2)$  and  $C(x_3, y_3)$  are collinear if and only if area of the triangle  $ABC = 0$

$$\text{iff } \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0 \quad \text{iff } \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0.$$

## Illustrative Examples

**Illustration 1 :** Using determinants find the area of the triangle whose vertices are  $(1, -1)$ ,  $(2, 4)$ ,  $(-3, 5)$ . Are the given points collinear?

**Solution :** We know the area of the triangle whose vertices are  $(x_1, y_1)$ ,  $(x_2, y_2)$  and  $(x_3, y_3)$  is

$$\frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}.$$

Thus the area of the triangle with vertices  $(1, -1)$ ,  $(2, 4)$ ,  $(-3, 5)$  is

$$\begin{aligned} \frac{1}{2} \begin{vmatrix} 1 & -1 & 1 \\ 2 & 4 & 1 \\ -3 & 5 & 1 \end{vmatrix} &= \frac{1}{2} \{(4 - 5) + (2 + 3) + (10 + 12)\} \\ &= \frac{1}{2} (-1 + 5 + 22) = 13 \text{ sq. units.} \end{aligned}$$

Since the area of the triangle is non-zero therefore the given three points are non-collinear.

**Illustration 2 :** Using determinants, show that the points  $(-2, 5)$ ,  $(-6, -7)$  and  $(-5, -4)$  are collinear.

**Solution :** We know that the three points are collinear if the area of the triangle formed by these points is zero.

Now area of the triangle formed by the given points

$$\begin{aligned}
 &= \frac{1}{2} \begin{vmatrix} -2 & 5 & 1 \\ -6 & -7 & 1 \\ -5 & -4 & 1 \end{vmatrix} = \frac{1}{2} \{-2(-7+4) - 5(-6+5) + (24-35)\} \\
 &= \frac{1}{2} (6+5-11) = 0.
 \end{aligned}$$

Hence the given points are collinear.

**Illustration 3 :** If the points  $(2, -3)$ ,  $(\lambda, -1)$  and  $(0, 4)$  are collinear, find the value of  $\lambda$ .

**Solution :** Since the given points are collinear therefore the area of the triangle formed by these points must be zero.

$$\therefore \frac{1}{2} \begin{vmatrix} 2 & -3 & 1 \\ \lambda & -1 & 1 \\ 0 & 4 & 1 \end{vmatrix} = 0$$

$$\text{or} \quad 2(-1-4) + 3(\lambda-0) + (4\lambda-0) = 0$$

$$\text{or} \quad -10 + 3\lambda + 4\lambda = 0 \quad \text{or} \quad 7\lambda = 10.$$

$$\therefore \lambda = \frac{10}{7}.$$

**Illustration 4 :** For what value of  $k$ , are the points  $(1, 5)$ ,  $(k, 1)$  and  $(1, 7)$  collinear? (Use determinants). [UPTU 2003]

**Solution :** If the three points are collinear then the area of the triangle formed by these points must be zero.

$$\therefore \frac{1}{2} \begin{vmatrix} 1 & 5 & 1 \\ k & 1 & 1 \\ 1 & 7 & 1 \end{vmatrix} = 0$$

$$\text{or} \quad 1(1-7) + 5(k-1) + 1(7k-1) = 0$$

$$\text{or} \quad -6 + 5k - 5 + 7k - 1 = 0$$

$$\text{or} \quad 12k - 12 = 0$$

$$\text{or} \quad 12k = 12$$

$$\text{or} \quad k = \frac{12}{12} = 1.$$

$$\therefore k = 1.$$

## Comprehensive Exercise 2

1. Using determinants find the area of the triangle with vertices

(i)  $(-2, -3)$ ,  $(3, 2)$ ,  $(-1, -8)$     (ii)  $(2, -7)$ ,  $(1, 3)$ ,  $(10, 8)$

(iii)  $(5, 4)$ ,  $(-2, 4)$ ,  $(2, -6)$     (iv)  $(-1, -8)$ ,  $(-2, -3)$ ,  $(3, 2)$ .



2. Using determinants, show that the following points are collinear:
  - (i)  $(3, 8), (-4, 2), (10, 14)$       (ii)  $(11, 7), (5, 5), (-1, 3)$
  - (iii)  $(1, -1), (2, 1), (4, 5)$       (iv)  $(2, 3), (-1, -2), (5, 8)$ .
3. For what value of  $k$ , the points  $(5, 5), (k, 1), (11, 7)$  are collinear?
4. Find the value of  $\lambda$  so that the points  $(1, -5), (-4, 5)$  and  $(\lambda, 7)$  are collinear.
5. If the points  $(a, b), (a', b')$  and  $(a - a', b - b')$  are collinear, prove that  $ab' = a'b$ . [UPTU 2006]
6. Prove that the points  $(a, b + c), (b, c + a), (c, a + b)$  are collinear. [UPTU 2001, 07]

## Answers

1. (i) 15 sq. units      (ii)  $\frac{101}{2}$  sq. units
- (iii) 35 sq. units      (iv) 15 sq. units.
3.  $k = -7$ .      4.  $\lambda = -5$ .

## 2.10 Application of Determinants in solving a System of Linear Equations

The equation  $ax + by + cz + d = 0$  is a linear non-homogeneous equation in unknowns  $x, y$  and  $z$ . If we have  $d = 0$ , the above equation becomes homogeneous linear equation in unknowns  $x, y$  and  $z$ .

Consider the system of  $n$  linear simultaneous equations in  $n$  unknowns

$$x_1, x_2, \dots, x_n$$

$$a_{11} x_1 + a_{12} x_2 + \dots + a_{1n} x_n = b_1$$

$$a_{21} x_1 + a_{22} x_2 + \dots + a_{2n} x_n = b_2$$

$$\dots \quad \dots \quad \dots \quad \dots$$

$$\dots \quad \dots \quad \dots \quad \dots$$

$$a_{n1} x_1 + a_{n2} x_2 + \dots + a_{nn} x_n = b_n.$$

If  $b$ 's are all zero, this system is called **homogeneous** and if at least one of  $b$ 's is non-zero then the above system is called **non-homogeneous**.

Any set of values of  $x_1, x_2, \dots, x_n$  (real or complex) which simultaneously satisfy all these equations is called a **solution** of this system.

When the system of equations has one or more solutions the equations are said to be **consistent** otherwise they are said to be **inconsistent**.

If the system of equations has a unique solution it is said to be **determinate** and it is said to be **indeterminate** if it has more than one solution :

## 2.11 System of Linear Non-homogeneous Equations in Two Unknowns. (Cramer's Rule)

*The solution of the system of non-homogeneous linear equations*

$$a_1 x + b_1 y = c_1, \quad a_2 x + b_2 y = c_2$$

is given by  $x = \frac{D_1}{D}$  and  $y = \frac{D_2}{D}$ , where

$$D = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}, \quad D_1 = \begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix} \text{ and } D_2 = \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}, \quad D \neq 0.$$

**Proof.** Multiplying first equation by  $b_2$  and second by  $b_1$  and subtracting, we get

$$(a_1 b_2 - a_2 b_1) x = b_2 c_1 - b_1 c_2.$$

$$\therefore x = \frac{b_2 c_1 - b_1 c_2}{a_1 b_2 - a_2 b_1}.$$

Substituting for  $x$  in the first equation, we get  $y = \frac{a_1 c_2 - a_2 c_1}{a_1 b_2 - a_2 b_1}.$

We can express the above solution in terms of determinants as follows :

$$x = \frac{\begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}} = \frac{D_1}{D} \quad \text{and} \quad y = \frac{\begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}} = \frac{D_2}{D}, \text{ provided } D \neq 0.$$

Here  $D$  is the determinant of the coefficients of  $x$  and  $y$  and  $D_1, D_2$  are the determinants obtained by replacing the coefficients of  $x$  and  $y$  respectively by constant terms in  $D$ .

*A system of non-homogeneous linear equations in two unknowns is*

(i) **Consistent** and has a unique solution given by

$$x = D_1 / D \text{ and } y = D_2 / D, \text{ if } D \neq 0.$$

(ii) **Inconsistent**, if  $D = 0$  but at least one of  $D_1, D_2$  is non-zero.

(iii) **Consistent and indeterminate** if  $D = 0 = D_1 = D_2$ .

### Homogeneous Linear Equations In Two Unknowns.

Consider the system  $a_1 x + b_1 y = 0, \quad a_2 x + b_2 y = 0$ .

This system is always consistent because the trivial solution (zero solution)  $x = 0, y = 0$  is always there.

The above system has a non-trivial solution iff  $D = 0$  and then the equations have an infinite number of solutions.

If  $D \neq 0$ , the only solution of this system is  $x = 0, y = 0$ .

## Illustrative Examples

**Illustration 1 :** (i) Solve by Cramer's rule

$$x - 2y = 4, \quad -3x + 5y = -7.$$

[UPTU 2001]

(ii) Using determinants solve the system of equations

$$2x - 4y = -3, \quad 4x + 2y = 9.$$

(iii) Using determinants solve the system of equations

$$2x - 2y = 1, \quad x + 2y = 2.$$

**Solution :** (i) We have  $D = \begin{vmatrix} 1 & -2 \\ -3 & 5 \end{vmatrix} = 5 - 6 = -1 \neq 0$ .

Also  $D_1 = \begin{vmatrix} 4 & -2 \\ -7 & 5 \end{vmatrix} = 20 - 14 = 6,$

$$D_2 = \begin{vmatrix} 1 & 4 \\ -3 & -7 \end{vmatrix} = -7 + 12 = 5.$$

$$\therefore x = \frac{D_1}{D} = \frac{6}{-1} = -6 \quad \text{and} \quad y = \frac{D_2}{D} = \frac{5}{-1} = -5.$$

Hence,  $x = -6, y = -5$ .

(ii) We have  $D = \begin{vmatrix} 2 & -4 \\ 4 & 2 \end{vmatrix} = 4 + 16 = 20 \neq 0$ .

Also  $D_1 = \begin{vmatrix} -3 & -4 \\ 9 & 2 \end{vmatrix} = -6 + 36 = 30,$

$$D_2 = \begin{vmatrix} 2 & -3 \\ 4 & 9 \end{vmatrix} = 18 + 12 = 30.$$

$$\therefore x = \frac{D_1}{D} = \frac{30}{20} = \frac{3}{2} \quad \text{and} \quad y = \frac{D_2}{D} = \frac{30}{20} = \frac{3}{2}.$$

Hence  $x = \frac{3}{2}, y = \frac{3}{2}$ .

(iii) We have  $D = \begin{vmatrix} 2 & -2 \\ 1 & 2 \end{vmatrix} = 4 + 2 = 6 \neq 0$ .

Also  $D_1 = \begin{vmatrix} 1 & -2 \\ 2 & 2 \end{vmatrix} = 2 + 4 = 6,$

$$D_2 = \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} = 4 - 1 = 3.$$

$$\therefore x = \frac{D_1}{D} = \frac{6}{6} = 1 \quad \text{and} \quad y = \frac{D_2}{D} = \frac{3}{6} = \frac{1}{2}.$$

Hence  $x = 1, y = 1/2$ .

**Illustration 2 :** Using determinants, solve the system of equations

$$3x - 5y = 0, 3x + 8y = 0.$$

**Solution :** We have  $D = \begin{vmatrix} 3 & -5 \\ 3 & 8 \end{vmatrix} = 24 + 15 = 39 \neq 0$ .

Since  $D \neq 0$ , therefore the given system of homogeneous equations has only zero solution :

Now  $D_1 = \begin{vmatrix} 0 & -5 \\ 0 & 5 \end{vmatrix} = 0, D_2 = \begin{vmatrix} 3 & 0 \\ 3 & 0 \end{vmatrix} = 0$ .

$$\therefore x = \frac{D_1}{D} = 0, y = \frac{D_2}{D} = 0.$$

## 2.12 System of Linear Non-homogeneous Equations in Three Unknowns. (Cramer's Rule)

The solution of the system of non-homogeneous linear equations

$$a_1 x + b_1 y + c_1 z = d_1, a_2 x + b_2 y + c_2 z = d_2,$$

$$a_3 x + b_3 y + c_3 z = d_3$$

is given by  $x = \frac{D_1}{D}, y = \frac{D_2}{D}, z = \frac{D_3}{D}$ , where

$$D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}, D_1 = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix},$$

$$D_2 = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}, D_3 = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}, D \neq 0.$$

**Proof.** Let  $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ .

Then  $x D = \begin{vmatrix} xa_1 & b_1 & c_1 \\ xa_2 & b_2 & c_2 \\ xa_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 x + b_1 y + c_1 z & b_1 & c_1 \\ a_2 x + b_2 y + c_2 z & b_2 & c_2 \\ a_3 x + b_3 y + c_3 z & b_3 & c_3 \end{vmatrix}$

applying  $C_1 \rightarrow C_1 + yC_2 + zC_3$

$$= \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix} = D_1, \text{ say.}$$

Similarly,  $yD = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix} = D_2$ ,  $zD = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix} = D_3$ .

$\therefore x = \frac{D_1}{D}, y = \frac{D_2}{D}, z = \frac{D_3}{D}$ , provided  $D \neq 0$ .

This method of solving a system of three non-homogeneous linear equations in three unknowns can be used in exactly the same way to solve a system of  $n$  simultaneous linear non-homogeneous equations in  $n$  unknowns.

A system of 3 simultaneous linear non-homogeneous equations in three unknowns is

(i) **consistent** and has a unique solution given by

$$x = \frac{D_1}{D}, y = \frac{D_2}{D}, z = \frac{D_3}{D} \text{ if } D \neq 0.$$

(ii) **consistent** and has infinitely many solutions if

$$D = 0 = D_1 = D_2 = D_3.$$

(iii) **inconsistent** if  $D = 0$  but at least one of  $D_1, D_2$  and  $D_3$  is non-zero.

### System Of Homogeneous Linear Equations In Three Unknowns.

Consider the system

$$\begin{aligned} a_1 x + b_1 y + c_1 z &= 0 \\ a_2 x + b_2 y + c_2 z &= 0 \\ a_3 x + b_3 y + c_3 z &= 0. \end{aligned}$$

This system is always **consistent** because  $x = 0, y = 0, z = 0$  is always a solution of this system.

If  $D = 0$ , this system has a non-zero solution and in this case infinite number of solutions exist.

If  $D \neq 0$ , zero solution  $x = 0, y = 0, z = 0$  is the only solution of the system.

## Illustrative Examples

**Illustration 1 :** Solve using Cramer's rule

$$x + y = 5, \quad y + z = 3, \quad z + x = 4. \quad [\text{UPTU 2001, 07}]$$

**Solution :** Let  $D = \begin{vmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{vmatrix} = 1(1 - 0) - 1(0 - 1) = 1 + 1 = 2$ .

Since  $D \neq 0$ , therefore the given system has a unique solution given by

$$\frac{x}{D_1} = \frac{y}{D_2} = \frac{z}{D_3} = \frac{1}{D}.$$

$$\text{Now } D_1 = \begin{vmatrix} 5 & 1 & 0 \\ 3 & 1 & 1 \\ 4 & 0 & 1 \end{vmatrix} = 5(1-0) - 1(3-4) = 5 + 1 = 6,$$

$$D_2 = \begin{vmatrix} 1 & 5 & 0 \\ 0 & 3 & 1 \\ 1 & 4 & 1 \end{vmatrix} = 1(3-4) - 5(0-1) = -1 + 5 = 4$$

$$\text{and } D_3 = \begin{vmatrix} 1 & 1 & 5 \\ 0 & 1 & 3 \\ 1 & 0 & 4 \end{vmatrix} = 1(4-0) - 1(0-3) + 5(0-1) = 4 + 3 - 5 = 2.$$

∴ The solution of the given system is

$$x = \frac{D_1}{D} = \frac{6}{2} = 3, y = \frac{D_2}{D} = \frac{4}{2} = 2, z = \frac{D_3}{D} = \frac{2}{2} = 1.$$

Hence, the required solution is  $x = 3, y = 2, z = 1$ .

**Illustration 2 :** Solve the following by using Cramer's rule

$$x - 2y + 3z = 2, 2x - 3z = 3, x + y + z = 6. \quad [\text{UPTU 2002}]$$

**Solution :** Let  $D = \begin{vmatrix} 1 & -2 & 3 \\ 2 & 0 & -3 \\ 1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 2 & 4 & -9 \\ 1 & 3 & -2 \end{vmatrix}$ , by  $R_2 + 2R_1, R_3 - 3R_1$

$$= \begin{vmatrix} 4 & -9 \\ 3 & -2 \end{vmatrix} = -8 + 27 = 19 \neq 0.$$

Since  $D \neq 0$ , therefore the given system has a unique solution given by

$$\frac{x}{D_1} = \frac{y}{D_2} = \frac{z}{D_3} = \frac{1}{D}.$$

Now,  $D_1 = \begin{vmatrix} 2 & -2 & 3 \\ 3 & 0 & -3 \\ 6 & 1 & 1 \end{vmatrix} = \begin{vmatrix} 2 & 0 & 5 \\ 3 & 3 & 0 \\ 6 & 7 & 7 \end{vmatrix}$ , by  $R_2 + R_1, R_3 + R_1$

$$= 2 \begin{vmatrix} 3 & 0 \\ 7 & 7 \end{vmatrix} + 0 + 5 \begin{vmatrix} 3 & 3 \\ 6 & 7 \end{vmatrix}$$

$$= 2(21-0) + 5(21-18) = 42 + 15 = 57,$$

$$D_2 = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & -3 \\ 1 & 6 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 2 & -1 & -9 \\ 1 & 4 & -2 \end{vmatrix}, \text{ by } R_2 - 2R_1, R_3 - 3R_1$$

$$= 1 \begin{vmatrix} -1 & -9 \\ 4 & -2 \end{vmatrix} = 2 + 36 = 38,$$

and 
$$D_3 = \begin{vmatrix} 1 & -2 & 2 \\ 2 & 0 & 3 \\ 1 & 1 & 6 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 2 & 4 & -1 \\ 1 & 3 & 4 \end{vmatrix}, \text{ by } R_2 + 2R_1, R_3 - 2R_1$$

$$= 1 \begin{vmatrix} 4 & -1 \\ 3 & 4 \end{vmatrix} = 16 + 3 = 19.$$

∴ The solution of the given system is

$$x = \frac{D_1}{D} = \frac{57}{19} = 3, y = \frac{D_2}{D} = \frac{38}{19} = 2, z = \frac{D_3}{D} = \frac{19}{19} = 1.$$

Hence, the required solution is  $x = 3, y = 2, z = 1$ .

**Illustration 3 :** Solve the following system of linear equations with the help of Cramer's rule :

$$x + 2y + 3z = 6, 2x + 4y + z = 7, 2x + 2y + 9z = 14. \quad [\text{UPTU 2008}]$$

**Solution :** Let  $D = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 1 \\ 2 & 2 & 9 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 3 \\ 0 & 0 & -5 \\ 0 & -2 & 3 \end{vmatrix} \text{ by } R_2 - 2R_1, R_3 - 2R_1$

$$= \begin{vmatrix} 0 & -5 \\ -2 & 3 \end{vmatrix} = -10 \neq 0.$$

Since  $D \neq 0$ , therefore the given system has a unique solution given by

$$\frac{x}{D_1} = \frac{y}{D_2} = \frac{z}{D_3} = \frac{1}{D}.$$

Now  $D_1 = \begin{vmatrix} 6 & 2 & 3 \\ 7 & 4 & 1 \\ 14 & 2 & 9 \end{vmatrix} = \begin{vmatrix} 6 & 2 & 3 \\ -5 & 0 & -5 \\ 8 & 0 & 6 \end{vmatrix} \text{ by } R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1$

$$= -2 \begin{vmatrix} -5 & -5 \\ 8 & 6 \end{vmatrix} = -2(-30 + 40) = -20,$$

$$D_2 = \begin{vmatrix} 1 & 6 & 3 \\ 2 & 7 & 1 \\ 2 & 14 & 9 \end{vmatrix} = \begin{vmatrix} 1 & 6 & 3 \\ 0 & -5 & -5 \\ 0 & 2 & 3 \end{vmatrix} \text{ by } R_2 \rightarrow R_2 - 2R_1, R_3 \rightarrow R_3 - 2R_1$$

$$= \begin{vmatrix} -5 & -5 \\ 2 & 3 \end{vmatrix} = -15 + 10 = -5,$$

and  $D_3 = \begin{vmatrix} 1 & 2 & 6 \\ 2 & 4 & 7 \\ 2 & 2 & 14 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 6 \\ 0 & 0 & -5 \\ 0 & -2 & 2 \end{vmatrix} \text{ by } R_2 \rightarrow R_2 - 2R_1, R_3 \rightarrow R_3 - 2R_1$

$$= \begin{vmatrix} 0 & -5 \\ -2 & 2 \end{vmatrix} = -10.$$

∴ The solution of the given system is

$$x = \frac{D_1}{D} = \frac{-20}{-10} = 2, y = \frac{D_2}{D} = \frac{-5}{-10} = \frac{1}{2}, z = \frac{D_3}{D} = \frac{-10}{-10} = 1.$$

Hence, the required solution is  $x = 2, y = \frac{1}{2}, z = 1$ .

**Illustration 4 :** Solve the following system of linear equations with the help of Cramer's rule :

$$3x - 4y + 5z = -6, x + y - 2z = -1, 2x + 3y + z = 5. [\text{UPTU 2004}]$$

**Solution :** Let  $D = \begin{vmatrix} 3 & -4 & 5 \\ 1 & 1 & -2 \\ 2 & 3 & 1 \end{vmatrix} = 3(1 + 6) + 4(1 + 4) + 5(3 - 2)$

$$= 3 \times 7 + 4 \times 5 + 5 \times 1 = 21 + 20 + 5 = 46 \neq 0.$$

Since  $D \neq 0$ , therefore the given system has a unique solution given by

$$\frac{x}{D_1} = \frac{y}{D_2} = \frac{z}{D_3} = \frac{1}{D}.$$

Now,  $D_1 = \begin{vmatrix} -6 & -4 & 5 \\ -1 & 1 & -2 \\ 5 & 3 & 1 \end{vmatrix}$ , by  $R_1 \rightarrow R_1 + 4R_2, R_3 \rightarrow R_3 - 3R_2$

$$= \begin{vmatrix} -10 & 0 & -3 \\ -1 & 1 & -2 \\ 8 & 0 & 7 \end{vmatrix} = \begin{vmatrix} -10 & -3 \\ 8 & 7 \end{vmatrix} = -70 + 24 = -46,$$

$$D_2 = \begin{vmatrix} 3 & -6 & 5 \\ 1 & -1 & -2 \\ 2 & 5 & 1 \end{vmatrix}, R_1 \rightarrow R_1 - 3R_2, R_3 \rightarrow R_3 - 2R_2$$

$$= \begin{vmatrix} 0 & -3 & 11 \\ 1 & -1 & -2 \\ 0 & 7 & 5 \end{vmatrix} = - \begin{vmatrix} -3 & 11 \\ 7 & 5 \end{vmatrix} = -(-15 - 77) = 92$$

and  $D_3 = \begin{vmatrix} 3 & -4 & -6 \\ 1 & 1 & -1 \\ 2 & 3 & 5 \end{vmatrix}, R_1 \rightarrow R_1 - 3R_2, R_3 \rightarrow R_3 - 2R_2$

$$= \begin{vmatrix} 0 & -7 & -3 \\ 1 & 1 & -1 \\ 0 & 1 & 7 \end{vmatrix} = - \begin{vmatrix} -7 & -3 \\ 1 & 7 \end{vmatrix} = -(-49 + 3) = 46.$$

$\therefore$  The solution of the given system is

$$x = \frac{D_1}{D} = \frac{-46}{46} = -1, y = \frac{D_2}{D} = \frac{92}{46} = 2, z = \frac{D_3}{D} = \frac{46}{46} = 1.$$

Hence, the required solution is  $x = -1, y = 2, z = 1$ .

**Illustration 5 :** If the system of equations  $x = cy + bz, y = az + cx, z = bx + ay$  are consistent, having non-zero solution, show that  $a^2 + b^2 + c^2 + 2abc = 1$ .

**Solution :** The given system of equation can be written as

$$x - cy - bz = 0, cx - y + az = 0, bx + ay - z = 0.$$



If the system of equations are consistent and have non-zero solution, then

$$D = 0 \quad \text{or} \quad \begin{vmatrix} 1 & -c & -b \\ c & -1 & a \\ b & a & -1 \end{vmatrix} = 0$$

or  $1(1 - a^2) + c(-c - ab) - b(ac + b) = 0$

or  $1 - a^2 - c^2 - abc - abc - b^2 = 0$

or  $a^2 + b^2 + c^2 + 2abc = 1.$

**Illustration 6 :** If  $a, b, c$  are all different, solve the system of equations:

$$x + y + z = 1, ax + by + cz = k, a^2x + b^2y + c^2z = k^2.$$

**Solution :** Let  $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}$

$$= \begin{vmatrix} 1 & 0 & 0 \\ a & b-a & c-a \\ a^2 & b^2-a^2 & c^2-a^2 \end{vmatrix} \begin{array}{l} \text{by } C_2 \rightarrow C_2 - C_1, \\ C_3 \rightarrow C_3 - C_1 \end{array}$$

$$= \begin{vmatrix} b-a & c-a \\ b^2-a^2 & c^2-a^2 \end{vmatrix} = (b-a)(c-a) \begin{vmatrix} 1 & 1 \\ b+a & c+a \end{vmatrix}$$

$$= (b-a)(c-a) \{c+a - (b+a)\} = (b-a)(c-a)(c-b)$$

$$= (a-b)(b-c)(c-a) \neq 0, \text{ since } a, b, c \text{ are all different.}$$

$\therefore$  The given system has a unique solution given by

$$\frac{x}{D_1} = \frac{y}{D_2} = \frac{z}{D_3} = \frac{1}{D}.$$

Now  $D_1 = \begin{vmatrix} 1 & 1 & 1 \\ k & b & c \\ k^2 & b^2 & c^2 \end{vmatrix} = (k-b)(b-c)(c-k),$

$$D_2 = \begin{vmatrix} 1 & 1 & 1 \\ a & k & c \\ a^2 & k^2 & c^2 \end{vmatrix} = (a-k)(k-c)(c-a),$$

and  $D_3 = \begin{vmatrix} 1 & 1 & 1 \\ a & b & k \\ a^2 & b^2 & k^2 \end{vmatrix} = (a-b)(b-k)(k-a).$

Hence 
$$x = \frac{(k-b)(b-c)(c-k)}{(a-b)(b-c)(c-a)} = \frac{(k-b)(c-k)}{(a-b)(c-a)},$$
$$y = \frac{(a-k)(k-c)}{(a-b)(b-c)}, \quad z = \frac{(b-k)(k-a)}{(b-c)(c-a)}.$$

## Comprehensive Exercise 3

Solve the following systems of linear equations by Cramer's rule :

1.  $5x + 7y = -2, 4x + 6y = -3.$
2.  $x + 2y = 1, 3x + y = 4.$
3.  $2x + 3y = 10, x + 6y = 4.$
4.  $2x - y = 1, 7x - 2y = -7.$

Solve the following systems of linear equations in three unknowns by Cramer's rule :

5.  $5x - 7y + z = 11, 6x - 8y - z = 15, 3x + 2y - 6z = 7.$
6.  $3x + y + z = 2, 2x - 4y + 3z = -1, 4x + y - 3z = -11.$
7.  $x - 4y - z = 11, 2x - 5y + 2z = 39, -3x + 2y + z = 1.$
8.  $2x - 3y + 4z = -9, -3x + 4y + 2z = -12, 4x - 2y - 3z = -3.$

## Answers

1.  $x = \frac{9}{2}, y = -\frac{7}{2}.$
2.  $x = \frac{7}{5}, y = -\frac{1}{5}.$
3.  $x = \frac{16}{3}, y = -\frac{2}{9}.$
3.  $x = -3, y = -7.$
5.  $x = 1, y = -1, z = -1.$
6.  $x = -1, y = 2, z = 3.$
7.  $x = -1, y = -5, z = 8.$
8.  $x = -\frac{342}{53}, y = -\frac{321}{53}, z = -\frac{189}{53}.$

## Comprehensive Exercise 4

### Fill in the Blanks

Fill in the blanks ".....", so that the following statements are complete and correct.

1. If  $\begin{vmatrix} 3x & 7 \\ 2 & 4 \end{vmatrix} = 10$ , then the value of  $x$  is .....

2. Let  $AX = B$  be a system of linear equations in  $n$  unknowns. Then, the system has a consistent and unique solution if .....
3. A determinant of order two has ..... rows and ..... columns.
4. The solution of the system of linear equations;  $x - 2y = 4$ ,  $3x + 4y = 5$  is  $x = \dots\dots\dots$  and  $y = \dots\dots\dots$ .
5. If two rows or two columns of a determinant are identical, then the value of the determinant is .....
6. The points  $(1, 5)$ ,  $(k, 1)$  and  $(11, 7)$  are collinear if  $k = \dots\dots\dots$ .
7. The value of a determinant  $\begin{vmatrix} x+y & 1 \\ x^3 & x^2 - xy + y^2 \end{vmatrix}$  is .....
8. If all the elements of any row or any column of a determinant are zero, the value of the determinant is .....
9. The value of a determinant  $\begin{vmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{vmatrix}$  is .....

### Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

10. The minor of the element 5 in the determinant  $\begin{vmatrix} 5 & 7 \\ 2 & 4 \end{vmatrix}$  is  
 (a) 4 (b) -4  
 (c) 7 (d) 2
11. The cofactor of the element 2 in the determinant  $\begin{vmatrix} 6 & 7 \\ 2 & 4 \end{vmatrix}$  is  
 (a) 7 (b) -7  
 (c) 4 (d) 6
12. The value of the determinant  $\begin{vmatrix} \sin 10^\circ & -\cos 10^\circ \\ \sin 80^\circ & \cos 80^\circ \end{vmatrix}$  is  
 (a) 1 (b) -1  
 (c) 0 (d) none of these
13. The value of the determinant  $\begin{vmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \\ \omega^2 & 1 & \omega \end{vmatrix}$ , where  $\omega$  is a complex cube root of unity is  
 (a) 1 (b) 0  
 (c)  $\omega^2$  (d)  $\omega$

14. The value of the determinant  $\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$  is
- (a)  $a_{12} a_{21} - a_{11} a_{22}$  (b)  $a_{11} a_{22} - a_{12} a_{21}$   
 (c)  $a_{11} a_{21} - a_{12} a_{22}$  (d)  $a_{11} a_{12} - a_{21} a_{22}$
15. The order and value of  $\begin{vmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{vmatrix}$  are respectively
- (a) 2, 1 (b) 1, 2  
 (c) 0, 2 (d) 1, 0
16. If  $\omega$  is a cube root of unity and  $\Delta = \begin{vmatrix} 1 & 2\omega \\ \omega & \omega^2 \end{vmatrix}$ , then  $\Delta$  is
- (a)  $\omega^2$  (b)  $-\omega^2$   
 (c) 1 (d)  $\omega$

### True or False

Write 'T' for true and 'F' for false statement.

17. The order and value of  $\begin{vmatrix} 4 & -3 \\ 2 & -1 \end{vmatrix}$  are  $2 \times 2$  and 2 respectively. [UPTU 2009]
18. The value of a determinant does not change when rows and columns are interchanged.
19. If a determinant  $\Delta$  becomes zero when we put  $x = \alpha$  in it, then  $(x - \alpha)$  is a factor of  $\Delta$ .
20. If  $\Delta_1 = \begin{vmatrix} 1 & 0 \\ a & b \end{vmatrix}$  and  $\Delta_2 = \begin{vmatrix} 1 & 0 \\ c & d \end{vmatrix}$  then  $\Delta_1 \Delta_2 = bd$ .
21. The value of the determinant  $\begin{vmatrix} 0 & a & -b \\ -a & 0 & c \\ b & -c & 0 \end{vmatrix} = 0$ .
22. The system of equations  $8x + 6y = 1$ ,  $4x + 3y = 5$  is inconsistent.
23. In a determinant the number of rows and columns may be different.
24. The co-factors of the elements in the 3rd row of the determinant  $\begin{vmatrix} 1 & 2 & -3 \\ 2 & 3 & 2 \\ 3 & -3 & -4 \end{vmatrix}$  are 13, -8, -1.

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## Answers

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- |         |                                     |          |           |
|---------|-------------------------------------|----------|-----------|
| 1. 2    | 2. $ A  \neq 0$                     | 3. 2, 2  | 4. -6, -5 |
| 5. zero | 6. 6                                | 7. $y^3$ | 8. zero   |
| 9. 0    | 10. (a)                             | 11. (b)  | 12. (a)   |
| 13. (b) | 14. $a_{11} a_{22} - a_{12} a_{21}$ | 15. (a)  |           |
| 16. (b) | 17. $T$                             | 18. $T$  | 19. $T$   |
| 20. $T$ | 21. $T$                             | 22. $T$  | 23. $F$   |
| 24. $T$ |                                     |          |           |



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## Chapter

# 3



# Matrices

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## 3.1 Matrix

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**Definition.** A set of  $mn$  numbers (real or complex) arranged in the form of a rectangular array having  $m$  rows and  $n$  columns is called an  $m \times n$  matrix [to be read as ‘ $m$  by  $n$ ’ matrix].

Thus a rectangular array of  $mn$  numbers of the form

$$\mathbf{A} = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ a_{31} & a_{32} & \cdots & a_{3n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{vmatrix}$$

is called an  $m \times n$  matrix.

The numbers  $a_{11}, a_{12}, \dots, a_{mn}$  etc. are called the elements of the matrix. The horizontal lines are called **rows** or **row vectors** and the vertical lines are called **columns** or **column vectors** of the matrix. The element  $a_{ij}$  belongs to the  $i$ th row

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and the  $j$ th column and is called the  $(i, j)^{\text{th}}$  element of the matrix. Thus in the element  $a_{ij}$  the first suffix  $i$  will always denote the number of the row and the second suffix  $j$ , the number of the column in which the element occurs.

In a compact form the above matrix is represented by  $\mathbf{A} = [a_{ij}]$ ,  $i = 1, 2, \dots, m$ ,  $j = 1, 2, \dots, n$  or simply by  $\mathbf{A} = [a_{ij}]_{m \times n}$ . Besides the square brackets  $[ ]$ , sometimes we also use parentheses  $( )$  or double bars  $|| \ ||$  to represent a matrix.

We use capital bold or italic letters to denote matrices.

In a matrix, the number of rows and columns need not be equal. Further, a matrix has no numerical value. It is just an ordered collection of numbers arranged in the form of a rectangular array.

## 3.2 Special Types of Matrices

**Square Matrix.** An  $m \times n$  matrix for which  $m = n$  (i.e., the number of rows is equal to the number of columns) is called a square matrix of order  $n$ .

A square matrix of order  $n$  is also called an  $n$ -rowed square matrix. Thus in a square matrix, we have the same number of rows and columns.

The elements  $a_{ij}$  of a square matrix  $\mathbf{A} = [a_{ij}]_{n \times n}$  for which  $i = j$  i.e., the elements  $a_{11}, a_{22}, a_{33}, \dots, a_{nn}$  are called the **diagonal elements** and the line along which they lie is called the **principal diagonal** of the matrix.

**Example.** The matrix  $\mathbf{A} = \begin{bmatrix} 3 & 0 & 1 & 2 \\ 4 & 1 & 7 & 0 \\ 0 & 0 & 5 & 1 \\ 1 & 3 & 9 & 1 \end{bmatrix}_{4 \times 4}$

is a square matrix of order 4. The elements 3, 1, 5, 1 constitute the principal diagonal of the matrix  $\mathbf{A}$ .

**Diagonal Matrix.** A square matrix each of whose non-diagonal elements is equal to zero is called a **diagonal matrix**.

Thus the square matrix  $\mathbf{A} = [a_{ij}]_{n \times n}$  is a diagonal matrix if  $a_{ij} = 0$  for  $i \neq j$ .

A diagonal matrix of order  $n$  is denoted by

$$\text{diag } [a_{11}, a_{22}, \dots, a_{nn}].$$

For example,  $\begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}$  and  $\begin{bmatrix} 3 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 1 \end{bmatrix}$  are the diagonal matrices of order 2

and 3 respectively.

**Scalar Matrix.** A square matrix each of whose non-diagonal elements is equal to zero and all the diagonal elements are equal is called a **scalar matrix**.

For example,  $\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$ ,  $\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$  are the scalar matrices of order 2 and 3

respectively.

**Unit Matrix.** A square matrix each of whose diagonal elements is 1 and each of whose non-diagonal elements is equal to zero is called a **unit matrix** or an **identity matrix** and is denoted by **I**.

The unit matrix of order  $n$  is denoted by  $\mathbf{I}_n$ . Thus a square matrix  $\mathbf{A} = [a_{ij}]$  is a unit matrix if  $a_{ij} = 1$  when  $i = j$  and  $a_{ij} = 0$  when  $i \neq j$ .

For example,  $\mathbf{I}_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$  and  $\mathbf{I}_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

are identity matrices of order 4 and 3 respectively.

**Row Matrix.** Any  $1 \times n$  matrix which has only one row and  $n$  columns is called a **row matrix** or a **row vector**.

For example,  $[1 \ 2 \ 7]$  and  $[0 \ 3 \ 1 \ 4 \ 9]$  are the row matrices of order  $1 \times 3$  and  $1 \times 5$  respectively.

**Column Matrix.** Any  $m \times 1$  matrix which has  $m$  rows and only one column is a **column matrix** or a **column vector**.

For example,  $\begin{bmatrix} 4 \\ -1 \\ 0 \end{bmatrix}$  and  $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$  are column matrices of order  $3 \times 1$  and  $2 \times 1$

respectively.

**Null Matrix.** The  $m \times n$  matrix whose elements are all 0 is called the **null matrix** or a **zero matrix** of the type  $m \times n$ .

The null matrix of order  $m \times n$  is usually denoted by **O** or by  $\mathbf{O}_{m,n}$ .

For example,  $\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$  and  $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

are zero matrices of order  $3 \times 4$  and  $2 \times 2$  respectively.

**Upper Triangular Matrix.** A square matrix  $\mathbf{A} = [a_{ij}]$  is called an **upper triangular matrix** if  $a_{ij} = 0$  whenever  $i > j$ .

Thus in an upper triangular matrix all the elements below the principal diagonal are zero.



For example, 
$$\begin{bmatrix} 5 & 0 & 1 & 2 \\ 0 & 4 & -3 & 1 \\ 0 & 0 & 1 & 7 \\ 0 & 0 & 0 & -2 \end{bmatrix} \text{ and } \begin{bmatrix} 1 & 7 & 4 \\ 0 & 6 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

are upper triangular matrices of order  $4 \times 4$  and  $3 \times 3$  respectively.

**Lower Triangular Matrix.** A square matrix  $\mathbf{A} = [a_{ij}]$  is called a lower triangular matrix if  $a_{ij} = 0$  whenever  $i < j$ .

Thus in a lower triangular matrix all the elements above the principal diagonal are zero.

For example, 
$$\begin{bmatrix} 1 & 0 \\ 7 & 2 \end{bmatrix} \text{ and } \begin{bmatrix} 2 & 0 & 0 & 0 \\ 3 & 1 & 0 & 0 \\ 0 & -2 & 9 & 0 \\ 4 & 5 & 1 & 1 \end{bmatrix}$$

are the lower triangular matrices of order 2 and 4 respectively.

### 3.3 Submatrices of a Matrix

**Definition.** Any matrix obtained by omitting some rows and columns from a given matrix  $\mathbf{A}$  is called a submatrix of  $\mathbf{A}$ .

The matrix  $\mathbf{A}$  itself is a submatrix of  $\mathbf{A}$  as it can be obtained from  $\mathbf{A}$  by omitting no rows or columns.

For example, 
$$\begin{bmatrix} 1 \\ 7 \\ 0 \end{bmatrix} \text{ and } \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 1 \end{bmatrix} \text{ are the submatrices of the matrix}$$

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 & 9 \\ 7 & 11 & 6 & 5 \\ 0 & 2 & 1 & 8 \end{bmatrix}.$$

### 3.4 Equality of Two Matrices

Two matrices  $\mathbf{A} = [a_{ij}]_{m \times n}$  and  $\mathbf{B} = [b_{ij}]_{p \times q}$  are said to be equal if

- (i) they are of the same order {i.e., if number of rows in  $\mathbf{A}$  equals the number of rows in  $\mathbf{B}$  ( $m = p$ ) and the number of columns in  $\mathbf{A}$  equals the number of columns in  $\mathbf{B}$  ( $n = q$ )}
- (ii) their corresponding elements are equal i.e.,  $a_{ij} = b_{ij}$  for each pair of subscripts  $i$  and  $j$ .

## Illustrative Examples

**Illustration 1 :** (i) Construct a  $2 \times 3$  matrix  $\mathbf{A}$  whose elements in the  $i$ th row and the  $j$ th column are given by  $a_{ij} = 2i - j$

(ii) Construct a  $2 \times 2$  matrix  $\mathbf{A} = [a_{ij}]$  whose elements are given by

$$a_{ij} = \frac{i-j}{i+j}$$

**Solution :** (i) We have  $a_{ij} = 2i - j$ ,  $1 \leq i \leq 2$  and  $1 \leq j \leq 3$ .

$$\therefore a_{11} = 2 \times 1 - 1 = 2 - 1 = 1, a_{12} = 2 \times 1 - 2 = 2 - 2 = 0$$

$$a_{13} = 2 - 3 = -1, a_{21} = 4 - 1 = 3, a_{22} = 4 - 2 = 2, a_{23} = 4 - 3 = 1$$

Hence the required matrix  $\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$

$$= \begin{bmatrix} 1 & 0 & -1 \\ 3 & 2 & 1 \end{bmatrix}.$$

(ii) We have  $a_{ij} = \frac{i-j}{1+j}$ ,  $1 \leq i \leq 2$  and  $1 \leq j \leq 2$ .

Now  $a_{11} = \frac{1-1}{1+1} = 0, a_{12} = \frac{1-2}{1+2} = -\frac{1}{3},$

$$a_{21} = \frac{2-1}{2+1} = \frac{1}{3}, a_{22} = \frac{2-2}{2+2} = 0$$

Hence the required matrix  $\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$

$$= \begin{bmatrix} 0 & -1/3 \\ 1/3 & 0 \end{bmatrix}.$$

**Illustration 2 :** If a matrix has 18 elements, what are the possible order it can have ?  
What, if it has 11 elements ?

**Solution :** Let the given matrix  $\mathbf{A}$  be of order  $m \times n$ .

Given  $mn = 18$

Factorising the possible values of  $(m, n)$  are  $(1, 18), (2, 9), (3, 6), (6, 3), (9, 2), (1, 8).$

$\therefore$  The possible orders matrix containing 18 elements are

$$1 \times 18, 2 \times 9, 3 \times 6, 6 \times 3, 9 \times 2, 1 \times 18.$$

Let the matrix  $\mathbf{B} = [b_{ij}]_{m \times n}$  contain 11 elements.

$\therefore m \times n = 11$ , (a prime number)

Thus the possible orders of  $\mathbf{B}$  are  $1 \times 11$  or  $11 \times 1$ .

**Illustration 3 :** Find the values of  $a, b, c$  and  $d$  so that the matrices **A** and **B** may be equal, where

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 1 & -1 \\ 0 & 3 \end{bmatrix}.$$

**Solution :** The matrices **A** and **B** are of the same order  $2 \times 2$ . If  $\mathbf{A} = \mathbf{B}$ , the corresponding elements of **A** and **B** must be same.

Hence, we have  $a = 1, b = -1, c = 0, d = 3$ .

**Illustration 4 :** If matrices  $\begin{bmatrix} a+b & 2 \\ 5 & ab \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 5 & 8 \end{bmatrix}$ , find the values of  $a$  and  $b$ .

**Solution :** We know the corresponding elements of two equal matrices are equal. Therefore we have

$$a + b = 6 \quad \dots(1)$$

$$ab = 8. \quad \dots(2)$$

Substituting for  $b$  from (2) in (1), we get

$$a + \frac{8}{a} = 6 \quad \text{or} \quad a^2 + 8 = 6a \quad \text{or} \quad a^2 - 6a + 8 = 0$$

or  $(a - 4)(a - 2) = 0 \quad \text{or} \quad a = 2, 4$ .

From (2) we have

$$b = 4, \text{ when } a = 2 \text{ and } b = 2 \text{ when } a = 4.$$

**Illustration 5 :** If  $\begin{bmatrix} x-y & 2x+z \\ 2x-y & 3z+w \end{bmatrix} = \begin{bmatrix} -1 & 5 \\ 0 & 13 \end{bmatrix}$ , find  $x, y, z$  and  $w$ .

**Solution.** We know the corresponding elements of equal matrices are equal. Therefore we have

$$x - y = -1, \quad 2x + z = 5, \quad 2x - y = 0, \quad 3z + w = 13.$$

Solving first and third equations simultaneously, we get

$$x = 1, \quad y = 2.$$

Substituting  $x = 1$  in  $2x + z = 5$ , we get  $z = 3$ .

Again putting  $z = 3$  in  $3z + w = 13$ , we get  $w = 4$ .

Hence  $x = 1, y = 2, z = 3, w = 4$ .

**Illustration 6 :** For what values of  $x$  and  $y$  are the following matrices equal

$$\mathbf{A} = \begin{bmatrix} 2x+1 & 3y \\ 0 & y^2-5y \end{bmatrix}, \mathbf{B} = \begin{bmatrix} x+3 & y^2+2 \\ 0 & -6 \end{bmatrix}.$$

**Solution.** For matrices **A** and **B** to be equal their corresponding elements must be same. Thus we have

$$2x + 1 = x + 3, \quad 3y = y^2 + 2, \quad y^2 - 5y = -6.$$

From first equation, we get  $x = 2$ .

From second equation we get  $y^2 - 3y + 2 = 0$

or  $(y - 2)(y - 1) = 0$  or  $y = 1, 2$ .

Substituting these values of  $y$  in the third equation we observe that only  $y = 2$  satisfies it. Therefore  $y = 1$  is inadmissible.

Hence we have  $x = 2, y = 2$ .

## 3.5 Addition of Matrices

**Definition.** Let  $\mathbf{A}$  and  $\mathbf{B}$  be two matrices of the same type  $m \times n$ . Then their sum (to be denoted by  $\mathbf{A} + \mathbf{B}$ ) is defined to be the matrix of the type  $m \times n$  obtained by adding the corresponding elements of  $\mathbf{A}$  and  $\mathbf{B}$ .

Thus if  $\mathbf{A} = [a_{ij}]_{m \times n}$  and  $\mathbf{B} = [b_{ij}]_{m \times n}$  then  $\mathbf{A} + \mathbf{B} = [a_{ij} + b_{ij}]_{m \times n}$ .

We observe that addition is defined only for the matrices which are of the same order.

Two matrices  $\mathbf{A}$  and  $\mathbf{B}$  are said to be **conformable for addition** if they are of the same type.

If 
$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}_{m \times n}$$

and 
$$\mathbf{B} = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ b_{m1} & b_{m2} & \cdots & b_{mn} \end{bmatrix}_{m \times n},$$

then 
$$\mathbf{A} + \mathbf{B} = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \cdots & a_{1n} + b_{1n} \\ a_{21} + b_{21} & a_{22} + b_{22} & \cdots & a_{2n} + b_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{m1} + b_{m1} & a_{m2} + b_{m2} & \cdots & a_{mn} + b_{mn} \end{bmatrix}_{m \times n}.$$

**Important Note.** If the matrices  $\mathbf{A}$  and  $\mathbf{B}$  are not of the same order, we cannot define  $\mathbf{A} + \mathbf{B}$ .

## 3.6 Properties of Matrix Addition

(i) **Matrix Addition Is Commutative.**

If  $\mathbf{A}$  and  $\mathbf{B}$  be two  $m \times n$  matrices then  $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$ .

(ii) **Matrix Addition Is Associative.**

If  $\mathbf{A}, \mathbf{B}, \mathbf{C}$  be three matrices each of the type  $m \times n$ , then

$$\mathbf{A} + (\mathbf{B} + \mathbf{C}) = (\mathbf{A} + \mathbf{B}) + \mathbf{C}.$$

**(iii) Existence Of Additive Identity.**

If  $\mathbf{O}$  be the  $m \times n$  matrix each of whose elements is zero, then

$$\mathbf{A} + \mathbf{O} = \mathbf{A} = \mathbf{O} + \mathbf{A}, \text{ for every } m \times n \text{ matrix } \mathbf{A}.$$

**(iv) Existence Of The Additive Inverse.**

**Negative Of A Matrix. Definition.** Let  $\mathbf{A} = [a_{ij}]_{m \times n}$ . Then the negative of the matrix  $\mathbf{A}$  is defined as the matrix  $[-a_{ij}]_{m \times n}$  and is denoted by  $-\mathbf{A}$ .

If  $\mathbf{A} = [a_{ij}]_{m \times n}$  then there exists a matrix  $(-\mathbf{A}) = [-a_{ij}]_{m \times n}$  such that

$$\mathbf{A} + (-\mathbf{A}) = \mathbf{O} = -\mathbf{A} + \mathbf{A}.$$

**Subtraction Of Two Matrices. Definition.** If  $\mathbf{A}$  and  $\mathbf{B}$  are two  $m \times n$  matrices, then we define

$$\mathbf{A} - \mathbf{B} = \mathbf{A} + (-\mathbf{B}).$$

The difference  $\mathbf{A} - \mathbf{B}$  is obtained by subtracting from each element of  $\mathbf{A}$  the corresponding element of  $\mathbf{B}$ .

**(v) Cancellation Laws Hold Good In The Case Of Addition Of Matrices.**

If  $\mathbf{A}, \mathbf{B}, \mathbf{C}$  are three  $m \times n$  matrices, then

$$\mathbf{A} + \mathbf{B} = \mathbf{A} + \mathbf{C} \rightarrow \mathbf{B} = \mathbf{C} \quad (\text{Left cancellation law})$$

$$\text{and} \quad \mathbf{B} + \mathbf{A} = \mathbf{C} + \mathbf{A} \rightarrow \mathbf{B} = \mathbf{C} \quad (\text{Right cancellation law})$$

### 3.7 Multiplication of a Matrix by a Scalar

**Definition.** Let  $\mathbf{A}$  be any  $m \times n$  matrix and  $k$  any complex number called scalar. The  $m \times n$  matrix obtained by multiplying every element of the matrix  $\mathbf{A}$  by  $k$  is called the scalar multiple of  $\mathbf{A}$  by  $k$  and is denoted by  $k\mathbf{A}$  or  $\mathbf{A}k$ .

Thus if  $\mathbf{A} = [a_{ij}]_{m \times n}$  and  $k$  is any scalar, then

$$k\mathbf{A} = \begin{bmatrix} k a_{11} & k a_{12} & \cdots & k a_{1n} \\ k a_{21} & k a_{22} & \cdots & k a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ k a_{m1} & k a_{m2} & \cdots & k a_{mn} \end{bmatrix} = [k a_{ij}]_{m \times n}.$$

**Example :** If  $k = 2$  and  $\mathbf{A} = \begin{bmatrix} 3 & 2 & -1 \\ 4 & -3 & 1 \end{bmatrix}_{2 \times 3}$ ,

$$\text{then} \quad 2\mathbf{A} = \begin{bmatrix} 2 \times 3 & 2 \times 2 & 2 \times -1 \\ 2 \times 4 & 2 \times -3 & 2 \times 1 \end{bmatrix} = \begin{bmatrix} 6 & 4 & -2 \\ 8 & -6 & 2 \end{bmatrix}_{2 \times 3}.$$

### 3.8 Properties of Multiplication of a Matrix by a Scalar

If  $\mathbf{A} = [a_{ij}]_{m \times n}$  and  $\mathbf{B} = [b_{ij}]_{m \times n}$ ,  $k$  and  $l$  are two scalars, then

$$(i) \quad k(\mathbf{A} + \mathbf{B}) = k\mathbf{A} + k\mathbf{B} \quad (ii) \quad (k + l)\mathbf{A} = k\mathbf{A} + l\mathbf{A}$$

- (iii)  $(kl) \mathbf{A} = k(l\mathbf{A}) = l(k\mathbf{A})$  (iv)  $(-k) \mathbf{A} = -(k\mathbf{A}) = k(-\mathbf{A})$   
 (v)  $(-1) \mathbf{A} = -\mathbf{A}$ .

## Illustrative Examples

**Illustration 1 :** (i) If  $\mathbf{A} = \begin{bmatrix} 3 & 5 \\ 7 & -9 \end{bmatrix}$  and  $\mathbf{B} = \begin{bmatrix} 2 & -4 \\ 5 & 3 \end{bmatrix}$ , find  $3\mathbf{A} - 2\mathbf{B}$ .

(ii) If  $\mathbf{A} = \begin{bmatrix} 2 & 1 & 4 \\ 5 & -3 & 7 \end{bmatrix}$ ,  $\mathbf{B} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & -4 & 6 \end{bmatrix}$ , find  $2\mathbf{A} - 3\mathbf{B}$  and  $\mathbf{B} - \frac{\mathbf{A}}{2}$ .

**Solution :** (i) We have  $3\mathbf{A} = \begin{bmatrix} 3 \times 3 & 3 \times 5 \\ 3 \times 7 & 3 \times -9 \end{bmatrix} = \begin{bmatrix} 9 & 15 \\ 21 & -27 \end{bmatrix}$

and  $-2\mathbf{B} = \begin{bmatrix} -2 \times 2 & -2 \times -4 \\ -2 \times 5 & -2 \times 3 \end{bmatrix} = \begin{bmatrix} -4 & 8 \\ -10 & -6 \end{bmatrix}$ .

Now  $3\mathbf{A} - 2\mathbf{B} = 3\mathbf{A} + (-2\mathbf{B}) = \begin{bmatrix} 9-4 & 15-8 \\ 21-10 & -27-6 \end{bmatrix} = \begin{bmatrix} 5 & 7 \\ 11 & -33 \end{bmatrix}$ .

(ii) We have  $2\mathbf{A} = \begin{bmatrix} 2 \times 2 & 2 \times 1 & 2 \times 4 \\ 2 \times 5 & 2 \times (-3) & 2 \times 7 \end{bmatrix} = \begin{bmatrix} 4 & 2 & 8 \\ 10 & -6 & 14 \end{bmatrix}$ ,

$$-3\mathbf{B} = \begin{bmatrix} -3 \times 1 & -3 \times 2 & -3 \times 3 \\ -3 \times 2 & -3 \times (-4) & -3 \times 6 \end{bmatrix} = \begin{bmatrix} -3 & -6 & -9 \\ -6 & 12 & -18 \end{bmatrix}$$

and  $-\frac{\mathbf{A}}{2} = \begin{bmatrix} -\frac{2}{2} & -\frac{1}{2} & -\frac{4}{2} \\ -\frac{5}{2} & \frac{3}{2} & -\frac{7}{2} \end{bmatrix} = \begin{bmatrix} -1 & -\frac{1}{2} & -2 \\ -\frac{5}{2} & \frac{3}{2} & -\frac{7}{2} \end{bmatrix}$ .

Now  $2\mathbf{A} - 3\mathbf{B} = 2\mathbf{A} + (-3\mathbf{B})$   
 $= \begin{bmatrix} 4-3 & 2-6 & 8-9 \\ 10-6 & -6+12 & 14-18 \end{bmatrix} = \begin{bmatrix} 1 & -4 & -1 \\ 4 & 6 & -4 \end{bmatrix}$

and  $\mathbf{B} - \frac{\mathbf{A}}{2} = \mathbf{B} + \left(-\frac{\mathbf{A}}{2}\right) = \begin{bmatrix} 1-1 & 2-\frac{1}{2} & 3-2 \\ 2-\frac{5}{2} & -4+\frac{3}{2} & 6-\frac{7}{2} \end{bmatrix}$   
 $= \begin{bmatrix} 0 & 3/2 & 1 \\ -1/2 & -5/2 & 5/2 \end{bmatrix}$ .

**Illustration 2 :** If  $\mathbf{A} = \begin{bmatrix} 1 & -3 & 2 \\ 2 & 0 & 2 \end{bmatrix}$ ,  $\mathbf{B} = \begin{bmatrix} 2 & -1 & -1 \\ 1 & 0 & -1 \end{bmatrix}$ , find the matrix  $\mathbf{C}$

such that  $\mathbf{A} + \mathbf{B} + \mathbf{C}$  is a zero matrix.

**Solution :**  $\mathbf{A} + \mathbf{B} + \mathbf{C} = \mathbf{O} \Rightarrow \mathbf{A} + \mathbf{B} + \mathbf{C} + (-\mathbf{C}) = \mathbf{O} + (-\mathbf{C})$

$\Rightarrow \mathbf{A} + \mathbf{B} + \mathbf{O} = -\mathbf{C} \Rightarrow \mathbf{A} + \mathbf{B} = -\mathbf{C} \Rightarrow \mathbf{C} = -(\mathbf{A} + \mathbf{B})$ .

Now  $\mathbf{A} + \mathbf{B} = \begin{bmatrix} 1+2 & -3-1 & 2-1 \\ 2+1 & 0+0 & 2-1 \end{bmatrix} = \begin{bmatrix} 3 & -4 & 1 \\ 3 & 0 & 1 \end{bmatrix}.$

$\therefore \mathbf{C} = -(\mathbf{A} + \mathbf{B}) = \begin{bmatrix} -3 & 4 & -1 \\ -3 & 0 & -1 \end{bmatrix}$  is the required matrix.

**Illustration 3 :** Find  $x, y, z, t$  if

$$2 \begin{bmatrix} x & z \\ y & t \end{bmatrix} + 3 \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} = 3 \begin{bmatrix} 3 & 5 \\ 4 & 6 \end{bmatrix}.$$

**Solution :** We have

$$2 \begin{bmatrix} x & z \\ y & t \end{bmatrix} + 3 \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} = 3 \begin{bmatrix} 3 & 5 \\ 4 & 6 \end{bmatrix}$$

or  $\begin{bmatrix} 2x & 2z \\ 2y & 2t \end{bmatrix} + \begin{bmatrix} 3 & -3 \\ 0 & 6 \end{bmatrix} = \begin{bmatrix} 9 & 15 \\ 12 & 18 \end{bmatrix}$

or  $\begin{bmatrix} 2x+3 & 2z-3 \\ 2y & 2t+6 \end{bmatrix} = \begin{bmatrix} 9 & 15 \\ 12 & 18 \end{bmatrix}$

or  $2x+3=9, 2z-3=15, 2y=12, 2t+6=18$

or  $x=3, z=9, y=6, t=6.$

**Illustration 4 :** Find  $\mathbf{X}$  and  $\mathbf{Y}$  if  $\mathbf{X} + \mathbf{Y} = \begin{bmatrix} 7 & 0 \\ 2 & 5 \end{bmatrix}$  and  $\mathbf{X} - \mathbf{Y} = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}.$

**Solution :** We have

$$\mathbf{X} + \mathbf{Y} + (\mathbf{X} - \mathbf{Y}) = \begin{bmatrix} 7 & 0 \\ 2 & 5 \end{bmatrix} + \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 7+3 & 0+0 \\ 2+0 & 5+3 \end{bmatrix}$$

or  $2\mathbf{X} = \begin{bmatrix} 10 & 0 \\ 2 & 8 \end{bmatrix}$

or  $\mathbf{X} = \frac{1}{2} \begin{bmatrix} 10 & 0 \\ 2 & 8 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ 1 & 4 \end{bmatrix}.$

Now  $\mathbf{X} + \mathbf{Y} = \begin{bmatrix} 7 & 0 \\ 2 & 5 \end{bmatrix} \rightarrow \mathbf{Y} = -\mathbf{X} + \begin{bmatrix} 7 & 0 \\ 2 & 5 \end{bmatrix}$

$\Rightarrow \mathbf{Y} = \begin{bmatrix} -5 & 0 \\ -1 & -4 \end{bmatrix} + \begin{bmatrix} 7 & 0 \\ 2 & 5 \end{bmatrix} = \begin{bmatrix} -5+7 & 0+0 \\ -1+2 & -4+5 \end{bmatrix}$   
 $= \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}.$

**Illustration 5 :** If  $\mathbf{A} + 2\mathbf{B} = \begin{bmatrix} 4 & 2 & 3 \\ 2 & 1 & 5 \end{bmatrix}$  and  $2\mathbf{A} - \mathbf{B} = \begin{bmatrix} 7 & 6 & 4 \\ 2 & 3 & 1 \end{bmatrix},$

determine the matrices  $\mathbf{A}$  and  $\mathbf{B}.$

**Solution :** We have

$$2(2\mathbf{A} - \mathbf{B}) + \mathbf{A} + 2\mathbf{B} = 2 \begin{bmatrix} 7 & 6 & 4 \\ 2 & 3 & 1 \end{bmatrix} + \begin{bmatrix} 4 & 2 & 3 \\ 2 & 1 & 5 \end{bmatrix}$$

$$\text{or} \quad 5\mathbf{A} = \begin{bmatrix} 14+4 & 12+2 & 8+3 \\ 4+2 & 6+1 & 2+5 \end{bmatrix}$$

$$\text{or} \quad \mathbf{A} = \frac{1}{5} \begin{bmatrix} 18 & 14 & 11 \\ 6 & 7 & 7 \end{bmatrix}.$$

$$\text{Now} \quad 2\mathbf{A} - \mathbf{B} - 2(\mathbf{A} + 2\mathbf{B}) = \begin{bmatrix} 7 & 6 & 4 \\ 2 & 3 & 1 \end{bmatrix} - 2 \begin{bmatrix} 4 & 2 & 3 \\ 2 & 1 & 5 \end{bmatrix}$$

$$\text{or} \quad -5\mathbf{B} = \begin{bmatrix} 7-8 & 6-4 & 4-6 \\ 2-4 & 3-2 & 1-10 \end{bmatrix}$$

$$\text{or} \quad \mathbf{B} = -\frac{1}{5} \begin{bmatrix} -1 & 2 & -2 \\ -2 & 1 & -9 \end{bmatrix}.$$

## Comprehensive Exercise 1

1. A matrix has 24 elements. What are the possible order or dimensions it can have? What if it has 13 elements?
2. Construct a  $m \times n$  matrix  $\mathbf{A} = [a_{ij}]$  whose element  $a_{ij}$  is given by

$$(i) \ a_{ij} = \frac{3i-j}{2}, \ m=2, \ n=3$$

$$(ii) \ a_{ij} = \frac{(i+2j)^2}{2}, \ m=3, \ n=2$$

$$(iii) \ a_{ij} = \frac{(i+j)^2}{2}, \ m=3, \ n=4$$

3. For what values of  $x$  and  $y$  are the following matrices equal?

$$(i) \ \mathbf{A} = \begin{bmatrix} 5 & x^2 \\ 2y^3 & -17 \end{bmatrix}, \ \mathbf{B} = \begin{bmatrix} 5 & 49 \\ 128 & -17 \end{bmatrix}$$

$$(ii) \ \mathbf{A} = \begin{bmatrix} 2x+1 & 2y \\ 0 & y^2-5y \end{bmatrix}, \ \mathbf{B} = \begin{bmatrix} x+3 & y^2+2 \\ 0 & -6 \end{bmatrix}.$$

4. Find  $a, b, c, d$  if

$$(i) \ \begin{bmatrix} 2a-3b & c-d & 3 \\ 1 & a+4b & 3c+4d \end{bmatrix} = \begin{bmatrix} 1 & -2 & 3 \\ 1 & 6 & 29 \end{bmatrix}$$

$$(ii) \ \begin{bmatrix} 3a+4b & 2 & a-2b \\ c+d & 2c-d & -1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 4 \\ 5 & -5 & -1 \end{bmatrix}.$$

5. Find the values of  $x$  and  $y$  from the following equation :

$$2 \begin{bmatrix} x & 5 \\ 7 & y-3 \end{bmatrix} + \begin{bmatrix} 3 & -4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}.$$



6. Find  $x$ ,  $y$  and  $z$  so that  $\mathbf{A} = \mathbf{B}$ , where

$$\mathbf{A} = \begin{bmatrix} x-2 & 3 & 2z \\ 18z & y+2 & 6z \end{bmatrix}, \mathbf{B} = \begin{bmatrix} y & z & 6 \\ 6y & x & 2y \end{bmatrix}.$$

7. If  $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$ ,  $\mathbf{B} = \begin{bmatrix} 0 & 3 \\ -1 & 7 \end{bmatrix}$ ,  $\mathbf{C} = \begin{bmatrix} 2 & 9 \\ -4 & 7 \end{bmatrix}$ , find

$$\mathbf{A} + \mathbf{B}, \mathbf{A} - \mathbf{C}, 2\mathbf{A} + 3\mathbf{B}, \mathbf{B} - 5\mathbf{C}.$$

8. If  $\mathbf{A} = \begin{bmatrix} 2 & 2 \\ -3 & 1 \\ 4 & 0 \end{bmatrix}$ ,  $\mathbf{B} = \begin{bmatrix} 6 & 2 \\ 1 & 3 \\ 0 & 4 \end{bmatrix}$ , find the matrix  $\mathbf{C}$  such that

$$\mathbf{A} + \mathbf{B} + \mathbf{C} \text{ is a zero matrix.}$$

9. If  $\mathbf{A} = \begin{bmatrix} 1 & 0 & 3 \\ 2 & -1 & 2 \\ 7 & 1 & 5 \end{bmatrix}$ ,  $\mathbf{B} = \begin{bmatrix} 3 & 1 & 5 \\ 1 & 2 & 3 \\ 0 & 4 & 7 \end{bmatrix}$ ,  $\mathbf{C} = \begin{bmatrix} 4 & 6 & 1 \\ 0 & 1 & 7 \\ 0 & 2 & 1 \end{bmatrix}$ , verify that

$$\mathbf{A} + (\mathbf{B} + \mathbf{C}) = (\mathbf{A} + \mathbf{B}) + \mathbf{C}.$$

10. Find matrices  $\mathbf{A}$  and  $\mathbf{B}$  if  $\mathbf{A} + \mathbf{B} = \begin{bmatrix} 5 & 2 \\ 0 & 9 \end{bmatrix}$ , and  $\mathbf{A} - \mathbf{B} = \begin{bmatrix} 3 & 6 \\ 0 & -1 \end{bmatrix}$ .

## Answers

1.  $1 \times 24, 2 \times 12, 3 \times 8, 4 \times 6, 6 \times 4, 8 \times 3, 12 \times 2, 24 \times 1; 1 \times 13, 13 \times 1$

2. (i)  $\begin{bmatrix} 1 & 1/2 & 0 \\ 5/2 & 2 & 3/2 \end{bmatrix}$  (ii)  $\begin{bmatrix} 9/2 & 25/2 \\ 8 & 18 \\ 25/2 & 49/2 \end{bmatrix}$

$$(iii) \begin{bmatrix} 2 & 9/2 & 8 & 25/2 \\ 9/2 & 8 & 25/2 & 18 \\ 8 & 25/2 & 18 & 49/2 \end{bmatrix}$$

3. (i)  $x = \pm 7, y = 4$ .

(ii) There do not exist such  $x$  and  $y$  for which  $\mathbf{A} = \mathbf{B}$ .

4. (i)  $a = 2, b = 1, c = 3, d = 5$ . (ii)  $a = 2, b = -1, c = 0, d = 5$ .

5.  $x = 2, y = 9$ .

6.  $x = 11, y = 9, z = 3$ .

7.  $\mathbf{A} + \mathbf{B} = \begin{bmatrix} 1 & 5 \\ 4 & 11 \end{bmatrix}$ ,  $\mathbf{A} - \mathbf{C} = \begin{bmatrix} -1 & -7 \\ 9 & -3 \end{bmatrix}$ ,

$$2\mathbf{A} + 3\mathbf{B} = \begin{bmatrix} 2 & 13 \\ 7 & 29 \end{bmatrix}, \mathbf{B} - 5\mathbf{C} = \begin{bmatrix} -10 & -42 \\ 19 & -28 \end{bmatrix}.$$

$$8. \quad C = \begin{bmatrix} -8 & -4 \\ 2 & -4 \\ -4 & -4 \end{bmatrix}.$$

$$10. \quad A = \begin{bmatrix} 4 & 4 \\ 0 & 4 \end{bmatrix}, B = \begin{bmatrix} 1 & -2 \\ 0 & 5 \end{bmatrix}.$$

### 3.9 Multiplication of Matrices

**Definition.** Let  $\mathbf{A} = [a_{ij}]_{m \times n}$  and  $\mathbf{B} = [b_{jk}]_{n \times p}$  be two matrices such that the number of columns in  $\mathbf{A}$  is equal to the number of rows in  $\mathbf{B}$ . Then the  $m \times p$  matrix  $\mathbf{C} = [c_{ik}]_{m \times p}$  such that

$$c_{ik} = \sum_{j=1}^n a_{ij} b_{jk}$$

is called the product of the matrices  $\mathbf{A}$  and  $\mathbf{B}$  in that order and we write  $\mathbf{C} = \mathbf{AB}$ .

Thus the  $(i, k)^{\text{th}}$  element  $c_{ik}$  of the matrix  $\mathbf{AB}$  is obtained by multiplying the corresponding elements of  $i^{\text{th}}$  row of  $\mathbf{A}$  and  $k^{\text{th}}$  column of  $\mathbf{B}$  and then adding the products.

The rule of multiplication is **row by column multiplication**.

In the product  $\mathbf{AB}$ ,  $\mathbf{A}$  is called the **pre-factor** (pre-multiplier) and  $\mathbf{B}$  the **post-factor** (post-multiplier).

It is important to note that the product  $\mathbf{AB}$  of two matrices  $\mathbf{A}$  and  $\mathbf{B}$  exists if and only if the number of columns in  $\mathbf{A}$  is equal to the number of rows in  $\mathbf{B}$ . Two such matrices are said to be *conformable for multiplication*.

For example, if  $\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}_{3 \times 2}$ ,  $\mathbf{B} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}_{2 \times 2}$

$$\text{then } \mathbf{AB} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \\ a_{31}b_{11} + a_{32}b_{21} & a_{31}b_{12} + a_{32}b_{22} \end{bmatrix}_{3 \times 2}.$$

If the product  $\mathbf{AB}$  exists, then it is not necessary that the product  $\mathbf{BA}$  will also exist. In the above example the number of columns in  $\mathbf{B}$  is not equal to the number of rows in  $\mathbf{A}$  therefore  $\mathbf{BA}$  does not exist.

### Illustrative Examples

**Illustration 1 :** If  $\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 2 & 1 \\ 2 & 3 & 0 \end{bmatrix}$  and  $\mathbf{B} = \begin{bmatrix} 1 & 2 \\ 3 & 0 \\ 4 & 1 \end{bmatrix}$ , find  $\mathbf{AB}$ . Does  $\mathbf{BA}$  exist ?

**Solution :** The matrix **A** has 3 columns and the matrix **B** has 3 rows therefore the product **AB** is defined. By row by column rule of multiplication, we have

$$\begin{aligned} \mathbf{AB} &= \begin{bmatrix} 0 & 1 & 0 \\ 1 & 2 & 1 \\ 2 & 3 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 0 \\ 4 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0.1 + 1.3 + 0.4 & 0.2 + 1.0 + 0.1 \\ 1.1 + 2.3 + 1.4 & 1.2 + 2.0 + 1.1 \\ 2.1 + 3.3 + 0.4 & 2.2 + 3.0 + 0.1 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 11 & 3 \\ 11 & 4 \end{bmatrix}_{3 \times 2}. \end{aligned}$$

The matrix **BA** does not exist since the number of columns in **B** is not equal to the number of rows in **A**.

**Illustration 2 :** If  $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 3 & 2 \end{bmatrix}$ ,  $\mathbf{B} = \begin{bmatrix} 3 & 5 \\ 2 & 1 \end{bmatrix}$ , find **AB** and **BA**. Show that **AB**  $\neq$  **BA**.

**Solution :** Since **A** and **B** are square matrices of same order 2, therefore both **AB** and **BA** are defined.

$$\text{We have } \mathbf{AB} = \begin{bmatrix} 1 & 0 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 3 & 5 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1.3 + 0.2 & 1.5 + 0.1 \\ 3.3 + 2.2 & 3.5 + 2.1 \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ 13 & 17 \end{bmatrix}$$

$$\text{and } \mathbf{BA} = \begin{bmatrix} 3 & 5 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 3.1 + 5.3 & 3.0 + 5.2 \\ 2.1 + 1.3 & 2.0 + 1.2 \end{bmatrix} = \begin{bmatrix} 18 & 10 \\ 5 & 2 \end{bmatrix}.$$

Both **AB** and **BA** are square matrices of the type  $2 \times 2$  but the corresponding elements of these matrices are not equal, therefore **AB**  $\neq$  **BA**.

### 3.10 Properties of Matrix Multiplication

(i) **Matrix Multiplication Is Associative.**

If **A**, **B**, **C** are three matrices conformable for multiplication, then

$$\mathbf{A}(\mathbf{BC}) = (\mathbf{AB})\mathbf{C}.$$

(ii) **Multiplication Of Matrices Is Distributive With Respect To Addition Of Matrices.**

If **A**, **B** and **C** are matrices of the type  $m \times n$ ,  $n \times p$ ,  $n \times p$  respectively then

$$\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}.$$

(iii) **Matrix Multiplication Is Not Always Commutative.**

(iv) **The Product Of Two Matrices May Be A Zero Matrix When Neither Of Them Is A Zero Matrix.**

### Illustrative Examples

**Illustration 1 :** Find the product matrix of the matrices

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix} \text{ and } \mathbf{B} = \begin{bmatrix} 2 & -2 & 0 \\ 0 & 1 & 1 \\ -2 & 3 & 0 \end{bmatrix}.$$

**Solution :** The matrix  $\mathbf{A}$  is of the type  $2 \times 3$  and  $\mathbf{B}$  is of the type  $3 \times 3$ . Therefore the product  $\mathbf{AB}$  is defined and it will be a matrix of the type  $2 \times 3$ .

$$\begin{aligned} \mathbf{AB} &= \begin{bmatrix} 2 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix} \times \begin{bmatrix} 2 & -2 & 0 \\ 0 & 1 & 1 \\ -2 & 3 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 2 \cdot 2 + 1 \cdot 0 + 2 \cdot (-2) & 2 \cdot (-2) + 1 \cdot 1 + 2 \cdot 3 & 2 \cdot 0 + 1 \cdot 1 + 2 \cdot 0 \\ 1 \cdot 2 + 1 \cdot 0 + 1 \cdot (-2) & 1 \cdot (-2) + 1 \cdot 1 + 1 \cdot 3 & 1 \cdot 0 + 1 \cdot 1 + 1 \cdot 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 3 & 1 \\ 0 & 2 & 1 \end{bmatrix}. \end{aligned}$$

**Illustration 2 :** If  $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & -1 & 4 \\ 2 & 6 & 1 \end{bmatrix}$  and  $\mathbf{B} = \begin{bmatrix} 1 & 1 & -1 \\ 2 & -2 & 2 \\ 3 & 2 & 1 \end{bmatrix}$  find  $\mathbf{AB}$  and

$\mathbf{BA}$ .

**Solution :** Since  $\mathbf{A}$  and  $\mathbf{B}$  are square matrices of same order 3, therefore both  $\mathbf{AB}$  and  $\mathbf{BA}$  are defined.

$$\begin{aligned} \text{We have } \mathbf{AB} &= \begin{bmatrix} 1 & 2 & 3 \\ 3 & -1 & 4 \\ 2 & 6 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & -1 \\ 2 & -2 & 2 \\ 3 & 2 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 \cdot 1 + 2 \cdot 2 + 3 \cdot 3 & 1 \cdot 1 + 2 \cdot (-2) + 3 \cdot 2 & 1 \cdot (-1) + 2 \cdot 2 + 3 \cdot 1 \\ 3 \cdot 1 + (-1) \cdot 2 + 4 \cdot 3 & 3 \cdot 1 + (-1) \cdot (-2) + 4 \cdot 2 & 3 \cdot (-1) + (-1) \cdot 2 + 4 \cdot 1 \\ 2 \cdot 1 + 6 \cdot 2 + 1 \cdot 3 & 2 \cdot 1 + 6 \cdot (-2) + 1 \cdot 2 & 2 \cdot (-1) + 6 \cdot 2 + 1 \cdot 1 \end{bmatrix} \\ &= \begin{bmatrix} 14 & 3 & 6 \\ 13 & 13 & -1 \\ 17 & -8 & 11 \end{bmatrix} \\ \text{and } \mathbf{BA} &= \begin{bmatrix} 1 & 1 & -1 \\ 2 & -2 & 2 \\ 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 3 & -1 & 4 \\ 2 & 6 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 \cdot 1 + 1 \cdot 3 + (-1) \cdot 2 & 1 \cdot 2 + 1 \cdot (-1) + (-1) \cdot 6 & 1 \cdot 3 + 1 \cdot 4 + (-1) \cdot 1 \\ 2 \cdot 1 + (-2) \cdot 3 + 2 \cdot 2 & 2 \cdot 2 + (-2) \cdot (-1) + 2 \cdot 6 & 2 \cdot 3 + (-2) \cdot 4 + 2 \cdot 1 \\ 3 \cdot 1 + 2 \cdot 3 + 1 \cdot 2 & 3 \cdot 2 + 2 \cdot (-1) + 1 \cdot 6 & 3 \cdot 3 + 2 \cdot 4 + 1 \cdot 1 \end{bmatrix} \\ &= \begin{bmatrix} 2 & -5 & 6 \\ 0 & 18 & 0 \\ 11 & 10 & 18 \end{bmatrix}. \end{aligned}$$

**Illustration 3 :** If  $A = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$ ,  $B = \begin{bmatrix} -1 & 0 \\ 2 & 3 \end{bmatrix}$  and  $C = \begin{bmatrix} 0 & -1 \\ 3 & 1 \end{bmatrix}$ ,

verify that  $A (B + C) = AB + AC$ .

**Solution :** Obviously both  $A (B + C)$  and  $AB + AC$  are square matrices of the same order 2.

Now  $B + C = \begin{bmatrix} -1+0 & 0-1 \\ 2+3 & 3+1 \end{bmatrix} = \begin{bmatrix} -1 & -1 \\ 5 & 4 \end{bmatrix}$ .

$\therefore A (B + C) = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} \times \begin{bmatrix} -1 & -1 \\ 5 & 4 \end{bmatrix}$   
 $= \begin{bmatrix} 1(-1) + (-1).5 & 1(-1) + (-1).4 \\ 0.(-1) + 2.5 & 0.(-1) + 2.4 \end{bmatrix} = \begin{bmatrix} -6 & -5 \\ 10 & 8 \end{bmatrix}$ .

Again  $AB = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} \times \begin{bmatrix} -1 & 0 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 1(-1) + (-1).2 & 1.0 + (-1).3 \\ 0.(-1) + 2.2 & 0.0 + 2.3 \end{bmatrix}$   
 $= \begin{bmatrix} -3 & -3 \\ 4 & 6 \end{bmatrix}$

and  $AC = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} \times \begin{bmatrix} 0 & -1 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 1.0 + (-1).3 & 1(-1) + (-1).1 \\ 0.0 + 2.3 & 0.(-1) + 2.1 \end{bmatrix}$   
 $= \begin{bmatrix} -3 & -2 \\ 6 & 2 \end{bmatrix}$ .

$\therefore AB + AC = \begin{bmatrix} -3-3 & -3-2 \\ 4+6 & 6+2 \end{bmatrix} = \begin{bmatrix} -6 & -5 \\ 10 & 8 \end{bmatrix}$ .

Thus,  $A (B + C) = AB + AC$ .

**Illustration 4 :** Find the value of  $x$  such that

$$\begin{bmatrix} 1 & 1 & x \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \mathbf{O}.$$

[UPTU 2004, 06]

**Solution :** The given matrix equation is

$$\begin{bmatrix} 1 & 1 & x \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \mathbf{O}$$

or  $\begin{bmatrix} 1+0+2x & 0+2+x & 2+1+0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \mathbf{O}$

or  $\begin{bmatrix} 1+2x & 2+x & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \mathbf{O}$

or  $\begin{bmatrix} 1+2x+2+x+3 \end{bmatrix} = \mathbf{O}$

or  $3x + 6 = 0$  or  $x = -2$ .

**Illustration 5 :** Find the value of  $x$  such that

$$\begin{bmatrix} 1 & x & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 2 \\ 2 & 5 & 1 \\ 15 & 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ x \end{bmatrix} = \mathbf{O}.$$

**Solution :** The given matrix equation is

$$\begin{bmatrix} 1 & x & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 2 \\ 2 & 5 & 1 \\ 15 & 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ x \end{bmatrix} = \mathbf{O}$$

or  $\begin{bmatrix} 1 + 2x + 15 & 3 + 5x + 3 & 2 + x + 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ x \end{bmatrix} = \mathbf{O}$

or  $\begin{bmatrix} 2x + 16 & 5x + 6 & x + 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ x \end{bmatrix} = \mathbf{O}$

or  $\begin{bmatrix} 2x + 16 + 10x + 12 + x^2 + 4x \end{bmatrix} = \mathbf{O}$

or  $x^2 + 16x + 28 = 0$

or  $(x + 2)(x + 14) = 0$  or  $x = -2, -14$ .

**Illustration 6 :** If  $\mathbf{A} = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix}$ ,  $\mathbf{B} = \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix}$  and  $(\mathbf{A} + \mathbf{B})^2 = \mathbf{A}^2 + \mathbf{B}^2$ , find  $a$

and  $b$ .

**Solution :** Given  $(\mathbf{A} + \mathbf{B})^2 = \mathbf{A}^2 + \mathbf{B}^2$

or  $(\mathbf{A} + \mathbf{B})(\mathbf{A} + \mathbf{B}) = \mathbf{A}^2 + \mathbf{B}^2$

or  $\mathbf{A}(\mathbf{A} + \mathbf{B}) + \mathbf{B}(\mathbf{A} + \mathbf{B}) = \mathbf{A}^2 + \mathbf{B}^2$

or  $\mathbf{A}^2 + \mathbf{AB} + \mathbf{BA} + \mathbf{B}^2 = \mathbf{A}^2 + \mathbf{B}^2$

or  $\mathbf{AB} + \mathbf{BA} = \mathbf{O}$

or  $\begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix} + \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix} = \mathbf{O}$

or  $\begin{bmatrix} a - b & 2 \\ 2a - b & 3 \end{bmatrix} + \begin{bmatrix} a + 2 & -a - 1 \\ b - 2 & -b + 1 \end{bmatrix} = \mathbf{O}$

or  $\begin{bmatrix} 2a - b + 2 & -a + 1 \\ 2a - 2 & -b + 4 \end{bmatrix} = \mathbf{O}$

or  $2a - b + 2 = 0, -a + 1 = 0, 2a - 2 = 0, -b + 4 = 0$

or  $a = 1, b = 4$ .

**Illustration 7 :** If  $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ -1 & 7 \end{bmatrix}$  and  $\mathbf{I} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ , find  $k$  so that  $\mathbf{A}^2 = 8\mathbf{A} + k\mathbf{I}$ .

**Solution :** We have

$$A^2 = A A = \begin{bmatrix} 1 & 0 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -8 & 49 \end{bmatrix}$$

and  $8A + kI = 8 \begin{bmatrix} 1 & 0 \\ -1 & 7 \end{bmatrix} + k \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 8 & 0 \\ -8 & 56 \end{bmatrix} + \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}$

$$= \begin{bmatrix} 8+k & 0 \\ -8 & 56+k \end{bmatrix}.$$

Now  $A^2 = 8A + kI \Rightarrow \begin{bmatrix} 1 & 0 \\ -8 & 49 \end{bmatrix} = \begin{bmatrix} 8+k & 0 \\ -8 & 56+k \end{bmatrix}.$

$$\Rightarrow 1 = 8 + k, 49 = 56 + k \Rightarrow k = -7.$$

**Illustration 8 :** Prove that the product of two matrices

$$\begin{bmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{bmatrix} \text{ and } \begin{bmatrix} \cos^2 \phi & \cos \phi \sin \phi \\ \cos \phi \sin \phi & \sin^2 \phi \end{bmatrix}$$

is a zero matrix when  $\theta$  and  $\phi$  differ by an odd multiple of  $\pi / 2$ .

**Solution :** We have

$$\begin{aligned} & \begin{bmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{bmatrix} \begin{bmatrix} \cos^2 \phi & \cos \phi \sin \phi \\ \cos \phi \sin \phi & \sin^2 \phi \end{bmatrix} \\ &= \begin{bmatrix} \cos^2 \theta \cos^2 \phi + \cos \theta \sin \theta \cos \phi \sin \phi & \cos^2 \theta \cos \phi \sin \phi + \cos \theta \sin \theta \sin^2 \phi \\ \cos \theta \sin \theta \cos^2 \phi + \sin^2 \theta \cos \phi \sin \phi & \cos \theta \sin \theta \cos \phi \sin \phi + \sin^2 \theta \sin^2 \phi \end{bmatrix} \\ &= \begin{bmatrix} \cos \theta \cos \phi (\cos \theta \cos \phi + \sin \theta \sin \phi) & \cos \theta \sin \phi (\cos \theta \cos \phi + \sin \theta \sin \phi) \\ \sin \theta \cos \phi (\cos \theta \cos \phi + \sin \theta \sin \phi) & \sin \theta \sin \phi (\cos \theta \cos \phi + \sin \theta \sin \phi) \end{bmatrix} \\ &= \begin{bmatrix} \cos \theta \cos \phi \cos (\theta - \phi) & \cos \theta \sin \phi \cos (\theta - \phi) \\ \sin \theta \cos \phi \cos (\theta - \phi) & \sin \theta \sin \phi \cos (\theta - \phi) \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \end{aligned}$$

since  $\theta$  and  $\phi$  differ by an odd multiple of  $\pi / 2$  therefore  $\cos (\theta - \phi) = 0$ .

**Illustration 9 :** Let  $A = \begin{bmatrix} 0 & -\tan (\alpha / 2) \\ \tan (\alpha / 2) & 0 \end{bmatrix}$  and  $I$  be the identity matrix of order 2. Show that  $I + A = (I - A) \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}.$

**Solution :** Let  $\tan \frac{\alpha}{2} = t.$

Then  $A = \begin{bmatrix} 0 & -t \\ t & 0 \end{bmatrix},$

$$\sin \alpha = \frac{2 \tan (\alpha / 2)}{1 + \tan^2 (\alpha / 2)} = \frac{2t}{1 + t^2}$$

and  $\cos \alpha = \frac{1 - \tan^2 (\alpha / 2)}{1 + \tan^2 (\alpha / 2)} = \frac{1 - t^2}{1 + t^2}.$

Now  $I + A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & -t \\ t & 0 \end{bmatrix} = \begin{bmatrix} 1 & -t \\ t & 1 \end{bmatrix}.$

and  $I - A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & -t \\ t & 0 \end{bmatrix} = \begin{bmatrix} 1 & t \\ -t & 1 \end{bmatrix}.$

$$\begin{aligned} \therefore (I - A) \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} &= \begin{bmatrix} 1 & t \\ -t & 1 \end{bmatrix} \begin{bmatrix} \frac{1-t^2}{1+t^2} & -\frac{2t}{1+t^2} \\ \frac{2t}{1+t^2} & \frac{1-t^2}{1+t^2} \end{bmatrix} \\ &= \begin{bmatrix} \frac{1-t^2}{1+t^2} + \frac{2t^2}{1+t^2} & -\frac{2t}{1+t^2} + \frac{t(1-t^2)}{1+t^2} \\ -\frac{t(1-t^2)}{1+t^2} + \frac{2t}{1+t^2} & \frac{2t^2}{1+t^2} + \frac{1-t^2}{1+t^2} \end{bmatrix} \\ &= \begin{bmatrix} 1 & -t \\ t & 1 \end{bmatrix} = I + A. \end{aligned}$$

Thus  $(I - A) \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} = I + A.$

**Illustration 10 :** If  $A = \begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix}$ , then prove that  $A^2 - 6A + 5I = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$  where

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

[UPTU 2002]

**Solution :** We have  $A^2 = A \cdot A = \begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 19 & 6 \\ 18 & 7 \end{bmatrix}.$

Now

$$A^2 - 6A + 5I = \begin{bmatrix} 19 & 6 \\ 18 & 7 \end{bmatrix} + \begin{bmatrix} -24 & -6 \\ -18 & -12 \end{bmatrix} + \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

## Comprehensive Exercise 2

1. For the following matrices verify  $A(BC) = (AB)C$  i.e., matrix multiplication is associative

(i)  $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & -1 & 0 \end{bmatrix}, B = \begin{bmatrix} 1 & 1 \\ -1 & 3 \\ 0 & 5 \end{bmatrix}$  and  $C = \begin{bmatrix} 3 & 2 \\ 1 & -2 \end{bmatrix}.$



(ii)  $\mathbf{A} = \begin{bmatrix} 5 & 1 & 0 \\ 3 & 4 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ ,  $\mathbf{B} = \begin{bmatrix} 0 & -1 & 3 \\ 1 & 2 & 9 \\ 0 & 3 & 7 \end{bmatrix}$  and  $\mathbf{C} = \begin{bmatrix} 2 & 1 \\ 0 & -1 \\ 3 & 5 \end{bmatrix}$ .

2. If  $\mathbf{A} = \begin{bmatrix} 3 & -1 \\ 1 & 0 \\ 2 & 1 \end{bmatrix}$ ,  $\mathbf{B} = \begin{bmatrix} 2 & 7 \\ 1 & 1 \end{bmatrix}$  and  $\mathbf{C} = \begin{bmatrix} 0 & 1 \\ -1 & 3 \end{bmatrix}$ ,

verify that  $\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$ .

3. If  $\mathbf{A} = \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix}$  and  $\mathbf{B} = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$ , show that

$\mathbf{AB} = \mathbf{A}$  and  $\mathbf{BA} = \mathbf{B}$ .

4. If  $\mathbf{A} = \begin{bmatrix} -1 & 1 & 0 \\ 3 & -3 & 3 \\ 5 & -5 & 5 \end{bmatrix}$ ,  $\mathbf{B} = \begin{bmatrix} 0 & 4 & 3 \\ 2 & 4 & 6 \\ -1 & 4 & 4 \end{bmatrix}$ , show that  $\mathbf{A}^2 \mathbf{B}^2 = \mathbf{A}^2$ .

[UPTU 2005]

5. Let  $f(x) = x^2 - 5x + 6$ , find  $f(\mathbf{A})$  i.e.,  $\mathbf{A}^2 - 5\mathbf{A} + 6\mathbf{I}$ , if  $\mathbf{A} = \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix}$ .

6. If  $\mathbf{A} = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$ , find  $(\mathbf{A} - 2\mathbf{I})(\mathbf{A} - 3\mathbf{I})$ .

7. If  $\mathbf{A} = \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$  and  $\mathbf{B} = \begin{bmatrix} 0 & 4 \\ -1 & 7 \end{bmatrix}$ , find  $3\mathbf{A}^2 - 2\mathbf{B}$ .

8. (i) If  $\mathbf{A} = \begin{bmatrix} 2 & 3 \\ 4 & -1 \end{bmatrix}$ ,  $\mathbf{B} = \begin{bmatrix} 3 & -2 \\ 2 & 1 \end{bmatrix}$ , and  $\mathbf{C} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ , verify

$(\mathbf{A} + \mathbf{B})\mathbf{C} = \mathbf{AC} + \mathbf{BC}$

[UPTU 2010]

(ii) If  $\mathbf{A} = \begin{bmatrix} 1 & 0 & -2 \\ 3 & -1 & 0 \\ -2 & 1 & 1 \end{bmatrix}$ ,  $\mathbf{B} = \begin{bmatrix} 0 & 5 & -4 \\ -2 & 1 & 3 \\ -1 & 0 & 2 \end{bmatrix}$

and  $\mathbf{C} = \begin{bmatrix} 1 & 5 & 2 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$ , verify that  $\mathbf{A}(\mathbf{B} - \mathbf{C}) = \mathbf{AB} - \mathbf{AC}$ .

9. Show that the matrix  $\mathbf{A} = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$  satisfies the equation

$\mathbf{A}^3 - 4\mathbf{A}^2 + \mathbf{A} = \mathbf{O}$ .

10. If  $\mathbf{A} = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$  and  $\mathbf{I} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ , then find  $\lambda$  so that  $\mathbf{A}^2 = 5\mathbf{A} + \lambda\mathbf{I}$ .

11. If  $\mathbf{A} = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$ , show that  $\mathbf{A}^2 - 5\mathbf{A} + 7\mathbf{I} = \mathbf{O}$ .

## Answers

$$5. \begin{bmatrix} 1 & -1 & -3 \\ -1 & -1 & -10 \\ -5 & 4 & 4 \end{bmatrix}.$$

$$6. \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

$$7. \begin{bmatrix} 3 & -20 \\ 38 & -11 \end{bmatrix}.$$

$$10. \lambda = -7.$$

### 3.11 Transpose of a Matrix

**Definition.** Let  $\mathbf{A}$  be an  $m \times n$  matrix. Then the  $n \times m$  matrix obtained from  $\mathbf{A}$  by interchanging its rows into columns and columns into rows is called the transpose of  $\mathbf{A}$  and is denoted by  $\mathbf{A}'$  or  $\mathbf{A}^T$ .

Thus if  $\mathbf{A} = [a_{ij}]_{m \times n}$  then  $\mathbf{A}' = [b_{ji}]_{n \times m}$ , where  $b_{ji} = a_{ij}$   
i.e., the  $(j, i)^{\text{th}}$  element of  $\mathbf{A}' = (i, j)^{\text{th}}$  element of  $\mathbf{A}$ .

For example, if  $\mathbf{A} = \begin{bmatrix} 1 & 3 & 2 & 5 \\ 0 & 1 & 1 & 1 \\ 2 & -1 & 3 & 0 \end{bmatrix}_{3 \times 4}$  then

$$\mathbf{A}' = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & -1 \\ 2 & 1 & 3 \\ 5 & 1 & 0 \end{bmatrix}_{4 \times 3}.$$

### 3.12 Properties of Transpose of a Matrix

If  $\mathbf{A}'$  and  $\mathbf{B}'$  are the transposes of  $\mathbf{A}$  and  $\mathbf{B}$  respectively then

- (i)  $(\mathbf{A}')' = \mathbf{A}$ ,
- (ii)  $(\mathbf{A} + \mathbf{B})' = \mathbf{A}' + \mathbf{B}'$ ,  $\mathbf{A}$  and  $\mathbf{B}$  being of the same size (order),
- (iii)  $(k \mathbf{A})' = k \mathbf{A}'$ ,  $k$  being any complex number,
- (iv)  $(\mathbf{AB})' = \mathbf{B}' \mathbf{A}'$ ,  $\mathbf{A}$  and  $\mathbf{B}$  being conformable for multiplication.

### 3.13 Symmetric and Skew-symmetric Matrices

**Symmetric Matrix:** A square matrix  $\mathbf{A} = [a_{ij}]$  is said to be symmetric if its  $(i, j)^{\text{th}}$  element is the same as its  $(j, i)^{\text{th}}$  element, i.e., if  $a_{ij} = a_{ji}$  for all  $i$  and  $j$ .

Thus a square matrix  $A$  is symmetric if  $A = A'$ .

**Example:**  $\begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}, \begin{bmatrix} a_1 & a_2 & a_3 \\ a_2 & b_1 & b_2 \\ a_3 & b_2 & c_1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix}$  are symmetric

matrices.

**Skew-symmetric Matrix :** A square matrix  $A = [a_{ij}]$  is said to be skew-symmetric if the  $(i, j)^{\text{th}}$  element of  $A$  is the negative of the  $(j, i)^{\text{th}}$  element of  $A$  i.e., if  $a_{ij} = -a_{ji}$  for all  $i$  and  $j$ .

Thus a square matrix  $A$  is skew-symmetric if  $A = -A'$ .

**Note:** For a skew-symmetric matrix, we have

$$a_{ij} = -a_{ji}, \text{ for all } i, j.$$

$$\therefore a_{ii} = -a_{ii}, \text{ for all values of } i$$

$$\text{or } 2a_{ii} = 0 \text{ or } a_{ii} = 0.$$

Thus the diagonal elements of a skew-symmetric matrix are all zero.

**Example :**

$$\begin{bmatrix} 0 & h & g \\ -h & 0 & f \\ -g & -f & 0 \end{bmatrix}, \begin{bmatrix} 0 & i & -3 \\ -i & 0 & -1 \\ 3 & 1 & 0 \end{bmatrix} \text{ are skew-symmetric matrices.}$$

## Illustrative Examples

**Illustration 1 :** If  $A, B$  are symmetric (skew-symmetric), then so is also  $A + B$ .

**Solution :** Let  $A$  and  $B$  be symmetric matrices of the same order. Then, we have  $A = A'$  and  $B = B'$ .

$$\text{Now } (A + B)' = A' + B' = A + B.$$

$\therefore A + B$  is a symmetric matrix.

Again suppose  $A$  and  $B$  are skew-symmetric matrices of the same order.

Then we have  $A' = -A, B' = -B$ .

$$\text{Now } (A + B)' = A' + B' = -A - B = -(A + B).$$

Since  $(A + B)' = -(A + B)$ , therefore  $A + B$  is a skew-symmetric matrix.

**Illustration 2 :** If  $A$  and  $B$  are symmetric matrices and  $AB = BA$ , then show that  $AB$  is symmetric.

**Solution :** Suppose  $A$  and  $B$  are symmetric matrices. Then we have  $A = A'$  and  $B = B'$ .

Also suppose that  $AB = BA$ .

$$\text{Now } (AB)' = B' A'$$

$$= BA$$

$$= AB.$$

$$[\because A = A', B = B']$$

$$[\because AB = BA]$$

Since  $(AB)' = AB$ , therefore  $AB$  is a symmetric matrix.

**Illustration 3 :** Show that every square matrix is uniquely expressible as the sum of a symmetric matrix and a skew-symmetric matrix.

**Solution :** Let  $A$  be any square matrix. Then we can write

$$A = \frac{1}{2}(A + A') + \frac{1}{2}(A - A') = P + Q, \text{ say,}$$

where  $P = \frac{1}{2}(A + A')$  and  $Q = \frac{1}{2}(A - A')$ .

We have 
$$P' = \left\{ \frac{1}{2}(A + A') \right\}' = \frac{1}{2}(A + A')' \quad [\because (kA)' = kA']$$

$$= \frac{1}{2}\{A' + (A')'\} = \frac{1}{2}(A' + A) \quad [(A')' = A]$$

$$= P.$$

$\therefore P$  is a symmetric matrix.

Again 
$$Q' = \left\{ \frac{1}{2}(A - A') \right\}' = \frac{1}{2}\{A - (A')'\} = \frac{1}{2}\{A' - (A')'\}$$

$$= \frac{1}{2}(A' - A) = -\frac{1}{2}(A - A') = -Q$$

$\therefore Q$  is a skew-symmetric matrix.

Thus we have expressed the square matrix  $A$  as the sum of a symmetric and a skew-symmetric matrix.

To prove that the representation is unique, let  $A = R + S$  be another such representation of  $A$ , where  $R$  is symmetric and  $S$  is skew-symmetric.

Then we have to show that  $R = P$ ,  $S = Q$ .

We have  $A' = (R + S)' = R' + S' = R - S.$

$[\because R = R', S = -S']$

$\therefore A + A' = 2R \quad \text{and} \quad A - A' = 2S,$

so that  $R = \frac{1}{2}(A + A') \quad \text{and} \quad S = \frac{1}{2}(A - A').$

Thus  $R = P \quad \text{and} \quad S = Q.$

Hence every square matrix is uniquely expressible as the sum of a symmetric and a skew-symmetric matrix.

**Illustration 4 :** Show that every square matrix is uniquely expressible as the sum of a symmetric matrix and a skew-symmetric matrix.

**Solution :** Let  $A$  be any square matrix. We can write

$$A = \frac{1}{2}(A + A') + \frac{1}{2}(A - A') = P + Q, \text{ say}$$

where  $P = \frac{1}{2}(A + A')$  and  $Q = \frac{1}{2}(A - A')$

We have 
$$P' = \left\{ \frac{1}{2}(A + A') \right\}' = \frac{1}{2}(A + A')' \quad [\because (kA)' = kA']$$

$$\begin{aligned}
 &= \frac{1}{2} \{ \mathbf{A}' + (\mathbf{A}')' \} & [\because (\mathbf{A} + \mathbf{B})' = \mathbf{A}' + \mathbf{B}'] \\
 &= \frac{1}{2} (\mathbf{A}' + \mathbf{A}) & [\because (\mathbf{A}')' = \mathbf{A}] \\
 &= \frac{1}{2} (\mathbf{A} + \mathbf{A}') = \mathbf{P}.
 \end{aligned}$$

Therefore  $\mathbf{P}$  is a symmetric matrix.

Again 
$$\begin{aligned}
 \mathbf{Q}' &= \left\{ \frac{1}{2} (\mathbf{A} - \mathbf{A}') \right\}' = \frac{1}{2} (\mathbf{A} - \mathbf{A}')' = \frac{1}{2} \{ \mathbf{A}' - (\mathbf{A}')' \} \\
 &= \frac{1}{2} (\mathbf{A}' - \mathbf{A}) = -\frac{1}{2} (\mathbf{A} - \mathbf{A}') = -\mathbf{Q}.
 \end{aligned}$$

Therefore  $\mathbf{Q}$  is a skew-symmetric matrix.

Thus we have expressed the square matrix  $\mathbf{A}$  as the sum of a symmetric and a skew-symmetric matrix.

To prove that the representation is unique, let  $\mathbf{A} = \mathbf{R} + \mathbf{S}$  be another such representation of  $\mathbf{A}$ , where  $\mathbf{R}$  is symmetric and  $\mathbf{S}$  skew-symmetric. Then to prove that  $\mathbf{R} = \mathbf{P}$  and  $\mathbf{S} = \mathbf{Q}$ .

We have 
$$\mathbf{A}' = (\mathbf{R} + \mathbf{S})' = \mathbf{R}' + \mathbf{S}' = \mathbf{R} - \mathbf{S} \quad [\because \mathbf{R}' = \mathbf{R} \text{ and } \mathbf{S}' = -\mathbf{S}]$$

$\therefore \mathbf{A} + \mathbf{A}' = 2\mathbf{R} \text{ and } \mathbf{A} - \mathbf{A}' = 2\mathbf{S}.$

This gives 
$$\mathbf{R} = \frac{1}{2} (\mathbf{A} + \mathbf{A}') \text{ and } \mathbf{S} = \frac{1}{2} (\mathbf{A} - \mathbf{A}').$$

Thus 
$$\mathbf{R} = \mathbf{P} \text{ and } \mathbf{S} = \mathbf{Q}.$$

Therefore the representation is unique.

**Illustration 5 :** If  $\mathbf{A} = \begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix}$ ,  $\mathbf{B} = \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix}$ , verify that

$$(\mathbf{A} + \mathbf{B})' = \mathbf{A}' + \mathbf{B}' \text{ and } (\mathbf{AB})' = \mathbf{B}' \mathbf{A}'.$$

**Solution :** We have

$$\mathbf{A}' = \begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix}^T = \begin{bmatrix} 2 & 0 \\ 3 & 1 \end{bmatrix} \text{ and } \mathbf{B}' = \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix}^T = \begin{bmatrix} 3 & 2 \\ 4 & 1 \end{bmatrix}.$$

Now 
$$\mathbf{A} + \mathbf{B} = \begin{bmatrix} 2+3 & 3+4 \\ 0+2 & 1+1 \end{bmatrix} = \begin{bmatrix} 5 & 7 \\ 2 & 2 \end{bmatrix}.$$

$\therefore (\mathbf{A} + \mathbf{B})' = \begin{bmatrix} 5 & 7 \\ 2 & 2 \end{bmatrix}^T = \begin{bmatrix} 5 & 2 \\ 7 & 2 \end{bmatrix}$

and 
$$\mathbf{A}' + \mathbf{B}' = \begin{bmatrix} 2+3 & 0+2 \\ 3+4 & 1+1 \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 7 & 2 \end{bmatrix}.$$

Thus, 
$$(\mathbf{A} + \mathbf{B})' = \mathbf{A}' + \mathbf{B}'.$$

Again 
$$\mathbf{AB} = \begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 2.3 + 3.2 & 2.4 + 3.1 \\ 0.3 + 1.2 & 0.4 + 1.1 \end{bmatrix} = \begin{bmatrix} 12 & 11 \\ 2 & 1 \end{bmatrix}.$$

$$\therefore (\mathbf{AB})' = \begin{bmatrix} 12 & 2 \\ 11 & 1 \end{bmatrix}.$$

$$\text{Also } \mathbf{B}' \mathbf{A}' = \begin{bmatrix} 3 & 2 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 3 \cdot 2 + 2 \cdot 3 & 3 \cdot 0 + 2 \cdot 1 \\ 4 \cdot 2 + 1 \cdot 3 & 4 \cdot 0 + 1 \cdot 1 \end{bmatrix} = \begin{bmatrix} 12 & 2 \\ 11 & 1 \end{bmatrix}.$$

$$\text{Thus, } (\mathbf{AB})' = \mathbf{B}' \mathbf{A}'.$$

### Comprehensive Exercise 3

1. If  $\mathbf{A} = \begin{bmatrix} 3 & 1 & -1 \\ 0 & 1 & 2 \end{bmatrix}$ , find  $\mathbf{AA}'$  and  $\mathbf{A}'\mathbf{A}$ .

Also show that  $\mathbf{AA}'$  and  $\mathbf{A}'\mathbf{A}$  are both symmetric matrices.

2. Express the following matrices as the sum of a symmetric and a skew symmetric matrix.

$$(i) \mathbf{A} = \begin{bmatrix} 2 & -1 & 5 \\ 3 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}, (ii) \begin{bmatrix} 2 & -1 \\ 4 & 2 \end{bmatrix}.$$

3. If  $\mathbf{A} = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$ , show that  $\mathbf{AA}' = \mathbf{I} = \mathbf{A}'\mathbf{A}$ .

4. For the matrices  $\mathbf{A}$  and  $\mathbf{B}$  verify that  $(\mathbf{AB})' = \mathbf{B}'\mathbf{A}'$ , where

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 4 & 5 \\ 1 & 2 \end{bmatrix}$$

5. If  $\mathbf{A} = \begin{bmatrix} 3 & 2 & 5 \\ 10 & 1 & -2 \end{bmatrix}$  and  $\mathbf{B} = \begin{bmatrix} 3 & 4 & 7 \\ -2 & 0 & 5 \end{bmatrix}$ , show that

$$(\mathbf{A} + \mathbf{B})' = \mathbf{A}' + \mathbf{B}'.$$

6. Show that  $(\mathbf{AB})' = \mathbf{B}'\mathbf{A}'$ , when  $\mathbf{A} = \begin{bmatrix} -1 & 3 & 0 \\ -7 & 2 & 8 \end{bmatrix}$ ,  $\mathbf{B} = \begin{bmatrix} -5 & 0 \\ 0 & 3 \\ 1 & -8 \end{bmatrix}$ .

7. If  $\mathbf{A} = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$ ,  $\mathbf{B} = \begin{bmatrix} 1 & 4 \\ 2 & 5 \end{bmatrix}$ , verify that  $(\mathbf{AB})' = \mathbf{B}'\mathbf{A}'$ .

### Answers

$$1. \mathbf{AA}' = \begin{bmatrix} 11 & -1 \\ -1 & 5 \end{bmatrix}, \mathbf{A}'\mathbf{A} = \begin{bmatrix} 9 & 3 & -3 \\ 3 & 2 & 1 \\ -3 & 1 & 5 \end{bmatrix}.$$

$$2. \quad (i) \mathbf{A} = \mathbf{P} + \mathbf{Q} = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 0 & 1 \\ 3 & 1 & 0 \end{bmatrix} + \begin{bmatrix} 0 & -2 & 2 \\ 2 & 0 & 0 \\ -2 & 0 & 0 \end{bmatrix}$$

$$(ii) \mathbf{A} = \mathbf{P} + \mathbf{Q} = \begin{bmatrix} 2 & 3/2 \\ 3/2 & 2 \end{bmatrix} + \begin{bmatrix} 2 & -5/2 \\ 5/2 & 0 \end{bmatrix}.$$

### 3.14 Adjoint of a Square Matrix

**Definition.** Let  $\mathbf{A} = [a_{ij}]_{n \times n}$  be any  $n \times n$  matrix. The transpose  $\mathbf{B}'$  of the matrix  $\mathbf{B} = [A_{ij}]_{n \times n}$ , where  $A_{ij}$  denotes the cofactor of the element  $a_{ij}$  in the determinant  $|\mathbf{A}|$ , is called the adjoint of the matrix  $\mathbf{A}$  and is denoted by the symbol  $\text{Adj } \mathbf{A}$ .

Thus the adjoint of a matrix  $\mathbf{A}$  is the transpose of the matrix formed by the cofactors of  $\mathbf{A}$ . Thus if

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix},$$

then

$$\text{adj } \mathbf{A} = \begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22} & \cdots & A_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ A_{n1} & A_{n2} & \cdots & A_{nn} \end{bmatrix}^T = \begin{bmatrix} A_{11} & A_{21} & \cdots & A_{n1} \\ A_{12} & A_{22} & \cdots & A_{n2} \\ \cdots & \cdots & \cdots & \cdots \\ A_{1n} & A_{2n} & \cdots & A_{nn} \end{bmatrix}.$$

The adjoint of a matrix is also called the **adjugate** of the matrix.

**Theorem.** If  $\mathbf{A}$  be any  $n$ -rowed square matrix, then

$$(\text{adj } \mathbf{A}) \mathbf{A} = \mathbf{A}(\text{adj } \mathbf{A}) = |\mathbf{A}| \mathbf{I}_n, \text{ where } \mathbf{I}_n \text{ is the } n\text{-rowed unit matrix.}$$

### Illustrative Examples

**Illustration 1 :** If  $\mathbf{A} = \begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix}$ , then find  $\text{adj } \mathbf{A}$ .

**Solution :** In  $|\mathbf{A}|$ , the cofactor of  $\alpha$  is  $\delta$  and the cofactor of  $\beta$  is  $-\gamma$ .

The cofactor of  $\gamma$  is  $-\beta$  and the cofactor of  $\delta$  is  $\alpha$ .

Therefore the matrix  $\mathbf{B}$  formed of the cofactors of the elements of  $|\mathbf{A}|$  is

$$\mathbf{B} = \begin{bmatrix} \delta & -\gamma \\ -\beta & \alpha \end{bmatrix}. \text{ Now } \text{adj } \mathbf{A} = \mathbf{B}^T = \begin{bmatrix} \delta & -\beta \\ -\gamma & \alpha \end{bmatrix}.$$

**Illustration 2 :** Find the adjoint of the matrix  $\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & 0 & 1 \end{bmatrix}$  and verify that

$$\mathbf{A} (\text{adj } \mathbf{A}) = |\mathbf{A}| \mathbf{I}_3 = (\text{adj } \mathbf{A}) \mathbf{A}.$$

**Solution :** We have  $|\mathbf{A}| = 1 \cdot (2 - 0) - 1 \cdot (1 + 6) + 1 \cdot (0 - 4) = -9$ .

Let us find the cofactors  $\mathbf{A}_{11}$ ,  $\mathbf{A}_{12}$  etc. of the elements of  $|\mathbf{A}|$ .

$$\text{We have } \mathbf{A}_{11} = \begin{vmatrix} 2 & -3 \\ 0 & 1 \end{vmatrix} = 2, \mathbf{A}_{12} = -\begin{vmatrix} 1 & -3 \\ 2 & 1 \end{vmatrix} = -7, \mathbf{A}_{13} = \begin{vmatrix} 1 & 2 \\ 2 & 0 \end{vmatrix} = -4,$$

$$\mathbf{A}_{21} = -\begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} = -1, \mathbf{A}_{22} = \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} = -1, \mathbf{A}_{23} = -\begin{vmatrix} 1 & 1 \\ 2 & 0 \end{vmatrix} = 2,$$

$$\mathbf{A}_{31} = \begin{vmatrix} 1 & 1 \\ 2 & -3 \end{vmatrix} = -5, \mathbf{A}_{32} = -\begin{vmatrix} 1 & 1 \\ 1 & -3 \end{vmatrix} = 4, \mathbf{A}_{33} = \begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix} = 1.$$

The matrix  $\mathbf{B}$  formed of the cofactors of the elements of  $|\mathbf{A}|$  is

$$\mathbf{B} = \begin{bmatrix} 2 & -7 & -4 \\ -1 & -1 & 2 \\ -5 & 4 & 1 \end{bmatrix}.$$

$$\text{Now } \text{adj } \mathbf{A} = \mathbf{B}^T = \begin{bmatrix} 2 & -1 & -5 \\ -7 & -1 & 4 \\ -4 & 2 & 1 \end{bmatrix}.$$

$$\begin{aligned} \therefore \mathbf{A} (\text{adj } \mathbf{A}) &= \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & -1 & -5 \\ -7 & -1 & 4 \\ -4 & 2 & 1 \end{bmatrix} = \begin{bmatrix} -9 & 0 & 0 \\ 0 & -9 & 0 \\ 0 & 0 & -9 \end{bmatrix} \\ &= -9 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = |\mathbf{A}| \mathbf{I}_3, \quad \text{since } |\mathbf{A}| = -9. \end{aligned}$$

$$\begin{aligned} \text{Also } (\text{adj } \mathbf{A}) \mathbf{A} &= \begin{bmatrix} 2 & -1 & -5 \\ -7 & -1 & 4 \\ -4 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} -9 & 0 & 0 \\ 0 & -9 & 0 \\ 0 & 0 & -9 \end{bmatrix} = |\mathbf{A}| \mathbf{I}_3. \end{aligned}$$

Hence,  $\mathbf{A} (\text{adj } \mathbf{A}) = |\mathbf{A}| \mathbf{I}_3 = (\text{adj } \mathbf{A}) \mathbf{A}.$

### 3.15 Inverse or Reciprocal of a Matrix

**Definition.** Let  $\mathbf{A}$  be any  $n$ -rowed square matrix. Then a matrix  $\mathbf{B}$ , if it exists, such that  $\mathbf{AB} = \mathbf{BA} = \mathbf{I}_n$



is called the inverse of  $\mathbf{A}$  i.e.,  $\mathbf{A}^{-1} = \mathbf{B}$ .

The inverse of a matrix  $\mathbf{A}$  is usually denoted by  $\mathbf{A}^{-1}$ .

**Theorem 1. (Uniqueness of the inverse of a matrix).** Every invertible matrix possesses a unique inverse.

**Note.** For the products  $\mathbf{AB}$ ,  $\mathbf{BA}$  to be both defined and be equal, it is necessary that  $\mathbf{A}$  and  $\mathbf{B}$  are both square matrices of the same order. Thus non-square matrices cannot possess inverse.

**Theorem 2. (Existence of the Inverse).** The necessary and sufficient condition for a square matrix  $\mathbf{A}$  to possess the inverse is that  $|\mathbf{A}| \neq 0$ .

OR

The necessary and sufficient condition for a matrix  $\mathbf{A}$  to be invertible is that it is non-singular.

**Important.** If  $\mathbf{A}$  be an invertible matrix then

$$\mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \text{adj } \mathbf{A}.$$

**Theorem 3. (Reversal law).** If  $\mathbf{A}$ ,  $\mathbf{B}$  be two  $n$ -rowed non-singular matrices then  $\mathbf{AB}$  is also non-singular and  $(\mathbf{AB})^{-1} = \mathbf{B}^{-1} \mathbf{A}^{-1}$ , i.e., the inverse of a product is the product of the inverses taken in the reverse order.

**Theorem 4.** If  $\mathbf{A}$  be an  $n \times n$  non-singular matrix, then  $(\mathbf{A}')^{-1} = (\mathbf{A}^{-1})'$ .

## Illustrative Examples

**Illustration 1 :** Find the inverse of the matrix  $\mathbf{A} = \begin{bmatrix} 1 & 2 & 5 \\ 1 & -1 & -1 \\ 2 & 3 & -1 \end{bmatrix}$ . [UPTU 2001]

**Solution :** We have  $|\mathbf{A}| = (1 + 3) - 2(-1 + 2) + 5(3 + 2) = 4 - 2 + 25 = 27$ .

Since  $|\mathbf{A}| \neq 0$ , therefore  $\mathbf{A}^{-1}$  exists.

Let  $A_{ij}$  be the cofactor of  $a_{ij}$  in  $|\mathbf{A}|$ . Then we have

$$A_{11} = 4, A_{12} = -1, A_{13} = 5, A_{21} = 17, A_{22} = -11,$$

$$A_{23} = 1, A_{31} = 3, A_{32} = 1, A_{33} = -3.$$

$$\text{Now adj } \mathbf{A} = \begin{bmatrix} 4 & -1 & 5 \\ 17 & -11 & 1 \\ 3 & 1 & -3 \end{bmatrix}^T = \begin{bmatrix} 4 & 17 & 3 \\ -1 & -11 & 1 \\ 5 & 1 & -3 \end{bmatrix}.$$

$$\therefore \mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \text{adj } \mathbf{A} = \frac{1}{27} \begin{bmatrix} 4 & 17 & 3 \\ -1 & -11 & 1 \\ 5 & 1 & -3 \end{bmatrix}.$$

**Illustration 2 :** Find the inverse of  $\mathbf{A} = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$  and verify that  $\mathbf{A}^{-1} \mathbf{A} = \mathbf{I}_3$ .

**Solution :** We have  $|\mathbf{A}| = (16 - 9) - 3(4 - 3) + 3(3 - 4) = 7 - 3 - 3 = 1$ .

Since  $|\mathbf{A}| \neq 0$ , therefore  $\mathbf{A}^{-1}$  exists.

Let  $A_{ij}$  be the cofactor of  $a_{ij}$  in  $|\mathbf{A}|$ . Then we have

$$A_{11} = 7, \quad A_{12} = -1, \quad A_{13} = -1, \quad A_{21} = -3, \quad A_{22} = 1, \quad A_{23} = 0, \\ A_{31} = -3, \quad A_{32} = 0, \quad A_{33} = 1.$$

Now  $\text{adj } \mathbf{A} = \begin{bmatrix} 7 & -1 & -1 \\ -3 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}^T = \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}.$

$\therefore \mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \text{adj } \mathbf{A} = \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}.$

Now  $\mathbf{A}^{-1} \mathbf{A} = \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$

$$= \begin{bmatrix} 7-3-3 & 21-12-9 & 21-9-12 \\ -1+1+0 & -3+4+0 & -3+3+0 \\ -1+0+1 & -3+0+3 & -3+0+4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{I}_3.$$

**Illustration 3 :** Find the inverse of the matrix  $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{bmatrix}$ . [UPTU 2001]

**Solution :** We have  $|\mathbf{A}| = 1(6 - 1) - 2(4 - 3) + 3(2 - 9) = 5 - 2 - 21 = -18$

Since  $|\mathbf{A}| \neq 0$ , therefore  $\mathbf{A}^{-1}$  exists.

Let  $A_{ij}$  be the cofactor of  $a_{ij}$  in  $|\mathbf{A}|$ . Then we have

$$A_{11} = 5, \quad A_{12} = -1, \quad A_{13} = -7, \quad A_{21} = -1, \quad A_{22} = -7, \quad A_{23} = 5, \\ A_{31} = -7, \quad A_{32} = 4, \quad A_{33} = -1.$$

Now  $\text{adj } \mathbf{A} = \begin{bmatrix} 5 & -1 & -7 \\ -1 & -7 & 5 \\ -7 & 5 & -1 \end{bmatrix}^T = \begin{bmatrix} 5 & -1 & -7 \\ -1 & -7 & 5 \\ -7 & 5 & -1 \end{bmatrix}.$

$\therefore \mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \text{adj } \mathbf{A} = \frac{-1}{18} \begin{bmatrix} 5 & -1 & -7 \\ -1 & -7 & 5 \\ -7 & 5 & -1 \end{bmatrix}.$

**Illustration 4 :** Find the inverse of the matrix  $\mathbf{A} = \begin{bmatrix} 2 & 3 & 4 \\ 4 & 3 & 1 \\ 1 & 2 & 4 \end{bmatrix}$ . [UPTU 2002]

**Solution :** We have

$$|\mathbf{A}| = 2(12 - 2) - 3(16 - 1) + 4(8 - 3) = 20 - 45 + 20 = -5.$$

Since  $|\mathbf{A}| \neq 0$ , Therefore  $\mathbf{A}^{-1}$  exists.

Let  $A_{ij}$  be the cofactor of  $a_{ij}$  in  $|\mathbf{A}|$ . Then we have

$$A_{11} = 10, A_{12} = -15, A_{13} = 5, A_{21} = -4, A_{22} = 4, A_{23} = -1,$$

$$A_{31} = -9, A_{32} = 14, A_{33} = -6$$

Now  $\text{adj } \mathbf{A} = \begin{bmatrix} 10 & -15 & 5 \\ -4 & 4 & -1 \\ -9 & 14 & -6 \end{bmatrix}^T = \begin{bmatrix} 10 & -4 & -9 \\ -15 & 4 & 14 \\ 5 & -1 & -6 \end{bmatrix}.$

$$\therefore \mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \text{adj } \mathbf{A} = -\frac{1}{5} \begin{bmatrix} 10 & -4 & -9 \\ -15 & 4 & 14 \\ 5 & -1 & -6 \end{bmatrix}.$$

**Illustration 5 :** If  $\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{bmatrix}$ , find  $\mathbf{A}^{-1}$ .

**Solution :** We have  $|\mathbf{A}| = 1(18 - 12) - 1(9 - 3) + 1(4 - 2)$   
 $= 6 - 6 + 2 = 2.$

Since  $|\mathbf{A}| \neq 0$ , therefore  $\mathbf{A}^{-1}$  exists.

Let  $A_{ij}$  be the cofactor of  $a_{ij}$  in  $|\mathbf{A}|$ . Then we have

$$A_{11} = 6, A_{12} = -6, A_{13} = 2, A_{21} = -5, A_{22} = 8,$$

$$A_{23} = -3, A_{31} = 1, A_{32} = -2, A_{33} = 1.$$

Now  $\text{adj } \mathbf{A} = \begin{bmatrix} 6 & -6 & 2 \\ -5 & 8 & -3 \\ 1 & -2 & 1 \end{bmatrix}^T = \begin{bmatrix} 6 & -5 & 1 \\ -6 & 8 & -2 \\ 2 & -3 & 1 \end{bmatrix}$

$$\therefore \mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \text{adj } \mathbf{A} = \frac{1}{2} \begin{bmatrix} 6 & -5 & 1 \\ -6 & 8 & -2 \\ 2 & -3 & 1 \end{bmatrix}.$$

**Illustration 6 :** If  $\mathbf{A} = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$ , using  $\mathbf{A}^2 - 4\mathbf{A} - 5\mathbf{I} = \mathbf{O}$  and hence find  $\mathbf{A}^{-1}$ . [UPTU 2003; 08]

**Solution :** We have  $\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix},$

$$\mathbf{A}^2 = \mathbf{A} \mathbf{A} = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix}.$$

$$\begin{aligned} \text{Now } \mathbf{A}^2 - 4\mathbf{A} - 5\mathbf{I} &= \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix} - 4 \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} - 5 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 9-4-5 & 8-8-0 & 8-8-0 \\ 8-8-0 & 9-4-5 & 8-8-0 \\ 8-8-0 & 8-8-0 & 9-8-0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \end{aligned}$$

Now we have to find  $\mathbf{A}^{-1}$ .

Multiplying both sides of  $\mathbf{A}^2 - 4\mathbf{A} - 5\mathbf{I} = \mathbf{O}$  by  $\mathbf{A}^{-1}$ , we get

$$\mathbf{A}^{-1}(\mathbf{A}^2 - 4\mathbf{A} - 5\mathbf{I}) = \mathbf{O}$$

$$\text{or } \mathbf{A}^{-1}\mathbf{A}^2 - 4\mathbf{A}^{-1}\mathbf{A} - 5\mathbf{A}^{-1}\mathbf{I} = \mathbf{O} \quad \text{or} \quad \mathbf{A} - 4\mathbf{I} - 5\mathbf{A}^{-1} = \mathbf{O}$$

$$\begin{aligned} \text{or } 5\mathbf{A}^{-1} = \mathbf{A} - 4\mathbf{I} &= \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} - 4 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} -3 & 2 & 2 \\ 2 & -3 & 2 \\ 2 & 2 & -3 \end{bmatrix}. \end{aligned}$$

$$\therefore \mathbf{A}^{-1} = \frac{1}{5} \begin{bmatrix} -3 & 2 & 2 \\ 2 & -3 & 2 \\ 2 & 2 & -3 \end{bmatrix}.$$

**Illustration 7 :** Show that  $\mathbf{A} = \begin{bmatrix} -8 & 5 \\ 2 & 4 \end{bmatrix}$  satisfies the equation

$x^2 + 4x - 42 = 0$ . Hence find  $\mathbf{A}^{-1}$ .

**Solution :** We have  $\mathbf{A} = \begin{bmatrix} -8 & 5 \\ 2 & 4 \end{bmatrix}$

$$\text{and } \mathbf{A}^2 = \mathbf{A} \mathbf{A} = \begin{bmatrix} -8 & 5 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} -8 & 5 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 74 & -20 \\ -8 & 26 \end{bmatrix}.$$

In order to show that the matrix  $\mathbf{A}$  satisfies the given equation, we have to show that  $\mathbf{A}^2 + 4\mathbf{A} - 42\mathbf{I} = \mathbf{O}$ .

$$\begin{aligned} \text{Now } \mathbf{A}^2 + 4\mathbf{A} - 42\mathbf{I} &= \begin{bmatrix} 74 & -20 \\ -8 & 26 \end{bmatrix} + 4 \begin{bmatrix} -8 & 5 \\ 2 & 4 \end{bmatrix} - 42 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 74-32-42 & -20+20 \\ -8+8+0 & 26+16-42 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \mathbf{O}. \end{aligned}$$

Thus the matrix  $\mathbf{A}$  satisfies the equation  $\mathbf{A}^2 + 4\mathbf{A} - 42\mathbf{I} = \mathbf{O}$ .

Now we have to find  $\mathbf{A}^{-1}$ .

Multiplying  $\mathbf{A}^2 + 4\mathbf{A} - 42\mathbf{I} = \mathbf{O}$  by  $\mathbf{A}^{-1}$ , we get

$$\mathbf{A}^{-1} (\mathbf{A}^2 + 4\mathbf{A} - 42\mathbf{I}) = \mathbf{A}^{-1} \mathbf{O}$$

$$\text{or } \mathbf{A}^{-1} \mathbf{A}^2 + 4\mathbf{A}^{-1} \mathbf{A} - 42\mathbf{A}^{-1} \mathbf{I} = \mathbf{O}$$

$$\text{or } \mathbf{A} + 4\mathbf{I} - 42\mathbf{A}^{-1} = \mathbf{O}$$

$$\text{or } 42\mathbf{A}^{-1} = \mathbf{A} + 4\mathbf{I} = \begin{bmatrix} -8 & 5 \\ 2 & 4 \end{bmatrix} + 4 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -4 & 5 \\ 6 & 8 \end{bmatrix}.$$

$$\therefore \mathbf{A}^{-1} = \frac{1}{42} \begin{bmatrix} -4 & 5 \\ 6 & 8 \end{bmatrix}.$$

**Illustration 8 :** For the matrix  $\mathbf{A} = \begin{bmatrix} 3 & 1 \\ 7 & 5 \end{bmatrix}$ , find  $x$  and  $y$ , so that  $\mathbf{A}^2 + x\mathbf{I} = y\mathbf{A}$ .

Hence find  $\mathbf{A}^{-1}$ .

**Solution :** Given  $\mathbf{A} = \begin{bmatrix} 3 & 1 \\ 7 & 5 \end{bmatrix}$ .

$$\mathbf{A}^2 = \mathbf{A} \mathbf{A} = \begin{bmatrix} 3 & 1 \\ 7 & 5 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 7 & 5 \end{bmatrix} = \begin{bmatrix} 16 & 8 \\ 56 & 32 \end{bmatrix}.$$

$$\text{We have } \mathbf{A}^2 + x\mathbf{I} = y\mathbf{A} \Rightarrow \begin{bmatrix} 16 & 8 \\ 56 & 32 \end{bmatrix} + x \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = y \begin{bmatrix} 3 & 1 \\ 7 & 5 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 16+x & 8 \\ 56 & 32+x \end{bmatrix} = \begin{bmatrix} 3y & y \\ 7y & 5y \end{bmatrix}$$

$$\Rightarrow 16+x=3y, 8=y, 56=7y, 32+x=5y$$

$$\Rightarrow y=8, x=8.$$

Substituting these values in the given equation, we get  $\mathbf{A}^2 + 8\mathbf{I} = 8\mathbf{A}$ .

Pre-multiplying both sides by  $\mathbf{A}^{-1}$ , we get

$$\mathbf{A}^{-1} (\mathbf{A}^2 + 8\mathbf{I}) = \mathbf{A}^{-1} (8\mathbf{A})$$

$$\text{or } \mathbf{A}^{-1} \mathbf{A}^2 + 8\mathbf{A}^{-1} \mathbf{I} = 8\mathbf{A}^{-1} \mathbf{A}$$

$$\text{or } \mathbf{A} + 8\mathbf{A}^{-1} = 8\mathbf{I} \quad [\because \mathbf{A}^{-1} \mathbf{A}^2 = (\mathbf{A}^{-1} \mathbf{A}) \mathbf{A} = \mathbf{I} \mathbf{A} = \mathbf{A}]$$

$$\text{or } 8\mathbf{A}^{-1} = 8\mathbf{I} - \mathbf{A} = 8 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 3 & 1 \\ 7 & 5 \end{bmatrix} = \begin{bmatrix} 5 & -1 \\ -7 & 3 \end{bmatrix}.$$

$$\therefore \mathbf{A}^{-1} = \frac{1}{8} \begin{bmatrix} 5 & -1 \\ -7 & 3 \end{bmatrix}.$$

**Illustration 9 :** Show that the matrix  $\mathbf{A} = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$  satisfies the equation

$$\mathbf{A}^3 - 6\mathbf{A}^2 + 9\mathbf{A} - 4\mathbf{I} = \mathbf{O}. \text{ Hence find } \mathbf{A}^{-1}.$$

[UPTU 2004]

**Solution :** We have  $\mathbf{A} = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$ ,

$$\begin{aligned} \mathbf{A}^2 = \mathbf{A} \mathbf{A} &= \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix}, \end{aligned}$$

$$\begin{aligned} \mathbf{A}^3 = \mathbf{A} \mathbf{A}^2 &= \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} \\ &= \begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix}. \end{aligned}$$

Now

$$\begin{aligned} &\mathbf{A}^3 - 6\mathbf{A}^2 + 9\mathbf{A} - 4\mathbf{I} \\ &= \begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix} - 6 \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} \\ &\quad + 9 \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} - 4 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \mathbf{O}. \end{aligned}$$

Now we have to find  $\mathbf{A}^{-1}$ .

Multiplying both sides of  $\mathbf{A}^3 - 6\mathbf{A}^2 + 9\mathbf{A} - 4\mathbf{I} = \mathbf{O}$  by  $\mathbf{A}^{-1}$ , we get

$$\mathbf{A}^{-1} (\mathbf{A}^3 - 6\mathbf{A}^2 + 9\mathbf{A} - 4\mathbf{I}) = \mathbf{O}$$

$$\text{or} \quad \mathbf{A}^{-1} \mathbf{A}^3 - 6\mathbf{A}^{-1} \mathbf{A}^2 + 9\mathbf{A}^{-1} \mathbf{A} - 4\mathbf{A}^{-1} \mathbf{I} = \mathbf{O}$$

$$\text{or} \quad \mathbf{A}^2 - 6\mathbf{A} + 9\mathbf{I} - 4\mathbf{A}^{-1} = \mathbf{O}$$

$$\text{or} \quad 4\mathbf{A}^{-1} = \mathbf{A}^2 - 6\mathbf{A} + 9\mathbf{I}$$

$$= \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} - 6 \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} + 9 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}.$$

$$\therefore \mathbf{A}^{-1} = \frac{1}{4} \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}.$$

## Comprehensive Exercise 4

1. Find the adjoint of the following matrices :

(i)  $\begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$  (ii)  $\begin{bmatrix} 2 & 3 \\ 5 & 1 \end{bmatrix}$

(iii)  $\begin{bmatrix} 1 & \tan(\alpha/2) \\ -\tan(\alpha/2) & 1 \end{bmatrix}.$

2. Find the adjoint of the following matrices :

(i)  $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 2 & 0 \\ 3 & -1 & 4 \end{bmatrix}$  (ii)  $\begin{bmatrix} 1 & 2 & 5 \\ 2 & 3 & 1 \\ -1 & 1 & 1 \end{bmatrix}.$

3. For the matrix  $\mathbf{A} = \begin{bmatrix} 1 & -2 & 3 \\ 0 & 2 & -1 \\ -4 & 5 & 2 \end{bmatrix}$  find  $\mathbf{A} (\text{adj } \mathbf{A})$ .

Also verify that  $\mathbf{A} (\text{adj } \mathbf{A}) = |\mathbf{A}| \mathbf{I}_3 = (\text{adj } \mathbf{A}) \mathbf{A}.$

4. If  $\mathbf{A} = \begin{bmatrix} -4 & -3 & -3 \\ 1 & 0 & 1 \\ 4 & 4 & 3 \end{bmatrix}$ , show that  $\text{adj } \mathbf{A} = \mathbf{A}.$

5. Find the inverse of each of the following matrices :

(i)  $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$  (ii)  $\mathbf{A} = \begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix}$  (iii)  $\begin{bmatrix} 2 & 5 \\ -3 & 1 \end{bmatrix}.$

6. Find the inverse of each of the following matrices :

(i)  $\begin{bmatrix} 0 & 0 & -1 \\ 3 & 4 & 5 \\ -2 & -4 & -7 \end{bmatrix}$  (ii)  $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{bmatrix}$

(iii)  $\begin{bmatrix} 2 & 3 & 4 \\ 4 & 3 & 1 \\ 1 & 2 & 4 \end{bmatrix}.$

7. If  $\mathbf{A} = \begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix}$ , find  $x$  and  $y$  such that  $\mathbf{A}^2 - x\mathbf{A} + y\mathbf{I} = \mathbf{O}$ . Hence evaluate  $\mathbf{A}^{-1}$ .

## Answers

1. (i)  $\begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$  (ii)  $\begin{bmatrix} 1 & -3 \\ -5 & 2 \end{bmatrix}$   
 (iii)  $\begin{bmatrix} 1 & -\tan(\alpha/2) \\ \tan(\alpha/2) & 1 \end{bmatrix}$ .
2. (i)  $\begin{bmatrix} 8 & -5 & -2 \\ -4 & -3 & 1 \\ -7 & 3 & -1 \end{bmatrix}$  (ii)  $\begin{bmatrix} 2 & 3 & -13 \\ -3 & 6 & 9 \\ 5 & -3 & -1 \end{bmatrix}$ .
3.  $|\mathbf{A}| \mathbf{I}_3$ , where  $|\mathbf{A}| = 25$ .
5. (i)  $\frac{1}{(ad-bc)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$  (ii)  $\frac{1}{19} \begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix} \mathbf{A}$   
 (iii)  $\frac{1}{17} \begin{bmatrix} 1 & -5 \\ 3 & 2 \end{bmatrix}$ .
6. (i)  $\frac{1}{4} \begin{bmatrix} -8 & 4 & 4 \\ 11 & -2 & -3 \\ -4 & 0 & 0 \end{bmatrix}$  (ii)  $-\frac{1}{18} \begin{bmatrix} 5 & -1 & -7 \\ -1 & -7 & 5 \\ -7 & 5 & -1 \end{bmatrix}$   
 (iii)  $-\frac{1}{5} \begin{bmatrix} 10 & -4 & -9 \\ -15 & 4 & 14 \\ 5 & -1 & -6 \end{bmatrix}$ .
7.  $\frac{1}{14} \begin{bmatrix} 5 & -3 \\ -2 & 4 \end{bmatrix}$ .

### 3.16 Solving Systems of Linear Equations using Inverse of a Matrix

Consider a system of  $n$  linear equations in  $n$  unknowns  $x_1, x_2, \dots, x_n$

$$\begin{aligned} a_{11} x_1 + a_{12} x_2 + \dots + a_{1n} x_n &= b_1 \\ a_{21} x_1 + a_{22} x_2 + \dots + a_{2n} x_n &= b_2 \\ \dots & \dots \dots \dots \dots \\ \dots & \dots \dots \dots \dots \\ a_{n1} x_1 + a_{n2} x_2 + \dots + a_{nn} x_n &= b_n. \end{aligned}$$

These equations can be written in the form of a single matrix equation



$$\mathbf{A} \mathbf{X} = \mathbf{B} \quad \dots(1)$$

where  $\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}_{n \times n}, \quad \mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix}_{n \times 1}, \quad \mathbf{B} = \begin{bmatrix} b_1 \\ b_2 \\ \dots \\ b_n \end{bmatrix}_{n \times 1}.$

Suppose  $\mathbf{A}$  is a non-singular matrix i.e.,  $|\mathbf{A}| \neq 0$ .

Then  $\mathbf{A}^{-1}$  exists. Therefore pre-multiplying (1) by  $\mathbf{A}^{-1}$ , we get

$$\mathbf{A}^{-1} (\mathbf{A} \mathbf{X}) = \mathbf{A}^{-1} \mathbf{B} \Rightarrow (\mathbf{A}^{-1} \mathbf{A}) \mathbf{X} = \mathbf{A}^{-1} \mathbf{B} \Rightarrow \mathbf{I} \mathbf{X} = \mathbf{A}^{-1} \mathbf{B}$$

$\Rightarrow \mathbf{X} = \mathbf{A}^{-1} \mathbf{B}$ , which gives the solution of the given equations.

This solution of the given system will be unique as shown below.

Let  $x_1, x_2$  be two solutions of  $\mathbf{A} \mathbf{X} = \mathbf{B}$ . Then  $\mathbf{A} \mathbf{X}_1 = \mathbf{B}$  and  $\mathbf{A} \mathbf{X}_2 = \mathbf{B}$ .

Now  $\mathbf{A} \mathbf{X}_1 = \mathbf{A} \mathbf{X}_2 \Rightarrow \mathbf{A}^{-1} (\mathbf{A} \mathbf{X}_1) = \mathbf{A}^{-1} (\mathbf{A} \mathbf{X}_2)$

$\Rightarrow (\mathbf{A}^{-1} \mathbf{A}) \mathbf{X}_1 = (\mathbf{A}^{-1} \mathbf{A}) \mathbf{X}_2 \Rightarrow \mathbf{I} \mathbf{X}_1 = \mathbf{I} \mathbf{X}_2 \Rightarrow \mathbf{X}_1 = \mathbf{X}_2$ .

Hence the solution is unique.

Thus, if  $\mathbf{A}$  is a non-singular matrix, then the system  $\mathbf{A} \mathbf{X} = \mathbf{B}$ , of linear equations, has a unique solution given by  $\mathbf{X} = \mathbf{A}^{-1} \mathbf{B}$ .

### Criterion For Consistency Of A System Of Linear Equations.

Let  $\mathbf{A} \mathbf{X} = \mathbf{B}$  be a system of  $n$  linear equations in  $n$  unknowns.

(i) if  $|\mathbf{A}| \neq 0$  i.e.,  $\mathbf{A}$  is non-singular, then the given system is consistent and has a unique solution given by  $\mathbf{X} = \mathbf{A}^{-1} \mathbf{B}$ .

(ii) If  $|\mathbf{A}| = 0$  and  $(\text{adj } \mathbf{A}) \mathbf{B} = 0$ , the system is consistent and has infinite number of solutions.

(iii) If  $|\mathbf{A}| = 0$  and  $(\text{adj } \mathbf{A}) \mathbf{B} \neq 0$ , the given system is inconsistent.

## Illustrative Examples

**Illustration 1 :** Solve the following system of equations using matrix method :

$$5x + 2y = 3, 3x + 2y = 5.$$

**Solution.** The given system of equations can be written in matrix form as

$$\mathbf{A} \mathbf{X} = \mathbf{B}, \quad \dots(1)$$

where  $\mathbf{A} = \begin{bmatrix} 5 & 2 \\ 3 & 2 \end{bmatrix}, \mathbf{X} = \begin{bmatrix} x \\ y \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}.$

We have  $|\mathbf{A}| = 10 - 6 = 4 \neq 0$ . Therefore  $\mathbf{A}^{-1}$  exists.

Cofactors of the elements of the first row of  $|\mathbf{A}|$  are 2, -3.

Cofactors of the elements of the second row of  $|\mathbf{A}|$  are -2, 5.

Now  $\text{adj } \mathbf{A} = \begin{bmatrix} 2 & -3 \\ -2 & 5 \end{bmatrix}^T = \begin{bmatrix} 2 & -2 \\ -3 & 5 \end{bmatrix}.$

$\therefore \mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \text{adj } \mathbf{A} = \frac{1}{4} \begin{bmatrix} 2 & -2 \\ -3 & 5 \end{bmatrix}.$

$\therefore \mathbf{X} = \mathbf{A}^{-1} \mathbf{B} = \frac{1}{4} \begin{bmatrix} 2 & -2 \\ -3 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}.$

Hence  $x = -1, y = 4.$

**Illustration 2 :** Solve for  $x$  and  $y$  by inverting the matrix in the following

$$\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}.$$

[UPTU 2001]

**Solution :** We have  $|\mathbf{A}| = 6 - 1 = 5 \neq 0.$  Therefore  $\mathbf{A}^{-1}$  exists.

Cofactors of the elements of the first row of  $|\mathbf{A}|$  are 3, -1.

Cofactors of the elements of the second row of  $|\mathbf{A}|$  are -1, 2.

Now,  $\text{adj } \mathbf{A} = \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}^T = \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}.$

$\therefore \mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \text{adj } \mathbf{A} = \frac{1}{5} \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}.$

$\therefore \mathbf{X} = \mathbf{A}^{-1} \mathbf{B} = \frac{1}{5} \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 7 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$

Hence  $x = 1, y = 2.$

**Illustration 3 :** Find  $\mathbf{A}^{-1}$ , if possible, where  $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$  and hence solve the equations

$ax + by + 1 = 0$  and  $cx + dy + 1 = 0.$

**Solution :** The given system of equations can be written in matrix form as

$$\mathbf{AX} = \mathbf{B} \quad \dots(1)$$

where  $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \mathbf{X} = \begin{bmatrix} x \\ y \end{bmatrix}, \mathbf{B} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}.$

We have  $|\mathbf{A}| = ad - bc \neq 0.$  Therefore  $\mathbf{A}^{-1}$  exists.

Cofactors of the elements of the first row of  $|\mathbf{A}|$  are  $d, -c.$

Cofactors of the elements of the second row of  $|\mathbf{A}|$  are  $-b, a.$

Now  $\text{adj } \mathbf{A} = \begin{bmatrix} d & -c \\ -b & a \end{bmatrix}^T = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$

$\therefore \mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \text{adj } \mathbf{A} = \frac{1}{(ad - bc)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$

$\therefore \mathbf{X} = \mathbf{A}^{-1} \mathbf{B} = \frac{1}{(ad - bc)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \begin{bmatrix} -1 \\ -1 \end{bmatrix}$

or 
$$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{(ad - bc)} \begin{bmatrix} -d + b \\ c - a \end{bmatrix}.$$

Hence,  $x = \frac{b - d}{ad - bc}, y = \frac{c - a}{ad - bc}.$

**Illustration 4 :** If  $\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{bmatrix}$ , find  $\mathbf{A}^{-1}$ . Hence solve the equations

$$x + y + z = 6, x - y + z = 2, 2x + y - z = 1$$

[UPTU 2007]

**Solution :** We have

$$|\mathbf{A}| = 1(1 - 1) - 1(-1 - 2) + 1(1 + 1) = 0 + 3 + 3 = 6.$$

Since,  $|\mathbf{A}| \neq 0$ , therefore  $\mathbf{A}$  is invertible.

Let  $A_{ij}$  be the cofactor of  $a_{ij}$  in  $|\mathbf{A}|$ .

Then we have

$$A_{11} = (-1)^{1+1} \begin{vmatrix} -1 & 1 \\ 1 & -1 \end{vmatrix} = 1 - 1 = 0,$$

$$A_{12} = (-1)^{1+2} \begin{vmatrix} 1 & 1 \\ 2 & -1 \end{vmatrix} = -(-1 - 2) = 3$$

$$A_{13} = 3, A_{21} = 2, A_{22} = -3, A_{23} = 1, A_{31} = 2, A_{32} = 0, A_{33} = -2.$$

Now, 
$$\text{adj } \mathbf{A} = \begin{bmatrix} 0 & 3 & 3 \\ 2 & -3 & 1 \\ 2 & 0 & -2 \end{bmatrix}^T = \begin{bmatrix} 0 & 2 & 2 \\ 3 & -3 & 0 \\ 3 & 1 & -2 \end{bmatrix}.$$

$$\therefore \mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \text{adj } \mathbf{A} = \frac{1}{6} \begin{bmatrix} 0 & 2 & 2 \\ 3 & -3 & 0 \\ 3 & 1 & -2 \end{bmatrix}.$$

Writing the given system as a single matrix equation,  $\mathbf{AX} = \mathbf{B}$ , we get

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix}.$$

$$\therefore \mathbf{X} = \mathbf{A}^{-1} \mathbf{B} = \frac{1}{6} \begin{bmatrix} 0 & 2 & 2 \\ 3 & -3 & 0 \\ 3 & 1 & -2 \end{bmatrix} \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

Hence,  $x = 1, y = 2, z = 3.$

**Illustration 5 :** Solve using matrix method, the following equations

$$x + y = 0, y + z = 1, z + x = 3.$$

[UPTU 2005]

**Solution :** The given system of equations can be written in matrix form as

$$\mathbf{AX} = \mathbf{B}$$

where  $\mathbf{A} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}, \mathbf{X} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}.$

We have  $|\mathbf{A}| = 1(1 - 0) - 1(0 - 1) = 1 + 1 = 2 \neq 0$

Let  $A_{ij}$  be the cofactor of  $a_{ij}$  in  $|\mathbf{A}|$ . Then we have

$$A_{11} = 1, A_{12} = 1, A_{13} = -1, A_{21} = -1, A_{22} = 1, \\ A_{23} = 1, A_{31} = 1, A_{32} = -1, A_{33} = 1.$$

Now  $\text{adj } \mathbf{A} = \begin{bmatrix} 1 & 1 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix}^T = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{bmatrix}.$

$\therefore \mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \text{adj } \mathbf{A} = \frac{1}{2} \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{bmatrix}.$

We have  $\mathbf{X} = \mathbf{A}^{-1} \mathbf{B} = \frac{1}{2} \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2 \\ -2 \\ 4 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}.$

Hence,  $x = 1, y = -1, z = 2.$

**Illustration 6 :** Solve the following equations by matrix method.

$$2x - y + 3z = 1, x + 2y - z = 2, 5y - 5z = 3. \quad [\text{UPTU 2003, 06}]$$

**Solution :** The given system can be written as a single matrix equation  $\mathbf{AX} = \mathbf{B}$ , where

$$\mathbf{A} = \begin{bmatrix} 2 & -1 & 3 \\ 1 & 2 & -1 \\ 0 & 5 & -5 \end{bmatrix}, \mathbf{X} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

We have  $|\mathbf{A}| = 2(-10 + 5) + 1(-5 + 0) + 3(5 - 0) = -10 - 5 + 15 = 0.$

Since  $|\mathbf{A}| = 0$ , therefore to judge whether the given system is consistent or inconsistent, we shall calculate  $(\text{adj } \mathbf{A}) \mathbf{B}$ .

Let  $A_{ij}$  be the cofactor of  $a_{ij}$  in  $|\mathbf{A}|$ .

Then we have

$$A_{11} = -5, A_{12} = 5, A_{13} = 5, A_{21} = 10, A_{22} = -10, A_{23} = -10, \\ A_{31} = -5, A_{32} = 5, A_{33} = 5.$$

Now  $\text{adj } \mathbf{A} = \begin{bmatrix} -5 & 5 & 5 \\ 10 & -10 & -10 \\ -5 & 5 & 5 \end{bmatrix}^T = \begin{bmatrix} -5 & 10 & -5 \\ 5 & -10 & 5 \\ 5 & -10 & 5 \end{bmatrix},$

$$(\text{adj } \mathbf{A}) \mathbf{B} = \begin{bmatrix} -5 & 10 & -5 \\ 5 & -10 & 5 \\ 5 & -10 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -5 + 20 - 15 \\ 5 - 20 + 15 \\ 5 - 20 + 15 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Since  $|\mathbf{A}| = 0$ ,  $(\text{adj } \mathbf{A}) \mathbf{B} = 0$ , therefore the given system of linear equations is consistent and possesses infinite number of solutions.

Taking  $z = k$ , where  $k$  is real.

Substituting for  $z$  in the first and second equations of the given system, we get

$$2x - y = 1 - 3k, \quad x + 2y = 2 + k.$$

Writing these equations as a single matrix equation  $\mathbf{A} \mathbf{X} = \mathbf{B}$ , we get

$$\begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 - 3k \\ 2 + k \end{bmatrix}.$$

Here  $|\mathbf{A}| = 4 - 1 = 5$ .

Cofactors of the elements of the first row of  $|\mathbf{A}|$  are 2, -1.

Cofactors of the elements of the second row of  $|\mathbf{A}|$  are 1, 2.

Now  $\text{adj } \mathbf{A} = \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}^T = \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix}.$

$$\therefore \mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \text{adj } \mathbf{A} = \frac{1}{5} \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix}.$$

We have,  $\mathbf{X} = \mathbf{A}^{-1} \mathbf{B} = \frac{1}{5} \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 - 3k \\ 2 + k \end{bmatrix}$

$$= \frac{1}{5} \begin{bmatrix} 2 - 6k + 2 + k \\ -1 + 3k + 4 + 2k \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 4 - 5k \\ 3 + 5k \end{bmatrix}.$$

$$\therefore x = \frac{1}{5} (4 - 5k), \quad y = \frac{1}{5} (3 + 5k).$$

Since  $x = \frac{1}{5} (4 - 5k)$ ,  $y = \frac{1}{5} (3 + 5k)$ ,  $z = k$ ,  $k$  is real, also satisfy the third equation of the given system, therefore  $x = \frac{1}{5} (4 - 5k)$ ,  $y = \frac{1}{5} (3 + 5k)$ ,  $z = k$  form infinite solutions of the given system for all real values of  $k$ .

## Comprehensive Exercise 5

**Solve the following systems of equations using matrix method :**

1.  $3x + 4y - 5 = 0, x - y + 3 = 0.$
2.  $3x + y = 7, 5x + 3y = 12.$
3.  $x + 2y + z = 7, x + 3z = 11, 2x - 3y = 1.$
4.  $3x - 4y + 2z = -1, 2x + 3y + 5z = 7, x + z = 2.$

**Show that each one of the following systems of linear equations is inconsistent :**

5.  $4x - 2y = 3, 6x - 3y = 5.$
6.  $2x + 5y = 7, 6x + 15y = 13.$

7.  $x + y - 2z = 5, x - 2y + z = -2, -2x + y + z = 4.$

Show that each of the following systems of linear equations is consistent. Also find their solutions :

8.  $2x + 3y = 5, 6x + 9y = 15.$

9.  $x + y + z = 6, x + 2y + 3z = 14, x + 4y + 7z = 30.$

10.  $2x - y + 3z = 5, 3x + 2y - z = 7, 4x + 5y - 5z = 9.$

11. Find  $\mathbf{A}^{-1}$  if  $\mathbf{A} = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 3 & 2 \\ 3 & -3 & -4 \end{bmatrix}.$

Hence solve the system of linear equations

$$x + 2y - 3z = -4, 2x + 3y + 2z = 2, 3x - 3y - 4z = 11.$$

## Answers

1.  $x = -1, y = 2.$

2.  $x = \frac{9}{4}, y = \frac{1}{4}.$

3.  $x = 2, y = 1, z = 3.$

4.  $x = 3, y = 2, z = -1.$

8.  $x = \frac{1}{2}(5 - 3k), y = k$ , where  $k$  is any real number.

9.  $x = -2 + k, y = 8 - 2k, z = k, k$  is real.

10.  $x = \frac{1}{7}(17 - 5k), y = \frac{1}{7}(11k - 1), z = k, k$  is real.

11.  $\mathbf{A}^{-1} = -\frac{1}{67} \begin{bmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{bmatrix}; x = 3, y = 2, z = 1.$

## Comprehensive Exercise 6

### Fill in the Blanks

Fill in the blanks “.....”, so that the following statements are complete and correct.

1. If  $\mathbf{A} = \begin{bmatrix} 6 & 3 \\ x & 8 \end{bmatrix}$  and  $\mathbf{B} = \begin{bmatrix} y & z \\ 7 & 8 \end{bmatrix}$  are equal matrices, then the respective values of  $x, y$  and  $z$  are .....

2. If  $\mathbf{X} + \mathbf{Y} = \begin{bmatrix} 5 & 2 \\ 0 & 9 \end{bmatrix}$  and  $\mathbf{X} - \mathbf{Y} = \begin{bmatrix} 3 & 6 \\ 0 & -1 \end{bmatrix}$ , then the matrix  $\mathbf{X}$  is .....

3. If  $\mathbf{Y} = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$  and  $2\mathbf{X} + \mathbf{Y} = \begin{bmatrix} 1 & 0 \\ -3 & 2 \end{bmatrix}$  then  $\mathbf{X} = \dots\dots\dots$

4. The order of the matrix  $\begin{bmatrix} 3 \\ 8 \end{bmatrix}$  is ..... .
5. A square matrix each of whose non-diagonal elements is equal to zero is called a ..... matrix.
6. The necessary and sufficient condition for a square matrix  $\mathbf{A}$  to possess the inverse is that ..... .
7. If  $\mathbf{A} = \begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix}$ , then  $(\text{adj } \mathbf{A}) = \dots\dots\dots$  .
8. A square matrix  $\mathbf{A} = [a_{ij}]$  is called an ..... triangular matrix if  $a_{ij} = 0$  whenever  $i > j$ .
9. If  $\mathbf{A}$  is any matrix and  $\mathbf{O}$  is a null matrix, then  $\mathbf{A} + \mathbf{O}$  is ..... .

### Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

10. If  $\mathbf{A} = \begin{bmatrix} 2 & 7 \\ 9 & 8 \end{bmatrix}$ ,  $\mathbf{B} = \begin{bmatrix} 1 & 2 \\ 0 & 8 \end{bmatrix}$ , then  $\mathbf{A} - \mathbf{B}$  is
    - (a)  $\begin{bmatrix} 1 & 5 \\ 9 & 5 \end{bmatrix}$
    - (b)  $\begin{bmatrix} 9 & 5 \\ 1 & 5 \end{bmatrix}$
    - (c)  $\begin{bmatrix} 9 & 5 \\ 5 & 1 \end{bmatrix}$
    - (d)  $\begin{bmatrix} 5 & 9 \\ 5 & 1 \end{bmatrix}$
  11. The values of  $x$  and  $y$  such that  $\begin{bmatrix} x - y & 4 \\ x + y & 7 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 6 & 7 \end{bmatrix}$  is
    - (a)  $x = 2, y = 4$
    - (b)  $x = 2, y = 2$
    - (c)  $x = 4, y = 4$
    - (d)  $x = 4, y = 2$
  12. If  $\mathbf{A} = \begin{bmatrix} 2 & -1 \\ 0 & 1 \end{bmatrix}$  and  $\mathbf{B} = \begin{bmatrix} 2 & 4 & -3 \\ 6 & 0 & 3 \end{bmatrix}$ , then  $2\mathbf{A} + \mathbf{B}$  is
    - (a)  $\begin{bmatrix} 4 & -4 & 2 & 4 & -3 \\ 0 & 2 & 6 & 0 & 3 \end{bmatrix}$
    - (b)  $\begin{bmatrix} 2 & 4 & -3 \\ 6 & 0 & 3 \end{bmatrix}$
    - (c) does not exist
    - (d) none of these.
  13. The inverse of the matrix  $\mathbf{A} = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$  is
    - (a)  $\begin{bmatrix} -\cos \alpha & \sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$
    - (b)  $\begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$
    - (c)  $\begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$
    - (d)  $\begin{bmatrix} \cos \alpha & \sin \alpha \\ \sin \alpha & -\cos \alpha \end{bmatrix}$
-

14. The value of  $\cos \alpha \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} + \sin \alpha \begin{bmatrix} \sin \alpha & -\cos \alpha \\ \cos \alpha & \sin \alpha \end{bmatrix}$  is
- (a)  $\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$  (b)  $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$
- (c)  $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$  (d)  $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
15. If  $\begin{bmatrix} 1 & y \\ 0 & 1 \end{bmatrix} + 2 \begin{bmatrix} 3 & 4 \\ 5 & x \end{bmatrix} = \begin{bmatrix} 7 & 0 \\ 10 & 5 \end{bmatrix}$ , then  $x$  and  $y$  is
- (a)  $x = 8, y = 2$ , (b)  $x = -8, y = 2$
- (c)  $x = 2, y = -8$  (d)  $x = -2, y = -8$
16. The inverse of a square matrix  $\mathbf{A}$ , i.e.,  $\mathbf{A}^{-1}$  is
- (a)  $(adj \mathbf{A})^{-1}$  (b)  $\frac{(adj \mathbf{A})}{|\mathbf{A}|}$
- (c)  $|\mathbf{A}| (adj \mathbf{A})$  (d) none of these
17. The product of two matrices  $\begin{bmatrix} 1 & 5 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 6 \\ -1 \end{bmatrix}$  is
- (a)  $\begin{bmatrix} 9 \\ 1 \end{bmatrix}$  (b)  $[1 \ 9]$
- (c)  $\begin{bmatrix} 1 \\ 9 \end{bmatrix}$  (d)  $[9 \ 1]$
18. If  $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}$  and  $\mathbf{B} = \begin{bmatrix} 3 & 0 & 2 \\ 1 & -2 & 3 \end{bmatrix}$ , then  $\mathbf{AB}$  is
- (a) undefined (b)  $\begin{bmatrix} 7 & -2 & 7 \\ 3 & 0 & 2 \end{bmatrix}$
- (c)  $\begin{bmatrix} 7 & 3 \\ -2 & 0 \\ 7 & 2 \end{bmatrix}$  (d) none of these
19. The inverse of a matrix  $\begin{bmatrix} 1 & 3 & 4 \\ 2 & 0 & 5 \end{bmatrix}$  is
- (a)  $\begin{bmatrix} 1 & 4 \\ 2 & 5 \end{bmatrix}$  (b)  $\begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$
- (c) does not exist (d) none of these

[UPTU 2011]

**True or False***Write 'T' for true and 'F' for false statement.*

20. In a matrix the number of rows and columns may be different.
21. Matrix multiplication is always commutative.



22. If  $\begin{bmatrix} a+b & 2 \\ 5 & a-b \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 5 & 8 \end{bmatrix}$ , then the values of  $a$  and  $b$  are 7 and  $-1$  respectively.
23. The inverse of the matrix  $\begin{bmatrix} 5 & 2 \\ 7 & 3 \end{bmatrix}$  is  $\begin{bmatrix} 3 & -2 \\ -7 & 5 \end{bmatrix}$ .
24. If the matrices  $\mathbf{A}$  and  $\mathbf{B}$  are not of the same order then we cannot define  $\mathbf{A} + \mathbf{B}$ .
25. If  $\mathbf{A}$  is a square matrix and  $|\mathbf{A}| \neq 0$ , then  $\mathbf{A}$  is a singular matrix. [UPTU 2010]
26. A square matrix  $\mathbf{A}$  is invertible only if  $|\mathbf{A}| = 1$ .
27. Let  $\mathbf{AX} = \mathbf{B}$  be a system of  $n$  linear equations in  $n$  unknowns. If  $|\mathbf{A}| \neq 0$ , then the given system is consistent and has infinite number of solutions.
28. If  $2 \begin{bmatrix} 1 & 2 \\ 3 & x \end{bmatrix} + 3 \begin{bmatrix} 2 & y \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 8 & 10 \\ 6 & 21 \end{bmatrix}$ , then the respective values of  $x$  and  $y$  are 3 and 2.
29. If  $\mathbf{A} = \begin{bmatrix} 3 & 0 & 2 \\ 4 & 2 & 1 \end{bmatrix}$  and  $\mathbf{B} = \begin{bmatrix} 4 & 2 & 1 \\ 2 & 0 & -1 \end{bmatrix}$ , then  $\mathbf{A} + \mathbf{B}$  is  $\begin{bmatrix} 3 & 0 & 2 \\ 0 & 0 & -1 \end{bmatrix}$ .
30. If  $\begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix}$ , then  $(adj \mathbf{A})$  is  $\begin{bmatrix} 3 & -5 \\ -1 & 2 \end{bmatrix}$ .

## Answers

- |                 |                                                   |                                                                        |                 |
|-----------------|---------------------------------------------------|------------------------------------------------------------------------|-----------------|
| 1. 7, 6, 3      | 2. $\begin{bmatrix} 4 & 4 \\ 0 & 4 \end{bmatrix}$ | 3. $\begin{bmatrix} -1 & -1 \\ -2 & -1 \end{bmatrix}$                  | 4. $3 \times 1$ |
| 5. diagonal     | 6. $ \mathbf{A}  \neq 0$                          | 7. $\begin{bmatrix} \delta & -\beta \\ -\gamma & \alpha \end{bmatrix}$ | 8. upper        |
| 9. $\mathbf{A}$ | 10. (a)                                           | 11. (d)                                                                | 12. (c)         |
| 13. (b)         | 14. (c)                                           | 15. (c)                                                                | 16. (b)         |
| 17. (c)         | 18. (b)                                           | 19. (c)                                                                | 20. $T$         |
| 21. $F$         | 22. $T$                                           | 23. $T$                                                                | 24. $T$         |
| 25. $F$         | 26. $F$                                           | 27. $F$                                                                | 28. $T$         |
| 29. $F$         | 30. $T$                                           |                                                                        |                 |



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# Chapter

## 4



# Trigonometry

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## 4.1 Angles

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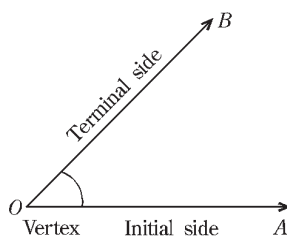
**A**n **angle**. If a ray  $\vec{OA}$  starting from its initial position  $OA$  rotates about its end point  $O$  and takes the final position  $OB$ , we say that the  $\angle AOB$  has been generated.

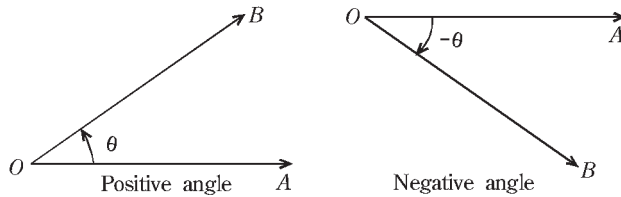
The revolving ray is called the *generating line* of the angle. Its initial position  $OA$  is called the *initial side* and the final position  $OB$  is called the *terminal side* of the angle. The end point  $O$  about which the ray rotates is called the *vertex* of the angle.

**Measure of an Angle.** *The measure of an angle is the amount of rotation of the revolving line from the initial side to the terminal side.*

**Positive and Negative Angles.** An angle described by a revolving ray is said to be *positive* or *negative* according as its rotation from the initial side to the terminal side is anticlockwise or clockwise respectively.

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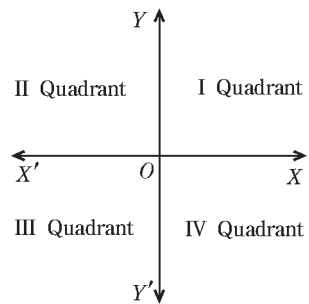




**Right Angle.** If the revolving ray starting from its initial position to the final position describes one quarter of a circle, we say that the measure of the angle so formed is a *right angle*.

## 4.2 Quadrants

Let  $X'OX$  and  $Y'OY$  be two coplanar lines at right angles to each other. These lines divide the plane into four equal parts, called *quadrants*. The lines  $X'OX$  and  $Y'OY$  are known as  $x$ -axis and  $y$ -axis respectively. The regions  $XOY$ ,  $YOX'$ ,  $X'OY'$  and  $Y'OX$  are known as the first, the second, the third and the fourth quadrants respectively.



## 4.3 Systems of Measurement of Angles

There are three systems for measuring angles, *viz.*, (i) Sexagesimal or English system (ii) Circular system and (iii) Centesimal or French system.

## 4.4 Sexagesimal System (English System or Degree Measure)

*The number of degrees on the circumference of the circle between the initial and terminal sides of the angle is its degree measure.*

Each degree is subdivided into 60 equal parts, called *minutes* and each minute is subdivided into 60 equal parts, called *seconds*. The symbol  $'$  is used to denote one minute and the symbol  $''$  is used to denote one second.

Thus,  $60^\circ 45' 25''$  means 60 degrees, 45 minutes and 25 seconds.

We have 1 right angle = 90 degrees ( $= 90^\circ$ )

$1^\circ = 60$  minutes ( $= 60'$ ) and  $1' = 60$  seconds ( $= 60''$ ).

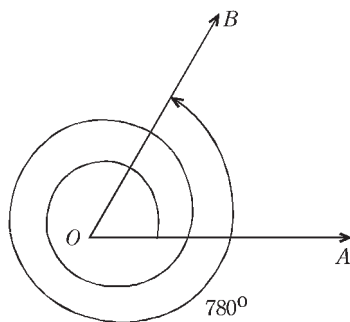
**Remark 1.** If there is no rotation, *i.e.*, the initial and terminal sides of the angle coincide, the measure of the angle is  $0^\circ$ .

**Remark 2.** If a ray  $\vec{OA}$  after having made  $n$  complete rotations arrives at the final position  $\vec{OB}$ , then the measure of the angle so formed is  $(n \cdot 360 + \theta)^\circ$ , where  $\angle AOB = \theta^\circ$ . In the adjoining figure, the ray  $\vec{OA}$  makes two complete rotations and then further moves through  $60^\circ$ . The measure of the angle so formed is

$$(2 \times 360 + 60)^\circ = 780^\circ.$$

**Different Types of Angles.** An angle  $\theta$  is said to be :

- (i) an acute angle, if  $0^\circ < \theta < 90^\circ$
- (ii) a right angle, if  $\theta = 90^\circ$
- (iii) an obtuse angle, if  $90^\circ < \theta < 180^\circ$
- (iv) a straight angle, if  $\theta = 180^\circ$
- (v) a reflex angle, if  $180^\circ < \theta < 360^\circ$ .



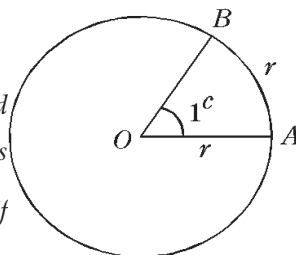
## 4.5 Circular System (or Radian Measure)

One radian, written as  $1^c$ , is the measure of an angle subtended at the centre of a circle of radius  $r$  by an arc of length  $r$ .

**Theorem 1.** Radian is a constant angle.

**Theorem 2.** The number of radians in an angle subtended by an arc of a circle at the centre  $= \frac{\text{arc}}{\text{radius}}$  i.e.,  $\theta = \frac{s}{r}$ , where  $\theta^c$  is

the angle subtended at the centre of a circle of radius  $r$  by an arc of length  $s$ .



## 4.6 Relation between Degrees and Radians

Suppose a ray  $\vec{OA}$  revolves about its end point  $O$  and generates a circle of radius  $r$ .

Then the radian measure of the angle formed by one complete revolution of the revolving ray

$$= \frac{2\pi r}{r} = 2\pi. \quad \left[ \because \theta = \frac{s}{r} = \frac{2\pi r}{r} \right]$$

But one complete revolution of the revolving ray generates an angle of  $360^\circ$ .

$\therefore 2\pi \text{ radians} = 360^\circ \text{ or } \pi \text{ radians} = 180^\circ.$

**Corollary 1.** Since  $\pi \text{ radians} = 180^\circ$ , therefore

$$1 \text{ radian} = \frac{180^\circ}{\pi} = \left( 180 \times \frac{7}{22} \right)^\circ = 57^\circ 16' 45'' \text{ (approx.)}$$

**Corollary 2.** Since  $180^\circ = \pi$  radians, therefore

$$1^\circ = \frac{\pi}{180} \text{ radians} = \left( \frac{22}{7 \times 180} \right) \text{ radians} = 0.01746 \text{ radian.}$$

## 4.7 Centesimal System (French System or Grade Measure)

In this system a right angle is divided into 100 equal parts, called *grades*. Each grade is subdivided into 100 equal parts, called *minutes* and each minute is subdivided into 100 equal parts, called *seconds*.

Thus, 1 right angle = 100 grades = 100<sup>g</sup>

1 grade = 100 minutes = 100' and 1 minute = 100 seconds = 100''.

Obviously,  $\pi$  radians =  $180^\circ = 200^\circ$ .

**Some Useful Facts About the Hands of a Clock.**

- (i) The angle between two consecutive digits in a clock is  $30^\circ$ , i.e.,  $\frac{\pi}{6}$  radians.
- (ii) The minute hand rotates through an angle of  $6^\circ$  in one minute.
- (iii) The hour hand rotates through an angle of  $30^\circ$  in one hour.

## Illustrative Examples

**Illustration 1 :** Find the degree measure corresponding to the following radian measures :

- (i)  $\left(\frac{7\pi}{12}\right)^c$
- (ii)  $\left(\frac{\pi}{8}\right)^c$
- (iii)  $\left(\frac{1}{4}\right)^c$
- (iv)  $-2^c$ .

**Solution :** We know that  $\pi$  radians =  $180^\circ$ . Therefore,  $1^c = \left(\frac{180}{\pi}\right)^\circ$ .

Hence,

- (i)  $\left(\frac{7\pi}{12}\right)^c = \left(\frac{7\pi}{12} \times \frac{180}{\pi}\right)^\circ = 105^\circ$ .
- (ii)  $\left(\frac{\pi}{8}\right)^c = \left(\frac{\pi}{8} \times \frac{180}{\pi}\right)^\circ = \left(\frac{45}{2}\right)^\circ = \left(22 \frac{1}{2}\right)^\circ = 22^\circ 30'$ .
- (iii)  $\left(\frac{1}{4}\right)^c = \left(\frac{1}{4} \times \frac{180}{\pi}\right)^\circ = \left(\frac{1}{4} \times 180 \times \frac{7}{22}\right)^\circ = \left(\frac{315}{22}\right)^\circ$   
 $= \left(14 \frac{7}{22}\right)^\circ = 14^\circ \left(\frac{7}{22} \times 60\right)' = 14^\circ \left(19 \frac{1}{11}\right)'$   
 $= 14^\circ 19' \left(\frac{1}{11} \times 60\right)'' = 14^\circ 19' 5''.$

$$\begin{aligned}
 \text{(iv)} \quad -2^\circ &= \left( \frac{180}{\pi} \times (-2) \right)^\circ = \left( 180 \times \frac{7}{22} \times (-2) \right)^\circ \\
 &= \left( -114 \frac{6}{11} \right)^\circ = -114^\circ 32' 44''.
 \end{aligned}$$

**Illustration 2 :** Find the radian measures corresponding to the following degree measures :

- (i)  $240^\circ$       (ii)  $75^\circ$       (iii)  $-22^\circ 30'$       (iv)  $5^\circ 37' 30''$ .

**Solution :** We know that  $180^\circ = \pi$  radians  $= \pi^c$ .

Therefore,  $1^\circ = \left( \frac{\pi}{180} \right)^c$ .

Hence,

$$\text{(i)} \quad 240^\circ = \left( \frac{\pi}{180} \times 240 \right)^c = \left( \frac{4\pi}{3} \right)^c.$$

$$\text{(ii)} \quad 75^\circ = \left( \frac{\pi}{180} \times 75 \right)^c = \left( \frac{5\pi}{12} \right)^c.$$

$$\text{(iii)} \quad -22^\circ 30' = \left( -22 \frac{1}{2} \right)^\circ = -\left( \frac{45}{2} \right)^\circ = -\left( \frac{\pi}{180} \times \frac{45}{2} \right)^c = -\left( \frac{\pi}{8} \right)^c.$$

$$\text{(iv)} \quad \text{We have, } 30'' = \left( \frac{30}{60} \right)' = \frac{1}{2}'.$$

$$\therefore 37^\circ 30'' = \left( 37 \frac{1}{2} \right)' = \left( \frac{75}{2} \right)' = \left( \frac{75}{2} \times \frac{1}{60} \right)^\circ = \left( \frac{5}{8} \right)^\circ.$$

$$\text{So, } 5^\circ 37' 30'' = \left( 5 \frac{5}{8} \right)^\circ = \left( \frac{45}{8} \right)^\circ = \left( \frac{45}{8} \times \frac{\pi}{180} \right)^c = \left( \frac{\pi}{32} \right)^c.$$

**Illustration 3 :** Find the length of an arc of a circle of radius 5 cm subtending a central angle  $15^\circ$ . [UPTU 2007]

**Solution :** Let  $s$  be the length of the arc subtending an angle  $\theta^c$  at the centre of a circle of radius  $r$ . Then,  $\theta = s / r$ .

$$\text{Here, } r = 5 \text{ cm and } \theta = 15^\circ = \left( 15 \times \frac{\pi}{180} \right)^c = \left( \frac{\pi}{12} \right)^c.$$

$$\therefore \theta = \frac{s}{r} \rightarrow s = \theta \times r = \left( \frac{\pi}{12} \times 5 \right) \text{ cm} = \frac{5\pi}{12} \text{ cm}.$$

**Illustration 4 :** Find the radius of a circle in which a central angle of  $45^\circ$  intercepts an arc of 187 cm.

**Solution :** Let  $r$  be the radius of the circle.

$$\text{Here, } s = 187 \text{ cm and } \theta = 45^\circ = \left( \frac{\pi}{180} \times 45 \right)^c = \left( \frac{\pi}{4} \right)^c.$$

$$\text{Now } \theta = \frac{s}{r} \rightarrow r = \frac{s}{\theta} = \left( 187 \times \frac{4}{\pi} \right) \text{ cm} = \left( 187 \times 4 \times \frac{7}{22} \right) \text{ cm} = 238 \text{ cm}.$$

**Illustration 5 :** Find in degrees the angle subtended at the centre of a circle of radius 10 ft by an arc of length 20 ft. [UPTU 2005]

**Solution :** Here,  $r = 10$  ft and  $s = 20$  ft.

$$\begin{aligned}\therefore \theta &= \left(\frac{s}{r}\right)^c = \left(\frac{20}{10}\right)^c = \left(2 \times \frac{180}{\pi}\right)^\circ \\ &= \left(2 \times \frac{7}{22} \times 180\right)^\circ = \left(\frac{1260}{11}\right)^\circ = \left(114 \frac{6}{11}\right)^\circ \\ &= 114^\circ \left(\frac{6}{11} \times 60\right)' = 114^\circ \left(32 \frac{8}{11}\right)' \\ &= 114^\circ 32' \left(\frac{8}{11} \times 60\right)'' = 114^\circ 32' 43''.\end{aligned}$$

**Illustration 6 :** A horse is tied to a post by a rope. If the horse moves along a circular path always keeping the rope tight and describes 88 metres when it has traced out  $72^\circ$  at the centre, find the length of the rope.

**Solution :** Let  $P$  be the position of the post and let  $PA$  be the length of the rope in tight position. Suppose the horse moves along the arc  $AB$  so that  $\angle APB = 72^\circ$  and arc  $AB = 88$  m.

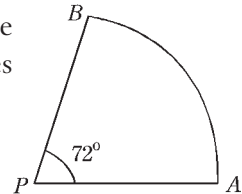
Let  $r$  be the length of the rope i.e.,  $PA = r$  metres.

$$\text{Now, } \theta = 72^\circ = \left(\frac{\pi}{180} \times 72\right)^c = \left(\frac{2\pi}{5}\right)^c$$

$$\text{and } s = \text{arc } AB = 88 \text{ metres.}$$

$$\text{We have, } \theta = \frac{s}{r} \Rightarrow \frac{2\pi}{5} = \frac{88}{r}$$

$$\Rightarrow r = \left(88 \times \frac{5}{2\pi}\right) \text{ m} = \left(88 \times \frac{5}{2} \times \frac{7}{22}\right) \text{ m} = 70 \text{ metres.}$$



**Illustration 7 :** The minute hand of a big clock is 70 cm long. How many centimetres does its extremity move in 6 minutes time ?

**Solution :** Angle traced by the minute hand in 60 minutes =  $360^\circ$ .

$$\therefore \text{Angle traced by it in 6 minutes} = \left(\frac{360}{60} \times 6\right)^\circ = 36^\circ.$$

$$\text{Thus, } \theta = 36^\circ = \left(\frac{\pi}{180} \times 36\right)^c = \left(\frac{\pi}{5}\right)^c \text{ and } r = 70 \text{ cm.}$$

Let  $s$  be the required arc length moved by the tip of the minute hand.

$$\begin{aligned}\text{Then, } \theta = \frac{s}{r} \Rightarrow s &= r \theta = \left(70 \times \frac{\pi}{5}\right) \text{ cm} \\ &= \left(70 \times \frac{22}{7} \times \frac{1}{5}\right) \text{ cm} = 44 \text{ cm.}\end{aligned}$$

**Illustration 8 :** Find the angle between the minute hand and the hour hand of a clock when the time is 7:20.

**Solution :** The angle traced by the hour hand in 12 hours =  $360^\circ$ .

$\therefore$  The angle traced by it in 7 hours 20 minutes, i.e.,  $\frac{22}{3}$  hrs

$$= \left( \frac{360}{12} \times \frac{22}{3} \right)^\circ = 220^\circ.$$

Also, the angle traced by the minute hand in 60 minutes =  $360^\circ$ .

$\therefore$  The angle traced by it in 20 minutes =  $\left( \frac{360}{60} \times 20 \right)^\circ = 120^\circ$ .

Hence, the required angle between the two hands =  $220^\circ - 120^\circ = 100^\circ$ .

**Illustration 9 :** The moon's distance from the earth is 360000 kms and its diameter subtends an angle of  $31'$  at the eye of the observer. Find the diameter of the moon.

**Solution :** Let  $AB$  be the diameter of the moon and let  $E$  be the eye of the observer. Since the distance between the earth and the moon is quite large, so we may take diameter  $AB$  of the moon as arc  $AB$  of a circle whose centre is  $E$ .

If the diameter of the moon is  $d$  kilometres, then arc  $AB = d$  km.

Now, 
$$\theta = 31' = \left( \frac{31}{60} \right)^\circ = \left( \frac{\pi}{180} \times \frac{31}{60} \right)^c$$

and 
$$r = 360000 \text{ km.}$$

$$\therefore \theta = \frac{\text{arc}}{\text{radius}}$$

$$\Rightarrow \frac{\pi}{180} \times \frac{31}{60} = \frac{d}{360000}$$

$$\Rightarrow d = \left( 360000 \times \frac{31}{60} \times \frac{22}{7} \times \frac{1}{180} \right) \text{ km} = 3247.62 \text{ km.}$$

Hence, the diameter of the moon is 3247.62 km.

**Illustration 10 :** The angles of a triangle are in A.P. The number of degrees in the least is to the number of radians in the greatest as  $60 : \pi$ . Find the angles in degrees.

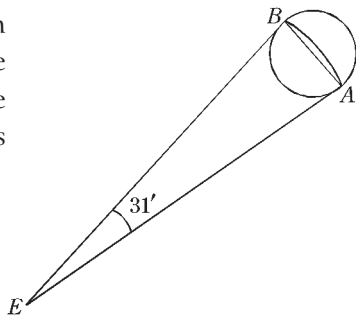
**Solution :** Let the angles be  $(a - d)^\circ$ ,  $a^\circ$  and  $(a + d)^\circ$ .

Then, 
$$(a - d) + a + (a + d) = 180$$

$$\Rightarrow 3a = 180 \Rightarrow a = 60.$$

$\therefore$  The angles are  $(60 - d)^\circ$ ,  $60^\circ$  and  $(60 + d)^\circ$ .

Obviously, the least angle is  $(60 - d)^\circ$  and the greatest is  $(60 + d)^\circ$ .





Now  $(60 + d)^\circ = \left[ \frac{\pi}{180} \times (60 + d) \right]^c$ .

It is given that,  $\frac{\text{Number of degrees in the least angle}}{\text{Number of radians in the greatest angle}} = \frac{60}{\pi}$ .

$$\therefore \frac{60 - d}{(60 + d) \cdot \frac{\pi}{180}} = \frac{60}{\pi}$$

$$\Rightarrow 3(60 - d) = 60 + d$$

$$\Rightarrow 120 = 4d \Rightarrow d = 30.$$

Hence, the angles are  $(60 - 30)^\circ$ ,  $60^\circ$  and  $(60 + 30)^\circ$  i.e.,  $30^\circ$ ,  $60^\circ$  and  $90^\circ$ .

## Comprehensive Exercise 1

1. Find the degree measure corresponding to the following radian measures :

(i)  $\left(\frac{2\pi}{15}\right)^c$

(ii)  $\left(-\frac{5\pi}{6}\right)^c$

(iii)  $\left(\frac{18\pi}{5}\right)^c$

(iv)  $11^c$

(v)  $-3^c$ .

2. Find the radian measure corresponding to the following degree measures :

(i)  $340^\circ$

(ii)  $15^\circ$

(iii)  $420^\circ$

(iv)  $-270^\circ$

(v)  $-300^\circ$

(vi)  $7^\circ 30'$ .

3. The difference between the two acute angles of a right angled triangle is

$\frac{2\pi}{5}$  radians. Express the angles in degrees.

4. The angles of a triangle are in A.P. such that the greatest is 5 times the least. Find the angles in radians.

5. The angles of a triangle are in A.P. The number of grades in the least is to the number of radians in the greatest as  $40 : \pi$ . Find the angles in degrees.

6. The circular measure of an angle of a triangle is  $\frac{3\pi}{10}$ , the number of grades in the second angle is 70. Find the number of degrees in the third angle.

7. The minute hand of a clock is 10 cm long. How far does the tip of the hand move in 20 minutes ?
8. A wheel makes 180 revolutions in 1 minute. Through how many radians does it turn in 1 second ?
9. A railway train is travelling on a circular curve of 1500 metres radius at the rate of 66 km per hour. Through what angle has it turned in 10 seconds ?
10. A railroad curve is to be laid out on a circle. What radius should be used if the track is to change direction of  $25^\circ$  in a distance of 40 metres ?
11. A wire 121 cm long is bent so as to lie along the arc of a circle of radius 180 cm. Find in degrees the angle subtended at the centre by the arc.
12. The number of sides in two regular polygons are 5 : 4 and the difference between their angles is  $9^\circ$ . Find the number of sides in the polygons.

## Answers

1. (i)  $24^\circ$       (ii)  $-150^\circ$       (iii)  $648^\circ$       (iv)  $630^\circ$   
 (v)  $-(171^\circ 49' 5'')$
2. (i)  $\left(\frac{17\pi}{9}\right)^c$       (ii)  $\left(\frac{\pi}{12}\right)^c$       (iii)  $\left(\frac{7\pi}{3}\right)^c$       (iv)  $\left(-\frac{3\pi}{2}\right)^c$   
 (v)  $\left(-\frac{5\pi}{3}\right)^c$       (vi)  $\left(\frac{\pi}{24}\right)^c$ .
3.  $81^\circ, 9^\circ$ .      4.  $\left(\frac{\pi}{9}\right)^c, \left(\frac{\pi}{3}\right)^c, \left(\frac{5\pi}{9}\right)^c$ .      5.  $20^\circ, 60^\circ$  and  $100^\circ$ .
6.  $63^\circ$ .      7.  $\frac{20\pi}{3}$  cm.      8.  $(6\pi)^c$ .      9.  $\left(\frac{11}{90}\right)^c$ .
10. 91.64 metres.      11.  $38^\circ 30'$ .      12. 10 and 8.

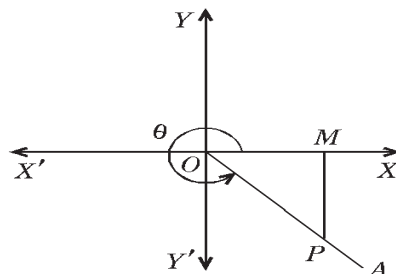
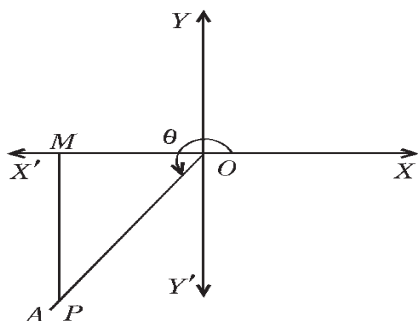
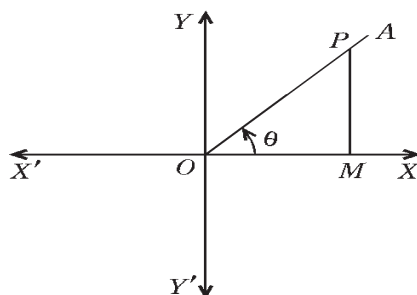
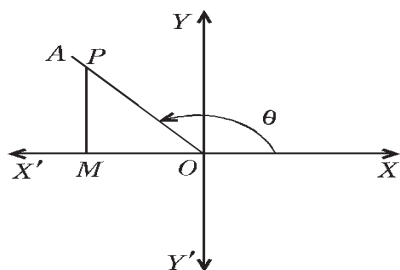
## 4.8 Trigonometric Ratios or Functions

Let a revolving ray start from  $OX$  and move about  $O$  so as to trace out an angle  $\theta = \angle XO A$ . Let  $P(x, y)$  be any point other than  $O$  on the terminal side  $OA$  of this angle  $\theta$ .

From  $P$  draw  $PM$  perpendicular to  $x$ -axis. Let length  $OP = r$ ,  $OM = x$  and  $MP = y$ . We take the length  $OP = r$  always positive while  $x$  and  $y$  can be positive or negative depending upon the position of the terminal side  $OA$  of  $\angle XO A$ .

In the right angled triangle  $OMP$  (known as the *triangle of reference*) we have base  $OM = x$ , perpendicular  $PM = y$  and hypotenuse  $OP = r$ .

We define the six *trigonometric ratios* which are also known as *trigonometric functions* as given below.



- (i) Sine  $\theta = \frac{\text{Perpendicular}}{\text{Hypotenuse}} = \frac{y}{r}$ , and is written as  $\sin \theta$  ;
- (ii) Cosine  $\theta = \frac{\text{Base}}{\text{Hypotenuse}} = \frac{x}{r}$ , and is written as  $\cos \theta$  ;
- (iii) Tangent  $\theta = \frac{\text{Perpendicular}}{\text{Base}} = \frac{y}{x}$ , and is written as  $\tan \theta$  ;
- (iv) Cosecant  $\theta = \frac{\text{Hypotenuse}}{\text{Perpendicular}} = \frac{r}{y}$ , and is written as  $\text{cosec } \theta$  ;
- (v) Secant  $\theta = \frac{\text{Hypotenuse}}{\text{Base}} = \frac{r}{x}$ , and is written as  $\sec \theta$  ;
- (vi) Cotangent  $\theta = \frac{\text{Base}}{\text{Perpendicular}} = \frac{x}{y}$ , and is written as  $\cot \theta$ .

**Remark 1.** When the terminal side coincides with  $x$ -axis, we have  $y = 0$ . So in this case  $\text{cosec } \theta$  and  $\cot \theta$  are not defined. When the terminal side coincides with  $y$ -axis, we have  $x = 0$ . So in this case  $\sec \theta$  and  $\tan \theta$  are not defined.

**Remark 2.** The trigonometric ratios may be positive or negative depending upon  $x$  and / or  $y$ .

**Remark 3.** The following relations are obvious from the definitions of trigonometric ratios :

$$(i) \sin \theta \operatorname{cosec} \theta = 1 \rightarrow \sin \theta = \frac{1}{\operatorname{cosec} \theta} \text{ and } \operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

$$(ii) \cos \theta \sec \theta = 1 \rightarrow \cos \theta = \frac{1}{\sec \theta} \text{ and } \sec \theta = \frac{1}{\cos \theta}$$

$$(iii) \tan \theta \cot \theta = 1 \rightarrow \tan \theta = \frac{1}{\cot \theta} \text{ and } \cot \theta = \frac{1}{\tan \theta}$$

$$(iv) \tan \theta = \frac{\sin \theta}{\cos \theta} \text{ and } \cot \theta = \frac{\cos \theta}{\sin \theta}.$$

## 4.9 Fundamental Trigonometric Identities

For all values of  $\theta$  for which the trigonometric functions involved are defined, we have

$$(i) \sin^2 \theta + \cos^2 \theta = 1 \quad (ii) 1 + \tan^2 \theta = \sec^2 \theta$$

$$(iii) 1 + \cot^2 \theta = \operatorname{cosec}^2 \theta \quad (iv) \tan \theta \cdot \cot \theta = 1$$

$$(v) \sec \theta \cdot \cos \theta = 1 \quad (vi) \sin \theta \cdot \operatorname{cosec} \theta = 1$$

$$(vii) \tan \theta = \frac{\sin \theta}{\cos \theta} \quad (viii) \cot \theta = \frac{\cos \theta}{\sin \theta}.$$

**Remark 1.** For convenience, we write  $(\sin \theta)^2$  as  $\sin^2 \theta$ ,  $(\cos \theta)^2$  as  $\cos^2 \theta$ ,  $(\tan \theta)^2$  as  $\tan^2 \theta$ , etc.

**Remark 2.**  $1 + \tan^2 \theta = \sec^2 \theta \rightarrow \sec^2 \theta - \tan^2 \theta = 1$

$$\Rightarrow (\sec \theta - \tan \theta)(\sec \theta + \tan \theta) = 1$$

$$\Rightarrow \sec \theta - \tan \theta = \frac{1}{\sec \theta + \tan \theta}$$

$$\text{and} \quad \sec \theta + \tan \theta = \frac{1}{\sec \theta - \tan \theta}.$$

## Illustrative Examples

**Illustration 1 :** Prove the following identities :

$$(i) \tan^4 \theta + \tan^2 \theta = \sec^4 \theta - \sec^2 \theta$$

$$(ii) \sin^8 \theta - \cos^8 \theta = (\sin^2 \theta - \cos^2 \theta)(1 - 2 \sin^2 \theta \cos^2 \theta)$$

$$(iii) \tan^2 \theta - \sin^2 \theta = \tan^2 \theta \sin^2 \theta$$

$$(iv) (\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2 = \tan^2 \theta + \cot^2 \theta + 7.$$

$$(v) \tan^2 \theta + \cot^2 \theta + 2 = \sec^2 \theta \cdot \operatorname{cosec}^2 \theta$$

[UPTU 2001 (Special)]

**Solution.** (i) We have L.H.S. =  $\tan^4 \theta + \tan^2 \theta$

$$= \tan^2 \theta (\tan^2 \theta + 1) = \tan^2 \theta \sec^2 \theta \quad [\because \sec^2 \theta = 1 + \tan^2 \theta]$$

$$\begin{aligned}
 &= (\sec^2 \theta - 1) \sec^2 \theta \quad [\because \sec^2 \theta = 1 + \tan^2 \theta \Rightarrow \tan^2 \theta = \sec^2 \theta - 1] \\
 &= \sec^4 \theta - \sec^2 \theta = \text{R.H.S.}
 \end{aligned}$$

(ii) We have L.H.S. =  $\sin^8 \theta - \cos^8 \theta$

$$\begin{aligned}
 &= (\sin^4 \theta)^2 - (\cos^4 \theta)^2 = (\sin^4 \theta - \cos^4 \theta)(\sin^4 \theta + \cos^4 \theta) \\
 &= (\sin^2 \theta - \cos^2 \theta)(\sin^2 \theta + \cos^2 \theta)[(\sin^2 \theta + \cos^2 \theta)^2 \\
 &\quad - 2 \sin^2 \theta \cos^2 \theta] \\
 &= (\sin^2 \theta - \cos^2 \theta)(1 - 2 \sin^2 \theta \cos^2 \theta) = \text{R.H.S.} \\
 &\quad [\because \sin^2 \theta + \cos^2 \theta = 1]
 \end{aligned}$$

(iii) We have L.H.S. =  $\tan^2 \theta - \sin^2 \theta$

$$\begin{aligned}
 &= \frac{\sin^2 \theta}{\cos^2 \theta} - \sin^2 \theta = \frac{\sin^2 \theta - \sin^2 \theta \cos^2 \theta}{\cos^2 \theta} \\
 &= \frac{\sin^2 \theta (1 - \cos^2 \theta)}{\cos^2 \theta} = \frac{\sin^2 \theta}{\cos^2 \theta} \cdot \sin^2 \theta = \tan^2 \theta \sin^2 \theta = \text{R.H.S.}
 \end{aligned}$$

(iv) We have L.H.S. =  $(\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2$

$$\begin{aligned}
 &= \sin^2 \theta + \operatorname{cosec}^2 \theta + 2 \sin \theta \operatorname{cosec} \theta + \cos^2 \theta + \sec^2 \theta \\
 &\quad + 2 \cos \theta \sec \theta \\
 &= (\sin^2 \theta + \cos^2 \theta) + \operatorname{cosec}^2 \theta + \sec^2 \theta + 2 + 2 \\
 &= 1 + (1 + \cot^2 \theta) + (1 + \tan^2 \theta) + 2 + 2 \\
 &= \tan^2 \theta + \cot^2 \theta + 7 = \text{R.H.S.}
 \end{aligned}$$

(v) We have L.H.S. =  $\tan^2 \theta + \cot^2 \theta + 2$

$$\begin{aligned}
 &= \frac{\sin^2 \theta}{\cos^2 \theta} + \frac{\cos^2 \theta}{\sin^2 \theta} + 2 = \frac{\sin^4 \theta + \cos^4 \theta}{\sin^2 \theta \cos^2 \theta} + 2 \\
 &= \frac{(\sin^2 \theta + \cos^2 \theta)^2 - 2 \sin^2 \theta \cos^2 \theta}{\sin^2 \theta \cos^2 \theta} + 2 \\
 &\quad [\because a^2 + b^2 = (a + b)^2 - 2ab] \\
 &= \frac{1 - 2 \sin^2 \theta \cos^2 \theta}{\sin^2 \theta \cos^2 \theta} + 2 = \frac{1}{\sin^2 \theta \cos^2 \theta} = \sec^2 \theta \operatorname{cosec}^2 \theta.
 \end{aligned}$$

**Illustration 2 :** Prove the following identities :

(i)  $\sin^6 \theta + \cos^6 \theta = 1 - 3 \sin^2 \theta \cos^2 \theta$

[UPTU 2007]

(ii)  $2 \sec^2 \theta - \sec^4 \theta - 2 \operatorname{cosec}^2 \theta + \operatorname{cosec}^4 \theta = \cot^4 \theta - \tan^4 \theta$

(iii)  $(1 + \cot \theta - \operatorname{cosec} \theta)(1 + \tan \theta + \sec \theta) = 2$

(iv)  $\frac{\tan \theta + \sec \theta - 1}{\tan \theta - \sec \theta + 1} = \frac{1 + \sin \theta}{\cos \theta}$

[UPTU 2004]

**Solution :** (i) We have L.H.S. =  $\sin^6 \theta + \cos^6 \theta = (\sin^2 \theta)^3 + (\cos^2 \theta)^3$   
 $= (\sin^2 \theta + \cos^2 \theta)(\sin^4 \theta + \cos^4 \theta - \sin^2 \theta \cos^2 \theta)$   
 $\quad [\because a^3 + b^3 = (a + b)(a^2 + b^2 - ab)]$   
 $= \sin^4 \theta + \cos^4 \theta - \sin^2 \theta \cos^2 \theta \quad [\because \sin^2 \theta + \cos^2 \theta = 1]$   
 $= [(\sin^2 \theta + \cos^2 \theta)^2 - 2 \sin^2 \theta \cos^2 \theta] - \sin^2 \theta \cos^2 \theta$   
 $= 1 - 3 \sin^2 \theta \cos^2 \theta = \text{R.H.S.}$

(ii) We have L.H.S. =  $2 \sec^2 \theta - \sec^4 \theta - 2 \operatorname{cosec}^2 \theta + \operatorname{cosec}^4 \theta$   
 $= 2(1 + \tan^2 \theta) - (\sec^2 \theta)^2 - 2(1 + \cot^2 \theta) + (\operatorname{cosec}^2 \theta)^2$   
 $= 2 + 2 \tan^2 \theta - (1 + \tan^2 \theta)^2 - 2 - 2 \cot^2 \theta + (1 + \cot^2 \theta)^2$   
 $= 2 \tan^2 \theta - (1 + \tan^4 \theta + 2 \tan^2 \theta) - 2 \cot^2 \theta$   
 $\quad + (1 + \cot^4 \theta + 2 \cot^2 \theta)$   
 $= \cot^4 \theta - \tan^4 \theta = \text{R.H.S.}$

(iii) We have L.H.S. =  $(1 + \cot \theta - \operatorname{cosec} \theta)(1 + \tan \theta + \sec \theta)$   
 $= \left(1 + \frac{\cos \theta}{\sin \theta} - \frac{1}{\sin \theta}\right) \left(1 + \frac{\sin \theta}{\cos \theta} + \frac{1}{\cos \theta}\right)$   
 $= \frac{\sin \theta + \cos \theta - 1}{\sin \theta} \cdot \frac{\cos \theta + \sin \theta + 1}{\cos \theta}$   
 $= \frac{(\sin \theta + \cos \theta)^2 - 1}{\sin \theta \cos \theta} = \frac{\sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta - 1}{\sin \theta \cos \theta}$   
 $= \frac{1 + 2 \sin \theta \cos \theta - 1}{\sin \theta \cos \theta} = \frac{2 \sin \theta \cos \theta}{\sin \theta \cos \theta} = 2 = \text{R.H.S.}$

(iv) We have L.H.S. =  $\frac{\tan \theta + \sec \theta - 1}{\tan \theta - \sec \theta + 1}$   
 $= \frac{(\tan \theta + \sec \theta) - (\sec^2 \theta - \tan^2 \theta)}{\tan \theta - \sec \theta + 1} \quad [\because \sec^2 \theta - \tan^2 \theta = 1]$   
 $= \frac{(\sec \theta + \tan \theta) - (\sec \theta + \tan \theta)(\sec \theta - \tan \theta)}{\tan \theta - \sec \theta + 1}$   
 $= \frac{(\sec \theta + \tan \theta) [1 - (\sec \theta - \tan \theta)]}{\tan \theta - \sec \theta + 1}$   
 $= \frac{(\sec \theta + \tan \theta)(\tan \theta - \sec \theta + 1)}{\tan \theta - \sec \theta + 1}$   
 $= \sec \theta + \tan \theta = \frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta} = \frac{1 + \sin \theta}{\cos \theta} = \text{R.H.S.}$

**Illustration 3 :** Prove that the expression  $2 (\sin^6 \theta + \cos^6 \theta) - 3 (\sin^4 \theta + \cos^4 \theta)$  is independent of the angle  $\theta$ . [UPTU 2005]

**Solution :** We have  $2 (\sin^6 \theta + \cos^6 \theta) - 3 (\sin^4 \theta + \cos^4 \theta)$   
 $= 2 [(\sin^2 \theta)^3 + (\cos^2 \theta)^3] - 3 (\sin^4 \theta + \cos^4 \theta)$   
 $= 2 [(\sin^2 \theta + \cos^2 \theta) (\sin^4 \theta + \cos^4 \theta - \sin^2 \theta \cos^2 \theta)]$   
 $\quad - 3 (\sin^4 \theta + \cos^4 \theta) \quad [\because a^3 + b^3 = (a + b) (a^2 + b^2 - ab)]$   
 $= 2 \sin^4 \theta + 2 \cos^4 \theta - 2 \sin^2 \theta \cos^2 \theta - 3 \sin^4 \theta - 3 \cos^4 \theta$   
 $\quad [\because \sin^2 \theta + \cos^2 \theta = 1]$   
 $= - (\sin^4 \theta + \cos^4 \theta + 2 \sin^2 \theta \cos^2 \theta)$   
 $= - (\sin^2 \theta + \cos^2 \theta)^2$   
 $= -1$ , which is independent of the angle  $\theta$ .

**Illustration 4 :** Prove the following identities :

$$(i) \frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta} = \sec \theta \operatorname{cosec} \theta + 1$$

$$(ii) \sqrt{\frac{1 + \cos \theta}{1 - \cos \theta}} = \operatorname{cosec} \theta + \cot \theta.$$

**Solution :** (i) L.H.S.  $= \frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta}$   
 $= \frac{\frac{\sin \theta}{\cos \theta}}{1 - \frac{\cos \theta}{\sin \theta}} + \frac{\frac{\cos \theta}{\sin \theta}}{1 - \frac{\sin \theta}{\cos \theta}}$   
 $= \frac{\sin \theta}{\cos \theta} \cdot \frac{\sin \theta}{\sin \theta - \cos \theta} + \frac{\cos \theta}{\sin \theta} \cdot \frac{\cos \theta}{\cos \theta - \sin \theta}$   
 $= \frac{\sin^2 \theta}{\cos \theta (\sin \theta - \cos \theta)} + \frac{\cos^2 \theta}{\sin \theta (\cos \theta - \sin \theta)}$   
 $= \frac{\sin^3 \theta - \cos^3 \theta}{\cos \theta \sin \theta (\sin \theta - \cos \theta)}$   
 $= \frac{(\sin \theta - \cos \theta)(\sin^2 \theta + \cos^2 \theta + \sin \theta \cos \theta)}{\cos \theta \sin \theta (\sin \theta - \cos \theta)}$   
 $\quad [\because a^3 - b^3 = (a - b) (a^2 + b^2 + ab)]$   
 $= \frac{1 + \sin \theta \cos \theta}{\sin \theta \cos \theta} = \frac{1}{\sin \theta \cos \theta} + 1 = \sec \theta \operatorname{cosec} \theta + 1 = \text{R.H.S.}$

$$(ii) \text{ We have L.H.S. } = \sqrt{\frac{1 + \cos \theta}{1 - \cos \theta}}$$

$$\begin{aligned}
 &= \frac{\sqrt{1 + \cos \theta}}{\sqrt{1 - \cos \theta}} \times \frac{\sqrt{1 + \cos \theta}}{\sqrt{1 + \cos \theta}} \\
 &= \frac{1 + \cos \theta}{\sqrt{(1 - \cos \theta)(1 + \cos \theta)}} = \frac{1 + \cos \theta}{\sqrt{1 - \cos^2 \theta}} \\
 &= \frac{1 + \cos \theta}{\sqrt{\sin^2 \theta}} = \frac{1 + \cos \theta}{\sin \theta} \\
 &= \frac{1}{\sin \theta} + \frac{\cos \theta}{\sin \theta} = \operatorname{cosec} \theta + \cot \theta = \text{R. H. S.}
 \end{aligned}$$

**Illustration 5 :** If  $\tan \theta + \sin \theta = m$  and  $\tan \theta - \sin \theta = n$ , show that

$$m^2 - n^2 = 4 \sqrt{mn}.$$

**Solution :** Given that  $\tan \theta + \sin \theta = m$  ... (1)

and  $\tan \theta - \sin \theta = n$  ... (2)

Adding (1) and (2), we have  $2 \tan \theta = m + n$

$$\text{or} \quad \tan \theta = \frac{m + n}{2} \quad \text{or} \quad \cot \theta = \frac{2}{m + n}.$$

Subtracting (2) from (1), we have

$$2 \sin \theta = m - n \quad \text{or} \quad \sin \theta = \frac{m - n}{2}$$

$$\text{or} \quad \operatorname{cosec} \theta = \frac{2}{m - n}.$$

Now we know that  $\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$ .

$$\therefore \quad \frac{4}{(m - n)^2} - \frac{4}{(m + n)^2} = 1$$

$$\text{or} \quad \frac{4[(m + n)^2 - (m - n)^2]}{(m - n)^2 (m + n)^2} = 1$$

$$\text{or} \quad \frac{4(4mn)}{(m^2 - n^2)^2} = 1$$

$$\text{or} \quad 16mn = (m^2 - n^2)^2$$

$$\text{or} \quad m^2 - n^2 = 4 \sqrt{mn}.$$

**Illustration 6 :** If  $\cos \theta + \sin \theta = \sqrt{2} \cos \theta$ , show that  $\cos \theta - \sin \theta = \sqrt{2} \sin \theta$ .

[UPTU 2003]

**Solution :** We have  $\cos \theta + \sin \theta = \sqrt{2} \cos \theta$

$$\Rightarrow \quad \sin \theta = (\sqrt{2} - 1) \cos \theta$$

$$\Rightarrow \quad (\sqrt{2} + 1) \sin \theta = (\sqrt{2} + 1)(\sqrt{2} - 1) \cos \theta,$$

[Multiplying both sides by  $(\sqrt{2} + 1)$ ]

$$\Rightarrow \quad (\sqrt{2} + 1) \sin \theta = (2 - 1) \cos \theta = \cos \theta$$



$$\Rightarrow \cos \theta - \sin \theta = \sqrt{2} \sin \theta.$$

**Illustration 7 :** If  $a \cos \theta - b \sin \theta = c$ , show that

$$a \sin \theta + b \cos \theta = \pm \sqrt{a^2 + b^2 - c^2}.$$

**Solution :** We have  $a \cos \theta - b \sin \theta = c$

$$\Rightarrow (a \cos \theta - b \sin \theta)^2 = c^2$$

$$\Rightarrow a^2 \cos^2 \theta - 2ab \cos \theta \sin \theta + b^2 \sin^2 \theta = c^2$$

$$\Rightarrow a^2 (1 - \sin^2 \theta) - 2ab \cos \theta \sin \theta + b^2 (1 - \cos^2 \theta) = c^2$$

$$\Rightarrow a^2 \sin^2 \theta + b^2 \cos^2 \theta + 2ab \cos \theta \sin \theta = a^2 + b^2 - c^2$$

$$\Rightarrow (a \sin \theta + b \cos \theta)^2 = a^2 + b^2 - c^2$$

$$\Rightarrow a \sin \theta + b \cos \theta = \pm \sqrt{a^2 + b^2 - c^2}.$$

**Illustration 8 :** If  $a \cos^3 \theta + 3a \cos \theta \sin^2 \theta = m$  and  $a \sin^3 \theta + 3a \cos^2 \theta \sin \theta = n$ , then prove that  $(m+n)^{2/3} + (m-n)^{2/3} = 2a^{2/3}$ .

**Solution :** We have

$$a \cos^3 \theta + 3a \cos \theta \sin^2 \theta = m$$

and  $a \sin^3 \theta + 3a \cos^2 \theta \sin \theta = n$

$$\Rightarrow a \cos^3 \theta + 3a \cos \theta \sin^2 \theta + a \sin^3 \theta + 3a \cos^2 \theta \sin \theta = m + n$$

and  $a \cos^3 \theta + 3a \cos \theta \sin^2 \theta - a \sin^3 \theta - 3a \cos^2 \theta \sin \theta = m - n$

$$\Rightarrow a (\cos^3 \theta + 3 \cos^2 \theta \sin \theta + 3 \cos \theta \sin^2 \theta + \sin^3 \theta) = m + n$$

and  $a (\cos^3 \theta - 3 \cos^2 \theta \sin \theta + 3 \cos \theta \sin^2 \theta - \sin^3 \theta) = m - n$

$$\Rightarrow a (\cos \theta + \sin \theta)^3 = m + n$$

and  $a (\cos \theta - \sin \theta)^3 = m - n$

$$\Rightarrow \cos \theta + \sin \theta = \left( \frac{m+n}{a} \right)^{1/3}$$

and  $\cos \theta - \sin \theta = \left( \frac{m-n}{a} \right)^{1/3}$

$$\Rightarrow (\cos \theta + \sin \theta)^2 + (\cos \theta - \sin \theta)^2 = \left( \frac{m+n}{a} \right)^{2/3} + \left( \frac{m-n}{a} \right)^{2/3}$$

$$\Rightarrow 2 (\cos^2 \theta + \sin^2 \theta) = \left( \frac{m+n}{a} \right)^{2/3} + \left( \frac{m-n}{a} \right)^{2/3}$$

$$\Rightarrow (m+n)^{2/3} + (m-n)^{2/3} = 2a^{2/3}.$$

**Illustration 9 :** If  $\frac{ax}{\cos \theta} + \frac{by}{\sin \theta} = a^2 - b^2$  and  $\frac{ax \sin \theta}{\cos^2 \theta} - \frac{by \cos \theta}{\sin^2 \theta} = 0$ , prove that

$$(ax)^{2/3} + (by)^{2/3} = (a^2 - b^2)^{2/3}.$$

**Solution :** We have  $\frac{a x \sin \theta}{\cos^2 \theta} - \frac{b y \cos \theta}{\sin^2 \theta} = 0$

$$\Rightarrow \frac{a x \sin \theta}{\cos^2 \theta} = \frac{b y \cos \theta}{\sin^2 \theta}$$

$$\Rightarrow \frac{\sin^3 \theta}{\cos^3 \theta} = \frac{b y}{a x} \Rightarrow \tan^3 \theta = \frac{b y}{a x}$$

$$\Rightarrow \tan \theta = \frac{(b y)^{1/3}}{(a x)^{1/3}}$$

$$\Rightarrow \sin \theta = \frac{(b y)^{1/3}}{\sqrt{(a x)^{2/3} + (b y)^{2/3}}}$$

and  $\cos \theta = \frac{(a x)^{1/3}}{\sqrt{(a x)^{2/3} + (b y)^{2/3}}}.$

Putting these values of  $\sin \theta$  and  $\cos \theta$  in  $\frac{a x}{\cos \theta} + \frac{b y}{\sin \theta} = a^2 - b^2$ , we get

$$(a x) \frac{[(a x)^{2/3} + (b y)^{2/3}]^{1/2}}{(a x)^{1/3}} + (b y) \frac{[(a x)^{2/3} + (b y)^{2/3}]^{1/2}}{(b y)^{1/3}} = a^2 - b^2$$

or  $[(a x)^{2/3} + (b y)^{2/3}]^{1/2} [(a x)^{2/3} + (b y)^{2/3}] = a^2 - b^2$

or  $[(a x)^{2/3} + (b y)^{2/3}]^{3/2} = a^2 - b^2$

or  $(a x)^{2/3} + (b y)^{2/3} = (a^2 - b^2)^{2/3}.$

## Comprehensive Exercise 2

**Prove the following trigonometric identities :**

1.  $\cot^4 \theta + \cot^2 \theta = \operatorname{cosec}^4 \theta - \operatorname{cosec}^2 \theta.$

2.  $(\operatorname{cosec} \theta - \sin \theta)(\sec \theta - \cos \theta)(\tan \theta + \cot \theta) = 1.$

3.  $\operatorname{cosec} \theta (\sec \theta - 1) - \cot \theta (1 - \cos \theta) = \tan \theta - \sin \theta.$

4.  $\sec^2 \theta + \operatorname{cosec}^2 \theta = \sec^2 \theta \operatorname{cosec}^2 \theta.$

5.  $\sec^4 \theta (1 - \sin^4 \theta) - 2 \tan^2 \theta = 1.$

6.  $\frac{1 - \tan \theta}{1 + \tan \theta} = \frac{\cot \theta - 1}{\cot \theta + 1}.$

7.  $\frac{1 + \cos \theta}{1 - \cos \theta} = (\operatorname{cosec} \theta + \cot \theta)^2.$

8.  $\frac{1}{1 + \sin \theta} + \frac{1}{1 - \sin \theta} = 2 \sec^2 \theta.$

9.  $\frac{\tan^2 \theta}{1 + \tan^2 \theta} + \frac{\cot^2 \theta}{1 + \cot^2 \theta} = 1.$

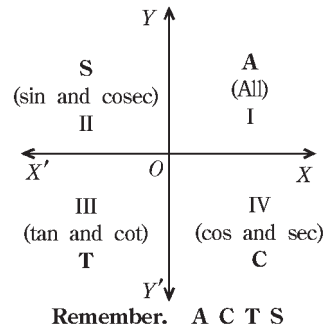
10.  $\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} = \cos^2 \theta - \sin^2 \theta$ .    11.  $\frac{\cot A}{\operatorname{cosec} A + 1} + \frac{\operatorname{cosec} A + 1}{\cot A} = 2 \sec A$ .
12.  $\cot^2 \theta \left( \frac{\sec \theta - 1}{1 + \sin \theta} \right) + \sec^2 \theta \left( \frac{\sin \theta - 1}{1 + \sec \theta} \right) = 0$ .
13.  $(\sec A - \operatorname{cosec} A)(1 + \tan A + \cot A) = \tan A \sec A - \cot A \operatorname{cosec} A$ .
14.  $\cos \theta (\tan \theta + 2)(2 \tan \theta + 1) = 2 \sec \theta + 5 \sin \theta$ .
15.  $1 - \frac{\sin^2 \theta}{1 + \cot \theta} - \frac{\cos^2 \theta}{1 + \tan \theta} = \sin \theta \cos \theta$ .
16.  $\frac{\tan A + \tan B}{\cot A + \cot B} = \tan A \tan B$ .
17.  $(1 + \tan \alpha \tan \beta)^2 + (\tan \alpha - \tan \beta)^2 = \sec^2 \alpha \sec^2 \beta$ .
18.  $\sin^2 A \cos^2 B - \cos^2 A \sin^2 B = \sin^2 A - \sin^2 B$ .
19.  $a \cos \theta + b \sin \theta = x$  and  $a \sin \theta - b \cos \theta = y$ , prove that  $a^2 + b^2 = x^2 + y^2$ .
20. If  $\operatorname{cosec} \theta - \sin \theta = a^3$ ,  $\sec \theta - \cos \theta = b^3$ , then prove that  $a^2 b^2 (a^2 + b^2) = 1$ .
21. If  $\cot \theta (1 + \sin \theta) = 4n$  and  $\cot \theta (1 - \sin \theta) = 4n$ , prove that  $(m^2 - n^2)^2 = mn$ .

## 4.10 Values of Trigonometric Functions at some Special Angles

| Measure of $\theta$ |                  | $\sin \theta$        | $\cos \theta$        | $\tan \theta$        | $\cot \theta$        | $\operatorname{cosec} \theta$ | $\sec \theta$        |
|---------------------|------------------|----------------------|----------------------|----------------------|----------------------|-------------------------------|----------------------|
| In degrees          | In radians       |                      |                      |                      |                      |                               |                      |
| $0^\circ$           | 0                | 0                    | 1                    | 0                    | undefined            | undefined                     | 1                    |
| $30^\circ$          | $\frac{\pi}{6}$  | $\frac{1}{2}$        | $\frac{\sqrt{3}}{2}$ | $\frac{1}{\sqrt{3}}$ | $\sqrt{3}$           | 2                             | $\frac{2}{\sqrt{3}}$ |
| $45^\circ$          | $\frac{\pi}{4}$  | $\frac{1}{\sqrt{2}}$ | $\frac{1}{\sqrt{2}}$ | 1                    | 1                    | $\sqrt{2}$                    | $\sqrt{2}$           |
| $60^\circ$          | $\frac{\pi}{3}$  | $\frac{\sqrt{3}}{2}$ | $\frac{1}{2}$        | $\sqrt{3}$           | $\frac{1}{\sqrt{3}}$ | $\frac{2}{\sqrt{3}}$          | 2                    |
| $90^\circ$          | $\frac{\pi}{2}$  | 1                    | 0                    | undefined            | 0                    | 1                             | undefined            |
| $180^\circ$         | $\pi$            | 0                    | -1                   | 0                    | undefined            | undefined                     | -1                   |
| $270^\circ$         | $\frac{3\pi}{2}$ | -1                   | 0                    | undefined            | 0                    | -1                            | undefined            |

# 4.11 Signs of the Values of Trigonometric Functions in Different Quadrants

- (i) In I quadrant, all trigonometric ratios are positive ;
- (ii) In II quadrant  $\sin \theta$  and  $\operatorname{cosec} \theta$  are positive while  $\cos \theta$ ,  $\sec \theta$ ,  $\tan \theta$  and  $\cot \theta$  are negative ;
- (iii) In III quadrant  $\tan \theta$  and  $\cot \theta$  are positive while  $\sin \theta$ ,  $\operatorname{cosec} \theta$ ,  $\cos \theta$  and  $\sec \theta$  are negative ;
- (iv) In IV quadrant  $\cos \theta$  and  $\sec \theta$  are positive while  $\sin \theta$ ,  $\operatorname{cosec} \theta$ ,  $\tan \theta$  and  $\cot \theta$  are negative.



| Quadrant:                               | I   | II                                | III               | IV                |
|-----------------------------------------|-----|-----------------------------------|-------------------|-------------------|
| Trigonometric ratios which are positive | All | $\sin$ and $\operatorname{cosec}$ | $\tan$ and $\cot$ | $\cos$ and $\sec$ |

# 4.12 Trigonometric Ratios of Allied Angles

The angles  $-\theta$ ,  $90^\circ \pm \theta$ ,  $180^\circ \pm \theta$ ,  $270^\circ \pm \theta$ ,  $360^\circ \pm \theta$ , etc., are called allied angles of  $\theta$ . Remember the following relations :

|                        | $-\theta$                      | $90^\circ - \theta$           | $90^\circ + \theta$            | $180^\circ - \theta$          | $180^\circ + \theta$           | $270^\circ - \theta$           | $270^\circ + \theta$          |
|------------------------|--------------------------------|-------------------------------|--------------------------------|-------------------------------|--------------------------------|--------------------------------|-------------------------------|
| $\sin$                 | $-\sin \theta$                 | $\cos \theta$                 | $\cos \theta$                  | $\sin \theta$                 | $-\sin \theta$                 | $-\cos \theta$                 | $-\cos \theta$                |
| $\cos$                 | $\cos \theta$                  | $\sin \theta$                 | $-\sin \theta$                 | $-\cos \theta$                | $-\cos \theta$                 | $-\sin \theta$                 | $\sin \theta$                 |
| $\tan$                 | $-\tan \theta$                 | $\cot \theta$                 | $-\cot \theta$                 | $-\tan \theta$                | $\tan \theta$                  | $\cot \theta$                  | $-\cot \theta$                |
| $\cot$                 | $-\cot \theta$                 | $\tan \theta$                 | $-\tan \theta$                 | $-\cot \theta$                | $\cot \theta$                  | $\tan \theta$                  | $-\tan \theta$                |
| $\operatorname{cosec}$ | $-\operatorname{cosec} \theta$ | $\sec \theta$                 | $\sec \theta$                  | $\operatorname{cosec} \theta$ | $-\operatorname{cosec} \theta$ | $-\sec \theta$                 | $-\sec \theta$                |
| $\sec$                 | $\sec \theta$                  | $\operatorname{cosec} \theta$ | $-\operatorname{cosec} \theta$ | $-\sec \theta$                | $-\sec \theta$                 | $-\operatorname{cosec} \theta$ | $\operatorname{cosec} \theta$ |

Trigonometric ratios for  $360^\circ + \theta$  and  $n \times 360^\circ + \theta$  are the same as those for  $\theta$ .

Also, the trigonometric ratios for  $360^\circ - \theta$  and  $n \times 360^\circ - \theta$  are the same as those for  $-\theta$ .

## Illustrative Examples

**Illustration 1 :** If  $\sin \theta = -\frac{3}{5}$  and  $\tan \theta$  is positive, find all other trigonometric ratios. In which quadrant does  $\theta$  lie?

**Solution :** Since  $\sin \theta$  is negative and  $\tan \theta$  is positive, therefore  $\theta$  lies in the third quadrant.

So,  $\cos \theta$  is negative.

$$\begin{aligned}\therefore \cos \theta &= -\sqrt{1 - \sin^2 \theta} \\ &= -\sqrt{1 - \left(-\frac{3}{5}\right)^2} \\ &= -\sqrt{1 - \frac{9}{25}} \\ &= -\sqrt{\frac{16}{25}} = -\frac{4}{5}.\end{aligned}$$

$$\text{Now, } \operatorname{cosec} \theta = \frac{1}{\sin \theta} = -\frac{5}{3}, \sec \theta = \frac{1}{\cos \theta} = -\frac{5}{4},$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \left(-\frac{3}{5}\right)\left(-\frac{5}{4}\right) = \frac{3}{4}$$

$$\text{and } \cot \theta = \frac{1}{\tan \theta} = \frac{4}{3}.$$

**Illustration 2 :** Find the trigonometric ratios of

- (i)  $135^\circ$  (ii)  $180^\circ$

**Solution :** (i) We have  $\sin 135^\circ = \sin (180^\circ - 45^\circ) = \sin 45^\circ = \frac{1}{\sqrt{2}},$

$$\cos 135^\circ = \cos (180^\circ - 45^\circ) = -\cos 45^\circ = -\frac{1}{\sqrt{2}}$$

$$\tan 135^\circ = \tan (180^\circ - 45^\circ) = -\tan 45^\circ = -1. \quad [\text{UPTU 2001}]$$

$$\therefore \operatorname{cosec} 135^\circ = \frac{1}{\sin 135^\circ} = \sqrt{2}, \sec 135^\circ = \frac{1}{\cos 135^\circ} = -\sqrt{2}$$

$$\text{and } \cot 135^\circ = \frac{1}{\tan 135^\circ} = \frac{1}{-1} = -1.$$

(ii) We have  $\sin 180^\circ = \sin (180^\circ - 0^\circ) = \sin 0^\circ = 0,$

$$\cos 180^\circ = \cos (180^\circ - 0^\circ) = -\cos 0^\circ = -1,$$

$$\tan 180^\circ = \tan (180^\circ - 0^\circ) = -\tan 0^\circ = 0.$$

$$\therefore \sec 180^\circ = \frac{1}{\cos 180^\circ} = \frac{1}{-1} = -1.$$

Obviously,  $\operatorname{cosec} 180^\circ$  and  $\cot 180^\circ$  are not defined because  $\sin 180^\circ = 0.$

**Illustration 3 :** If  $\sec \theta + \tan \theta = x$ , prove that  $\tan \theta = \frac{x^2 - 1}{2x}$  and  $\sin \theta = \frac{x^2 - 1}{x^2 + 1}$ .

**Solution :** We have  $\sec^2 \theta - \tan^2 \theta = 1$

$$\Rightarrow (\sec \theta - \tan \theta)(\sec \theta + \tan \theta) = 1$$

$$\Rightarrow \sec \theta - \tan \theta = \frac{1}{\sec \theta + \tan \theta}.$$

It is given that  $\sec \theta + \tan \theta = x$ .

...(1)

$$\therefore \sec \theta - \tan \theta = \frac{1}{x}.$$

...(2)

Subtracting (2) from (1), we get

$$2 \tan \theta = x - \frac{1}{x} = \frac{x^2 - 1}{x} \Rightarrow \tan \theta = \frac{x^2 - 1}{2x}.$$

Adding (1) and (2), we get

$$2 \sec \theta = x + \frac{1}{x} = \frac{x^2 + 1}{x} \Rightarrow \sec \theta = \frac{x^2 + 1}{2x}.$$

Now, 
$$\sin \theta = \frac{\tan \theta}{\sec \theta} = \frac{x^2 - 1}{2x} \cdot \frac{2x}{x^2 + 1} = \frac{x^2 - 1}{x^2 + 1}.$$

**Illustration 4 :** Find the values of

(i)  $\cos 855^\circ$

(ii)  $\tan(-1575^\circ)$

**Solution :** (i) We have,  $\cos 855^\circ = \cos(2 \times 360^\circ + 135^\circ)$

$$= \cos 135^\circ = \cos(180^\circ - 45^\circ) = -\cos 45^\circ = -\frac{1}{\sqrt{2}}.$$

(ii) We have  $\tan(-1575^\circ) = -\tan 1575^\circ$

$$= -\tan(4 \times 360^\circ + 135^\circ) = -\tan 135^\circ$$

$$= -\tan(180^\circ - 45^\circ) = -(-\tan 45^\circ) = \tan 45^\circ = 1.$$

**Illustration 5 :** Show that

$$3 \sin^2 30^\circ + \frac{1}{4} \operatorname{cosec}^2 60^\circ - 2 \tan^2 45^\circ + \frac{2}{3} \sin^2 60^\circ = -\frac{5}{12}.$$

**Solution :** We have  $3 \sin^2 30^\circ + \frac{1}{4} \operatorname{cosec}^2 60^\circ - 2 \tan^2 45^\circ + \frac{2}{3} \sin^2 60^\circ$

$$= 3 \cdot \left(\frac{1}{2}\right)^2 + \frac{1}{4} \cdot \left(\frac{2}{\sqrt{3}}\right)^2 - 2 \cdot (1)^2 + \frac{2}{3} \cdot \left(\frac{\sqrt{3}}{2}\right)^2$$

$$= \frac{3}{4} + \frac{1}{3} - 2 + \frac{1}{2} = \frac{9 + 4 - 24 + 6}{12} = -\frac{5}{12}.$$

**Illustration 6 :** Prove that

(i)  $\tan 225^\circ \cot 405^\circ + \tan 765^\circ \cot 675^\circ = 0.$

(ii)  $\sin \theta + \sin(90^\circ + \theta) + \sin(180^\circ + \theta) + \sin(270^\circ + \theta) = 0.$

**Solution :** (i) We have  $\tan 225^\circ \cot 405^\circ + \tan 765^\circ \cot 675^\circ$   
 $= \tan (180^\circ + 45^\circ) \cot (360^\circ + 45^\circ) + \tan (2 \times 360^\circ + 45^\circ)$   
 $\cot (2 \times 360^\circ - 45^\circ)$   
 $= \tan 45^\circ \cot 45^\circ + (\tan 45^\circ) (-\cot 45^\circ)$   
 $= 1 \cdot 1 + 1 \cdot (-1) = 1 - 1 = 0.$

(ii) We have  $\sin \theta + \sin (90^\circ + \theta) + \sin (180^\circ + \theta) + \sin (270^\circ + \theta)$   
 $= \sin \theta + \cos \theta - \sin \theta - \cos \theta = 0.$

**Illustration 7 :** Find the values of  $\theta$  for which  $\sec \theta$  and  $\operatorname{cosec} \theta$  are equal, where  $0^\circ \leq \theta \leq 360^\circ$ .

**Solution :** We have  $\sec \theta = \operatorname{cosec} \theta$

$$\Rightarrow \frac{1}{\cos \theta} = \frac{1}{\sin \theta} \Rightarrow \frac{\sin \theta}{\cos \theta} = 1 \Rightarrow \tan \theta = 1$$

$$\Rightarrow \theta = 45^\circ \text{ or } 180^\circ + 45^\circ \quad [\because \tan \theta \text{ is +ve in I and III quadrants}]$$

$$\Rightarrow \theta = 45^\circ \text{ or } 225^\circ.$$

Hence, the required values of  $\theta$  are  $45^\circ$  and  $225^\circ$ .

**Illustration 8 :** Find all values of  $\theta$  between  $0^\circ$  and  $360^\circ$  for which

(i)  $\sin \theta = \frac{1}{\sqrt{2}}$

(ii)  $\operatorname{cosec} \theta = -\frac{2}{\sqrt{3}}$

**Solution :** (i) We know that  $\sin \theta$  is positive in I and II quadrants.

Now,  $\sin \theta = \frac{1}{\sqrt{2}} = \sin 45^\circ.$

So,  $\theta = 45^\circ.$

Also,  $\sin 45^\circ = \sin (180^\circ - 45^\circ) = \sin 135^\circ. \quad [\because \sin (180^\circ - \theta) = \sin \theta]$

Hence,  $\theta = 45^\circ, 135^\circ.$

(ii) We have  $\operatorname{cosec} \theta = -\frac{2}{\sqrt{3}} \Rightarrow \sin \theta = -\frac{\sqrt{3}}{2}.$

Now,  $\sin \theta$  is negative in III and IV quadrants.

We have  $\sin \theta = -\frac{\sqrt{3}}{2} = -\sin 60^\circ$

$$= \sin (180^\circ + 60^\circ) \text{ or } = \sin (360^\circ - 60^\circ)$$

$$[\because \sin (180^\circ + \theta) = -\sin \theta \text{ and } \sin (360^\circ - \theta) = -\sin \theta]$$

$$= \sin 240^\circ \text{ or } = \sin 300^\circ.$$

Hence,  $\theta = 240^\circ \text{ or } 300^\circ.$

**Illustration 9 :** Find the acute angle  $\theta$  such that

(i)  $\tan \theta = 2 \sin \theta$

(ii)  $\cot \theta + \tan \theta = 2 \operatorname{cosec} \theta.$

**Solution :** (i) We have  $\tan \theta = 2 \sin \theta$

$$\Rightarrow \frac{\sin \theta}{\cos \theta} = 2 \sin \theta \Rightarrow \sin \theta = 2 \sin \theta \cos \theta$$

$$\Rightarrow \sin \theta - 2 \sin \theta \cos \theta = 0$$

$$\Rightarrow \sin \theta (1 - 2 \cos \theta) = 0$$

$$\Rightarrow \sin \theta = 0 \quad \text{or} \quad \cos \theta = \frac{1}{2}$$

$$\Rightarrow \theta = 0^\circ \quad \text{or} \quad \theta = 60^\circ.$$

(ii) We have  $\cot \theta + \tan \theta = 2 \operatorname{cosec} \theta$

$$\Rightarrow \frac{1}{\tan \theta} + \tan \theta = 2 \operatorname{cosec} \theta$$

$$\Rightarrow 1 + \tan^2 \theta = 2 \operatorname{cosec} \theta \tan \theta$$

$$\Rightarrow \sec^2 \theta = 2 \sec \theta$$

$$\Rightarrow \sec \theta (\sec \theta - 2) = 0$$

$$\Rightarrow \sec \theta = 2 \quad \text{i.e.,} \quad \cos \theta = \frac{1}{2} \quad \text{i.e.,} \quad \theta = 60^\circ. \quad [\because \sec \theta = 0 \text{ is not possible}]$$

**Illustration 10:** If  $A, B, C$  and  $D$  are angles of a cyclic quadrilateral taken in order, prove that

$$\cos A + \cos B + \cos C + \cos D = 0.$$

**Solution :** We know that the opposite angles of a cyclic quadrilateral are supplementary i.e.,  $A + C = 180^\circ$  and  $B + D = 180^\circ$ .

We have  $\cos A + \cos B + \cos C + \cos D$

$$= \cos A + \cos B + \cos (180^\circ - A) + \cos (180^\circ - B)$$

$$[\because C = 180^\circ - A \quad \text{and} \quad D = 180^\circ - B]$$

$$= \cos A + \cos B - \cos A - \cos B = 0.$$

$$[\because \cos (180^\circ - \theta) = -\cos \theta]$$

### Comprehensive Exercise 3

1. If  $\theta$  is acute, find all other trigonometric ratios when

$$(i) \sin \theta = \frac{24}{25}$$

$$(ii) \cos \theta = \frac{1}{\sqrt{2}}$$

$$(iii) \cot \theta = \frac{9}{40}$$

$$(iv) \operatorname{cosec} \theta = \frac{17}{8}.$$

2. In which quadrant does  $\theta$  lie, if

$$(i) \cos \theta = -\frac{5}{12}$$

$$(ii) \tan \theta = \frac{5}{4}$$

$$(iii) \operatorname{cosec} \theta = -4.$$

3. Find the values of other five trigonometric ratios, when

$$(i) \sin \theta = -\frac{8}{17} \text{ and } \theta \text{ lies in the third quadrant ;}$$

$$(ii) \cos \theta = -\frac{4}{5} \text{ and } \theta \text{ lies in the second quadrant ;}$$

$$(iii) \tan \theta = -\frac{1}{3} \text{ and } \theta \text{ lies in the fourth quadrant.}$$



4. If  $\cos \theta = \frac{5}{13}$  and  $\theta$  lies in the fourth quadrant, find the value of 
$$\frac{2 - 3 \cot \theta}{4 + 9 \sqrt{\sec^2 \theta - 1}}.$$
5. Evaluate  $\frac{4}{3} \cot^2 30^\circ + 3 \sin^2 60^\circ - 2 \operatorname{cosec}^2 45^\circ + \frac{1}{3} \tan^2 60^\circ - \frac{1}{4} \cos 0^\circ.$
6. Find  $x$  from the following equations :  
 (i)  $\operatorname{cosec} (90^\circ + \theta) + x \cos \theta \cot (90^\circ + \theta) = \sin (90^\circ + \theta)$   
 (ii)  $x \cot (90^\circ + \theta) + \tan (90^\circ + \theta) \sin \theta + \operatorname{cosec} (90^\circ + \theta) = 0$
7. Find the trigonometric ratios of  
 (i)  $225^\circ$  (ii)  $330^\circ$
8. Verify that : (i)  $\cos 2\theta = 1 - 2 \sin^2 \theta$ , when  $\theta = 30^\circ$   
 (ii)  $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$ , when  $\theta = 60^\circ$   
 (iii)  $\cos 2\theta = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$ , when  $\theta = 30^\circ$   
 (iv)  $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$ , when  $\theta = 60^\circ.$
9. Show that the value of  $(\sin^2 \theta + \operatorname{cosec}^2 \theta)$  can never be less than 2.
10. Prove that : (i)  $\cos 510^\circ \cos 330^\circ + \sin 390^\circ \cos 120^\circ = -1$   
 (ii)  $\sin \frac{8\pi}{3} \cos \frac{23\pi}{6} + \cos \frac{13\pi}{3} \sin \frac{35\pi}{6} = \frac{1}{2}$   
 (iii)  $\sin 180^\circ + 2 \cos 180^\circ + 3 \sin 270^\circ + 4 \cos 270^\circ$   
 $- 5 \sec 180^\circ - 6 \operatorname{cosec} 270^\circ = 6.$
11. Prove that  
 (i)  $\sin (270^\circ - \theta) \sin (90^\circ - \theta) - \cos (270^\circ - \theta) \cos (90^\circ + \theta) = -1$   
 (ii)  $\frac{\tan (90^\circ - \theta) \sec (180^\circ - \theta) \sin (-\theta)}{\sin (180^\circ + \theta) \cot (360^\circ - \theta) \operatorname{cosec} (90^\circ - \theta)} = 1$   
 (iii)  $\sec \left( \frac{3\pi}{2} - \theta \right) \sec \left( \theta - \frac{5\pi}{2} \right) + \tan \left( \frac{5\pi}{2} + \theta \right) \tan \left( \theta - \frac{3\pi}{2} \right) = -1$   
 (iv)  $\frac{\sin (180^\circ + \theta) \cos (90^\circ + \theta) \tan (270^\circ - \theta) \cot (360^\circ - \theta)}{\sin (360^\circ - \theta) \cos (360^\circ + \theta) \operatorname{cosec} (-\theta) \sin (270^\circ + \theta)} = 1.$
12. Prove that  
 (i)  $\tan 720^\circ - \cos 270^\circ - \sin 150^\circ \cos 120^\circ = \frac{1}{4}$   
 (ii)  $\sin 780^\circ \cos 30^\circ + \cos 120^\circ \sin 390^\circ = \frac{1}{2}$   
 (iii)  $\sin 600^\circ \cos 390^\circ + \cos 480^\circ \sin 150^\circ = -1$   
 (iv)  $\frac{\sin 300^\circ \tan 330^\circ \sec 420^\circ}{\cot 135^\circ \cos 210^\circ \operatorname{cosec} 315^\circ} = -\sqrt{\frac{2}{3}}.$

13. In any quadrilateral  $ABCD$ , prove that
- $\sin(A + B) + \sin(C + D) = 0$
  - $\cos(A + B) = \cos(C + D)$ .
14. In a  $\Delta ABC$ , prove that
- $\cos(A + B) + \cos C = 0$
  - $\cos\left(\frac{A + B}{2}\right) = \sin\frac{C}{2}$
  - $\tan\left(\frac{A + B}{2}\right) = \cot\frac{C}{2}$ .
15. If  $A, B, C, D$  be the angles of a cyclic quadrilateral, taken in order, prove that
- $$\cos(180^\circ - A) + \cos(180^\circ + B) + \cos(180^\circ + C) - \sin(90^\circ + D) = 0.$$

## Answers

- $\cos \theta = \frac{7}{25}, \tan \theta = \frac{24}{7}, \cot \theta = \frac{7}{24}, \sec \theta = \frac{25}{7}, \operatorname{cosec} \theta = \frac{25}{24}$ .
  - $\sin \theta = \frac{1}{\sqrt{2}}, \tan \theta = 1, \cot \theta = 1, \operatorname{cosec} \theta = \sqrt{2}, \sec \theta = \sqrt{2}$ .
  - $\operatorname{cosec} \theta = \frac{41}{40}, \sec \theta = \frac{41}{9}, \sin \theta = \frac{40}{41}, \cos \theta = \frac{9}{41}$  and  $\tan \theta = \frac{40}{9}$ .
  - $\sin \theta = \frac{8}{17}, \cot \theta = \frac{15}{8}, \tan \theta = \frac{8}{15}, \cos \theta = \frac{15}{17}$  and  $\sec \theta = \frac{17}{15}$ .
- II or III
  - I or III
  - III or IV
- $\cos \theta = -\frac{15}{17}, \sec \theta = -\frac{17}{15}, \operatorname{cosec} \theta = -\frac{17}{8}, \tan \theta = \frac{8}{15}$  and  $\cot \theta = \frac{15}{8}$ .
  - $\sin \theta = \frac{3}{5}, \operatorname{cosec} \theta = \frac{5}{3}, \sec \theta = -\frac{5}{4}, \tan \theta = -\frac{3}{4}$  and  $\cot \theta = -\frac{4}{3}$ .
  - $\cot \theta = -3, \operatorname{cosec} \theta = -\sqrt{10}, \sec \theta = \frac{\sqrt{10}}{3}, \cos \theta = \frac{3}{\sqrt{10}}$   
and  $\sin \theta = -\frac{1}{\sqrt{10}}$ .
- $\frac{65}{512}$ .
- 3.
- $\tan \theta$
  - $\sin \theta$ .
- $\sin 225^\circ = -\frac{1}{\sqrt{2}}, \cos 225^\circ = -\frac{1}{\sqrt{2}}, \operatorname{cosec} 225^\circ = -\sqrt{2},$   
 $\sec 225^\circ = -\sqrt{2}, \tan 225^\circ = 1$  and  $\cot 225^\circ = 1$ .
  - $\sin 330^\circ = -\frac{1}{2}, \cos 330^\circ = \frac{\sqrt{3}}{2}, \operatorname{cosec} 330^\circ = -2,$   
 $\sec 330^\circ = \frac{2}{\sqrt{3}}, \tan 330^\circ = -\frac{1}{\sqrt{3}}$  and  $\cot 330^\circ = -\sqrt{3}$ .

## 4.13 Trigonometric Ratios of Compound Angles

### (a) Trigonometric Ratios of the Sum of Two Angles

For any two angles  $A$  and  $B$ , we have

$$(i) \sin(A + B) = \sin A \cos B + \cos A \sin B,$$

$$(ii) \cos(A + B) = \cos A \cos B - \sin A \sin B,$$

$$\text{and } (iii) \tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}.$$

### (b) Trigonometric Ratios of the Difference of Two Angles

For any two angles  $A$  and  $B$ , we have

$$(i) \sin(A - B) = \sin A \cos B - \cos A \sin B,$$

$$(ii) \cos(A - B) = \cos A \cos B + \sin A \sin B,$$

$$\text{and } (iii) \tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}.$$

### (c) For any two angles $A$ and $B$ , we have

$$(i) \sin(A + B) \sin(A - B) = \sin^2 A - \sin^2 B \quad [\text{UPTU 2001}]$$

$$\text{and } (ii) \cos(A + B) \cos(A - B) = \cos^2 A - \sin^2 B.$$

**Proof.** (i) We have  $\sin(A + B) \sin(A - B)$

$$\begin{aligned} &= (\sin A \cos B + \cos A \sin B) (\sin A \cos B - \cos A \sin B) \\ &= \sin^2 A \cos^2 B - \cos^2 A \sin^2 B \\ &= \sin^2 A (1 - \sin^2 B) - (1 - \sin^2 A) \sin^2 B \\ &= \sin^2 A - \sin^2 A \sin^2 B - \sin^2 B + \sin^2 A \sin^2 B \\ &= \sin^2 A - \sin^2 B. \end{aligned}$$

(ii) We have  $\cos(A + B) \cos(A - B)$

$$\begin{aligned} &= (\cos A \cos B - \sin A \sin B) (\cos A \cos B + \sin A \sin B) \\ &= \cos^2 A \cos^2 B - \sin^2 A \sin^2 B \\ &= \cos^2 A (1 - \sin^2 B) - (1 - \cos^2 A) \sin^2 B \\ &= \cos^2 A - \cos^2 A \sin^2 B - \sin^2 B + \cos^2 A \sin^2 B \\ &= \cos^2 A - \sin^2 B. \end{aligned}$$

### (d) For any three angles $A$ , $B$ and $C$ , we have

$$(i) \sin(A + B + C) = \sin A \cos B \cos C + \cos A \sin B \cos C + \cos A \cos B \sin C - \sin A \sin B \sin C,$$

$$(ii) \cos(A + B + C) = \cos A \cos B \cos C - \cos A \sin B \sin C - \sin A \cos B \sin C - \sin A \sin B \cos C,$$

$$\text{and } (iii) \tan(A + B + C) = \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A}.$$

## Illustrative Examples

**Illustration 1 :** Evaluate :

(i)  $\sin 78^\circ \cos 18^\circ - \cos 78^\circ \sin 18^\circ$

[UPTU 2001]

(ii)  $\cos 68^\circ \cos 8^\circ + \sin 68^\circ \sin 8^\circ$

**Solution.** (i) We have  $\sin 78^\circ \cos 18^\circ - \cos 78^\circ \sin 18^\circ$

$$= \sin (78^\circ - 18^\circ) = \sin 60^\circ = \frac{\sqrt{3}}{2}.$$

$$[\because \sin A \cos B - \cos A \sin B = \sin (A - B)]$$

(ii) We have  $\cos 68^\circ \cos 8^\circ + \sin 68^\circ \sin 8^\circ$

$$= \cos (68^\circ - 8^\circ) = \cos 60^\circ = \frac{1}{2}.$$

$$[\because \cos A \cos B + \sin A \sin B = \cos (A - B)]$$

**Illustration 2 :** Prove that

(i)  $\cos \frac{5\pi}{3} \cos \frac{\pi}{4} + \sin \frac{5\pi}{3} \sin \frac{\pi}{4} = \frac{(1 - \sqrt{3})}{2\sqrt{2}}$

(ii)  $\sin \frac{\pi}{4} \cos \frac{\pi}{12} + \cos \frac{\pi}{4} \sin \frac{\pi}{12} = \frac{\sqrt{3}}{2}$

(iii)  $\sin \frac{7\pi}{12} \cos \frac{\pi}{4} - \cos \frac{7\pi}{12} \sin \frac{\pi}{4} = \frac{\sqrt{3}}{2}.$

**Solution :** (i) We have  $\cos \frac{5\pi}{3} \cos \frac{\pi}{4} + \sin \frac{5\pi}{3} \sin \frac{\pi}{4}$

$$= \cos \left( 2\pi - \frac{\pi}{3} \right) \cos \frac{\pi}{4} + \sin \left( 2\pi - \frac{\pi}{3} \right) \sin \frac{\pi}{4}$$

$$= \cos \frac{\pi}{3} \cos \frac{\pi}{4} - \sin \frac{\pi}{3} \sin \frac{\pi}{4}$$

$$[\because \cos (2\pi - \theta) = \cos \theta \text{ and } \sin (2\pi - \theta) = -\sin \theta]$$

$$= \frac{1}{2} \cdot \frac{1}{\sqrt{2}} - \frac{\sqrt{3}}{2} \cdot \frac{1}{2} = \frac{(1 - \sqrt{3})}{2\sqrt{2}}.$$

(ii) We have  $\sin \frac{\pi}{4} \cos \frac{\pi}{12} + \cos \frac{\pi}{4} \sin \frac{\pi}{12} = \sin \left( \frac{\pi}{4} + \frac{\pi}{12} \right)$

$$[\because \sin A \cos B + \cos A \sin B = \sin (A + B)]$$

$$= \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}.$$

(iii) We have  $\sin \frac{7\pi}{12} \cos \frac{\pi}{4} - \cos \frac{7\pi}{12} \sin \frac{\pi}{4} = \sin \left( \frac{7\pi}{12} - \frac{\pi}{4} \right)$

$$[\because \sin A \cos B - \cos A \sin B = \sin (A - B)]$$

$$= \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}.$$

**Illustration 3 :** Prove that  $\cos 105^\circ + \cos 15^\circ = \sin 75^\circ - \sin 15^\circ$ .

**Solution :** We have  $\cos 105^\circ + \cos 15^\circ = \cos (90^\circ + 15^\circ) + \cos (90^\circ - 75^\circ)$   
 $= -\sin 15^\circ + \sin 75^\circ$

$$\begin{aligned} & [\because \cos (90^\circ + \theta) = -\sin \theta \text{ and } \cos (90^\circ - \theta) = \sin \theta] \\ & = \sin 75^\circ - \sin 15^\circ. \end{aligned}$$

**Illustration 4 :** Obtain the values of

- (i)  $\sin 15^\circ$  [UPTU 2004]      (ii)  $\cos 15^\circ$       (iii)  $\sin 75^\circ$   
 (iv)  $\cos 75^\circ$       (v)  $\cos 105^\circ$       (vi)  $\cot 75^\circ$  [UPTU 2002]  
 (vii)  $\tan 15^\circ$ . [UPTU 2006]

**Solution :** (i) We have  $\sin 15^\circ = \sin (45^\circ - 30^\circ)$

$$\begin{aligned} & = \sin 45^\circ \cos 30^\circ - \cos 45^\circ \sin 30^\circ \\ & = \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \cdot \frac{1}{2} = \frac{\sqrt{3} - 1}{2\sqrt{2}}. \end{aligned}$$

$$\begin{aligned} \text{(ii) We have } \cos 15^\circ & = \cos (45^\circ - 30^\circ) \\ & = \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ \\ & = \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2} = \frac{\sqrt{3} + 1}{2\sqrt{2}}. \end{aligned}$$

$$\begin{aligned} \text{(iii) We have } \sin 75^\circ & = \sin (45^\circ + 30^\circ) \\ & = \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ \\ & = \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2} = \frac{\sqrt{3} + 1}{2\sqrt{2}}. \end{aligned}$$

$$\begin{aligned} \text{(iv) We have } \cos 75^\circ & = \cos (45^\circ + 30^\circ) \\ & = \cos 45^\circ \cos 30^\circ - \sin 45^\circ \sin 30^\circ \\ & = \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \cdot \frac{1}{2} = \frac{\sqrt{3} - 1}{2\sqrt{2}}. \end{aligned}$$

$$\begin{aligned} \text{(v) We have } \cos 105^\circ & = \cos (60^\circ + 45^\circ) \\ & = \cos 60^\circ \cos 45^\circ - \sin 60^\circ \sin 45^\circ \\ & = \frac{1}{2} \cdot \frac{1}{\sqrt{2}} - \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}} = \frac{1 - \sqrt{3}}{2\sqrt{2}} = -\frac{\sqrt{3} - 1}{2\sqrt{2}}. \end{aligned}$$

$$\begin{aligned} \text{(vi) We have } \cot 75^\circ & = \cot (45^\circ + 30^\circ) \\ & = \frac{\cot 45^\circ \cot 30^\circ - 1}{\cot 45^\circ + \cot 30^\circ} = \frac{1 \cdot \sqrt{3} - 1}{1 + \sqrt{3}} = \frac{\sqrt{3} - 1}{\sqrt{3} + 1}. \end{aligned}$$

$$\text{(vi) We have } \tan 15^\circ = \tan (45^\circ - 30^\circ) = \frac{\tan 45^\circ - \tan 30^\circ}{1 + \tan 45^\circ \tan 30^\circ}$$

$$\begin{aligned} & = \frac{1 - \frac{1}{\sqrt{3}}}{1 + \frac{1}{\sqrt{3}}} = \frac{\sqrt{3} - 1}{\sqrt{3} + 1}. \end{aligned}$$

**Illustration 5 :** If  $\tan A = \frac{5}{6}$  and  $\tan B = \frac{1}{11}$ , prove that  $A + B = 45^\circ$ .

**Solution :** We have  $\tan (A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$

$$= \frac{\frac{5}{6} + \frac{1}{11}}{1 - \frac{5}{6} \times \frac{1}{11}} = \frac{\frac{55 + 6}{6 \times 11}}{\frac{66 - 5}{6 \times 11}} = \frac{61}{61} = 1 = \tan 45^\circ.$$

$$\therefore A + B = 45^\circ.$$

**Illustration 6 :** If  $A + B = \pi / 4$ , prove that  $(1 + \tan A)(1 + \tan B) = 2$ .

[UPTU 2005]

**Solution :** It is given that  $A + B = \pi / 4$ .

$$\therefore \tan (A + B) = \tan (\pi / 4) = 1 \Rightarrow \frac{\tan A + \tan B}{1 - \tan A \tan B} = 1$$

$$\Rightarrow \tan A + \tan B = 1 - \tan A \tan B$$

$$\Rightarrow \tan A + \tan B + \tan A \tan B = 1$$

$$\Rightarrow 1 + \tan A + \tan B + \tan A \tan B = 2$$

$$\Rightarrow (1 + \tan A)(1 + \tan B) = 2.$$

**Illustration 7 :** Prove that  $\frac{\cos 8^\circ + \sin 8^\circ}{\cos 8^\circ - \sin 8^\circ} = \cot 37^\circ$ .

[UPTU 2002]

**Solution :** We have  $\frac{\cos 8^\circ + \sin 8^\circ}{\cos 8^\circ - \sin 8^\circ}$

$$= \frac{1 + \tan 8^\circ}{1 - \tan 8^\circ} \quad [\text{Dividing the Nr and the Dr by } \cos 8^\circ]$$

$$= \frac{\tan 45^\circ + \tan 8^\circ}{1 - \tan 45^\circ \tan 8^\circ} \quad [\because \tan 45^\circ = 1]$$

$$= \tan (45^\circ + 8^\circ) = \tan 53^\circ$$

$$\left[ \because \frac{\tan A + \tan B}{1 - \tan A \tan B} = \tan (A + B) \right]$$

$$= \tan (90^\circ - 37^\circ) = \cot 37^\circ.$$

**Illustration 8 :** Prove that  $\tan 70^\circ = \tan 20^\circ + 2 \tan 50^\circ$ .

**Solution :** We have  $\tan 70^\circ = \tan (20^\circ + 50^\circ)$

$$\Rightarrow \tan 70^\circ = \frac{\tan 20^\circ + \tan 50^\circ}{1 - \tan 20^\circ \tan 50^\circ}$$

$$\Rightarrow \tan 70^\circ - \tan 70^\circ \tan 20^\circ \tan 50^\circ = \tan 20^\circ + \tan 50^\circ$$

$$\Rightarrow \tan 70^\circ - \tan (90^\circ - 20^\circ) \tan 20^\circ \tan 50^\circ = \tan 20^\circ + \tan 50^\circ$$

$$\Rightarrow \tan 70^\circ - \cot 20^\circ \tan 20^\circ \tan 50^\circ = \tan 20^\circ + \tan 50^\circ$$

$$[\because \tan (90^\circ - \theta) = \cot \theta]$$

$$\Rightarrow \tan 70^\circ - \tan 50^\circ = \tan 20^\circ + \tan 50^\circ \quad [\because \cot A \tan A = 1]$$

$$\Rightarrow \tan 70^\circ = \tan 20^\circ + \tan 50^\circ + \tan 50^\circ = \tan 20^\circ + 2 \tan 50^\circ.$$

**Illustration 9 :** Prove that  $\tan 13\theta - \tan 9\theta - \tan 4\theta = \tan 13\theta \tan 9\theta \tan 4\theta$ .

[UPTU 2005]

**Solution :** We have  $\tan 13\theta = \tan (9\theta + 4\theta)$

$$\Rightarrow \tan 13\theta = \frac{\tan 9\theta + \tan 4\theta}{1 - \tan 9\theta \tan 4\theta}$$

$$\Rightarrow \tan 13\theta (1 - \tan 9\theta \tan 4\theta) = \tan 9\theta + \tan 4\theta$$

$$\Rightarrow \tan 13\theta - \tan 13\theta \tan 9\theta \tan 4\theta = \tan 9\theta + \tan 4\theta$$

$$\Rightarrow \tan 13\theta - \tan 9\theta - \tan 4\theta = \tan 13\theta \tan 9\theta \tan 4\theta.$$

**Illustration 10 :** Prove that  $\cos^2 45^\circ - \sin^2 15^\circ = \frac{\sqrt{3}}{4}$ .

**Solution :** We have  $\cos^2 45^\circ - \sin^2 15^\circ = \cos (45^\circ + 15^\circ) \cos (45^\circ - 15^\circ)$

$$[\because \cos^2 A - \sin^2 B = \cos (A + B) \cos (A - B)]$$

$$= \cos 60^\circ \cos 30^\circ = \frac{1}{2} \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{4}.$$

**Illustration 11 :** If  $\sin A = \frac{4}{5}$  and  $\cos B = \frac{5}{13}$ , find the values of  $\sin (A - B)$  and

$\cos (A + B)$ , where  $0 < A < \frac{\pi}{2}$  and  $0 < B < \frac{\pi}{2}$ .

[UPTU 2001]

**Solution :** We have  $\sin A = \frac{4}{5}$  and  $0 < A < \frac{\pi}{2}$

$$\Rightarrow \cos A = \sqrt{1 - \sin^2 A} = \sqrt{1 - \left(\frac{4}{5}\right)^2} = \sqrt{1 - \frac{16}{25}} = \sqrt{\frac{9}{25}} = \frac{3}{5}.$$

Again  $\cos B = \frac{5}{13}$  and  $0 < B < \frac{\pi}{2}$

$$\Rightarrow \sin B = \sqrt{1 - \cos^2 B} = \sqrt{1 - \left(\frac{5}{13}\right)^2} = \sqrt{\frac{144}{169}} = \frac{12}{13}.$$

$$\therefore \sin (A - B) = \sin A \cos B - \cos A \sin B$$

$$= \left(\frac{4}{5} \times \frac{5}{13}\right) - \left(\frac{3}{5} \times \frac{12}{13}\right) = \frac{20}{65} - \frac{36}{65} = -\frac{16}{65}.$$

Also  $\cos (A + B) = \cos A \cos B - \sin A \sin B$

$$= \left(\frac{3}{5} \times \frac{5}{13}\right) - \left(\frac{4}{5} \times \frac{12}{13}\right) = \frac{15}{65} - \frac{48}{65} = -\frac{33}{65}.$$

**Illustration 12 :** If  $\sin A = \frac{3}{5}$ ,  $0 < A < \frac{\pi}{2}$  and  $\cos B = -\frac{12}{13}$ ,  $\pi < B < \frac{3\pi}{2}$ , find the following :

(i)  $\sin (A - B)$

(ii)  $\cos (A + B)$

(iii)  $\tan (A - B).$

**Solution :** We have  $\sin A = \frac{3}{5}$ , where  $0 < A < \frac{\pi}{2}$ .

Since,  $A$  lies in the first quadrant, therefore  $\sin A$  is positive and  $\cos A$  is positive.

$$\therefore \cos A = \sqrt{1 - \sin^2 A} = \sqrt{1 - \left(\frac{3}{5}\right)^2} = \sqrt{1 - \frac{9}{25}} = \sqrt{\frac{16}{25}} = \frac{4}{5}.$$

Also,  $\tan A = \frac{\sin A}{\cos A} = \frac{3}{5} \times \frac{5}{4} = \frac{3}{4}.$

Again,  $\cos B = -\frac{12}{13}, \pi < B < \frac{3\pi}{2}.$

Since,  $B$  lies in the third quadrant, therefore  $\cos B$  is negative and  $\sin B$  is also negative.

$$\begin{aligned} \therefore \sin B &= -\sqrt{1 - \cos^2 B} = -\sqrt{1 - \left(-\frac{12}{13}\right)^2} \\ &= -\sqrt{1 - \frac{144}{169}} = -\sqrt{\frac{25}{169}} = -\frac{5}{13}. \end{aligned}$$

Also,  $\tan B = \frac{\sin B}{\cos B} = \left(-\frac{5}{13}\right) \times \left(-\frac{13}{12}\right) = \frac{5}{12}.$

Now, (i)  $\sin(A - B) = \sin A \cos B - \cos A \sin B$

$$= \frac{3}{5} \times \left(-\frac{12}{13}\right) - \left(\frac{4}{5}\right) \left(-\frac{5}{13}\right) = -\frac{36}{65} + \frac{20}{65} = \frac{-16}{65}.$$

(ii)  $\cos(A + B) = \cos A \cos B - \sin A \sin B$

$$= \frac{4}{5} \times \left(-\frac{12}{13}\right) - \frac{3}{5} \times \left(-\frac{5}{13}\right) = -\frac{48}{65} + \frac{15}{65} = -\frac{33}{65}.$$

$$(iii) \tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B} = \frac{\frac{3}{4} - \frac{5}{12}}{1 + \frac{3}{4} \times \frac{5}{12}} = \frac{16}{63}.$$

**Illustration 13:** If  $\cos(\alpha + \beta) = \frac{4}{5}$ ,  $\sin(\alpha - \beta) = \frac{5}{13}$  and  $\alpha, \beta$  lie between 0 and  $\frac{\pi}{4}$ , prove that  $\tan 2\alpha = \frac{56}{33}.$

**Solution :** Since  $\alpha, \beta$  lie between 0 and  $\frac{\pi}{4}$ , therefore  $-\frac{\pi}{4} < \alpha - \beta < \frac{\pi}{4}$  and  $0 < \alpha + \beta < \frac{\pi}{2}$ . Consequently  $\sin(\alpha + \beta)$  and  $\cos(\alpha - \beta)$  are both positive.

Now,  $\sin(\alpha + \beta) = \sqrt{1 - \cos^2(\alpha + \beta)} = \sqrt{1 - \frac{16}{25}} = \frac{3}{5}$

and  $\cos(\alpha - \beta) = \sqrt{1 - \sin^2(\alpha - \beta)} = \sqrt{1 - \left(\frac{5}{13}\right)^2} = \frac{12}{13}.$



$$\therefore \tan (\alpha + \beta) = \frac{\sin (\alpha + \beta)}{\cos (\alpha + \beta)} = \frac{3 / 5}{4 / 5} = \frac{3}{4}$$

$$\text{and} \quad \tan (\alpha - \beta) = \frac{\sin (\alpha - \beta)}{\cos (\alpha - \beta)} = \frac{5 / 13}{12 / 13} = \frac{5}{12}.$$

$$\begin{aligned} \text{Hence,} \quad \tan 2 \alpha &= \tan [(\alpha + \beta) + (\alpha - \beta)] \\ &= \frac{\tan (\alpha + \beta) + \tan (\alpha - \beta)}{1 - \tan (\alpha + \beta) \tan (\alpha - \beta)} = \frac{\frac{3}{4} + \frac{5}{12}}{1 - \frac{3}{4} \times \frac{5}{12}} = \frac{36 + 20}{48 - 15} = \frac{56}{33}. \end{aligned}$$

**Illustration 14 :** If  $\tan A + \tan B = a$  and  $\cot A + \cot B = b$ , prove that

$$\cot (A + B) = \frac{1}{a} - \frac{1}{b}.$$

**Solution :** We have  $\cot (A + B) = \frac{1}{\tan (A + B)}$

$$\begin{aligned} &= \frac{1 - \tan A \tan B}{\tan A + \tan B} = \frac{1}{\tan A + \tan B} - \frac{\tan A \tan B}{\tan A + \tan B} \\ &= \frac{1}{\tan A + \tan B} - \frac{1}{\frac{\tan A + \tan B}{\tan A \tan B}} \\ &= \frac{1}{\tan A + \tan B} - \frac{1}{\frac{\tan A}{\tan A \tan B} + \frac{\tan B}{\tan A \tan B}} \\ &= \frac{1}{\tan A + \tan B} - \frac{1}{\frac{1}{\tan B} + \frac{1}{\tan A}} \\ &= \frac{1}{\tan A + \tan B} - \frac{1}{\cot A + \cot B} = \frac{1}{a} - \frac{1}{b}. \end{aligned}$$

**Illustration 15 :** If  $\tan A = m \tan B$ , prove that  $\frac{\sin (A - B)}{\sin (A + B)} = \frac{m - 1}{m + 1}$ .

**Solution :** We have  $\tan A = m \tan B$

$$\Rightarrow \frac{\tan A}{\tan B} = \frac{m}{1} \Rightarrow \frac{\sin A \cos B}{\cos A \sin B} = \frac{m}{1}$$

$$\Rightarrow \frac{\sin A \cos B - \cos A \sin B}{\sin A \cos B + \cos A \sin B} = \frac{m - 1}{m + 1}$$

[By componendo and dividendo]

$$\Rightarrow \frac{\sin (A - B)}{\sin (A + B)} = \frac{m - 1}{m + 1}.$$

## Comprehensive Exercise 4

1. Evaluate :

- (i)  $\sin 36^\circ \cos 9^\circ + \cos 36^\circ \sin 9^\circ$
- (ii)  $\sin 76^\circ \cos 16^\circ - \cos 76^\circ \sin 16^\circ$
- (iii)  $\cos 70^\circ \cos 10^\circ + \sin 70^\circ \sin 10^\circ$
- (iv)  $\cos 44^\circ \cos 16^\circ - \sin 44^\circ \sin 16^\circ$ .

2. Prove that

- (i)  $\frac{\tan 69^\circ + \tan 66^\circ}{1 - \tan 69^\circ \tan 66^\circ} = -1$
- (ii)  $\frac{\cos 9^\circ + \sin 9^\circ}{\cos 9^\circ - \sin 9^\circ} = \tan 54^\circ$
- (iii)  $\frac{\cos 8^\circ - \sin 8^\circ}{\cos 8^\circ + \sin 8^\circ} = \tan 37^\circ$
- (iv)  $\frac{\cos 11^\circ + \sin 11^\circ}{\cos 11^\circ - \sin 11^\circ} = \tan 56^\circ$ .

3. Prove that

- (i)  $\cos (45^\circ - A) \cos (45^\circ - B) - \sin (45^\circ - A) \sin (45^\circ - B) = \sin (A + B)$
- (ii)  $\sin (60^\circ - A) \cos (30^\circ + A) + \cos (60^\circ - A) \sin (30^\circ + A) = 1$
- (iii)  $\cos A \cos (A - B) + \sin A \sin (A - B) = \cos B$

4. Prove that

- (i)  $\tan 3\theta - \tan 2\theta - \tan \theta = \tan 3\theta \tan 2\theta \tan \theta$
- (ii)  $\tan 7\theta - \tan 5\theta - \tan 2\theta = \tan 7\theta \tan 5\theta \tan 2\theta$ .

5. Prove that

- (i)  $\frac{\sin (A + B) + \sin (A - B)}{\cos (A + B) + \cos (A - B)} = \tan A$
- (ii)  $\frac{\sin (A - B)}{\cos A \cos B} + \frac{\sin (B - C)}{\cos B \cos C} + \frac{\sin (C - A)}{\cos C \cos A} = 0$
- (iii)  $\frac{\sin (A - B)}{\sin A \sin B} + \frac{\sin (B - C)}{\sin B \sin C} + \frac{\sin (C - A)}{\sin C \sin A} = 0$ .

6. Prove that

- (i)  $\tan (45^\circ + A) = \frac{1 + \tan A}{1 - \tan A}$
- (ii)  $\tan (45^\circ - A) = \frac{1 - \tan A}{1 + \tan A}$ .

7. If  $\tan A = \frac{m}{m+1}$  and  $\tan B = \frac{1}{2m+1}$ , show that  $A + B = \frac{\pi}{4}$ . [UPTU 2007]
8. If  $\tan (A + B) = m$  and  $\tan (A - B) = n$ , find the values of  
 (i)  $\tan 2A$  (ii)  $\tan 2B$ .
9. If  $A + B = 45^\circ$ , prove that  
 (i)  $(1 + \tan A)(1 + \tan B) = 2$   
 (ii)  $(\cot A - 1)(\cot B - 1) = 2$ .
10. Prove that  
 (i)  $\sin^2 A + \sin^2 (A - B) - 2 \sin A \cos B \sin (A - B) = \sin^2 B$   
 (ii)  $\frac{\tan (A + B)}{\cot (A - B)} = \frac{\tan^2 A - \tan^2 B}{1 - \tan^2 A \tan^2 B}$   
 (iii)  $\frac{\tan^2 2\theta - \tan^2 \theta}{1 - \tan^2 2\theta \tan^2 \theta} = \tan 3\theta \tan \theta$ .
11. If  $\cos \theta + \sin \phi = m$  and  $\sin \theta + \cos \phi = n$ , prove that  

$$\sin (\theta + \phi) = \frac{1}{2} (m^2 + n^2 - 2).$$
12. If  $\sin A = \frac{3}{5}$  and  $\cos B = \frac{9}{41}$ ,  $0^\circ < A < 90^\circ$  and  $0^\circ < B < 90^\circ$ , find the values  
 of (i)  $\sin (A - B)$  (ii)  $\cos (A + B)$ .
13. If  $\cos \theta = \frac{1}{7}$  and  $\cos \phi = \frac{13}{14}$ , where  $\theta, \phi$  are acute angles, show that  
 $\theta - \phi = 60^\circ$ .
14. If  $\cos A = \frac{4}{5}$ ,  $\cos B = \frac{12}{13}$ , where  $\frac{3\pi}{2} < A < 2\pi$  and  $\frac{3\pi}{2} < B < 2\pi$ , find the  
 values of  
 (i)  $\cos (A + B)$  (ii)  $\sin (A - B)$
15. If  $\tan (\alpha + \theta) = n \tan (\alpha - \theta)$ , show that  $(n + 1) \sin 2\theta = (n - 1) \sin 2\alpha$ .
16. Evaluate the following :  
 (i)  $\sin \frac{7\pi}{12} \cos \frac{\pi}{4} - \cos \frac{7\pi}{12} \sin \frac{\pi}{4}$   
 (ii)  $\cos \frac{2\pi}{3} \cos \frac{\pi}{4} - \sin \frac{2\pi}{3} \sin \frac{\pi}{4}$ .
17. Prove that  
 (i)  $\tan 15^\circ + \tan 30^\circ + \tan 15^\circ \tan 30^\circ = 1$   
 (ii)  $\tan 36^\circ + \tan 9^\circ + \tan 36^\circ \tan 9^\circ = 1$ .

## Answers

- |                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>1. (i) <math>\frac{1}{\sqrt{2}}</math></p> <p>(iii) <math>\frac{1}{2}</math></p> <p>2. (i) <math>-\frac{133}{205}</math></p> <p>8. (i) <math>\frac{m+n}{1-mn}</math></p> <p>14. (i) <math>\frac{33}{65}</math></p> <p>16. (i) <math>\frac{\sqrt{3}}{2}</math></p> | <p>(ii) <math>\frac{\sqrt{3}}{2}</math></p> <p>(iv) <math>\frac{1}{2}</math></p> <p>(ii) <math>-\frac{84}{205}</math></p> <p>(ii) <math>\frac{m-n}{1+mn}</math></p> <p>(ii) <math>-\frac{16}{65}</math></p> <p>(ii) <math>\frac{-(\sqrt{3}+1)}{2\sqrt{2}}</math></p> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 4.14 Transformation Formulae

#### (a) Formulae to Transform the Product into Sum or Difference.

If  $A$  and  $B$  are any two angles, then

$$(i) \quad 2 \sin A \cos B = \sin (A+B) + \sin (A-B)$$

$$(ii) \quad 2 \cos A \sin B = \sin (A+B) - \sin (A-B)$$

$$(iii) \quad 2 \cos A \cos B = \cos (A+B) + \cos (A-B)$$

and  $(iv) \quad 2 \sin A \sin B = \cos (A-B) - \cos (A+B).$

#### (b) Formulae to Transform the Sum or Difference into Product.

If  $C$  and  $D$  are any two angles, then

$$(i) \quad \sin C + \sin D = 2 \sin \left( \frac{C+D}{2} \right) \cos \left( \frac{C-D}{2} \right)$$

$$(ii) \quad \sin C - \sin D = 2 \cos \left( \frac{C+D}{2} \right) \sin \left( \frac{C-D}{2} \right)$$

$$(iii) \quad \cos C + \cos D = 2 \cos \left( \frac{C+D}{2} \right) \cos \left( \frac{C-D}{2} \right)$$

and  $(iv) \quad \cos C - \cos D = 2 \sin \frac{C+D}{2} \sin \left( \frac{D-C}{2} \right).$

## Illustrative Examples

**Illustration 1 :** Express the following as the algebraic sum of sines and cosines :

$$(i) \quad 2 \sin 3A \cos A$$

$$(ii) \quad 2 \cos 3A \cos 4A$$

$$(iii) \quad 2 \cos 4A \sin 2A$$

$$(iv) \quad 2 \sin 3A \sin 4A.$$

**Solution :** (i) We have  $2 \sin 3A \cos A$

$$\begin{aligned} &= \sin (3A + A) + \sin (3A - A) \\ &\quad [ \because 2 \sin A \cos B = \sin (A + B) + \sin (A - B) ] \\ &= \sin 4A + \sin 2A. \end{aligned}$$

(ii) We have  $2 \cos 3A \cos 4A = \cos (3A + 4A) + \cos (3A - 4A)$   
 $[ \because 2 \cos A \cos B = \cos (A + B) + \cos (A - B) ]$   
 $= \cos 7A + \cos (-A) = \cos 7A + \cos A.$

(iii) We have  $2 \cos 4A \sin 2A = \sin (4A + 2A) - \sin (4A - 2A)$   
 $[ \because 2 \cos A \sin B = \sin (A + B) - \sin (A - B) ]$   
 $= \sin 6A - \sin 2A.$

(iv) We have  $2 \sin 3A \sin 4A = \cos (3A - 4A) - \cos (3A + 4A)$   
 $[ \because 2 \sin A \sin B = \cos (A - B) - \cos (A + B) ]$   
 $= \cos (-A) - \cos 7A = \cos A - \cos 7A.$

**Illustration 2 :** Prove that

(i)  $2 \sin \frac{5\pi}{12} \sin \frac{\pi}{12} = \frac{1}{2}$                       (ii)  $2 \sin \frac{5\pi}{12} \cos \frac{\pi}{12} = \frac{2 + \sqrt{3}}{2}.$

**Solution :** (i) We have  $2 \sin \frac{5\pi}{12} \sin \frac{\pi}{12} = \cos \left( \frac{5\pi}{12} - \frac{\pi}{12} \right) - \cos \left( \frac{5\pi}{12} + \frac{\pi}{12} \right)$   
 $[ \because 2 \sin A \sin B = \cos (A - B) - \cos (A + B) ]$   
 $= \cos \frac{\pi}{3} - \cos \frac{\pi}{2} = \frac{1}{2} - 0 = \frac{1}{2}.$

(ii) We have  $2 \sin \frac{5\pi}{12} \cos \frac{\pi}{12} = \sin \left( \frac{5\pi}{12} + \frac{\pi}{12} \right) + \sin \left( \frac{5\pi}{12} - \frac{\pi}{12} \right)$   
 $[ \because 2 \sin A \cos B = \sin (A + B) + \sin (A - B) ]$   
 $= \sin \frac{\pi}{2} + \sin \frac{\pi}{3} = 1 + \frac{\sqrt{3}}{2} = \frac{2 + \sqrt{3}}{2}.$

**Illustration 3 :** Prove that (i)  $\sin 40^\circ + \sin 20^\circ = \cos 10^\circ$

(ii)  $\sin 105^\circ + \cos 105^\circ = \cos 45^\circ$                       (iii)  $\cos 20^\circ + \cos 100^\circ + \cos 140^\circ = 0.$

**Solution :** (i) We have  $\sin 40^\circ + \sin 20^\circ$

$$\begin{aligned} &= 2 \sin \left( \frac{40^\circ + 20^\circ}{2} \right) \cos \left( \frac{40^\circ - 20^\circ}{2} \right) = 2 \sin 30^\circ \cos 10^\circ \\ &= 2 \cdot \frac{1}{2} \cdot \cos 10^\circ = \cos 10^\circ. \end{aligned}$$

(ii) We have  $\sin 105^\circ + \cos 105^\circ = \sin (90^\circ + 15^\circ) + \cos 105^\circ$   
 $= \cos 15^\circ + \cos 105^\circ \quad [ \because \sin (90^\circ + A) = \cos A ]$

$$\begin{aligned}
 &= 2 \cos \left( \frac{105^\circ + 15^\circ}{2} \right) \cos \left( \frac{105^\circ - 15^\circ}{2} \right) \\
 &\quad \left[ \because \cos C + \cos D = 2 \cos \left( \frac{C + D}{2} \right) \cos \left( \frac{C - D}{2} \right) \right] \\
 &= 2 \cos 60^\circ \cos 45^\circ = 2 \times \frac{1}{2} \times \cos 45^\circ = \cos 45^\circ.
 \end{aligned}$$

(iii) We have  $\cos 20^\circ + \cos 100^\circ + \cos 140^\circ$

$$\begin{aligned}
 &= \cos 20^\circ + (\cos 140^\circ + \cos 100^\circ) \\
 &= \cos 20^\circ + 2 \cos \left( \frac{140^\circ + 100^\circ}{2} \right) \cos \left( \frac{140^\circ - 100^\circ}{2} \right) \\
 &= \cos 20^\circ + 2 \cos 120^\circ \cos 20^\circ \\
 &= \cos 20^\circ + 2 \cos 20^\circ \cos (180^\circ - 60^\circ) \\
 &= \cos 20^\circ + 2 \cos 20^\circ (-\cos 60^\circ) \left[ \because \cos (180^\circ - \theta) = -\cos \theta \right] \\
 &= \cos 20^\circ + 2 \times \left( -\frac{1}{2} \right) \cos 20^\circ = \cos 20^\circ - \cos 20^\circ = 0.
 \end{aligned}$$

**Illustration 4 :** Show that

(i)  $\sin \frac{5\pi}{18} - \cos \frac{4\pi}{9} = \sqrt{3} \sin \frac{\pi}{9}$

[UPTU 2003]

(ii)  $\cos \frac{\pi}{12} - \sin \frac{\pi}{12} = \frac{1}{\sqrt{2}}$

**Solution :** (i) We have  $\sin \frac{5\pi}{18} - \cos \frac{4\pi}{9}$

$$\begin{aligned}
 &= \sin \frac{5\pi}{18} - \sin \left( \frac{\pi}{2} - \frac{4\pi}{9} \right) \quad \left[ \because \cos \theta = \sin \left( \frac{\pi}{2} - \theta \right) \right] \\
 &= \sin \frac{5\pi}{18} - \sin \frac{\pi}{18} = 2 \cos \left( \frac{\frac{5\pi}{18} + \frac{\pi}{18}}{2} \right) \sin \left( \frac{\frac{5\pi}{18} - \frac{\pi}{18}}{2} \right) \\
 &= 2 \cos \frac{\pi}{6} \sin \frac{\pi}{9} = \left( 2 \times \frac{\sqrt{3}}{2} \right) \sin \frac{\pi}{9} = \sqrt{3} \sin \frac{\pi}{9}.
 \end{aligned}$$

(ii) We have  $\cos \frac{\pi}{12} - \sin \frac{\pi}{12}$

$$\begin{aligned}
 &= \cos \frac{\pi}{12} - \cos \left( \frac{\pi}{2} - \frac{\pi}{12} \right) \quad \left[ \because \cos \left( \frac{\pi}{2} - \theta \right) = \sin \theta \right] \\
 &= \cos \frac{\pi}{12} - \cos \frac{5\pi}{12} = 2 \sin \left( \frac{\frac{\pi}{12} + \frac{5\pi}{12}}{2} \right) \sin \left( \frac{\frac{5\pi}{12} - \frac{\pi}{12}}{2} \right) \\
 &= 2 \sin \frac{\pi}{4} \sin \frac{\pi}{6} = 2 \times \frac{1}{\sqrt{2}} \times \frac{1}{2} = \frac{1}{\sqrt{2}}.
 \end{aligned}$$

**Illustration 5 :** Prove that  $\frac{\sin A + \sin B}{\cos A + \cos B} = \tan \left( \frac{A+B}{2} \right)$ . [UPTU 2001 (Special)]

**Solution :** We have

$$\begin{aligned} \text{L.H.S.} &= \frac{\sin A + \sin B}{\cos A + \cos B} = \frac{2 \sin \left( \frac{A+B}{2} \right) \cos \left( \frac{A-B}{2} \right)}{2 \cos \left( \frac{A+B}{2} \right) \cos \left( \frac{A-B}{2} \right)} \\ &\left[ \begin{array}{l} \because \sin C + \sin D = 2 \sin \left( \frac{C+D}{2} \right) \cos \left( \frac{C-D}{2} \right) \\ \cos C + \cos D = 2 \cos \left( \frac{C+D}{2} \right) \cos \left( \frac{C-D}{2} \right) \end{array} \right] \\ &= \frac{\sin \left( \frac{A+B}{2} \right)}{\cos \left( \frac{A+B}{2} \right)} = \tan \left( \frac{A+B}{2} \right) = \text{R.H.S.} \end{aligned}$$

**Illustration 6 :** Prove that  $\frac{\sin \theta + \sin 3\theta + \sin 5\theta + \sin 7\theta}{\cos \theta + \cos 3\theta + \cos 5\theta + \cos 7\theta} = \tan 4\theta$ .

**Solution :** We have

$$\begin{aligned} \text{L.H.S.} &= \frac{(\sin 7\theta + \sin \theta) + (\sin 5\theta + \sin 3\theta)}{(\cos 7\theta + \cos \theta) + (\cos 5\theta + \cos 3\theta)} \\ &= \frac{2 \sin \left( \frac{7\theta + \theta}{2} \right) \cos \left( \frac{7\theta - \theta}{2} \right) + 2 \sin \left( \frac{5\theta + 3\theta}{2} \right) \cos \left( \frac{5\theta - 3\theta}{2} \right)}{2 \cos \left( \frac{7\theta + \theta}{2} \right) \cos \left( \frac{7\theta - \theta}{2} \right) + 2 \cos \left( \frac{5\theta + 3\theta}{2} \right) \cos \left( \frac{5\theta - 3\theta}{2} \right)} \\ &= \frac{2 \sin 4\theta \cos 3\theta + 2 \sin 4\theta \cos \theta}{2 \cos 4\theta \cos 3\theta + 2 \cos 4\theta \cos \theta} \\ &= \frac{2 \sin 4\theta (\cos 3\theta + \cos \theta)}{2 \cos 4\theta (\cos 3\theta + \cos \theta)} = \frac{\sin 4\theta}{\cos 4\theta} = \tan 4\theta = \text{R.H.S.} \end{aligned}$$

**Illustration 7 :** Prove that  $2 \cos \frac{\pi}{13} \cos \frac{9\pi}{13} + \cos \frac{3\pi}{13} + \cos \frac{5\pi}{13} = 0$ . [UPTU 2007]

**Solution :** We have  $2 \cos \frac{\pi}{13} \cos \frac{9\pi}{13} + \cos \frac{3\pi}{13} + \cos \frac{5\pi}{13}$

$$= \cos \left( \frac{\pi}{13} + \frac{9\pi}{13} \right) + \cos \left( \frac{\pi}{13} - \frac{9\pi}{13} \right) + \cos \frac{3\pi}{13} + \cos \frac{5\pi}{13}$$

$$[ \because 2 \cos A \cos B = \cos (A+B) + \cos (A-B) ]$$

$$\begin{aligned}
 &= \cos \frac{10\pi}{13} + \cos \frac{8\pi}{13} + \cos \frac{3\pi}{13} + \cos \frac{5\pi}{13} \quad [\because \cos(-A) = \cos A] \\
 &= \cos \left( \pi - \frac{3\pi}{13} \right) + \cos \left( \pi - \frac{5\pi}{13} \right) + \cos \frac{3\pi}{13} + \cos \frac{5\pi}{13} \\
 &= -\cos \frac{3\pi}{13} - \cos \frac{5\pi}{13} + \cos \frac{3\pi}{13} + \cos \frac{5\pi}{13} = 0. \\
 &\quad [\because \cos(\pi - \theta) = -\cos \theta]
 \end{aligned}$$

**Illustration 8 :** Prove that  $(\cos \alpha - \cos \beta)^2 + (\sin \alpha - \sin \beta)^2 = 4 \sin^2 \left( \frac{\alpha - \beta}{2} \right)$ .

**Solution :** We have L.H.S.  $= (\cos \alpha - \cos \beta)^2 + (\sin \alpha - \sin \beta)^2$

$$\begin{aligned}
 &= \left[ 2 \sin \left( \frac{\alpha + \beta}{2} \right) \sin \left( \frac{\beta - \alpha}{2} \right) \right]^2 + \left[ 2 \cos \left( \frac{\alpha + \beta}{2} \right) \sin \left( \frac{\alpha - \beta}{2} \right) \right]^2 \\
 &= 4 \sin^2 \left( \frac{\alpha + \beta}{2} \right) \sin^2 \left( \frac{\alpha - \beta}{2} \right) + 4 \cos^2 \left( \frac{\alpha + \beta}{2} \right) \sin^2 \left( \frac{\alpha - \beta}{2} \right) \\
 &\quad \left[ \because \sin \left( \frac{\beta - \alpha}{2} \right) = -\sin \left( \frac{\alpha - \beta}{2} \right) \right] \\
 &= 4 \sin^2 \left( \frac{\alpha - \beta}{2} \right) \left[ \sin^2 \left( \frac{\alpha + \beta}{2} \right) + \cos^2 \left( \frac{\alpha + \beta}{2} \right) \right] \\
 &= \left[ 4 \sin^2 \left( \frac{\alpha - \beta}{2} \right) \right] \cdot 1 = 4 \sin^2 \left( \frac{\alpha - \beta}{2} \right) = \text{R.H.S.}
 \end{aligned}$$

**Illustration 9 :** Prove that  $\sin 10^\circ \sin 30^\circ \sin 50^\circ \sin 70^\circ = \frac{1}{16}$ . [UPTU 2003]

**Solution :** We have L.H.S.  $= (\sin 30^\circ) (\sin 50^\circ \sin 10^\circ) \sin 70^\circ$

$$\begin{aligned}
 &= \frac{1}{2} \cdot \frac{1}{2} [2 \sin 50^\circ \sin 10^\circ] \sin 70^\circ \\
 &= \frac{1}{4} (\cos 40^\circ - \cos 60^\circ) \sin 70^\circ \\
 &\quad [\because 2 \sin A \sin B = \cos(A - B) - \cos(A + B)] \\
 &= \frac{1}{4} \left( \cos 40^\circ - \frac{1}{2} \right) \sin 70^\circ = \frac{1}{8} (2 \sin 70^\circ \cos 40^\circ) - \frac{1}{8} \sin 70^\circ \\
 &= \frac{1}{8} (\sin 110^\circ + \sin 30^\circ) - \frac{1}{8} \sin 70^\circ \\
 &\quad [\because 2 \sin A \cos B = \sin(A + B) + \sin(A - B)] \\
 &= \frac{1}{8} \sin 110^\circ + \frac{1}{8} \sin 30^\circ - \frac{1}{8} \sin 70^\circ \\
 &= \frac{1}{8} \sin 70^\circ + \frac{1}{8} \cdot \frac{1}{2} - \frac{1}{8} \sin 70^\circ \\
 &\quad [\because \sin 110^\circ = \sin(180^\circ - 70^\circ) = \sin 70^\circ] \\
 &= \frac{1}{16} = \text{R.H.S.}
 \end{aligned}$$



**Illustration 10 :** Prove that  $\cos 20^\circ \cos 40^\circ \cos 80^\circ = \frac{1}{8}$ .

[UPTU 2005]

**Solution :** We have L.H.S. =  $(\cos 40^\circ \cos 20^\circ) \cos 80^\circ$

$$= \frac{1}{2} [2 \cos 40^\circ \cos 20^\circ] \cos 80^\circ = \frac{1}{2} (\cos 60^\circ + \cos 20^\circ) \cos 80^\circ$$

$$= \frac{1}{2} \left( \frac{1}{2} + \cos 20^\circ \right) \cos 80^\circ = \frac{1}{4} \cos 80^\circ + \frac{1}{4} (2 \cos 80^\circ \cos 20^\circ)$$

$$= \frac{1}{4} \cos 80^\circ + \frac{1}{4} (\cos 100^\circ + \cos 60^\circ)$$

$$= \frac{1}{4} \cos 80^\circ - \frac{1}{4} \cos 80^\circ + \frac{1}{4} \cos 60^\circ$$

$$[\because \cos 100^\circ = \cos (180^\circ - 80^\circ) = -\cos 80^\circ]$$

$$= \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8} = \text{R.H.S.}$$

## Comprehensive Exercise 5

1. Prove that

(i)  $\sin 50^\circ - \sin 70^\circ + \sin 10^\circ = 0$

(ii)  $\cos \frac{\pi}{12} - \sin \frac{\pi}{12} = \frac{1}{\sqrt{2}}$

(iii)  $\cos 52^\circ + \cos 68^\circ + \cos 172^\circ = 0$

(iv)  $\sin 80^\circ - \cos 70^\circ = \cos 50^\circ$ .

2. Prove that

(i)  $2 \sin \frac{5\pi}{12} \sin \frac{\pi}{12} = \frac{1}{2}$

(ii)  $2 \cos 45^\circ \cos 15^\circ = \frac{\sqrt{3} + 1}{2}$ .

3. Prove that

(i)  $\frac{\cos 20^\circ - \cos 70^\circ}{\sin 70^\circ - \sin 20^\circ} = 1$

(ii)  $\frac{\sin 50^\circ - \sin 40^\circ}{\cos 50^\circ - \cos 40^\circ} = -1$

(iii)  $2 \cos 75^\circ \cos 15^\circ = \frac{1}{2}$

(iv)  $2 \sin 75^\circ \cos 15^\circ = \frac{\sqrt{3} + 2}{2}$ .

4. Prove that

(i)  $\cos 3A + \cos 5A + \cos 7A + \cos 15A = 4 \cos 4A \cos 5A \cos 6A$

- (ii)  $\cos 20^\circ \cos 100^\circ + \cos 100^\circ \cos 140^\circ - \cos 140^\circ \cos 200^\circ = -\frac{3}{4}$ .
5. Prove that
- (i)  $\frac{\sin \theta + \sin 3\theta}{\cos \theta - \cos 3\theta} = \cot \theta$
- (ii)  $\frac{\sin A + \sin B}{\sin A - \sin B} = \tan \left( \frac{A+B}{2} \right) \cot \left( \frac{A-B}{2} \right)$ .
6. Prove that
- (i)  $\frac{\cos 4\theta + \cos 3\theta + \cos 2\theta}{\sin 4\theta + \sin 3\theta + \sin 2\theta} = \cot 3\theta$
- (ii)  $\frac{\sin \theta + \sin 2\theta + \sin 4\theta + \sin 5\theta}{\cos \theta + \cos 2\theta + \cos 4\theta + \cos 5\theta} = \tan 3\theta$ .
7. Prove that  $\frac{\sin 3\theta \cos 4\theta - \sin \theta \cos 2\theta}{\sin 4\theta \sin \theta + \cos \theta \cos 6\theta} = \tan 2\theta$ .
8. Prove that  $\frac{(\cos \theta - \cos 3\theta)(\sin 8\theta + \sin 2\theta)}{(\sin 5\theta - \sin \theta)(\cos 4\theta - \cos 6\theta)} = 1$ .
9. Prove that  $(\cos \alpha + \cos \beta)^2 + (\sin \alpha + \sin \beta)^2 = 4 \cos^2 \left( \frac{\alpha + \beta}{2} \right)$ .
10. Prove that  $\cos 10^\circ \cos 30^\circ \cos 50^\circ \cos 70^\circ = \frac{3}{16}$ .
11. Prove that  $\sin 20^\circ \sin 40^\circ \sin 60^\circ \sin 80^\circ = \frac{3}{16}$ . [UPTU 2004, 06]
12. Prove that  $\sin 10^\circ \sin 50^\circ \sin 60^\circ \sin 70^\circ = \frac{\sqrt{3}}{16}$ .
13. Prove that  $\cos 40^\circ \cos 80^\circ \cos 160^\circ = -\frac{1}{8}$ .
14. Prove that  $\frac{\cos 8A \cos 5A - \cos 12A \cos 9A}{\sin 8A \cos 5A + \cos 12A \sin 9A} = \tan 4A$ .
15. If  $\cos A + \cos B = \frac{1}{3}$  and  $\sin A + \sin B = \frac{1}{4}$ , prove that  $\tan \left( \frac{A+B}{2} \right) = \frac{3}{4}$ .
16. Prove that  $\frac{\sin(A-C) + 2 \sin A + \sin(A+C)}{\sin(B-C) + 2 \sin B + \sin(B+C)} = \frac{\sin A}{\sin B}$ .

## 4.15 Trigonometric Ratios of Multiple Angles

For all values of angle  $A$ , we have

(i)  $\sin 2A = 2 \sin A \cos A$

(ii)  $\cos 2A = \cos^2 A - \sin^2 A = 1 - 2 \sin^2 A = 2 \cos^2 A - 1$

(iii)  $1 + \cos 2A = 2 \cos^2 A$

$$(iv) 1 - \cos 2A = 2 \sin^2 A$$

$$(v) \tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$$(vi) \sin 2A = \frac{2 \tan A}{1 + \tan^2 A}$$

$$(vii) \cos 2A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$$

$$(viii) \sin 3A = 3 \sin A - 4 \sin^3 A$$

$$(ix) \cos 3A = 4 \cos^3 A - 3 \cos A$$

$$(x) \tan 3A = \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A}.$$

## 4.16 Trigonometric Ratios of Sub-multiple Angles

Replacing  $A$  by  $A/2$  in the above results (i) to (vii) and  $A$  by  $A/3$  in the above results (viii) to (x), given in 4.15, we get

$$(i) \sin A = 2 \sin \frac{A}{2} \cos \frac{A}{2}$$

$$(ii) \cos A = \cos^2 \frac{A}{2} - \sin^2 \frac{A}{2} = 1 - 2 \sin^2 \frac{A}{2} = 2 \cos^2 \frac{A}{2} - 1$$

$$(iii) 1 + \cos A = 2 \cos^2 \frac{A}{2}$$

$$(iv) 1 - \cos A = 2 \sin^2 \frac{A}{2}$$

$$(v) \tan A = \frac{2 \tan (A/2)}{1 - \tan^2 (A/2)}$$

$$(vi) \sin A = \frac{2 \tan (A/2)}{1 + \tan^2 (A/2)}$$

$$(vii) \cos A = \frac{1 - \tan^2 (A/2)}{1 + \tan^2 (A/2)}$$

$$(viii) \sin A = 3 \sin \frac{A}{3} - 4 \sin^3 \frac{A}{3}$$

$$(ix) \cos A = 4 \cos^3 \frac{A}{3} - 3 \cos \frac{A}{3}$$

$$(x) \tan A = \frac{3 \tan (A/3) - \tan^3 (A/3)}{1 - 3 \tan^2 (A/3)}.$$

**Trigonometric Ratios of the Angle  $A/2$  in Terms of  $\cos A$** 

$$\sin \frac{A}{2} = \pm \sqrt{\frac{1 - \cos A}{2}}, \quad \cos \frac{A}{2} = \pm \sqrt{\frac{1 + \cos A}{2}}, \quad \tan \frac{A}{2} = \pm \sqrt{\frac{1 - \cos A}{1 + \cos A}}.$$

These relations are very useful to find the trigonometric ratios of the angles  $22\frac{1}{2}^\circ, 7\frac{1}{2}^\circ, 11\frac{1}{4}^\circ$  etc.

**4.17 Trigonometric Ratios of Angles of  $18^\circ, 36^\circ, 54^\circ$  and  $72^\circ$ .****(i) To find  $\sin 18^\circ$ .**

[UPTU 2001]

Let  $\theta = 18^\circ$ .Then,  $5\theta = 90^\circ \Rightarrow 2\theta = 90^\circ - 3\theta$ 

$$\Rightarrow \sin 2\theta = \sin (90^\circ - 3\theta) = \cos 3\theta$$

$$\Rightarrow 2 \sin \theta \cos \theta = 4 \cos^3 \theta - 3 \cos \theta$$

$$\Rightarrow \cos \theta (2 \sin \theta - 4 \cos^2 \theta + 3) = 0$$

$$\Rightarrow 2 \sin \theta - 4 \cos^2 \theta + 3 = 0 \quad [\because \cos \theta = \cos 18^\circ \neq 0]$$

$$\Rightarrow 2 \sin \theta - 4 (1 - \sin^2 \theta) + 3 = 0$$

$$\Rightarrow 4 \sin^2 \theta + 2 \sin \theta - 1 = 0$$

$$\Rightarrow \sin \theta = \frac{-2 \pm \sqrt{4 + 16}}{8} = \frac{-2 \pm \sqrt{20}}{8} = \frac{-1 \pm \sqrt{5}}{4}$$

$$\Rightarrow \sin 18^\circ = \frac{\sqrt{5} - 1}{4}.$$

[ $\because \theta = 18^\circ$  lies in I quadrant and so  $\sin 18^\circ$  is positive]

Hence,  $\sin 18^\circ = \frac{\sqrt{5} - 1}{4}.$

**(ii) To find  $\cos 72^\circ$ .**

We have  $\cos 72^\circ = \cos (90^\circ - 18^\circ) = \sin 18^\circ = \frac{\sqrt{5} - 1}{4}.$

**(iii) To find  $\cos 18^\circ$ .**If  $\theta$  is an acute angle, then  $\cos \theta = \sqrt{1 - \sin^2 \theta}.$ 

$$\begin{aligned} \therefore \cos 18^\circ &= \sqrt{1 - \sin^2 18^\circ} = \sqrt{1 - \left(\frac{\sqrt{5} - 1}{4}\right)^2} \\ &= \sqrt{\frac{16 - (5 + 1 - 2\sqrt{5})}{16}} = \frac{\sqrt{10 + 2\sqrt{5}}}{4}. \end{aligned}$$

(iv) **To find  $\sin 72^\circ$ .**

$$\text{We have } \sin 72^\circ = \sin (90^\circ - 18^\circ) = \cos 18^\circ = \frac{\sqrt{10 + 2\sqrt{5}}}{4}.$$

(v) **To find  $\cos 36^\circ$ .**

We have  $\cos 2\theta = 1 - 2\sin^2 \theta$ . Putting  $\theta = 18^\circ$ , we get

$$\begin{aligned}\cos 36^\circ &= 1 - 2\sin^2 18^\circ = 1 - 2\left(\frac{\sqrt{5} - 1}{4}\right)^2 \\ &= 1 - \frac{5 + 1 - 2\sqrt{5}}{8} = \frac{2 + 2\sqrt{5}}{8} = \frac{\sqrt{5} + 1}{4}.\end{aligned}$$

$$\text{Hence, } \cos 36^\circ = \frac{\sqrt{5} + 1}{4}.$$

(vi) **To find  $\sin 54^\circ$**

$$\text{We have } \sin 54^\circ = \sin (90^\circ - 36^\circ) = \cos 36^\circ = \frac{\sqrt{5} + 1}{4}.$$

(vii) **To find  $\sin 36^\circ$ .**

[UPTU 2006]

$$\begin{aligned}\text{We have } \sin 36^\circ &= \sqrt{1 - \cos^2 36^\circ} = \sqrt{1 - \left(\frac{\sqrt{5} + 1}{4}\right)^2} \\ &= \frac{\sqrt{10 - 2\sqrt{5}}}{4}.\end{aligned}$$

$$\text{Hence, } \sin 36^\circ = \frac{\sqrt{10 - 2\sqrt{5}}}{4}.$$

(viii) **To find  $\cos 54^\circ$ .**

$$\text{We have } \cos 54^\circ = \cos (90^\circ - 36^\circ) = \sin 36^\circ = \frac{\sqrt{10 - 2\sqrt{5}}}{4}.$$

## Illustrative Examples

**Illustration 1 :** Find the values of

(i)  $\cos 22 \frac{1}{2}^\circ$  [UPTU 2001]

(ii)  $\tan 22 \frac{1}{2}^\circ$

[UPTU 2002]

(iii)  $\sin 7 \frac{1}{2}^\circ$

(iv)  $\tan 11 \frac{1}{4}^\circ$ .

**Solution :** (i) We have  $\cos^2 A = \frac{1 + \cos 2A}{2}$ .

Putting  $2A = 45^\circ$  i.e.,  $A = 22 \frac{1}{2}^\circ$ , we get

$$\cos^2 22 \frac{1}{2}^\circ = \frac{1 + \cos 45^\circ}{2} = \frac{1 + (1/\sqrt{2})}{2} = \frac{\sqrt{2} + 1}{2\sqrt{2}}.$$

$$\therefore \cos 22 \frac{1}{2}^\circ = \sqrt{\left(\frac{\sqrt{2} + 1}{2\sqrt{2}}\right)} \quad [\because \cos 22 \frac{1}{2}^\circ \text{ is +ve}]$$

$$(ii) \text{ We have } \tan^2 \frac{A}{2} = \frac{2 \sin^2 (A/2)}{2 \cos^2 (A/2)} = \frac{1 - \cos A}{1 + \cos A}.$$

$$\text{Putting } A = 45^\circ \text{ i.e., } \frac{A}{2} = 22 \frac{1}{2}^\circ, \text{ we get}$$

$$\begin{aligned} \tan^2 22 \frac{1}{2}^\circ &= \frac{1 - \cos 45^\circ}{1 + \cos 45^\circ} = \frac{1 - (1/\sqrt{2})}{1 + (1/\sqrt{2})} = \frac{\sqrt{2} - 1}{\sqrt{2} + 1} \\ &= \frac{(\sqrt{2} - 1)^2}{(\sqrt{2} + 1)(\sqrt{2} - 1)} = \frac{(\sqrt{2} - 1)^2}{2 - 1} = (\sqrt{2} - 1)^2 \end{aligned}$$

$$\therefore \tan 22 \frac{1}{2}^\circ = \sqrt{2} - 1. \quad [\because \tan 22 \frac{1}{2}^\circ \text{ is +ve}]$$

$$\begin{aligned} (iii) \text{ We have } \cos 15^\circ &= \cos (45^\circ - 30^\circ) \\ &= \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ \\ &= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2} = \frac{\sqrt{3} + 1}{2\sqrt{2}}. \end{aligned}$$

$$\text{Now } \sin^2 A = \frac{1 - \cos 2A}{2}.$$

$$\text{Putting } 2A = 15^\circ \text{ i.e., } A = 7 \frac{1}{2}^\circ, \text{ we get}$$

$$\begin{aligned} \sin^2 7 \frac{1}{2}^\circ &= \frac{1 - \cos 15^\circ}{2} = \frac{1 - \frac{\sqrt{3} + 1}{2\sqrt{2}}}{2} = \frac{2\sqrt{2} - \sqrt{3} - 1}{4\sqrt{2}} \\ &= \frac{4 - \sqrt{6} - \sqrt{2}}{8} \end{aligned}$$

[Multiplying Nr and Dr by  $\sqrt{2}$ ]

$$\therefore \sin 7 \frac{1}{2}^\circ = \frac{\sqrt{4 - \sqrt{6} - \sqrt{2}}}{2\sqrt{2}}. \quad \left[ \because \sin 7 \frac{1}{2}^\circ \text{ is +ve} \right]$$

$$(iv) \text{ We have } \tan A = \frac{2 \tan (A/2)}{1 - \tan^2 (A/2)}$$

$$\Rightarrow \tan 22 \frac{1}{2}^\circ = \frac{2 \tan 11 \frac{1}{4}^\circ}{1 - \tan^2 11 \frac{1}{4}^\circ} = \frac{2x}{1 - x^2},$$

$$\text{where } x = \tan 11 \frac{1}{4}^\circ$$

$$\Rightarrow \frac{2x}{1-x^2} = \sqrt{2} - 1 \quad \left[ \because \tan 22 \frac{1}{2}^\circ = \sqrt{2} - 1 \right]$$

$$\Rightarrow (\sqrt{2} - 1)x^2 + 2x - (\sqrt{2} - 1) = 0$$

$$\Rightarrow x = \frac{-2 \pm \sqrt{4 + 4(\sqrt{2} - 1)^2}}{2(\sqrt{2} - 1)}$$

$$= \frac{-1 \pm \sqrt{1 + (\sqrt{2} - 1)^2}}{\sqrt{2} - 1} = \frac{-1 \pm \sqrt{4 - 2\sqrt{2}}}{\sqrt{2} - 1}.$$

But  $x = \tan 11 \frac{1}{4}^\circ$  is +ve.

$$\therefore \tan 11 \frac{1}{4}^\circ = \frac{-1 + \sqrt{4 - 2\sqrt{2}}}{\sqrt{2} - 1} = \frac{(-1 + \sqrt{4 - 2\sqrt{2}})(\sqrt{2} + 1)}{(\sqrt{2} - 1)(\sqrt{2} + 1)}$$

$$= \frac{\sqrt{(4 - 2\sqrt{2})(\sqrt{2} + 1)^2} - (\sqrt{2} + 1)}{1}$$

$$= \sqrt{(4 - 2\sqrt{2})(3 + 2\sqrt{2})} - (\sqrt{2} + 1)$$

$$= \sqrt{4 + 2\sqrt{2}} - (\sqrt{2} + 1).$$

**Illustration 2 :** Calculate  $\sin \frac{\pi}{10}$  and  $\cos \frac{\pi}{5}$ .

[UPTU 2003]

**Solution :** We have  $\cos \frac{3\pi}{10} = \sin \left( \frac{\pi}{2} - \frac{3\pi}{10} \right) = \sin \left( \frac{2\pi}{10} \right).$

$$\left[ \because \cos A = \sin \left( \frac{\pi}{2} - A \right) \right]$$

Now,  $\cos \frac{3\pi}{10} = \sin \frac{2\pi}{10}$

$$\Rightarrow 4 \cos^3 \frac{\pi}{10} - 3 \cos \frac{\pi}{10} = 2 \sin \frac{\pi}{10} \cos \frac{\pi}{10}$$

$$\left[ \because \cos 3A = 4 \cos^3 A - 3 \cos A \text{ and } \sin 2A = 2 \sin A \cos A \right]$$

$$\Rightarrow \left( \cos \frac{\pi}{10} \right) \left[ 4 \cos^2 \frac{\pi}{10} - 3 - 2 \sin \frac{\pi}{10} \right] = 0$$

$$\Rightarrow 4 \cos^2 \frac{\pi}{10} - 3 - 2 \sin \frac{\pi}{10} = 0 \quad \left[ \because \cos \frac{\pi}{10} \neq 0 \right]$$

$$\Rightarrow 4 \left( 1 - \sin^2 \frac{\pi}{10} \right) - 3 - 2 \sin \frac{\pi}{10} = 0$$

$$\Rightarrow 4 \sin^2 \frac{\pi}{10} + 2 \sin \frac{\pi}{10} - 1 = 0$$

$$\Rightarrow \sin \frac{\pi}{10} = \frac{-2 \pm \sqrt{4 + 16}}{8}$$

$$\left[ \text{Solving the quadratic equation in } \sin \frac{\pi}{10} \right]$$

$$= \frac{-1 \pm \sqrt{5}}{4}.$$

Since  $0 < \frac{\pi}{10} < \frac{\pi}{2}$ , therefore  $\sin \frac{\pi}{10}$  is positive.

$$\text{Hence, } \sin \frac{\pi}{10} = \sin 18^\circ = \frac{-1 + \sqrt{5}}{4} = \frac{\sqrt{5} - 1}{4}.$$

$$\begin{aligned} \text{Now, } \cos \frac{\pi}{5} &= 1 - 2 \sin^2 \frac{\pi}{10} & \left[ \because \cos A = 1 - 2 \sin^2 \frac{A}{2} \right] \\ &= 1 - 2 \left( \frac{\sqrt{5} - 1}{4} \right)^2 = 1 - \frac{5 + 1 - 2\sqrt{5}}{8} = \frac{8 - 6 + 2\sqrt{5}}{8} \\ &= \frac{2(\sqrt{5} + 1)}{8} = \frac{\sqrt{5} + 1}{4}. \end{aligned}$$

$$\text{Hence, } \cos \frac{\pi}{5} = \cos 36^\circ = \frac{\sqrt{5} + 1}{4}.$$

**Illustration 3 :** Prove that  $\cot 7 \frac{1}{2}^\circ = \sqrt{2} + \sqrt{3} + \sqrt{4} + \sqrt{6}$

$$\text{or } \tan 82 \frac{1}{2}^\circ = (\sqrt{3} + \sqrt{2})(\sqrt{2} + 1).$$

$$\begin{aligned} \text{Solution : } \text{We have } \cot 7 \frac{1}{2}^\circ &= \frac{\cos 7 \frac{1}{2}^\circ}{\sin 7 \frac{1}{2}^\circ} \\ &= \frac{2 \cos^2 7 \frac{1}{2}^\circ}{2 \sin 7 \frac{1}{2}^\circ \cos 7 \frac{1}{2}^\circ} = \frac{1 + \cos 15^\circ}{\sin 15^\circ} \\ &= \frac{1 + \cos (45^\circ - 30^\circ)}{\sin (45^\circ - 30^\circ)} = \frac{1 + \frac{\sqrt{3} + 1}{2\sqrt{2}}}{\frac{\sqrt{3} - 1}{2\sqrt{2}}} = \frac{2\sqrt{2} + \sqrt{3} + 1}{\sqrt{3} - 1} \\ &= \frac{(2\sqrt{2} + \sqrt{3} + 1)(\sqrt{3} + 1)}{(\sqrt{3} - 1)(\sqrt{3} + 1)} = \frac{2\sqrt{6} + 2\sqrt{2} + 3 + \sqrt{3} + \sqrt{3} + 1}{3 - 1} \\ &= \frac{2\sqrt{6} + 2\sqrt{2} + 2\sqrt{3} + 4}{2} \\ &= \sqrt{2} + \sqrt{3} + 2 + \sqrt{6} = \sqrt{2} + \sqrt{3} + \sqrt{4} + \sqrt{6}. \end{aligned}$$

$$\begin{aligned} \text{Further } \cot 7 \frac{1}{2}^\circ &= 2 + \sqrt{2} + \sqrt{6} + \sqrt{3} = \sqrt{2}(\sqrt{2} + 1) + \sqrt{3}(\sqrt{2} + 1) \\ &= (\sqrt{2} + 1)(\sqrt{2} + \sqrt{3}) \end{aligned}$$

$$\text{or } \cot \left( 90^\circ - 82 \frac{1}{2}^\circ \right) = (\sqrt{2} + 1)(\sqrt{2} + \sqrt{3})$$



or  $\tan 82 \frac{1^\circ}{2} = (\sqrt{3} + \sqrt{2}) (\sqrt{2} + 1).$

**Illustration 4 :** Prove that

(i)  $\tan \theta - \cot \theta = -2 \cot 2 \theta$

(ii)  $\frac{\cos \theta}{1 + \sin \theta} = \tan \left( 45^\circ - \frac{\theta}{2} \right)$

[UPTU 2001 (Special)]

(iii)  $\tan \left( \frac{\pi}{4} + \frac{\theta}{2} \right) + \tan \left( \frac{\pi}{4} - \frac{\theta}{2} \right) = 2 \sec \theta$

(iv)  $\sqrt{\frac{1 - \sin \theta}{1 + \sin \theta}} = \tan \left( \frac{\pi}{4} - \frac{\theta}{2} \right)$

[UPTU 2003, 06]

(v)  $\sec \alpha + \tan \alpha = \tan \left( \frac{\pi}{4} + \frac{\alpha}{2} \right).$

[UPTU 2002, 06]

**Solution :** (i) We have  $\tan \theta - \cot \theta = \frac{\sin \theta}{\cos \theta} - \frac{\cos \theta}{\sin \theta} = \frac{\sin^2 \theta - \cos^2 \theta}{\sin \theta \cos \theta}$

$$= \frac{-2 (\cos^2 \theta - \sin^2 \theta)}{2 \sin \theta \cos \theta} = \frac{-2 \cos 2 \theta}{\sin 2 \theta} = -2 \cot 2 \theta.$$

(ii) We have  $\frac{\cos \theta}{1 + \sin \theta} = \frac{\sin (90^\circ - \theta)}{1 + \cos (90^\circ - \theta)}$

$$= \frac{2 \sin \left( \frac{90^\circ - \theta}{2} \right) \cos \left( \frac{90^\circ - \theta}{2} \right)}{2 \cos^2 \left( \frac{90^\circ - \theta}{2} \right)} = \frac{\sin \left( 45^\circ - \frac{\theta}{2} \right)}{\cos \left( 45^\circ - \frac{\theta}{2} \right)}$$

$$= \tan \left( 45^\circ - \frac{\theta}{2} \right).$$

(iii) We have  $\tan \left( \frac{\pi}{4} + \frac{\theta}{2} \right) + \tan \left( \frac{\pi}{4} - \frac{\theta}{2} \right)$

$$= \frac{\tan \frac{\pi}{4} + \tan \frac{\theta}{2}}{1 - \tan \frac{\pi}{4} \tan \frac{\theta}{2}} + \frac{\tan \frac{\pi}{4} - \tan \frac{\theta}{2}}{1 + \tan \frac{\pi}{4} \tan \frac{\theta}{2}}$$

$$= \frac{1 + \tan \frac{\theta}{2}}{1 - \tan \frac{\theta}{2}} + \frac{1 - \tan \frac{\theta}{2}}{1 + \tan \frac{\theta}{2}}$$

$$= \frac{\left( 1 + \tan \frac{\theta}{2} \right)^2 + \left( 1 - \tan \frac{\theta}{2} \right)^2}{\left( 1 - \tan \frac{\theta}{2} \right) \left( 1 + \tan \frac{\theta}{2} \right)}$$

$$= 2 \left( \frac{1 + \tan^2 \frac{\theta}{2}}{1 - \tan^2 \frac{\theta}{2}} \right) = \frac{2}{\cos \theta} = 2 \sec \theta.$$

$$\begin{aligned} \text{(iv) We have } \sqrt{\frac{1 - \sin \theta}{1 + \sin \theta}} &= \sqrt{\frac{1 - \cos (\pi / 2 - \theta)}{1 + \cos (\pi / 2 - \theta)}} \\ &= \sqrt{\frac{2 \sin^2 (\pi / 4 - \theta / 2)}{2 \cos^2 (\pi / 4 - \theta / 2)}} = \sqrt{\tan^2 \left( \frac{\pi}{4} - \frac{\theta}{2} \right)} = \tan \left( \frac{\pi}{4} - \frac{\theta}{2} \right). \end{aligned}$$

$$\begin{aligned} \text{(v) We have } \sec \alpha + \tan \alpha &= \frac{1}{\cos \alpha} + \frac{\sin \alpha}{\cos \alpha} = \frac{1 + \sin \alpha}{\cos \alpha} \\ &= \frac{1 - \cos (\pi / 2 + \alpha)}{\sin (\pi / 2 + \alpha)} = \frac{2 \sin^2 (\pi / 4 + \alpha / 2)}{2 \sin (\pi / 4 + \alpha / 2) \cos (\pi / 4 + \alpha / 2)} \\ &= \tan \left( \frac{\pi}{4} + \frac{\alpha}{2} \right). \end{aligned}$$

**Illustration 5 :** Prove that  $\cos A \cos (60^\circ - A) \cos (60^\circ + A) = \frac{1}{4} \cos 3A$ .

**Solution :** We have  $\cos A \cos (60^\circ - A) \cos (60^\circ + A)$

$$\begin{aligned} &= \cos A (\cos^2 60^\circ - \sin^2 A) \\ &\quad [\because \cos (A + B) \cos (A - B) = \cos^2 A - \sin^2 B] \\ &= \cos A \left( \frac{1}{4} - \sin^2 A \right) = \cos A \left[ \frac{1}{4} - (1 - \cos^2 A) \right] \\ &= \cos A \left( -\frac{3}{4} + \cos^2 A \right) = \frac{1}{4} \cos A (-3 + 4 \cos^2 A) \\ &= \frac{1}{4} (4 \cos^3 A - 3 \cos A) = \frac{1}{4} \cos 3A. \end{aligned}$$

**Illustration 6 :** Prove that  $\sin^2 72^\circ - \sin^2 60^\circ = \frac{\sqrt{5} - 1}{8}$ . [UPTU 2001]

**Solution :** We have  $\sin^2 72^\circ - \sin^2 60^\circ = \left( \frac{\sqrt{10 + 2\sqrt{5}}}{4} \right)^2 - \left( \frac{\sqrt{3}}{2} \right)^2$

$$\begin{aligned} &= \frac{10 + 2\sqrt{5}}{16} - \frac{3}{4} = \frac{10 + 2\sqrt{5} - 12}{16} \\ &= \frac{2\sqrt{5} - 2}{16} = \frac{\sqrt{5} - 1}{8}. \end{aligned}$$

**Illustration 7 :** Prove that  $\sin \frac{\pi}{5} \sin \frac{2\pi}{5} \sin \frac{3\pi}{5} \sin \frac{4\pi}{5} = \frac{5}{16}$ .

**Solution :** We have  $\sin \frac{\pi}{5} \sin \frac{2\pi}{5} \sin \frac{3\pi}{5} \sin \frac{4\pi}{5}$

$$\begin{aligned}
&= \sin \frac{\pi}{5} \sin \frac{2\pi}{5} \sin \left( \pi - \frac{2\pi}{5} \right) \sin \left( \pi - \frac{\pi}{5} \right) \\
&= \sin \frac{\pi}{5} \sin \frac{2\pi}{5} \sin \frac{2\pi}{5} \sin \frac{\pi}{5} \\
&= \sin^2 \frac{\pi}{5} \sin^2 \frac{2\pi}{5} = \sin^2 36^\circ \sin^2 72^\circ \\
&= \sin^2 36^\circ \cos^2 18^\circ \quad [\because \sin 72^\circ = \cos 18^\circ] \\
&= \frac{10 - 2\sqrt{5}}{16} \cdot \frac{10 + 2\sqrt{5}}{16} = \frac{(10)^2 - (2\sqrt{5})^2}{256} \\
&= \frac{100 - 20}{256} = \frac{80}{256} = \frac{5}{16}.
\end{aligned}$$

**Illustration 8 :** Prove that  $\sin^2 \left( \frac{\pi}{8} + \frac{\theta}{2} \right) - \sin^2 \left( \frac{\pi}{8} - \frac{\theta}{2} \right) = \frac{1}{\sqrt{2}} \sin \theta$ . [UPTU 2003]

**Solution :** We have

$$\begin{aligned}
&\sin^2 \left( \frac{\pi}{8} + \frac{\theta}{2} \right) - \sin^2 \left( \frac{\pi}{8} - \frac{\theta}{2} \right) \\
&= \frac{1}{2} \left[ 2 \sin^2 \left( \frac{\pi}{8} + \frac{\theta}{2} \right) - 2 \sin^2 \left( \frac{\pi}{8} - \frac{\theta}{2} \right) \right] \\
&= \frac{1}{2} \left[ \left\{ 1 - \cos \left( \frac{\pi}{4} + \theta \right) \right\} - \left\{ 1 - \cos \left( \frac{\pi}{4} - \theta \right) \right\} \right] \\
&\quad [\because 2 \sin^2 \theta = 1 - \cos 2\theta] \\
&= \frac{1}{2} \left[ \cos \left( \frac{\pi}{4} - \theta \right) - \cos \left( \frac{\pi}{4} + \theta \right) \right] \\
&= \frac{1}{2} \left[ 2 \sin \frac{1}{2} \left( \frac{\pi}{4} - \theta + \frac{\pi}{4} + \theta \right) \sin \frac{1}{2} \left( \frac{\pi}{4} + \theta - \frac{\pi}{4} + \theta \right) \right] \\
&\quad \left[ \because \cos A - \cos B = 2 \sin \frac{A+B}{2} \cos \frac{B-A}{2} \right] \\
&= \sin \frac{\pi}{4} \sin \theta = \frac{1}{\sqrt{2}} \sin \theta. \quad \left[ \because \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}} \right]
\end{aligned}$$

**Illustration 9 :** Prove that  $\cos \frac{2\pi}{15} \cos \frac{4\pi}{15} \cos \frac{8\pi}{15} \cos \frac{14\pi}{15} = \frac{1}{16}$ .

**Solution :** We have  $\cos \frac{2\pi}{15} \cos \frac{4\pi}{15} \cos \frac{8\pi}{15} \cos \frac{14\pi}{15}$

$$\begin{aligned}
&= \cos 24^\circ \cos 48^\circ \cos 96^\circ \cos 168^\circ \\
&= \frac{1}{4} (2 \cos 24^\circ \cos 96^\circ) (2 \cos 48^\circ \cos 168^\circ) \\
&= \frac{1}{4} (\cos 120^\circ + \cos 72^\circ) (\cos 216^\circ + \cos 120^\circ)
\end{aligned}$$


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$$= \frac{1}{4} \left( -\frac{1}{2} + \sin 18^\circ \right) \left( -\cos 36^\circ - \frac{1}{2} \right)$$

$$[ \because \cos 120^\circ = \cos (180^\circ - 60^\circ) = -\cos 60^\circ = -\frac{1}{2}, \cos 72^\circ = \sin 18^\circ$$

$$\text{and } \cos 216^\circ = \cos (180^\circ + 36^\circ) = -\cos 36^\circ ]$$

$$= \frac{1}{4} \left( \frac{1}{2} - \sin 18^\circ \right) \left( \frac{1}{2} + \cos 36^\circ \right)$$

$$= \frac{1}{4} \left( \frac{1}{2} - \frac{\sqrt{5}-1}{4} \right) \left( \frac{1}{2} + \frac{\sqrt{5}+1}{4} \right)$$

$$= \frac{1}{4} \left( \frac{3-\sqrt{5}}{4} \right) \left( \frac{3+\sqrt{5}}{4} \right)$$

$$= \frac{1}{4} \cdot \frac{9-5}{16} = \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16}.$$

**Illustration 10:** Prove that  $\left(1 + \cos \frac{\pi}{8}\right) \left(1 + \cos \frac{3\pi}{8}\right) \left(1 + \cos \frac{5\pi}{8}\right) \left(1 + \cos \frac{7\pi}{8}\right) = \frac{1}{8}.$

**Solution :** We have  $\left(1 + \cos \frac{\pi}{8}\right) \left(1 + \cos \frac{3\pi}{8}\right) \left(1 + \cos \frac{5\pi}{8}\right) \left(1 + \cos \frac{7\pi}{8}\right)$   
 $= \left(1 + \cos \frac{\pi}{8}\right) \left(1 + \cos \frac{3\pi}{8}\right) \left(1 - \cos \frac{3\pi}{8}\right) \left(1 - \cos \frac{\pi}{8}\right)$   
 $\left[ \because \cos \frac{5\pi}{8} = \cos \left(\pi - \frac{3\pi}{8}\right) = -\cos \frac{3\pi}{8} \right.$   
 $\left. \text{and } \cos \frac{7\pi}{8} = \cos \left(\pi - \frac{\pi}{8}\right) = -\cos \frac{\pi}{8} \right]$

$$= \left(1 - \cos^2 \frac{\pi}{8}\right) \left(1 - \cos^2 \frac{3\pi}{8}\right) = \sin^2 \frac{\pi}{8} \sin^2 \frac{3\pi}{8}$$

$$= \frac{1}{4} \left(2 \sin^2 \frac{\pi}{8}\right) \left(2 \sin^2 \frac{3\pi}{8}\right) = \frac{1}{4} \left(1 - \cos \frac{\pi}{4}\right) \left(1 - \cos \frac{3\pi}{4}\right)$$

$$= \frac{1}{4} \left(1 - \frac{1}{\sqrt{2}}\right) \left(1 + \frac{1}{\sqrt{2}}\right) = \frac{1}{4} \left(1 - \frac{1}{2}\right) = \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}.$$

**Illustration 11:** Prove that  $\frac{\tan 3x}{\tan x}$  never lies between  $\frac{1}{3}$  and 3.

**Solution :** Let  $y = \frac{\tan 3x}{\tan x}.$

$$\begin{aligned} \text{Then } y &= \frac{3 \tan x - \tan^3 x}{\tan x (1 - 3 \tan^2 x)} = \frac{\tan x (3 - \tan^2 x)}{\tan x (1 - 3 \tan^2 x)} \\ &= \frac{3 - \tan^2 x}{1 - 3 \tan^2 x}. \end{aligned}$$

$$\therefore y (1 - 3 \tan^2 x) = 3 - \tan^2 x$$

$$\text{or} \quad \tan^2 x (3y - 1) = y - 3$$

$$\text{or} \quad \tan^2 x = \frac{y - 3}{3y - 1}.$$

Now,  $\tan^2 x \geq 0$  for all  $x$ .

$$\therefore \quad \frac{y - 3}{3y - 1} \geq 0 \Rightarrow \frac{(y - 3)(3y - 1)}{(3y - 1)^2} \geq 0$$

$$\Rightarrow (y - 3)(3y - 1) \geq 0 \quad [\because (3y - 1)^2 \geq 0]$$

$$\Rightarrow \left(y - \frac{1}{3}\right)(y - 3) \geq 0$$

$$\Rightarrow y \leq \frac{1}{3} \quad \text{or} \quad y \geq 3$$

$$\Rightarrow y \text{ does not lie between } \frac{1}{3} \text{ and } 3.$$

**Illustration 12 :** If  $\sin 2A = \frac{4}{5}$ , find the values of  $\tan A \cdot \left(0 \leq A \leq \frac{\pi}{4}\right)$ .

[UPTU 2002]

**Solution :** We have  $\sin 2A = \frac{2 \tan A}{1 + \tan^2 A}$

$$\Rightarrow \frac{4}{5} = \frac{2 \tan A}{1 + \tan^2 A} \quad \left[ \text{Given } \sin 2A = \frac{4}{5} \right]$$

$$\Rightarrow 2 + 2 \tan^2 A = 5 \tan A$$

$$\Rightarrow 2 \tan^2 A - 5 \tan A + 2 = 0$$

$$\Rightarrow (\tan A - 2)(2 \tan A - 1) = 0$$

$$\Rightarrow \tan A = 2 \quad \text{or} \quad \tan A = \frac{1}{2}.$$

Since  $0 \leq A \leq \frac{\pi}{4}$ , therefore  $0 \leq \tan A \leq 1$ .

Hence,  $\tan A = \frac{1}{2}$ .

**Illustration 13 :** If  $\tan \frac{\theta}{2} = \frac{p}{q}$ , find the value of  $(p \sin \theta + q \cos \theta)$ .

[UPTU 2007]

**Solution :** We have

$$\begin{aligned} p \sin \theta + q \cos \theta &= p \frac{2 \tan (\theta / 2)}{1 + \tan^2 (\theta / 2)} + q \frac{1 - \tan^2 (\theta / 2)}{1 + \tan^2 (\theta / 2)} \\ &= p \frac{2 (p / q)}{1 + p^2 / q^2} + q \frac{1 - p^2 / q^2}{1 + p^2 / q^2} \end{aligned}$$

$$\begin{aligned}
 &= p \frac{2pq}{q^2 + p^2} + q \frac{q^2 - p^2}{q^2 + p^2} \\
 &= \frac{2p^2q + q^3 - p^2q}{q^2 + p^2} \\
 &= \frac{q^3 + p^2q}{q^2 + p^2} = \frac{q(q^2 + p^2)}{q^2 + p^2} = q.
 \end{aligned}$$

## Comprehensive Exercise 6

1. Prove that

(i)  $\frac{\sin 2\theta}{1 - \cos 2\theta} = \cot \theta$

(ii)  $\frac{1 - \cos 2\theta + \sin 2\theta}{1 + \cos 2\theta + \sin 2\theta} = \tan \theta$

(iii)  $\frac{1 + \sin \theta - \cos \theta}{1 + \sin \theta + \cos \theta} = \tan \frac{\theta}{2}$

(iv)  $\sqrt{\frac{1 - \cos 2x}{1 + \cos 2x}} = \tan x$

(v)  $\frac{\cos 2\theta}{1 + \sin 2\theta} = \tan \left( \frac{\pi}{4} - \theta \right)$

(vi)  $\frac{\cos \theta}{1 - \sin \theta} = \tan \left( \frac{\pi}{4} + \frac{\theta}{2} \right).$

2. Prove that

(i)  $(\cos A + \cos B)^2 + (\sin A - \sin B)^2 = 4 \cos^2 \left( \frac{A+B}{2} \right)$

(ii)  $\sin^4 \frac{\pi}{8} + \sin^4 \frac{3\pi}{8} + \sin^4 \frac{5\pi}{8} + \sin^4 \frac{7\pi}{8} = \frac{3}{2}$

(iii)  $(\sin 3A + \sin A) \sin A + (\cos 3A - \cos A) \cos A = 0.$

3. Prove that  $\cos \frac{2\pi}{15} \cos \frac{4\pi}{15} \cos \frac{8\pi}{15} \cos \frac{14\pi}{15} = \frac{1}{16}.$

4. Prove that  $\tan 4\theta = \frac{4 \tan \theta (1 - \tan^2 \theta)}{1 - 6 \tan^2 \theta + \tan^4 \theta}.$

5. Prove that

(i)  $2 \sin 15^\circ \cos 15^\circ = \frac{1}{2}$

(ii)  $\sin \frac{\pi}{8} \cos \frac{\pi}{8} = \frac{1}{2\sqrt{2}}$

$$(iii) 1 - 2 \sin^2 \left( 22 \frac{1}{2}^\circ \right) = \frac{1}{\sqrt{2}}$$

$$(iv) \frac{2 \tan 22 \frac{1}{2}^\circ}{1 - \tan^2 22 \frac{1}{2}^\circ} = 1$$

$$(v) \sqrt{\frac{1 + \cos 120^\circ}{2}} = \frac{1}{2}$$

$$(vi) \sqrt{\frac{1 - \cos 300^\circ}{2}} = \frac{1}{2}.$$

6. Prove that

$$(i) (\cos A + \cos B)(\cos A - \cos B) = \frac{1}{2}(\cos 2A - \cos 2B)$$

$$(ii) \cos^2 \left( \frac{\pi}{4} - \theta \right) - \sin^2 \left( \frac{\pi}{4} - \theta \right) = \sin 2\theta$$

$$(iii) \sin 3\theta + \sin 2\theta - \sin \theta = 4 \sin \theta \cos \frac{\theta}{2} \cos \frac{3\theta}{2}.$$

7. Prove that

$$(i) \frac{\sin \theta + \sin 2\theta}{1 + \cos \theta + \cos 2\theta} = \tan \theta$$

$$(ii) \frac{\cot \theta - \tan \theta}{\cot \theta + \tan \theta} = \cos 2\theta$$

$$(iii) \frac{\tan^2 \theta}{\tan^2 \theta - 1} + \frac{\operatorname{cosec}^2 \theta}{\sec^2 \theta - \operatorname{cosec}^2 \theta} = -\sec 2\theta$$

$$(iv) \frac{\cos \theta + \sin \theta}{\cos \theta - \sin \theta} - \frac{\cos \theta - \sin \theta}{\cos \theta + \sin \theta} = 2 \tan 2\theta.$$

8. Prove that

$$(i) \tan A + \tan (60^\circ + A) + \tan (60^\circ - A) = 3 \tan 3A$$

$$(ii) \sin^3 A + \sin^3 \left( \frac{2\pi}{3} + A \right) + \sin^3 \left( \frac{4\pi}{3} + A \right) = -\frac{3}{4} \sin 3A.$$

9. Prove that

$$(i) \cos 12^\circ \cos 24^\circ \cos 48^\circ \cos 84^\circ = \frac{1}{16}$$

$$(ii) \sin 36^\circ \sin 72^\circ \sin 108^\circ \sin 144^\circ = \frac{5}{16}.$$

10. Prove that

$$(i) \sin \frac{\pi}{10} \sin \frac{13\pi}{10} = -\frac{1}{4}$$

$$(ii) \sin 12^\circ \sin 48^\circ \sin 54^\circ = \frac{1}{8}.$$

## 4.18 Some Important Trigonometric Identities Involving the Angles of a Triangle.

If  $A + B + C = 180^\circ$ , then

$$(i) \quad \tan A + \tan B + \tan C = \tan A \tan B \tan C$$

$$(ii) \quad \cot B \cot C + \cot C \cot A + \cot A \cot B = 1$$

$$(iii) \quad \tan \frac{B}{2} \tan \frac{C}{2} + \tan \frac{C}{2} \tan \frac{A}{2} + \tan \frac{A}{2} \tan \frac{B}{2} = 1$$

$$(iv) \quad \cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2} = \cot \frac{A}{2} \cot \frac{B}{2} \cot \frac{C}{2}$$

$$(v) \quad \sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$$

$$(vi) \quad \sin A + \sin B + \sin C = 4 \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}$$

$$(vii) \quad \cos 2A + \cos 2B + \cos 2C = -1 - 4 \cos A \cos B \cos C$$

$$(viii) \quad \cos A + \cos B + \cos C = 1 + 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$$

$$(ix) \quad \Sigma \sin A \sin B \cos C = 1 + \cos A \cos B \cos C$$

$$\text{i.e.,} \quad \sin B \sin C \cos A + \sin C \sin A \cos B + \sin A \sin B \cos C \\ = 1 + \cos A \cos B \cos C$$

$$(x) \quad \Sigma \sin A \cos B \cos C = \sin A \sin B \sin C$$

$$\text{i.e.,} \quad \sin A \cos B \cos C + \sin B \cos C \cos A + \sin C \cos A \cos B \\ = \sin A \sin B \sin C.$$

**Type I. Identities Involving Sines and Cosines of Multiples or Sub-multiples of the Angles Involved.**

**Procedure :** (i) Express the sum of first two terms (or any two terms) as a product. In this product express the sum of two angles in terms of the third by the given relation.

(ii) Expand the remaining third term by using the formulae

$$\sin 2\theta = 2 \sin \theta \cos \theta, \cos 2\theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta, \text{ etc.}$$

(iii) Take the common factor outside .

(iv) Express the trigonometric ratio of a single angle into that of the sum of two angles and then use the suitable formula to express the sum or difference of two sines or cosines as the product of sines, cosines or sines and cosines.

## Illustrative Examples

**Illustration 1 :** If  $A + B + C = \pi$ , prove that

$$(i) \quad \sin 2A + \sin 2B - \sin 2C = 4 \cos A \cos B \sin C$$

$$(ii) \quad \cos 2A + \cos 2B + \cos 2C = -1 - 4 \cos A \cos B \cos C.$$



**Solution :** (i) We have

$$\begin{aligned}
 \text{L.H.S.} &= \sin 2A + \sin 2B - \sin 2C = (\sin 2A + \sin 2B) - \sin 2C \\
 &= 2 \sin \left( \frac{2A + 2B}{2} \right) \cos \left( \frac{2A - 2B}{2} \right) - \sin 2C \\
 &= 2 \sin (A + B) \cos (A - B) - \sin 2C \\
 &= 2 \sin C \cos (A - B) - 2 \sin C \cos C \\
 &\quad [\because \sin (A + B) = \sin (\pi - C) = \sin C] \\
 &= 2 \sin C [\cos (A - B) - \cos C] \\
 &= 2 \sin C [\cos (A - B) + \cos (A + B)] \\
 &\quad [\because \cos C = \cos \{\pi - (A + B)\} = -\cos (A + B)] \\
 &= 2 \sin C [\cos (A + B) + \cos (A - B)] \\
 &= 2 \sin C \left[ 2 \cos \left\{ \frac{(A + B) + (A - B)}{2} \right\} \cos \left\{ \frac{(A + B) - (A - B)}{2} \right\} \right] \\
 &= 2 \sin C [2 \cos A \cos B] = 4 \cos A \cos B \sin C = \text{R.H.S.}
 \end{aligned}$$

(ii) We have L.H.S. =  $\cos 2A + \cos 2B + \cos 2C$

$$\begin{aligned}
 &= (\cos 2A + \cos 2B) + \cos 2C \\
 &= 2 \cos \left( \frac{2A + 2B}{2} \right) \cos \left( \frac{2A - 2B}{2} \right) + \cos 2C \\
 &= 2 \cos (A + B) \cos (A - B) + \cos 2C \\
 &= 2 \cos (\pi - C) \cos (A - B) + \cos 2C \quad [\because A + B + C = \pi] \\
 &= -2 \cos C \cos (A - B) + 2 \cos^2 C - 1 \\
 &= -1 - 2 \cos C [\cos (A - B) - \cos C] \\
 &= -1 - 2 \cos C [\cos (A - B) - \cos \{\pi - (A + B)\}] \\
 &= -1 - 2 \cos C [\cos (A - B) + \cos (A + B)] \\
 &= -1 - 2 \cos C [\cos (A + B) + \cos (A - B)] \\
 &= -1 - 2 \cos C \left[ 2 \cos \left\{ \frac{(A + B) + (A - B)}{2} \right\} \right. \\
 &\quad \left. \cos \left\{ \frac{(A + B) - (A - B)}{2} \right\} \right] \\
 &= -1 - 2 \cos C [2 \cos A \cos B] \\
 &= -1 - 4 \cos A \cos B \cos C = \text{R.H.S.}
 \end{aligned}$$

**Illustration 2 :** If  $A + B + C = \pi$ , prove that

$$(i) \sin A + \sin B + \sin C = 4 \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}$$

$$(ii) \cos A + \cos B - \cos C = -1 + 4 \cos \frac{A}{2} \cos \frac{B}{2} \sin \frac{C}{2}.$$

**Solution :** (i) We have L.H.S. =  $\sin A + \sin B + \sin C$

$$= (\sin A + \sin B) + \sin C$$


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$$\begin{aligned}
&= 2 \sin \left( \frac{A+B}{2} \right) \cos \left( \frac{A-B}{2} \right) + \sin C \\
&= 2 \sin \left( \frac{\pi}{2} - \frac{C}{2} \right) \cos \left( \frac{A-B}{2} \right) + \sin C \\
&\quad \left[ \because A+B+C=\pi \Rightarrow \frac{A+B}{2} = \frac{\pi}{2} - \frac{C}{2} \right] \\
&= 2 \cos \frac{C}{2} \cos \left( \frac{A-B}{2} \right) + 2 \sin \frac{C}{2} \cos \frac{C}{2} \\
&= 2 \cos \frac{C}{2} \left[ \cos \left( \frac{A-B}{2} \right) + \sin \frac{C}{2} \right] \\
&= 2 \cos \frac{C}{2} \left[ \cos \left( \frac{A-B}{2} \right) + \sin \left\{ \frac{\pi}{2} - \left( \frac{A+B}{2} \right) \right\} \right] \\
&= 2 \cos \frac{C}{2} \left[ \cos \left( \frac{A-B}{2} \right) + \cos \left( \frac{A+B}{2} \right) \right] \\
&= 2 \cos \frac{C}{2} \left[ 2 \cos \left\{ \frac{\left( \frac{A-B}{2} \right) + \left( \frac{A+B}{2} \right)}{2} \right\} \right. \\
&\quad \left. \cos \left\{ \frac{\left( \frac{A+B}{2} \right) - \left( \frac{A-B}{2} \right)}{2} \right\} \right] \\
&= 2 \cos \frac{C}{2} \left[ 2 \cos \frac{A}{2} \cos \frac{B}{2} \right] = 4 \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2} = \text{R.H.S.}
\end{aligned}$$

(ii) We have L.H.S. =  $\cos A + \cos B - \cos C = (\cos A + \cos B) - \cos C$

$$\begin{aligned}
&= 2 \cos \left( \frac{A+B}{2} \right) \cos \left( \frac{A-B}{2} \right) - \cos C \\
&= 2 \cos \left( \frac{\pi}{2} - \frac{C}{2} \right) \cos \left( \frac{A-B}{2} \right) - \cos C \\
&= 2 \sin \frac{C}{2} \cos \left( \frac{A-B}{2} \right) - \left( 1 - 2 \sin^2 \frac{C}{2} \right) \\
&= -1 + 2 \sin \frac{C}{2} \left[ \cos \left( \frac{A-B}{2} \right) + \sin \frac{C}{2} \right] \\
&= -1 + 2 \sin \frac{C}{2} \left[ \cos \left( \frac{A-B}{2} \right) + \sin \left\{ \frac{\pi}{2} - \left( \frac{A+B}{2} \right) \right\} \right] \\
&= -1 + 2 \sin \frac{C}{2} \left[ \cos \left( \frac{A-B}{2} \right) + \cos \left( \frac{A+B}{2} \right) \right]
\end{aligned}$$


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$$\begin{aligned}
&= -1 + 2 \sin \frac{C}{2} \left[ \cos \left( \frac{A}{2} + \frac{B}{2} \right) + \cos \left( \frac{A}{2} - \frac{B}{2} \right) \right] \\
&= -1 + 2 \sin \frac{C}{2} \left[ 2 \cos \frac{A}{2} \cos \frac{B}{2} \right] \\
&\quad \left[ \because \cos C + \cos D = 2 \cos \left( \frac{C+D}{2} \right) \cos \left( \frac{C-D}{2} \right) \right] \\
&= -1 + 4 \cos \frac{A}{2} \cos \frac{B}{2} \sin \frac{C}{2} = \text{R.H.S.}
\end{aligned}$$

**Illustration 3 :** If  $A + B + C = \pi$ , prove that

$$\begin{aligned}
\cos \frac{A}{2} + \cos \frac{B}{2} + \cos \frac{C}{2} &= 4 \cos \left( \frac{\pi - A}{4} \right) \cos \left( \frac{\pi - B}{4} \right) \cos \left( \frac{\pi - C}{4} \right) \\
&= 4 \cos \left( \frac{B+C}{4} \right) \cos \left( \frac{C+A}{4} \right) \cos \left( \frac{A+B}{4} \right).
\end{aligned}$$

**Solution :** We have L.H.S. =  $\cos \frac{A}{2} + \cos \frac{B}{2} + \cos \frac{C}{2}$

$$\begin{aligned}
&= \left( \cos \frac{A}{2} + \cos \frac{B}{2} \right) + \cos \frac{C}{2} \\
&= 2 \cos \left( \frac{A+B}{4} \right) \cos \left( \frac{A-B}{4} \right) + \cos \frac{C}{2} \\
&= 2 \cos \left( \frac{A+B}{4} \right) \cos \left( \frac{A-B}{4} \right) + \sin \left( \frac{A+B}{2} \right) \\
&\quad \left[ \because \cos \frac{C}{2} = \cos \left\{ \frac{\pi}{2} - \left( \frac{A+B}{2} \right) \right\} = \sin \left( \frac{A+B}{2} \right) \right] \\
&= 2 \cos \left( \frac{A+B}{4} \right) \cos \left( \frac{A-B}{4} \right) + 2 \sin \left( \frac{A+B}{4} \right) \cos \left( \frac{A+B}{4} \right) \\
&= 2 \cos \left( \frac{A+B}{4} \right) \left[ \sin \left( \frac{A+B}{4} \right) + \cos \left( \frac{A-B}{4} \right) \right] \\
&= 2 \cos \left( \frac{A+B}{4} \right) \left[ \cos \left( \frac{\pi}{2} - \frac{A+B}{4} \right) + \cos \left( \frac{A-B}{4} \right) \right] \\
&= 2 \cos \left( \frac{A+B}{4} \right) \left[ 2 \cos \left( \frac{\pi - B}{4} \right) \cos \left( \frac{\pi - A}{4} \right) \right] \\
&\quad \left[ \because \cos C + \cos D = 2 \cos \left( \frac{C+D}{2} \right) \cos \left( \frac{C-D}{2} \right) \right] \\
&= 4 \cos \left( \frac{\pi - A}{4} \right) \cos \left( \frac{\pi - B}{4} \right) \cos \left( \frac{\pi - C}{4} \right) \\
&\quad [ \because A + B + C = \pi \Rightarrow A + B = \pi - C ]
\end{aligned}$$

$$= 4 \cos \left( \frac{B+C}{4} \right) \cos \left( \frac{C+A}{4} \right) \cos \left( \frac{A+B}{4} \right) = \text{R.H.S.}$$

$$[ \because \pi - A = B + C, \pi - B = C + A, \pi - C = A + B ]$$

## Type II. Identities Involving Squares of Sines and Cosines.

**Procedure :** Change the squares of sines or cosines into cosines of double the angles by using the formulae

$$\cos^2 \theta = \frac{1}{2} (1 + \cos 2\theta) \text{ and } \sin^2 \theta = \frac{1}{2} (1 - \cos 2\theta).$$

## Illustrative Examples

**Illustration 1 :** If  $A + B + C = \pi$ , prove that

$$(i) \sin^2 A + \sin^2 B + \sin^2 C = 2 (1 + \cos A \cos B \cos C)$$

$$(ii) \cos^2 A + \cos^2 B - \cos^2 C = 1 - 2 \sin A \sin B \cos C.$$

**Solution :** (i) We have L.H.S. =  $\sin^2 A + \sin^2 B + \sin^2 C$

$$\begin{aligned} &= \left( \frac{1 - \cos 2A}{2} \right) + \left( \frac{1 - \cos 2B}{2} \right) + \left( \frac{1 - \cos 2C}{2} \right) \\ &= \frac{3}{2} - \frac{1}{2} (\cos 2A + \cos 2B + \cos 2C) \\ &= \frac{3}{2} - \frac{1}{2} (-1 - 4 \cos A \cos B \cos C) \end{aligned}$$

[Proceeding as in Illustration 1, part (ii)]

$$= 2 + 2 \cos A \cos B \cos C = 2 (1 + \cos A \cos B \cos C) = \text{R.H.S.}$$

(ii) We have L.H.S. =  $\cos^2 A + \cos^2 B - \cos^2 C$

$$\begin{aligned} &= (1 - \sin^2 A) + \cos^2 B - \cos^2 C \\ &= 1 + (\cos^2 B - \sin^2 A) - \cos^2 C \\ &= 1 + \cos (B + A) \cos (B - A) - \cos^2 C \\ &= 1 - \cos C \cos (B - A) - \cos^2 C \end{aligned}$$

$$[ \because \cos (B + A) = \cos (\pi - C) = -\cos C ]$$

$$\begin{aligned} &= 1 - \cos C [\cos (B - A) + \cos C] \\ &= 1 - \cos C [\cos (B - A) - \cos (A + B)] \\ &= 1 - \cos C [2 \sin B \sin A] = 1 - 2 \sin A \sin B \cos C = \text{R.H.S.} \end{aligned}$$

**Illustration 2 :** If  $A + B + C = \pi$ , prove that

$$\sin^2 \frac{A}{2} + \sin^2 \frac{B}{2} - \sin^2 \frac{C}{2} = 1 - 2 \cos \frac{A}{2} \cos \frac{B}{2} \sin \frac{C}{2}.$$

**Solution :** We have L.H.S. =  $\sin^2 \frac{A}{2} + \sin^2 \frac{B}{2} - \sin^2 \frac{C}{2}$

$$\begin{aligned}
&= \left(1 - \cos^2 \frac{A}{2}\right) + \sin^2 \frac{B}{2} - \sin^2 \frac{C}{2} \\
&= 1 - \left(\cos^2 \frac{A}{2} - \sin^2 \frac{B}{2}\right) - \sin^2 \frac{C}{2} \\
&= 1 - \cos \left(\frac{A+B}{2}\right) \cos \left(\frac{A-B}{2}\right) - \sin^2 \frac{C}{2} \\
&= 1 - \cos \left(\frac{\pi}{2} - \frac{C}{2}\right) \cos \left(\frac{A-B}{2}\right) - \sin^2 \frac{C}{2} \\
&\quad \left[ \because \frac{A+B}{2} = \frac{\pi-C}{2} = \frac{\pi}{2} - \frac{C}{2} \right] \\
&= 1 - \sin \frac{C}{2} \cos \left(\frac{A-B}{2}\right) - \sin^2 \frac{C}{2} \\
&= 1 - \sin \frac{C}{2} \left[ \cos \left(\frac{A-B}{2}\right) + \sin \frac{C}{2} \right] \\
&= 1 - \sin \frac{C}{2} \left[ \cos \left(\frac{A-B}{2}\right) + \sin \left(\frac{\pi}{2} - \frac{A+B}{2}\right) \right] \\
&= 1 - \sin \frac{C}{2} \left[ \cos \left(\frac{A-B}{2}\right) + \cos \left(\frac{A+B}{2}\right) \right] \\
&= 1 - \sin \frac{C}{2} \left[ 2 \cos \frac{A}{2} \cos \frac{B}{2} \right] \\
&= 1 - 2 \cos \frac{A}{2} \cos \frac{B}{2} \sin \frac{C}{2} = \text{R.H.S.}
\end{aligned}$$

### Type III. Identities Involving Tangents and Cotangents.

**Procedure :** Express the sum of two angles in terms of the third by using the given relation. Take tangents of both sides and expand the L.H.S. Now, cross multiply and transpose.

In case the identity involves cotangents, divide both sides of the identity obtained in the above procedure by the product of tangents of all angles.

## Illustrative Examples

**Illustration 1 :** If  $A + B + C = \pi$ , prove that

(i)  $\tan A + \tan B + \tan C = \tan A \tan B \tan C$

[UPTU 2004, 07]

(ii)  $\cot B \cot C + \cot C \cot A + \cot A \cot B = 1$ .

**Solution :** (i) We have  $A + B + C = \pi \rightarrow A + B = \pi - C$

$$\Rightarrow \tan(A + B) = \tan(\pi - C) = -\tan C$$

$$\Rightarrow \frac{\tan A + \tan B}{1 - \tan A \tan B} = -\tan C$$

- $$\Rightarrow \tan A + \tan B = -\tan C (1 - \tan A \tan B)$$
- $$\Rightarrow \tan A + \tan B = -\tan C + \tan A \tan B \tan C$$
- $$\Rightarrow \tan A + \tan B + \tan C = \tan A \tan B \tan C.$$
- (ii) Dividing both sides of (i) by  $\tan A \tan B \tan C$ , we get

$$\frac{\tan A + \tan B + \tan C}{\tan A \tan B \tan C} = 1$$

- $$\Rightarrow \frac{1}{\tan B \tan C} + \frac{1}{\tan C \tan A} + \frac{1}{\tan A \tan B} = 1$$
- $$\Rightarrow \cot B \cot C + \cot C \cot A + \cot A \cot B = 1.$$

**Illustration 2 :** If  $A + B + C = \pi$ , prove that

$$\tan \frac{A}{2} \tan \frac{B}{2} + \tan \frac{B}{2} \tan \frac{C}{2} + \tan \frac{C}{2} \tan \frac{A}{2} = 1$$

**Solution :** We have  $A + B + C = \pi \Rightarrow A + B = \pi - C$

- $$\Rightarrow \frac{A}{2} + \frac{B}{2} = \frac{\pi}{2} - \frac{C}{2}$$
- $$\Rightarrow \tan \left( \frac{A}{2} + \frac{B}{2} \right) = \tan \left( \frac{\pi}{2} - \frac{C}{2} \right)$$
- $$\Rightarrow \frac{\tan \frac{A}{2} + \tan \frac{B}{2}}{1 - \tan \frac{A}{2} \tan \frac{B}{2}} = \cot \frac{C}{2} = \frac{1}{\tan \frac{C}{2}}$$
- $$\Rightarrow \tan \frac{C}{2} \left( \tan \frac{A}{2} + \tan \frac{B}{2} \right) = 1 - \tan \frac{A}{2} \tan \frac{B}{2}$$
- $$\Rightarrow \tan \frac{A}{2} \tan \frac{B}{2} + \tan \frac{B}{2} \tan \frac{C}{2} + \tan \frac{C}{2} \tan \frac{A}{2} = 1.$$

**Illustration 3 :** If  $A + B + C = \pi$ , prove that

$$\frac{\cot A + \cot B}{\tan A + \tan B} + \frac{\cot B + \cot C}{\tan B + \tan C} + \frac{\cot C + \cot A}{\tan C + \tan A} = 1.$$

**Solution :** We have  $A + B + C = \pi$

- $$\Rightarrow \tan (A + B + C) = \tan \pi = 0$$
- $$\Rightarrow \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A} = 0$$
- $$\Rightarrow \tan A + \tan B + \tan C - \tan A \tan B \tan C = 0$$
- $$\Rightarrow \tan A + \tan B + \tan C = \tan A \tan B \tan C$$
- $$\Rightarrow \frac{\tan A + \tan B + \tan C}{\tan A \tan B \tan C} = \frac{\tan A \tan B \tan C}{\tan A \tan B \tan C}$$
- $$\Rightarrow \frac{1}{\tan B \tan C} + \frac{1}{\tan C \tan A} + \frac{1}{\tan A \tan B} = 1$$
- $$\Rightarrow \cot B \cot C + \cot C \cot A + \cot A \cot B = 1$$

$$\Rightarrow \cot B \cot C \left( \frac{\tan B + \tan C}{\tan B + \tan C} \right) + \cot C \cot A \left( \frac{\tan C + \tan A}{\tan C + \tan A} \right) + \cot A \cot B \left( \frac{\tan A + \tan B}{\tan A + \tan B} \right) = 1$$

$$\Rightarrow \frac{\cot B + \cot C}{\tan B + \tan C} + \frac{\cot C + \cot A}{\tan C + \tan A} + \frac{\cot A + \cot B}{\tan A + \tan B} = 1.$$

**Illustration 4 :** Prove that

$$\tan (x - y) + \tan (y - z) + \tan (z - x) = \tan (x - y) \tan (y - z) \tan (z - x).$$

**Solution :** Let  $\alpha = x - y$ ,  $\beta = y - z$  and  $\gamma = z - x$ .

$$\text{Then } \alpha + \beta + \gamma = (x - y) + (y - z) + (z - x) = 0$$

$$\Rightarrow \tan (\alpha + \beta + \gamma) = \tan 0$$

$$\Rightarrow \frac{\tan \alpha + \tan \beta + \tan \gamma - \tan \alpha \tan \beta \tan \gamma}{1 - \tan \alpha \tan \beta - \tan \beta \tan \gamma - \tan \gamma \tan \alpha} = 0$$

$$\Rightarrow \tan \alpha + \tan \beta + \tan \gamma - \tan \alpha \tan \beta \tan \gamma = 0$$

$$\Rightarrow \tan \alpha + \tan \beta + \tan \gamma = \tan \alpha \tan \beta \tan \gamma$$

$$\Rightarrow \tan (x - y) + \tan (y - z) + \tan (z - x) = \tan (x - y) \tan (y - z) \tan (z - x).$$

**Type IV. Some Miscellaneous Conditional Trigonometric Identities.**

## Illustrative Examples

**Illustration 1 :** If  $\alpha + \beta - \gamma = \pi$ , prove that

$$\sin^2 \alpha + \sin^2 \beta - \sin^2 \gamma = 2 \sin \alpha \sin \beta \cos \gamma.$$

**Solution :** We have L.H.S. =  $\sin^2 \alpha + (\sin^2 \beta - \sin^2 \gamma)$

$$= \sin^2 \alpha + \sin (\beta + \gamma) \sin (\beta - \gamma)$$

$$= \sin^2 \alpha + \sin (\beta + \gamma) \sin (\pi - \alpha)$$

$$[ \because \alpha + \beta - \gamma = \pi \Rightarrow \beta - \gamma = \pi - \alpha ]$$

$$= \sin^2 \alpha + \sin (\beta + \gamma) \sin \alpha = \sin \alpha [\sin \alpha + \sin (\beta + \gamma)]$$

$$= \sin \alpha [\sin (\beta - \gamma) + \sin (\beta + \gamma)]$$

$$[ \because \sin \alpha = \sin \{\pi - (\beta - \gamma)\} = \sin (\beta - \gamma) ]$$

$$= \sin \alpha [2 \sin \beta \cos \gamma] = 2 \sin \alpha \sin \beta \cos \gamma = \text{R.H.S.}$$

**Illustration 2 :** If  $A + B + C = 2S$ , show that

$$\cos^2 S + \cos^2 (S - A) + \cos^2 (S - B) + \cos^2 (S - C) = 2 + 2 \cos A \cos B \cos C.$$

**Solution :** We have L.H.S.

$$= \cos^2 S + \cos^2 (S - A) + \cos^2 (S - B) + \cos^2 (S - C)$$

$$\begin{aligned}
&= \frac{1}{2} [2 \cos^2 S + 2 \cos^2 (S - A) + 2 \cos^2 (S - B) + 2 \cos^2 (S - C)] \\
&= \frac{1}{2} [(1 + \cos 2S) + \{1 + \cos (2S - 2A)\} + \{1 + \cos (2S - 2B)\} \\
&\quad + \{1 + \cos (2S - 2C)\}] \\
&= 2 + \frac{1}{2} [\{\cos 2S + \cos (2S - 2A)\} \\
&\quad + \{\cos (2S - 2B) + \cos (2S - 2C)\}] \\
&= 2 + \frac{1}{2} [2 \cos (2S - A) \cos A + 2 \cos (2S - B - C) \cos (B - C)] \\
&= 2 + [\cos (B + C) \cos A + \cos A \cos (B - C)] \\
&\quad [\because A + B + C = 2S \Rightarrow 2S - A = B + C \\
&\quad \text{and } 2S - B - C = A] \\
&= 2 + \cos A [\cos (B + C) + \cos (B - C)] \\
&= 2 + \cos A [2 \cos B \cos C] = 2 + 2 \cos A \cos B \cos C = \text{R.H.S.}
\end{aligned}$$

## Comprehensive Exercise 7

If  $A + B + C = \pi$ , prove that :

1.  $\sin 2A - \sin 2B + \sin 2C = 4 \cos A \sin B \cos C$ .
2.  $\sin (B + C - A) + \sin (C + A - B) + \sin (A + B - C) = 4 \sin A \sin B \sin C$ .
3.  $\cos 2A - \cos 2B - \cos 2C = -1 + 4 \cos A \sin B \sin C$ .
4.  $\sin A + \sin B - \sin C = 4 \sin \frac{A}{2} \sin \frac{B}{2} \cos \frac{C}{2}$ .
5.  $\frac{\sin 2A + \sin 2B + \sin 2C}{\sin A + \sin B + \sin C} = 8 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$ .
6.  $\sin^2 A + \sin^2 B - \sin^2 C = 2 \sin A \sin B \cos C$ .
7.  $\cos^2 A - \cos^2 B - \cos^2 C = 2 \cos A \sin B \sin C - 1$ .
8.  $\cos^2 A + \cos^2 B - \sin^2 C = -2 \cos A \cos B \cos C$ .
9.  $\cos^4 A + \cos^4 B + \cos^4 C$   
 $= \frac{1}{2} - 2 \cos A \cos B \cos C + \frac{1}{2} \cos 2A \cos 2B \cos 2C$ .
10. If  $A + B + C = \frac{\pi}{2}$ , prove that
  - (i)  $\sin 2A + \sin 2B + \sin 2C = 4 \cos A \cos B \cos C$
  - (ii)  $\sin^2 A + \sin^2 B + \sin^2 C = 1 - 2 \sin A \sin B \sin C$ .
  - (iii)  $\cos^2 A + \cos^2 B + \cos^2 C = 2 + 2 \sin A \sin B \sin C$ .

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11. If  $A + B + C = 2S$ , show that

$$\sin (S - A) + \sin (S - B) + \sin (S - C) - \sin C = 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}.$$



12. If  $\alpha + \beta = \gamma$ , prove that  $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1 + 2 \cos \alpha \cos \beta \cos \gamma$ .

## Comprehensive Exercise 8

### Fill in the Blanks

Fill in the blanks “.....”, so that the following statements are complete and correct.

- The value of  $\cos 150^\circ$  is .....
- If  $\sin \theta + \cos \theta = m$  and  $\sec \theta + \operatorname{cosec} \theta = n$ , then  $n(m+1)(m-1) = \dots$
- The value of  $\cos 15^\circ$  is .....
- The value of  $\frac{1 - \tan 15^\circ}{1 + \tan 15^\circ}$  is .....

### Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

- The value of  $\cos A + \sin (270^\circ + A) - \sin (270^\circ - A) + \cos (180^\circ + A)$  is  
 (a)  $-1$  (b)  $1$   
 (c)  $0$  (d) None of these
- If  $x = \sec \theta + \tan \theta$ ; then  $x + \frac{1}{x}$  is  
 (a)  $1$  (b)  $2 \sec \theta$   
 (c)  $2$  (d)  $2 \tan \theta$
- The value of  $\sin 50^\circ - \sin 70^\circ + \sin 10^\circ$  is equal to  
 (a)  $1$  (b)  $0$   
 (c)  $1/2$  (d)  $2$

### True or False

Write 'T' for true and 'F' for false statement.

- The value of  $\operatorname{cosec} (-750^\circ)$  is  $-2$ .
- The degree measure to the radian measure  $\frac{7\pi}{6}$  is  $210^\circ$ .
- The radian measure to the degree measure  $420^\circ$  is  $\frac{\pi}{3}$  radian.

## Answers

- |                  |         |                               |                         |        |
|------------------|---------|-------------------------------|-------------------------|--------|
| 1. $-\sqrt{3}/2$ | 2. $2m$ | 3. $(\sqrt{3} + 1)/2\sqrt{2}$ | 4. $\frac{1}{\sqrt{3}}$ | 5. (c) |
| 6. (b)           | 7. (b)  | 8. T                          | 9. T                    | 10. F  |



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## Chapter

# 5



# Logarithms

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## 5.1 Logarithm

**Definition.** The logarithm of any number,  $y > 0$  to a given base  $a > 0$  and  $a \neq 1$  is the index (power) to which the base must be raised in order to become equal to the given number.

Thus if  $a^x = y$ , then  $x$  is called the **logarithm** of  $y$  to the base  $a$ . The logarithm of  $y$  to the base  $a$  is usually written as  **$\log_a y$** . Thus the following equations express the same meaning  $a^x = y$  and  $x = \log_a y$ .

The first equation is called the **exponential form**, whereas the second equation is called the corresponding **logarithmic form**.

Thus, we have

$$1. \quad 4^3 = 64, \quad \therefore \quad \log_4 64 = 3$$

$$2. \quad 2^{-3} = \frac{1}{8} = 0.125, \quad \therefore \quad \log_2 0.125 = -3.$$

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## 5.2 Properties of Logarithms

(i) *The logarithm of 1 is 0 at every base.*

We have  $a^0 = 1$ , for all values of  $a$ ; therefore  **$\log 1 = 0$** , whatever the base may be.

(ii) *Logarithm of  $a$  to the base  $a$  itself is 1.*

We have  $a^1 = a$ .

$\therefore \log_a a = 1$ .

**Note.** Since  $1^1 = 1, 1^2 = 1, 1^3 = 1$ , so by definition of logarithm, we have

$$\log_1 1 = 1, \log_1 1 = 2, \log_1 1 = 3$$

$\Rightarrow 1 = 2 = 3 = \dots$ ; which is not possible.

$\therefore \log_1 1$  is meaningless.

(iii)  $a^{\log_a n} = n$ .

Let  $\log_a n = x$ .

Then  $a^x = n$  or  $a^{\log_a n} = n$ . [  $\because x = \log_a n$  ]

## 5.3 Fundamental Laws of Logarithms

(i) **Logarithm of a product.** *If  $m$  and  $n$  are any two numbers, then*

$$\log_a mn = \log_a m + \log_a n.$$

*i.e., the log of the product of two numbers is equal to the sum of their logs.*

**Proof.** Let  $\log_a m = x$  and  $\log_a n = y$ .

Then, by definition of logarithm,  $m = a^x$  and  $n = a^y$ .

Now,  $mn = a^x \cdot a^y = a^{x+y}$  [By law of indices]

$\therefore \log_a mn = x + y = \log_a m + \log_a n$ .

Similarly, if  $m_1, m_2, \dots, m_n$  are  $n$  numbers, then

$$\log_a (m_1 m_2 \dots m_n) = \log_a m_1 + \log_a m_2 + \dots + \log_a m_n.$$

(ii) **Logarithm of ratio or quotient of two numbers.** *If  $m$  and  $n$  are any two numbers, then*

$$\log_a \frac{m}{n} = \log_a m - \log_a n$$

*i.e., the log of the quotient of two numbers is equal to the difference of their logs.*

**Proof.** Let  $\log_a m = x$  and  $\log_a n = y$ .

Then, by definition of logarithm,  $m = a^x$  and  $n = a^y$ .

Now  $\frac{m}{n} = \frac{a^x}{a^y} = a^{x-y}$  [By law of indices]

$\therefore \log_a \frac{m}{n} = x - y = \log_a m - \log_a n$ .

(iii) **Logarithm of power of a number.** If  $m$  and  $n$  are any two numbers, then

$$\log_a m^n = n \log_a m$$

i.e., the logarithm of the power of a number is the product of the logarithm of the number and the index.

**Proof.** Let  $\log_a m = x$ .

Then, by definition of logarithm,  $m = a^x$ .

Now  $m^n = (a^x)^n = a^{nx}$  [By law of indices]

$$\therefore \log_a m^n = nx = n \log_a m.$$

## 5.4 Transformation of Logarithm from one Base to another

**Formula :**  $\log_b a = (\log_c a) \times \frac{1}{\log_c b} = \frac{\log_c a}{\log_c b}.$

**Proof.** Let  $y = \log_b a$ . Then  $b^y = a$ .

$$\therefore \log_c a = \log_c (b^y) = y \log_c b.$$

$$\therefore y = \log_c a \cdot \frac{1}{\log_c b}$$

$$\text{or} \quad \log_b a = \log_c a \cdot \frac{1}{\log_c b}. \quad \dots(1)$$

**Corollary.** Putting  $c$  for  $a$  in (1), we get

$$\log_b c = \log_c c \cdot \frac{1}{\log_c b} = 1 \cdot \frac{1}{\log_c b}$$

$$\text{or} \quad (\log_b c) (\log_c b) = 1.$$

$$\therefore (1) \text{ can also be written as } \log_b a = \log_c a \cdot \log_b c.$$

## 5.5 Common Logarithms

Logarithms to the base 10 are called common logarithms. We shall denote  $\log_{10} m$  as  $\log m$  only i.e., if no base is mentioned, the base is to be taken as 10.

### Illustrative Examples

**Illustration 1 :** Write the following in the logarithmic form :

$$(i) 10^{-1} = 0.1 \quad (ii) 9^{-5/2} = \frac{1}{243}.$$

**Solution :** (i) By the definition of logarithm,

$$10^{-1} = 0.1 \Rightarrow \log_{10} 0.1 = -1.$$

(ii) By the definition of logarithm,  $9^{-5/2} = \frac{1}{243} \Rightarrow \log_9 \frac{1}{243} = -\frac{5}{2}$ .

**Illustration 2 :** Write the following in the exponential form :

(i)  $\log_{10} 0.01 = -2$  (ii)  $\log_2 \frac{1}{64} = -6$ .

**Solution :** (i) By the definition of logarithm,  
 $\log_{10} 0.01 = -2 \Rightarrow 10^{-2} = 0.01$ .

(ii) By the definition of logarithm,  
 $\log_2 \frac{1}{64} = -6 \Rightarrow 2^{-6} = \frac{1}{64}$ .

**Illustration 3 :** Find the value of each of the following :

(i)  $\log_{\sqrt{2}} 8$  (ii)  $\log_9 3$  (iii)  $\log_{10} 0.000001$ .

**Solution :** (i) Let  $\log_{\sqrt{2}} 8 = x$ .

Then  $(\sqrt{2})^x = 8 \Rightarrow 2^{x/2} = 2^3 \Rightarrow \frac{x}{2} = 3 \Rightarrow x = 6$ .

$\therefore \log_{\sqrt{2}} 8 = 6$ .

(ii) Let  $\log_9 3 = x$ .

Then  $9^x = 3 \Rightarrow (3^2)^x = 3 \Rightarrow 3^{2x} = 3^1 \Rightarrow 2x = 1 \Rightarrow x = \frac{1}{2}$ .

$\therefore \log_9 3 = \frac{1}{2}$ .

(iii) Let  $\log_{10} 0.000001 = x$ .

Then  $10^x = 0.000001 = \frac{1}{10^6} = 10^{-6} \Rightarrow x = -6$ .

$\therefore \log_{10} 0.000001 = -6$ .

**Illustration 4 :** Find the value of  $x$  in each of the following :

(i)  $\log_{10} x = -3$  (ii)  $\log_{\sqrt{3}} x = 6$  (iii)  $\log_{16} x = 1.5$ .

**Solution :** (i) We have  $\log_{10} x = -3$

$\Rightarrow x = 10^{-3} = \frac{1}{10^3} = \frac{1}{1000} = 0.001$ .

(ii) We have  $\log_{\sqrt{3}} x = 6$

$\Rightarrow x = (\sqrt{3})^6 = (3^{1/2})^6 = 3^3 = 27$ .

(iii) We have  $\log_{16} x = 1.5$

$\Rightarrow x = (16)^{1.5} = (16)^{3/2} = (4^2)^{3/2} = 4^3 = 64$ .

**Illustration 5 :** If  $\log_2 m = x$  and  $\log_3 n = x$ , find  $108^x$  in terms of  $m$  and  $n$ .

**Solution :** We have  $\log_2 m = x \Rightarrow 2^x = m$

and  $\log_3 n = x \Rightarrow 3^x = n$ .

Now  $108^x = (4 \times 27)^x = (2^2 \times 3^3)^x = 2^{2x} \times 3^{3x}$   
 $= (2^x)^2 \times (3^x)^3 = m^2 n^3$ .

**Illustration 6 :** Simplify the following :

$$(i) \log_a 81 \qquad (ii) \log_a \frac{3^4 \times 2^{1/5}}{\sqrt{7}}.$$

**Solution :** (i) We have  $\log_a 81 = \log_a 3^4 = 4 \log_a 3$ .

$$\begin{aligned} (ii) \text{ We have } \log_a \frac{3^4 \times 2^{1/5}}{\sqrt{7}} &= \log_a (3^4 \times 2^{1/5}) - \log_a 7^{1/2} \\ &= \log_a 3^4 + \log_a 2^{1/5} - \frac{1}{2} \log_a 7 \\ &= 4 \log_a 3 + \frac{1}{5} \log_a 2 - \frac{1}{2} \log_a 7. \end{aligned}$$

**Illustration 7 :** Prove that  $\log \frac{15}{16} + \log \frac{2}{5} - \log \frac{3}{8} = 0$ .

$$\begin{aligned} \text{Solution :} \quad \text{We have } \log \frac{15}{16} + \log \frac{2}{5} - \log \frac{3}{8} \\ &= \log 15 - \log 16 + \log 2 - \log 5 - (\log 3 - \log 8) \\ &= \log (3 \times 5) - \log 2^4 + \log 2 - \log 5 - \log 3 + \log 8 \\ &= \log 3 + \log 5 - 4 \log 2 + \log 2 - \log 5 - \log 3 + \log 2^3 \\ &= -3 \log 2 + 3 \log 2 = 0. \end{aligned}$$

**Alternative Method.**

$$\begin{aligned} \text{We have } \log \frac{15}{16} + \log \frac{2}{5} - \log \frac{3}{8} \\ &= \log \left\{ \left( \frac{15}{16} \times \frac{2}{5} \right) \div \frac{3}{8} \right\} = \log \left( \frac{15}{16} \times \frac{2}{5} \times \frac{8}{3} \right) \end{aligned}$$

**Illustration 8 :** Express  $\log \sqrt{972}$  in terms of  $\log 2$  and  $\log 3$ .

$$\begin{aligned} \text{Solution :} \quad \text{We have } \log \sqrt{972} &= \log (972)^{1/2} \\ &= \log (9 \times 9 \times 3 \times 4)^{1/2} = \log (3^5 \times 2^2)^{1/2} \\ &= \log (3^{5/2} \times 2) = \log 3^{5/2} + \log 2 = \frac{5}{2} \log 3 + \log 2. \end{aligned}$$

**Illustration 9 :** Prove that

$$2 \log \frac{15}{8} - \log \frac{25}{162} + 3 \log \frac{4}{9} = \log 2.$$

$$\begin{aligned} \text{Solution :} \quad \text{We have } 2 \log \frac{15}{8} - \log \frac{25}{162} + 3 \log \frac{4}{9} \\ &= \log \left( \frac{15}{8} \right)^2 - \log \frac{25}{162} + \log \left( \frac{4}{9} \right)^3 \\ &= \log \left( \frac{15}{8} \right)^2 + \log \left( \frac{4}{9} \right)^3 - \log \frac{25}{162} = \log \left[ \left\{ \left( \frac{15}{8} \right)^2 \times \left( \frac{4}{9} \right)^3 \right\} \div \frac{25}{162} \right] \\ &= \log \left( \frac{225}{64} \times \frac{64}{729} \times \frac{162}{25} \right) = \log \frac{162}{81} = \log 2. \end{aligned}$$

**Illustration 10 :** Prove that

$$\log_b a \times \log_c b \times \log_a c = 1.$$

**Solution :** We have  $\log_b a \times \log_c b \times \log_a c$

$$= \frac{\log_m a}{\log_m b} \times \frac{\log_m b}{\log_m c} \times \frac{\log_m c}{\log_m a}, \text{ changing all logarithms to the base } m$$

$$= 1.$$

**Illustration 11 :** If  $x = \log_{2a} a$ ,  $y = \log_{3a} 2a$ ,  $z = \log_{4a} 3a$ , prove that  $xyz + 1 = 2yz$ .

**Solution :** We have  $x = \log_{2a} a \Rightarrow (2a)^x = a$ .

Taking log, we get  $\log (2a)^x = \log a$

$$\Rightarrow x \log 2a = \log a \Rightarrow x = \frac{\log a}{\log 2a}.$$

Similarly,  $y = \frac{\log 2a}{\log 3a}$  and  $z = \frac{\log 3a}{\log 4a}.$

$$\begin{aligned} \therefore x y z + 1 &= \frac{\log a}{\log 2a} \times \frac{\log 2a}{\log 3a} \times \frac{\log 3a}{\log 4a} + 1 \\ &= \frac{\log a}{\log 4a} + 1 = \frac{\log a + \log 4a}{\log 4a} \\ &= \frac{\log 4a^2}{\log 4a} = \frac{\log (2a)^2}{\log 4a} = \frac{2 \log 2a}{\log 4a} \\ &= 2 \frac{\log 2a}{\log 3a} \times \frac{\log 3a}{\log 4a} = 2 yz. \end{aligned}$$

**Illustration 12 :** Solve :  $\log (4x - 3) - \log (x + 1) = \log 3$ .

**Solution :** We have  $\log (4x - 3) - \log (x + 1) = \log 3$

$$\Rightarrow \log \left( \frac{4x - 3}{x + 1} \right) = \log 3 \Rightarrow \frac{4x - 3}{x + 1} = 3$$

$$\Rightarrow 4x - 3 = 3x + 3 \Rightarrow 4x - 3x = 3 + 3 \Rightarrow x = 6.$$

## Comprehensive Exercise 1

1. Write the following in the logarithmic form :

(i)  $4^3 = 64$

(ii)  $3^5 = 243$

(iii)  $5^{-2} = 0.04$

(iv)  $10^{-3} = 0.001$

(v)  $x^m = n$ .

2. Write the following in the exponential form :

(i)  $\log_2 64 = 6$

(ii)  $\log_2 0.125 = -3$ .

3. Find the value of each of the following :
- (i)  $\log_2 16$  (ii)  $\log_5 5^{-2}$   
(iii)  $\log_4 256$  (iv)  $\log_3 (3\sqrt{3})$ .
4. Find the value of  $\log_2 \sqrt{3} 144$ .
5. If  $\log_5 x = y$ , find the value of  $5^{3y}$  in terms of  $x$ .
6. If  $\log_{10} x = a$ , find the value of  $10^{2a-1}$  in terms of  $x$ .
7. If  $\log_2 x = a$  and  $\log_5 y = a$ , find the value of  $2000^{2a-1}$  in terms of  $x$  and  $y$ .
8. Simplify each of the following :
- (i)  $\log 24 - \log 8$  (ii)  $\log 16 - \log 6 + \log 3$   
(iii)  $\log \frac{a}{b} + \log \frac{b}{c} + \log \frac{c}{a}$  (iv)  $2 + \log 5$ .
9. Express  $\log \frac{a^2 b^{1/3}}{c^{2/3}}$  in terms of logarithms of  $a, b, c$ .
10. Express the following in terms of  $\log 2$  and  $\log 3$  :
- (i)  $\log 108$  (ii)  $\log 144$   
(iii)  $\log (\sqrt{54} \times \sqrt[3]{243})$  (iv)  $\log \frac{16}{27}$ .
11. Prove the following :
- (i)  $\log_{10} 25 = 2 (1 - \log_{10} 2)$ .  
(ii)  $\log (1 + 2 + 3) = \log 1 + \log 2 + \log 3$ .  
(iii)  $\log \frac{10}{13} + \log \frac{65}{68} - \log \frac{25}{34} = 0$ .  
(iv)  $7 \log \frac{16}{15} + 5 \log \frac{25}{24} + 3 \log \frac{81}{80} = \log 2$ .
12. Simplify the following :
- (i)  $\log \frac{65}{19} + \log \frac{133}{143} + \log \frac{11}{35}$ .  
(ii)  $\log \frac{75}{16} - 2 \log \frac{5}{9} + \log \frac{32}{243}$ .  
(iii)  $\log \frac{16}{15} + \log \frac{25}{24} + \log \frac{81}{80}$ .
13. Prove that :  $\log (m n p q) = \log m + \log n + \log p + \log q$ .
14. If  $\log (a + b) = \log a + \log b$ , show that  $b = \frac{a}{a-1}$ .
15. If  $\frac{1}{2} \log_{10} a + \log_{10} b = 1$ , then prove that  $ab^2 = 100$ .
16. If  $\log \left( \frac{a+b}{2} \right) = \frac{1}{2} (\log a + \log b)$ , show that  $a = b$ .
17. If  $\log \left( \frac{a+b}{3} \right) = \frac{1}{2} (\log a + \log b)$ , prove that  $a^2 + b^2 = 7ab$ .
-



## Answers

1. (i)  $\log_4 64 = 3$  (ii)  $\log_3 243 = 5$   
 (iii)  $\log_5 0.04 = -2$  (iv)  $\log_{10} 0.001 = -3$   
 (v)  $\log_x n = m$ .
2. (i)  $2^6 = 64$  (ii)  $2^{-3} = 0.125$ .
3. (i) 4 (ii)  $-2$  (iii) 4 (iv)  $\frac{3}{2}$ .
4. 4. 5.  $x^3$ .
6.  $\frac{x^2}{10}$ . 7.  $\frac{x^8 y^6}{2000}$ .
8. (i)  $\log 3$  (ii)  $\log 8$   
 (iii) 0 (iv)  $\log 500$ .
9.  $2 \log a + \frac{1}{3} \log b - \frac{2}{3} \log c$ .
10. (i)  $2 \log 2 + 3 \log 3$  (ii)  $4 \log 2 + 2 \log 3$   
 (iii)  $\frac{1}{2} \log 2 + \frac{19}{6} \log 3$  (iv)  $4 \log 2 - 3 \log 3$ .
12. (i) 0. (ii)  $\log 2$ . (iii)  $2 \log 3 - 3 \log 2$ .

### 5.6 Standard Form of Decimal

We can express any number in decimal form as the product of a number between 1 and 10 and an integral power of 10. Thus, any positive decimal number  $n$  can be written in this form as  $n = m \times 10^p$ ,

where  $p$  is an integer (positive, zero or negative) and  $1 \leq m < 10$ .

This is known as the **standard form** of the decimal number.

**Working Rule to convert a given decimal number into standard form.**

Move the decimal point to the left or to the right, as may be necessary, to bring one non-zero digit to the left of decimal point.

If the decimal is shifted  $p$  places to the left, multiply by  $10^p$ .

If the decimal is shifted  $p$  places to the right, multiply by  $10^{-p}$ .

If the decimal is not moved at all, multiply by  $10^0$ .

**Illustration :** Write each of the following in standard form :

- |                 |            |              |
|-----------------|------------|--------------|
| (i) 1.234       | (ii) 12.34 | (iii) 123.4  |
| (iv) 1234       | (v) 0.1234 | (vi) 0.01234 |
| (vii) 0.001234. |            |              |

**Solution :** (i)  $1.234 = 1.234 \times 10^0$

- (ii)  $12 \cdot 34 = 1 \cdot 234 \times 10^1$       (iii)  $123 \cdot 4 = 1 \cdot 234 \times 10^2$   
 (iv)  $1234 = 1 \cdot 234 \times 10^3$       (v)  $0 \cdot 1234 = 1 \cdot 234 \times 10^{-1}$   
 (vi)  $0 \cdot 01234 = 1 \cdot 234 \times 10^{-2}$       (vii)  $0 \cdot 001234 = 1 \cdot 234 \times 10^{-3}$ .

## 5.7 Characteristic and Mantissa of a Common Logarithm

Let the standard form of  $n$  be

$$n = m \times 10^p, \text{ where } 1 \leq m < 10 \text{ and } p \text{ is an integer.}$$

Then  $\log n = \log (m \times 10^p) = \log m + \log 10^p$

$$= \log m + p \log 10 = p + \log m. \quad [\because \log_{10} 10 = 1]$$

Since  $1 \leq m < 10$ , so  $0 \leq \log m < 1$  i.e.,  $\log m$  lies between 0 and 1.

Thus, the logarithm of a positive real number  $n$  has two parts :

(i) the integral part  $p$  known as the **characteristic**

and (ii) the fractional part  $\log m$  known as the **mantissa**.

Hence,  $\log n = \text{characteristic} + \text{mantissa}$ .

It follows that characteristic is an integer which may be positive, negative or zero and mantissa is always a positive number less than 1.

## 5.8 To find the Characteristic

Put the number  $n$  in the standard form as  $n = m \times 10^p$ . Then, the characteristic is  $p$ .

**Alternative Method (Computation of Characteristic by Inspection).**

|                 |              |                       |
|-----------------|--------------|-----------------------|
| (a) $10^0 = 1,$ | $\therefore$ | $\log_{10} 1 = 0,$    |
| $10^1 = 10,$    | $\therefore$ | $\log_{10} 10 = 1,$   |
| $10^2 = 100,$   | $\therefore$ | $\log_{10} 100 = 2,$  |
| $10^3 = 1000,$  | $\therefore$ | $\log_{10} 1000 = 3,$ |
| .....           |              | .....                 |
| .....           |              | .....                 |

It is clear from the above that logarithms of all the numbers lying between 1 and 10 will lie between 0 and 1 i.e., the integral part of the logarithm is zero.

Logarithms of all the numbers lying between 10 and 100 will lie between 1 and 2 i.e., the integral part of the logarithm is 1.

Similarly, the integral part of the logarithm of a number lying between 100 and 1000 is 2 ; the integral part of the logarithm of a number lying between 1000 and 10,000 is 3, etc.

Thus, we obtain the following law :

**Law 1 :** *The characteristic of the logarithm of a number greater than unity is less by one than the number of digits in its integral part, and is positive i.e.,  $> 0$ .*

**Example :** The characteristics of  $\log 312$ ,  $\log 64 \cdot 372$  and  $\log 2250$  are 2, 1 and 3 respectively.

$$\begin{aligned}
 \text{(b)} \quad 10^0 &= 1, & \therefore \log_{10} 1 &= 0, \\
 10^{-1} &= \frac{1}{10} = 0 \cdot 1, & \therefore \log_{10} 0 \cdot 1 &= -1, \\
 10^{-2} &= \frac{1}{100} = 0 \cdot 01, & \therefore \log_{10} 0 \cdot 01 &= -2, \\
 10^{-3} &= \frac{1}{1000} = 0 \cdot 001, & \therefore \log_{10} 0 \cdot 001 &= -3, \\
 \dots & \dots \dots \dots & \dots & \dots \dots \dots
 \end{aligned}$$

Thus, we observe that the logarithm of a number lying between  $0 \cdot 1$  and  $1$  lies between  $-1$  and  $0$  and hence it can be written as  $-1 + \text{decimal part}$ .

Thus, the characteristic of the logarithm of a number smaller than unity and having no zero after the decimal point, is  $-1$  and we shall write it as  $\bar{1}$  in place of  $-1$ .

The logarithm of a number lying between  $0 \cdot 01$  and  $0 \cdot 1$  lies between  $-2$  and  $-1$  and it can be written as  $-2 + \text{decimal part}$ . Thus, the characteristic of the logarithm of such number is  $-2$  and we shall write it as  $\bar{2}$ .

Similarly, the characteristic of the logarithm of a number lying between  $0 \cdot 001$  and  $0 \cdot 01$  is  $-3$  i.e.,  $\bar{3}$ .

Thus, we obtain the following law :

**Law 2 :** The characteristic of the logarithm of a positive decimal fraction less than one is negative and one more than the number of consecutive zeros immediately after the decimal point.

**Example :** The characteristics of  $\log 0 \cdot 532$ ,  $\log 0 \cdot 0417$  and  $\log 0 \cdot 000187$  are  $-1$ ,  $-2$  and  $-4$  respectively.

## 5.9 To find the Mantissa

**Example :** If  $\log 7513 = 3 \cdot 8758$ , then find  $\log 75 \cdot 13$ ,  $\log 751 \cdot 3$ ,  $\log 0 \cdot 7513$  and  $\log 0 \cdot 007513$ .

**Solution :** We have  $\log 7513 = 3 \cdot 8758$ .

$$\begin{aligned}
 \text{Now} \quad \log 75 \cdot 13 &= \log (7513 \times 10^{-2}) = \log 7513 + \log 10^{-2} \\
 &= 3 \cdot 8758 - 2 = 1 \cdot 8758 \\
 \log 751 \cdot 3 &= \log (7513 \times 10^{-1}) = \log 7513 + \log 10^{-1} \\
 &= 3 \cdot 8758 - 1 = 2 \cdot 8758 \\
 \log 0 \cdot 7513 &= (\log 7513 \times 10^{-4}) = \log 7513 + \log 10^{-4} \\
 &= 3 \cdot 8758 - 4 = \bar{1} \cdot 8758 \\
 \log 0 \cdot 007513 &= \log (7513 \times 10^{-6}) = \log 7513 + \log 10^{-6} \\
 &= 3 \cdot 8758 - 6 = \bar{3} \cdot 8758.
 \end{aligned}$$

For the above example, we conclude the following law :

**Law 3 :** *The mantissa of logarithms of all the numbers having the same digits in the same order is the same.*

The mantissa is found from the log tables. For this purpose the position of the decimal is immaterial. We consider the first four significant digits from the left side of the number.

## 5.10 Use of Log Tables

**(a) To find the logarithm of a number of one digit.** Suppose we have to find  $\log 7$ . See the log table in the end of the book. We find that the number in the row headed by 70 and under the column of 0 is 8451.

So mantissa of  $\log 70$  is  $0.8451$ . According to Law 3, the mantissa of  $\log 7$  will be the same. Again, since 7 is a number of one digit, so by Law 1 the characteristic of  $\log 7$  will be zero.

$$\therefore \log 7 = 0.8451.$$

**(b) To find the logarithm of a number of two digits.** Suppose we have to find  $\log 72$ . We find that in the log table the number in the row headed by 72 and under the column of 0 is 8573. So mantissa of  $\log 72$  is  $0.8573$ . Since 72 is a number of two digits, so the characteristic of its logarithm will be 1.

$$\therefore \log 72 = 1.8573.$$

**(c) To find the logarithm of a number of three digits.** Suppose we have to find  $\log 734$ . We find that the number in the row headed by 73 and under the column of 4 is 8657.

$$\therefore \log 734 = 2.8657.$$

Similarly,  $\log 73.4 = 1.8657$ ,  $\log .00734 = \bar{3}.8657$ , etc.

**(d) To find the logarithm of a number of four digits.** Suppose we have to find  $\log 71.56$ . As explained in (c), we have  $\log 71.5 = 1.8543$ . Now locate the number in the row headed by 71 and under the column headed by 6 of mean differences. We find the number 4 there.

$$\text{Thus } \log 71.50 = 1.8543$$

$$\text{difference of } 6 = 4$$

$$\therefore \log 71.56 = 1.8547.$$

[On adding]

## 5.11 Anti-logarithm

If  $\log n = m$ , then  $n = \text{antilog of } m$ .

For example,  $\log 71.56 = 1.8547$ .

$$\therefore \text{antilog } 1.8547 = 71.56.$$

**To find the Antilog of a Number from the table.**

First take just the decimal part (mantissa) of the number. In the first column of antilogarithm table the first two digits in the decimal part are given.

Thus, in the antilog table locate the number formed by the first two digits in the decimal part. In the row against this number, find the number in the column headed by the third digit and to this number add the mean difference under the

column headed by the fourth digit.

To find the position of the decimal point, we consider the characteristic. If the characteristic is non-negative and is equal to  $n$ , then insert the decimal point after  $(n + 1)$  digits from the left in the number seen from the antilog table. If  $n > 4$ , then put zeros on the right of the number seen from the antilog table so that the total number of digits becomes  $n + 1$ .

If the characteristic is negative and is equal to  $-n$  i.e.,  $\bar{n}$ , then write  $(n - 1)$  zeros between the decimal point and the number seen from the antilog table.

For example, suppose we have to find that number whose logarithm is  $1 \cdot 3156$ . See 31 in the first column of the antilogarithm table. In the row against 31, the number in the column headed by 5 is 2065. In the same row, the mean difference of 6 is 3. Adding this to 2065, we get 2068. Now since the characteristic of logarithm is 1, so in the required number the decimal point will be inserted after two digits. Thus, the number whose logarithm being  $1 \cdot 3156$  is  $20 \cdot 68$ .

If we have to find the antilog of  $\bar{2} \cdot 9875$ , then we shall use the table in this way :

$$\begin{array}{rcl} \text{antilog } \bar{2} \cdot 987 & = & 0 \cdot 09705 \\ \text{difference of } 5 & = & 11 \\ \therefore \text{antilog } \bar{2} \cdot 9875 & = & 0 \cdot 09716. \end{array} \quad [\text{On adding}]$$

## Illustrative Examples

**Illustration 1 :** How many digits are there in the integral part of a number whose logarithm is  $13 \cdot 7993$ ?

**Solution :** Since the characteristic of the logarithm of the number is 13, so the number of digits in the integral part of the number =  $13 + 1 = 14$ .

**Illustration 2 :** The logarithm of a number is  $\bar{5} \cdot 5798$ . How many consecutive zeros are in the number after the decimal point?

**Solution :** Since the characteristic of the logarithm of the number is  $\bar{5}$ , so the number of consecutive zeros immediately after the decimal point in the number =  $5 - 1 = 4$ .

**Illustration 3 :** Find the number of digits in  $2^{100}$  if

$$\log_{10} 2 = 0 \cdot 30103.$$

**Solution :** Let  $x = 2^{100}$ .

Then  $\log_{10} x = 100 \log_{10} 2 = 100 \times 0 \cdot 30103 = 30 \cdot 103$ .

Since the characteristic of  $\log x$  is 30, so the number of digits in  $2^{100} = 30 + 1 = 31$ .

**Illustration 4 :** Evaluate  $\frac{3 \cdot 274 \times 0 \cdot 0059}{14 \cdot 83 \times 0 \cdot 077}$ , correct to 4 decimal places.

**Solution :** Let  $x = \frac{3 \cdot 274 \times 0 \cdot 0059}{14 \cdot 83 \times 0 \cdot 077}$ .

Then  $\log x = \log 3 \cdot 274 + \log 0 \cdot 0059 - \log 14 \cdot 83 - \log 0 \cdot 077 \quad \dots(1)$

Using log tables, we have

$$\log 3 \cdot 27 = 0 \cdot 5145$$

$$\text{diff. of } 4 = 5$$

$$\log 14 \cdot 8 = 1 \cdot 1703$$

$$\text{diff. of } 3 = 9$$

$$\therefore \log 3 \cdot 274 = \overline{0 \cdot 5150}$$

$$\therefore \log 14 \cdot 83 = \overline{1 \cdot 1712}$$

Again,  $\log 0 \cdot 0059 = \overline{3 \cdot 7709}$  and  $\log 0 \cdot 077 = \overline{2 \cdot 8865}$

Putting these values in (1), we get

$$\begin{aligned}\log x &= 0 \cdot 5150 + \overline{3 \cdot 7709} - 1 \cdot 1712 - \overline{2 \cdot 8865} \\ &= \overline{2 \cdot 2859} - 0 \cdot 0577 = \overline{2 \cdot 2282}.\end{aligned}$$

Again from antilog table, we have

$$\text{antilog } \overline{2 \cdot 2282} = 0 \cdot 01691.$$

$$\therefore x = 0 \cdot 0169.$$

**Illustration 5 :** Find the cube root of 11, correct to four decimal places.

**Solution :** Let  $x = 11^{1/3}$ .

$$\begin{aligned}\text{Then } \log x &= \frac{1}{3} \log 11 = \frac{1}{3} \times 1 \cdot 0414, \text{ from table} \\ &= 0 \cdot 3471.\end{aligned}$$

Now  $\text{antilog } 0 \cdot 347 = 2 \cdot 223$ , from the table

$$\text{diff. of } 1 = \underline{\quad 1 \quad}$$

$$\therefore \text{antilog } 0 \cdot 3471 = \overline{2 \cdot 224}$$

$$\therefore (11)^{1/3} = 2 \cdot 224.$$

**Illustration 6 :** If  $\log 2 = 0 \cdot 3010$  and  $\log 3 = 0 \cdot 4771$ , then find the number of digits in  $6^{17}$ .

**Solution :** Let  $x = 6^{17}$ .

$$\begin{aligned}\text{Then } \log x &= 17 \log 6 = 17 \log (2 \times 3) \\ &= 17 (\log 2 + \log 3) = 17 (0 \cdot 3010 + 0 \cdot 4771) \\ &= 17 \times 0 \cdot 7781 = 13 \cdot 2277.\end{aligned}$$

Since the characteristic of  $\log x$  is 13, so the number of digits in  $6^{17} = 13 + 1 = 14$ .

**Illustration 7 :** If  $\log 7 = 0 \cdot 8451$ , then how many consecutive zeros are after the decimal

point in converting  $\left(\frac{1}{49}\right)^{11}$  into a decimal fraction ?

**Solution :** Let  $x = \left(\frac{1}{49}\right)^{11} = (7^{-2})^{11} = 7^{-22}$ .

$$\begin{aligned}\text{Then } \log x &= -22 \log 7 = -22 \times 0 \cdot 8451 \\ &= -18 \cdot 5922 = -18 - 1 + (1 - 0 \cdot 5922) \\ &= -19 + 0 \cdot 4078 = \overline{19 \cdot 4078}.\end{aligned}$$

Since the characteristic of  $\log x$  is  $-19$ , so number of consecutive zeros after the decimal point in  $x = 19 - 1 = 18$ .

**Illustration 8 :** Solve :  $2^x = 25$ , if  $\log 2 = 0 \cdot 30103$ .

**Solution :** We have  $2^x = 25$ .

$$\therefore \log 2^x = \log 25 = \log \frac{100}{4} = \log \frac{100}{2^2}$$

$$\text{or } x \log 2 = \log 100 - 2 \log 2$$

$$\text{or } x \times 0.30103 = 2 - 2 \times 0.30103 = 1.39794$$

$$\text{or } x = \frac{1.39794}{0.30103} = 4.64386.$$

## Comprehensive Exercise 2

- Find, only by inspection, the characteristic of logarithm of the following numbers :  
 (i) 19                      (ii) 4173                      (iii)  $375.57$   
 (iv)  $0.0205$               (v)  $0.000137$
- If  $\log_{10} 2 = 0.3010$ , then find the value of  $\log_{10} 200$ .
- If  $\log 6732 = 3.8281$ , then find logarithms of the following numbers :  
 (i)  $67.32$                       (ii)  $6732000$                       (iii)  $0.6732$   
 (iv)  $6.7320$ .
- If  $\log 8.057 = 0.9062$ , then find the numbers having following logarithms:  
 (i)  $4.9062$                       (ii)  $1.9062$                       (iii)  $\bar{1}.9062$ .
- How many digits are there in the integral part of a number whose logarithm is  $17.3212$ ?
- How many digits are there in  $2^{1000}$ , if  $\log_{10} 2 = 0.30103$ ?
- If  $\log 2 = .3010$ ,  $\log 3 = .4771$ , then find the number of digits in the following :  
 $2^{30}, 2^{64}, 3^{42}, (540)^{10}, (1296)^8$ .
- Solve the following equations :  
 (i)  $6^{x-2} = 10$                       (ii)  $2^x \times 3^{2-x} = 5$ .
- If  $\log 3 = 0.47712$ , then find the value of  $2 \log \frac{4}{5} + \log \frac{25}{24} + \log \frac{9}{2}$ .
- If  $\log 3 = 0.4771$ , find the number of digits in  $3^{15}$  and the position of the first significant digit in  $3^{-15}$ .
- Evaluate the following :  
 (i)  $\sqrt{215.9}$  if  $\log 2159 = 3.3342$   
 (ii)  $\sqrt[3]{0.0007652}$  if  $\log 76.52 = 1.8838$ .
- Evaluate by using log tables  $\frac{35.8 \times \sqrt{375.2 + 12.8}}{29 \times 0.0015}$ .

13. Using the log tables evaluate the following :

$$\frac{(53 \cdot 2)^2 (0 \cdot 126)^{1/3}}{1 - (0 \cdot 234)^2}.$$

## Answers

1. (i) 1 (ii) 3 (iii) 2 (iv)  $\bar{2}$   
 (v)  $\bar{4}$   
 2.  $2 \cdot 3010$ .  
 3. (i)  $1 \cdot 8281$  (ii)  $6 \cdot 8281$  (iii)  $\bar{1} \cdot 8281$   
 (iv)  $0 \cdot 8281$ .  
 4. (i)  $80570$  (ii)  $80 \cdot 57$  (iii)  $0 \cdot 8057$ .  
 5. 18. 6. 302.  
 7. 10, 20, 21, 28, 25. 8. (i)  $3 \cdot 2851$  (ii)  $1 \cdot 4496$ .  
 9.  $0 \cdot 47712$ . 10. 8, 8th.  
 11. (i)  $14 \cdot 51$  (ii)  $0 \cdot 09147$ . 12. 16210.  
 13. 1501.

## 5.12 Applications of Logarithms

**(I) Determination of Half Life.** The **half-life period** of a radioactive element is the time during which half the amount of a given sample of the element disintegrates. It is denoted by  $t_{1/2}$ .

$$t_{1/2} = \frac{\log 2}{\lambda} = \frac{0 \cdot 693}{\lambda},$$

where  $\lambda$  is the decay constant or disintegration constant.

### Illustrative Examples

**Illustration 1 :** Half life of radium is 1600 years. Calculate the disintegration constant.

**Solution :** We have  $t_{1/2} = \frac{\log 2}{\lambda}$

or  $\lambda = \frac{\log 2}{t_{1/2}}$

or  $\lambda = \frac{0 \cdot 693}{1600 \times 365 \times 24 \times 60 \times 60 \text{ s}} = 1 \cdot 3736 \times 10^{-11} \text{ s}^{-1}.$

**Illustration 2 :** Calculate the half life period of a nucleus if at the end of 4.2 days,  $N = 0 \cdot 798 N_0$ . Given  $N = N_0 e^{-\lambda t}$ .



**Solution :** Given  $N = N_0 e^{-\lambda t}$

$$\Rightarrow \lambda = -\frac{\log(N/N_0)}{t} = -\frac{\log(0.798)}{4.2 \text{ days}} = 0.054 \text{ day}^{-1}.$$

Half life,  $t_{1/2} = \frac{0.693}{\lambda} = \frac{0.693}{0.054 \text{ day}^{-1}} = 12.83 \text{ days}$

**Illustration 3 :** Find the age of death of an organism from the following data :  $t_{1/2}$  of  $^{14}_6\text{C} = 5600$  years, and ratio of amounts of  $^{14}_6\text{C}$  at the death and present time is  $10^8$ .

The formula used for carbon dating is

$$t = \frac{2.303}{\lambda} \log \frac{N_0}{N}, \text{ where } \lambda = \frac{0.693}{t_{1/2}} = \frac{0.693}{5600} \text{ year}^{-1}.$$

**Solution :** We have  $\frac{N_0}{N} = 10^8$

$$\begin{aligned} \Rightarrow t &= \frac{2.303}{0.693/5600} \log 10^8 = \frac{2.303 \times 5600}{0.693} \times 8 \log 10 \\ &= \frac{2.303 \times 5600}{0.693} \times 8 = 1.488 \times 10^5 \text{ years.} \end{aligned}$$

**Illustration 4 :** A piece of wood recorded in an excavation has 25.6% as much  $\text{C}^{14}$  (half-life = 5760 years) as ordinary wood today has. When did the piece get buried ?

**Solution :** Here  $N_0 = 100, N_t = 25.6, t_{1/2} = 5760$  years.

Also  $\lambda = \frac{0.693}{5760}$  and  $\lambda = \frac{2.303}{t} \log \frac{100}{25.6}.$

$$\therefore \frac{0.693}{5760} = \frac{2.303}{t} \log \frac{100}{25.6}$$

$$\Rightarrow t = \frac{2.303 \times 5760}{0.693} \log \frac{100}{25.6} = 11330 \text{ years.}$$

**Illustration 5 :** A piece of wood was found to have  $\text{C}^{14} / \text{C}^{12}$  ratio 0.7 times that in a living plant. Calculate the period when the plant died. Half-life of  $\text{C}^{14} = 5760$  years.

**Solution :** Here  $N_0 = 1, N_t = 0.70, t_{1/2} = 5760$  years.

We have,  $\lambda = \frac{2.303}{t} \log \frac{N_0}{N_t}$  and  $\lambda = \frac{0.693}{t_{1/2}} = \frac{0.693}{5760}.$

$$\therefore \frac{0.693}{5760} = \frac{2.303}{t} \log \frac{1}{0.70}$$

$$\Rightarrow t = \frac{2.303 \times 5760 \times 0.155}{0.693} = 2970 \text{ years.}$$

**Illustration 6 :** One microgram of  $\text{Na}-24$  is injected into the blood of a patient. How long will it take the radioactivity to fall to 10% of initial value ?

( $t_{1/2}$  for  $\text{Na}-24$  is 14.8 hours).

**Solution :** We know that  $t_{1/2} = \frac{0.693}{\lambda}$  and  $t_{1/2} = 14.8$  hours (given).

$$\therefore \lambda = \frac{0.693}{14.8}.$$

Also  $t = \frac{1}{\lambda} \times 2.303 \log \frac{N_0}{N}$  and  $\frac{N_0}{N} = \frac{100}{10} = 10.$

$$\therefore t = \frac{14.8}{0.693} \times 2.303 \log 10 = \frac{14.8}{0.693} \times 2.303 \times 1$$

$$= 49.2 \text{ hrs} = 2.05 \text{ days}.$$

## (II) Computation of Equilibrium Constant.

**Illustration 7 :** Calculate  $K_p$  for the reaction



$\Delta G^\circ$  for the reaction is  $163.43 \text{ kJ mol}^{-1}$ . Given  $\Delta G^\circ = -RT \log K_p$ .

**Solution :**  $\Delta G^\circ = -RT \log K_p$

$$\Rightarrow \log K_p = -\frac{\Delta G^\circ}{RT} = -\frac{163.43 \times 10^3 \text{ J mol}^{-1}}{(8.314 \text{ JK}^{-1} \text{ mol}^{-1})(298 \text{ K})}$$

$$\therefore K_p = 2 \times 10^{-29}.$$

## (III) Computation of Avogadro's Number.

**Illustration 8 :** Using Stirling's approximation,

$$\log N_A! \approx N_A \log N_A - N_A$$

calculate  $\log N_A!$ , where  $N_A$  is Avogadro's Number. ( $N_A = 6.022 \times 10^{23}$ ).

**Solution :** By Stirling's approximation

$$\begin{aligned} \log N_A! &\approx N_A \log N_A - N_A \\ &= (6.022 \times 10^{23}) \log (6.022 \times 10^{23}) - (6.022 \times 10^{23}) \\ &= 323.77 \times 10^{23} = 3.2377 \times 10^{25}. \end{aligned}$$

## (IV) Chemical Kinetics Problems.

**Illustration 9 :** In a first order reaction, it takes the reactant 40.5 minutes to be 25% decomposed. Calculate the rate constant of the reaction.

Given  $k_1 = \frac{1}{t} \log \frac{a}{a-x}.$

**Solution :** The reactant gets 25% decomposed in 40.5 min, thus the remaining reactant is 75%.

Now at  $t = 40.5 \text{ min}$ , we have  $a - x = 0.75 a$ .

$$\therefore k_1 = \frac{1}{t} \log \frac{a}{a-x} = \frac{1}{40.5 \text{ min}} \log \frac{a}{0.75 a} = 7.11 \times 10^{-3} \text{ min}^{-1}.$$

**Illustration 10 :** Show that for a first order reaction,  $t = \frac{1}{k_1} \log \frac{a}{a-x}$ , the time required

for 99.9% completion of the reaction is 10 times that required for 50% completion.

**Solution :** We have  $t = \frac{1}{k_1} \log \frac{a}{a-x}$

$$\Rightarrow \frac{t(99.9\%)}{t(50\%)} = \frac{\frac{1}{k_1} \log \frac{100}{100 - 99.9}}{\frac{1}{k_1} \log \frac{100}{100 - 50}} = \frac{3.00}{0.3010} \approx 10.$$

**Illustration 11 :** Decomposition of diazobenzene chloride was followed at constant temperature by measuring the volume of nitrogen evolved at suitable intervals. The readings thus, obtained are given below :

|                       |   |    |    |    |          |
|-----------------------|---|----|----|----|----------|
| Time from start (min) | 0 | 20 | 50 | 70 | $\infty$ |
| Volume of $N_2$ (ml)  | 0 | 10 | 25 | 33 | 162      |

Calculate the order of the reaction.

**Solution :** Here  $a = V_\infty = 162$  ml and  $a - x = V_\infty - V_t$ .

For the first order reaction

$$k_1 = \frac{2.303}{t} \log \frac{a}{a-x} = \frac{2.303}{t} \log \frac{V_\infty}{V_\infty - V_t}.$$

|     |                  |                                                                                  |
|-----|------------------|----------------------------------------------------------------------------------|
| $t$ | $V_\infty - V_t$ | $k_1 = \frac{2.303}{t} \log \frac{V_\infty}{V_\infty - V_t}$                     |
| 20  | $162 - 10 = 152$ | $k_1 = \frac{2.303}{20} \log \frac{162}{152} = 3.22 \times 10^{-3} / \text{min}$ |
| 50  | $162 - 25 = 137$ | $k_1 = \frac{2.303}{50} \log \frac{162}{137} = 3.36 \times 10^{-3} / \text{min}$ |
| 70  | $162 - 33 = 129$ | $k_1 = \frac{2.303}{70} \log \frac{162}{129} = 3.26 \times 10^{-3} / \text{min}$ |

Thus, constant value of  $k_1$  shows that the reaction is of the first order.

**Illustration 12 :** Half change time for the first order decomposition.



is 30 minutes. Calculate :

- (i) What fraction of azo-methane will be decomposed in 1.5 hours ?  
 (ii) How long will it take to be 60% completed ?

**Solution :** (i) The first order reaction equation is  $k_1 = \frac{2.303}{t} \log \frac{a}{a-x}$ .

For half change,  $x = a/2$ , then

$$k = \frac{2.303}{t_{1/2}} \log \frac{a}{a-(a/2)} = \frac{2.303}{30} \log 2 = \frac{0.693}{30} = 0.0231 / \text{min}.$$

Let  $y\%$  of azo-methane decompose in 1.5 hours. Then  $x = \frac{y}{100} a$ .

$$\therefore k_1 = \frac{2.303}{t} \log \frac{a}{a - \frac{ya}{100}} = \frac{2.303}{t} \log \frac{100}{100 - y}$$

$$\text{or } 0.0231 = \frac{2.303}{90} \log \frac{100}{100 - y} \quad \text{or } y = 87.4.$$

So, the fraction decomposed is  $0.874$ .

(ii) For 60% decomposition,  $x = \frac{60}{100} a$ .

Here 
$$k_1 = \frac{2.303}{90} \log \frac{a}{a - \frac{60}{100} a}$$

or 
$$0.0231 = \frac{2.303}{t} \log \frac{100}{40} \quad [\because k_1 = 0.0231]$$

$\therefore t = 39.7$  minutes.

## Comprehensive Exercise 3

- The half life of  $^{238}_{92}\text{U}$  is  $4.51 \times 10^9$  years. What percentage of  $^{238}_{92}\text{U}$  that existed  $10^{10}$  years ago still survives ?
- Calculate the age in the following cases :
  - A piece of hair in which C-14 activity is 60% of the activity found today ( $t_{1/2} = 5760$  years).
  - A vegetarian beverage whose tritium content is only 5% of level in living plants ( $t_{1/2} = 12.3$  years).
  - A uranium rock has uranium-238 and lead-206 in mass ratio of 1.50 to 1.00 ( $\lambda = 1.52 \times 10^{-10} \text{ year}^{-1}$ ).
- 100 mg of Co – 60 is stocked in a laboratory as a  $\gamma$ -ray source. It has a half-life of 5.26 years. Calculate the percentage decrease in its activity after 1 year.
- Following observations were made during hydrolysis of methyl acetate at  $25^\circ\text{C}$ , using 0.05 N HCl as catalysts :
 

|                             |      |       |       |       |          |
|-----------------------------|------|-------|-------|-------|----------|
| $t$ (sec.) :                | 0    | 75    | 119   | 183   | $\infty$ |
| Volume of the alkali used : | 9.62 | 12.10 | 13.10 | 14.75 | 21.05    |

Show that the reaction is of the first order.
- The decomposition of  $\text{H}_2\text{O}_2$  was studied by titrating it at different intervals of time with potassium permanganate. Calculate the velocity constant from the following data, if the reaction is of the first order.

|                         |      |      |      |     |
|-------------------------|------|------|------|-----|
| $t$ (min.) :            | 0    | 10   | 30   | 40  |
| $\text{KMnO}_4$ (ml.) : | 25.0 | 20.0 | 12.5 | 9.6 |

## Answers

- 21.5%.
- (i) 4245 years (ii) 53.17 years (iii)  $3.76 \times 10^9$  years.
- 12.36%. 5. 0.0231 per min.

## Comprehensive Exercise 4

### Fill in the Blanks

Fill in the blanks “.....”, so that the following statements are complete and correct.

- The value of  $\log_3 27$  is .....
- The logarithmic form of  $2^5 = 32$  is .....
- If  $\log(k^2 - 4k + 5) = 0$ , then the value of  $k$  is .....
- If  $\log_m x + \log_m y + \log_m z = 0$ , then  $xyz = \dots\dots\dots$

### Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

- The value of  $\log_b a \log_c b \log_a c$  is  
 (a) 0 (b)  $\log(abc)$  (c) 1 (d) 10
- If  $x = \log_6 216$ , then the value of  $x$  is  
 (a) 3 (b) 2 (c) 1 (d) 0
- If  $x = 0.0768$ , then  $\log x$  is  
 (a) 0.8854 (b)  $\bar{2}.8854$  (c)  $\bar{1}.8854$  (d)  $\bar{3}.8854$
- The characteristic of  $\log 37295$  is  
 (a) 1 (b) 2 (c) 3 (d) 4

### True or False

Write ‘T’ for true and ‘F’ for false statement.

- The logarithm of 1 is 0 at every base.
- The logarithm of  $a$  to the base  $a$  itself is 0.
- $\log_b a = \frac{\log_c a}{\log_c b}$ .
- The mantissa of 21.34 is 0.3292.

## Answers

- |        |                      |        |        |
|--------|----------------------|--------|--------|
| 1. 3   | 2. $\log_2 32 = 5$ . | 3. 2   | 4. 1   |
| 5. (c) | 6. (a)               | 7. (b) | 8. (d) |
| 9. T   | 10. F                | 11. T  | 12. T  |



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## Chapter

# 6



# System Of Coordinates

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## 6.1 Introduction

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The French Mathematician and Philosopher Descartes (1596–1650) was the first, who established relation between Geometry and Algebra. He invented **Analytical Geometry** or **Coordinate Geometry**. In this branch of Geometry, the first important relation is established between the points in a plane and the ordered pairs of real numbers.

## 6.2 Rectangular Cartesian Coordinates

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Let  $X'OX$  and  $Y'OY$  be two mutually perpendicular lines, intersecting at  $O$ . We fix up a convenient unit of length and starting from  $O$ , mark off distances on  $x$ -axis as well as  $y$ -axis.

The two lines  $X'OX$ ,  $Y'OY$  are called **axes of reference**.  $X'OX$  is called the  $x$ -axis and  $Y'OY$  is called the  $y$ -axis. The point of intersection of axes,  $O$ , is called the **origin**.

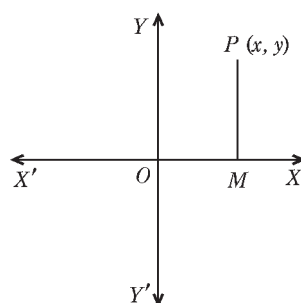
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Let  $P$  be any point in the plane of lines. Draw  $PM$  perpendicular from  $P$  on  $OX$ . Then the lengths  $OM$  and  $MP$  are called the **rectangular cartesian coordinates** or in short only coordinates and generally denoted by  $x$  and  $y$  respectively.

The length  $OM$  or  $x$  is called the  **$x$ -coordinate** or **abscissa** of the point  $P$  and the length  $MP$  or  $y$  is called the  **$y$ -coordinate** or **ordinate** of the point  $P$ . The point  $P$  is represented by  $(x, y)$ .

The lines  $X'OX$  and  $Y'OY$  are infinite. They divide the plane into 4 regions, called the **quadrants**.

The regions  $XOY$ ,  $YOX'$ ,  $X'OY'$  and  $Y'OX$  are called the first, the second, the third and the fourth quadrants respectively.

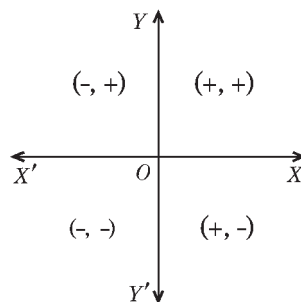


### 6.3 Sign Convention

The distances measured in the direction  $OX$  are taken positive while distances measured in the opposite direction  $OX'$  are taken negative.

Similarly, distances measured in the direction  $OY$  are taken positive and in the opposite direction  $OY'$  are taken negative.

From the above sign convention, it is clear that if a point  $(x, y)$  lies in the first quadrant then both  $x$  and  $y$  are positive; if a point lies in the second quadrant then  $x$  is negative and  $y$  is positive; if a point lies in the third quadrant then both  $x$  and  $y$  are negative and if the point lies in the fourth quadrant then  $x$  is positive and  $y$  is negative.



### 6.4 Distance between two Points

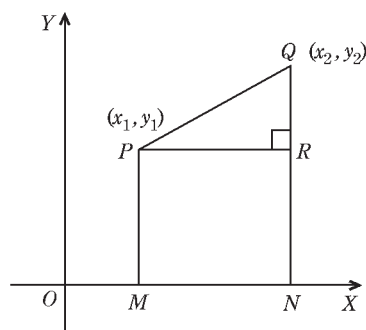
Let  $P(x_1, y_1)$  and  $Q(x_2, y_2)$  be two given points and let  $PQ = d$ .

Draw  $PM$  and  $QN$  perpendiculars from  $P$  and  $Q$  respectively on the  $x$ -axis. Draw  $PR$  perpendicular from  $P$  on  $QN$ . Then

$$OM = x_1, MP = y_1, ON = x_2, NQ = y_2.$$

$$\therefore PR = MN = ON - OM = x_2 - x_1,$$

$$\text{and } RQ = NQ - NR = NQ - MP = y_2 - y_1.$$



Now, in  $\Delta PRQ$ ,  $\angle R$  is right angle, so

$$PQ^2 = PR^2 + RQ^2,$$

$$\text{i.e.,} \quad d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

$$\therefore d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(\text{difference of } x\text{-coordinates})^2 + (\text{difference of } y\text{-coordinates})^2}.$$

**Corollary.** The distance of the point  $(x, y)$  from the origin is  $\sqrt{x^2 + y^2}$ .

## 6.5 Properties of some Geometrical Figures

(i) **Equilateral triangle.** All sides are equal and each angle is  $60^\circ$ .

(ii) **Isosceles triangle.** Two sides (or two angles) are equal.

(iii) **Parallelogram.** Opposite sides are parallel, or a pair of opposite sides are parallel and equal, or the diagonals bisect each other.

(iv) **Rectangle.** Opposite sides are equal and each angle is a right angle. The diagonals are equal.

(v) **Rhombus.** All the four sides are equal. The diagonals bisect each other at right angles and are in general not equal.

(vi) **Square.** All the four sides are equal and each angle is a right angle. The diagonals are also equal.

(vii) **Circum-centre of a triangle.** It is the centre of the circle which passes through the vertices of the triangle and so it is equidistant from the three vertices of the triangle. It is the point of intersection of the right bisectors of the sides of the triangle (i.e., the lines through the mid-point of each side and perpendicular to it).

## Illustrative Examples

**Illustration 1.** Find the distance between the points

(i)  $A(3, -1)$  and  $B(-2, 4)$  (ii)

$A(a \cos \alpha, a \sin \alpha)$  and  $B(a \cos \beta, a \sin \beta)$ .

**Solution :** (i) The distance between the given points =  $AB$

$$= \sqrt{\{-2 - 3\}^2 + \{4 - (-1)\}^2} = \sqrt{25 + 25} = 5\sqrt{2}.$$

(ii) The distance between the given points =  $AB$

$$\begin{aligned} &= \sqrt{(a \cos \beta - a \cos \alpha)^2 + (a \sin \beta - a \sin \alpha)^2} \\ &= \sqrt{a^2 (\cos \beta - \cos \alpha)^2 + a^2 (\sin \beta - \sin \alpha)^2} \end{aligned}$$



$$\begin{aligned}
 &= a \sqrt{(\cos \beta - \cos \alpha)^2 + (\sin \beta - \sin \alpha)^2} \\
 &= a \sqrt{(\cos^2 \beta + \sin^2 \beta) + (\cos^2 \alpha + \sin^2 \alpha) - 2(\cos \alpha \cos \beta + \sin \alpha \sin \beta)} \\
 &= a \sqrt{2 - 2 \cos (\alpha - \beta)} = a \sqrt{2 \{1 - \cos (\alpha - \beta)\}} \\
 &= a \sqrt{4 \sin^2 \left( \frac{\alpha - \beta}{2} \right)} = 2 a \sin \left( \frac{\alpha - \beta}{2} \right).
 \end{aligned}$$

**Illustration 2 :** Find the value of  $a$  if the distance between the points  $(3, a)$  and  $(4, 1)$  is  $\sqrt{10}$ .

[UPTU 2003]

**Solution :** Here,  $\sqrt{(4 - 3)^2 + (1 - a)^2} = \sqrt{10}$

$$\text{or} \quad 1 + (1 - a)^2 = 10 \quad \text{or} \quad (1 - a)^2 = 9 \quad \text{or} \quad 1 - a = \pm 3.$$

$$\therefore \quad a = 1 \pm 3 = -2, 4.$$

**Illustration 3 :** Show that the points  $A(2, -2)$ ,  $B(8, 4)$ ,  $C(5, 7)$ ,  $D(-1, 1)$  are the vertices of a rectangle.

[UPTU 2007]

**Solution :** Here,

$$AB = \sqrt{(8 - 2)^2 + (4 + 2)^2} = \sqrt{6^2 + 6^2} = \sqrt{72},$$

$$BC = \sqrt{(5 - 8)^2 + (7 - 4)^2} = \sqrt{(-3)^2 + 3^2} = \sqrt{18},$$

$$CD = \sqrt{(-1 - 5)^2 + (1 - 7)^2} = \sqrt{(-6)^2 + (-6)^2} = \sqrt{72},$$

$$DA = \sqrt{(2 + 1)^2 + (-2 - 1)^2} = \sqrt{3^2 + (-3)^2} = \sqrt{18},$$

$$AC = \sqrt{(5 - 2)^2 + (7 + 2)^2} = \sqrt{3^2 + 9^2} = \sqrt{90},$$

$$\text{and} \quad BD = \sqrt{(-1 - 8)^2 + (1 - 4)^2} = \sqrt{(-9)^2 + (-3)^2} = \sqrt{90}.$$

$$\therefore \quad AB = CD, BC = DA \text{ and } AC = BD.$$

Thus, the opposite sides of the quadrilateral  $ABCD$  are equal and its diagonals are also equal. Hence, it is a rectangle.

**Illustration 4 :** Show that the points  $(12, 8)$ ,  $(-2, 6)$  and  $(6, 0)$  are the vertices of a right angled triangle.

**Solution :** Let  $A$ ,  $B$  and  $C$  be the points  $(12, 8)$ ,  $(-2, 6)$  and  $(6, 0)$  respectively.

$$\text{Now} \quad AB^2 = (12 + 2)^2 + (8 - 6)^2 = 14^2 + 2^2 = 200,$$

$$BC^2 = (-2 - 6)^2 + (6 - 0)^2 = (-8)^2 + 6^2 = 100$$

$$\text{and} \quad CA^2 = (12 - 6)^2 + (8 - 0)^2 = 6^2 + 8^2 = 100.$$

$$\text{Thus,} \quad BC^2 + CA^2 = AB^2,$$

i.e., the sum of the squares of two sides is equal to the square of the third side. This shows that  $ABC$  is a right angled triangle.

Hence, the given points are the vertices of a right angled triangle.

**Illustration 5 :** The vertices  $A, B, C$  of a triangle are  $(2, 1), (5, 2)$  and  $(3, 4)$  respectively. Find the coordinates of the circum-centre and also the radius of the circum-circle.

**Solution :** Let  $(x, y)$  be the coordinates of the centre  $P$  of the circum-circle of the triangle  $ABC$ . Then  $PA^2 = PB^2$

$$\begin{aligned} \Rightarrow (x-2)^2 + (y-1)^2 &= (x-5)^2 + (y-2)^2 \\ \Rightarrow -4x + 4 - 2y + 1 &= -10x + 25 - 4y + 4 \\ \Rightarrow 6x + 2y &= 24 \\ \Rightarrow 3x + y &= 12. \end{aligned} \quad \dots(1)$$

Also,  $PB^2 = PC^2$

$$\begin{aligned} \Rightarrow (x-5)^2 + (y-2)^2 &= (x-3)^2 + (y-4)^2 \\ \Rightarrow -10x + 25 - 4y + 4 &= -6x + 9 - 8y + 16 \\ \Rightarrow -4x + 4y + 4 &= 0 \\ \Rightarrow x - y &= 1. \end{aligned} \quad \dots(2)$$

Solving the equations (1) and (2), we get the centre  $P$  as the point  $\left(\frac{13}{4}, \frac{9}{4}\right)$ .

Now if  $r$  be the radius of the circum-circle, then

$$\begin{aligned} r^2 = PA^2 &= \left(\frac{13}{4} - 2\right)^2 + \left(\frac{9}{4} - 1\right)^2 = \frac{25}{16} + \frac{25}{16} = \frac{50}{16} \\ \therefore r &= \frac{5\sqrt{2}}{4}. \end{aligned}$$

**Illustration 6 :** The vertices of a triangle  $ABC$  are  $A(0, 0), B(2, -1)$  and  $C(9, 2)$ . Evaluate  $\cos B$ .

**Solution :** We know that

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac} = \frac{BC^2 + AB^2 - AC^2}{2BC \cdot AB}. \quad \dots(1)$$

Now  $BC = \sqrt{(9-2)^2 + (2+1)^2} = \sqrt{49+9} = \sqrt{58},$

$$AC = \sqrt{(9-0)^2 + (2-0)^2} = \sqrt{81+4} = \sqrt{85},$$

and  $AB = \sqrt{(2-0)^2 + (-1-0)^2} = \sqrt{4+1} = \sqrt{5}.$

Substituting these values in (1), we get

$$\cos B = \frac{58 + 5 - 85}{2(\sqrt{58})(\sqrt{5})} = \frac{-11}{\sqrt{290}}.$$

## Comprehensive Exercise 1

- respectively and  $S$  is the point  $(a, 0)$ . Show that  $\frac{1}{SP} + \frac{1}{SQ}$  is independent of  $t$ .

# Answers

- $$\text{(iv) } a(t_1 - t_2) \sqrt{(t_1 + t_2)^2 + 4}.$$

2.  $3 \pm \sqrt{60}$ .  
 11.  $(5, 2); 5$ .  
 4.  $x = 0, y = \pm \sqrt{7}$ .  
 12.  $\frac{63}{65}$ .

## 6.6 Section Formulae.

(i) The coordinates of a point which divides the line joining the two given points  $(x_1, y_1)$  and  $(x_2, y_2)$  internally in the ratio  $m_1 : m_2$  is

$$x = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \quad y = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}.$$

**Corollary.** (i) The coordinates of the mid-point of the line joining the points  $(x_1, y_1)$  and  $(x_2, y_2)$  are  $\left( \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$ .

(ii) The coordinates of a point which divides the line joining the two given points  $(x_1, y_1)$  and  $(x_2, y_2)$  externally in the ratio  $p : q$  is

$$x = \frac{p x_2 - q x_1}{p - q}, \quad y = \frac{p y_2 - q y_1}{p - q}.$$

## Illustrative Examples

**Illustration 1 :** Find the coordinates of the points which divide internally and externally the line segment joining the points  $(1, 7)$  and  $(6, -3)$  in the ratio  $2 : 3$ .

**Solution :** **Internal division.** Let the coordinates of the required point be  $(x_1, y_1)$ . Then

$$x_1 = \frac{2 \cdot 6 + 3 \cdot 1}{2 + 3} = 3, \quad y_1 = \frac{2 \cdot (-3) + 3 \cdot 7}{2 + 3} = 3.$$

$\therefore$  The required point is  $(3, 3)$ .

**External division.** If the coordinates of the required point be  $(x_2, y_2)$ , then

$$x_2 = \frac{2 \cdot 6 - 3 \cdot 1}{2 - 3} = -9, \quad y_2 = \frac{2 \cdot (-3) - 3 \cdot 7}{2 - 3} = 27.$$

$\therefore$  The required point is  $(-9, 27)$ .

**Illustration 2 :** If the coordinates of the ends of a diameter of a circle are  $(-4, 7)$  and  $(2, 5)$ , then find the centre of the circle.

**Solution :** Let  $(x, y)$  be the centre of the circle. Then  $(x, y)$  is the mid-point of  $(-4, 7)$  and  $(2, 5)$ .

$$\therefore \quad x = \frac{-4 + 2}{2} = -1 \quad \text{and} \quad y = \frac{7 + 5}{2} = 6.$$

Hence, the coordinates of the centre are  $(-1, 6)$ .

**Illustration 3 :** In what ratio the line  $x + y = 4$  divides the line joining the points  $(-1, 1)$  and  $(5, 7)$  ? [UPTU 2008]

**Solution :** Suppose the line  $x + y = 4$  ... (1)

divides the line joining the points  $(-1, 1)$  and  $(5, 7)$  in the ratio  $m : n$ . If  $(x_1, y_1)$  are the coordinates of the point dividing the line joining  $(-1, 1)$  and  $(5, 7)$  in the ratio  $m : n$ , then

$$x_1 = \frac{m \cdot 5 + n \cdot (-1)}{m + n} = \frac{5m - n}{m + n}$$

and

$$y_1 = \frac{m \cdot 7 + n \cdot 1}{m + n} = \frac{7m + n}{m + n}.$$

Since the point  $(x_1, y_1)$  lies on the line (1), therefore,  $x_1 + y_1 = 4$

$$\text{or} \quad \frac{5m - n}{m + n} + \frac{7m + n}{m + n} = 4$$

$$\text{or} \quad 5m - n + 7m + n = 4m + 4n$$

$$\text{or} \quad 8m = 4n \quad \text{or} \quad \frac{m}{n} = \frac{4}{8} = \frac{1}{2}$$

$$\text{or} \quad m : n = 1 : 2.$$

**Illustration 4 :** The extremities of the diagonal of a parallelogram are the points  $(3, -4)$  and  $(-6, 5)$ . Third vertex is the point  $(-2, 1)$ . Find the coordinates of the fourth vertex.

**Solution :** Let  $(x, y)$  be the coordinates of the fourth vertex  $D$ . If  $M$  is the point of intersection of the diagonals  $AC$  and  $BD$ , then  $M$  is the middle point of both the diagonals  $AC$  and  $BD$ .

$$\therefore \quad \frac{x - 2}{2} = \frac{3 - 6}{2}$$

$$\text{and} \quad \frac{y + 1}{2} = \frac{-4 + 5}{2}$$

$$\text{or} \quad x - 2 = -3 \quad \text{and} \quad y + 1 = 1$$

$$\text{or} \quad x = -1 \quad \text{and} \quad y = 0.$$

Hence, the fourth vertex is the point  $(-1, 0)$ .

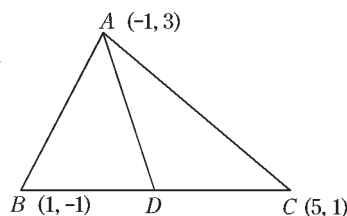
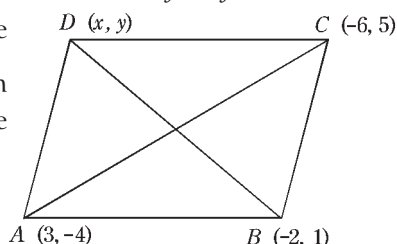
**Illustration 5 :** If  $A(-1, 3)$ ,  $B(1, -1)$  and  $C(5, 1)$  are the vertices of a triangle  $ABC$ , find the length of the median through  $A$ . [UPTU 2005]

**Solution :** Let  $D$  be the middle point of  $BC$ .

Then  $AD$  is the median through  $A$ , and  $D$  is the point

$$\left( \frac{1+5}{2}, \frac{-1+1}{2} \right) \text{ i.e. } (3, 0)$$

$$\therefore \quad AD = \sqrt{\{3 - (-1)\}^2 + \{0 - 3\}^2}$$



$$= \sqrt{4^2 + (-3)^2} = 5.$$

**Illustration 6 :** The coordinates of the mid-points of the sides of a triangle are  $(1, 2)$ ,  $(0, -1)$  and  $(2, -1)$ . Find the coordinates of its vertices. [UPTU 2007]

**Solution :** Let  $(1, 2)$ ,  $(0, -1)$  and  $(2, -1)$  be the

mid-points of  $AB$ ,  $BC$  and  $CA$  respectively.

Let the coordinates of  $A, B, C$  be  $(x_1, y_1)$ ,  $(x_2, y_2)$ ,  $(1, 2)$  and  $(x_3, y_3)$  respectively.

Then

$$\frac{x_1 + x_2}{2} = 1, \quad \frac{x_2 + x_3}{2} = 0, \quad \frac{x_3 + x_1}{2} = 2$$

$$\text{and } \frac{y_1 + y_2}{2} = 2, \quad \frac{y_2 + y_3}{2} = -1, \quad \frac{y_3 + y_1}{2} = -1$$

$$\text{or } x_1 + x_2 = 2, \quad x_2 + x_3 = 0, \quad x_3 + x_1 = 4 \quad \dots(1)$$

$$\text{and } y_1 + y_2 = 4, \quad y_2 + y_3 = -2, \quad y_3 + y_1 = -2 \quad \dots(2)$$

From (1) and (2), we get

$$2(x_1 + x_2 + x_3) = 6$$

$$\text{and } 2(y_1 + y_2 + y_3) = 0$$

$$\text{or } x_1 + x_2 + x_3 = 3$$

$$\text{and } y_1 + y_2 + y_3 = 0.$$

$$\therefore x_1 = 3, x_2 = -1, x_3 = 1; y_1 = 2, y_2 = 2, y_3 = -4.$$

Hence, the vertices  $A, B, C$  of the triangle are  $(3, 2)$ ,  $(-1, 2)$  and  $(1, -4)$  respectively.

**Illustration 7 :** A quadrilateral has the vertices at the points  $(-4, 2)$ ,  $(2, 6)$ ,  $(8, 5)$  and  $(9, -7)$ . Show that the mid-points of the sides of this quadrilateral are the vertices of a parallelogram. [UPTU 2004]

**Solution :** Let  $ABCD$  be the given quadrilateral with vertices  $A(-4, 2)$ ,  $B(2, 6)$ ,  $C(8, 5)$  and  $D(9, -7)$ . Let  $E, F, G$  and  $H$  be the mid-points of the sides  $AB, BC, CD$  and  $AD$  respectively. Then the co-ordinates of  $E$  are  $\left(\frac{-4+2}{2}, \frac{2+6}{2}\right)$  i.e.,  $(-1, 4)$ ;

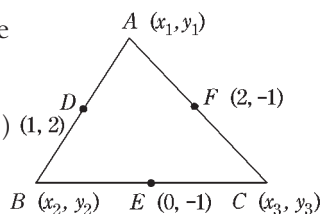
the coordinates of  $F$  are  $\left(\frac{2+8}{2}, \frac{6+5}{2}\right)$  i.e.,  $\left(5, \frac{11}{2}\right)$ ;

the coordinates of  $G$  are  $\left(\frac{8+9}{2}, \frac{5-7}{2}\right)$  i.e.,  $\left(\frac{17}{2}, -1\right)$ ;

and the coordinates of  $H$  are  $\left(\frac{-4+9}{2}, \frac{2-7}{2}\right)$  i.e.,  $\left(\frac{5}{2}, -\frac{5}{2}\right)$ .

Now the coordinates of the mid-point of  $EG$  are

$$\left(\frac{-1 + \frac{17}{2}}{2}, \frac{4 - 1}{2}\right) \text{ i.e., } \left(\frac{15}{4}, \frac{3}{2}\right)$$



and the coordinates of the mid-point of  $FH$  are  $\left(\frac{5 + \frac{5}{2}}{2}, \frac{\frac{11}{2} - \frac{5}{2}}{2}\right)$  i.e.,  $\left(\frac{15}{4}, \frac{3}{2}\right)$ .

Thus, we see that the diagonals  $EG$  and  $FH$  of the quadrilateral  $EFGH$  bisect each other. Hence  $EFGH$  is a parallelogram.

## Comprehensive Exercise 2

- Find the coordinates of the point which divides the line segment joining the points  $(6, 3)$  and  $(-4, 5)$  in the ratio  $3 : 2$  : (i) internally and (ii) externally.
- Find the coordinates of the mid-point of the line segment joining the points  $(-2, -5)$  and  $(3, -1)$ .
- Find the points on the line through  $A(5, -4)$  and  $B(-3, 2)$ , which are, twice as far from  $A$  as from  $B$ .
- Find the coordinates of the point which divides the line segment joining the points  $(5, -2)$  and  $(9, 6)$  in the ratio  $3 : 1$ .
- The coordinates of one end point of a diameter of a circle are  $(3, 5)$ . If the coordinates of the centre be  $(6, 6)$ , find the coordinates of the other end point of the diameter.
- Find the ratio in which the point  $(2, y)$  divides the join of  $(-4, 3)$  and  $(6, 3)$ , and hence, find the value of  $y$ .
- In what ratio does the point  $(-1, -1)$  divide the line segment joining the points  $(4, 4)$  and  $(7, 7)$ ?
- $A(1, 1)$  and  $B(2, -3)$  are two points and  $D$  is a point on  $AB$  produced such that  $AD = 3AB$ . Find the coordinates of  $D$ .
- If the point  $C(-1, 2)$  divides externally the line segment joining  $A(2, 5)$  and  $B$  in the ratio  $3 : 4$ , find the coordinates of  $B$ .
- Find the lengths of the medians of a triangle whose vertices are  $A(-1, 3)$ ,  $B(1, -1)$  and  $C(5, 1)$ .
- The three vertices of a parallelogram taken in order are  $(-1, 0)$ ,  $(3, 1)$  and  $(2, 2)$  respectively. Find the coordinates of the fourth vertex.
- Show that the points  $A(-2, -1)$ ,  $B(1, 0)$ ,  $C(4, 3)$  and  $D(1, 2)$  are the vertices of a parallelogram.
- If the points  $(-2, -1)$ ,  $(1, 0)$ ,  $(x, 3)$ ,  $(1, y)$  form a parallelogram, find the values of  $x$  and  $y$ .
- The three vertices of a rhombus, taken in order are  $(2, -1)$ ,  $(3, 4)$  and  $(-2, 3)$ . Find the fourth vertex.
- Show that the points  $(-4, -1)$ ,  $(-2, -4)$ ,  $(4, 0)$  and  $(2, 3)$  are the vertices of a rectangle.

## Answers

1. (i)  $\left(0, \frac{21}{5}\right)$  (ii)  $(-24, 9)$ .
2.  $\left(\frac{1}{2}, -3\right)$ .
3.  $\left(-\frac{1}{3}, 0\right), (-11, 8)$ .
4.  $(8, 4)$ .
5.  $(9, 7)$ .
6.  $3 : 2 ; 3$ .
7. externally  $5 : 8$ .
8.  $(4, -11)$ .
9.  $(3, 6)$ .
10.  $AD = 5, BE = \sqrt{10}, CF = 5$ .
11.  $(-2, 1)$ .
13.  $x = 4, y = 2$ .
14.  $(-3, -2)$ .

### 6.7 Area of a Triangle.

The area of a triangle when the coordinates of its three vertices  $A, B$  and  $C$  are  $(x_1, y_1), (x_2, y_2)$  and  $(x_3, y_3)$  respectively is given by

$$\begin{aligned}
 \text{area of } \Delta ABC &= \frac{1}{2} \{x_1 (y_2 - y_3) + x_2 (y_3 - y_1) + x_3 (y_1 - y_2)\} \\
 &= \frac{1}{2} \{(x_1 y_2 + x_2 y_3 + x_3 y_1) - (y_1 x_2 + y_2 x_3 + y_3 x_1)\} \\
 &= \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}.
 \end{aligned}$$

**Condition for collinearity of three points.** The points  $(x_1, y_1), (x_2, y_2), (x_3, y_3)$  will be collinear if  $x_1 (y_2 - y_3) + x_2 (y_3 - y_1) + x_3 (y_1 - y_2) = 0$ .

**Corollary.** If one vertex of a triangle is  $(0, 0)$  and the other two vertices are  $(x_1, y_1)$  and  $(x_2, y_2)$ , then its area  $= \frac{1}{2} (x_1 y_2 - x_2 y_1)$ .

**Note:** If the area of  $\Delta ABC$  is zero, then the three points  $A, B, C$  will be **collinear**.

### 6.8 Area of a Quadrilateral

If  $(x_1, y_1), (x_2, y_2), (x_3, y_3)$  and  $(x_4, y_4)$  be the four vertices  $A, B, C$  and  $D$  respectively of a quadrilateral, then

**area of quadrilateral  $ABCD$**

$$= \frac{1}{2} \{(x_1 y_2 + x_2 y_3 + x_3 y_4 + x_4 y_1) - (y_1 x_2 + y_2 x_3 + y_3 x_4 + y_4 x_1)\}.$$



## Illustrative Examples

**Illustration 1 :** Find the area of the triangle whose vertices are  $(0, 5)$ ,  $(2, 3)$  and  $(4, 5)$ .

[UPTU 2002, 09]

**Solution :** Here,  $x_1 = 0$ ,  $y_1 = 5$ ,  $x_2 = 2$ ,  $y_2 = 3$ ,  $x_3 = 4$ ,  $y_3 = 5$ .

$\therefore$  area of the given triangle

$$\begin{aligned} &= \frac{1}{2} [(x_1 y_2 + x_2 y_3 + x_3 y_1) - (y_1 x_2 + y_2 x_3 + y_3 x_1)] \\ &= \frac{1}{2} [(0 \cdot 3) + (2 \cdot 5) + (4 \cdot 5) - \{(5 \cdot 2) + (3 \cdot 4) + (5 \cdot 0)\}] \\ &= \frac{1}{2} [(0 + 10 + 20) - (10 + 12 + 0)] = \frac{1}{2} [30 - 22] = \frac{1}{2} \cdot 8 = 4 \text{ sq. units.} \end{aligned}$$

**Illustration 2 :** Prove that the points  $(a, b + c)$ ,  $(b, c + a)$ ,  $(c, a + b)$  are collinear.

[UPTU 2001, 07]

**Solution :** Here  $x_1 = a$ ,  $y_1 = b + c$ ,  $x_2 = b$ ,  $y_2 = c + a$ ,  $x_3 = c$ ,  $y_3 = a + b$ .

Now, area of the triangle formed by the given points

$$\begin{aligned} &= \frac{1}{2} [(x_1 y_2 + x_2 y_3 + x_3 y_1) - (y_1 x_2 + y_2 x_3 + y_3 x_1)] \\ &= \frac{1}{2} [a(c + a) + b(a + b) + c(b + c) - \{(b + c)b + (c + a)c + (a + b)a\}] \\ &= \frac{1}{2} [ac + a^2 + ba + b^2 + cb + c^2 - b^2 - cb - c^2 - ac - a^2 - ba] = 0. \end{aligned}$$

Hence, the given points are collinear.

**Illustration 3 :** For what value of  $k$  are the points  $(k, 2 - 2k)$ ,  $(1 - k, 2k)$  and  $(-4 - k, 6 - 2k)$  collinear?

**Solution :** If the points  $A(k, 2 - 2k)$ ,  $B(1 - k, 2k)$  and  $C(-4 - k, 6 - 2k)$  are collinear, then the area of  $\Delta ABC$  must be zero,

$$\begin{aligned} \text{i.e., } \frac{1}{2} [k \{2k - (6 - 2k)\} + (1 - k) \{(6 - 2k) - (2 - 2k)\} \\ + (-4 - k) \{(2 - 2k) - 2k\}] = 0 \end{aligned}$$

$$\text{or } 2k^2 + k - 1 = 0$$

$$\text{or } (2k - 1)(k + 1) = 0.$$

$$\therefore k = -1 \text{ or } \frac{1}{2}.$$

**Illustration 4 :** Find the area of a quadrilateral whose vertices are  $(1, 1)$ ,  $(3, 4)$ ,  $(5, -2)$  and  $(4, -7)$ .

**Solution :** Here,  $x_1 = 1$ ,  $y_1 = 1$ ,  $x_2 = 3$ ,  $y_2 = 4$ ,  $x_3 = 5$ ,  $y_3 = -2$ ,  $x_4 = 4$ ,  $y_4 = -7$ .

Required area

$$\begin{aligned}
 &= \frac{1}{2} [(x_1 y_2 + x_2 y_3 + x_3 y_4 + x_4 y_1) \\
 &\quad - (y_1 x_2 + y_2 x_3 + y_3 x_4 + y_4 x_1)] \\
 &= \frac{1}{2} [1 \cdot 4 + 3 \cdot (-2) + 5 \cdot (-7) + 4 \cdot 1] \\
 &\quad - [1 \cdot 3 + 4 \cdot 5 + (-2) \cdot 4 + (-7) \cdot 1] \\
 &= \frac{1}{2} [4 - 6 - 35 + 4] - [3 + 20 - 8 - 7] = \frac{1}{2} [(-33) - 8] \\
 &= \frac{41}{2}, \text{ numerically.}
 \end{aligned}$$

**Illustration 5 :** If the four points  $(1, 2)$ ,  $(-5, 6)$ ,  $(7, -4)$  and  $(a, -2)$  are collinear, find  $a$ .

[UPTU 2008]

**Solution :** Let  $A, B, C$  and  $D$  be the points  $(1, 2)$ ,  $(-5, 6)$ ,  $(7, -4)$  and  $(a, -2)$  respectively. If these points are collinear, then the area of the quadrilateral  $ABCD$  is zero.

$$\begin{aligned}
 \therefore \quad &\frac{1}{2} [1 \cdot 6 + (-5) \cdot (-4) + 7 \cdot (-2) + a \cdot 2] \\
 &\quad - \{(-5) \cdot 2 + 7 \cdot 6 + a \cdot (-4) + 1 \cdot (-2)\} = 0
 \end{aligned}$$

$$\text{or} \quad [(12 + 2a) - (30 - 4a)] = 0 \quad \text{or} \quad 6a - 18 = 0$$

$$\therefore \quad a = 3.$$

**Illustration 6 :** Find the area of a triangle formed by the lines

$$y = 2x, \quad y = x \quad \text{and} \quad y = 3x + 4. \quad [\text{UPTU 2001}]$$

**Solution :** Let the equations of the sides  $AB, BC$  and  $CA$  of  $\Delta ABC$  be  $y - x = 0$ ,  $y - 2x = 0$  and  $y - 3x - 4 = 0$  respectively.

Solving these equations in pairs, the coordinates of  $A, B$  and  $C$  are  $(-2, -2)$ ,  $(0, 0)$  and  $(-4, -8)$  respectively.

$$\begin{aligned}
 \therefore \quad \text{Area of } \Delta ABC &= \frac{1}{2} \{x_1 (y_2 - y_3) + x_2 (y_3 - y_1) + x_3 (y_1 - y_2)\} \\
 &= \frac{1}{2} \{(-2)(0 + 8) + 0(-8 + 2) + (-4)(-2 - 0)\} = \frac{1}{2} \{-16 + 8\} \\
 &= \frac{1}{2} (-8) = 4 \text{ sq. units, neglecting the negative sign.}
 \end{aligned}$$

**Illustration 7 :** Four points  $A(6, 3)$ ,  $B(-3, 5)$ ,  $C(4, -2)$  and  $D(x, 3x)$  are given in

such a way that  $\frac{\Delta DBC}{\Delta ABC} = \frac{1}{2}$ , find  $x$ .

[UPTU 2006]

$$\begin{aligned}
 \text{Solution :} \quad \frac{\text{Area of } \Delta DBC}{\text{Area of } \Delta ABC} &= \frac{\frac{1}{2} [x(5 + 2) - 3(-2 - 3x) + 4(3x - 5)]}{\frac{1}{2} [6(5 + 2) - 3(-2 - 3) + 4(3 - 5)]} \\
 \Rightarrow \quad \frac{1}{2} &= \frac{7x + 6 + 9x + 12x - 20}{42 + 15 - 8}
 \end{aligned}$$

$$\begin{aligned} \Rightarrow \quad \frac{1}{2} &= \frac{28x - 14}{49} \\ \Rightarrow \quad \frac{1}{2} &= \frac{4x - 2}{7} \\ \Rightarrow \quad \frac{7}{2} &= 4x - 2 \\ \Rightarrow \quad x &= \frac{11}{8}. \end{aligned}$$

### Comprehensive Exercise 3

- Find the area of a triangle whose vertices are :
  - (4, 4), (3, -16) and (3, -2)
  - (3, 2), (11, 8) and (8, 12)
  - (a, c + a), (a, c) and (-a, c - a)
  - (at<sub>1</sub><sup>2</sup>, 2at<sub>1</sub>), (at<sub>2</sub><sup>2</sup>, 2at<sub>2</sub>) and (at<sub>3</sub><sup>2</sup>, 2at<sub>3</sub>).
- Find the area of the quadrilateral whose vertices are :
  - (1, 1), (7, -3), (12, 2) and (7, 21)
  - (1, 2), (-1, 4), (-5, 1) and (0, -3).
- If (1, -1), (6, -6) and (4, k) are the vertices of a triangle whose area is 10 square units, find k.
- Find whether the following points are collinear or not :  
(-2, 1), (0, 5), (-1, 2).
- Show that the points (a<sup>2</sup>, 0), (0, b<sup>2</sup>) and (1, 1) are collinear, if  

$$\frac{1}{a^2} + \frac{1}{b^2} = 1.$$
- If a ≠ b ≠ c, prove that the points (a, a<sup>2</sup>), (b, b<sup>2</sup>), (c, c<sup>2</sup>) can never be collinear.
- Find the condition that the point (x, y) may lie on the line joining the points (3, 4) and (-5, -6).
- For what value of k, the following points are collinear :
  - (k, -1), (2, 1) and (4, 5) (UPTU 2010)
  - (2, 3), (5, k) and (6, 7).
- Find the value of p if the points (p + 1, 1), (2p + 1, 3) and (2p + 2, 2p) are collinear.
- Find whether the points (0, 0), (a, b), (-a, -b), (a<sup>2</sup>, b<sup>2</sup>) are collinear or not.
- Find the area of a triangle ABC the coordinates of the mid-points of whose sides are D (-1, -2), E (6, 1) and F (3, 5) respectively.

## Answers

- |                                                                                                                                                                               |                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>1. (i) 7 sq. units<br/>(iii) <math>a^2</math></p> <p>2. (i) 132 sq. units</p> <p>3. 0.</p> <p>7. <math>5x - 4y + 1 = 0</math>.</p> <p>9. <math>2, -\frac{1}{2}</math>.</p> | <p>(ii) 25 sq. units<br/>(iv) <math>a^2 (t_1 - t_2) (t_2 - t_3) (t_3 - t_1)</math>.</p> <p>(ii) 21.5 sq. units.</p> <p>4. not.</p> <p>8. (i) 1 (ii) 6.</p> <p>10. Not.      11. 74 sq. units.</p> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 6.9 Locus and its Equation

**Definition.** The locus of a moving point is the path traced out by it under certain geometrical condition or conditions.

**To find the equation of the locus of a point,** we should proceed as follows :

Take the coordinates of the moving point whose locus is to be found as  $(h, k)$ . Use the given geometrical condition or conditions and find a relation between  $h, k$  by eliminating the variable quantity for different values of which the locus of  $(h, k)$  is to be found. Finally, generalise  $(h, k)$ , i.e., put  $x$  for  $h$  and  $y$  for  $k$  to obtain the required equation of the locus of  $(h, k)$ .

### Illustrative Examples

**Illustration 1 :** Find the locus of the moving point  $P$ , such that  $2PA = 3PB$ , where  $A$  is  $(0, 0)$  and  $B$  is  $(4, -3)$ .

**Solution :** Let the coordinates of  $P$  be  $(h, k)$ .

Given  $2PA = 3PB$ .

$$\therefore 4PA^2 = 9PB^2$$

$$\text{or } 4(h^2 + k^2) = 9[(h - 4)^2 + (k + 3)^2]$$

$$\text{or } 4(h^2 + k^2) = 9[h^2 + k^2 - 8h + 6k + 25]$$

$$\text{or } 5h^2 + 5k^2 - 72h + 54k + 225 = 0.$$

$$\therefore \text{ locus of } P(h, k) \text{ on generalising, is } 5x^2 + 5y^2 - 72x + 54y + 225 = 0.$$

**Illustration 2 :** A point moves so that its distance from  $(3, 0)$  is twice the distance from  $(-3, 0)$ . Find the equation of the locus. [UPTU 2005]

**Solution :** Let  $A$  represent the point  $(3, 0)$ ;  $B$  the point  $(-3, 0)$  and  $P(h, k)$  be the moving point. According to the question,

$$PA = 2PB \quad \text{or} \quad (PA)^2 = 4(PB)^2$$

$$\begin{aligned} \text{or} \quad & [(h-3)^2 + (k-0)^2] = 4[(h+3)^2 + (k-0)^2] \\ \text{or} \quad & h^2 + 9 - 6h + k^2 = 4h^2 + 36 + 24h + 4k^2 \\ \text{or} \quad & 3h^2 + 3k^2 + 30h + 27 = 0. \end{aligned}$$

Generalising, the required locus is  $3x^2 + 3y^2 + 30x + 27 = 0$ .

**Illustration 3 :** Find the locus of a point such that the line segments having end points  $(2, 0)$  and  $(-2, 0)$  subtend a right angle at that point. [UPTU 2006]

**Solution :** Let  $A(2, 0)$  and  $B(-2, 0)$  be the given points and  $P(h, k)$  be the variable point.

According to the question,

$$\angle APB = 90^\circ,$$

i.e.,  $\Delta APB$  is a right triangle

$$\therefore AB^2 = PA^2 + PB^2$$

$$\Rightarrow \{2 - (-2)\}^2 + \{0 - 0\}^2 = \{(2 - h)^2 + (0 - k)^2\} + \{(-2 - h)^2 + (0 - k)^2\}$$

$$\Rightarrow 16 = (2 - h)^2 + k^2 + (-2 - h)^2 + k^2$$

$$\Rightarrow 16 = 4 + h^2 - 4h + 2k^2 + 4 + h^2 + 4h$$

$$\Rightarrow 16 = 2h^2 + 2k^2 + 8$$

$$\Rightarrow h^2 + k^2 = 4.$$

Generalising, the required locus is  $x^2 + y^2 = 4$ .

**Illustration 4 :** A point moves so that the sum of the squares of its distances from two fixed points  $A(a, 0)$  and  $B(-a, 0)$  is constant and equal to  $2c^2$ ; find the locus of the point.

**Solution :** Let  $P(h, k)$  be the moving point. According to the question,

$$PA^2 + PB^2 = 2c^2$$

$$\text{or} \quad [(h-a)^2 + (k-0)^2] + [(h+a)^2 + (k-0)^2] = 2c^2$$

$$\text{or} \quad h^2 + k^2 = c^2 - a^2.$$

$\therefore$  the required locus of the point  $P(h, k)$  is  $x^2 + y^2 = c^2 - a^2$ .

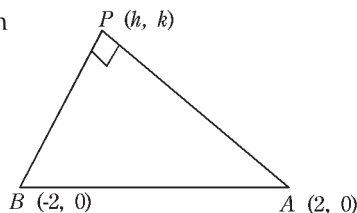
**Illustration 5 :** Find the equation to the locus of a point which moves so that the sum of its distances from  $(3, 0)$  and  $(-3, 0)$  is less than 9. [UPTU 2003, 04]

**Solution :** Let  $A(3, 0)$  and  $B(-3, 0)$  be the two given points and  $(h, k)$  be the coordinates of the moving point  $P$  whose locus is to be found. According to question

$$PA + PB < 9$$

$$\text{or} \quad \sqrt{(h-3)^2 + k^2} + \sqrt{(h+3)^2 + k^2} < 9$$

$$\text{or} \quad \sqrt{(h-3)^2 + k^2} < 9 - \sqrt{(h+3)^2 + k^2}$$



Squaring both the sides, we get

$$(h-3)^2 + k^2 < 81 - 18\sqrt{(h+3)^2 + k^2} + (h+3)^2 + k^2$$

or  $-12h - 81 < -18\sqrt{(h+3)^2 + k^2}$

or  $4h + 27 > 6\sqrt{(h+3)^2 + k^2}$ .

Again squaring both the sides, we get

$$16h^2 + 729 + 216h > 36[h^2 + 9 + 6h + k^2]$$

or  $20h^2 + 36k^2 < 405$ .

$\therefore$  the required locus of the point  $(h, k)$  is  $20x^2 + 36y^2 < 405$ .

## Comprehensive Exercise 4

1. Find the equation to the locus of a point equidistant from the points  $A(1, 3)$  and  $B(-2, 1)$ .
2. Find the locus of a point, so that the join of  $(-5, 1)$  and  $(3, 2)$  subtends a right angle at the moving point.
3. A point moves so that its distance from the axis of  $y$  is half its distance from the origin. Find the equation of its locus. [UPTU 2008]
4. A point  $P$  moves in such a way that the area of the triangle formed with  $A(1, -1)$  and  $B(5, 2)$  is of magnitude 5 units. Find the locus of  $P$ .
5. Find the locus of a point such that the sum of its distances from the points  $(0, 2)$  and  $(0, -2)$  is 6.
6. Find the equation of the locus of a point which moves so that its distance from  $(a, 0)$  is equal to its distance from the  $y$ -axis. [UPTU 2007]
7. Find the locus of the point of intersection of the lines  $x \cos \alpha + y \sin \alpha = a$  and  $x \sin \alpha - y \cos \alpha = b$ , where  $\alpha$  is a variable.
8. A rod of length  $l$  slides with its ends on two perpendicular lines. Find the locus of its mid-point.
9. Find the locus of the mid-point of the portion of the line  $x \cos \alpha + y \sin \alpha = p$  which is intercepted between the axes.
10. If  $O$  is the origin and  $Q$  is a variable point on  $x^2 = 4y$ , find the locus of the mid-point of  $OQ$ .
11. If  $O$  is the origin and  $Q$  is a variable point on  $y^2 = x$ , find the locus of the mid-point of  $OQ$ .

1.  $6x + 4y = 5$ .
2.  $x^2 + y^2 + 2x - 3y - 13 = 0$ .
3.  $x^2 - 3y^2 = 0$ .
4.  $3x - 4y + 3 = 0, 3x - 4y - 17 = 0$ .
5.  $9x^2 + 5y^2 = 45$ .
6.  $y^2 - 2ax + a^2 = 0$ .
7.  $x^2 + y^2 = a^2 + b^2$ .
8.  $4x^2 + 4y^2 = l^2$ .
9.  $p^2(x^2 + y^2) = 4x^2y^2$ .
10.  $x^2 = 2y$ .
11.  $2y^2 = x$ .

## Fill in the Blanks

1. If  $(3, a)$  is the mid-point of the line joining the points  $(4, 3)$  and  $(2, 5)$ , the value of  $a$  is .....
2. The distance of the point  $(x, y)$  from the origin is .....
3. The condition for collinearity of three points  $(x_1, y_1), (x_2, y_2), (x_3, y_3)$  is .....
4. If one vertex of a triangle is  $(0, 0)$  and the other two vertices are  $(x_1, y_1)$  and  $(x_2, y_2)$ , then its area is .....
5. If the points  $(-2, -5), (2, -2), (8, a)$  are collinear, then the value of  $a$  is .....
6. The area of a triangle whose vertices are  $(4, 4), (3, -16)$  and  $(3, -2)$  is .....
7. The co-ordinates of the points which divides the join of  $(3, -4)$  and  $(-5, -3)$  externally in the ratio  $3 : 4$  are .....

## Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

8. The area of the triangle whose vertices are  $(2, 2)$ ,  $(-4, 0)$  and  $(1, -1)$  is
- (a) 8 sq. units                      (b) 6 sq. units

- (c) 7 sq. units (d) 5 sq. units
9. If the four points  $(1, 2)$ ,  $(-5, 6)$ ,  $(7, -4)$  and  $(a, -2)$  are collinear, then  $a$  is  
(a) 0 (b) 1  
(c) 2 (d) 3
10. The co-ordinates of a point which divides the line joining the two given points  $(x_1, y_1)$  and  $(x_2, y_2)$  externally in the ratio  $p : q$  is  
(a)  $x = \frac{py_2 - qy_1}{p - q}$ ,  $y = \frac{px_2 - qx_1}{p - q}$   
(b)  $x = \frac{px_2 - qx_1}{p - q}$ ,  $y = \frac{py_2 - qy_1}{p - q}$   
(c)  $x = \frac{px_2 + qx_1}{p - q}$ ,  $y = \frac{py_2 + qy_1}{p - q}$   
(d)  $x = \frac{px_2 - qx_1}{p + q}$ ,  $y = \frac{py_2 - qy_1}{p + q}$
11. The distance between the points  $(-1, -4)$  and  $(3, 5)$  is  
(a) 97 (b)  $\sqrt{97}$   
(c) 13 (d)  $\sqrt{39}$
12. The points  $(-2, 2)$ ,  $(8, -2)$  and  $(-4, -3)$  are the vertices of  
(a) a right angled triangle  
(b) an isosceles triangle  
(c) an equilateral triangle  
(d) none of these

### True or False

Write 'T' for true and 'F' for false statement.

13. If the area of  $\triangle ABC$  is one, then the three points  $A, B, C$  will be collinear.
14. The points  $(-5, 7)$ ,  $(-4, 5)$  and  $(1, -5)$  are collinear.
15. The coordinates of a point which divides the line joining the two given points  $(x_1, y_1)$  and  $(x_2, y_2)$  internally in the ratio  $m_1 : m_2$  are  
$$x = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, y = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}.$$
16. The distance between the points  $(3, -1)$  and  $(-2, 4)$  is  $5\sqrt{2}$ .
17. The points  $(6, 2)$ ,  $(3, -1)$  and  $(-2, 4)$  are the vertices of a right angle triangle.
18. If the points  $(x, -1)$ ,  $(2, 1)$  and  $(4, 5)$  are collinear, then  $x$  is 2.
19. The co-ordinates of the point which divides the join of  $(2, 3)$  and  $(5, -3)$  internally in the ratio  $1 : 2$  are  $(1, 1)$ .



## Answers

- |                                                              |                         |              |               |
|--------------------------------------------------------------|-------------------------|--------------|---------------|
| 1. 4.                                                        | 2. $\sqrt{(x^2 + y^2)}$ |              |               |
| 3. $x_1 (y_2 - y_3) + x_2 (y_3 - y_1) + x_3 (y_1 - y_2) = 0$ |                         |              |               |
| 4. $\frac{1}{2} (x_1 y_2 - x_2 y_1)$                         | 5. $5/2$                | 6. $-7$      | 7. $(27, -7)$ |
| 8. (a)                                                       | 9. (d)                  | 10. (b)      | 11. (b)       |
| 12. (a)                                                      | 13. <i>F</i>            | 14. <i>T</i> | 15. <i>T</i>  |
| 16. <i>T</i>                                                 | 17. <i>T</i>            | 18. <i>F</i> | 19. <i>F</i>  |



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## Chapter

# 7



## The Straight Line

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### 7.1 Straight Line

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A straight line is a curve such that all the points of the line segment joining any two points on it lie on it.

### 7.2 Equations of Lines Parallel to the Coordinate Axes

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(i) Equation of a line parallel to the  $x$ -axis.

The equation of the straight line parallel to the  $x$ -axis and at a distance  $b$  from it is  $y = b$ .

If the line parallel to  $x$ -axis lies below it at a distance  $b$ , then its equation is  $y = -b$ .

**Corollary.** The equation of the  $x$ -axis is  $y = 0$ .

(ii) Equation of a line parallel to the  $y$ -axis.

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The equation of the straight line parallel to the  $y$ -axis and at a distance  $a$  from it is  $x = a$ .

If the line parallel to  $y$ -axis lies to the left of it at a distance  $a$ , then its equation is  $x = -a$ .

**Corollary.** The equation of the  $y$ -axis is  $x = 0$ .

## 7.3 Slope or Gradient of a Non-vertical Line

The tangent of the angle  $\theta$  which the line  $AB$  makes with the positive direction of the  $x$ -axis is called the slope or gradient of the line  $AB$  and is denoted by  $m$ .

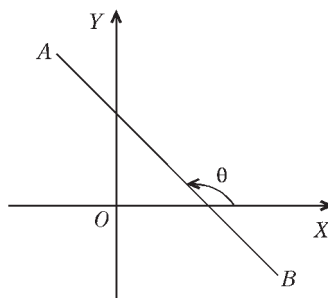
$$\text{slope } m = \tan \theta.$$

The angle  $\theta$  is positive if it is measured in anticlockwise direction from  $OX$  and it is negative if measured in clockwise direction.

**Note.** The angle of inclination of a line with the positive direction of  $x$ -axis in anticlockwise direction always lies between  $0^\circ$  and  $180^\circ$ .

The slope of  $x$ -axis or a line parallel to  $x$ -axis is 0.

The slope of  $y$ -axis or a line parallel to  $y$ -axis is not defined.



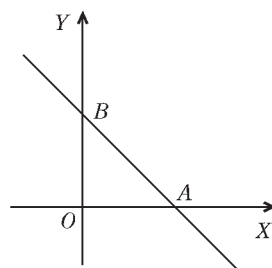
## 7.4 Intercepts of a Line on the Axes

If a line intersects the  $x$ -axis at  $A$ , then  $OA$  is called the **intercept of the line on  $x$ -axis** or  **$x$ -intercept of the line**.

Similarly, if a line intersects the  $y$ -axis at  $B$ , then  $OB$  is called the **intercept of the line on  $y$ -axis** or  **$y$ -intercept of the line**.

The intercepts are positive or negative according as the line meets with positive or negative directions of the coordinate axes.

If a line intersects the axes at  $A$  and  $B$ , then  $AB$  is called the **portion of the line intercepted between the axes**.



## 7.5 Various Forms of Equations of a Line

(i) The slope-intercept form of a line.

To find the equation of a straight line which cuts off an intercept  $c$  on  $y$ -axis and whose slope is  $m$ .

Let the line  $AB$  make angle  $BHX = \theta$  with  $OX$ . Then  $m = \tan \theta$ . Suppose that  $AB$  cuts off from the  $y$ -axis an intercept  $OK$ , equal to  $c$ .

Take any point  $P(x, y)$  on  $AB$ .

To find the equation of  $AB$ , we have to find a relation between  $x, y, m$  and  $c$ .

From  $P$  draw  $PM$  perpendicular to the  $x$ -axis and from  $K$  draw  $KL$  perpendicular to  $PM$ .

Then, in  $\triangle PKL$

$$\angle PKL = \angle BHX = \theta, \quad LK = x,$$

$$LP = MP - ML = MP - OK = y - c$$

and

$$\angle PLK = 1 \text{ right angle.}$$

$$\therefore m = \tan \theta = \frac{LP}{LK} = \frac{y - c}{x} \quad \text{or} \quad y = mx + c,$$

which is the required equation of the line  $AB$ .

**Corollary.** If the straight line passes through the origin then  $c = 0$ . Hence, the equation of the line passing through the origin is  $y = mx$ .

**Note.** The equation  $y = mx + c$  represents any straight line except those whose equations are of the form  $x = a$ , i.e., which are parallel to the  $y$ -axis.

### (ii) The double intercept form of a line.

To find the equation of a straight line which cuts off intercepts of lengths  $a$  and  $b$  from  $x$ -axis and  $y$ -axis respectively.

Let the line  $AB$  meet  $x$ -axis and  $y$ -axis at  $A$  and  $B$  respectively. Then  $OA = a$  and  $OB = b$ .

Let  $P(x, y)$  be any point on  $AB$ .

To find the equation of  $AB$ , we have to find a relation between  $x, y, a, b$ .

From  $P$ , draw  $PM$  and  $PN$  perpendiculars on the axes. Join  $OP$ .

Evidently, in area

$$\triangle OAB = \triangle OAP + \triangle OPB$$

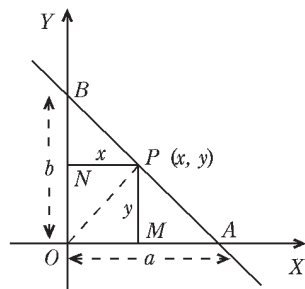
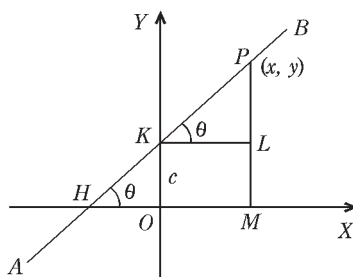
$$\therefore \frac{1}{2} OA \cdot OB = \frac{1}{2} OA \cdot MP + \frac{1}{2} OB \cdot NP$$

$$\text{or} \quad \frac{1}{2} ab = \frac{1}{2} ay + \frac{1}{2} bx.$$

Dividing both sides by  $\frac{1}{2} ab$ , we get

$$1 = \frac{y}{b} + \frac{x}{a} \quad \text{or} \quad \frac{x}{a} + \frac{y}{b} = 1,$$

which is the required equation of the line  $AB$ .



**(iii) The normal (perpendicular) form of a line.**

To find the equation of a straight line upon which the length of the perpendicular from the origin is  $p$  and this perpendicular makes an angle  $\alpha$  with positive direction of  $x$ -axis.

Let the required line be  $AB$  and  $OL$  be the perpendicular on it from  $O$ . Then  $OL = p$  and  $\angle LOX = \alpha$ .

To find the equation of  $AB$ , we have to find a relation between  $x$ ,  $y$ ,  $p$  and  $\alpha$ .

Let  $P(x, y)$  be any point on  $AB$ .

From  $P$ , draw  $PM$  perpendicular on the  $x$ -axis. From  $M$ , draw  $MK$  perpendicular on  $OL$  and lastly draw  $PN$  perpendicular on  $MK$ .

Then  $\angle PMN = 90^\circ - \angle KMO = \angle MOK = \alpha$ ,

and  $OM = x$ ,  $MP = y$ ,  $PN = LK$ .

Now  $OL = OK + KL = OK + NP$ .

But  $\frac{OK}{OM} = \cos \alpha$  and  $\frac{NP}{MP} = \sin \alpha$ .

$\therefore OL = OM \cos \alpha + MP \sin \alpha$ ,

i.e.,  $p = x \cos \alpha + y \sin \alpha$ .

Hence, the required equation is  $x \cos \alpha + y \sin \alpha = p$ .

**Note.** The convention is to regard the length of the perpendicular from the origin on the line as positive, so that  $p$  is positive. The value of  $\alpha$  is taken positive or negative according as the line making an angle  $\alpha$  with  $OX$  takes the position  $OL$  after rotating anticlockwise or clockwise respectively. The angle  $\alpha$  will be taken to lie between  $-\pi$  and  $\pi$ .

**(iv) The two-points form of a line.**

To find the equation of a straight line which passes through two given points  $(x_1, y_1)$  and  $(x_2, y_2)$ .

Let  $A$  and  $B$  be two points whose coordinates are  $(x_1, y_1)$  and  $(x_2, y_2)$ .

To find the equation of  $AB$ , we have to find a relation between  $x$ ,  $y$ ,  $x_1$ ,  $y_1$ ,  $x_2$  and  $y_2$ .

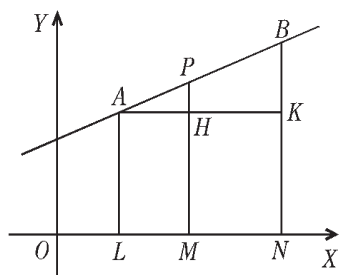
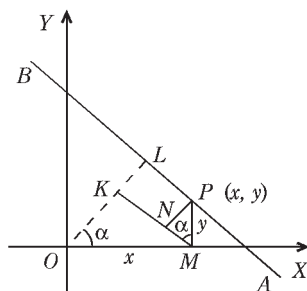
Let  $P(x, y)$  be any point on  $AB$ .

From  $A$ ,  $P$  and  $B$  draw  $AL$ ,  $PM$  and  $BN$  respectively perpendiculars on  $OX$ .

From  $A$ , draw  $AK$  perpendicular to  $BN$  which meets  $PM$  at the point  $H$ . Then triangles  $AHP$  and  $AKB$  are similar triangles.

$\therefore \frac{AH}{AK} = \frac{HP}{KB}$  ... (1)

Now  $AH = LM = OM - OL = x - x_1$ ,  $AK = LN = ON - OL = x_2 - x_1$ ,



$$HP = MP - MH = MP - LA = y - y_1$$

and  $KB = NB - NK = NB - LA = y_2 - y_1$ .

Substituting these values in (1), we get

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1} \quad \text{or} \quad y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1),$$

which is the required equation of the line  $AB$ .

**Note.** The slope of the line joining the two points  $(x_1, y_1)$  and  $(x_2, y_2)$  is

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{\text{difference of ordinates}}{\text{difference of abscissae}}.$$

**Corollary 1.** The equation of the straight line having slope  $m$  and passing through the point  $(x_1, y_1)$  is  $y - y_1 = m(x - x_1)$ .

**Corollary 2.** The equation of any straight line joining the origin and any point  $(x_1, y_1)$  is  $\frac{x}{x_1} = \frac{y}{y_1}$ .

## Illustrative Examples

**Illustration 1 :** Find the equation of a straight line which cuts off an intercept of 3 units on negative direction of  $y$ -axis and makes an angle of  $120^\circ$  with the positive direction of  $x$ -axis.

**Solution :** We have,  $m = \tan 120^\circ = \tan (180^\circ - 60^\circ) = -\tan 60^\circ = -\sqrt{3}$

and  $c = -3$ .

So the equation of the line is  $y = mx + c$

i.e.,  $y = -\sqrt{3}x - 3$  or  $\sqrt{3}x + y + 3 = 0$ .

**Illustration 2 :** Find the equation of the line which makes intercepts of 2 and  $-3$  with  $x$  axis and  $y$ -axis respectively.

**Solution :** Here,  $a = 2$  and  $b = -3$ .

Hence, the required equation of the line is

$$\frac{x}{2} + \frac{y}{(-3)} = 1 \quad \text{or} \quad \frac{x}{2} - \frac{y}{3} = 1.$$

**Illustration 3 :** Find the equation of straight line which makes equal intercepts on the axes and passes through the point  $(3, -5)$ . [UPTU 2002]

**Solution :** Let the equation of the straight line be

$$\frac{x}{a} + \frac{y}{b} = 1 \quad \dots(1)$$

The line (1) makes equal intercepts on the axes i.e.,  $a = b$ .

$$\therefore \frac{x}{a} + \frac{y}{a} = 1 \quad \text{or} \quad x + y = a.$$

If this line passes through the point  $(3, -5)$ , then

$$3 - 5 = a \quad \text{or} \quad a = -2$$

Hence, the required equation is

$$x + y = -2 \quad \text{or} \quad x + y + 2 = 0.$$

**Illustration 4 :** A line passes through (3, 4) and the sum of its intercepts on the axes is 14.

Find the equation of the line.

[UPTU 2009]

**Solution :** Let the intercept made by the line on  $x$ -axis be  $a$ .

Then the intercept made by the line on  $y$ -axis is  $14 - a$ .

$\therefore$  The equation of the line in intercept form is

$$\frac{x}{a} + \frac{y}{14 - a} = 1. \quad \dots(1)$$

If the line (1) passes through the point (3, 4), then

$$\frac{3}{a} + \frac{4}{14 - a} = 1 \quad \text{or} \quad 42 - 3a + 4a = 14a - a^2$$

$$\text{or} \quad a^2 - 13a + 42 = 0 \quad \text{or} \quad (a - 7)(a - 6) = 0$$

$$\Rightarrow \quad a = 7 \quad \text{or} \quad 6.$$

Hence, the required equation is

$$\frac{x}{7} + \frac{y}{7} = 1 \quad \text{or} \quad \frac{x}{6} + \frac{y}{8} = 1$$

$$\text{i.e.,} \quad x + y = 7 \quad \text{or} \quad 4x + 3y = 24.$$

**Illustration 5 :** Find the equation of the straight line the portion of which intercepted between the axes is divided by the point  $(-2, 6)$  in the ratio  $3 : 2$ . [UPTU 2007]

**Solution :** Let the equation of the straight line be

$$\frac{x}{a} + \frac{y}{b} = 1. \quad \dots(1)$$

The line (1) meets  $x$ -axis at the point  $A(a, 0)$  and  $y$ -axis at the point  $B(0, b)$ .

According to question, the point  $(-2, 6)$  divides the line segment  $AB$  in the ratio  $3 : 2$ .

$$\therefore \quad -2 = \frac{2 \cdot a + 3 \cdot 0}{2 + 3} = \frac{2a}{5} \quad \text{or} \quad a = -5$$

$$\text{and} \quad 6 = \frac{2 \cdot 0 + 3 \cdot b}{2 + 3} = \frac{3b}{5} \quad \text{or} \quad b = 10.$$

Putting the values of  $a$  and  $b$  in (1), the required equation of the line is

$$\frac{x}{-5} + \frac{y}{10} = 1 \quad \text{or} \quad y - 2x = 10.$$

**Illustration 6 :** A straight line, drawn through the point  $A(2, 1)$ , makes an angle  $\frac{\pi}{4}$  with

+ive  $x$ -axis and intersects another line  $x + 2y + 1 = 0$  at point  $B$ . Find the length  $AB$ .

[UPTU 2003]

**Solution :** The equation of any line passing through the given point  $A(2, 1)$  and making an angle  $\frac{\pi}{4}$  with  $x$ -axis is

$$\frac{x-2}{\cos 45^\circ} = \frac{y-1}{\sin 45^\circ} = r \text{ (say),} \quad \dots(1)$$

where  $r$  represents the distance of any point  $B$  on this line from the given point  $A(2, 1)$ .

The coordinates  $(x, y)$  of any point  $B$  on the line (1) are

$$(2 + r \cos 45^\circ, 1 + r \sin 45^\circ) \text{ i.e., } \left(2 + r \cdot \frac{1}{\sqrt{2}}, 1 + r \cdot \frac{1}{\sqrt{2}}\right).$$

If the point  $B$  lies on the line  $x + 2y + 1 = 0$ , then

$$\left(2 + r \cdot \frac{1}{\sqrt{2}}\right) + 2\left(1 + r \cdot \frac{1}{\sqrt{2}}\right) + 1 = 0$$

$$\text{or} \quad 5 + r \frac{3}{\sqrt{2}} = 0 \quad \text{or} \quad r = -\frac{5}{3} \sqrt{2}.$$

Hence, the length  $AB = 5\sqrt{2} / 3$ .

**Illustration 7 :** Find the equation of the line passing through the points  $(4, 3)$  and  $(7, 8)$ .

[UPTU 2001]

**Solution :** The two points are  $(x_1, y_1) = (4, 3)$  and  $(x_2, y_2) = (7, 8)$ .

Using  $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$ , the required equation is

$$y - 3 = \frac{8 - 3}{7 - 4} (x - 4)$$

$$\text{or} \quad y - 3 = \frac{5}{3} (x - 4) \quad \text{or} \quad 5x - 3y - 11 = 0.$$

**Illustration 8 :** Find the slope and the equation of the straight line joining the points  $(2, -5)$  and  $(4, 1)$ .

[UPTU 2002]

**Solution :** The slope of the line joining the points  $(2, -5)$  and  $(4, 1)$  is

$$= \frac{1 - (-5)}{4 - 2} = \frac{6}{2} = 3.$$

Now the equation of the straight line joining the points  $(2, -5)$  and  $(4, 1)$  and whose slope is 3 is

$$y - (-5) = 3(x - 2) \quad \text{or} \quad y + 5 = 3x - 6$$

$$\text{or} \quad 3x - y = 11.$$

**Illustration 9 :** Find the equation of the straight line which divides the line joining points  $(2, 3)$  and  $(-5, 8)$  in the ratio  $3 : 4$  and is also perpendicular to it. [UPTU 2006]

**Solution :** The equation of the line joining the points  $(2, 3)$  and  $(-5, 8)$  is

$$y - 3 = \frac{8 - 3}{-5 - 2} (x - 2) \quad \text{or} \quad y - 3 = \frac{5}{-7} (x - 2)$$



$$\text{or} \quad -7y + 21 = 5x - 10$$

$$\text{or} \quad 5x + 7y = 31. \quad \dots(1)$$

The slope of line (1) is  $-\frac{5}{7}$  and so the slope of the line perpendicular to it will be  $7/5$ .

The coordinates  $(h, k)$  of the point dividing line (1) in the ratio 3 : 4 are given as

$$h = \frac{3 \times (-5) + 4 \times 2}{3 + 4}$$

$$\text{and} \quad k = \frac{3 \times 8 + 4 \times 3}{3 + 4}$$

$$\text{i.e.,} \quad h = -1 \quad \text{and} \quad k = \frac{36}{7}.$$

Hence the equation of the line passing through  $(h, k)$  and having slope  $\frac{7}{5}$  is

$$y - k = \frac{7}{5}(x - h)$$

$$\text{or} \quad y - \frac{36}{7} = \frac{7}{5}[x - (-1)]$$

$$\text{or} \quad 35y - 180 = 49x + 49$$

$$\text{or} \quad 49x - 35y + 229 = 0.$$

## Comprehensive Exercise 1

1. Write the equation of a straight line
  - (i) which is parallel to  $x$ -axis and lies at a distance  $-5$  from it.
  - (ii) which is parallel to  $y$ -axis and lies at a distance 3 from it.
  - (iii) which is parallel to  $x$ -axis and passes through the point  $(3, -5)$ .
  - (iv) which is parallel to  $y$ -axis and passes through the point  $(-4, 7)$ .
  - (v) which is equidistant from the lines  $x = -4$  and  $x = 8$ .
  - (vi) which is equidistant from the lines  $y = 10$  and  $y = -2$ .
2. Find the equation of a straight line
  - (i) with slope 2 and  $y$ -intercept 3;
  - (ii) with slope  $-\frac{1}{3}$  and  $y$ -intercept  $-4$ .
3. Find the equation of a line with slope  $-1$  and cutting off an intercept of 4 units on negative direction of  $y$ -axis.
4. Find the equation of a line which makes an angle of  $\tan^{-1}(3)$  with the  $x$ -axis and cuts off an intercept of 4 units on negative direction of  $y$ -axis.
5. Find the equation of a straight line cutting off an intercept  $-1$  from  $y$ -axis and being equally inclined to the axes.

6. Find the equation of a straight line which cuts off intercepts
  - (i) 2 and  $-3$  from  $x$ -axis and  $y$ -axis respectively;
  - (ii)  $-5$  and 6 from  $x$ -axis and  $y$ -axis respectively.
7. Find the equation of the line which cuts off an intercept 4 on the positive direction of  $x$ -axis and an intercept 3 on the negative direction of  $y$ -axis.
8. Find the equation of a line which passes through  $(1, -2)$  and cuts off equal intercepts on the axes.
9. Find the equation of the line which passes through  $(3, -5)$  and cuts off intercepts on the axes which are equal in magnitude but opposite in sign.
10. Find the equation of a line which passes through the point  $(-3, 8)$  and cuts off positive intercepts on the coordinate axes whose sum is 7.
11. Find the equation of the line which passes through the point  $(-4, 3)$  and is such that the portion of it between the axes is divided by the point in the ratio 5 : 3.
12. Find the equation of a line which passes through the point  $(22, -6)$  and is such that the intercept on  $x$ -axis exceeds the intercept on  $y$ -axis by 5.
13. Find the equation of a straight line
  - (i) which passes through  $(5, -7)$  and having slope 4;
  - (ii) which passes through  $(-2, 3)$  and having slope  $-3$ .
14. Determine the equation of the line through the point  $(3, -4)$  and parallel to  $x$ -axis.
15. Find the equation of the line passing through :
  - (i)  $(-1, 3)$  and  $(4, -2)$
  - (ii)  $(0, -4)$  and  $(-6, 2)$
  - (iii)  $(1, 2)$  and  $(7, 8)$ . [UPTU 2001 (Special)]
16. Show that the points  $(5, 1)$ ,  $(1, -1)$  and  $(1, 4)$  are collinear. Also find the equation of the straight line on which these points lie.
17. Find the equation of the straight line which passes through the points  $(3, 3)$  and  $(7, 6)$  and find the length of the intercept cut off by the axes.
18. The vertices of a quadrilateral are  $A(-2, 6)$ ;  $B(1, 2)$ ;  $C(10, 4)$  and  $D(7, 8)$ . Find the equations of its diagonals.
19. Find the equations of the sides of a triangle whose vertices are the points  $(0, 0)$ ,  $(2, 4)$  and  $(6, 4)$ .
20. The mid-points of the sides of a triangle are  $(2, 1)$ ,  $(-5, 7)$  and  $(-5, -5)$ . Find the equations of the sides.

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## Answers

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- |                     |                          |                |
|---------------------|--------------------------|----------------|
| 1. (i) $y = -5$     | (ii) $x = 3$             | (iii) $y = -5$ |
| (iv) $x = -4$       | (v) $x = 2$              | (vi) $y = 4$ . |
| 2. (i) $y = 2x + 3$ | (ii) $x + 3y + 12 = 0$ . |                |
-

3.  $x + y + 4 = 0$ .                      4.  $y = 3x - 4$ .
5.  $x - y - 1 = 0$       or       $x + y + 1 = 0$ .
6. (i)  $3x - 2y = 6$                       (ii)  $-6x + 5y = 30$ .
7.  $3x - 4y = 12$ .                      8.  $x + y = -1$ .
9.  $x - y = 8$ .                      10.  $4x + 3y = 12$ .
11.  $9x - 20y + 96 = 0$ .                      12.  $6x + 11y - 66 = 0$ .
13. (i)  $4x - y - 27 = 0$                       (ii)  $3x + y + 3 = 0$ .
14.  $y + 4 = 0$ .
15. (i)  $x + y - 2 = 0$                       (ii)  $x + y + 4 = 0$       (iii)  $x - y + 1 = 0$ .
16.  $x - 2y - 3 = 0$ .                      17.  $3x - 4y + 3 = 0 ; \frac{5}{4}$ .
18.  $x + 6y - 34 = 0 ; x - y + 1 = 0$ .
19.  $y = 4, 2x - 3y = 0, 2x - y = 0$ .
20.  $x = 2, 7y - 6x - 79 = 0, 7y + 6x + 65 = 0$ .

## 7.6 Reduction of General Form to Standard Forms

Let the general equation of the line be

$$ax + by + c = 0. \quad \dots(1)$$

### (i) Slope-intercept form.

The equation (1) can be written as

$$by = -ax - c \quad \text{or} \quad y = -\frac{a}{b}x - \frac{c}{b}, \text{ if } b \neq 0.$$

This is the equation of a line in slope-intercept form.

Here,  $m = \text{slope} = -\frac{a}{b}$  and  $y\text{-intercept} = -\frac{c}{b}$ .

### (ii) Intercept form.

The equation (1) can be written as

$$ax + by = -c \quad \text{or} \quad \frac{ax}{-c} + \frac{by}{-c} = 1,$$

$$\text{if } c \neq 0 \quad \text{or} \quad \frac{x}{-\frac{c}{a}} + \frac{y}{-\frac{c}{b}} = 1.$$

This is the equation of a line in intercept form.

Here,  $x\text{-intercept} = -\frac{c}{a}$  and  $y\text{-intercept} = -\frac{c}{b}$ .

### (iii) Normal form.

First, write the equation (1) so that the constant term  $c$  is negative (if necessary, then multiply by  $-1$ ). Dividing (1) by  $\sqrt{a^2 + b^2}$ , we get

$$\frac{a}{\sqrt{a^2 + b^2}}x + \frac{b}{\sqrt{a^2 + b^2}}y = \frac{-c}{\sqrt{a^2 + b^2}}, \quad \dots(2)$$

where the term on the R.H.S. is positive.

$$\text{Since, } \left( \frac{a}{\sqrt{a^2 + b^2}} \right)^2 + \left( \frac{b}{\sqrt{a^2 + b^2}} \right)^2 = \frac{a^2 + b^2}{a^2 + b^2} = 1,$$

we can find an angle  $\alpha$  such that

$$\cos \alpha = \frac{a}{\sqrt{a^2 + b^2}}$$

$$\text{and } \sin \alpha = \frac{b}{\sqrt{a^2 + b^2}}.$$

Putting these values in (2), we get

$$x \cos \alpha + y \sin \alpha = - \frac{c}{\sqrt{a^2 + b^2}}.$$

This is the equation of a line in normal form.

$$\text{Here, } \cos \alpha = \frac{a}{\sqrt{a^2 + b^2}}, \sin \alpha = \frac{b}{\sqrt{a^2 + b^2}}, p = - \frac{c}{\sqrt{a^2 + b^2}}.$$

## Illustrative Examples

**Illustration 1 :** Reduce the equation  $\sqrt{3}x + y + 2 = 0$  to

- (i) slope-intercept form and find its slope and y-intercept;
- (ii) intercept form and find intercepts on the axes;
- (iii) normal form and find  $p$  and  $\alpha$ .

**Solution :** The given equation of the line is

$$\sqrt{3}x + y + 2 = 0. \quad \dots(1)$$

- (i) The equation (1) can be written as

$$y = -\sqrt{3}x - 2, \text{ which is the slope-intercept form.}$$

Here, slope =  $-\sqrt{3}$  and y-intercept =  $-2$ .

- (ii) The equation (1) can be written as

$$\sqrt{3}x + y = -2 \quad \text{or} \quad \frac{\sqrt{3}x}{-2} + \frac{y}{-2} = 1$$

$$\text{or } \frac{x}{-\frac{2}{\sqrt{3}}} + \frac{y}{-2} = 1, \text{ which is the intercept form.}$$

Here, x-intercept =  $-\frac{2}{\sqrt{3}}$  and y-intercept =  $-2$ .

- (iii) The equation (1) can be written as

$$-\sqrt{3}x - y - 2 = 0 \quad \text{or} \quad -\sqrt{3}x - y = 2.$$

Dividing both sides by  $\sqrt{(-\sqrt{3})^2 + (-1)^2} = 2$ , we get

$$-\frac{\sqrt{3}}{2}x - \frac{1}{2}y = 1, \text{ which is the normal form.}$$

Here,  $\cos \alpha = -\frac{\sqrt{3}}{2}$ ,  $\sin \alpha = -\frac{1}{2}$  and  $p = 1$ .

Since  $\sin \alpha$  and  $\cos \alpha$  both are negative, therefore,  $\alpha$  is in the third quadrant. But  $\alpha$  lies between  $-\pi$  and  $\pi$  so we take  $\alpha = -\frac{5\pi}{6}$ .

Hence,  $\alpha = -\frac{5\pi}{6}$  and  $p = 1$ .

**Illustration 2 :** Find the equation of a line at a distance of 3 units from the origin such that the perpendicular from the origin to the line makes an angle  $\tan^{-1}\left(\frac{3}{4}\right)$  with the positive direction of  $x$ -axis. [UPTU 2006]

**Solution :** We have  $p = 3$  and  $\alpha = \tan^{-1}\left(\frac{3}{4}\right)$ .

$$\therefore \tan \alpha = \frac{3}{4}$$

$$\Rightarrow \cos \alpha = \frac{4}{5} \quad \text{and} \quad \sin \alpha = \frac{3}{5}.$$

Hence, the equation of the line in normal form is

$$x \cos \alpha + y \sin \alpha = p$$

$$\text{or} \quad x \times \frac{4}{5} + y \times \frac{3}{5} = 3$$

$$\text{or} \quad 4x + 3y = 15.$$

**Illustration 3 :** Reduce  $4x + 3y - 9 = 0$  to the normal form and find the distance (perpendicular distance  $p$ ) from origin. [UPTU 2004]

**Solution :** We have  $4x + 3y - 9 = 0$  or  $4x + 3y = 9$ .

Dividing both sides by  $\sqrt{(4)^2 + (3)^2} = 5$ , we get

$$\frac{4}{5}x + \frac{3}{5}y = \frac{9}{5}, \text{ which is the normal form.}$$

Hence, the length of perpendicular from the origin to the line is  $p = \frac{9}{5}$ .

**Illustration 4 :** Find the angle made by the line

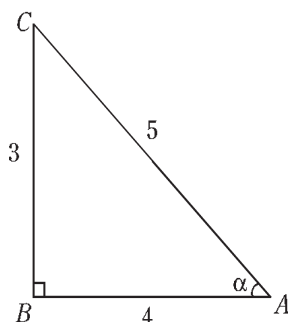
$x \cos 30^\circ + y \sin 30^\circ + \sin 120^\circ = 0$  with the positive direction of  $x$ -axis.

[UPTU 2007]

**Solution :** The equation of the given line is

$$x \cos 30^\circ + y \sin 30^\circ + \sin 120^\circ = 0 \quad \dots (1)$$

$$\text{or} \quad y = -\frac{\cos 30^\circ x}{\sin 30^\circ} - \frac{\sin 120^\circ}{\sin 30^\circ}$$



or  $y = -\cot 30^\circ x - \frac{\sqrt{3}/2}{1/2}$

or  $y = \tan 120^\circ x - \sqrt{3}$ , which is the slope-intercept form.

Hence, the angle made by the given line with the positive direction of  $x$ -axis is  $120^\circ$ .

## Comprehensive Exercise 2

- Reduce the equation  $\sqrt{3}x + y - 8 = 0$  to :
  - slope-intercept form and find slope and  $y$ -intercept;
  - intercept form and find intercepts on the axes;
  - the normal form and find  $p$  and  $\alpha$ .
- Find the angle made by the line  $x + \sqrt{3}y - 6 = 0$  with the positive direction of  $x$ -axis.
- Find the intercepts cut off by the line  $2x - y + 16 = 0$  on the coordinate axes.
- Find the intercepts of the line  $x \sin \alpha + y \cos \alpha = \sin 2\alpha$  on the coordinate axes and obtain the mid-point of the line-segment intercepted between the axes.
- Find the length of perpendicular from origin to the line  $x + y = \sqrt{2}$ . Also, find the inclination of this perpendicular with the  $x$ -axis.
- Which of the lines  $2x - y + 3 = 0$  and  $x - 4y - 7 = 0$  is farther from the origin?

## Answers

- $y = -\sqrt{3}x + 8$ ; slope  $= -\sqrt{3}$  and  $y$ -intercept  $= 8$
  - $\frac{x}{8/\sqrt{3}} + \frac{y}{8} = 1$ ;  $x$ -intercept  $= \frac{8}{\sqrt{3}}$  and  $y$ -intercept  $= 8$
  - $\frac{\sqrt{3}}{2}x + \frac{1}{2}y = 4$ ;  $\alpha = \frac{\pi}{6}$  and  $p = 4$ .
- $150^\circ$ .
- $x$ -intercept  $= -8$ ,  $y$ -intercept  $= 16$ .
- $2 \cos \alpha, 2 \sin \alpha$ ;  $(\cos \alpha, \sin \alpha)$ .
- $p = 1, \alpha = 45^\circ$ .
- $x - 4y - 7 = 0$ .

### 7.7 Angle between two Lines.

Prove that the angle  $\theta$  between two lines whose slopes are  $m_1$  and  $m_2$  is given by

$$\tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2}.$$

Let  $y = m_1 x + c_1$  and  $y = m_2 x + c_2$  be the equations of two given lines  $AB$  and  $CD$  respectively which intersect at the point  $K$ .

Then, we have to find the  $\angle BKD$ .

Let  $\angle BKD = \theta$ . If the lines  $AB$  and  $CD$  make angles  $\theta_1$  and  $\theta_2$  with the  $x$ -axis, then

$$\tan \theta_1 = m_1 \text{ and } \tan \theta_2 = m_2.$$

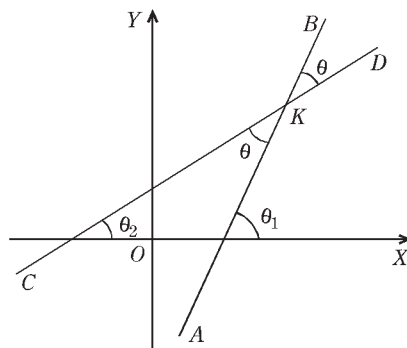
It is clear from the figure, that

$$\angle \theta = \angle \theta_1 - \angle \theta_2$$

$$\begin{aligned} \therefore \tan \theta &= \tan (\theta_1 - \theta_2) \\ &= \frac{\tan \theta_1 - \tan \theta_2}{1 + \tan \theta_1 \tan \theta_2} = \frac{m_1 - m_2}{1 + m_1 m_2} \end{aligned}$$

*i.e.*, the angle between the lines  $y = m_1 x + c_1$  and  $y = m_2 x + c_2$  is given by

$$\theta = \tan^{-1} \frac{m_1 - m_2}{1 + m_1 m_2}.$$



## 7.8 Condition for two given Lines to be Parallel

Two lines  $y = m_1 x + c_1$  and  $y = m_2 x + c_2$  are parallel if the angle  $\theta$  between them is  $0^\circ$  or  $180^\circ$  *i.e.*, if  $\tan \theta = 0$ , *i.e.*, if  $m_1 - m_2 = 0$

*i.e.*, if  $m_1 = m_2$ , *i.e.*, their slopes are equal.

Thus, the lines  $a_1 x + b_1 y + c_1 = 0$  and  $a_2 x + b_2 y + c_2 = 0$  are parallel if

$$-\frac{a_1}{b_1} = -\frac{a_2}{b_2}, \quad \text{i.e.,} \quad \frac{a_1}{a_2} = \frac{b_1}{b_2}.$$

*i.e.*, coefficients of  $x$  and  $y$  in their equations are proportional.

## 7.9 Condition for two given Lines to be Perpendicular

Two lines  $y = m_1 x + c_1$  and  $y = m_2 x + c_2$  are perpendicular if the angle  $\theta$  between them is  $90^\circ$

$$\text{i.e.,} \quad \text{if } \tan \theta = \tan 90^\circ = \infty, \quad \text{i.e.,} \quad \text{if } 1 + m_1 m_2 = 0$$

$$\text{i.e.,} \quad \text{if } m_1 m_2 = -1 \quad (\text{UPTU 2011})$$

*i.e.*, the product of their slopes is equal to  $-1$ .

If the slope of a line is  $m$ , then the slope of any line perpendicular to it is  $-\frac{1}{m}$ .

The lines  $a_1 x + b_1 y + c_1 = 0$  and  $a_2 x + b_2 y + c_2 = 0$  are perpendicular if

$$\left(-\frac{a_1}{b_1}\right) \cdot \left(-\frac{a_2}{b_2}\right) = -1$$

or  $a_1 a_2 + b_1 b_2 = 0$

i.e., product of the coefficients of  $x$  + product of the coefficients of  $y = 0$ .

## 7.10 Parallel Lines

The equation of any straight line parallel to a given line  $a x + b y + c = 0$  is

$$a x + b y + \lambda = 0.$$

**Rule.** In order to write the equation of any line parallel to a given line, we do not change the terms of  $x$  and  $y$  but we only change the constant term. The changed constant will be found by an additional given condition.

## 7.11 Perpendicular Lines

The equation of any straight line perpendicular to a given line  $a x + b y + c = 0$  is  $b x - a y + \lambda = 0$ .

**Rule.** In order to write the equation of any line perpendicular to a given line, we interchange the coefficients of  $x$  and  $y$  in the given equation and change the sign of one of these coefficients. Also we change the constant term. The value of the new constant is to be found by an additional given condition.

## Illustrative Examples

**Illustration 1 :** Find the angle between the lines

(i)  $x - y \sqrt{3} - 5 = 0$  and  $\sqrt{3}x + y - 7 = 0$  ;

[UPTU 2001 (Special)]

(ii)  $y = (2 - \sqrt{3})x + 5$  and  $y = (2 + \sqrt{3})x - 7$ .

**Solution :** (i) The equations of two lines are

$$x - y \sqrt{3} - 5 = 0 \quad \dots(1)$$

and  $\sqrt{3}x + y - 7 = 0. \quad \dots(2)$

Here,  $m_1 = \text{slope of the line (1)} = -\frac{1}{-\sqrt{3}} = \frac{1}{\sqrt{3}}$

and  $m_2 = \text{slope of the line (2)} = -\frac{\sqrt{3}}{1} = -\sqrt{3}.$

Clearly,  $m_1 m_2 = -1$ .

Hence, the two lines are at right angle.



(ii) The equations of two lines are

$$y = (2 - \sqrt{3})x + 5 \quad \dots(1)$$

and

$$y = (2 + \sqrt{3})x - 7. \quad \dots(2)$$

Here,

$$m_1 = \text{slope of the line (1)} = 2 - \sqrt{3}$$

and

$$m_2 = \text{slope of the line (2)} = 2 + \sqrt{3}.$$

Let  $\theta$  be the acute angle between the lines. Then

$$\tan \theta = \left| \frac{(2 - \sqrt{3}) - (2 + \sqrt{3})}{1 + (2 - \sqrt{3})(2 + \sqrt{3})} \right| = \sqrt{3}.$$

$$\therefore \theta = 60^\circ.$$

Hence, the acute angle between the lines is  $60^\circ$ .

**Illustration 2 :** Show that the line passing through the points  $(3, -4)$  and  $(-2, 6)$  is parallel to the line passing through  $(-3, 6)$  and  $(9, -18)$ .

**Solution :** Here,

$$\begin{aligned} m_1 &= \text{slope of the line joining the points } (3, -4) \text{ and } (-2, 6) \\ &= \frac{6 - (-4)}{-2 - 3} = \frac{10}{-5} = -2 \end{aligned}$$

and

$$\begin{aligned} m_2 &= \text{slope of the line joining the points } (-3, 6) \text{ and } (9, -18) \\ &= \frac{-18 - 6}{9 - (-3)} = \frac{-24}{12} = -2. \end{aligned}$$

Since

$$m_1 = m_2, \text{ therefore, the two lines are parallel.}$$

**Illustration 3 :** The line joining  $(-5, 7)$  and  $(0, -2)$  is perpendicular to the line joining  $(1, 3)$  and  $(4, x)$ , then find  $x$ . [UPTU 2003]

**Solution :** Here,  $m_1$  = slope of line joining the points  $(-5, 7)$  and  $(0, -2)$

$$= \frac{-2 - 7}{0 - (-5)} = -\frac{9}{5},$$

and

$$\begin{aligned} m_2 &= \text{slope of the line joining the points } (1, 3) \text{ and } (4, x). \\ &= \frac{x - 3}{4 - 1} = \frac{x - 3}{3}. \end{aligned}$$

If the given two lines are perpendicular, then

$$\begin{aligned} m_1 m_2 &= -1 \\ \Rightarrow \left(-\frac{9}{5}\right) \left(\frac{x-3}{3}\right) &= -1 \\ \Rightarrow -9(x-3) &= -15 \\ \Rightarrow -9x + 27 &= -15 \quad \Rightarrow \quad -9x = -42 \\ \Rightarrow x &= \frac{14}{3}. \end{aligned}$$

**Illustration 4 :** Find the equation of the straight line which passes through  $(-1, 2)$  and is parallel to the line  $3x - 4y + 5 = 0$ .

**Solution :** The equation of any straight line parallel to the line  $3x - 4y + 5 = 0$  is

$$3x - 4y + \lambda = 0. \quad \dots(1)$$

If the line (1) passes through the point  $(-1, 2)$ , then

$$-3 - 8 + \lambda = 0 \quad \text{or} \quad \lambda = 11.$$

Putting  $\lambda = 11$  in (1), the required equation of the line is

$$3x - 4y + 11 = 0.$$

**Illustration 5 :** Find the equation of the straight line which passes through  $(1, 2)$  and is perpendicular to the line  $4x - 3y = 8$ . [UPTU 2001]

**Solution :** The equation of any straight line perpendicular to the line

$$4x - 3y - 8 = 0 \text{ is}$$

$$3x + 4y + \lambda = 0. \quad \dots(1)$$

If the line (1) passes through the point  $(1, 2)$ , then

$$3 + 8 + \lambda = 0 \quad \text{or} \quad \lambda = -11.$$

Putting  $\lambda = -11$  in (1), the required equation of the line is

$$3x + 4y - 11 = 0.$$

**Illustration 6 :** Prove that the equation to the straight line passing through the point  $(a \cos^3 \theta, a \sin^3 \theta)$  and perpendicular to the line  $x \sec \theta + y \operatorname{cosec} \theta = a$  is

$$x \cos \theta - y \sin \theta = a \cos 2 \theta. \quad [\text{UPTU 2007}]$$

**Solution :** The slope of the given line  $x \sec \theta + y \operatorname{cosec} \theta = a$  is

$$-\frac{\sec \theta}{\operatorname{cosec} \theta}, \quad \text{i.e.,} \quad -\frac{\sin \theta}{\cos \theta}.$$

$\therefore$  the slope of a line perpendicular to the given line  $= \frac{\cos \theta}{\sin \theta}.$

Now the equation of the straight line which passes through the point  $(a \cos^3 \theta, a \sin^3 \theta)$  and whose slope is  $\frac{\cos \theta}{\sin \theta}$  is

$$y - a \sin^3 \theta = \frac{\cos \theta}{\sin \theta} (x - a \cos^3 \theta)$$

$$\text{or} \quad x \cos \theta - y \sin \theta = a (\cos^4 \theta - \sin^4 \theta)$$

$$\text{or} \quad x \cos \theta - y \sin \theta = a (\cos^2 \theta + \sin^2 \theta)(\cos^2 \theta - \sin^2 \theta)$$

$$\text{or} \quad x \cos \theta - y \sin \theta = a \cos 2 \theta.$$

**Illustration 7 :** Find the equation of the line which is perpendicular to the line  $3x - 2y + 4 = 0$  and passes through the point of intersection of the lines  $x + 2y + 1 = 0$  and  $y = x + 7$ . [UPTU 2009]

**Solution :** The point of intersection of the lines  $x + 2y + 1 = 0$  and  $y = x + 7$  is  $(-5, 2)$ .

The equation of any straight line perpendicular to the line  $3x - 2y + 4 = 0$  is

$$2x + 3y + \lambda = 0 \quad \dots(1)$$

If the line (1) passes through the point  $(-5, 2)$ , then

$$2(-5) + 3(2) + \lambda = 0 \quad \text{or} \quad \lambda = 4.$$

Putting  $\lambda = 4$  in (1), the required equation of the line is  $2x + 3y + 4 = 0$ .

**Illustration 8 :** Find the equation of perpendicular bisector of the line segment joining the points  $A(2, 3)$  and  $B(6, -5)$ . [UPTU 2004]

**Solution :** The slope of the line joining the points  $A(2, 3)$  and  $B(6, -5)$  is

$$= \frac{-5 - 3}{6 - 2} = -\frac{8}{4} = -2.$$

$\therefore$  slope of a line perpendicular to the line  $AB = \frac{1}{2}$ .

The coordinates of the middle point  $M$  of  $AB$  are

$$\left( \frac{2+6}{2}, \frac{3+(-5)}{2} \right) \quad \text{i.e.,} \quad (4, -1).$$

Hence, the equation of the perpendicular bisector of  $AB$ , i.e., the equation of the line passing through  $M$  and perpendicular to  $AB$  is

$$y + 1 = \frac{1}{2}(x - 4) \quad \text{or} \quad x - 2y = 6.$$

### Comprehensive Exercise 3

1. Find the angle between the lines whose equations are :

(i)  $y - \sqrt{3}x - 5 = 0$  and  $\sqrt{3}y - x + 6 = 0$ ;

(ii)  $x - 2y + 3 = 0$  and  $3x + y - 1 = 0$ ;

(iii)  $2x - y + 3 = 0$  and  $x + y + 2 = 0$ ;

(iv)  $\frac{x}{a} + \frac{y}{b} = 1$  and  $\frac{x}{b} - \frac{y}{a} = 1$ .

2. Show that

(i) the line joining  $(2, -3)$  and  $(-5, 1)$  is parallel to the line joining  $(7, -1)$  and  $(0, 3)$ ;

(ii) the line through  $(2, -5)$  and  $(-2, 5)$  is perpendicular to the line through  $(6, 3)$  and  $(1, 1)$ .

3. What is the value of  $y$  so that the line through  $(3, y)$  and  $(2, 7)$  is parallel to the line through  $(-1, 4)$  and  $(0, 6)$ ? [UPTU 2006]

4. Without using Pythagoras theorem, show that the points  $A(0, 4)$ ,  $B(1, 2)$  and  $C(3, 3)$  are the vertices of a right angled triangle.

5. Show that the points  $A(2, -1)$ ,  $B(0, 2)$ ,  $C(2, 3)$  and  $D(4, 0)$  are the coordinates of the vertices of a parallelogram and find the angle between its diagonals.

6. Show that  $A(4, 3)$ ,  $B(6, 4)$ ,  $C(5, 6)$  and  $D(3, 5)$  are the angular points of a square.
7. Find the angles of a triangle whose vertices are  $A(3, 4)$ ,  $B(4, 4)$  and  $C(4, 5)$ .
8. Find the equation of the perpendicular bisector of the line joining the points  $A(2, 3)$  and  $B(6, -5)$ .
9. Find the equation of a straight line passing through the origin and parallel to the line  $3x - 2y + 1 = 0$ .
10. Find the equation of a line passing through  $(1, 1)$  and perpendicular to the straight line  $3x - 4y = 6$ . [UPTU 2001 (Special)]
11. Find the equation of the line passing through  $(-2, -4)$  and perpendicular to the line  $3x - y + 5 = 0$ . [UPTU 2007]
12. Find the equation of a line passing through  $(7, 1)$  and parallel to the line joining the points  $(2, -2)$  and  $(4, 8)$ .
13. Let  $A(1, 1)$ ,  $B(2, -1)$  and  $C(-1, 3)$  be the vertices of a  $\Delta ABC$ . Find the equations of the perpendiculars drawn from vertices to the opposite sides.
14. Find the equation of a line which is perpendicular to the line  $\sqrt{3}x - y + 5 = 0$  and cuts off  $-4$  units from the  $y$ -axis.
15. Find the equation of a line perpendicular to the line  $\sqrt{3}x - y + 5 = 0$  and at a distance of 3 units from the origin.

## Answers

- |                                                        |                                 |
|--------------------------------------------------------|---------------------------------|
| 1. (i) $30^\circ$                                      | (ii) $\tan^{-1}(7)$             |
| (iii) $\tan^{-1}(3)$                                   | (iv) $90^\circ$ .               |
| 3. 9.                                                  | 5. $\tan^{-1} 2$ .              |
| 7. $45^\circ, 90^\circ, 45^\circ$ .                    | 8. $x - 2y - 6 = 0$ .           |
| 9. $3x - 2y = 0$ .                                     | 10. $4x + 3y - 7 = 0$ .         |
| 11. $x + 3y + 14 = 0$ .                                | 12. $5x - y - 34 = 0$ .         |
| 13. $3x - 4y + 1 = 0, x - y - 3 = 0, x - 2y + 7 = 0$ . |                                 |
| 14. $x + \sqrt{3}y + 4\sqrt{3} = 0$ .                  | 15. $x + \sqrt{3}y \pm 6 = 0$ . |

### 7.12 The Length of Perpendicular

The length of the perpendicular drawn from the point  $(x_1, y_1)$  to the line  $x \cos \alpha + y \sin \alpha = p$  is  $x_1 \cos \alpha + y_1 \sin \alpha - p$ .

The length of the perpendicular drawn from the point  $(x_1, y_1)$  to the line

$$ax + by + c = 0 \text{ is } \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}}.$$

In particular, the length of perpendicular from origin  $(0, 0)$  to the line  $ax + by + c = 0$  is  $\frac{c}{\sqrt{a^2 + b^2}}$ .

**Rule.** Bring all the terms in the equation of the line to the L.H.S. and make the R.H.S. 0, substitute for  $x$  and  $y$  the coordinates of the point  $(x_1, y_1)$  and divide by the square root of the sum of the squares of the coefficients of  $x$  and  $y$ .

**Note.** If the length of perpendicular obtained from the above formula is negative, we can ignore the sign and give the positive value in answer, unless there is some special reason.

## 7.13 Distance between two Parallel Lines

**First method.** Choose a convenient point on any of the lines (put  $x = 0$  and find the value of  $y$  or put  $y = 0$  and find the value of  $x$ ). Now the perpendicular of this point from the other line will give the required distance between the given parallel lines.

**Second method.** Write the equations of the two parallel lines in such a way that coefficients of  $x$  are of the same sign. Then the two lines will lie on the same side of the origin if  $c_1$  and  $c_2$  are of the same sign and on opposite sides of the origin if  $c_1$  and  $c_2$  are of opposite signs.

Find the perpendicular distances  $p_1$  and  $p_2$  of both the lines from the origin and retain their signs. The required distance between the parallel lines is  $|p_1 - p_2|$ .

## Illustrative Examples

**Illustration 1 :** If  $p$  is the length of perpendicular from origin on the line whose intercepts on the axes are  $a$  and  $b$ , show that

$$\frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2}.$$

**Solution :** The equation of the line making intercepts  $a$  and  $b$  on the coordinate axes is

$$\frac{x}{a} + \frac{y}{b} = 1 \quad \text{or} \quad \frac{x}{a} + \frac{y}{b} - 1 = 0 \quad \dots(1)$$

If  $p$  be the length of the perpendicular drawn from origin  $(0, 0)$  to the line (1), then

$$p = \frac{-1}{\sqrt{\left(\frac{1}{a}\right)^2 + \left(\frac{1}{b}\right)^2}}$$

or 
$$\frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2}.$$

**Illustration 2 :** Find the equations of two lines through  $(0, a)$  which are at a distance  $a$  from the point  $(2a, 2a)$ .

**Solution :** Equation of any line through  $(0, a)$  is

$$y - a = m(x - 0)$$

$$\text{or} \quad mx - y + a = 0. \quad \dots(1)$$

If the length of perpendicular from  $(2a, 2a)$  to the line (1) is  $a$ , then

$$a = \left| \frac{m(2a) - 2a + a}{\sqrt{m^2 + 1}} \right|$$

$$\text{or} \quad 2m - 1 = \pm \sqrt{m^2 + 1}.$$

Squaring, we get  $(2m - 1)^2 = m^2 + 1$

$$\text{or} \quad 3m^2 - 4m = 0 \Rightarrow m = 0, \frac{4}{3}.$$

Putting these values of  $m$  in (1), the required equations are

$$y - a = 0 \quad \text{and} \quad 4x - 3y + 3a = 0.$$

**Illustration 3 :** Find the distance between the parallel lines  $3x + 4y = 12$  and  $3x + 4y = 3$ . [UPTU 2004]

**Solution :** The given lines are  $3x + 4y = 12$  ...(1)

and  $3x + 4y = 3$ . ...(2)

Putting  $x = 0$  in (1), we get  $y = 3$ . Thus  $(0, 3)$  is a point on the line (1). The perpendicular distance between the lines (1) and (2) is

$$\begin{aligned} &= \text{the length of perpendicular from the point } (0, 3) \text{ to the line (2)} \\ &= \frac{3 \cdot 0 + 4 \cdot 3 - 3}{\sqrt{9 + 16}} = \frac{9}{5}. \end{aligned}$$

**Illustration 4 :** Show that the lines  $2x + 3y = 19$  and  $2x + 3y = -7$  are equidistant from the line  $2x + 3y = 6$ .

**Solution :** All the three lines are parallel. Putting  $x = 0$  in the equation  $2x + 3y = 6$  of the third line, we get  $y = 2$ . Thus,  $(0, 2)$  is a point on the third line. The perpendicular distances of this point from the first and second lines are

$$\frac{0 + 6 - 19}{\sqrt{13}} = -\sqrt{13}$$

$$\text{and} \quad \frac{0 + 6 + 7}{\sqrt{13}} = \sqrt{13}.$$

These two perpendiculars are equal in magnitude and opposite in sign. Hence, the first two lines are equidistant from the third line on opposite sides of it.

**Illustration 5 :** Are the points  $(3, 4)$  and  $(2, -6)$  on the same or opposite sides of the line  $3x - 4y = 8$ ?

**Solution :** The equation of the given line is

$$3x - 4y - 8 = 0. \quad \dots(1)$$

Putting  $x = 3$ ,  $y = 4$  on the L.H.S. of (1), we get

$$3 \cdot (3) - 4 \cdot (4) - 8 = 9 - 16 - 8 = -15, \text{ which is negative.}$$

Again putting  $x = 2$ ,  $y = -6$  on the L.H.S. of (1), we get

$$3 \cdot (2) - 4 \cdot (-6) - 8 = 6 + 24 - 8 = 22, \text{ which is positive.}$$

Since the values of the L.H.S. of the equation of the line (1) are of the opposite signs when the coordinates of the points (3, 4) and (2, -6) are substituted in it, therefore, these points lie on the opposite sides of the line (1).

## Comprehensive Exercise 4

- Find the distance of the point (4, 5) from the line  $3x - 5y + 7 = 0$ .
- Find the length of perpendicular drawn from (2, -1) to the line  $4(x - 3) = 3(y + 4)$ .
- Find the length of perpendicular drawn from the origin to the line  $a(x + a) + b(y + b) = 0$ .
- If the length of the perpendicular from the point (1, 1) to the line  $ax - by + c = 0$  be unity, show that  $\frac{1}{c} + \frac{1}{a} - \frac{1}{b} = \frac{c}{2ab}$ .
- If  $p$  and  $p'$  be the perpendiculars from the origin to the straight lines  $x \sec \theta - y \operatorname{cosec} \theta = a$  and  $x \cos \theta - y \sin \theta = a \cos 2\theta$ , prove that  $4p^2 + p'^2 = a^2$ .
- Find the distance between the parallel lines :
  - $3x - 4y = 8$  and  $3x - 4y + 12 = 0$  ;
  - $8x + 15y + 20 = 0$  and  $8x + 15y = 14$  ;
  - $2x - 3y + 9 = 0$  and  $4x - 6y + 1 = 0$ . [UPTU 2005]
- Are the points (2, 5) and (-3, 7) on the same or opposite sides of the line  $2x - 3y = 4$  ?
- Show that the origin and the point (-2, 5) lie on the opposite sides of the line  $4x - 3y + 9 = 0$ .

## Answers

- $\frac{6}{\sqrt{34}}$
- $\frac{13}{5}$
- $\sqrt{a^2 + b^2}$
- (i) 4      (ii) 2      (iii)  $-\frac{17}{2\sqrt{13}}$
- same side.

## 7.14 Equations of the Bisectors of the Angles between two given Lines

The equations of the bisectors of the angles between the lines

$$a_1x + b_1y + c_1 = 0$$

and

$$a_2x + b_2y + c_2 = 0$$

are

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \pm \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}.$$

### Particular Cases :

The equations of the lines should be so written that  $c_1$  and  $c_2$  are either both positive or both negative.

(i) Equation of the bisector of the angle in which the origin lies is

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = + \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$$

and the equation of the bisector of the angle which does not contain the origin is

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = - \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}.$$

(ii) **Bisectors of acute and obtuse angles.** If  $\theta$  be the angle between one of the given lines and one bisector, then if the absolute value of  $\tan \theta$ , i.e.,  $|\tan \theta| < 1$ , then it is the **bisector of acute angle** and if  $|\tan \theta| > 1$ , then it is the **bisector of obtuse angle**.

(iii) If the constant terms  $c_1$  and  $c_2$  in the two equations

$$a_1x + b_1y + c_1 = 0$$

and

$$a_2x + b_2y + c_2 = 0$$

are of the same sign, then the angle between these two lines in which the origin lies is **acute** if  $a_1a_2 + b_1b_2 < 0$  and **obtuse** if  $a_1a_2 + b_1b_2 > 0$ .

## Illustrative Examples

**Illustration 1 :** Find the equations of the bisectors of the angles between the lines  $3x - 4y + 7 = 0$  and  $12x - 5y - 8 = 0$ .

**Solution :** The equations of the bisectors of the angles between the lines  $3x - 4y + 7 = 0$  and  $12x - 5y - 8 = 0$  are



$$\frac{3x - 4y + 7}{\sqrt{3^2 + (-4)^2}} = \pm \frac{12x - 5y - 8}{\sqrt{12^2 + (-5)^2}}$$

or 
$$\frac{3x - 4y + 7}{5} = \pm \frac{12x - 5y - 8}{13}$$

or 
$$39x - 52y + 91 = \pm (60x - 25y - 40).$$

Taking positive sign, we get one bisector as  $21x + 27y - 131 = 0$ .

Taking negative sign, we get the other bisector as  $99x - 77y + 51 = 0$ .

**Illustration 2 :** Find the equation of the bisector of the acute angle between the lines  $3x - 4y + 7 = 0$  and  $12x + 5y - 2 = 0$ .

**Solution :** Making the constant terms of the **same sign**, the equations of the given lines are

$$3x - 4y + 7 = 0 \quad \dots(1)$$

and 
$$-12x - 5y + 2 = 0. \quad \dots(2)$$

The equations of the bisectors of the angles between the lines (1) and (2) are

$$\frac{3x - 4y + 7}{\sqrt{3^2 + 4^2}} = \pm \frac{-12x - 5y + 2}{\sqrt{12^2 + 5^2}} \quad \dots(3)$$

i.e., 
$$13(3x - 4y + 7) = \pm 5(-12x - 5y + 2)$$

i.e., 
$$11x - 3y + 9 = 0 \quad \dots(4)$$

and 
$$21x + 77y - 101 = 0. \quad \dots(5)$$

Let  $\theta$  be the acute angle between the line (1) and the bisector (4). Then

$$\tan \theta = \left| \frac{\frac{11}{3} - \frac{3}{4}}{1 + \frac{11}{3} \cdot \frac{3}{4}} \right| = \frac{35}{45} < 1$$

i.e., 
$$\theta < 45^\circ \quad \text{or} \quad 2\theta < 90^\circ.$$

$\therefore 11x - 3y + 9 = 0$  is the bisector of the acute angle between the given lines. The line (5) bisects the obtuse angle between the given lines.

**Illustration 3 :** Find the equation of the line which bisects the obtuse angle between the lines  $x - 2y + 4 = 0$  and  $4x - 3y + 2 = 0$ .

**Solution :** Keeping the constant terms of the **same sign**, the equations of the given lines are

$$x - 2y + 4 = 0 \quad \dots(1)$$

and  $4x - 3y + 2 = 0$ . ...(2)

The equations of the bisectors of the angles between the lines (1) and (2) are

$$\frac{x - 2y + 4}{\sqrt{1 + 4}} = \pm \frac{4x - 3y + 2}{\sqrt{16 + 9}},$$

*i.e.*,  $\sqrt{5}(x - 2y + 4) = \pm(4x - 3y + 2)$ . ...(3)

The positive sign gives  $(4 - \sqrt{5})x + (2\sqrt{5} - 3)y - (4\sqrt{5} - 2) = 0$  ...(4)

and the negative sign gives

$$(4 + \sqrt{5})x - (2\sqrt{5} + 3)y + (4\sqrt{5} + 2) = 0. \quad \text{...(5)}$$

Let  $\theta$  be the acute angle between the line (1) and the bisector (4). Then

$$\begin{aligned} \tan \theta &= \left| \frac{\frac{1}{2} - \frac{4 - \sqrt{5}}{3 - 2\sqrt{5}}}{1 + \frac{1}{2} \cdot \frac{4 - \sqrt{5}}{3 - 2\sqrt{5}}} \right| = \left| \frac{-5}{10 - 5\sqrt{5}} \right| = \left| \frac{-1}{2 - \sqrt{5}} \right| \\ &= \frac{1}{\sqrt{5} - 2} = \frac{(\sqrt{5} + 2)}{(\sqrt{5} - 2)(\sqrt{5} + 2)} = \sqrt{5} + 2 > 1 \end{aligned}$$

*i.e.*,  $\theta > 45^\circ$  or  $2\theta > 90^\circ$ .

$\therefore$  the line (4) bisects the obtuse angle between the given lines.

## Comprehensive Exercise 5

1. For the lines  $4x + 3y - 6 = 0$  and  $5x + 12y + 9 = 0$ , find the equation of the bisector of the angle which contains the origin.
2. Find the equation of the obtuse angle bisector of the lines  $12x - 5y + 7 = 0$  and  $3y - 4x - 1 = 0$ .
3. Find the bisector of the acute angle between the lines  $3x + 4y - 11 = 0$  and  $12x - 5y - 2 = 0$ .

## Answers

1.  $7x + 9y - 3 = 0$ .
2.  $4x + 7y + 11 = 0$ .
3.  $11x + 3y - 17 = 0$ .

## Comprehensive Exercise 6

### Fill in the Blanks

Fill in the blanks “.....”, so that the following statements are complete and correct.

1. The slope of the line joining the points  $(2, -5)$  and  $(4, 1)$  is .....  
[UPTU 2009]
2. The slope of the line  $y = \frac{1}{3}x\sqrt{3} + 3$  is ..... .
3. The equation of the line passing through the points  $(4, 2)$  and  $(-2, 8)$  is ..... .
4. The equation of the line with slope 4 and passing through  $(4, 3)$  is ..... .
5. The line passing through the point  $(2, 1)$  and parallel to the join of the points  $(1, 3)$  and  $(-3, 1)$  is ..... .
6. The angle  $\theta$  between two lines whose slopes are  $m_1$  and  $m_2$  is given by ..... .
7. The lines  $a_1x + b_1y + c_1 = 0$  and  $a_2x + b_2y + c_2 = 0$  are parallel if ..... .
8. The lines  $a_1x + b_1y + c_1 = 0$  and  $a_2x + b_2y + c_2 = 0$  are perpendicular if ..... .

### Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

9. The slope of the line passing through the points  $(3, 4)$  and  $(-3, 6)$  is
  - (a)  $\frac{1}{3}$
  - (b)  $\frac{-1}{3}$
  - (c)  $\frac{2}{3}$
  - (d)  $\frac{-2}{3}$
10. The equation of a straight line which cuts off intercepts of lengths  $a$  and  $b$  from  $x$ -axis and  $y$ -axis respectively is
  - (a)  $\frac{x}{a} + \frac{y}{b} = 1$
  - (b)  $\frac{x}{b} + \frac{y}{a} = 1$
  - (c)  $x + y = a$
  - (d)  $x + y = b$

11. The equation of the line passing through the points (1, 1) and (5, 4) is  
(a)  $y = 3x + 1$  (b)  $4y = 3x + 1$   
(c)  $4y = 3x - 1$  (d)  $4y = 4x - 3$
12. The equation of the straight line having slope  $m$  and passing through the point  $(x_1, y_1)$  is  
(a)  $y_1 - y = m(x - x_1)$  (b)  $y - y_1 = m(x - x_1)$   
(c)  $x - x_1 = m(y - y_1)$  (d) none of these
13. The equation of a line passing through (2, -3) and inclined at an angle of  $135^\circ$  with the positive direction of  $x$ -axis is  
(a)  $x + y + 1 = 0$  (b)  $x + y - 1 = 0$   
(c)  $x + y = 1/2$  (d)  $x + y = 0$
14. Condition for two given lines to be parallel is  
(a)  $m_1 m_2 = 1$  (b)  $\frac{m_1}{m_2} = \frac{1}{2}$   
(c)  $m_1 = m_2$  (d)  $m_1 m_2 = -1$
15. Condition for two given lines to be perpendicular is  
(a)  $m_1 = m_2$  (b)  $m_1 = -m_2$   
(c)  $m_1 m_2 = 1$  (d)  $m_1 m_2 = -1$  [UPTU 2011]

### True or False

Write 'T' for true and 'F' for false statement

16. The equation of a straight line upon which the length of the perpendicular from the origin is  $p$  and this perpendicular makes an angle  $\alpha$  with positive direction of  $x$ -axis is  $p = x \cos \alpha + y \sin \alpha$
17. The slope of the line joining the two points  $(x_1, y_1)$  and  $(x_2, y_2)$  is  $\frac{y_2 - y_1}{x_2 - x_1}$ .
18. The length of the intercept on  $x$ -axis cut by the line  $4x + 3y = 20$  is 4.
19. The length of the intercept of the line  $2x + 3y = 4$  from  $x$ -axis is 3.
20. The angle between the lines joining the points (0, 0) and (2, 3) and the points (2, -2) and (3, 5) is  $\theta = \tan^{-1} \frac{11}{23}$ .
21. The equation of the line cutting equal intercepts on the axes and passing through the point (6, 9) is  $x + y = 6$ .
-

22. The equation of any straight line perpendicular to a given line  $ax + by + c = 0$  is  $bx + ay + \lambda = 0$ .
23. The equation of any straight line parallel to a given line  $ax + by + c = 0$  is  $ax + by + \lambda = 0$ .
24. The length of the perpendicular drawn from the point  $(x_1, y_1)$  to the line  $x \cos \alpha + y \sin \alpha = p$  is  $x_1 \cos \alpha - y_1 \sin \alpha - p$ .

## Answers

- |                            |                                                  |                                        |                 |
|----------------------------|--------------------------------------------------|----------------------------------------|-----------------|
| 1. 3                       | 2. $30^\circ$                                    | 3. $x + y - 6 = 0$                     | 4. $y = 4x - 3$ |
| 5. $x - 2y = 0$            | 6. $\tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2}$ | 7. $\frac{a_1}{a_2} = \frac{b_1}{b_2}$ |                 |
| 8. $a_1 a_2 + b_1 b_2 = 0$ | 9. (b)                                           | 10. (a)                                | 11. (b)         |
| 12. (b)                    | 13. (a)                                          | 14. (c)                                | 15. (d)         |
| 16. T                      | 17. F                                            | 18. F                                  | 19. F           |
| 20. T                      | 21. F                                            | 22. F                                  | 23. T           |
| 24. F                      |                                                  |                                        |                 |



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## Chapter

# 8



## Functions And Limits

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### 8.1 Function

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A **constant** is a quantity which, during any set of mathematical operations, retains the same value.

A **variable** is a quantity, or a symbol representing a number, which is capable of assuming different values.

A **continuous variable** is a variable that can take all the numerical values between two given numbers.

An **independent variable** is one which may take up any arbitrary value that may be assigned to it.

A **dependent variable** is a variable which can assume its value as a result of some other variable taking some assigned value.

A **set** is any well-defined class or collection of objects. The objects constituting the set are also called **elements** or **members** of the set.

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**Set Theoretic Definition Of A Function.** Let  $A$  and  $B$  be any two given non-empty sets. Suppose there exists a correspondence denoted by  $f$ , which associates to **each** member of  $A$  a **unique** member of  $B$ . Then  $f$  is called a **function** or a **mapping** from  $A$  to  $B$ .

The mapping  $f$  of  $A$  to  $B$  is denoted by  $f : A \rightarrow B$ . The set  $A$  is called the **domain** of the function  $f$ , and  $B$  is called the **co-domain** of  $f$ . The element  $y \in B$  which the mapping  $f$  associates to an element  $x \in A$  is symbolically denoted by  $y = f(x)$  and is called the  **$f$ -image** of  $x$  or the **value** of the function  $f$  at  $x$  or  $y$  is a function of  $x$  or  $y$  equals  $f$  of  $x$ . Each element of  $A$  has a unique image and each element of  $B$  need not appear as the image of an element in  $A$ . We define the **range** of  $f$  to consist of those elements in  $B$  which appear as the image of at least one element in  $A$ . It is denoted by  $f(A)$ . This symbolic representation was invented by Leonhard Euler.

Thus,  $\text{range } f = f(A) = \{f(x) : x \in A\}$ . The range of a mapping  $f : A \rightarrow B$  is always a subset of the co-domain  $B$  of  $f$  i.e.,  $f(A) \subseteq B$ .

### How to find the domain of a function ?

The domain of a function  $f(x)$  is the set of those real numbers  $x$  for which  $f(x)$  is defined i.e., for which  $f(x)$  is a unique real number. Sometimes a function is described only by a formula and the domain of the function is not explicitly stated. In such cases we consider the set of all real numbers  $x$  for which  $f(x)$  is meaningful i.e., for which  $f(x)$  is a unique real number, as the domain of  $f$ .

### How to find the range of a function ?

The range of a function  $f$  is the set of all values taken by  $f(x)$  when  $x$  takes values in the domain of  $f$ .

Thus,  $\text{range } f = \{y \in R : y = f(x) \text{ for some } x \in \text{domain } f\}$ .

So to find the range of a function  $f(x)$  we put  $y = f(x)$  and try to solve this equation to get  $x$  in terms of  $y$  in the form  $x = \phi(y)$ . Then we find those real numbers  $y$  for which  $\phi(y)$  is in the domain of  $f$ . The set of these real numbers  $y$  constitute the range of  $f$ .

## Illustrative Examples

**Illustration 1 :** State the domain of the function  $f(x) = \frac{x^2 + 3x + 1}{x^2 - 5x + 6}$ . [UPTU 2007]

**Solution :** Obviously  $f(x)$  is not defined when  $x^2 - 5x + 6 = 0$ .

Now  $x^2 - 5x + 6 = 0 \Rightarrow (x - 2)(x - 3) = 0 \Rightarrow x = 2 \text{ or } x = 3$ .

So  $f(x)$  is not defined at  $x = 2, 3$ . Hence,  $\text{domain } f = R - \{2, 3\}$ .

**Illustration 2 :** Find the domain of the function  $f(x) = \sqrt{x^2 - 9}$ .

**Solution :** Obviously  $f(x)$  is a real number if and only if  $x^2 - 9 \geq 0$ . So the domain of  $f(x)$  is the set of those real numbers  $x$  for which  $x^2 - 9 \geq 0$ .

$$\text{Now } x^2 - 9 \geq 0 \Rightarrow x^2 \geq 9 \Rightarrow |x|^2 \geq 9$$

$$\Rightarrow |x| \geq 3 \Rightarrow x \geq 3 \text{ or } x \leq -3 \Rightarrow x \in (-\infty, -3] \cup [3, \infty).$$

Hence, domain  $f = (-\infty, -3] \cup [3, \infty)$ .

**Illustration 3 :** Find the domain and range of the function  $f(x) = \frac{1+x}{1-x}$ .

**Solution :** Obviously  $f(x)$  is not defined when  $1-x=0$ , i.e., when  $x=1$  and at every other real number  $x$ ,  $f(x)$  is a unique real number. So

$$\text{domain } f = R - \{1\} = (-\infty, 1) \cup (1, \infty).$$

Now to find range  $f(x)$ , we put  $y = f(x)$ .

$$\text{Then } y = \frac{1+x}{1-x} \Rightarrow y(1-x) = 1+x \Rightarrow x(y+1) = y-1$$

$$\Rightarrow x = \frac{y-1}{y+1}, \text{ provided } y+1 \neq 0.$$

Obviously when  $y+1=0$  i.e., when  $y=-1$ ,  $x$  is not a real number and so it is not in the domain of  $f$ . For all real numbers  $y$  except  $y=-1$ ,  $x$  is in the domain of  $f$ . Hence, range  $f = R - \{-1\}$ .

**Illustration 4 :** If  $f(x) = x^2 + 3x + 2$ , find that value of  $x$  for which  $f(x+1) = f(x)$ .

**Solution :** We have  $f(x) = x^2 + 3x + 2$ .

$$\therefore f(x+1) = (x+1)^2 + 3(x+1) + 2 = x^2 + 5x + 6.$$

$$\text{Now } f(x+1) = f(x) \Rightarrow x^2 + 5x + 6 = x^2 + 3x + 2$$

$$\Rightarrow 2x = -4 \Rightarrow x = -2.$$

**Illustration 5 :** If  $f(x) = \frac{1-x}{1+x}$ , show that  $f(\cos \theta) = \tan^2 \frac{\theta}{2}$ .

**Solution :** We have  $f(x) = \frac{1-x}{1+x}$ .

$$\therefore f(\cos \theta) = \frac{1-\cos \theta}{1+\cos \theta} = \frac{2 \sin^2 \frac{1}{2} \theta}{2 \cos^2 \frac{1}{2} \theta} = \tan^2 \frac{\theta}{2}.$$

**Illustration 6 :** If  $f(x) = \log \left( \frac{1+x}{1-x} \right)$ , show that  $f\left(\frac{2x}{1+x^2}\right) = 2f(x)$ .

**Solution :** We have  $f(x) = \log \left( \frac{1+x}{1-x} \right)$ .



$$\begin{aligned} \therefore f\left(\frac{2x}{1+x^2}\right) &= \log \left\{ \frac{1 + \frac{2x}{1+x^2}}{1 - \frac{2x}{1+x^2}} \right\} = \log \frac{1+x^2+2x}{1+x^2-2x} \\ &= \log \frac{(1+x)^2}{(1-x)^2} = \log \left( \frac{1+x}{1-x} \right)^2 = 2 \log \left( \frac{1+x}{1-x} \right) = 2f(x). \end{aligned}$$

**Illustration 7 :** If  $f(x) = \log \left( \frac{1+x}{1-x} \right)$  show that  $f(x) + f(y) = f\left(\frac{x+y}{1+xy}\right)$ .

[UPTU 2005]

**Solution :** Given  $f(x) = \log \left( \frac{1+x}{1-x} \right)$  and  $f(y) = \log \left( \frac{1+y}{1-y} \right)$ .

$$\begin{aligned} \text{We have } f\left(\frac{x+y}{1+xy}\right) &= \log \left( \frac{1 + \left( \frac{x+y}{1+xy} \right)}{1 - \left( \frac{x+y}{1+xy} \right)} \right) \\ &= \log \left[ \frac{1+xy+x+y}{1+xy-x-y} \right] = \log \left( \frac{1+x+y+xy}{1-x-y+xy} \right) \quad \dots(1) \end{aligned}$$

$$\begin{aligned} \text{and } f(x) + f(y) &= \log \left( \frac{1+x}{1-x} \right) + \log \left( \frac{1+y}{1-y} \right) \\ &= \log \left[ \left( \frac{1+x}{1-x} \right) \left( \frac{1+y}{1-y} \right) \right] = \log \left[ \frac{1+x+y+xy}{1-x-y+xy} \right]. \quad \dots(2) \end{aligned}$$

From (1) and (2), we get

$$f(x) + f(y) = f\left(\frac{x+y}{1+xy}\right).$$

**Illustration 8 :** Find the domain of the following functions :

$$(i) f(x) = \frac{1}{\sqrt{9-x^2}} \quad (ii) f(x) = \sqrt{(x-2)(x+5)}.$$

**Solution :** (i) We have  $f(x) = \frac{1}{\sqrt{9-x^2}}$ .

Obviously  $f(x)$  is not defined for those real numbers  $x$  for which  $9-x^2 \leq 0$  i.e.,  $x^2 \geq 9$ .

Thus,  $f(x)$  is defined only when  $x^2 < 9$ .

$$\text{Now } x^2 < 9 \Rightarrow |x|^2 < 9 \Rightarrow |x| < 3 \Rightarrow -3 < x < 3.$$

$$\therefore \text{Dom } f = ]-3, 3[.$$

(ii) We have  $f(x) = \sqrt{(x-2)(x+5)}$ .

Obviously  $f(x)$  is defined only for those real numbers  $x$  for which

$$(x-2)(x+5) \geq 0.$$

Now  $(x-2)(x+5) \geq 0 \Rightarrow [x - (-5)](x-2) \geq 0$

$\Rightarrow x \leq -5 \text{ or } x \geq 2 \Rightarrow x \in ]-\infty, -5] \cup [2, \infty[.$

Hence,  $\text{dom } f = ]-\infty, -5] \cup [2, \infty[.$

## Comprehensive Exercise 1

1. If  $f(x) = \frac{\sin x}{1 + \sin x}$ , find  $f\left(\frac{\pi}{2}\right)$ .
2. If  $f(x) = (x-a)^2(x-b)^2$ , find  $f(a+b)$ .
3. If  $f(x) = x^2 - 3x + 4$ , find the values of  $x$  for which  $f(x) = f(2x+1)$ .
4. If  $f(x) = x + \frac{1}{x}$ , prove that  $[f(x)]^3 = f(x^3) + 3f\left(\frac{1}{x}\right)$ .
5. If  $f(x) = x^3 - \frac{1}{x^3}$ , find the value of  $f(x) + f\left(\frac{1}{x}\right)$ .
6. If  $f(x) = \frac{2x}{1+x^2}$ , prove that  $f(\tan \theta) = \sin 2\theta$ .
7. If  $f(x) = \frac{1-x^2}{1+x^2}$ , prove that  $f(\tan \theta) = \cos 2\theta$ .
8. If  $f(x) = \log\left(\frac{x}{x-1}\right)$ , show that  $f(x) + f(x+1) = \log\left(\frac{x+1}{x-1}\right)$ .
9. Find the domain and the range of the constant function  $f(x) = 5$ .
10. For what value or values of  $x$ , if any, are the following functions not defined?  

|                                     |                                           |
|-------------------------------------|-------------------------------------------|
| (i) $f(x) = \frac{\sin x}{x}$       | (ii) $f(x) = \sin \frac{1}{x}$            |
| (iii) $f(x) = \frac{1}{\sqrt{x-7}}$ | (iv) $f(x) = \frac{1}{\sqrt{(1-x)(x-2)}}$ |
| (v) $f(x) = \frac{1}{x^3 + x}$      | (vi) $f(x) = \sqrt{1-x^2}$ .              |
11. Find the domain and range of the following functions :  

|                                  |                                       |
|----------------------------------|---------------------------------------|
| (i) $f(x) = \frac{1}{x^2 - 1}$   | (ii) $f(x) = \frac{x}{1-x}$           |
| (iii) $f(x) = \frac{x^2}{1+x^2}$ | (iv) $f(x) = \frac{1}{2 - \cos 3x}$ . |

12. Find the domain and the range of the following functions.

$$(i) f(x) = \frac{|x-3|}{x-3} \quad (ii) f(x) = \frac{3x-5}{4x-7}$$

$$(iii) f(x) = 3 \sin \sqrt{(\pi^2/16) - x^2}.$$

## Answers

1.  $\frac{1}{2}$ .      2.  $a^2 b^2$ .      3.  $-1, 2/3$ .      5. 0.
9. Dom  $f = R$ , range  $f = \{5\}$ .
10. (i)  $x = 0$  (ii)  $x = 0$  (iii)  $x \leq 7$   
 (iv)  $x \leq 1$  or  $x \geq 2$  (v)  $x = 0$  (vi)  $x < -1$  or  $x > 1$ .
11. (i) dom  $f = R - \{-1, 1\}$ , range  $f = R - ]-1, 0]$   
 (ii) dom  $f = R - \{1\}$ , range  $f = R - \{-1\}$   
 (iii) dom  $f = R$ , range  $f = [0, 1[$   
 (iv) dom  $f = R$ , range  $f = \left[ \frac{1}{3}, 1 \right]$ .
12. (i) dom  $f = R - \{3\}$ , range  $f = \{-1, 1\}$ .  
 (ii) dom  $f = R - \{7/4\}$ , range  $f = R - \{3/4\}$ .  
 (iii) dom  $f = \left[ -\frac{\pi}{4}, \frac{\pi}{4} \right]$ , range  $f = \left[ 0, \frac{3}{\sqrt{2}} \right]$ .

## 8.2 Types of Functions

**Even and odd functions.** A function  $f(x)$  is said to be an **even function**, if  $f(-x) = f(x)$  for all  $x \in \text{domain } f$ .

A function  $f(x)$  is said to be an **odd function**, if

$$f(-x) = -f(x) \text{ for all } x \in \text{domain } f.$$

In the above definitions it is implied that if  $x$  is in domain  $f$ , then  $-x$  is also in domain  $f$ .

For example,  $f(x) = \cos x$  is an even function because

$$f(-x) = \cos(-x) = \cos x = f(x) \text{ for all } x \in R.$$

Similarly  $f(x) = \sin^2 x$  and  $f(x) = x^2 + 1$  are also even functions.

The function  $f(x) = \sin x$  is an odd function because  $f(-x) = \sin(-x) = -\sin x$  for all  $x \in R$ . Similarly  $f(x) = x^3$  is an odd function.

**Bounded functions.** A function  $f(x)$  defined on a domain  $D$  is said to be **bounded on**  $D$ , if there exists a positive real number  $k$  such that  $|f(x)| \leq k$  for all  $x \in D$ .

For example, the function  $f(x) = \sin x$  is bounded on  $R$  because

$$|\sin x| \leq 1 \text{ for all } x \in R.$$

**‘Increasing’ or ‘decreasing’ functions.** A function  $f(x)$  is said to be an **increasing function** on a domain  $D$ , if  $x_1 > x_2 \Rightarrow f(x_1) \geq f(x_2)$  for all  $x_1, x_2 \in D$ .

Similarly, a function  $f(x)$  is said to be a **decreasing function** on a domain  $D$  if  $x_1 > x_2 \Rightarrow f(x_1) \leq f(x_2)$  for all  $x_1, x_2 \in D$ .

### 8.3 Composite Functions

Let  $f$  and  $g$  be two functions with domains  $D_1$  and  $D_2$  respectively.

If range  $f \subseteq \text{domain } g$ , then the **composite mapping**  $g \circ f$  of the mappings  $f$  and  $g$  is defined by the formula

$$(g \circ f)(x) = g(f(x)) \text{ for all } x \in D_1.$$

Also, if range  $g \subseteq \text{domain } f$ , the composite mapping  $f \circ g$  of the mappings  $f$  and  $g$  is defined by the formula  $(f \circ g)(x) = f(g(x))$  for all  $x \in D_2$ .

### Illustrative Examples

**Illustration 1 :** Let  $f(x) = \sin x$ ,  $g(x) = 2x$  and  $h(x) = \cos x$ .

Prove that  $f \circ g = g \circ (f \circ h)$ .

**Solution :** Obviously  $\text{dom}(f \circ g) = \text{dom } g = R$

and  $\text{dom}[g \circ (f \circ h)] = \text{dom}(f \circ h) = \text{dom } f \cap \text{dom } h = R \cap R = R$ .

Thus,  $\text{dom}(f \circ g) = \text{dom}[g \circ (f \circ h)] = R$ .

Now for all  $x \in R$ , we have  $(f \circ g)(x) = f(g(x)) = f(2x) = \sin 2x$  ... (1)

and  $[g \circ (f \circ h)](x) = g[(f \circ h)(x)] = g[f(x)h(x)]$   
 $= g(\sin x \cos x) = 2(\sin x \cos x) = \sin 2x$ . ... (2)

From (1) and (2), we have  $(f \circ g)(x) = [g \circ (f \circ h)](x)$ ,  $\forall x \in R$ .

So by definition of equality of two functions, we have  $f \circ g = g \circ (f \circ h)$ .

**Illustration 2 :** Let  $f(x) = x^2$  and  $g(x) = \cos x$  for all  $x \in R$ . Show that  $g \circ f \neq f \circ g$ .

**Solution :** Here obviously  $\text{dom}(g \circ f) = \text{dom } f = R$

and  $\text{dom}(f \circ g) = \text{dom } g = R$ .

We have  $(g \circ f)(x) = g(f(x)) = g(x^2) = \cos x^2$

and  $(f \circ g)(x) = f(g(x)) = f(\cos x) = (\cos x)^2 = \cos^2 x$ .

Since there exist real numbers  $x$  for which  $\cos x^2 \neq \cos^2 x$  i.e.,

$(g \circ f)(x) \neq (f \circ g)(x)$ , therefore  $g \circ f \neq f \circ g$ .

**Illustration 3 :** Let  $f(x) = \begin{cases} 1+x, & 0 \leq x \leq 2 \\ 3-x, & 2 < x \leq 3. \end{cases}$  Find  $f \circ f$ .

**Solution :** When  $0 \leq x \leq 1$ , we have

$$(f \circ f)(x) = f(f(x)) = f(1+x) = 1 + (1+x) = 2+x$$

$$[\because 0 \leq x \leq 1 \Rightarrow 1 \leq 1+x \leq 2]$$

When  $1 < x \leq 2$ , we have

$$(f \circ f)(x) = f(f(x)) = f(1+x) = 3 - (1+x) = 2-x$$

$$[\because 1 < x \leq 2 \Rightarrow 2 < 1+x \leq 3]$$

and when  $2 < x \leq 3$ , we have

$$(f \circ f)(x) = f(f(x)) = f(3-x) = 1 + (3-x) = 4-x.$$

$$[\because 2 < x \leq 3 \Rightarrow 0 \leq 3-x < 1]$$

$$\text{Hence, } (f \circ f)(x) = \begin{cases} 2+x, & 0 \leq x \leq 1 \\ 2-x, & 1 < x \leq 2 \\ 4-x, & 2 < x \leq 3. \end{cases}$$

**Illustration 4 :** Test the nature of the following functions for even or odd:

$$(i) f(x) = x^5 + 7x - 4 \sin^3 x \quad (ii) f(x) = e^x - e^{-x} + \sin x$$

$$(iii) f(x) = x + \frac{1}{x}, x \neq 0 \quad (iv) f(x) = x \frac{e^x - 1}{e^x + 1}$$

$$(v) f(x) = \frac{|x|}{x}, x \neq 0 \quad (vi) f(x) = \log(x + \sqrt{x^2 + 1}).$$

**Solution :** (i) We have  $f(x) = x^5 + 7x - 4 \sin^3 x$ .

$$\therefore f(-x) = (-x)^5 + 7(-x) - 4 \sin^3(-x)$$

$$= -x^5 - 7x + 4 \sin^3 x = -(x^5 + 7x - 4 \sin^3 x) = -f(x).$$

Hence,  $f(x)$  is an odd function.

(ii) We have  $f(x) = e^x - e^{-x} + \sin x$ .

$$\therefore f(-x) = e^{-x} - e^{-(-x)} + \sin(-x)$$

$$= e^{-x} - e^x - \sin x = -(e^x - e^{-x} + \sin x) = -f(x).$$

Hence,  $f(x)$  is an odd function.

(iii) We have  $f(x) = x + \frac{1}{x}, x \neq 0$ .

$$\therefore f(-x) = -x + \frac{1}{(-x)} = -x - \frac{1}{x} = -\left(x + \frac{1}{x}\right) = -f(x).$$

Hence,  $f(x)$  is an odd function.

(iv) We have  $f(x) = x \frac{e^x - 1}{e^x + 1} = x \frac{e^{x/2} - e^{-x/2}}{e^{x/2} + e^{-x/2}}.$

$$\therefore f(-x) = (-x) \frac{e^{-x/2} - e^{x/2}}{e^{-x/2} + e^{x/2}} = (-x)(-1) \frac{e^{x/2} - e^{-x/2}}{e^{x/2} + e^{-x/2}}$$

$$= x \frac{e^{x/2} - e^{-x/2}}{e^{x/2} + e^{-x/2}} = f(x).$$

Hence,  $f(x)$  is an even function.

(v) We have  $f(x) = \frac{|x|}{x}, x \neq 0$ .

$$\therefore f(-x) = \frac{|-x|}{-x} = \frac{|x|}{-x} = -\frac{|x|}{x} = -f(x).$$

Hence,  $f(x)$  is an odd function.

(vi) We have  $f(x) = \log(x + \sqrt{x^2 + 1}).$

$$\begin{aligned} \therefore f(-x) &= \log(-x + \sqrt{x^2 + 1}) \\ &= \log \left[ \frac{-x + \sqrt{x^2 + 1}}{x + \sqrt{x^2 + 1}} \cdot (x + \sqrt{x^2 + 1}) \right] \\ &= \log \left[ \frac{(x^2 + 1) - x^2}{x + \sqrt{x^2 + 1}} \right] = \log \left[ \frac{1}{x + \sqrt{x^2 + 1}} \right] \\ &= -\log(x + \sqrt{x^2 + 1}) = -f(x). \end{aligned}$$

Hence,  $f(x)$  is an odd function.

## Comprehensive Exercise 2

- If  $f(x) = e^x$  and  $g(x) = \log_e x$ , find
  - $(f + g)(1)$
  - $(fg)(1)$
  - $(f \circ g)(1)$
  - $(g \circ f)(1)$
  - $(4f)(1)$
- If  $f(x) = \cos^2 x$  and  $g(x) = 3 \sin x$ , find
  - $(f + g)(\pi/3)$
  - $(f \cdot g)(\pi/3)$
- Let  $f(x) = x^2 + x$  and  $g(x) = 3 - x$ . Describe
  - $f + g$
  - $f - g$
  - $fg$
  - $f / g$

Find the domain in each case.
- Test the nature of the following functions for even or odd :
  - $f(x) = x \sin x$
  - $f(x) = x \sqrt{x^2 - 1}$
  - $f(x) = \sin^2 x \cos x$
  - $f(x) = x^2 - |x|$

## Answers

- $e$
  - $0$
  - $1$
  - $1$
  - $4e$
- $\frac{6\sqrt{3} + 1}{4}$
  - $\frac{3\sqrt{3}}{8}$
- $(f + g)(x) = x^2 + 3, \text{dom}(f + g) = R$
  - $(f - g)(x) = x^2 + 2x - 3, \text{dom}(f - g) = R$

$$(iii) (fg)(x) = 3x + 2x^2 - x^3, \text{dom}(fg) = R$$

$$(iv) \left(\frac{f}{g}\right)(x) = \frac{x^2 + x}{3 - x}, \text{dom}\left(\frac{f}{g}\right) = R - \{3\}.$$

4. (i) even (ii) odd  
(iii) even (iv) even.

## 8.4 Graphs of Functions

To draw the graph of a function  $f(x)$ , we put  $y = f(x)$ . Then the aggregate of the points  $(x, y)$  satisfying the equation  $y = f(x)$  gives us the graph of the function  $f(x)$ . In practice we plot some of the points and join them by a smooth free hand curve to get the graph.

**An important property of the graph of an even function.** *The graph of an even function  $f(x)$  is symmetrical about the  $y$ -axis.* Since  $f(-x) = f(x)$ , therefore the equation  $y = f(x)$  remains unchanged on replacing  $x$  by  $-x$ . So, if the point  $(x, y)$  lies on the curve  $y = f(x)$ , the point  $(-x, y)$  also lies on it and consequently the curve is symmetrical about  $y$ -axis.

**An important property of the graph of an odd function.** *The graph of an odd function  $f(x)$  is symmetrical in opposite quadrants i.e., symmetrical about the origin.* Since  $f(-x) = -f(x)$ , therefore the equation  $y = f(x)$  remains unchanged on replacing  $x$  by  $-x$  and  $y$  by  $-y$ . So, if the point  $(x, y)$  lies on the curve  $y = f(x)$ , the point  $(-x, -y)$  also lies on it and consequently the curve is symmetrical in opposite quadrants.

## Illustrative Examples

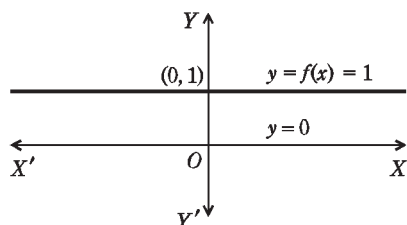
**Illustration 1 :** Draw the graphs of the constant functions

- (i)  $f(x) = 1$  (ii)  $f(x) = 0$ .

**Solution :** (i) Putting  $y = f(x)$ , the graph

of the function  $f(x) = 1$  is the straight line  $y = 1$ . It is parallel to  $x$ -axis and is at a distance 1 above  $x$ -axis.

(ii) The graph of the function  $f(x) = 0$  is the straight line  $y = 0$  which is  $x$ -axis.



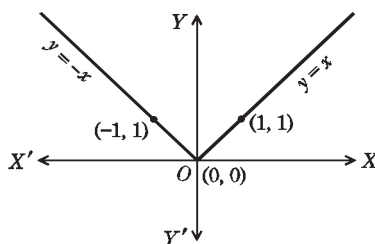
**Remark.** The graph of the constant function  $f(x) = c$  is always a straight line parallel to  $x$ -axis. It is above  $x$ -axis if  $c > 0$  and is below  $x$ -axis if  $c < 0$ . If  $c = 0$ , it coincides with  $x$ -axis.

**Illustration 2 :** Draw the graph of the modulus function  $f(x) = |x|$ .

**Solution :** We have  $f(x) = |x| = \begin{cases} x, & x \geq 0 \\ -x, & x \leq 0. \end{cases}$

Putting  $y = f(x)$ , the graph of the function  $f(x) = |x|$  consists of the straight lines  $y = x$ , when  $x \geq 0$  and  $y = -x$ , when  $x \leq 0$ .

We have  $f(-x) = |-x| = |x| = f(x)$  and so  $f(x)$  is an even function of  $x$ . Hence, the graph is symmetrical about  $y$ -axis. Also  $|x| \geq 0$  for all  $x \in \mathbb{R}$  and so the graph does not lie below  $x$ -axis.



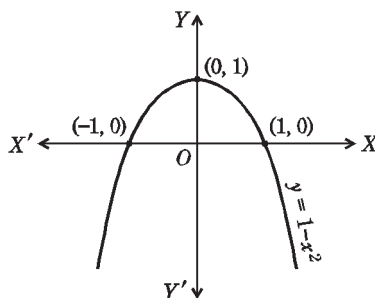
The line  $y = x$  has been drawn in the region where  $x \geq 0$ . It passes through the points  $(0, 0)$  and  $(1, 1)$ . Its slope is 1.

The line  $y = -x$  has been drawn in the region where  $x \leq 0$ . It passes through the points  $(0, 0)$  and  $(-1, 1)$  and its slope is  $-1$  so that it makes an angle of  $135^\circ$  with the positive direction of  $x$ -axis.

**Illustration 3 :** Draw the graph of the polynomial functions  $f(x) = 1 - x^2$ .

**Solution :** Putting  $y = f(x)$ , the graph of the function  $f(x) = 1 - x^2$  is the curve  $y = 1 - x^2$  which is the parabola  $x^2 = -(y - 1)$ .

Since  $f(x)$  is an even function of  $x$ , the graph is symmetrical about  $y$ -axis. The vertex of the parabola is the point  $(0, 1)$  and its axis is the straight line  $x = 0$  i.e., the  $y$ -axis.



The curve passes through the points  $(1, 0)$  and  $(-1, 0)$ . If  $y > 1$ , then  $x$  is not real and so the curve does not lie above the straight line  $y = 1$ .

**Remark.** If the function  $f(x) = ax^2 + bx + c$ ,  $a \neq 0$ , i.e.,  $f(x)$  is a quadratic polynomial, then its graph is always a parabola.

**Illustration 4 :** Draw the graph of the function  $f(x) = \frac{1}{2}x^2$  over the interval  $[-3, 3]$ .

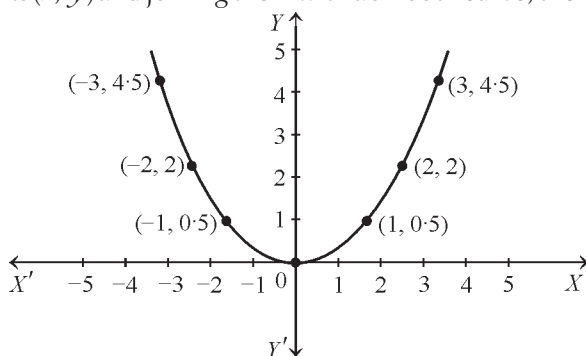
**Solution :** Putting  $y = f(x)$ , the graph of the function  $f(x) = \frac{1}{2}x^2$  is the curve  $y = \frac{1}{2}x^2$  which is a parabola. Since  $f(x)$  is an even function of  $x$ , the graph is symmetrical about  $y$ -axis.

The vertex of the parabola is at the origin and its axis coincides with the  $y$ -axis. Since  $y = f(x) \geq 0$  for all  $x \in \mathbb{R}$ , therefore the graph does not lie below  $x$ -axis. In the first quadrant as  $x$  increases,  $y$  also increases.

| $x$                  | -3  | -2 | -1  | 0 | 1   | 2 | 3   |
|----------------------|-----|----|-----|---|-----|---|-----|
| $y = \frac{1}{2}x^2$ | 4.5 | 2  | 0.5 | 0 | 0.5 | 2 | 4.5 |



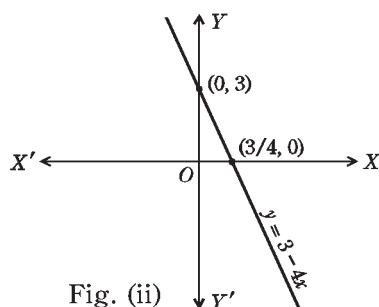
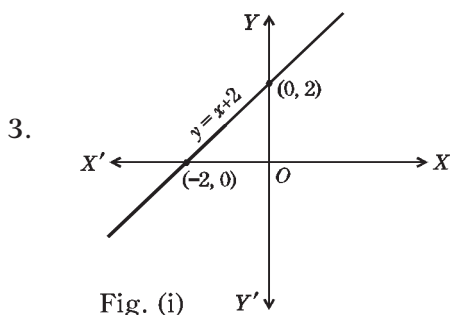
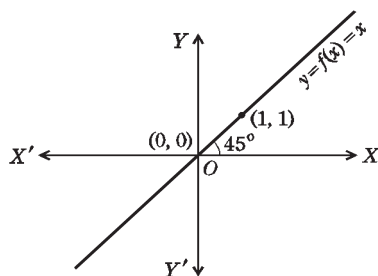
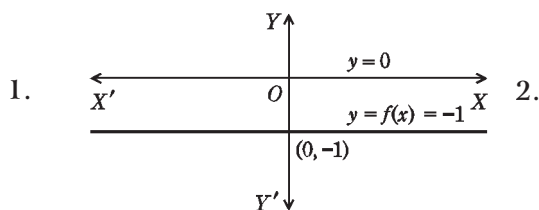
Plotting the points  $(x, y)$  and joining them with a smooth curve, the required graph is



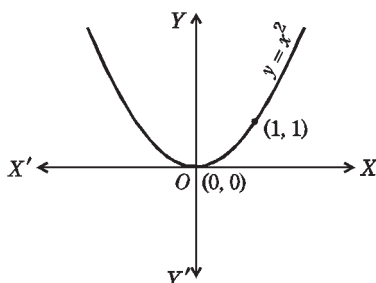
## Comprehensive Exercise 3

1. Draw the graph of the constant function  $f(x) = -1$ .
2. Draw the graph of the identity function  $f(x) = x$ .
3. Draw the graphs of the linear functions
  - (i)  $f(x) = 2 + x$
  - (ii)  $f(x) = 3 - 4x$ .
4. Draw the graph of the polynomial function  $f(x) = x^2$ .

## Answers



4.



## 8.5 Limit of a Function at a Point

Consider the function  $y = \frac{x^2 - 1}{x - 1}$ . The value of this function at  $x = 1$  is of the form  $\frac{0}{0}$  which is meaningless. In this case we cannot divide the numerator by the denominator since  $x - 1$  is zero. Now suppose  $x$  is not actually equal to 1 but tends to 1. Then  $x - 1$  is not equal to zero. Hence, in this case we can divide the numerator by the denominator.

$$\therefore \frac{x^2 - 1}{x - 1} = \frac{(x - 1)(x + 1)}{x - 1} = x + 1.$$

If  $x$  is little greater or smaller than 1, then the value of  $y$  will be little greater or smaller than 2 and as  $x$  gets nearer to 1,  $y$  comes nearer to 2. Now the difference between  $y$  and 2 is

$$\left| \frac{x^2 - 1}{x - 1} - 2 \right| = \left| \frac{x^2 - 2x + 1}{x - 1} \right| = \left| \frac{(x - 1)^2}{x - 1} \right| = |x - 1|.$$

This difference  $|x - 1|$  can be made as small as we please by letting  $x$  tend to 1.

Thus, we see that when  $x$  has a fixed value 1, the value of  $y$  is meaningless but when  $x$  tends to 1,  $y$  tends to 2 and we say that the limit of  $y$  is 2 when  $x$  tends to 1. Symbolically we write

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = 2.$$

**Definition Of Limit Of A Function At A Point.** Let  $f$  be a function defined on some neighbourhood of a point  $a$  except possibly at  $a$  itself so that  $x$  may tend to  $a$ . Then a real number  $l$  is said to be the **limit** of  $f$  at  $x = a$  or the limit of  $f$  as  $x$  tends to  $a$  if for any arbitrarily chosen positive number  $\epsilon$ , there exists a corresponding number  $\delta > 0$  such that

$$|f(x) - l| < \epsilon \quad \text{i.e.,} \quad l - \epsilon < f(x) < l + \epsilon$$

for all values of  $x$  for which  $0 < |x - a| < \delta$  i.e.,  $a - \delta < x < a$  or  $a < x < a + \delta$ .

Symbolically we write  $\lim_{x \rightarrow a} f(x) = l$ ,

which is read as limit of  $f(x)$ , as  $x$  tends to  $a$ , is  $l$ .

## 8.6 Algebra of Limits

**Theorem :** Let  $\lim_{x \rightarrow a} f(x) = l$  and  $\lim_{x \rightarrow a} g(x) = m$ . Then

$$(i) \lim_{x \rightarrow a} \{f(x) + g(x)\} = l + m = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x).$$

$$(ii) \lim_{x \rightarrow a} \{f(x) - g(x)\} = l - m = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x).$$

$$(iii) \lim_{x \rightarrow a} \{f(x) g(x)\} = lm = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x).$$

In particular,

$$\lim_{x \rightarrow a} \{f(x)\}^2 = \left[ \lim_{x \rightarrow a} f(x) \right]^2,$$

$$\lim_{x \rightarrow a} \{f(x)\}^3 = \left[ \lim_{x \rightarrow a} f(x) \right]^3, \text{ etc.}$$

$$(iv) \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{l}{m} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}, \text{ provided } m \neq 0.$$

$$(v) \lim_{x \rightarrow a} c f(x) = c l = c \lim_{x \rightarrow a} f(x), \text{ where } c \text{ is some scalar i.e., } c \text{ is any real number.}$$

$$(vi) \lim_{x \rightarrow a} |f(x)| = |l| = \left| \lim_{x \rightarrow a} f(x) \right|.$$

$$(vii) \lim_{x \rightarrow a} |f(x)| = 0 \Leftrightarrow \lim_{x \rightarrow a} f(x) = 0.$$

$$(viii) \lim_{x \rightarrow a} f(x)^{g(x)} = l^m.$$

## 8.7 Some Important Expansions

(i) **Binomial Expansion.**

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!} x^2 + \frac{n(n-1)(n-2)}{3!} x^3 + \dots$$

If  $n$  is a positive integer, the number of terms in the expansion is  $n + 1$ . If  $n$  is not a positive integer, the expansion is an infinite series and in that case we must have  $|x| < 1$ .

$$(ii) \quad e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots, \quad \forall x \in R.$$

$$(iii) \quad e^{-x} = 1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots, \quad \forall x \in R.$$

$$(iv) \quad a^x = 1 + x(\log a) + \frac{x^2}{2!}(\log a)^2 + \dots, \quad \forall x \in R.$$

$$(v) \quad \log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots, \quad -1 < x \leq 1.$$

$$(vi) \quad \log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \dots, \quad -1 \leq x < 1.$$

$$(vii) \quad \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots, \quad \forall x \in R.$$

$$(viii) \quad \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots, \quad \forall x \in R.$$

$$(ix) \quad \tan x = x + \frac{x^3}{3} + \frac{2}{15}x^5 + \dots$$

### Some Important Factors.

$$(i) \quad a^2 - b^2 = (a - b)(a + b)$$

$$(ii) \quad a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

$$(iii) \quad a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$(iv) \quad a^4 - b^4 = (a^2 - b^2)(a^2 + b^2) = (a - b)(a + b)(a^2 + b^2)$$

$$(v) \quad a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + a^{n-3}b^2 + \dots + ab^{n-2} + b^{n-1})$$

(vi) If  $f(x)$  is a polynomial, then  $f(a) = 0 \Leftrightarrow (x - a)$  is a factor of  $f(x)$ . For example, if  $f(x) = x^3 - 6x^2 + 11x - 6$ , then  $x - 1$  is a factor of  $f(x)$  because  $f(1) = 0$ .

## 8.8 Some Important Properties of Limits

$$(i) \quad \lim_{x \rightarrow a} f(x) = \lim_{h \rightarrow 0} f(a + h), \text{ where } h \text{ may be positive or negative.}$$

$$(ii) \quad \lim_{x \rightarrow a} f(x) = \lim_{h \rightarrow 0} f(a - h), \text{ where } h \text{ may be positive or negative.}$$

$$(iii) \quad \text{If } f \text{ and } g \text{ are two functions such that}$$

$$\lim_{x \rightarrow a} g(x) = l \quad \text{and} \quad \lim_{y \rightarrow l} f(y) = m, \quad \text{then} \quad \lim_{x \rightarrow a} f(g(x)) = m.$$

## 8.9 Some Standard Limits

- (i)  $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = n a^{n-1}$ , where  $n$  is a positive integer.
- (ii)  $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = n a^{n-1}$ , where  $a > 0$  and  $n$  is any real number.
- (iii)  $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$ .
- (iv)  $\lim_{x \rightarrow 0} \frac{e^{ax} - 1}{ax} = 1, a \neq 0$ .
- (v)  $\lim_{x \rightarrow 0} e^x = 1$ .
- (vi)  $\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log_e a$ .
- (vii)  $\lim_{x \rightarrow 0} \frac{a^{bx} - 1}{bx} = \log_e a, b \neq 0$ .
- (viii)  $\lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1$ .
- (ix)  $\lim_{x \rightarrow 0} \frac{\log(1+ax)}{ax} = 1, a \neq 0$ .
- (x)  $\lim_{x \rightarrow 0} \log(1+x) = 0$ .
- (xi)  $\lim_{x \rightarrow 0} (1+x)^{1/x} = e$ .
- (xii)  $\lim_{x \rightarrow 0} (1+ax)^{1/x} = e^a$ .

## 8.10 Direct Substitution Method

### Illustrative Examples

**Illustration 1 :** Evaluate (i)  $\lim_{x \rightarrow 3} x^2$ , (ii)  $\lim_{x \rightarrow a} (4x^2 - 6x + 7)$ ,

(iii)  $\lim_{x \rightarrow -1} (2x^3 - 3x^2 + 4x - 1)$ .

**Solution :** (i) We have  $\lim_{x \rightarrow 3} x^2 = \lim_{x \rightarrow 3} (x \cdot x) = \lim_{x \rightarrow 3} x \cdot \lim_{x \rightarrow 3} x$   
 $[\because \text{limit of a product} = \text{product of the limits}]$   
 $= 3 \cdot 3 = 9.$   $\left[ \because \lim_{x \rightarrow c} x = c \right]$

(ii) We have  $\lim_{x \rightarrow a} (4x^2 - 6x + 7)$   
 $= 4 \left( \lim_{x \rightarrow a} x^2 \right) - 6 \left( \lim_{x \rightarrow a} x \right) + \lim_{x \rightarrow a} 7 = 4a^2 - 6a + 7.$

(iii) We have  $\lim_{x \rightarrow -1} (2x^3 - 3x^2 + 4x - 1)$   
 $= 2 \left( \lim_{x \rightarrow -1} x^3 \right) - 3 \left( \lim_{x \rightarrow -1} x^2 \right) + 4 \left( \lim_{x \rightarrow -1} x \right) - \lim_{x \rightarrow -1} 1$   
 $= -2 - 3 - 4 - 1 = -10.$

**Illustration 2 :** Evaluate (i)  $\lim_{x \rightarrow 1} \frac{x-2}{x+3}$ , (ii)  $\lim_{x \rightarrow 1} \frac{x^2+2}{x-2}$ .

**Solution :** (i) We have  $\lim_{x \rightarrow 1} \frac{x-2}{x+3} = \frac{\lim_{x \rightarrow 1} (x-2)}{\lim_{x \rightarrow 1} (x+3)}$   
 $= \frac{\lim_{x \rightarrow 1} x - \lim_{x \rightarrow 1} 2}{\lim_{x \rightarrow 1} x + \lim_{x \rightarrow 1} 3} = \frac{1-2}{1+3} = -\frac{1}{4}.$

(ii) We have  $\lim_{x \rightarrow 1} \frac{x^2+2}{x-2} = \frac{\lim_{x \rightarrow 1} (x^2+2)}{\lim_{x \rightarrow 1} (x-2)}$   
 $= \frac{\lim_{x \rightarrow 1} x^2 + \lim_{x \rightarrow 1} 2}{\lim_{x \rightarrow 1} x - \lim_{x \rightarrow 1} 2} = \frac{1+2}{1-2} = -3.$

**Illustration 3 :** If  $\lim_{x \rightarrow 2} \frac{x^n - 2^n}{x - 2} = 80$  and if  $n$  is a positive integer, find  $n$ .

**Solution :** We have  $\lim_{x \rightarrow 2} \frac{x^n - 2^n}{x - 2} = n \cdot 2^{n-1} = 80 = 5 \cdot 2^{5-1} \Rightarrow n = 5.$

## 8.11 Factorisation Method

Suppose we are to evaluate  $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ , where  $f(x)$  and  $g(x)$  are polynomials

such that  $\lim_{x \rightarrow a} f(x) = 0$  and  $\lim_{x \rightarrow a} g(x) = 0$ . Here the method of direct substitution gives the indeterminate form  $0/0$ , which is meaningless. So we proceed as follows :

Since  $f(a) = 0$  and  $g(a) = 0$ , therefore  $x - a$  is a common factor of  $f(x)$  and  $g(x)$ . So we factorize the numerator and the denominator. Now  $x \rightarrow a \Rightarrow x - a \neq 0$  and so  $x - a$  can be cancelled from the numerator and the denominator. After cancelling  $x - a$ , we proceed to find the limit by direct substitution and if we get a meaningful value the solution is complete. But if we still get the indeterminate form  $0/0$ , we repeat the above process again till we get a meaningful value.

### Illustrative Examples

**Illustration 1 :** Evaluate (i)  $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x^2 - 3x + 2}$ , (ii)  $\lim_{x \rightarrow -4} \frac{x^2 - 16}{x + 4}$

(iii)  $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x}$ .

[UPTU 2005]

**Solution :** (i) We have  $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x^2 - 3x + 2}$  [ form  $\frac{0}{0}$  ]

$$= \lim_{x \rightarrow 1} \frac{(x-1)(x+1)}{(x-1)(x-2)}$$

$$= \lim_{x \rightarrow 1} \frac{x+1}{x-2} \quad [ \because x \rightarrow 1 \Rightarrow x-1 \neq 0 \text{ and so } x-1 \text{ can be}$$

cancelled from the numerator and the denominator]

$$= \frac{1+1}{1-2} = -2.$$

(ii) We have  $\lim_{x \rightarrow -4} \frac{x^2 - 16}{x + 4} = \lim_{x \rightarrow -4} \frac{x^2 - (4)^2}{x + 4}$

$$= \lim_{x \rightarrow -4} \frac{(x+4)(x-4)}{(x+4)}$$

$$= \lim_{x \rightarrow -4} (x-4) = \lim_{x \rightarrow -4} x - \lim_{x \rightarrow -4} 4$$

$$= (-4) - 4 = -8.$$

$$\begin{aligned}
 \text{(iii) We have } \lim_{x \rightarrow 0} \frac{\sqrt{(1+x)} - 1}{x} & \quad \left[ \text{form } \frac{0}{0} \right] \\
 &= \lim_{x \rightarrow 0} \frac{[\sqrt{(1+x)} - 1][\sqrt{(1+x)} + 1]}{x[\sqrt{(1+x)} + 1]} \\
 &= \lim_{x \rightarrow 0} \frac{[\sqrt{(1+x)}]^2 - (1)^2}{x[\sqrt{(1+x)} + 1]} \\
 &= \lim_{x \rightarrow 0} \frac{1 + x - 1}{x[\sqrt{(1+x)} + 1]} \\
 &= \lim_{x \rightarrow 0} \frac{1}{\sqrt{(1+x)} + 1} = \frac{1}{\sqrt{(1+0)} + 1} = \frac{1}{2}.
 \end{aligned}$$

**Illustration 2 :** Evaluate  $\lim_{x \rightarrow a} \frac{\cos x - \cos a}{x - a}$ . [UPTU 2004]

$$\begin{aligned}
 \text{Solution : We have } \lim_{x \rightarrow a} \frac{\cos x - \cos a}{x - a} & \quad \left[ \text{form } \frac{0}{0} \right] \\
 &= \lim_{x \rightarrow a} \frac{-\sin x - 0}{1 - 0} = \lim_{x \rightarrow a} -\sin x = -\sin a.
 \end{aligned}$$

**Illustration 3 :** Evaluate  $\lim_{x \rightarrow 1} \left( \frac{1}{x-1} - \frac{2}{x^2-1} \right)$ .

$$\begin{aligned}
 \text{Solution : We have } \lim_{x \rightarrow 1} \left( \frac{1}{x-1} - \frac{2}{x^2-1} \right) & \quad [\text{form } \infty - \infty] \\
 &= \lim_{x \rightarrow 1} \left[ \frac{1}{x-1} - \frac{2}{(x-1)(x+1)} \right] \\
 &= \lim_{x \rightarrow 1} \frac{(x+1) - 2}{(x-1)(x+1)} = \lim_{x \rightarrow 1} \frac{x-1}{(x-1)(x+1)} \quad \left[ \text{form } \frac{0}{0} \right] \\
 &= \lim_{x \rightarrow 1} \frac{1}{x+1} = \frac{1}{2}.
 \end{aligned}$$

**Illustration 4 :** Evaluate  $\lim_{x \rightarrow a} \frac{(x+2)^{5/3} - (a+2)^{5/3}}{x-a}$ .

$$\begin{aligned}
 \text{Solution : We have } \lim_{x \rightarrow a} \frac{(x+2)^{5/3} - (a+2)^{5/3}}{x-a} \\
 &= \lim_{x \rightarrow a} \frac{(x+2)^{5/3} - (a+2)^{5/3}}{(x+2) - (a+2)} \\
 &= \lim_{y \rightarrow b} \frac{y^{5/3} - b^{5/3}}{y-b}, \quad \text{putting } x+2 = y \text{ and } a+2 = b;
 \end{aligned}$$

obviously  $x \rightarrow a \Rightarrow y \rightarrow a+2$  i.e.,  $y \rightarrow b$



$$= \frac{5}{3} \cdot b^{(5/3)-1} = \frac{5}{3} \cdot b^{2/3} = \frac{5}{3} \cdot (a+2)^{2/3}.$$

**Illustration 5 :** Evaluate the following limits :

$$(i) \lim_{x \rightarrow 0} \frac{e^{4x} - 1}{x} \quad (ii) \lim_{x \rightarrow 0} \frac{2^x - 1}{x} \quad (iii) \lim_{x \rightarrow 0} \frac{a^x - 1}{x} \quad [\text{UPTU 2006}]$$

$$(iv) \lim_{x \rightarrow 0} \frac{a^x - b^x}{x} \quad (v) \lim_{x \rightarrow 0} \frac{e^x + e^{-x} - 2}{x^2}. \quad [\text{UPTU 2007}]$$

**Solution :** (i) We have  $\lim_{x \rightarrow 0} \frac{e^{4x} - 1}{x}$

$$= \lim_{x \rightarrow 0} \left( \frac{e^{4x} - 1}{4x} \cdot 4 \right) = 4 \lim_{y \rightarrow 0} \frac{e^y - 1}{y},$$

putting  $4x = y$  ; obviously  $y \rightarrow 0$  when  $x \rightarrow 0$

$$= 4 \cdot 1 = 4. \quad \left[ \because \lim_{y \rightarrow 0} \frac{e^y - 1}{y} = 1 \right]$$

$$(ii) \text{ We have } \lim_{x \rightarrow 0} \frac{2^x - 1}{x} = \lim_{x \rightarrow 0} \frac{e^{\log 2^x} - 1}{x} = \lim_{x \rightarrow 0} \frac{e^{x \log 2} - 1}{x}$$

$$= \lim_{x \rightarrow 0} \left( \frac{e^{x \log 2} - 1}{x \log 2} \cdot \log 2 \right) = (\log 2) \lim_{y \rightarrow 0} \frac{e^y - 1}{y},$$

putting  $x \log 2 = y$  ; obviously  $y \rightarrow 0$  when  $x \rightarrow 0$

$$= (\log 2) \cdot 1 = \log_e 2.$$

$$(iii) \text{ We have } \lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \lim_{x \rightarrow 0} \frac{e^{\log a^x} - 1}{x}$$

$$= \lim_{x \rightarrow 0} \frac{e^{x \log a} - 1}{x} = \lim_{x \rightarrow 0} \left( \frac{e^{x \log a} - 1}{x \log a} \cdot \log a \right)$$

$$= (\log a) \cdot \lim_{y \rightarrow 0} \frac{e^y - 1}{y},$$

putting  $x \log a = y$ , obviously  $x \rightarrow 0 \Rightarrow y \rightarrow 0$

$$= (\log a) \cdot 1 = \log_e a.$$

$$(iv) \text{ We have } \lim_{x \rightarrow 0} \frac{a^x - b^x}{x} = \lim_{x \rightarrow 0} \frac{(a^x - 1) - (b^x - 1)}{x}$$

$$= \lim_{x \rightarrow 0} \left( \frac{a^x - 1}{x} - \frac{b^x - 1}{x} \right)$$

$$= \left( \lim_{x \rightarrow 0} \frac{a^x - 1}{x} \right) - \left( \lim_{x \rightarrow 0} \frac{b^x - 1}{x} \right)$$

$$= \log a - \log b = \log (a/b).$$

$$\begin{aligned}
 \text{(v) We have } \lim_{x \rightarrow 0} \frac{e^x + e^{-x} - 2}{x^2} &= \lim_{x \rightarrow 0} \frac{e^{2x} - 2e^x + 1}{x^2 e^x} \quad \left[ \because e^{-x} = \frac{1}{e^x} \right] \\
 &= \lim_{x \rightarrow 0} \left[ \frac{1}{e^x} \cdot \left( \frac{e^x - 1}{x} \right)^2 \right] \\
 &= \left( \lim_{x \rightarrow 0} \frac{1}{e^x} \right) \left[ \lim_{x \rightarrow 0} \left( \frac{e^x - 1}{x} \right)^2 \right] \\
 &= \left( \frac{1}{1} \right) \cdot (1)^2 = 1. \quad \left[ \because \lim_{x \rightarrow 0} e^x = 1 \text{ and } \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1 \right]
 \end{aligned}$$

**Illustration 6 :** Evaluate the following limits :

$$\begin{aligned}
 \text{(i) } \lim_{x \rightarrow 1} \frac{x-1}{\log x} & \qquad \text{(ii) } \lim_{x \rightarrow 5} \frac{\log x - \log 5}{x-5} \\
 \text{(iii) } \lim_{x \rightarrow 0} \frac{e^x - e^{-x}}{x} & \qquad \text{(iv) } \lim_{x \rightarrow 0} \frac{\log(a+x) - \log(a-x)}{x}
 \end{aligned}$$

**Solution :** (i) We have  $\lim_{x \rightarrow 1} \frac{x-1}{\log x}$

$$\begin{aligned}
 &= \lim_{h \rightarrow 0} \frac{(1+h)-1}{\log(1+h)} \quad \left[ \because \lim_{x \rightarrow a} f(x) = \lim_{h \rightarrow 0} f(a+h) \right] \\
 &= \lim_{h \rightarrow 0} \frac{h}{\log(1+h)} \\
 &= \lim_{h \rightarrow 0} \frac{1}{\frac{\log(1+h)}{h}} = \frac{1}{\lim_{h \rightarrow 0} \frac{\log(1+h)}{h}} \\
 &= \frac{1}{1} = 1. \quad \left[ \because \lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1 \right]
 \end{aligned}$$

(ii) We have  $\lim_{x \rightarrow 5} \frac{\log x - \log 5}{x-5}$

$$\begin{aligned}
 &= \lim_{h \rightarrow 0} \frac{\log(5+h) - \log 5}{(5+h)-5} \quad \left[ \because \lim_{x \rightarrow a} f(x) = \lim_{h \rightarrow 0} f(a+h) \right] \\
 &= \lim_{h \rightarrow 0} \frac{\log\left(\frac{5+h}{5}\right)}{h} = \lim_{h \rightarrow 0} \frac{\log\left(1 + \frac{h}{5}\right)}{5 \cdot (h/5)} \\
 &= \frac{1}{5} \lim_{h \rightarrow 0} \frac{\log\{1 + (h/5)\}}{h/5} = \frac{1}{5} \cdot 1 = \frac{1}{5} \\
 &\quad \left[ \because \lim_{h \rightarrow 0} \frac{\log[1 + (h/5)]}{h/5} = \lim_{y \rightarrow 0} \frac{\log(1+y)}{y} = 1 \right]
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii) We have } & \lim_{x \rightarrow 0} \frac{e^x - e^{-x}}{x} \\
 &= \lim_{x \rightarrow 0} \frac{e^{2x} - 1}{x e^x} \quad \left[ \because e^{-x} = \frac{1}{e^x} \right] \\
 &= \lim_{x \rightarrow 0} \left( \frac{1}{e^x} \cdot \frac{e^{2x} - 1}{2x} \cdot 2 \right) = \frac{1}{1} \cdot 1 \cdot 2 = 2. \\
 & \quad \left[ \because \lim_{x \rightarrow 0} \frac{e^{2x} - 1}{2x} = \lim_{y \rightarrow 0} \frac{e^y - 1}{y} = 1 \right]
 \end{aligned}$$

$$\begin{aligned}
 \text{(iv) We have } & \lim_{x \rightarrow 0} \frac{\log(a+x) - \log(a-x)}{x} \\
 &= \lim_{x \rightarrow 0} \frac{\log[a\{1+(x/a)\}] - \log[a\{1-(x/a)\}]}{x} \\
 &= \lim_{x \rightarrow 0} \frac{\log a + \log\{1+(x/a)\} - \log a - \log\{1-(x/a)\}}{x} \\
 &= \lim_{x \rightarrow 0} \frac{\log\{1+(x/a)\}}{x} - \lim_{x \rightarrow 0} \frac{\log\{1-(x/a)\}}{x} \\
 &= \lim_{x \rightarrow 0} \frac{\log\{1+(x/a)\}}{a \cdot (x/a)} + \lim_{x \rightarrow 0} \frac{\log\{1+(-x/a)\}}{a(-x/a)} \\
 &= \frac{1}{a} \lim_{x \rightarrow 0} \frac{\log\{1+(x/a)\}}{x/a} + \frac{1}{a} \lim_{x \rightarrow 0} \frac{\log\{1+(-x/a)\}}{(-x/a)} \\
 &= \frac{1}{a} \cdot 1 + \frac{1}{a} \cdot 1 = \frac{2}{a}.
 \end{aligned}$$

## Comprehensive Exercise 4

Evaluate the following limits :

1. (i)  $\lim_{x \rightarrow 1} \{(x-1)^2 + 7\}$

(ii)  $\lim_{x \rightarrow 0} \{(x-1)^2 + 5\}$

2. (i)  $\lim_{x \rightarrow 1} \frac{x-1}{x+1}$

(ii)  $\lim_{x \rightarrow a} \frac{x^2 - a^2}{x - a}$

3. (i)  $\lim_{x \rightarrow 5} \frac{x^2 - 9x + 20}{x^2 - 6x + 5}$

(ii)  $\lim_{x \rightarrow -5} \frac{2x^2 + 9x - 5}{x + 5}$

4. (i)  $\lim_{x \rightarrow 5} \frac{x^5 - 3125}{x - 5}$

(ii)  $\lim_{x \rightarrow 4} \frac{x^3 - 64}{x^2 - 16}$

- (iii)  $\lim_{x \rightarrow 2} \frac{x-2}{x^{3/2} - 2^{3/2}}$       (iv)  $\lim_{x \rightarrow a} \frac{x^{5/7} - a^{5/7}}{x^{2/7} - a^{2/7}}$ .
5. (i)  $\lim_{x \rightarrow 0} \frac{e^{-x} - 1}{x}$       (ii)  $\lim_{x \rightarrow 1} \frac{e^x - e}{x - 1}$ .
6. (i)  $\lim_{x \rightarrow 1} \frac{\log_e x}{x - 1}$       (ii)  $\lim_{x \rightarrow 0} \frac{\log(5+x) - \log(5-x)}{x}$ .
7. If  $\lim_{x \rightarrow 3} \frac{x^n - 3^n}{x - 3} = 108$ , and  $n$  is a positive integer, find the value of  $n$ .
8. If  $\lim_{x \rightarrow -a} \frac{x^9 + a^9}{x + a} = 9$ , find all possible values of  $a$ .

## Answers

1. (i) 7      (ii) 6.  
 2. (i) 0      (ii)  $2a$ .  
 3. (i)  $\frac{1}{4}$       (ii)  $-11$ .  
 4. (i) 3125      (ii) 6  
 (iii)  $\frac{\sqrt{2}}{3}$       (iv)  $\frac{5}{2} a^{3/7}$ .  
 5. (i)  $-1$       (ii)  $e$ .  
 6. (i) 1      (ii)  $2/5$ .  
 7. 4      8.  $\pm 1$ .

### Some useful results.

1.  $\lim_{x \rightarrow 0} \frac{x}{\sin x} = \lim_{x \rightarrow 0} \frac{1}{\frac{\sin x}{x}} = \frac{1}{\lim_{x \rightarrow 0} \frac{\sin x}{x}} = \frac{1}{1} = 1$ .
2.  $\lim_{x \rightarrow 0} \frac{\tan x}{x} = \lim_{x \rightarrow 0} \left( \frac{\sin x}{x} \cdot \frac{1}{\cos x} \right) = \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \lim_{x \rightarrow 0} \frac{1}{\cos x} = 1 \cdot \frac{1}{1} = 1$ .
3.  $\lim_{x \rightarrow 0} \frac{x}{\tan x} = \lim_{x \rightarrow 0} \left( \frac{x}{\sin x} \cdot \cos x \right) = \lim_{x \rightarrow 0} \frac{x}{\sin x} \cdot \lim_{x \rightarrow 0} \cos x = 1 \cdot 1 = 1$ .
4.  $\lim_{x \rightarrow a} \frac{\sin(x-a)}{x-a} = \lim_{h \rightarrow 0} \frac{\sin(a+h-a)}{a+h-a}$
- $$\left[ \because \lim_{x \rightarrow a} f(x) = \lim_{h \rightarrow 0} f(a+h) \right]$$
- $$= \lim_{h \rightarrow 0} \frac{\sin h}{h} = 1$$

$$5. \lim_{x \rightarrow a} \frac{\tan(x-a)}{x-a} = \lim_{h \rightarrow 0} \frac{\tan(a+h-a)}{a+h-a} = \lim_{h \rightarrow 0} \frac{\tan h}{h} = 1.$$

$$6. \lim_{x \rightarrow 0} \sin ax = 0 \quad \text{and} \quad \lim_{x \rightarrow 0} \cos ax = 1.$$

$$7. \lim_{x \rightarrow 0} \frac{\sin ax}{ax} = 1, \quad \lim_{x \rightarrow 0} \frac{\tan ax}{ax} = 1, \quad a \neq 0.$$

$$8. \lim_{x \rightarrow 0} \frac{\sin^{-1} x}{x} = \lim_{\theta \rightarrow 0} \frac{\theta}{\sin \theta} = 1,$$

putting  $\sin^{-1} x = \theta$  or  $x = \sin \theta$ , obviously  $\theta \rightarrow 0$  when  $x \rightarrow 0$ .

$$9. \lim_{x \rightarrow 0} \frac{x}{\sin^{-1} x} = \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1,$$

putting  $\sin^{-1} x = \theta$  or  $x = \sin \theta$ ; obviously  $\theta \rightarrow 0$  when  $x \rightarrow 0$ .

$$10. \lim_{x \rightarrow 0} \frac{\tan^{-1} x}{x} = \lim_{\theta \rightarrow 0} \frac{\theta}{\tan \theta},$$

putting  $\tan^{-1} x = \theta$  or  $x = \tan \theta$ ; obviously  $\theta \rightarrow 0$  when  $x \rightarrow 0$ .

$$11. \lim_{x \rightarrow 0} \frac{x}{\tan^{-1} x} = \lim_{\theta \rightarrow 0} \frac{\tan \theta}{\theta},$$

putting  $\tan^{-1} x = \theta$  or  $x = \tan \theta$ ; obviously  $\theta \rightarrow 0$  when  $x \rightarrow 0 = 1$ .

All the above results can be directly applied while evaluating limits of trigonometric functions. However, these results can themselves be asked in the form of short questions.

## Illustrative Examples

**Illustration 1 :** Evaluate the following limits :

$$(i) \lim_{x \rightarrow 0} \frac{\sin 3x}{5x} \qquad (ii) \lim_{x \rightarrow 0} \frac{x^3}{\sin x^2}$$

$$(iii) \lim_{\theta \rightarrow 0} \frac{\sin a \theta}{\sin b \theta}, \text{ where } a \text{ and } b \text{ are fixed non-zero real numbers}$$

$$(iv) \lim_{\theta \rightarrow 0} \frac{\sin^2 a \theta}{\sin^2 b \theta} \qquad (v) \lim_{x \rightarrow 0} \frac{\sin 5x}{\tan 3x}.$$

**Solution :** (i) We have  $\lim_{x \rightarrow 0} \frac{\sin 3x}{5x}$

$$= \lim_{x \rightarrow 0} \left( \frac{\sin 3x}{3x} \cdot \frac{3}{5} \right) = \frac{3}{5} \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} = \frac{3}{5} \cdot 1 = \frac{3}{5}.$$

$$\begin{aligned}
 \text{(ii) We have } \lim_{x \rightarrow 0} \frac{x^3}{\sin x^2} &= \lim_{x \rightarrow 0} \left( \frac{x^2}{\sin x^2} \cdot x \right) \\
 &= \left( \lim_{x \rightarrow 0} \frac{x^2}{\sin x^2} \right) \cdot \lim_{x \rightarrow 0} x = 1 \cdot 0 = 0. \\
 &\left[ \because x \rightarrow 0 \Rightarrow x^2 \rightarrow 0 \text{ and so } \lim_{x \rightarrow 0} \frac{x^2}{\sin x^2} = \lim_{x^2 \rightarrow 0} \frac{x^2}{\sin x^2} = 1 \right]
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii) We have } \lim_{\theta \rightarrow 0} \frac{\sin a \theta}{\sin b \theta} &= \lim_{\theta \rightarrow 0} \left( \frac{\sin a \theta}{a \theta} \cdot \frac{b \theta}{\sin b \theta} \cdot \frac{a}{b} \right) \\
 &= \frac{a}{b} \lim_{\theta \rightarrow 0} \frac{\sin a \theta}{a \theta} \cdot \lim_{\theta \rightarrow 0} \frac{b \theta}{\sin b \theta} = \frac{a}{b} \cdot 1 \cdot 1 = \frac{a}{b}. \\
 &\quad [\because \theta \rightarrow 0 \Rightarrow a \theta \rightarrow 0 \text{ and } b \theta \rightarrow 0]
 \end{aligned}$$

$$\begin{aligned}
 \text{(iv) We have } \lim_{\theta \rightarrow 0} \frac{\sin^2 a \theta}{\sin^2 b \theta} &= \lim_{\theta \rightarrow 0} \left( \frac{\sin a \theta}{\sin b \theta} \right)^2 \\
 &= \left( \lim_{\theta \rightarrow 0} \frac{\sin a \theta}{\sin b \theta} \right)^2 = \left[ \lim_{\theta \rightarrow 0} \left( \frac{\sin a \theta}{a \theta} \cdot \frac{b \theta}{\sin b \theta} \cdot \frac{a}{b} \right) \right]^2 \\
 &= \left( 1 \cdot 1 \cdot \frac{a}{b} \right)^2 = \frac{a^2}{b^2}.
 \end{aligned}$$

$$\begin{aligned}
 \text{(v) We have } \lim_{x \rightarrow 0} \frac{\sin 5x}{\tan 3x} &= \lim_{x \rightarrow 0} \left( \frac{\sin 5x}{5x} \cdot \frac{3x}{\tan 3x} \cdot \frac{5}{3} \right) \\
 &= \frac{5}{3} \lim_{x \rightarrow 0} \frac{\sin 5x}{5x} \cdot \lim_{x \rightarrow 0} \frac{3x}{\tan 3x} = \frac{5}{3} \cdot 1 \cdot 1 = \frac{5}{3}.
 \end{aligned}$$

**Illustration 2 :** Evaluate the following limits :

$$\text{(i) } \lim_{\theta \rightarrow 0} \frac{1 - \cos m\theta}{1 - \cos n\theta}, \text{ where } m \text{ and } n \text{ are fixed non-zero real numbers.}$$

$$\text{(ii) } \lim_{x \rightarrow 0} \frac{1 - \cos 5x}{1 - \cos 6x} \qquad \text{(iii) } \lim_{x \rightarrow 0} \frac{\cos mx - \cos nx}{x^2}$$

$$\text{(iv) } \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3} \qquad \text{(v) } \lim_{x \rightarrow 0} \frac{\operatorname{cosec} x - \cot x}{x}$$

**Solution :** (i) We have  $\lim_{\theta \rightarrow 0} \frac{1 - \cos m\theta}{1 - \cos n\theta}$

$$\begin{aligned}
 &= \lim_{\theta \rightarrow 0} \frac{2 \sin^2 (m\theta / 2)}{2 \sin^2 (n\theta / 2)} = \lim_{\theta \rightarrow 0} \left[ \frac{\sin (m\theta / 2)}{\sin (n\theta / 2)} \right]^2 \\
 &= \lim_{\theta \rightarrow 0} \left[ \frac{\sin (m\theta / 2)}{(m\theta / 2)} \cdot \frac{(n\theta / 2)}{\sin (n\theta / 2)} \cdot \frac{m}{n} \right]^2 = \left( 1 \cdot 1 \cdot \frac{m}{n} \right)^2 = \frac{m^2}{n^2}.
 \end{aligned}$$

$$[\because \theta \rightarrow 0 \Rightarrow (m\theta / 2) \rightarrow 0 \text{ and } (n\theta / 2) \rightarrow 0]$$

(ii) We have  $\lim_{x \rightarrow 0} \frac{1 - \cos 5x}{1 - \cos 6x} = \lim_{x \rightarrow 0} \frac{2 \sin^2 (5x/2)}{2 \sin^2 3x}$

$$\begin{aligned}
 &= \lim_{x \rightarrow 0} \left[ \frac{\sin (5x/2)}{\sin 3x} \right]^2 \\
 &= \lim_{x \rightarrow 0} \left[ \frac{\sin (5x/2)}{(5x/2)} \cdot \frac{3x}{\sin 3x} \cdot \frac{5}{2} \cdot \frac{1}{3} \right]^2 \\
 &= \left( 1 \cdot 1 \cdot \frac{5}{6} \right)^2 = \frac{25}{36}.
 \end{aligned}$$

(iii)  $\lim_{x \rightarrow 0} \frac{\cos mx - \cos nx}{x^2} = \lim_{x \rightarrow 0} \frac{2 \sin \frac{1}{2} (m+n)x \sin \frac{1}{2} (n-m)x}{x^2}$

$$= 2 \lim_{x \rightarrow 0} \left[ \frac{\sin \frac{1}{2} (m+n)x}{\frac{1}{2} (m+n)x} \cdot \frac{\sin \frac{1}{2} (n-m)x}{\frac{1}{2} (n-m)x} \cdot \frac{(m+n)(n-m)}{4} \right]$$

$$= 2 \cdot \frac{n^2 - m^2}{4} \cdot 1 \cdot 1 = \frac{n^2 - m^2}{2}.$$

(iv) We have  $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3} = \lim_{x \rightarrow 0} \frac{\sin x (1 - \cos x)}{x^3 \cos x}$

$$= \lim_{x \rightarrow 0} \frac{\sin x (1 - \cos x)(1 + \cos x)}{x^3 \cos x (1 + \cos x)}$$

$$= \lim_{x \rightarrow 0} \frac{\sin x \cdot \sin^2 x}{x^3 \cos x (1 + \cos x)}$$

$$= \lim_{x \rightarrow 0} \left[ \left( \frac{\sin x}{x} \right)^3 \cdot \frac{1}{\cos x (1 + \cos x)} \right]$$

$$= (1)^3 \cdot \frac{1}{1(1+1)} = \frac{1}{2}.$$

(v) We have  $\lim_{x \rightarrow 0} \frac{\operatorname{cosec} x - \cot x}{x}$

$$= \lim_{x \rightarrow 0} \frac{1 - \cos x}{x \sin x} = \lim_{x \rightarrow 0} \frac{(1 - \cos x)(1 + \cos x)}{x \sin x (1 + \cos x)}$$

$$= \lim_{x \rightarrow 0} \frac{\sin^2 x}{x \sin x (1 + \cos x)}$$

$$= \lim_{x \rightarrow 0} \left( \frac{\sin x}{x} \cdot \frac{1}{1 + \cos x} \right) = 1 \cdot \frac{1}{2} = \frac{1}{2}.$$


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**Illustration 3 :** Evaluate the following limits :

$$(i) \lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x}$$

$$(ii) \lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{2\theta^2}$$

$$(iii) \lim_{x \rightarrow 0} \frac{\sin 3x + 7x}{4x + \sin 2x}$$

$$(iv) \lim_{x \rightarrow 0} \frac{\tan 3x - 2x}{3x - \sin^2 x}$$

$$(v) \lim_{x \rightarrow \pi} \frac{1 + \cos x}{\tan^2 x}$$

$$(vi) \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{\sin^3 x}.$$

**Solution :** (i) We have  $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x}$  [form  $\frac{0}{0}$ ]

$$= \lim_{x \rightarrow 0} \frac{2 \sin^2 x}{x} = 2 \lim_{x \rightarrow 0} \left[ \left( \frac{\sin x}{x} \right) \cdot \sin x \right] = 2 \cdot 1 \cdot 0 = 0.$$

(ii) We have  $\lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{2\theta^2}$  [form  $\frac{0}{0}$ ]

$$= \lim_{\theta \rightarrow 0} \frac{(1 - \cos \theta)(1 + \cos \theta)}{2\theta^2 (1 + \cos \theta)}$$

$$= \frac{1}{2} \lim_{\theta \rightarrow 0} \left[ \left( \frac{\sin \theta}{\theta} \right)^2 \cdot \frac{1}{1 + \cos \theta} \right]$$

$$= \frac{1}{2} \cdot (1)^2 \cdot \frac{1}{1 + 1} = \frac{1}{4}.$$

(iii) We have  $\lim_{x \rightarrow 0} \frac{\sin 3x + 7x}{4x + \sin 2x}$  [form  $\frac{0}{0}$ ]

$$= \lim_{x \rightarrow 0} \frac{\frac{\sin 3x}{x} + 7}{4 + \frac{\sin 2x}{x}}$$

[Dividing the numerator and the denominator by  $x$ ]

$$= \lim_{x \rightarrow 0} \frac{\left( \frac{\sin 3x}{3x} \cdot 3 \right) + 7}{4 + 2 \cdot \frac{\sin 2x}{2x}} = \frac{(1 \cdot 3) + 7}{4 + 2 \cdot 1} = \frac{10}{6} = \frac{5}{3}.$$

(iv) We have  $\lim_{x \rightarrow 0} \frac{\tan 3x - 2x}{3x - \sin^2 x}$  [form  $\frac{0}{0}$ ]

$$= \lim_{x \rightarrow 0} \frac{\frac{\tan 3x}{x} - 2}{3 - \frac{\sin^2 x}{x}}$$

[Dividing the numerator and the denominator by  $x$ ]



$$= \lim_{x \rightarrow 0} \frac{3 \cdot \frac{\tan 3x}{3x} - 2}{3 - \frac{\sin x}{x} \cdot \sin x} = \frac{3 \cdot 1 - 2}{3 - 1 \cdot 0} = \frac{1}{3}.$$

(v) We have  $\lim_{x \rightarrow \pi} \frac{1 + \cos x}{\tan^2 x} \quad \left[ \text{form } \frac{0}{0} \right]$

$$= \lim_{x \rightarrow \pi} \frac{\cos^2 x (1 + \cos x)}{\sin^2 x} = \lim_{x \rightarrow \pi} \frac{\cos^2 x (1 + \cos x)}{(1 - \cos^2 x)}$$

$$= \lim_{x \rightarrow \pi} \frac{\cos^2 x (1 + \cos x)}{(1 + \cos x)(1 - \cos x)} = \lim_{x \rightarrow \pi} \frac{\cos^2 x}{1 - \cos x}$$

$$= \frac{(-1)^2}{1 - (-1)} = \frac{1}{2}.$$

(vi) We have  $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{\sin^3 x} \quad \left[ \text{form } \frac{0}{0} \right]$

$$= \lim_{x \rightarrow 0} \frac{\sin x (1 - \cos x)}{\cos x \sin^3 x} = \lim_{x \rightarrow 0} \frac{1 - \cos x}{\cos x \sin^2 x}$$

$$= \lim_{x \rightarrow 0} \frac{1 - \cos x}{\cos x (1 - \cos^2 x)}$$

$$= \lim_{x \rightarrow 0} \frac{1 - \cos x}{\cos x (1 - \cos x)(1 + \cos x)}$$

$$= \lim_{x \rightarrow 0} \frac{1}{\cos x (1 + \cos x)} = \frac{1}{1(1+1)} = \frac{1}{2}.$$

**Illustration 4 :** Evaluate the following limits :

(i)  $\lim_{x \rightarrow 0} \frac{\sin x^\circ}{x}$

(ii)  $\lim_{x \rightarrow \pi/2} (\sec x - \tan x)$

(iii)  $\lim_{x \rightarrow \pi} \frac{1 + \sec^3 x}{\tan^2 x}$

(iv)  $\lim_{x \rightarrow 0} \frac{\sin 2x + \sin 6x}{\sin 5x - \sin 3x}$

(v)  $\lim_{y \rightarrow 0} \frac{(x+y) \sec(x+y) - x \sec x}{y}.$

**Solution :** (i) We have  $\lim_{x \rightarrow 0} \frac{\sin x^\circ}{x} \quad \left[ \text{form } \frac{0}{0} \right]$

$$= \lim_{x \rightarrow 0} \frac{\sin(\pi x / 180)}{x} \quad \left[ \because x^\circ = \frac{\pi}{180} x \text{ radians} \right]$$

$$= \lim_{x \rightarrow 0} \left[ \frac{\sin(\pi x / 180)}{(\pi x / 180)} \cdot \frac{\pi}{180} \right]$$

$$= \frac{\pi}{180} \lim_{x \rightarrow 0} \frac{\sin(\pi x / 180)}{(\pi x / 180)} = \frac{\pi}{180} \cdot 1 = \frac{\pi}{180}.$$

$$[\because x \rightarrow 0 \Rightarrow (\pi x / 180) \rightarrow 0]$$

(ii) We have  $\lim_{x \rightarrow \pi/2} (\sec x - \tan x)$  [form  $\infty - \infty$ ]

$$\begin{aligned} &= \lim_{x \rightarrow \pi/2} \left( \frac{1}{\cos x} - \frac{\sin x}{\cos x} \right) = \lim_{x \rightarrow \pi/2} \frac{1 - \sin x}{\cos x} \\ &= \lim_{x \rightarrow \pi/2} \frac{(1 - \sin x)(1 + \sin x)}{\cos x (1 + \sin x)} = \lim_{x \rightarrow \pi/2} \frac{1 - \sin^2 x}{\cos x (1 + \sin x)} \\ &= \lim_{x \rightarrow \pi/2} \frac{\cos^2 x}{\cos x (1 + \sin x)} \\ &= \lim_{x \rightarrow \pi/2} \frac{\cos x}{1 + \sin x} = \frac{0}{1 + 1} = \frac{0}{2} = 0. \end{aligned}$$

(iii) We have  $\lim_{x \rightarrow \pi} \frac{1 + \sec^3 x}{\tan^2 x}$  [form  $\frac{0}{0}$ ]

$$\begin{aligned} &= \lim_{x \rightarrow \pi} \frac{(1 + \sec x)(1 - \sec x + \sec^2 x)}{\sec^2 x - 1} \\ &= \lim_{x \rightarrow \pi} \frac{(\sec x + 1)(1 - \sec x + \sec^2 x)}{(\sec x + 1)(\sec x - 1)} \\ &= \lim_{x \rightarrow \pi} \frac{1 - \sec x + \sec^2 x}{\sec x - 1} = \frac{1 - (-1) + (-1)^2}{-1 - 1} = -\frac{3}{2}. \end{aligned}$$

(iv) We have  $\lim_{x \rightarrow 0} \frac{\sin 2x + \sin 6x}{\sin 5x - \sin 3x}$  [form  $\frac{0}{0}$ ]

$$\begin{aligned} &= \lim_{x \rightarrow 0} \frac{2 \sin 4x \cos 2x}{2 \cos 4x \sin x} \\ &= \lim_{x \rightarrow 0} \frac{\frac{\sin 4x}{4x} \cdot 4x \cdot \cos 2x}{\cos 4x \cdot \frac{\sin x}{x} \cdot x} \\ &= 4 \lim_{x \rightarrow 0} \frac{\frac{\sin 4x}{4x} \cdot \cos 2x}{\cos 4x \cdot \frac{\sin x}{x}} = 4 \cdot \frac{1 \cdot 1}{1 \cdot 1} = 4. \end{aligned}$$

(v) We have  $\lim_{y \rightarrow 0} \frac{(x + y) \sec(x + y) - x \sec x}{y}$  [form  $\frac{0}{0}$ ]

$$= \lim_{y \rightarrow 0} \frac{x \{\sec(x + y) - \sec x\} + y \sec(x + y)}{y}$$

$$\begin{aligned}
 &= \lim_{y \rightarrow 0} x \cdot \frac{\sec(x+y) - \sec x}{y} + \lim_{y \rightarrow 0} \sec(x+y) \\
 &= \left[ \lim_{y \rightarrow 0} x \frac{\cos x - \cos(x+y)}{y \cos x \cos(x+y)} \right] + \sec x \\
 &= \sec x + \lim_{y \rightarrow 0} \left[ x \cdot \frac{2 \sin\left(x + \frac{y}{2}\right) \sin \frac{y}{2}}{(y/2) \cdot 2} \cdot \frac{1}{\cos x \cos(x+y)} \right] \\
 &= \sec x + \lim_{y \rightarrow 0} \left[ \frac{\sin(y/2)}{(y/2)} \cdot \sin\left(x + \frac{y}{2}\right) \cdot \frac{x}{\cos x \cos(x+y)} \right] \\
 &= \sec x + 1 \cdot \sin x \cdot \frac{x}{\cos x \cdot \cos x} = \sec x + x \sec x \tan x.
 \end{aligned}$$

**Illustration 5 :** Evaluate the following limits.

$$\begin{array}{ll}
 (i) \quad \lim_{\theta \rightarrow \pi/2} \frac{\cot \theta}{(\pi/2) - \theta} & (ii) \quad \lim_{x \rightarrow \pi/2} \frac{e^{\cos x} - 1}{\cos x} \\
 (iii) \quad \lim_{x \rightarrow \pi/4} \frac{\sin x - \cos x}{x - (\pi/4)} & (iv) \quad \lim_{x \rightarrow \pi/2} \frac{2x - \pi}{\cos x} \\
 (v) \quad \lim_{x \rightarrow 1} (1-x) \tan \frac{\pi x}{2} & (vi) \quad \lim_{x \rightarrow \pi/2} \frac{1 + \cos 2x}{(\pi - 2x)^2}.
 \end{array}$$

**Solution :** (i) We have  $\lim_{\theta \rightarrow \pi/2} \frac{\cot \theta}{(\pi/2) - \theta}$

$$\begin{aligned}
 &= \lim_{h \rightarrow 0} \frac{\cot\left(\frac{1}{2}\pi + h\right)}{\frac{1}{2}\pi - \left(\frac{1}{2}\pi + h\right)} \left[ \because \lim_{x \rightarrow a} f(x) = \lim_{h \rightarrow 0} f(a+h) \right] \\
 &= \lim_{h \rightarrow 0} \frac{-\tan h}{-h} = \lim_{h \rightarrow 0} \frac{\tan h}{h} = 1.
 \end{aligned}$$

(ii) We have  $\lim_{x \rightarrow \pi/2} \frac{e^{\cos x} - 1}{\cos x}$

$$= \lim_{y \rightarrow 0} \frac{e^y - 1}{y},$$

putting  $\cos x = y$ , obviously  $y \rightarrow 0$  when  $x \rightarrow \pi/2$

= 1.

(iii) We have  $\lim_{x \rightarrow \pi/4} \frac{\sin x - \cos x}{x - (\pi/4)}$

$$\begin{aligned}
 &= \lim_{x \rightarrow \pi/4} \frac{\sqrt{2} \{ (1/\sqrt{2}) \sin x - (1/\sqrt{2}) \cos x \}}{x - (\pi/4)} \\
 &= \lim_{x \rightarrow \pi/4} \frac{\sin \left( x - \frac{1}{4} \pi \right)}{\sqrt{2} \cdot \frac{x - \frac{1}{4} \pi}{\sqrt{2}}} = \sqrt{2} \cdot \lim_{y \rightarrow 0} \frac{\sin y}{y}, \\
 &\quad \text{putting } x - \frac{1}{4} \pi = y; \text{ obviously } y \rightarrow 0 \text{ when } x \rightarrow \pi/4 \\
 &= (\sqrt{2}) \cdot 1 = \sqrt{2}.
 \end{aligned}$$

(iv) We have  $\lim_{x \rightarrow \pi/2} \frac{2x - \pi}{\cos x} = \lim_{h \rightarrow 0} \frac{2 \left( \frac{1}{2} \pi + h \right) - \pi}{\cos \left( \frac{1}{2} \pi + h \right)}$

$$\begin{aligned}
 &\left[ \because \lim_{x \rightarrow a} f(x) = \lim_{h \rightarrow 0} f(a+h) \right] \\
 &= \lim_{h \rightarrow 0} \frac{2h}{-\sin h} = -2 \lim_{h \rightarrow 0} \frac{h}{\sin h} = (-2) \cdot 1 = -2.
 \end{aligned}$$

(v)  $\lim_{x \rightarrow 1} (1-x) \tan \frac{\pi x}{2} = \lim_{h \rightarrow 0} \{1 - (1+h)\} \tan \left\{ \frac{\pi}{2} (1+h) \right\}$

$$\begin{aligned}
 &\left[ \because \lim_{x \rightarrow a} f(x) = \lim_{h \rightarrow 0} f(a+h) \right] \\
 &= \lim_{h \rightarrow 0} -h \tan \left\{ \frac{1}{2} \pi + (\pi h / 2) \right\} = \lim_{h \rightarrow 0} h \cot (\pi h / 2) \\
 &= \lim_{h \rightarrow 0} \left[ \frac{(\pi h / 2)}{\tan (\pi h / 2)} \cdot \frac{2}{\pi} \right] = 1 \cdot \frac{2}{\pi} = \frac{2}{\pi}.
 \end{aligned}$$

(vi)  $\lim_{x \rightarrow \pi/2} \frac{1 + \cos 2x}{(\pi - 2x)^2} = \lim_{h \rightarrow 0} \frac{1 + \cos \left\{ 2 \left( \frac{1}{2} \pi + h \right) \right\}}{\left\{ \pi - 2 \left( \frac{1}{2} \pi + h \right) \right\}^2}$

$$\begin{aligned}
 &\left[ \because \lim_{x \rightarrow a} f(x) = \lim_{h \rightarrow 0} f(a+h) \right] \\
 &= \lim_{h \rightarrow 0} \frac{1 + \cos (\pi + 2h)}{4h^2} = \lim_{h \rightarrow 0} \frac{1 - \cos 2h}{4h^2} \\
 &= \lim_{h \rightarrow 0} \frac{2 \sin^2 h}{4h^2} = \frac{1}{2} \lim_{h \rightarrow 0} \left( \frac{\sin h}{h} \right)^2 \\
 &= \frac{1}{2} \cdot (1)^2 = \frac{1}{2}.
 \end{aligned}$$

**Illustration 6 :** Evaluate the following limits :

$$(i) \lim_{x \rightarrow 0} \frac{e^x - \sin x - 1}{x} \quad (ii) \lim_{x \rightarrow 0} \frac{\log(1+x^3)}{\sin^3 x}$$

$$(iii) \lim_{x \rightarrow 0} \frac{e^{\tan x} - 1}{x} \quad (\text{UPTU 2011}) \quad (iv) \lim_{x \rightarrow 0} \frac{2^{3x} - 3^x}{\sin 3x}$$

$$(v) \lim_{x \rightarrow 0} \frac{e^{2x} - e^x}{\sin 2x} \quad (vi) \lim_{x \rightarrow 0} \frac{x(2^x - 1)}{1 - \cos x}$$

**Solution :** (i) We have  $\lim_{x \rightarrow 0} \frac{e^x - \sin x - 1}{x}$

$$= \lim_{x \rightarrow 0} \frac{e^x - 1}{x} - \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 - 1 = 0.$$

$$(ii) \lim_{x \rightarrow 0} \frac{\log(1+x^3)}{\sin^3 x} = \lim_{x \rightarrow 0} \frac{\log(1+x^3)}{x^3} \cdot \frac{x^3}{\sin^3 x}$$

$$= \lim_{x \rightarrow 0} \frac{\log(1+x^3)}{x^3} \cdot \lim_{x \rightarrow 0} \left( \frac{x}{\sin x} \right)^3 = 1 \cdot (1)^3 = 1.$$

$$(iii) \lim_{x \rightarrow 0} \frac{e^{\tan x} - 1}{x} = \lim_{x \rightarrow 0} \left[ \frac{e^{\tan x} - 1}{\tan x} \cdot \frac{\tan x}{x} \right]$$

$$= \lim_{x \rightarrow 0} \frac{e^{\tan x} - 1}{\tan x} \cdot \lim_{x \rightarrow 0} \frac{\tan x}{x} = \left( \lim_{y \rightarrow 0} \frac{e^y - 1}{y} \right) \cdot 1,$$

in the first limit putting  $\tan x = y$  ;  
obviously  $y \rightarrow 0$  when  $x \rightarrow 0$

$$= 1 \cdot 1 = 1.$$

$$(iv) \lim_{x \rightarrow 0} \frac{2^{3x} - 3^x}{\sin 3x} = \lim_{x \rightarrow 0} \left[ \frac{(2^{3x} - 1) - (3^x - 1)}{x} \cdot \frac{3x}{\sin 3x} \cdot \frac{1}{3} \right]$$

$$= \frac{1}{3} \lim_{x \rightarrow 0} \frac{3x}{\sin 3x} \cdot \left[ \lim_{x \rightarrow 0} \left( \frac{2^{3x} - 1}{3x} \cdot 3 \right) - \lim_{x \rightarrow 0} \frac{3^x - 1}{x} \right]$$

$$= \frac{1}{3} \cdot 1 \cdot (3 \log 2 - \log 3) = \log 2 - \frac{1}{3} \log 3.$$

(v) We have  $\lim_{x \rightarrow 0} \frac{e^{2x} - e^x}{\sin 2x} = \lim_{x \rightarrow 0} \frac{e^x (e^x - 1)}{\sin 2x}$

$$= \lim_{x \rightarrow 0} \left[ e^x \cdot \frac{e^x - 1}{x} \cdot \frac{2x}{\sin 2x} \cdot \frac{1}{2} \right] = 1 \cdot 1 \cdot 1 \cdot \frac{1}{2} = \frac{1}{2}.$$

$$(vi) \lim_{x \rightarrow 0} \frac{x(2^x - 1)}{1 - \cos x} = \lim_{x \rightarrow 0} \left[ \frac{2^x - 1}{x} \cdot \frac{x^2}{1 - \cos^2 x} \cdot (1 + \cos x) \right]$$

$$\begin{aligned}
 &= \lim_{x \rightarrow 0} \left[ \frac{2^x - 1}{x} \cdot \left( \frac{x}{\sin x} \right)^2 \cdot (1 + \cos x) \right] = (\log 2) \cdot (1)^2 \cdot (1 + 1) \\
 &= 2 \log 2 = \log 2^2 = \log 4.
 \end{aligned}$$

**Illustration 7 :** Evaluate the following limits :

$$\begin{aligned}
 (i) \quad \lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{\sin^{-1} x} & \quad (ii) \quad \lim_{x \rightarrow 0} \frac{\tan^{-1} 2x}{\sin 3x} \\
 (iii) \quad \lim_{x \rightarrow 0} \frac{1}{x} \cos^{-1} \left( \frac{1-x^2}{1+x^2} \right) & \quad (iv) \quad \lim_{x \rightarrow 1} \frac{1-\sqrt{x}}{(\cos^{-1} x)^2} \\
 (v) \quad \lim_{x \rightarrow 1/\sqrt{2}} \left[ \frac{x - \cos(\sin^{-1} x)}{1 - \tan(\sin^{-1} x)} \right] & \quad (vi) \quad \lim_{x \rightarrow 1} \frac{1-x}{\pi - 2 \sin^{-1} x}.
 \end{aligned}$$

**Solution :** (i) We have  $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{\sin^{-1} x} \quad \left[ \text{form } \frac{0}{0} \right]$

$$\begin{aligned}
 &= \lim_{x \rightarrow 0} \left[ \frac{(1+x) - (1-x)}{\sin^{-1} x} \cdot \frac{1}{\sqrt{1+x} + \sqrt{1-x}} \right] \\
 &\quad \text{[Rationalising the numerator]}
 \end{aligned}$$

$$\begin{aligned}
 &= \lim_{x \rightarrow 0} \left[ 2 \cdot \frac{x}{\sin^{-1} x} \cdot \frac{1}{\sqrt{1+x} + \sqrt{1-x}} \right] \\
 &= 2 \lim_{x \rightarrow 0} \frac{x}{\sin^{-1} x} \cdot \lim_{x \rightarrow 0} \frac{1}{\sqrt{1+x} + \sqrt{1-x}} = 2 \cdot 1 \cdot \frac{1}{2} = 1.
 \end{aligned}$$

$$\begin{aligned}
 (ii) \text{ We have } \lim_{x \rightarrow 0} \frac{\tan^{-1} 2x}{\sin 3x} &= \lim_{x \rightarrow 0} \left[ \frac{\tan^{-1} 2x}{2x} \cdot \frac{3x}{\sin 3x} \cdot \frac{2}{3} \right] \\
 &= \frac{2}{3} \lim_{x \rightarrow 0} \frac{\tan^{-1} 2x}{2x} \cdot \lim_{x \rightarrow 0} \frac{3x}{\sin 3x} = \frac{2}{3} \cdot 1 \cdot 1 = \frac{2}{3}.
 \end{aligned}$$

$$\begin{aligned}
 (iii) \text{ We have } \lim_{x \rightarrow 0} \frac{1}{x} \cos^{-1} \left( \frac{1-x^2}{1+x^2} \right) &= \lim_{x \rightarrow 0} \left[ \frac{1}{x} \cdot 2 \tan^{-1} x \right] \\
 &\quad \left[ \because \cos^{-1} \frac{1-x^2}{1+x^2} = 2 \tan^{-1} x \right] \\
 &= 2 \lim_{x \rightarrow 0} \frac{\tan^{-1} x}{x} = 2 \cdot 1 = 2.
 \end{aligned}$$

$$\begin{aligned}
 (iv) \text{ We have } \lim_{x \rightarrow 1} \frac{1-\sqrt{x}}{(\cos^{-1} x)^2} &= \lim_{x \rightarrow 1} \frac{1-x}{(1+\sqrt{x})(\cos^{-1} x)^2} \\
 &\quad \text{[Rationalizing the numerator]}
 \end{aligned}$$

$$= \lim_{x \rightarrow 1} \frac{1}{1 + \sqrt{x}} \cdot \lim_{x \rightarrow 1} \frac{1 - x}{(\cos^{-1} x)^2}$$

$$= \frac{1}{2} \lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta^2},$$

putting  $\cos^{-1} x = \theta$  or  $x = \cos \theta$ ; obviously  $\theta \rightarrow 0$  when  $x \rightarrow 1$

$$= \frac{1}{2} \lim_{\theta \rightarrow 0} \left( \frac{\sin^2 \theta}{\theta^2} \cdot \frac{1}{1 + \cos \theta} \right)$$

$$= \frac{1}{2} \lim_{\theta \rightarrow 0} \left[ \left( \frac{\sin \theta}{\theta} \right)^2 \cdot \frac{1}{1 + \cos \theta} \right]$$

$$= \frac{1}{2} \cdot (1)^2 \cdot \frac{1}{2} = \frac{1}{4}.$$

(v) We have  $\lim_{x \rightarrow 1/\sqrt{2}} \left[ \frac{x - \cos(\sin^{-1} x)}{1 - \tan(\sin^{-1} x)} \right]$

$$= \lim_{\theta \rightarrow \pi/4} \frac{\sin \theta - \cos \theta}{1 - \tan \theta}, \quad \text{putting } \sin^{-1} x = \theta \text{ or } x = \sin \theta;$$

obviously  $\theta \rightarrow \pi/4$  when  $x \rightarrow 1/\sqrt{2}$

$$= \lim_{\theta \rightarrow \pi/4} \left( \frac{\sin \theta - \cos \theta}{\cos \theta - \sin \theta} \right) \cdot \cos \theta = \lim_{\theta \rightarrow \pi/4} (-\cos \theta) = -\frac{1}{\sqrt{2}}.$$

(vi) We have  $\lim_{x \rightarrow 1} \frac{1 - x}{\pi - 2 \sin^{-1} x} = \lim_{\theta \rightarrow \pi/2} \frac{1 - \sin \theta}{\pi - 2\theta},$

putting  $\sin^{-1} x = \theta$  or  $x = \sin \theta$ ; obviously  $\theta \rightarrow \pi/2$  when  $x \rightarrow 1$

$$= \lim_{h \rightarrow 0} \frac{1 - \sin(\frac{1}{2}\pi + h)}{\pi - 2(\frac{1}{2}\pi + h)} \quad \left[ \because \lim_{x \rightarrow a} f(x) = \lim_{h \rightarrow 0} f(a + h) \right]$$

$$= \lim_{h \rightarrow 0} \frac{1 - \cos h}{-2h} = -\frac{1}{2} \lim_{h \rightarrow 0} \frac{(1 - \cos h)(1 + \cos h)}{h(1 + \cos h)}$$

$$= -\frac{1}{2} \lim_{h \rightarrow 0} \left[ \left( \frac{\sin h}{h} \right) \cdot \sin h \cdot \frac{1}{1 + \cos h} \right] = -\frac{1}{2} (1) \cdot 0 \cdot \frac{1}{2} = 0.$$

**Illustration 8 :** Evaluate the following limits :

(i)  $\lim_{x \rightarrow \infty} \frac{3x^2 + 5x - 4}{7x^2 - 8x + 3}$

(ii)  $\lim_{x \rightarrow \infty} \frac{3x - 7}{5x^2 + 8x - 5}$

(iii)  $\lim_{x \rightarrow \infty} \frac{7x^2 - 6x + 5}{5x + 8}$

(iv)  $\lim_{x \rightarrow \infty} x - \sqrt{x^2 + x}.$

[UPTU 2008]

**Solution :** (i) We have  $\lim_{x \rightarrow \infty} \frac{3x^2 + 5x - 4}{7x^2 - 8x + 3}$

$$= \lim_{x \rightarrow \infty} \frac{3 + 5 \cdot (1/x) - 4 \cdot (1/x^2)}{7 - 8 \cdot (1/x) + 3 \cdot (1/x^2)}$$

[Dividing the Nr. and the Dr. by  $x^2$ ]

$$= \frac{3 + 5 \cdot 0 - 4 \cdot 0}{7 - 8 \cdot 0 + 3 \cdot 0} = \frac{3}{7} \quad \left[ \because \lim_{x \rightarrow \infty} \frac{1}{x} = 0, \lim_{x \rightarrow \infty} \frac{1}{x^2} = 0 \right]$$

(ii) We have 
$$\lim_{x \rightarrow \infty} \frac{3x - 7}{5x^2 + 8x - 5} = \lim_{x \rightarrow \infty} \frac{x [3 - 7 \cdot (1/x)]}{x^2 [5 + 8 \cdot (1/x) - 5 \cdot (1/x^2)]}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{1}{x} \cdot [3 - 7 \cdot (1/x)]}{5 + 8 \cdot (1/x) - 5 \cdot (1/x^2)}$$

$$= 0 \cdot \frac{3 - 7 \cdot 0}{5 + 8 \cdot 0 - 5 \cdot 0} = 0 \cdot \frac{3}{5} = 0 \quad \left[ \because \lim_{x \rightarrow \infty} \frac{1}{x} = 0, \lim_{x \rightarrow \infty} \frac{1}{x^2} = 0 \right]$$

(iii) We have 
$$\lim_{x \rightarrow \infty} \frac{7x^2 - 6x + 5}{5x + 8}$$

$$= \lim_{x \rightarrow \infty} \frac{x^2 [7 - 6 \cdot (1/x) + 5 \cdot (1/x^2)]}{x [5 + 8 \cdot (1/x)]}$$

$$= \lim_{x \rightarrow \infty} x \cdot \frac{7 - 6 \cdot (1/x) + 5 \cdot (1/x^2)}{5 + 8 \cdot (1/x)} = \infty.$$

$$\left[ \because \lim_{x \rightarrow \infty} x = \infty, \lim_{x \rightarrow \infty} \frac{1}{x} = 0, \lim_{x \rightarrow \infty} \frac{1}{x^2} = 0 \right]$$

(iv) We have 
$$\lim_{x \rightarrow \infty} x - \sqrt{x^2 + x} = \lim_{x \rightarrow \infty} \frac{(x - \sqrt{x^2 + x})(x + \sqrt{x^2 + x})}{(x + \sqrt{x^2 + x})}$$

$$= \lim_{x \rightarrow \infty} \frac{x^2 - (x^2 + x)}{x + \sqrt{x^2 + x}} = \lim_{x \rightarrow \infty} \frac{-x}{x + \sqrt{x^2 + x}}$$

$$= \lim_{x \rightarrow \infty} \frac{-x}{x + x \sqrt{1 + \frac{1}{x}}} = - \lim_{x \rightarrow \infty} \frac{1}{1 + \sqrt{1 + \frac{1}{x}}}$$

$$= \frac{-1}{1 + \sqrt{1 + 0}} = \frac{-1}{2}.$$

**Illustration 9 :** Show that  $\lim_{x \rightarrow \infty} (\sqrt{x^2 + x + 1} - x) \neq \lim_{x \rightarrow \infty} (\sqrt{x^2 + 1} - x).$

**Solution :** We have 
$$\lim_{x \rightarrow \infty} (\sqrt{x^2 + x + 1} - x)$$

$$= \lim_{x \rightarrow \infty} \frac{(\sqrt{x^2 + x + 1} - x)(\sqrt{x^2 + x + 1} + x)}{(\sqrt{x^2 + x + 1} + x)}$$



$$\begin{aligned}
 &= \lim_{x \rightarrow \infty} \frac{(x^2 + x + 1) - x^2}{(\sqrt{x^2 + x + 1} + x)} \\
 &= \lim_{x \rightarrow \infty} \frac{x + 1}{\sqrt{x^2 + x + 1} + x} \\
 &= \lim_{x \rightarrow \infty} \frac{1 + (1/x)}{\sqrt{1 + (1/x) + (1/x)^2} + 1}
 \end{aligned}$$

[Dividing the Nr. and the Dr. by  $x$ ]

$$= \frac{1 + 0}{\sqrt{1 + 0 + 0} + 1} = \frac{1}{2}.$$

Again  $\lim_{x \rightarrow \infty} (\sqrt{x^2 + 1} - x) = \lim_{x \rightarrow \infty} \frac{(\sqrt{x^2 + 1} - x)(\sqrt{x^2 + 1} + x)}{\sqrt{x^2 + 1} + x}$

$$= \lim_{x \rightarrow \infty} \frac{(x^2 + 1) - x^2}{\sqrt{x^2 + 1} + x}$$

$$= \lim_{x \rightarrow \infty} \frac{1}{\sqrt{x^2 + 1} + x}$$

$$= \lim_{x \rightarrow \infty} \frac{1}{x} \cdot \frac{1}{\sqrt{1 + (1/x^2)} + 1} \quad [\text{Dividing the Nr. and the Dr. by } x]$$

$$= 0 \cdot \frac{1}{\sqrt{1 + 0} + 1} = 0 \cdot \frac{1}{2} = 0.$$

Hence,  $\lim_{x \rightarrow \infty} (\sqrt{x^2 + x + 1} - x) \neq \lim_{x \rightarrow \infty} (\sqrt{x^2 + 1} - x).$

## Comprehensive Exercise 5

Evaluate the following limits.

1. (i)  $\lim_{x \rightarrow 0} \frac{\sin 7x}{5x}$

(ii)  $\lim_{\theta \rightarrow 0} \frac{\sin 3\theta}{\sin 2\theta}$

(iii)  $\lim_{x \rightarrow 0} \frac{\tan 3x}{\sin 2x}$

(iv)  $\lim_{x \rightarrow 0} \frac{\sin x \cos x}{3x}$

2. (i)  $\lim_{x \rightarrow 0} \frac{3 \sin x - 4 \sin^3 x}{x}$

(ii)  $\lim_{x \rightarrow 0} \frac{x^2}{\sin x^2}$

(iii)  $\lim_{x \rightarrow 0} \frac{\cos 3x - \cos 7x}{x^2}$

(iv)  $\lim_{\theta \rightarrow 0} \frac{1 - \cos 4\theta}{1 - \cos 5\theta}$

3. (i)  $\lim_{x \rightarrow \pi} \frac{\sin x}{x - \pi}$  (ii)  $\lim_{\theta \rightarrow \pi/2} \frac{\cos \theta}{(\pi/2) - \theta}$
- (iii)  $\lim_{x \rightarrow 1} \frac{1 + \cos \pi x}{(1 - x)^2}$  (iv)  $\lim_{x \rightarrow \pi/4} \frac{1 - \tan x}{x - (\pi/4)}$
4. (i)  $\lim_{x \rightarrow \pi/4} \frac{\sin 2x - 1}{x - (\pi/4)}$  (ii)  $\lim_{x \rightarrow \pi/3} \frac{\sqrt{3} - \tan x}{\pi - 3x}$
- (iii)  $\lim_{x \rightarrow 0} \frac{e^x + \sin x - 1}{x}$  (iv)  $\lim_{x \rightarrow 0} \frac{e^{\sin x} - 1}{\sin x}$
5. (i)  $\lim_{x \rightarrow \infty} \frac{x + 3}{x - 3}$  (ii)  $\lim_{x \rightarrow \infty} \frac{8x^2 - 9x + 5}{4x^2 + 6x - 7}$
- (iii)  $\lim_{x \rightarrow \infty} \frac{3x^3 - 7x^2 + 5x - 1}{4x^3 + x^2 - 6x + 8}$
- (iv)  $\lim_{x \rightarrow \infty} \frac{(3x - 1)(4x - 2)}{(x + 8)(x - 1)}$
6. (i)  $\lim_{x \rightarrow \infty} \frac{2x^2 - 5x + 7}{3x^3 - 6x^2 + x - 4}$
- (ii)  $\lim_{x \rightarrow \infty} \frac{4x^2 - 7x + 11}{3 - 5x}$
- (iii)  $\lim_{x \rightarrow \infty} [\sqrt{x^2 + x + 1} - \sqrt{x^2 + 1}]$
- (iv)  $\lim_{x \rightarrow \infty} [\sqrt{x + 1} - \sqrt{x}]$

## Answers

1. (i)  $\frac{7}{5}$  (ii)  $\frac{3}{2}$  (iii)  $\frac{3}{2}$  (iv)  $\frac{1}{3}$
2. (i) 3 (ii) 1 (iii) 20 (iv)  $\frac{16}{25}$
3. (i) -1 (ii) 1 (iii)  $\frac{\pi^2}{2}$  (iv) -2
4. (i) 0 (ii)  $\frac{4}{3}$  (iii) 2 (iv) 1
5. (i) 1 (ii) 2 (iii)  $\frac{3}{4}$  (iv) 12
6. (i) 0 (ii)  $-\infty$  (iii)  $\frac{1}{2}$  (iv) 0

## Comprehensive Exercise 6

### Fill in the Blanks

Fill in the blanks “.....”, so that the following statements are complete and correct.

1. If  $f(x) = \frac{1-x}{1+x}$ , then  $f(\cos \theta) = \dots\dots\dots$ .
2. If  $f(x) = \frac{\sin x}{1 + \sin x}$ , then  $f(\pi / 2) = \dots\dots\dots$ .
3. The value of  $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$  is  $\dots\dots\dots$ . [UPTU 2009, 10]
4. The value of  $\lim_{x \rightarrow 0} e^x$  is  $\dots\dots\dots$ .
5. The value of  $\lim_{x \rightarrow 3} \frac{4x^2 - 17x + 15}{x^2 - x - 6}$  is  $\dots\dots\dots$ .
6. The domain of the real function  $f(x) = \frac{1}{1 - x^2}$  is  $\dots\dots\dots$ .
7. The value of  $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$  is  $\dots\dots\dots$ .
8. The value of  $\lim_{x \rightarrow 0} \frac{ax + b}{cx + d}$  is  $\dots\dots\dots$ .
9. The value of  $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = \dots\dots\dots$ , where  $a > 0$  and  $n$  is any real number.
10. The value of  $\lim_{x \rightarrow \infty} \frac{3x^{-1} + 4x^{-2}}{5x^{-1} + 6x^{-2}} = \dots\dots\dots$ .
11. The value of  $\lim_{x \rightarrow \infty} \left(1 + \frac{a}{x}\right)^x$  is  $\dots\dots\dots$ . [UPTU 2010]

### Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

12. The value of  $\lim_{x \rightarrow 0} \log (1 + x)$  is
 

|        |              |
|--------|--------------|
| (a) 1  | (b) 0        |
| (c) -1 | (d) $\infty$ |

13. The value of  $\lim_{x \rightarrow 0} (1+x)^{4x}$  is  
 (a)  $e^2$  (b)  $e^{-1}$   
 (c)  $e$  (d)  $-e$
14. The value of  $\lim_{x \rightarrow 0} \left( \frac{\sin 2x}{x} \right)$  is  
 (a) 0 (b) 1  
 (c)  $1/2$  (d) 2
15. The value of  $\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$  is  
 (a) 1 (b)  $-1$   
 (c) 0 (d)  $\frac{1}{2}$
16. The value of  $\lim_{x \rightarrow 0} \frac{a^{bx} - 1}{bx}$  is ....., where  $b \neq 0$   
 (a)  $\log_e b$  (b)  $\log_e a$   
 (c)  $\log_e \frac{b}{a}$  (d)  $\log_e \frac{a}{b}$
17. If  $\lim_{x \rightarrow a} \frac{x^9 - a^9}{x - a} = 9$ , the all possible values of  $a$  are  
 (a) 1 (b)  $-1$   
 (c)  $\pm 1$  (d) 0

[UPTU 2011]

### True or False

Write 'T' for true and 'F' for false statement.

18. The value of  $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$  is  $(x + 4)$ .
19. The value of  $\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log_e a$ .
20. The value of  $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ .
21. The value of  $\lim_{x \rightarrow \infty} \frac{x - 1}{x + 1}$  is 0.
22. The domain of the real function  $f(x) = \frac{x^2}{1 + x^2}$  is  $R$ .
23. The value of  $\lim_{\theta \rightarrow 0} \frac{\sin a\theta}{\sin b\theta} = \frac{a}{b}$ .
24. The value of  $\lim_{x \rightarrow \infty} \frac{x^2 + 2}{3x^2 - 4}$  is  $\frac{1}{3}$ .

[UPTU 2010]

[UPTU 2011]

25. The value of  $\lim_{x \rightarrow \infty} \frac{x^2 + ax + b}{x^2 + cx + d}$  is 1.

26. The value of  $\lim_{x \rightarrow \infty} \frac{ax^2 + bx + c}{dx^2 + ex + f} = \frac{d}{a}$ .

27. The value of  $\lim_{\theta \rightarrow 0} \frac{\sin 4\theta}{\tan 3\theta} = \frac{4}{3}$ .

## Answers

- |                          |                    |           |          |
|--------------------------|--------------------|-----------|----------|
| 1. $\tan^2 (\theta / 2)$ | 2. $\frac{1}{2}$   | 3. 2      | 4. 1     |
| 5. $\frac{7}{5}$         | 6. $R - \{-1, 1\}$ | 7. 8      | 8. $b/d$ |
| 9. $na^{n-1}$            | 10. $3/5$          | 11. $e^a$ | 12. (b)  |
| 13. (c)                  | 14. (d)            | 15. (a)   | 16. (b)  |
| 17. (c)                  | 18. $T$            | 19. $T$   | 20. $T$  |
| 21. $F$                  | 22. $T$            | 23. $T$   | 24. $F$  |
| 25. $T$                  | 26. $F$            | 27. $T$   |          |



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## Chapter

# 9



## Differentiation

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### 9.1 The Differential Coefficient

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The differential coefficient of a function  $y = f(x)$  with respect to  $x$  is defined as

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) - f(x)}{\delta x}$$

provided the limit exists. The differential coefficient is also called the **derivative**, or the **derived function**. The differential coefficient of  $y = f(x)$  with respect to  $x$  may be denoted by any of the symbols

$$\frac{d}{dx} y, \frac{dy}{dx}, y', Dy, \frac{d}{dx} f(x), f'(x), Df(x).$$

The process of finding the differential coefficient is called **differentiation**.

**The Differential Coefficient At A Point.** If  $y = f(x)$  is a function of  $x$ , then

$$\lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}$$

---

is called the differential coefficient of  $f(x)$  for  $x = a$ , provided the above limit exists. It is denoted by  $\left(\frac{dy}{dx}\right)_{x=a}$ ,  $(y')_a$ , or  $f'(a)$ . It gives us the rate of change of  $y$  with respect to  $x$  at  $x = a$ .

## 9.2 Some Standard Results

### (I) Differential Coefficient Of $x^n$ ( $n$ being real number).

Let  $y = x^n$ . Then  $y + \delta y = (x + \delta x)^n$ . Therefore  $\delta y = (x + \delta x)^n - x^n$ .

$$\therefore \frac{\delta y}{\delta x} = \frac{(x + \delta x)^n - x^n}{\delta x}.$$

Taking limit when  $\delta x \rightarrow 0$ , we get

$$\begin{aligned} \frac{dy}{dx} &= \lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} \frac{(x + \delta x)^n - x^n}{\delta x} \\ &= \lim_{\delta x \rightarrow 0} \frac{x^n \left(1 + \frac{\delta x}{x}\right)^n - x^n}{\delta x} = \lim_{\delta x \rightarrow 0} \frac{x^n \left[\left(1 + \frac{\delta x}{x}\right)^n - 1\right]}{\delta x} \\ &= \lim_{\delta x \rightarrow 0} \frac{x^n \left[1 + n\left(\frac{\delta x}{x}\right) + \frac{n(n-1)}{2!} \left(\frac{\delta x}{x}\right)^2 + \dots - 1\right]}{\delta x} \end{aligned}$$

[Expanding by binomial theorem since  $\delta x / x$  is numerically less than unity,  $\delta x$  being numerically small]

$$\begin{aligned} &= \lim_{\delta x \rightarrow 0} \frac{x^n \left[n\left(\frac{\delta x}{x}\right) + \frac{n(n-1)}{2!} \left(\frac{\delta x}{x}\right)^2 + \dots\right]}{\delta x} \\ &= \lim_{\delta x \rightarrow 0} x^n \left[\frac{n}{x} + \frac{n(n-1)}{2!} \frac{\delta x}{x^2} + \dots\right] = x^n \cdot \frac{n}{x} = nx^{n-1}. \end{aligned}$$

Thus  $\frac{d}{dx} x^n = nx^{n-1}$ .

### (II) Differential Coefficient Of $\sin x$ .

We have  $\frac{d}{dx} \sin x = \lim_{\delta x \rightarrow 0} \frac{\sin(x + \delta x) - \sin x}{\delta x}$ , by definition

$$\begin{aligned} &= \lim_{\delta x \rightarrow 0} \frac{2 \cos\left(x + \frac{\delta x}{2}\right) \sin \frac{\delta x}{2}}{\delta x} = \lim_{\delta x \rightarrow 0} \cos\left(x + \frac{\delta x}{2}\right) \frac{\sin \frac{\delta x}{2}}{\frac{\delta x}{2}} \\ &= \cos x, \text{ since } \lim_{\delta x \rightarrow 0} \frac{\sin(\delta x / 2)}{\delta x / 2} = 1. \end{aligned}$$

Thus,  $\frac{d}{dx} \sin x = \cos x$ .

Similarly, it can be shown that  $\frac{d}{dx} \cos x = -\sin x$ .

### (III) Differential Coefficient of $e^x$ .

We have  $\frac{d}{dx} e^x = \lim_{\delta x \rightarrow 0} \frac{e^{x+\delta x} - e^x}{\delta x}$ , by definition

$$\begin{aligned} &= \lim_{\delta x \rightarrow 0} \frac{e^x e^{\delta x} - e^x}{\delta x} = \lim_{\delta x \rightarrow 0} \frac{e^x (e^{\delta x} - 1)}{\delta x} \\ &= \lim_{\delta x \rightarrow 0} \frac{e^x \left[ 1 + \frac{\delta x}{1!} + \frac{(\delta x)^2}{2!} + \frac{(\delta x)^3}{3!} + \dots - 1 \right]}{\delta x} \\ &= \lim_{\delta x \rightarrow 0} \frac{e^x \left[ \frac{\delta x}{1!} + \frac{(\delta x)^2}{2!} + \frac{(\delta x)^3}{3!} + \dots \right]}{\delta x} \\ &= \lim_{\delta x \rightarrow 0} e^x \left[ 1 + \frac{\delta x}{2!} + \frac{(\delta x)^2}{3!} + \dots \right] = e^x. \end{aligned}$$

Thus  $\frac{d}{dx} (e^x) = e^x$ .

### (IV) Differential Coefficient of $a^x \log_e a$ .

$$\begin{aligned} \text{We have } \frac{d}{dx} a^x &= \lim_{h \rightarrow 0} \frac{a^{x+h} - a^x}{h} = \lim_{h \rightarrow 0} \frac{a^x a^h - a^x}{h} \\ &= \lim_{h \rightarrow 0} a^x \left( \frac{a^h - 1}{h} \right) = a^x \lim_{h \rightarrow 0} \frac{a^h - 1}{h} = a^x \cdot \log_e a \\ &\quad \left[ \because \lim_{h \rightarrow 0} \frac{a^h - 1}{h} = \log_e a \right] \end{aligned}$$

Thus,  $\frac{d}{dx} a^x = a^x \log_e a = a^x \log a$ .

### (V) Differential Coefficient of $\log_e x$ .

$$\begin{aligned} \text{We have } \frac{d}{dx} \log x &= \lim_{\delta x \rightarrow 0} \frac{\log(x + \delta x) - \log x}{\delta x}, \text{ by definition} \\ &= \lim_{\delta x \rightarrow 0} \frac{\log \left( \frac{x + \delta x}{x} \right)}{\delta x} = \lim_{\delta x \rightarrow 0} \frac{\log \left( 1 + \frac{\delta x}{x} \right)}{\delta x} \end{aligned}$$



$$= \lim_{\delta x \rightarrow 0} \frac{\frac{\delta x}{x} - \frac{(\delta x)^2}{2x^2} + \frac{(\delta x)^3}{3x^3} - \dots}{\delta x}$$

(the expansion is justified since  $\delta x / x$  is numerically less than unity,  $\delta x$  being numerically small)

$$= \lim_{\delta x \rightarrow 0} \left[ \frac{1}{x} - \frac{\delta x}{2x^2} + \frac{(\delta x)^2}{3x^3} - \dots \right] = \frac{1}{x}.$$

Thus  $\frac{d}{dx} (\log_e x) = \frac{1}{x}.$

#### (VI) Differential Coefficient of a Constant.

Let  $f(x) = c$ , where  $c$  is a constant.

Then 
$$\begin{aligned} \frac{d}{dx} f(x) &= \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) - f(x)}{\delta x}, \text{ by definition} \\ &= \lim_{\delta x \rightarrow 0} \frac{c - c}{\delta x}, \text{ since } f(x) = c \text{ for every value of } x \\ &= \lim_{\delta x \rightarrow 0} \frac{0}{\delta x} = 0. \end{aligned}$$

Thus, the differential coefficient of a constant is zero.

#### (VII) Differential coefficient of the product of a constant and a function.

Let  $c$  be a constant and  $f(x)$  be a function of  $x$ . Then by definition

$$\begin{aligned} \frac{d}{dx} \{c f(x)\} &= \lim_{\delta x \rightarrow 0} \frac{c f(x + \delta x) - c f(x)}{\delta x} \\ &= \lim_{\delta x \rightarrow 0} c \cdot \frac{f(x + \delta x) - f(x)}{\delta x} \\ &= c \cdot \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) - f(x)}{\delta x} = c \frac{d}{dx} f(x). \end{aligned}$$

Thus the differential coefficient of the product of a constant and a function is equal to the product of the constant and the differential coefficient of the function.

#### (VIII) Differential coefficient of $\log_a x$ .

We have  $\log_a x = \log_e x \log_a e = \log_a e \log_e x$ , where  $\log_a e$  is simply a constant.

$$\begin{aligned} \therefore \frac{d}{dx} (\log_a x) &= \log_a e \frac{d}{dx} (\log_e x) = (\log_a e) \cdot \frac{1}{x} = \frac{1}{x} \cdot \log_a e \\ &= \frac{1}{x \log_e a}, \text{ since } \log_a e \cdot \log_e a = 1. \end{aligned}$$

Thus, 
$$\frac{d}{dx} (\log_a x) = \frac{1}{x \log_e a} = \frac{1}{x \log a}.$$

## 9.3 List of Standard Results to be Committed to Memory

$$\frac{d}{dx} x^n = nx^{n-1}$$

$$\frac{d}{dx} \log_e x = \frac{1}{x}$$

$$\frac{d}{dx} \sin x = \cos x$$

$$\frac{d}{dx} \tan x = \sec^2 x$$

$$\frac{d}{dx} \sec x = \sec x \tan x$$

$$\frac{d}{dx} \sinh x = \cosh x$$

$$\frac{d}{dx} \tanh x = \operatorname{sech}^2 x$$

$$\frac{d}{dx} \operatorname{sech} x = -\operatorname{sech} x \tanh x$$

$$\frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2}$$

$$\frac{d}{dx} \sec^{-1} x = \frac{1}{x\sqrt{x^2-1}}$$

$$\frac{d}{dx} \sinh^{-1} x = \frac{1}{\sqrt{x^2+1}}$$

$$\frac{d}{dx} \tanh^{-1} x = \frac{1}{1-x^2}$$

$$\frac{d}{dx} e^x = e^x$$

$$\frac{d}{dx} \log_a x = \frac{1}{x} \log_a e \quad \text{or} \quad \frac{1}{x \log_e a}$$

$$\frac{d}{dx} \cos x = -\sin x$$

$$\frac{d}{dx} \cot x = -\operatorname{cosec}^2 x$$

$$\frac{d}{dx} \operatorname{cosec} x = -\operatorname{cosec} x \cot x$$

$$\frac{d}{dx} \cosh x = \sinh x$$

$$\frac{d}{dx} \coth x = -\operatorname{cosech}^2 x$$

$$\frac{d}{dx} \operatorname{cosech} x = -\operatorname{cosech} x \coth x$$

$$\frac{d}{dx} \cos^{-1} x = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx} \cot^{-1} x = -\frac{1}{1+x^2}$$

$$\frac{d}{dx} \operatorname{cosec}^{-1} x = -\frac{1}{x\sqrt{x^2-1}}$$

$$\frac{d}{dx} \cosh^{-1} x = \frac{1}{\sqrt{x^2-1}}$$

$$\frac{d}{dx} a^x = a^x \log_e a$$

## 9.4 Differential Coefficient of the Sum of two Functions

Let  $f(x) = f_1(x) + f_2(x)$ .

$$\begin{aligned} \text{Then } \frac{d}{dx} f(x) &= \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) - f(x)}{\delta x} \\ &= \lim_{\delta x \rightarrow 0} \frac{\{f_1(x + \delta x) + f_2(x + \delta x)\} - \{f_1(x) + f_2(x)\}}{\delta x} \\ &= \lim_{\delta x \rightarrow 0} \frac{\{f_1(x + \delta x) - f_1(x)\} + \{f_2(x + \delta x) - f_2(x)\}}{\delta x} \\ &= \lim_{\delta x \rightarrow 0} \left\{ \frac{f_1(x + \delta x) - f_1(x)}{\delta x} + \frac{f_2(x + \delta x) - f_2(x)}{\delta x} \right\} \end{aligned}$$

$$\begin{aligned}
 &= \lim_{\delta x \rightarrow 0} \frac{f_1(x + \delta x) - f_1(x)}{\delta x} + \lim_{\delta x \rightarrow 0} \frac{f_2(x + \delta x) - f_2(x)}{\delta x} \\
 &= \frac{d}{dx} f_1(x) + \frac{d}{dx} f_2(x).
 \end{aligned}$$

$$\therefore \frac{d}{dx} [f_1(x) + f_2(x)] = \frac{d}{dx} f_1(x) + \frac{d}{dx} f_2(x).$$

$$\begin{aligned}
 \text{Similarly, } \frac{d}{dx} \{f_1(x) + f_2(x) + \dots + f_n(x)\} \\
 = \frac{d}{dx} f_1(x) + \frac{d}{dx} f_2(x) + \dots + \frac{d}{dx} f_n(x)
 \end{aligned}$$

$$\begin{aligned}
 \text{and } \frac{d}{dx} \{f_1(x) - f_2(x) - \dots - f_n(x)\} \\
 = \frac{d}{dx} f_1(x) - \frac{d}{dx} f_2(x) - \dots - \frac{d}{dx} f_n(x).
 \end{aligned}$$

## Illustrative Examples

**Illustration 1 :** Find the derivatives of the following functions with respect to (w.r.t.)  $x$ .

(i)  $1/x$

(ii)  $x^{-8}$

(iii)  $5^{\cot x}$

(iv)  $\log_2 x$ .

[UPTU 2001]

**Solution :** We know that  $\frac{d}{dx} x^n = n x^{n-1}$ . Therefore

(i)  $\frac{d}{dx} \left(\frac{1}{x}\right) = \frac{d}{dx} x^{-1} = -1 x^{-1-1} = -x^{-2} = -\frac{1}{x^2}.$

(ii)  $\frac{d}{dx} x^{-8} = -8 x^{-8-1} = -8x^{-9}.$

(iii)  $\frac{d}{dx} 5^{\cot x} = 5^{\cot x} \log e^5.$

(iv)  $\frac{d}{dx} (\log_2 x) = \frac{1}{x} \log_2 e \quad \text{or} \quad \frac{1}{x \log_e 2}.$

**Illustration 2 :** Differentiate the following functions with respect to  $x$ :

(i)  $4x^3 - 7 - 6x^2$

(ii)  $5 \sin x - 2 \log x + 7 \sec x.$

**Solution :** (i)  $\frac{d}{dx} (4x^3 - 7 - 6x^2) = \frac{d}{dx} (4x^3) - \frac{d}{dx} (7) - \frac{d}{dx} (6x^2)$   
 $= 4 \frac{d}{dx} x^3 - 0 - 6 \frac{d}{dx} x^2 = 4 \cdot (3x^2) - 0 - 6 \cdot (2x) = 12x^2 - 12x.$

(ii)  $\frac{d}{dx} (5 \sin x - 2 \log x + 7 \sec x) = 5 \frac{d}{dx} \sin x - 2 \frac{d}{dx} \log x + 7 \frac{d}{dx} \sec x$   
 $= 5 \cos x - 2 \cdot \frac{1}{x} + 7 \sec x \tan x = 5 \cos x - \frac{2}{x} + 7 \sec x \tan x.$

## Comprehensive Exercise 1

Differentiate the following functions with respect to  $x$ .

1.  $x + \frac{1}{x}$ .

2.  $3 \sin \frac{x}{3} - 4 \sin^3 \frac{x}{3}$ .

3.  $5x^{-7}$ .

4.  $\log_e \sqrt{x}$ .

5.  $\tan^{-1} x + \cot^{-1} x$ .

6.  $\tan x + 2 \sin x + 3 \cos x - \frac{1}{2} \log x - e^x$ .

7.  $e^x + 4 \sec x + 5 \cos^{-1} x$ .

8.  $2^x + x^2 + 2^2$ .

## Answers

1.  $1 - \frac{1}{x^2}$ .

2.  $\cos x$ .

3.  $-35x^{-8}$ .

4.  $\frac{1}{2x}$ .

5. 0.

6.  $\sec^2 x + 2 \cos x - 3 \sin x - \frac{1}{2x} - e^x$ .

7.  $e^x + 4 \sec x \tan x - \frac{5}{\sqrt{1-x^2}}$ .

8.  $2^x \log_e 2 + 2x$ .

## 9.5 Differential Coefficient of the Product of two Functions

Let  $f(x) = f_1(x) f_2(x)$ .

Then  $\frac{d}{dx} [f_1(x) f_2(x)] = f_1(x) \frac{d}{dx} f_2(x) + f_2(x) \frac{d}{dx} f_1(x)$ .

(Product Rule)

## 9.6 Differential Coefficient of the Quotient of two Functions

Let  $f(x) = \frac{f_1(x)}{f_2(x)}$ .

Then  $\frac{d}{dx} \left\{ \frac{f_1(x)}{f_2(x)} \right\} = \frac{f_2(x) \cdot \frac{d}{dx} f_1(x) - f_1(x) \cdot \frac{d}{dx} f_2(x)}{[f_2(x)]^2}$ .

(Quotient Rule)

## Illustrative Examples

**Illustration 1 :** Find the derivatives of the following functions with respect to  $x$

(i)  $e^x \cos x$  [UPTU 2004] (ii)  $x^3 \log x$

(iii)  $x \log x - x$  (iv)  $(1 + x^2) \tan^{-1} x$

(v)  $\sin x \cdot \log x$  (vi)  $e^x + e^x \log x$ . [UPTU 2007]

**Solution :** (i)  $\frac{d}{dx} (e^x \cos x) = e^x \frac{d}{dx} \cos x + \cos x \frac{d}{dx} e^x$   
 $= e^x (-\sin x) + \cos x \cdot e^x = e^x (\cos x - \sin x).$

(ii)  $\frac{d}{dx} (x^3 \log x) = x^3 \frac{d}{dx} \log x + \log x \frac{d}{dx} x^3$   
 $= x^3 \cdot \frac{1}{x} + (\log x) \cdot 3x^2$   
 $= x^2 + 3x^2 \log x = x^2 (1 + 3 \log x).$

(iii)  $\frac{d}{dx} (x \log x - x) = \frac{d}{dx} (x \log x) - \frac{d}{dx} (x)$   
 $= x \cdot \frac{d}{dx} (\log x) + (\log x) \cdot \frac{d}{dx} (x) - 1$   
 $= x \cdot \frac{1}{x} + (\log x) \cdot 1 - 1 = 1 + \log x - 1 = \log x.$

(iv)  $\frac{d}{dx} \{(1 + x^2) \tan^{-1} x\} = (1 + x^2) \frac{d}{dx} \tan^{-1} x + (\tan^{-1} x) \frac{d}{dx} (1 + x^2)$   
 $= (1 + x^2) \cdot \frac{1}{1 + x^2} + (\tan^{-1} x) \cdot (0 + 2x)$   
 $= 1 + 2x \tan^{-1} x.$

(v)  $\frac{d}{dx} (\sin x \log x) = \sin x \frac{d}{dx} (\log x) + \log x \frac{d}{dx} (\sin x)$   
 $= \sin x \cdot \frac{1}{x} + \log x \cdot (\cos x)$   
 $= \frac{\sin x}{x} + \log x (\cos x).$

(vi)  $\frac{d}{dx} (e^x + e^x \log x) = \frac{d}{dx} (e^x) + \frac{d}{dx} (e^x \log x)$   
 $= e^x + e^x \frac{d}{dx} (\log x) + \log x \frac{d}{dx} (e^x)$   
 $= e^x + e^x \cdot \frac{1}{x} + \log x \cdot e^x$   
 $= e^x \left( 1 + \frac{1}{x} + \log x \right).$

**Illustration 2 :** Find the derivatives of the following functions with respect to  $x$ .

(i)  $\frac{3x^2 - 2}{x^2 + 7}$

(ii)  $\frac{e^x}{1 + \sin x}$

(iii)  $\frac{x^2 + 2x + 4}{x + 4}$

(iv)  $\frac{e^{ax} - e^{-ax}}{e^{ax} + e^{-ax}}$

**Solution :** (i)  $\frac{d}{dx} \left( \frac{3x^2 - 2}{x^2 + 7} \right) = \frac{(x^2 + 7) \frac{d}{dx} (3x^2 - 2) - (3x^2 - 2) \frac{d}{dx} (x^2 + 7)}{(x^2 + 7)^2}$

[By quotient rule for differentiation]

$$= \frac{(x^2 + 7)(6x) - (3x^2 - 2)(2x)}{(x^2 + 7)^2}$$

$$= \frac{6x^3 + 42x - 6x^3 + 4x}{(x^2 + 7)^2} = \frac{46x}{(x^2 + 7)^2}.$$

(ii)  $\frac{d}{dx} \left( \frac{e^x}{1 + \sin x} \right) = \frac{(1 + \sin x) \frac{d}{dx} e^x - e^x \frac{d}{dx} (1 + \sin x)}{(1 + \sin x)^2}$

$$= \frac{(1 + \sin x) e^x - e^x \cos x}{(1 + \sin x)^2} = \frac{e^x (1 + \sin x - \cos x)}{(1 + \sin x)^2}.$$

(iii)  $\frac{d}{dx} \left( \frac{x^2 + 2x + 4}{x + 4} \right) = \frac{(x + 4) \frac{d}{dx} (x^2 + 2x + 4) - (x^2 + 2x + 4) \frac{d}{dx} (x + 4)}{(x + 4)^2}$

$$= \frac{(x + 4)(2x + 2) - (x^2 + 2x + 4)(1)}{(x + 4)^2}$$

$$= \frac{2(x + 4)(x + 1) - (x + 2)^2}{(x + 4)^2}.$$

(iv)  $\frac{d}{dx} \left( \frac{e^{ax} - e^{-ax}}{e^{ax} + e^{-ax}} \right) = \frac{(e^{ax} + e^{-ax}) \frac{d}{dx} (e^{ax} - e^{-ax}) - (e^{ax} - e^{-ax}) \frac{d}{dx} (e^{ax} + e^{-ax})}{(e^{ax} + e^{-ax})^2}$

$$= \frac{(e^{ax} + e^{-ax})(ae^{ax} + ae^{-ax}) - (e^{ax} - e^{-ax})(ae^{ax} - ae^{-ax})}{(e^{ax} + e^{-ax})^2}$$

$$= \frac{a(e^{ax} + e^{-ax})^2 - a(e^{ax} - e^{-ax})^2}{(e^{ax} + e^{-ax})^2}$$

$$= \frac{a\{(e^{2ax} + e^{-2ax} + 2) - (e^{2ax} + e^{-2ax} - 2)\}}{(e^{ax} + e^{-ax})^2}$$

$$= \frac{4a}{(e^{ax} + e^{-ax})^2}$$

## Comprehensive Exercise 2

Differentiate the following functions with respect to  $x$ .

1.  $x^2 e^x$ .
2.  $x^3 \sin x$ .
3.  $\sec x \tan x$ .
4.  $(1 + x^4) \cos x$ .

Find the derivatives of the following functions with respect to  $x$ .

5.  $\frac{x + \sin x}{x + \cos x}$ .
6.  $\frac{2x + 3}{x^2 - 5}$ .
7.  $\frac{x}{1 + \tan x}$ .
8.  $\frac{e^x}{1 + x^2}$ .
9.  $\frac{\log x}{x}$ .
10.  $\frac{x^2 + 5x - 6}{4x^2 - x + 3}$ .

## Answers

1.  $x(x + 2)e^x$ .
2.  $x^2(3 \sin x + x \cos x)$ .
3.  $\sec x (\sec^2 x + \tan^2 x)$ .
4.  $4x^3 \cos x - (1 + x^4) \sin x$ .
5.  $\frac{x(\cos x + \sin x) + \cos x - \sin x + 1}{(x + \cos x)^2}$ .
6.  $\frac{-2(x^2 + 3x + 5)}{(x^2 - 5)^2}$ .
7.  $\frac{1 + \tan x - x \sec^2 x}{(1 + \tan x)^2}$ .
8.  $\frac{e^x(1 - x)^2}{(1 + x^2)^2}$ .
9.  $\frac{1 - \log x}{x^2}$ .
10.  $\frac{9 + 54x - 21x^2}{(4x^2 - x + 3)^2}$ .

## 9.7 Differential Coefficient of a Function of a Function (Chain Rule of Differentiation)

Consider the function  $\log \sin x$ . Here  $\log(\sin x)$  is a function of  $\sin x$  where  $\sin x$  is itself a function of  $x$ . Thus we have case of a function of a function.

If  $y$  is a function of  $t$  and  $t$  is a function of  $x$ , then  $y$  is also a function of  $x$  and we have

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}.$$

Similarly, if  $y$  is a function of  $u$ ,  $u$  is a function of  $v$  and  $v$  is a function of  $x$ , then  $y$  is also a function of  $x$  and we have,

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx}$$

i.e., derivative of  $y$  w.r.t.  $x$

= (derivative of  $y$  w.r.t.  $u$ ). (derivative of  $u$  w.r.t.  $v$ ). (derivative of  $v$  w.r.t.  $x$ ).

## Illustrative Examples

**Illustration 1 :** Find the derivatives of the following functions with respect to  $x$ .

(i)  $\operatorname{cosec}^5 x$

(ii)  $e^{\sin x}$

(iii)  $\tan(\sin x)$

(iv)  $e^{\tan^{-1} x}$

**Solution :** (i) Let  $y = \operatorname{cosec}^5 x$ .

Put  $\operatorname{cosec} x = t$ . Then  $y = t^5$  and  $t = \operatorname{cosec} x$ .

$$\therefore \frac{dy}{dt} = \frac{d}{dt} (t^5) = 5t^4$$

$$\text{and } \frac{dt}{dx} = \frac{d}{dx} \operatorname{cosec} x = -\operatorname{cosec} x \cot x.$$

$$\begin{aligned} \text{Now } \frac{dy}{dx} &= \frac{dy}{dt} \cdot \frac{dt}{dx} = 5t^4 \cdot (-\operatorname{cosec} x \cot x) \\ &= 5 \operatorname{cosec}^4 x \cdot (-\operatorname{cosec} x \cot x) = -5 \operatorname{cosec}^5 x \cot x. \end{aligned}$$

$$\text{Hence, } \frac{d}{dx} \operatorname{cosec}^5 x = -5 \operatorname{cosec}^5 x \cot x.$$

(ii) Let  $y = e^{\sin x}$ . Put  $\sin x = t$ . Then  $y = e^t$  and  $t = \sin x$ .

$$\therefore \frac{dy}{dt} = e^t \quad \text{and} \quad \frac{dt}{dx} = \cos x.$$

$$\text{Now } \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = e^t \cdot \cos x = e^{\sin x} \cdot \cos x.$$

(iii) Let  $y = \tan(\sin x)$ . Put  $\sin x = t$ . Then  $y = \tan t$ .

$$\therefore \frac{dy}{dt} = \sec^2 t \quad \text{and} \quad \frac{dt}{dx} = \cos x.$$

$$\text{Now } \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = (\sec^2 t) \cdot (\cos x) = (\cos x) \sec^2(\sin x).$$

(iv) Let  $y = e^{\tan^{-1} x}$ . Put  $\tan^{-1} x = t$ . Then  $y = e^t$  and  $t = \tan^{-1} x$ .

$$\therefore \frac{dy}{dt} = e^t \quad \text{and} \quad \frac{dt}{dx} = \frac{1}{1+x^2}.$$

$$\text{Now } \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = e^t \cdot \frac{1}{1+x^2} = \frac{e^{\tan^{-1} x}}{1+x^2}.$$



**Illustration 2 :** Find the derivatives of the following functions with respect to  $x$

(i)  $\sin^{-1}(\sqrt{x})$

(ii)  $\log_5(\log_5 x)$

(iii)  $\log_e \left(x + \frac{1}{x}\right)$

[UPTU 2002]

(iv)  $\log_3(x^2 + 3x)$ .

[UPTU 2001 (Special)]

**Solution :** (i)  $\frac{d}{dx} \sin^{-1}(\sqrt{x}) = \frac{1}{\sqrt{1-(\sqrt{x})^2}} \cdot \frac{d}{dx} x^{1/2}$   
 $= \frac{1}{\sqrt{1-x}} \cdot \frac{1}{2} x^{-1/2} = \frac{1}{2\sqrt{x(1-x)}}.$

(ii)  $\frac{d}{dx} \log_5(\log_5 x) = \frac{1}{\log_5 x \cdot \log_e 5} \cdot \frac{d}{dx} \log_5 x$   
 $= \frac{1}{\log_5 x \cdot \log_e 5} \cdot \frac{1}{x \log_e 5} = \frac{1}{(\log_e 5)^2 \cdot x \log_5 x}.$

(iii)  $\frac{d}{dx} \log_e \left(x + \frac{1}{x}\right) = \frac{1}{\left(x + \frac{1}{x}\right)} \cdot \frac{d}{dx} \left(x + \frac{1}{x}\right)$   
 $= \frac{1}{x + \frac{1}{x}} \left\{ 1 + \left(-\frac{1}{x^2}\right) \right\} = \frac{\left(1 - \frac{1}{x^2}\right)}{\frac{x^2 + 1}{x}} = \frac{(x^2 - 1)}{x(x^2 + 1)}.$

(iv)  $\frac{d}{dx} \log_3(x^2 + 3x) = \frac{1}{(x^2 + 3x) \log_e 3} \cdot \frac{d}{dx} (x^2 + 3x)$   
 $= \frac{1}{(x^2 + 3x) \log_e 3} (2x + 3).$

**Illustration 3 :** Find the derivatives of the following functions with respect to  $x$ .

(i)  $\log \tan \left(\frac{x}{2}\right)$

(ii)  $\sqrt{\frac{1 - \tan x}{1 + \tan x}}$

(iii)  $\log \log \log x^3$

[UPTU 2007]

(iv)  $\sin(\sqrt{\sin x + \cos x})$

(v)  $\log \sin(x^2 + 1)$ .

[UPTU 2004]

(vi)  $\sin(a \tan^{-1} x)$ .

[UPTU 2005]

**Solution :** (i)  $\frac{d}{dx} \log \tan \left(\frac{x}{2}\right) = \frac{1}{\tan(x/2)} \cdot \frac{d}{dx} \tan \left(\frac{x}{2}\right)$   
 $= \frac{1}{\tan(x/2)} \cdot \sec^2(x/2) \cdot \frac{d}{dx} \left(\frac{x}{2}\right)$   
 $= \frac{1}{\sin(x/2) \cos(x/2)} \cdot \frac{1}{2} \cdot \frac{d}{dx} (x)$   
 $= \frac{1}{2 \sin(x/2) \cos(x/2)} = \frac{1}{\sin x} = \operatorname{cosec} x.$

$$\begin{aligned}
 \text{(ii)} \quad \frac{d}{dx} \left( \frac{1 - \tan x}{1 + \tan x} \right)^{1/2} &= \frac{1}{2} \left( \frac{1 - \tan x}{1 + \tan x} \right)^{-1/2} \cdot \frac{d}{dx} \left( \frac{1 - \tan x}{1 + \tan x} \right) \\
 &= \frac{1}{2} \left( \frac{1 + \tan x}{1 - \tan x} \right)^{1/2} \cdot \frac{(1 + \tan x) \cdot (-\sec^2 x) - (1 - \tan x) \cdot \sec^2 x}{(1 + \tan x)^2} \\
 &= \frac{1}{2} \frac{(1 + \tan x)^{1/2}}{(1 - \tan x)^{1/2}} \cdot \frac{-2 \sec^2 x}{(1 + \tan x)^2} = \frac{-\sec^2 x}{(1 - \tan x)^{1/2} (1 + \tan x)^{3/2}}.
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii)} \quad \frac{d}{dx} \log \log \log x^3 &= \frac{1}{\log \log x^3} \cdot \frac{d}{dx} \log \log x^3 \\
 &= \frac{1}{\log \log x^3} \cdot \frac{1}{\log x^3} \cdot \frac{d}{dx} \log x^3 \\
 &= \frac{1}{\log \log x^3} \cdot \frac{1}{3 \log x} \cdot \frac{d}{dx} (3 \log x) \\
 &= \frac{1}{\log \log x^3} \cdot \frac{1}{3 \log x} \cdot 3 \cdot \frac{1}{x} = \frac{1}{(x \log x)(\log \log x^3)}.
 \end{aligned}$$

$$\begin{aligned}
 \text{(iv)} \quad \frac{d}{dx} \sin(\sqrt{\sin x + \cos x}) &= \cos(\sqrt{\sin x + \cos x}) \cdot \frac{d}{dx} \sqrt{\sin x + \cos x} \\
 &= \cos(\sqrt{\sin x + \cos x}) \cdot \frac{1}{2} (\sin x + \cos x)^{-1/2} \cdot \frac{d}{dx} (\sin x + \cos x) \\
 &= \frac{(\cos x - \sin x) \cos(\sqrt{\sin x + \cos x})}{2 \sqrt{\sin x + \cos x}}.
 \end{aligned}$$

$$\begin{aligned}
 \text{(v)} \quad \frac{d}{dx} \log \sin(x^2 + 1) &= \frac{1}{\sin(x^2 + 1)} \cdot \frac{d}{dx} \sin(x^2 + 1) \\
 &= \frac{1}{\sin(x^2 + 1)} \cos(x^2 + 1) \cdot \frac{d}{dx} (x^2 + 1) = 2x \cot(x^2 + 1).
 \end{aligned}$$

$$\begin{aligned}
 \text{(vi)} \quad \frac{d}{dx} \sin(a \tan^{-1} x) &= \cos(a \tan^{-1} x) \cdot \frac{d}{dx} a (\tan^{-1} x) \\
 &= \frac{a}{a + x^2} \cos(a \tan^{-1} x).
 \end{aligned}$$

**Illustration 4 :** Find  $\frac{dy}{dx}$  by using chain rule for the following functions

$$y = u^2 + u + 2, u = v^2 + 5, v = 6x + 7$$

**Solution :** We have

$$y = u^2 + u + 2, u = v^2 + 5, v = 6x + 7$$

$$\therefore \frac{dy}{du} = 2u + 1, \frac{du}{dv} = 2v, \frac{dv}{dx} = 6$$

Now by chain rule,

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx} = (2u + 1) \cdot (2v) \cdot (6)$$

$$\begin{aligned}
 &= \{2(v^2 + 5) + 1\} \cdot 12v \\
 &= (2u^2 + 11) \cdot 12v \\
 &= \{2(6x + 7)^2 + 11\} \cdot 12(6x + 7) \\
 &= 12(6x + 7)(72x^2 + 168x + 109)
 \end{aligned}$$

**Illustration 5:** If  $y = z + 1$ ,  $z = w + 2$ ,  $w = u^2$ ,  $u = 4x + 5$ , find  $\frac{dy}{dw}$ ,  $\frac{dw}{dx}$ ,  $\frac{dz}{dx}$  and  $\frac{dy}{dx}$ .

**Solution :** We have  $y = z + 1$ ,  $z = w + 2$ ,  $w = u^2$ ,  $u = 4x + 5$ .

$$\therefore \frac{dy}{dz} = 1, \frac{dz}{dw} = 1, \frac{dw}{du} = 2u, \frac{du}{dx} = 4$$

$$\text{Now } \frac{dy}{dw} = \frac{dy}{dz} \cdot \frac{dz}{dw} = 1 \cdot 1 = 1$$

$$\frac{dw}{dx} = \frac{dw}{du} \cdot \frac{du}{dx} = 2u \cdot 4 = 8u$$

$$= 8(4x + 5).$$

$$\frac{dz}{dx} = \frac{dz}{dw} \cdot \frac{dw}{du} \cdot \frac{du}{dx} = 1 \cdot 2u \cdot 4$$

$$= 8u = 8(4x + 5)$$

$$\frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{dz}{dw} \cdot \frac{dw}{du} \cdot \frac{du}{dx}$$

$$= 1 \cdot 1 \cdot 2u \cdot 4 = 8u = 8(4x + 5)$$

**Illustration 6:** If  $y = u^2 + 1$  and  $u = x^3 + x + 4$ , find  $\frac{dy}{dx}$  (i) Without using chain rule

(ii) Using chain rule. Verify that the derivatives are same.

**Solution :** We have  $y = u^2 + 1$  and  $u = x^3 + x + 4$

(i) Without using chain rule

$$\text{Now } y = u^2 + 1 = (x^3 + x + 4)^2 + 1 \quad [\because u = x^3 + x + 4]$$

$$= x^6 + x^2 + 16 + 2x^4 + 8x + 8x^3 + 1$$

$$= x^6 + 2x^4 + 8x^3 + x^2 + 8x + 17$$

$$\therefore \frac{dy}{dx} = 6x^5 + 8x^3 + 24x^2 + 2x + 8. \quad \dots(1)$$

(ii) Using chain rule

$$\text{Now } \frac{dy}{du} = 2u, \frac{du}{dx} = 3x^2 + 1$$

$$\therefore \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = (2u) \cdot (3x^2 + 1)$$

$$= 2(x^3 + x + 4)(3x^2 + 1)$$

$$= 6x^5 + 8x^3 + 24x^2 + 2x + 8 \quad \dots(2)$$

From (1) and (2) we see that derivatives are same.

## Comprehensive Exercise 3

### 1. Differentiate the following functions with respect to $x$ .

- (i)  $\cos(ax + b)$ . (ii)  $\tan(3x/2)$ .  
 (iii)  $\sin x^2$ . (iv)  $e^{\cos x}$ .  
 (v)  $\tan^{-1}(5x)$ . (vi)  $\sin^{-1} e^x$ .  
 (vii)  $\sin[\cos(\tan x)]$ . [UPTU 2006]

### 2. Find $\frac{dy}{dx}$ by using chain rule for the following functions :

- (i)  $y = u^2 + u + 5, u = x^2 - x + 1$   
 (ii)  $y = \sqrt{z}, z = 5x^4 + 3$   
 (iii)  $y = u^2, u = x^2 + 4$   
 (iv)  $y = v^2 + v + 1, v = u^2 + 1, u = 6x$   
 (v)  $y = v^3 + 2v^2 + 4, v = 3u + 7, u = 9x + 3$   
 (vi)  $y = 2u^2 + 3, u = 12v^2 + v, v = 3w + 8, w = 2x$

### 3. If $y = u^2$ and $u = 1 + x^3$ , find the derivative of $y$ w.r. to $x$ :

- (i) without using chain rule (ii) using chain rule.

## Answers

1. (i)  $-a \sin(ax + b)$ . (ii)  $\frac{3}{2} \sec^2(3x/2)$ .  
 (iii)  $2x \cos x^2$ . (iv)  $-\sin x \cdot e^{\cos x}$ .  
 (v)  $\frac{5}{1 + 25x^2}$ . (vi)  $\frac{e^x}{\sqrt{1 - e^{2x}}}$ .  
 (vii)  $-\cos\{\cos(\tan x)\}[\sin(\tan x)][\sec^2 x]$ .
2. (i)  $(2x - 1)(2x^2 - 2x + 3)$  (ii)  $10x^3 / \sqrt{5x^4 + 3}$   
 (iii)  $4x(x^2 + 4)$  (iv)  $72x(72x^2 + 3)$   
 (v)  $27(2187x^2 + 2700x + 832)$   
 (vi)  $24(62208x^3 + 250128x^2 + 335238x + 149768)$
3.  $6x^2(1 + x^3)$ .

## 9.8 Differential Coefficients Of Inverse Trigonometric Functions

$$\begin{aligned}\frac{d}{dx} \sin^{-1} x &= \frac{1}{\sqrt{1-x^2}}, \quad \frac{d}{dx} \cos^{-1} x = -\frac{1}{\sqrt{1-x^2}}; \quad \frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2}; \\ \frac{d}{dx} \cot^{-1} x &= -\frac{1}{1+x^2}; \quad \frac{d}{dx} \sec^{-1} x = \frac{1}{x\sqrt{x^2-1}}; \\ \frac{d}{dx} \operatorname{cosec}^{-1} x &= -\frac{1}{x\sqrt{x^2-1}}.\end{aligned}$$

### Illustrative Examples

**Illustration 1 :** Find the derivatives of the following functions with respect to  $x$ .

- (i)  $\tan^{-1} \log x$  (ii)  $e^{\sin^{-1}(x+1)}$   
(iii)  $(\cot^{-1} x)^2$  (iv)  $\cos^{-1}(\cot x)$ .

**Solution :** (i)  $\frac{d}{dx} \{\tan^{-1}(\log x)\} = \frac{1}{1+(\log x)^2} \cdot \frac{d}{dx} \log x$   

$$= \frac{1}{1+(\log x)^2} \cdot \frac{1}{x} = \frac{1}{x[1+(\log x)^2]}.$$

(ii)  $\frac{d}{dx} \{e^{\sin^{-1}(x+1)}\} = e^{\sin^{-1}(x+1)} \cdot \frac{d}{dx} \sin^{-1}(x+1)$   

$$= e^{\sin^{-1}(x+1)} \cdot \frac{1}{\sqrt{1-(x+1)^2}} \cdot \frac{d}{dx} (x+1) = \frac{e^{\sin^{-1}(x+1)}}{\sqrt{1-(x+1)^2}}.$$

(iii)  $\frac{d}{dx} (\cot^{-1} x)^2 = 2 \cot^{-1} x \cdot \frac{d}{dx} \cot^{-1} x = 2 \cot^{-1} x \cdot \left(-\frac{1}{1+x^2}\right)$   

$$= \frac{-2 \cot^{-1} x}{1+x^2}.$$

(iv)  $\frac{d}{dx} \{\cos^{-1}(\cot x)\} = \left(-\frac{1}{\sqrt{1-\cot^2 x}}\right) \cdot \frac{d}{dx} (\cot x)$   

$$= \left(-\frac{1}{\sqrt{1-\cot^2 x}}\right) \cdot (-\operatorname{cosec}^2 x)$$
  

$$= \frac{\operatorname{cosec}^2 x}{\sqrt{1-\cot^2 x}}.$$

**Illustration 2 :** Find the derivatives of the following functions with respect to  $x$

(i)  $\tan (\sin^{-1} x)$

(ii)  $\sin (\tan^{-1} x)$ .

**Solution :** (i) Let  $y = \tan (\sin^{-1} x)$ .

Putting  $\sin^{-1} x = t$ , we get  $y = \tan t$  and  $t = \sin^{-1} x$ .

$$\therefore \frac{dy}{dt} = \sec^2 t \text{ and } \frac{dt}{dx} = \frac{1}{\sqrt{1-x^2}}.$$

$$\begin{aligned} \text{Now } \frac{dy}{dx} &= \frac{dy}{dt} \cdot \frac{dt}{dx} = \sec^2 t \cdot \frac{1}{\sqrt{1-x^2}} = \frac{1}{\cos^2 t} \cdot \frac{1}{\sqrt{1-x^2}} \\ &= \frac{1}{1-\sin^2 t} \cdot \frac{1}{\sqrt{1-x^2}} = \frac{1}{1-x^2} \cdot \frac{1}{\sqrt{1-x^2}} \\ & \quad [\because t = \sin^{-1} x \Rightarrow \sin t = x] \\ &= \frac{1}{(1-x^2)^{3/2}}. \end{aligned}$$

(ii) Let  $y = \sin (\tan^{-1} x)$ .

Putting  $\tan^{-1} x = t$ , we get  $y = \sin t$  and  $t = \tan^{-1} x$ .

$$\therefore \frac{dy}{dt} = \cos t \quad \text{and} \quad \frac{dt}{dx} = \frac{1}{1+x^2}.$$

$$\begin{aligned} \text{Now } \frac{dy}{dx} &= \frac{dy}{dt} \cdot \frac{dt}{dx} = \cos t \cdot \frac{1}{1+x^2} \\ &= \frac{1}{\sec t} \cdot \frac{1}{1+x^2} = \frac{1}{\sqrt{1+\tan^2 t}} \cdot \frac{1}{1+x^2} \\ &= \frac{1}{\sqrt{1+x^2}} \cdot \frac{1}{1+x^2} \quad [\because t = \tan^{-1} x \Rightarrow \tan t = x] \\ &= \frac{1}{(1+x^2)^{3/2}}. \end{aligned}$$

## Comprehensive Exercise 4

Find the derivatives of the following functions with respect to  $x$ .

1.  $\tan^{-1} (\cot x)$ .

2.  $\sin^{-1} \sqrt{x}$ .

3.  $\sec^{-1} \sqrt{x}$ .

4.  $\cot^{-1} x^2$ .

5.  $\log (\tan^{-1} x)$ .

6.  $(\cot^{-1} x^2)^3$ .

## Answers

1.  $-1$ .                      2.  $\frac{1}{2\sqrt{x(1-x)}}$ .                      3.  $\frac{1}{2x\sqrt{x-1}}$ .
4.  $\frac{-2x}{1+x^4}$ .                      5.  $\frac{1}{(1+x^2)\tan^{-1}x}$ .                      6.  $\frac{-6x(\cot^{-1}x^2)^2}{1+x^4}$ .

## 9.9 Logarithmic Differentiation

Whenever we are required to differentiate a function of  $x$  in which a function of  $x$  is raised to a power which itself is a function of  $x$ , neither the formula for  $a^x$  nor that for  $x^n$  is applicable. In such cases we first take logarithm of the function and then differentiate. This process is called *logarithmic differentiation*. It is also helpful in the cases where we are to differentiate a function which consists of the product or the quotient of a number of functions.

## Illustrative Examples

**Illustration 1 :** Differentiate the following functions with respect to  $x$ .

- (i)  $x^x$                                               (ii)  $x^{\sin^{-1}x}$                                               [UPTU 2003]  
 (iii)  $x^{\sin x}$ . [UPTU 2004]                                              (iv)  $(\sin x)^{\log x}$

**Solution :** (i) Let  $y = x^x$ . Then  $\log y = \log x^x = x \cdot \log x$ .

Differentiating both sides w.r.t.  $x$ , we get

$$\frac{1}{y} \frac{dy}{dx} = x \cdot \frac{1}{x} + (\log x) \cdot 1 = 1 + \log x.$$

$$\therefore \frac{dy}{dx} = y (1 + \log x) = x^x (1 + \log x).$$

Thus,  $\frac{d}{dx} x^x = x^x (1 + \log x).$

(ii) Let  $y = x^{\sin^{-1}x}$ .

Then  $\log y = \log (x^{\sin^{-1}x}) = \sin^{-1}x \cdot \log x.$

Differentiating both sides w.r.t.  $x$ , we get

$$\frac{1}{y} \frac{dy}{dx} = (\sin^{-1}x) \cdot \frac{1}{x} + (\log x) \cdot \left( \frac{1}{\sqrt{1-x^2}} \right).$$

$$\begin{aligned}\therefore \frac{dy}{dx} &= y \left[ \frac{\sin^{-1} x}{x} + \frac{\log x}{\sqrt{1-x^2}} \right] \\ &= x^{\sin^{-1} x} \left[ \frac{\sin^{-1} x}{x} + \frac{\log x}{\sqrt{1-x^2}} \right].\end{aligned}$$

(iii) Let  $y = x^{\sin x}$ .

Then  $\log y = \log (x^{\sin x}) = \sin x \log x$ .

Differentiating both sides w.r.t.  $x$ , we get

$$\begin{aligned}\frac{1}{y} \frac{dy}{dx} &= \sin x \cdot \frac{1}{x} + \log x \cdot (\cos x). \\ \therefore \frac{dy}{dx} &= y \left[ \frac{\sin x}{x} + (\cos x) \log x \right] \\ &= x^{\sin x} \left[ \frac{\sin x}{x} + (\cos x) \log x \right].\end{aligned}$$

(iv) Let  $y = (\sin x)^{\log x}$ .

Then  $\log y = \log \{(\sin x)^{\log x}\} = \log x \cdot \log \sin x$ .

Differentiating both sides w.r.t.  $x$ , we get

$$\begin{aligned}\frac{1}{y} \frac{dy}{dx} &= (\log x) \cdot \left( \frac{1}{\sin x} \cdot \cos x \right) + (\log \sin x) \cdot \frac{1}{x}. \\ \therefore \frac{dy}{dx} &= y \left[ \cot x \log x + \frac{\log \sin x}{x} \right] \\ &= (\sin x)^{\log x} \left[ \cot x \log x + \frac{\log \sin x}{x} \right].\end{aligned}$$

**Illustration 2 :** If  $y = x^{(x^x)}$ , find  $\frac{dy}{dx}$ .

[UPTU 2008]

**Solution :** We have  $y = x^{(x^x)}$ .

$$\therefore \log y = x^x \cdot \log x.$$

Differentiating both sides w.r.t.  $x$ , we get

$$\begin{aligned}\frac{1}{y} \frac{dy}{dx} &= x^x \cdot \frac{1}{x} + (\log x) \cdot \frac{d}{dx} x^x \quad [\text{By the product rule}] \\ \therefore \frac{dy}{dx} &= y \left[ x^{x-1} + (\log x) \cdot \left\{ x \cdot x^{x-1} \cdot \frac{d}{dx} (x) + x^x \log x \cdot \frac{d}{dx} (x) \right\} \right] \\ &= y [x^{x-1} + (\log x) \cdot (x^x + x^x \log x)] \\ &= x^{(x^x)} [x^{x-1} + x^x (1 + \log x) \log x].\end{aligned}$$



**Illustration 3 :** If  $y = \log (x^x + \operatorname{cosec}^2 x)$ , find  $(dy / dx)$ .

**Solution :** We have  $y = \log (x^x + \operatorname{cosec}^2 x)$ .

$$\begin{aligned} \therefore \frac{dy}{dx} &= \frac{1}{x^x + \operatorname{cosec}^2 x} \cdot \frac{d}{dx} (x^x + \operatorname{cosec}^2 x) \\ &= \frac{1}{x^x + \operatorname{cosec}^2 x} \cdot \left\{ \frac{d}{dx} x^x + \frac{d}{dx} \operatorname{cosec}^2 x \right\} \\ &= \frac{1}{x^x + \operatorname{cosec}^2 x} \times \\ &\quad \left\{ x \cdot x^{x-1} \cdot \frac{d}{dx} (x) + x^x \log x \cdot \frac{d}{dx} (x) + 2 \operatorname{cosec} x \cdot (-\operatorname{cosec} x \cot x) \right\} \\ &= \frac{1}{x^x + \operatorname{cosec}^2 x} \cdot \{x^x (1 + \log x) - 2 \operatorname{cosec}^2 x \cot x\}. \end{aligned}$$

**Illustration 4 :** If  $y = (\tan x)^{\cot x} + (\cot x)^{\tan x}$ , find  $(dy / dx)$ . [UPTU 2009]

**Solution :** Let  $u = (\tan x)^{\cot x}$  and  $v = (\cot x)^{\tan x}$ .

Then  $y = u + v$ .

$$\therefore \frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx} \quad \dots(1)$$

Now  $u = (\tan x)^{\cot x}$

$$\Rightarrow \log u = \cot x \cdot \log \tan x$$

$$\Rightarrow \frac{1}{u} \frac{du}{dx} = \cot x \cdot \left( \frac{1}{\tan x} \cdot \sec^2 x \right) + (\log \tan x) \cdot (-\operatorname{cosec}^2 x)$$

[Differentiating both sides w.r.t.  $x$ ]

$$\begin{aligned} \Rightarrow \frac{du}{dx} &= u [\operatorname{cosec}^2 x - \operatorname{cosec}^2 x \log \tan x] \\ &= (\tan x)^{\cot x} \cdot \operatorname{cosec}^2 x (1 - \log \tan x). \end{aligned}$$

Again  $v = (\cot x)^{\tan x}$ .

$$\therefore \log v = \tan x \cdot \log \cot x.$$

Differentiating both sides w.r.t.  $x$ , we get

$$\frac{1}{v} \frac{dv}{dx} = (\tan x) \cdot \left\{ \frac{1}{\cot x} \cdot (-\operatorname{cosec}^2 x) \right\} + (\log \cot x) \cdot \sec^2 x$$

$$\begin{aligned} \text{or } \frac{dv}{dx} &= v [-\sec^2 x + \sec^2 x \log \cot x] \\ &= (\cot x)^{\tan x} \sec^2 x \{(\log \cot x) - 1\}. \end{aligned}$$

Putting the values of  $du / dx$  and  $dv / dx$  in (1), we get

$$\begin{aligned} \frac{dy}{dx} &= (\tan x)^{\cot x} \cdot \operatorname{cosec}^2 x (1 - \log \tan x) \\ &\quad + (\cot x)^{\tan x} \cdot \sec^2 x \{(\log \cot x) - 1\}. \end{aligned}$$

## 9.10 Implicit Functions

If  $y$  is a function of  $x$  given by a relation of the type  $y = f(x)$ , then  $y$  is said to be an **explicit function** of  $x$ . On the other hand, if the relation between  $x$  and  $y$  is given by an equation involving both  $x$  and  $y$  s.t.  $y$  cannot be expressed as  $y = f(x)$ , then  $y$  is said to be an **implicit function** of  $x$ . If we are given  $y$  implicitly in terms of  $x$ , we can find  $dy/dx$  without first expressing  $y$  explicitly in terms of  $x$ . Thinking of  $y$  as a function of  $x$ , we differentiate both sides of the given equation with respect to  $x$  and then solve the resulting relation for  $dy/dx$ .

### Illustrative Examples

**Illustration 1 :** If  $x^y = e^{x-y}$ , prove that  $\frac{dy}{dx} = \frac{\log x}{(1 + \log x)^2}$ .

**Solution :** We are given that  $x^y = e^{x-y}$ .

Taking logarithm of both sides, we get

$$y \log x = (x - y)$$

$$\text{or } y(1 + \log x) = x$$

$$\text{or } y = \frac{x}{1 + \log x}.$$

$$\begin{aligned} \therefore \frac{dy}{dx} &= \frac{(1 + \log x) \cdot 1 - x \cdot \{0 + (1/x)\}}{(1 + \log x)^2} \\ &= \frac{1 + \log x - 1}{(1 + \log x)^2} = \frac{\log x}{(1 + \log x)^2}. \end{aligned}$$

**Illustration 2 :** If  $\sin y = x \sin(a + y)$ , prove that

$$\frac{dy}{dx} = \frac{\sin(a + y)}{\cos y - x \cos(a + y)}.$$

[UPTU 2008]

**Solution :** Given that  $\sin y = x \sin(a + y)$ .

...(1)

Differentiating both sides of (1) with respect to  $x$ , we get

$$\cos y \cdot \frac{dy}{dx} = x \cdot \cos(a + y) \cdot \frac{d}{dx}(a + y) + \sin(a + y) \cdot \frac{d}{dx}(x)$$

$$\text{or } \cos y \cdot \frac{dy}{dx} = x \cos(a + y) \frac{dy}{dx} + \sin(a + y)$$

$$\text{or } \frac{dy}{dx} \{\cos y - x \cos(a + y)\} = \sin(a + y)$$

$$\text{or } \frac{dy}{dx} = \frac{\sin(a + y)}{\cos y - x \cos(a + y)}.$$

**Illustration 3 :** If  $x\sqrt{1+y} + y\sqrt{1+x} = 0$ , show that

$$\frac{dy}{dx} = -\frac{1}{(x+1)^2}.$$

**Solution :** Given that  $x\sqrt{1+y} + y\sqrt{1+x} = 0$

or  $x\sqrt{1+y} = -y\sqrt{1+x}$

or  $x^2(1+y) = -y^2(1+x)$  [On squaring both sides]

or  $x^2 + x^2y = y^2 + xy^2$

or  $x^2 - y^2 = xy^2 - x^2y$

or  $(x+y)(x-y) = -xy(x-y)$  [ $\because (x-y) \neq 0$ ]

or  $x+y = -xy$  or  $x+y+xy = 0$

or  $x+y(1+x)$  or  $y = -\frac{x}{1+x}$

$\therefore \frac{dy}{dx} = -\frac{(1+x)-x}{(1+x)^2} = \frac{-1}{(1+x)^2}.$

**Illustration 4 :** If  $e^x + e^y = e^{x+y}$ , show that  $\frac{dy}{dx} + e^{y-x} = 0$ .

**Solution :** Given that  $e^x + e^y = e^{x+y}$

or  $e^x + e^y = e^x e^y$  or  $e^x = e^y(e^x - 1)$

or  $e^y = \frac{e^x}{e^x - 1}$  ....(1)

Differentiating both sides of (1) w.r. to  $x$ , we get

$$e^y \frac{dy}{dx} = \frac{(e^x - 1)e^x - e^x(e^x - 0)}{(e^x - 1)^2}$$

$$= \frac{e^{2x} - e^x - e^{2x}}{(e^x - 1)^2} = \frac{-e^x}{(e^x - 1)^2}$$

or  $\frac{dy}{dx} = \frac{-e^x}{e^y(e^x - 1)^2} = \frac{-e^{x-y}}{(e^x - 1)^2}$

$$= \frac{-e^{x-y}}{(e^x / e^y)^2} \quad \left[ \because e^x - 1 = \frac{e^x}{e^y} \right]$$

$$= -\frac{e^{x-y}}{(e^{x-y})^2} = -\frac{1}{e^{x-y}} = -e^{-(x-y)}$$

$$= -e^{y-x}$$

or  $\frac{dy}{dx} + e^{y-x} = 0.$

**Illustration 5 :** Find  $\frac{dy}{dx}$  for the following implicit function  $xy + xe^{-y} + ye^x = x^2$ .

**Solution :** Given that  $xy + xe^{-y} + ye^x = x^2$

Differentiating both sides w.r. to  $x$ , we get

$$\left(x \frac{dy}{dx} + y \cdot 1\right) + \left(x(-e^{-y}) \frac{dy}{dx} + e^{-y}\right) + \left(ye^x + e^x \frac{dy}{dx}\right) = 2x$$

or  $\frac{dy}{dx}(x - xe^{-y} + e^x) = 2x - y - e^{-y} - ye^x$

or  $\frac{dy}{dx} = \frac{2x - y - e^{-y} - ye^x}{x - xe^{-y} + e^x}.$

**Illustration 6 :** Find  $\frac{dy}{dx}$  for the following implicit function  $\sin(xy) + \frac{x}{y} = x^2 - y$

**Solution :** Given that  $\sin(xy) + \frac{x}{y} = x^2 - y$

Differentiating both sides w.r.t.  $x$ , we get

$$\cos(xy) \left[ x \frac{dy}{dx} + y \cdot 1 \right] + \frac{y \cdot 1 - x \frac{dy}{dx}}{y^2} = 2x - \frac{dy}{dx}$$

or  $x \cos(xy) \frac{dy}{dx} + y \cos(xy) + \frac{1}{y} - \frac{x}{y^2} \frac{dy}{dx} = 2x - \frac{dy}{dx}$

or  $\frac{dy}{dx} \left[ x \cos(xy) - \frac{x}{y^2} + 1 \right] = 2x - \frac{1}{y} - y \cos(xy)$

or  $\frac{dy}{dx} \left[ \frac{xy^2 \cos(xy) - x + y^2}{y^2} \right] = \frac{2xy - 1 - y^2 \cos(xy)}{y}$

or  $\frac{dy}{dx} = \frac{y(2xy - 1 - y^2 \cos(xy))}{xy^2 \cos(xy) - x + y^2}$

## Comprehensive Exercise 5

1. Find  $(dy / dx)$  when  $x$  and  $y$  are connected by the following relations:

(i)  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$

(ii)  $y^2 = 4a \{x + a \sin(x/a)\}.$

(iii)  $x^{2/3} + y^{2/3} = a^{2/3}$

(iv)  $ax^2 + 2hxy + by^2 = 1.$

$$(v) 3x^2y - y^3 - 2 = 0.$$

$$(vi) \cos(x + y) = y \sin x.$$

$$(vii) x^3 + y^3 = 3axy$$

$$(viii) y \sec x + \tan y + xy = 0$$

2. Find the derivatives of the following functions with respect to  $x$ .

$$(i) (\sin x)^x.$$

$$(ii) x^{1/x}.$$

$$(iii) (2x + 3)^{x-5}.$$

$$(iv) (x + 1)^x.$$

$$(v) (\log x)^x.$$

$$(vi) x^{\cos^{-1} x}.$$

$$(vii) x^{\sin x + \cos x}.$$

$$(viii) x^{\sqrt{x}}.$$

$$(ix) e^x \cos x + x^{\sin x}. \quad [\text{UPTU 2004}]$$

$$(x) x^{\log x} + (\log x)^x$$

3. Find  $(dy/dx)$  of the following functions :

$$(i) y \tan x - y^2 \cos x + 2x = 0$$

$$(ii) \sin y + \log y = x^2 + 18x + 3$$

$$(iii) \tan(x + y) + \tan(x - y) = 1$$

$$(iv) y^2 \sin x + y \tan x + (1 + x^2) \cos x = 0$$

$$(v) \log \sqrt{(x^2 + y^2)} = \tan^{-1} \frac{y}{x}$$

4. (i) If  $y = x \sin y$ , show that  $x \frac{dy}{dx} = \frac{y}{1 - x \cos y}.$

(ii) If  $xy = \tan(xy)$ , show that  $\frac{dy}{dx} = -\frac{y}{x}.$

(iii) If  $\sec y = x^3$ , show that  $x \frac{dy}{dx} = \frac{3}{\sqrt{x^6 - 1}}.$

## Answers

1. (i)  $-\frac{b^2 x}{a^2 y}.$

(ii)  $\frac{2a [1 + \cos(x/a)]}{y}.$

(iii)  $-\frac{y^{1/3}}{x^{1/3}}.$

(iv)  $-\frac{ax + hy}{hx + by}.$

(v)  $\frac{2xy}{y^2 - x^2}.$

(vi)  $-\frac{\sin(x + y) + y \cos x}{\sin x + \sin(x + y)}.$

(vii)  $\frac{ay - x^2}{y^2 - ax}$

(viii)  $-\frac{y \sec x \tan x + y}{\sec x + \sec^2 y + x}$

2. (i)  $(\sin x)^x [x \cot x + \log \sin x]$ .
- (ii)  $\frac{x^{1/x} (1 - \log x)}{x^2}$ .
- (iii)  $(2x + 3)^{x-5} \left[ \frac{2(x-5)}{2x+3} + \log(2x+3) \right]$ .
- (iv)  $(x+1)^x \left[ \frac{x}{x+1} + \log(x+1) \right]$ .
- (v)  $(\log x)^x \left[ \frac{1}{\log x} + \log \log x \right]$ .
- (vi)  $x^{\cos^{-1} x} \left[ \frac{\cos^{-1} x}{x} - \frac{\log x}{\sqrt{1-x^2}} \right]$ .
- (vii)  $x^{\sin x + \cos x} \left[ \frac{\sin x + \cos x}{x} + (\cos x - \sin x) \log x \right]$ .
- (viii)  $x^{\sqrt{x}} \left[ \frac{\sqrt{x}}{x} + \frac{\log x}{2\sqrt{x}} \right]$ .
- (ix)  $e^x (\cos x - \sin x) + x^{\sin x} \left[ \frac{\sin x}{x} + (\cos x) \log x \right]$ .
- (x)  $x^{\log x} \left( \frac{2 \log x}{x} \right) + (\log x)^x \left( \frac{1}{\log x} + \log(\log x) \right)$
3. (i)  $\frac{y \sec^2 x + y^2 \sin x + 2}{2y \cos x - \tan x}$
- (ii)  $\frac{2y(x+9)}{y \cos y + 1}$
- (iii)  $-\frac{\sec^2(x+y) + \sec^2(x-y)}{\sec^2(x+y) - \sec^2(x-y)}$
- (iv)  $\frac{(1+x^2) \sin x - (2x+y^2) \cos x - y \sec^2 x}{2y \sin x + \tan x}$
- (v)  $\frac{x+y}{x-y}$ .

## 9.11 Parametric Equations

If  $x$  and  $y$  are both expressed in terms of a third variable, say  $t$ , then  $t$  is usually called a *parameter*. In the case of parametric equations we can always find  $dy/dx$ , without first eliminating the parameter.

Thus, if the parametric equations are  $x = \phi(t)$ ,  $y = \psi(t)$ , then

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} \quad \text{or} \quad \frac{dy}{dx} = \frac{dy}{dt} \bigg/ \frac{dx}{dt}.$$

## 9.12 Differentiation of a Function with Respect to a Function

Suppose we are to find the differential coefficient of the function  $u = f(x)$  with respect to another function, say,  $v = \phi(x)$ .

It means we are to find  $du/dv$ , where  $u$  and  $v$  are both given in terms of a third variable  $x$ . Therefore, as in the case of parametric equations, we have

$$\frac{du}{dv} = \frac{du/dx}{dv/dx}, \quad \text{i.e.,} \quad \frac{df(x)}{d\phi(x)} = \frac{df(x)}{dx} \bigg/ \frac{d\phi(x)}{dx}.$$

## Illustrative Examples

**Illustration 1 :** (i) If  $x = at^2$  and  $y = 2at$ , find  $dy/dx$ .

[UPTU 2005]

(ii) If  $x = a(\cos t + \log \tan \frac{1}{2}t)$ ,  $y = a \sin t$ , find  $dy/dx$ .

**Solution :** (i) Here  $\frac{dx}{dt} = 2at$  and  $\frac{dy}{dt} = 2a$ .

Now, 
$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{2a}{2at} = \frac{1}{t}.$$

(ii) Here 
$$\begin{aligned} \frac{dx}{dt} &= a \left\{ -\sin t + \frac{1}{\tan(t/2)} \cdot \left( \sec^2 \frac{t}{2} \right) \cdot \frac{1}{2} \right\} \\ &= a \left\{ -\sin t + \frac{1}{2 \sin \frac{1}{2}t \cos \frac{1}{2}t} \right\} = a \left( \frac{1 - \sin^2 t}{\sin t} \right) = a \frac{\cos^2 t}{\sin t}. \end{aligned}$$

Also  $\frac{dy}{dt} = a \cos t$ .

Now 
$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{a \cos t}{(a \cos^2 t) / \sin t} = \tan t.$$

**Illustration 2 :** If  $x = a \sec^3 \theta$  and  $y = a \tan^3 \theta$ , find  $dy/dx$  at  $\theta = \pi/4$ .

**Solution :** Here  $\frac{dx}{d\theta} = 3a \sec^2 \theta \cdot \sec \theta \tan \theta = 3a \sec^3 \theta \tan \theta$

and 
$$\frac{dy}{d\theta} = 3a \tan^2 \theta \cdot \sec^2 \theta = 3a \sec^2 \theta \tan^2 \theta.$$

$$\therefore \frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{3a \sec^2 \theta \tan^2 \theta}{3a \sec^3 \theta \tan \theta} = \frac{\tan \theta}{\sec \theta} = \sin \theta.$$

$$\therefore \left( \frac{dy}{dx} \right)_{\text{at } \theta = \pi/4} = \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}}.$$

**Illustration 3 :** Differentiate  $x^{\sin^{-1} x}$  with respect to  $\sin^{-1} x$ .

**Solution :** Let  $u = x^{\sin^{-1} x}$  and  $v = \sin^{-1} x$ .

Then  $\log u = \sin^{-1} x \cdot \log x$ .

$$\therefore \frac{1}{u} \frac{du}{dx} = \frac{1}{\sqrt{1-x^2}} \log x + \frac{1}{x} \sin^{-1} x$$

$$\text{or } \frac{du}{dx} = x^{\sin^{-1} x} \left[ \frac{\log x}{\sqrt{1-x^2}} + \frac{\sin^{-1} x}{x} \right].$$

$$\text{Again } \frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}.$$

$$\begin{aligned} \text{Now } \frac{du}{dv} &= \frac{du/dx}{dv/dx} = \frac{x^{\sin^{-1} x} \left[ \frac{\log x}{\sqrt{1-x^2}} + \frac{\sin^{-1} x}{x} \right]}{\frac{1}{\sqrt{1-x^2}}} \\ &= x^{\sin^{-1} x} \left[ \frac{x \log x + \sqrt{1-x^2} \sin^{-1} x}{x} \right]. \end{aligned}$$

**Illustration 4 :** Differentiate  $\log \sin x$  with respect to  $\sqrt{\cos x}$ .

**Solution :** Let  $u = \log \sin x$  and  $v = \sqrt{\cos x}$ . Then to find  $du/dv$ .

$$\text{We have } u = \log \sin x \Rightarrow \frac{du}{dx} = \frac{1}{\sin x} \cdot \cos x = \cot x$$

$$\text{and } v = \sqrt{\cos x} \Rightarrow \frac{dv}{dx} = \frac{1}{2\sqrt{\cos x}} \cdot (-\sin x) = \frac{-\sin x}{2\sqrt{\cos x}}.$$

$$\begin{aligned} \therefore \frac{du}{dv} &= \frac{du/dx}{dv/dx} = (\cot x) \div \left( \frac{-\sin x}{2\sqrt{\cos x}} \right) = \cot x \cdot \frac{2\sqrt{\cos x}}{(-\sin x)} \\ &= -2\sqrt{\cos x} \cdot \operatorname{cosec} x \cot x. \end{aligned}$$

**Illustration 5 :** Differentiate  $\sin 2x$  with respect to  $\cos x$ .

**Solution :** Let  $u = \sin 2x$  and  $v = \cos x$ .

Then to find  $\frac{du}{dv}$ .

$$\text{We have } u = \sin 2x \Rightarrow \frac{du}{dx} = 2 \cos 2x,$$

$$\text{and } v = \cos x \Rightarrow \frac{dv}{dx} = -\sin x$$

$$\begin{aligned} \therefore \frac{du}{dv} &= \frac{du/dx}{dv/dx} = \frac{2 \cos 2x}{-\sin x} \\ &= -2 \cos 2x \operatorname{cosec} x. \end{aligned}$$



## Comprehensive Exercise 6

### 1. Find $dy/dx$ , when

(i)  $x = \frac{a(1-t^2)}{1+t^2}, y = \frac{2bt}{1+t^2}.$

(ii)  $x = \frac{3}{1+t^3}, y = \frac{3t^2}{1+t^3}.$

(iii)  $x = a \cos t, y = b \sin t.$

(iv)  $x = a \sec \theta, y = b \tan \theta.$

(v)  $x = \sin 2t, y = 2 \cos t.$

(vi)  $x = \log t, y = \sin t.$

(vii)  $x = a(\theta - \sin \theta), y = a(1 - \cos \theta).$

(viii)  $x = \frac{3at}{1+t^2}, y = \frac{3at^2}{1+t^2}.$

2. (i)  $x = a e^{\theta} (\sin \theta - \cos \theta), y = a e^{\theta} (\sin \theta + \cos \theta).$

(ii)  $x = a(\cos \theta + \theta \sin \theta), y = a(\sin \theta - \theta \cos \theta).$

[UPTU 2009]

(iii)  $x = 4 \sin \theta, y = 7 \cos \theta$

(iv)  $x = a \cos^3 \theta, y = a \sin^3 \theta$

3. (i) Differentiate  $x^2$  with respect to  $x^3$ .

(ii) Differentiate  $\tan^{-1} \left( \frac{2x}{1-x^2} \right)$  with respect to  $\tan^{-1} x$ .

(iii) Differentiate  $\sin^{-1} \frac{2x}{1+x^2}$  with respect to  $\tan^{-1} x$ .

(iv) Differentiate  $\tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$  with respect to  $\tan^{-1} \frac{2x}{1-x^2}$ .

## Answers

1. (i)  $\frac{b(t^2-1)}{2at}$

(ii)  $\frac{t^2}{3} - \frac{2}{3t}$

(iii)  $-\frac{b}{a} \cot t.$

(iv)  $\frac{b}{a} \operatorname{cosec} \theta.$

(v)  $\frac{-\sin t}{\cos 2t}.$

(vi)  $t \cos t.$

(vii)  $\cot \theta / 2.$

(viii)  $\frac{2t}{1-t^2}.$

2. (i)  $\cot \theta$ . (ii)  $\tan \theta$ . (iii)  $\frac{-7}{4} \tan \theta$  (iv)  $-\tan \theta$
3. (i)  $\frac{2}{3x}$ . (ii) 2 (iii) 2 (iv)  $1/4$ .

## 9.13 Differentiation of Infinite Recurring Expressions

If from an infinite collection of terms we take out a single term or a finite number of terms, the collection still remains infinite. We use this fact to find the derivatives of infinite recurring expressions.

### Illustrative Examples

**Illustration 1 :** If  $y = \sqrt{\tan x + \sqrt{\tan x + \sqrt{\tan x + \dots \infty}}}$ ,

prove that  $\frac{dy}{dx} = \frac{\sec^2 x}{2y - 1}$ .

**Solution :** It is given that

$$y = \sqrt{\tan x + \sqrt{\tan x + \sqrt{\tan x + \dots \infty}}}.$$

$$\therefore y = \sqrt{\tan x + y}.$$

Squaring both sides, we get

$$y^2 = \tan x + y.$$

Now differentiating both sides w.r.t.  $x$ , we get

$$2y \frac{dy}{dx} = \sec^2 x + \frac{dy}{dx}$$

$$\text{or } \frac{dy}{dx} (2y - 1) = \sec^2 x$$

$$\text{or } \frac{dy}{dx} = \frac{\sec^2 x}{2y - 1}.$$

**Illustration 2 :** If  $y = (\sqrt{x})^{(\sqrt{x})^{(\sqrt{x}) \dots \infty}}$ , prove that  $\frac{dy}{dx} = \frac{y^2}{x(2 - y \log x)}$ .

**Solution :** It is given that  $y = (\sqrt{x})^{(\sqrt{x})^{(\sqrt{x}) \dots \infty}}$ .

$$\therefore y = (\sqrt{x})^y.$$

Taking logarithm of both sides, we get

$$\log y = y \log x^{1/2}$$

$$\text{or } \log y = \frac{y}{2} \log x.$$

Now differentiating both sides w.r.t.  $x$ , we get

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{2} \left[ \frac{dy}{dx} \cdot (\log x) + y \cdot \frac{1}{x} \right]$$

or  $\left[ \frac{1}{y} - \frac{1}{2} \log x \right] \frac{dy}{dx} = \frac{y}{2x}$

or  $\left[ \frac{2 - y \log x}{2y} \right] \frac{dy}{dx} = \frac{y}{2x}$

or  $\frac{dy}{dx} = \frac{y^2}{x(2 - y \log x)}$ .

**Illustration 3 :** If  $y = e^{x + e^{x + e^{x + \dots \text{to } \infty}}}$ , show that  $\frac{dy}{dx} = \frac{y}{1 - y}$ . [UPTU 2003]

**Solution :** From the given relation, we have  $y = e^{x + y}$ .

Taking logarithm of both sides, we get  $\log y = x + y$ .

Differentiating both sides w.r.t.  $x$ , we get

$$\frac{1}{y} \frac{dy}{dx} = 1 + \frac{dy}{dx} \quad \text{or} \quad \frac{dy}{dx} \left( \frac{1}{y} - 1 \right) = 1$$

or  $\frac{dy}{dx} \cdot \frac{1 - y}{y} = 1 \quad \text{or} \quad \frac{dy}{dx} = \frac{y}{1 - y}$ .

## Comprehensive Exercise 7

1. If  $y = x^{x^{x \dots \infty}}$ , prove that  $\frac{dy}{dx} = \frac{y^2}{x(1 - y \log x)}$ .
2. If  $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \dots \text{to } \infty}}}$ , prove that  $\frac{dy}{dx} = \frac{\cos x}{2y - 1}$ .
3. If  $y = (\sin x)^{(\sin x)^{(\sin x) \dots \infty}}$ , prove that  $\frac{dy}{dx} = \frac{y^2 \cot x}{1 - y \log \sin x}$ .
4. If  $y = \sqrt{x + \sqrt{x + \sqrt{x + \dots \text{to } \infty}}}$ , prove that  $\frac{dy}{dx} = \frac{1}{2y - 1}$ .
5. If  $y = \sqrt{\log x + \sqrt{\log x + \sqrt{\log x + \dots \text{to } \infty}}}$ , prove that  $(2y - 1) \frac{dy}{dx} = \frac{1}{x}$ .

## 9.14 Derivatives of Second and Higher Order (Successive Differentiation)

Let  $y = f(x)$  be a differentiable function of  $x$ ; then its derivative  $y' = \frac{dy}{dx}$  is called the *first derivative* of  $y$  with respect to  $x$ . If the first derivative  $\frac{dy}{dx}$  itself is differentiable, then its derivative i.e.,  $\frac{d}{dx} \left( \frac{dy}{dx} \right)$  is called the *second derivative* of  $y$  or *the derivative of  $y$  of second order* and is denoted by  $y''$  or  $\frac{d^2 y}{dx^2}$ . Similarly, the derivative of  $\frac{d^2 y}{dx^2}$  is called the *third derivative* of  $y$  and is written as  $y'''$  or  $\frac{d^3 y}{dx^3}$ . In general, the  $n$ th derivative of  $y$  or the *differential coefficient of  $y$  of order  $n$*  is denoted by  $y^{(n)} = \frac{d^n y}{dx^n}$  and is obtained by differentiating  $y$ ,  $n$  times successively with respect to  $x$ . The method of obtaining successive derivatives is called **successive differentiation**.

The successive derivatives of  $y = f(x)$  are also denoted as :

$$Dy, D^2 y, D^3 y, \dots, D^n y, \dots$$

$$y_1, y_2, y_3, \dots, y_n, \dots$$

$$f'(x), f''(x), f'''(x), \dots, f^{(n)}(x), \dots$$

The value of the  $n$ th derivative of  $y = f(x)$  at  $x = a$  is denoted by  $(y_n)_{x=a}$  or by  $(y_n)_{(a)}$  or by  $f^{(n)}(a)$ .

### Illustrative Examples

**Illustration 1 :** If  $y = e^x + \sin x$ , find  $\frac{d^2 y}{dx^2}$ .

[UPTU 2006]

**Solution :** We have  $y = e^x + \sin x$ .

$$\therefore \frac{dy}{dx} = \frac{d}{dx} (e^x) + \frac{d}{dx} (\sin x) = e^x + \cos x.$$

$$\therefore \frac{d^2 y}{dx^2} = \frac{d}{dx} \left( \frac{dy}{dx} \right) = \frac{d}{dx} (e^x + \cos x) = \frac{d}{dx} (e^x) + \frac{d}{dx} (\cos x) = e^x - \sin x.$$

**Illustration 2 :** Find the second derivative of  $e^{3x} \sin 4x$ .

**Solution :** Let  $y = e^{3x} \sin 4x$ .

Then  $\frac{dy}{dx} = 3e^{3x} \sin 4x + 4e^{3x} \cos 4x = e^{3x} (3 \sin 4x + 4 \cos 4x).$

$$\begin{aligned} \therefore \frac{d^2 y}{dx^2} &= \frac{d}{dx} \left( \frac{dy}{dx} \right) = \frac{d}{dx} \{e^{3x} (3 \sin 4x + 4 \cos 4x)\} \\ &= 3e^{3x} (3 \sin 4x + 4 \cos 4x) + e^{3x} (12 \cos 4x - 16 \sin 4x) \\ &= e^{3x} (24 \cos 4x - 7 \sin 4x). \end{aligned}$$

**Illustration 3 :** (i) If  $y = A \cos nx + B \sin nx$ , show that  $\frac{d^2 y}{dx^2} + n^2 y = 0.$

(ii) If  $y = \operatorname{cosec} x + \cot x$ , show that  $\sin x \frac{d^2 y}{dx^2} = y^2.$

**Solution :** (i) We have  $y = A \cos nx + \sin nx.$

$$\therefore \frac{dy}{dx} = -A n \sin nx + B n \cos nx$$

$$\begin{aligned} \text{and } \frac{d^2 y}{dx^2} &= -A n \cdot n \cos nx - B n \cdot n \sin nx \\ &= -n^2 (A \cos nx + B \sin nx) = -n^2 y \end{aligned}$$

$$\text{Hence, } \frac{d^2 y}{dx^2} + n^2 y = 0.$$

(ii) We have  $y = \operatorname{cosec} x + \cot x$

$$\therefore \frac{dy}{dx} = -\operatorname{cosec} x \cot x - \operatorname{cosec}^2 x.$$

$$\begin{aligned} \text{and } \frac{d^2 y}{dx^2} &= -\operatorname{cosec} x (-\operatorname{cosec}^2 x) - (\operatorname{cosec} x \cot x) \cot x \\ &\quad - 2 \operatorname{cosec} x (-\operatorname{cosec} x \cot x) \\ &= \operatorname{cosec}^3 x + \operatorname{cosec} x \cot^2 x + 2 \operatorname{cosec}^2 x \cot x \end{aligned}$$

$$\begin{aligned} \therefore \sin x \frac{d^2 y}{dx^2} &= \sin x \operatorname{cosec} x (\operatorname{cosec}^2 x + \cot^2 x + 2 \operatorname{cosec} x \cot x) \\ &= (\operatorname{cosec} x + \cot x)^2 = y^2 \end{aligned}$$

$$\text{Hence, } \sin x \frac{d^2 y}{dx^2} = y^2.$$

**Illustration 4 :** If  $y = (\sin^{-1} x)^2$ , prove that  $(1-x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} - 2 = 0.$

[UPTU 2011]

**Solution :** We have  $y = (\sin^{-1} x)^2.$

...(1)

$$\therefore \frac{dy}{dx} = 2 \sin^{-1} x \cdot \frac{d}{dx} (\sin^{-1} x) = \frac{2 \sin^{-1} x}{\sqrt{1-x^2}}$$

Squaring both sides, we get

$$(1 - x^2) \left( \frac{dy}{dx} \right)^2 = 4 (\sin^{-1} x)^2$$

$$\text{or} \quad (1 - x^2) \left( \frac{dy}{dx} \right)^2 = 4y \quad [\because \text{from (1)}]$$

$$\text{or} \quad (1 - x^2) \left( \frac{dy}{dx} \right)^2 - 4y = 0 \quad \dots(2)$$

Differentiating both sides again w.r.t.  $x$ , we have

$$(1 - x^2) 2 \frac{dy}{dx} \cdot \frac{d}{dx} \left( \frac{dy}{dx} \right) + \left( \frac{dy}{dx} \right)^2 (-2x) - 4 \frac{dy}{dx} = 0$$

$$\text{or} \quad 2(1 - x^2) \frac{dy}{dx} \cdot \frac{d^2 y}{dx^2} - 2x \left( \frac{dy}{dx} \right)^2 - 4 \frac{dy}{dx} = 0$$

$$\text{or} \quad 2 \frac{dy}{dx} \left[ (1 - x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} - 2 \right] = 0$$

$$\text{or} \quad (1 - x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} - 2 = 0. \quad \left[ \because 2 \frac{dy}{dx} \neq 0 \right]$$

**Illustration 5 :** If  $y = e^{a \sin^{-1} x}$ , prove that  $(1 - x^2) y_2 - xy_1 - a^2 y_1 = 0$ .

**Solution :** We have  $y = e^{\sin^{-1} x}$  ...(1)

$$\therefore y_1 = \frac{dy}{dx} = e^{a \sin^{-1} x} \cdot \frac{a}{\sqrt{1 - x^2}} = \frac{ay}{\sqrt{1 - x^2}}$$

$$\text{or} \quad y_1 \sqrt{1 - x^2} = ay \quad \dots(2)$$

Squaring both sides of (2), we get

$$(1 - x^2) y_1^2 = a^2 y^2$$

$$\text{or} \quad (1 - x^2) y_1^2 - a^2 y^2 = 0 \quad \dots(3)$$

Differentiating both sides of (3) w.r.t.  $x$ , we get

$$(1 - x^2) 2y_1 y_2 + y_1^2 (-2x) - a^2 \cdot 2yy_1 = 0$$

$$\text{or} \quad 2y_1 [(1 - x^2) y_2 - xy_1 - a^2 y] = 0$$

$$\text{or} \quad (1 - x^2) y_2 - xy_1 - a^2 y = 0 \quad [\because 2y_1 \neq 0]$$

**Illustration 6 :** Find  $\frac{d^2 y}{dx^2}$ , if  $x = at^2$  and  $y = 2at$ .

**Solution :** We have  $x = at^2$  and  $y = 2at$

$$\therefore \frac{dx}{dt} = 2at \text{ and } \frac{dy}{dt} = 2a$$

$$\therefore \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{2a}{2at} = \frac{1}{t}.$$

$$\begin{aligned}
 \therefore \frac{d^2 y}{dx^2} &= \frac{d}{dx} \left( \frac{dy}{dx} \right) = \frac{d}{dx} \left( \frac{1}{t} \right) = \left[ \frac{d}{dt} \left( \frac{1}{t} \right) \right] \frac{dt}{dx} \\
 &= \frac{-1}{t^2} \cdot \frac{1}{2at} \quad \left[ \because \frac{dx}{dt} = 2at \Rightarrow \frac{dt}{dx} = \frac{1}{2at} \right] \\
 &= -\frac{1}{2at^3}
 \end{aligned}$$

**Illustration 7 :** If  $x = a \cos^3 \theta$ ,  $y = a \sin^3 \theta$ , then find  $\frac{d^2 y}{dx^2}$ . [UPTU 2010]

**Solution :** We have  $x = a \cos^3 \theta$ ,  $y = a \sin^3 \theta$

$$\therefore \frac{dx}{d\theta} = -3a \cos^2 \theta \sin \theta \text{ and } \frac{dy}{d\theta} = 3a \sin^2 \theta \cos \theta$$

$$\therefore \frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{3a \sin^2 \theta \cos \theta}{-3a \cos^2 \theta \sin \theta} = -\tan \theta$$

$$\begin{aligned}
 \therefore \frac{d^2 y}{dx^2} &= \frac{d}{dx} \left( \frac{dy}{dx} \right) = \frac{d}{dx} (-\tan \theta) = \left[ \frac{d}{d\theta} (-\tan \theta) \right] \cdot \frac{d\theta}{dx} \\
 &= -\sec^2 \theta \cdot \frac{1}{-3a \cos^2 \theta \sin \theta}
 \end{aligned}$$

$$\begin{aligned}
 &\left[ \because \frac{dx}{d\theta} = -3a \cos^2 \theta \sin \theta \Rightarrow \frac{d\theta}{dx} = \frac{1}{-3a \cos^2 \theta \sin \theta} \right] \\
 &= \frac{1}{3a} \sec^4 \theta \operatorname{cosec} \theta.
 \end{aligned}$$

## Comprehensive Exercise 8

Find the second derivatives of the following functions.

1. (i)  $x \sin x$ . (ii)  $\log x - x$ .  
 (iii)  $\log \log x$ . (iv)  $e^x \sin x$ .  
 (v)  $e^{4x} \sin 3x$ . (vi)  $\cos^{-1} x$ .  
 (vii)  $\sin 3x \sin 5x$ .
2. If  $y = \tan^{-1} x$ , show that  $(1 + x^2) \frac{d^2 y}{dx^2} + 2x \frac{dy}{dx} = 0$ .
3. If  $y = e^m \cos^{-1} x$ , show that  $(1 - x^2) y_2 - xy_1 - m^2 y = 0$ .
4. If  $y = \frac{\sin^{-1} x}{\sqrt{1 - x^2}}$ , then show that  $(1 - x^2) \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} - y = 0$

5. If  $y = \tan x + \sec x$ , prove that  $\frac{d^2 y}{dx^2} = \frac{\cos x}{(1 - \sin x)^2}$
6. If  $x = a \cos \theta$ ,  $y = b \sin \theta$ , show that  $\frac{d^2 y}{dx^2} = -\frac{b^4}{a^2 y^3}$ .
7. If  $x = a(t - \sin t)$  and  $y = a(1 + \cos t)$ , prove that  $\frac{d^2 y}{dx^2} = \frac{1}{4a} \operatorname{cosec}^4 \left( \frac{t}{2} \right)$
8. If  $x = a(\cos \theta + \theta \sin \theta)$  and  $y = a(\sin \theta - \theta \cos \theta)$ , show that  $\frac{d^2 y}{dx^2} = \frac{\sec^2 \theta}{a \theta}$ .
9. If  $x = a(\theta + \sin \theta)$  and  $y = a(1 - \cos \theta)$ , find  $\frac{d^2 y}{dx^2}$  at  $\theta = \frac{\pi}{2}$ .

## Answers

1. (i)  $2 \cos x - x \sin x$ . (ii)  $-1/x^2$ .
- (iii)  $-\frac{1 + \log x}{(x \log x)^2}$ . (iv)  $2e^x \cos x$ .
- (v)  $e^{4x} (7 \sin 3x + 24 \cos 3x)$ .
- (vi)  $-\frac{x}{(1 - x^2)^{3/2}}$ .
- (vii)  $32 \cos 8x - 2 \cos 2x$ .
9.  $\frac{1}{a}$ .

## Comprehensive Exercise 9

### Fill in the Blanks

Fill in the blanks “.....”, so that the following statements are complete and correct.

1. If  $y = a^x$ , then  $\frac{dy}{dx} = \dots\dots\dots$ .
2. If  $y = \cos \sqrt{x}$ , then  $\frac{dy}{dx} = \dots\dots\dots$ .
3. If  $y = \log_a x$ , then  $\frac{dy}{dx} = \dots\dots\dots$ .

[UPTU 2011]



4. If  $y = x^2 + 2^x$ , then  $\frac{dy}{dx} = \dots\dots\dots$  . [UPTU 2009]
5. If  $y = \sec^{-1} x$ , then  $\frac{dy}{dx} = \dots\dots\dots$  .
6. If  $y = (2x + 3)^5$ , then  $\frac{dy}{dx} = \dots\dots\dots$  .
7. If  $y = \cos^2 x^2$ , then  $\frac{dy}{dx} = \dots\dots\dots$  .
8. If  $y = \cos^{-1} 2x$ , then  $\frac{dy}{dx} = \dots\dots\dots$  .
9. If  $x^3 + y^3 = 3axy$ , then  $\frac{dy}{dx} = \dots\dots\dots$  .
10. If  $y = \cosh^{-1} x$ , then  $\frac{dy}{dx} = \dots\dots\dots$  .
11. If  $y = e^{\cos x}$ , then  $\frac{dy}{dx} = \dots\dots\dots$  .

### Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

12. If  $y = e^{-3x}$ , then  $\frac{dy}{dx}$  is
- (a)  $3e^{-3x}$  (b)  $\frac{1}{3}e^{-3x}$   
 (c)  $-3e^{-3x}$  (d)  $\frac{-1}{3}e^{-3x}$
13. If  $y = \log \sec x$ , then  $\frac{dy}{dx}$  is
- (a)  $\cot x$  (b)  $\tan x$   
 (c)  $\tan x \sec x$  (d)  $\sec x$
14. If  $y = 3^{x+2}$ , then  $\frac{dy}{dx}$  is
- (a)  $\frac{1}{a} 3^x \log 3$  (b)  $9 (3^x \log 3)$   
 (c)  $\frac{1}{9} (3^x \log x)$  (d)  $9 (3^x \log x)$
15. If  $y = \sin^{-1} \frac{x}{a}$ , then  $\frac{dy}{dx}$  is
- (a)  $\frac{1}{\sqrt{x^2 - a^2}}$  (b)  $-\frac{1}{\sqrt{(a^2 - x^2)}}$   
 (c)  $\frac{1}{\sqrt{(a^2 - x^2)}}$  (d)  $-\frac{1}{\sqrt{(x^2 - a^2)}}$

16. If  $y = \cot^{-1} (1/x)$ , then  $\frac{dy}{dx}$  is

- (a)  $\frac{1}{x^2}$  (b)  $x^2$   
 (c)  $\frac{1}{1+x^2}$  (d)  $\frac{1}{1-x^2}$

17. If  $y = e^{\cot x}$ , then  $\frac{dy}{dx}$  is

- (a)  $e^{\cot x} (\operatorname{cosec}^2 x)$  (b)  $-e^{\cot x} (\operatorname{cosec}^2 x)$   
 (c)  $e^{-\cot x} (\operatorname{cosec}^2 x)$  (d)  $-e^{-\cot x} (\operatorname{cosec}^2 x)$

18. If  $y = \sin x^\circ$  then  $\frac{dy}{dx}$  is

- (a)  $\cos x^\circ$  (b)  $-\cos x^\circ$   
 (c)  $\frac{\pi}{180} \cos x^\circ$  (d) None of these

[UPTU 2011]

### True or False

Write 'T' for true and 'F' for false statement.

19. The derivative of the function  $(a^2 - x^2)^{5/2}$  with respect to  $x$  is  $5x(a^2 - x^2)^{1/2}$ .

20. The differentiation of  $(x-1)(x-2)$  with respect to  $x$  is  $2x-3$ .

21. If  $y = \log \sin 3x$ , then  $\frac{dy}{dx} = 3 \cot 3x$ .

22. If  $y$  is a function of  $t$  and  $t$  is a function of  $x$ , then  $\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$ .

23. If  $y = \sin (\tan^{-1} x)$ , then  $\frac{dy}{dx} = \frac{\cos (\tan^{-1} x)}{1-x^2}$ .

24. If  $y = e^x \cot x$ , then  $\frac{dy}{dx} = e^x (\cot x - \operatorname{cosec}^2 x)$ .

25. If  $y = \log \sin x$ , then  $\frac{dy}{dx} = \tan x$ .

26. If  $y = \sin (\log x)$ , then  $\frac{dy}{dx} = \frac{\sin (\log x)}{x}$ .

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## Answers

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1.  $a^x \log a$

2.  $-\frac{\sin \sqrt{x}}{2\sqrt{x}}$

3.  $\frac{1}{x \log a}$     4.  $2x + 2^x \log_e 2$

5.  $\frac{1}{x\sqrt{x^2-1}}$

6.  $10(2x+3)^4$

7.  $-4x \cos x^2 \sin x^2$

8.  $\frac{-2}{\sqrt{(1-4x^2)}}$

9.  $\frac{ay-x^2}{y^2-ax}$

10.  $\frac{1}{\sqrt{x^2-1}}$

11.  $-e^{\cos x} \cdot \sin x$

12. (c)

13. (b)    14. (b)

15. (c)

16. (c)

17. (b)    18. (c)

19. *F*

20. *T*

21. *T*    22. *T*

23. *F*

24. *T*

25. *F*    26. *F*



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## Chapter

# 10

# Linear Differential Equations

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## 10.1 Differential Equation and its Solution

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A **differential equation** is an equation containing the dependent and independent variables and different derivatives of the dependent variables w.r.t. one or more independent variables.

The **order of a differential equation** is the order of the highest derivative (or differential coefficient) occurring in the equation.

The **degree of a differential equation** is the degree of the highest derivative (or diff. coeff.) which occurs in it, after the differential equation has been rationalized (*i.e.*, made free from radicals and fractions so far as derivatives are concerned). A differential equation is called **ordinary**, if the unknown function depends on only one argument (independent variable).

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A differential equation is said to be **partial** if there are *two or more* independent variables.

A differential equation is said to be **linear** if the dependent variable, say, ' $y$ ' and all its derivatives occur in the first degree, otherwise it is **non-linear**.

A function  $y = f(x)$  is called a **Solution** (or the *primitive*) of a differential equation if, when substituted into the equation, it reduces the equation to an identity and the process of finding all the solutions is called integrating (or *solving*) the differential equation.

### General solution.

A solution of a differential equation, containing independent arbitrary constants equal in number to the order of the differential equation is called its *general solution*.

### Particular solution.

A solution obtained by giving particular values to the arbitrary constants in the general solution is called a *particular solution* or *particular integral*.

**Arbitrary Constants.** The solution of a differential equation may contain as many arbitrary constants as is the order of the differential equation *i.e.*, the solution of an  $n$ th order differential equation may contain  $n$  arbitrary constants.

## 10.2 Linear Differential Equation

A linear differential equation is an equation in which the dependent variable and its derivatives appear only in the first degree.

A linear differential equation of order  $n$  of the form

$$\frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + a_2 \frac{d^{n-2} y}{dx^{n-2}} + \dots + a_{n-1} \frac{dy}{dx} + a_n y = Q, \quad \dots(1)$$

where  $a_1, a_2, \dots, a_{n-1}, a_n$  are constants and  $Q$  is any function of  $x$  is called a **linear differential equation with constant coefficients**.

For convenience, the operators  $\frac{d}{dx}, \frac{d^2}{dx^2}, \frac{d^3}{dx^3}, \dots, \frac{d^n}{dx^n}$  are also denoted by  $D, D^2, D^3, \dots, D^n$  respectively.

Thus the equation (1) can also be written as

$$D^n y + a_1 D^{n-1} y + \dots + a_{n-1} Dy + a_n y = Q$$

$$\text{or} \quad [D^n + a_1 D^{n-1} + \dots + a_{n-1} D + a_n] y = Q. \quad \dots(1)$$

If  $y = f(x)$  is the general solution of

$$[D^n + a_1 D^{n-1} + \dots + a_{n-1} D + a_n] y = 0, \quad \dots(2)$$

and  $y = \phi(x)$  is any particular solution of the equation (1) not containing any arbitrary constant, then

$$y = f(x) + \phi(x),$$

is the general solution of (1).

Thus the method of solving a linear equation is divided into two parts :

First, we find the general solution of the equation (2).

It is called the *complementary function* (C.F.). It must contain as many arbitrary constants as is the order of the given differential equation.

Next, we find a solution of (1) which does not contain an arbitrary constant. This is called the *particular integral* (P.I.).

If we add (C.F.) and (P.I.), we get the general solution of (1). Thus the general solution of (1) is

$$y = \text{C.F.} + \text{P.I.}$$

### 10.3 Determination of Complementary Function (C.F.)

Consider a linear  $n$ th order differential equation with constant coefficients of the form  $f(D)y = 0$  i.e.,

$$[D^n + a_1 D^{n-1} + a_2 D^{n-2} + \dots + a_n] y = 0. \quad \dots(1)$$

This is equivalent to

$$[(D - m_1)(D - m_2) \dots (D - m_n)] y = 0. \quad \dots(2)$$

The solution of any one of the equations

$$(D - m_1)y = 0, (D - m_2)y = 0, \dots, (D - m_n)y = 0 \quad \dots(3)$$

is also a solution of (2) and we know that the general solution of  $(D - m_1)y = 0$  is  $y = Ae^{m_1 x}$ . Hence we can assume that a solution of the equation (2) is of the form

$y = e^{mx}$ . Then, substituting  $e^{mx}$  for  $y$  in (1), so that  $Dy = me^{mx}$ ,  $D^2 y = m^2 e^{mx}$ , ...,  $D^n y = m^n e^{mx}$ , we get

$$e^{mx} (m^n + a_1 m^{n-1} + a_2 m^{n-2} + \dots + a_n) = 0$$

or  $m^n + a_1 m^{n-1} + a_2 m^{n-2} + \dots + a_n = 0$ , because  $e^{mx} \neq 0$ .

Hence  $e^{mx}$  will be a solution of (1) if  $m$  has the value obtained from the equation

$$m^n + a_1 m^{n-1} + \dots + a_n = 0. \quad \dots(4)$$

The equation (4) is called the **auxiliary equation (A.E.)** and is obtained by putting  $D = m$  in  $f(D) = 0$ .

It will give in general  $n$  roots, say,  $m_1, m_2, m_3, \dots, m_n$ .

**Case I.** If all the roots of the Auxiliary equation (A.E.) are distinct.

If the roots  $m_1, m_2, m_3, \dots, m_n$  are all distinct, then  $e^{m_1 x}, e^{m_2 x}, \dots, e^{m_n x}$  are all distinct and linearly independent. So the general solution of (1) in this case is

$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x} + \dots + c_n e^{m_n x}. \quad \dots(5)$$

**Case II. Auxiliary equation having equal roots :**

If two roots are equal say  $m_1 = m_2$ , then the solution (5) becomes

$$y = c_1 e^{m_1 x} + c_2 e^{m_1 x} + c_3 e^{m_3 x} + \dots + c_n e^{m_n x}$$

or 
$$y = (c_1 + c_2) e^{m_1 x} + c_3 e^{m_3 x} + \dots + c_n e^{m_n x}.$$

Now  $(c_1 + c_2)$  can be replaced by single constant say  $c$ .

Similarly if three roots of the auxiliary equation are equal say,  $m_1 = m_2 = m_3$ , the general solution of  $f(D)y = 0$  will be

$$y = (c_1 + c_2 x + c_3 x^2) e^{m_1 x} + c_4 e^{m_4 x} + \dots + c_n e^{m_n x}$$

and so on.

**Case III. Auxiliary equation having complex roots.**

Let the two roots of the auxiliary equation be complex, say  $m_1 = \alpha + i\beta$  and  $m_2 = \alpha - i\beta$ , (where  $i = \sqrt{-1}$ ).

The solution corresponding to these two roots will be

$$y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$$

**Note.** The expression  $e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$  can also be written as

$$c_1 e^{\alpha x} \sin(\beta x + c_2) \quad \text{or} \quad c_1 e^{\alpha x} \cos(\beta x + c_2).$$

**Case IV.** If the imaginary roots are repeated, say  $\alpha + i\beta$  and  $\alpha - i\beta$  occur twice then the solution will be

$$y = e^{\alpha x} [(c_1 + c_2 x) \cos \beta x + (c_3 + c_4 x) \sin \beta x],$$

and so on.

**Case V.** If a pair of the roots of the auxiliary equation are irrational i.e., they are  $\alpha \pm \sqrt{\beta}$ , where  $\beta$  is positive, then the corresponding term in the C.F. will be

$$e^{\alpha x} (c_1 \cosh \sqrt{\beta} x + c_2 \sinh \sqrt{\beta} x),$$

or 
$$c_1 e^{\alpha x} \sinh(\sqrt{\beta} x + c_2) \quad \text{or} \quad c_1 e^{\alpha x} \cosh(\sqrt{\beta} x + c_2).$$

If these irrational roots are repeated twice, then the corresponding portion of the solution will be

$$e^{\alpha x} \{(c_1 + c_2 x) \cosh \sqrt{\beta} x + (c_3 + c_4 x) \sinh \sqrt{\beta} x\}.$$

## Illustrative Examples

**Illustration 1 :** Solve  $\frac{d^2 y}{dx^2} - 7 \frac{dy}{dx} + 12y = 0$ .

[UPTU 2001]

**Solution :** The given differential equation is  $(D^2 - 7D + 12)y = 0$ .

$\therefore$  the auxiliary equation is  $m^2 - 7m + 12 = 0$

or 
$$(m - 3)(m - 4) = 0.$$

$\therefore$  
$$m = 3, 4.$$

Hence the solution is

$$y = c_1 e^{3x} + c_2 e^{4x}.$$

**Illustration 2 :** Solve  $(D^2 \pm \mu^2) y = 0$ . [UPTU 2008]

**Solution :** The auxiliary equation is  $m^2 - \mu^2 = 0$  or  $m^2 + \mu^2 = 0$ .

When  $m^2 - \mu^2 = 0$ , we have  $m = \pm \mu$

and when  $m^2 + \mu^2 = 0$ , we have  $m = 0 \pm \mu i$ .

Hence the solution of  $(D^2 - \mu^2) y = 0$  is  $y = c_1 e^{\mu x} + c_2 e^{-\mu x}$

and the solution of  $(D^2 + \mu^2) y = 0$  is

$$y = e^{0x} (c_3 \cos \mu x + c_4 \sin \mu x)$$

or  $y = c_3 \cos \mu x + c_4 \sin \mu x$ .

**Illustration 3 :** Solve  $(D^3 + 6D^2 + 11D + 6) y = 0$ . [UPTU 2001]

**Solution :** The auxiliary equation is

$$m^3 + 6m^2 + 11m + 6 = 0$$

or  $(m + 1)(m^2 + 5m + 6) = 0$

or  $(m + 1)(m + 2)(m + 3) = 0$ .

$\therefore m = -1, -2, -3$ .

Hence the solution is

$$y = c_1 e^{-x} + c_2 e^{-2x} + c_3 e^{-3x}.$$

**Illustration 4 :** Solve  $\frac{d^2 x}{dt^2} - 3 \frac{dx}{dt} + 2x = 0$ , given that when  $t = 0, x = 0$  and  $dx/dt = 0$ .

**Solution :** The auxiliary equation is  $m^2 - 3m + 2 = 0$

or  $(m - 1)(m - 2) = 0$  or  $m = 1, 2$ .

Hence the solution is  $x = c_1 e^t + c_2 e^{2t}$ , ... (1)

where  $c_1$  and  $c_2$  are arbitrary constants.

Now  $x = 0$  when  $t = 0$ ;

$\therefore 0 = c_1 + c_2$ . ... (2)

Also  $dx/dt = c_1 e^t + 2c_2 e^{2t}$ , and  $dx/dt = 0$  when  $t = 0$ .

$\therefore 0 = c_1 + 2c_2$ . ... (3)

Solving (2) and (3), we get  $c_1 = 0, c_2 = 0$ .

Now putting the values of  $c_1$  and  $c_2$  in (1), we get the required solution as  $x = 0$ .

**Illustration 5 :** Solve  $(d^2 y / dx^2) + y = 0$ , given  $y = 2$  for  $x = 0$ , and  $y = -2$  for  $x = \frac{1}{2} \pi$ .

**Solution :** The auxiliary equation is  $m^2 + 1 = 0$ .

$\therefore m = \pm i = 0 \pm i$ .



Hence the solution is

$$y = c_1 \cos x + c_2 \sin x. \quad \dots(1)$$

Now if  $x = 0, y = 2$ ;

$$\therefore c_1 = 2, \text{ from (1).}$$

Also when  $x = \frac{1}{2} \pi, y = -2$ ;

$$\therefore c_2 = -2, \text{ from (1).}$$

Substituting these values of  $c_1$  and  $c_2$  in (1), the solution is

$$y = 2 \cos x - 2 \sin x = 2 (\cos x - \sin x)$$

$$\text{or } y = 2\sqrt{2} \left\{ \left( \frac{1}{\sqrt{2}} \right) \cos x - \left( \frac{1}{\sqrt{2}} \right) \sin x \right\} = 2\sqrt{2} \cos \left( x + \frac{1}{4} \pi \right).$$

**Illustration 6 :** Solve  $(D^2 + 1)(D - 1)y = 0$ .

**Solution :** Given differential equation is,

$$(D^2 + 1)(D - 1)y = 0 \quad \dots(1)$$

The A.E. is,  $(m^2 + 1)(m - 1) = 0 \Rightarrow m = \pm i, 1$ .

Hence the solution of (1) is,

$$y = \text{C.F.}$$

$$\text{i.e., } y = c_1 \cos x + c_2 \sin x + c_3 e^x$$

**Illustration 7 :** Solve  $\frac{d^2 y}{dx^2} - 2p \frac{dy}{dx} + (p^2 + q^2)y = 0$ .

**Solution :** The given differential equation is

$$\frac{d^2 y}{dx^2} - 2p \frac{dy}{dx} + (p^2 + q^2)y = 0.$$

$$\text{or } [D^2 - 2pD + (p^2 + q^2)]y = 0$$

The A.E. is,

$$m^2 - 2mp + p^2 + q^2 = 0 \quad \text{or} \quad (m - p)^2 = -q^2$$

$$\rightarrow m - p = \pm qi \quad \text{or} \quad m = p \pm qi.$$

Hence the required solution is,

$$y = \text{C.F.}$$

$$\text{i.e., } y = e^{px} (c_1 \cos qx + c_2 \sin qx)$$

**Illustration 8 :** Solve  $(D^3 - 3D + 2)y = 0$ .

**Solution :** The auxiliary equation is  $m^3 - 3m + 2 = 0$

$$\text{or } (m - 1)(m^2 + m - 2) = 0$$

$$\text{or } (m - 1)\{(m - 1)(m + 2)\} = 0.$$

$$\therefore m = 1, 1, -2.$$

Hence the solution is

$$y = (c_1 + c_2 x) e^x + c_3 e^{-2x}.$$

**Illustration 9 :** Solve  $\frac{d^3 y}{dx^3} - 8y = 0$ .

[UPTU 2008]

**Solution :** The auxiliary equation is  $m^3 - 8 = 0$

or  $(m - 2)(m^2 + 2m + 4) = 0$  i.e.,  $m - 2 = 0$  and  $m^2 + 2m + 4 = 0$ .

$\therefore m = 2$  and  $m = -1 \pm i\sqrt{3}$ .

Hence the solution is

$$y = e^{-x} (c_1 \cos \sqrt{3}x + c_2 \sin \sqrt{3}x) + c_3 e^{2x}.$$

**Illustration 10 :** Solve  $\frac{d^4 y}{dx^4} - 2\frac{d^3 y}{dx^3} - 2\frac{dy}{dx} - y = 0$ .

**Solution :** The auxiliary equation is

$$m^4 - 2m^3 - 2m - 1 = 0 \quad \text{or} \quad (m^4 - 1) - 2m(m^2 + 1) = 0$$

or  $(m^2 + 1)(m^2 - 1) - 2m(m^2 + 1) = 0$

or  $(m^2 + 1)(m^2 - 2m - 1) = 0$  i.e.,  $m^2 = -1$  or  $m^2 - 2m - 1 = 0$ .

$\therefore m = 0 \pm i, 1 \pm \sqrt{2}$ .

Hence the solution is

$$y = e^{0x} (c_1 \cos x + c_2 \sin x) + e^x (c_3 \cosh \sqrt{2}x + c_4 \sinh \sqrt{2}x)$$

or  $y = c_1 \cos x + c_2 \sin x + e^x \{c_3 \cosh \sqrt{2}x + c_4 \sinh \sqrt{2}x\}$ .

**Illustration 11 :** Write the complementary function for

$$(D + 1)^2 (D^2 + 4) y = 0.$$

**Solution :** Given differential equation is

$$(D + 1)^2 (D^2 + 4) y = 0$$

The A.E. is,

$$(m + 1)^2 (m^2 + 4) = 0 \quad \text{or} \quad m + 1 = 0, m + 1 = 0, m^2 + 4 = 0 \text{ (Note)}$$

or  $m = -1, -1, \pm 2i$

Hence the required solution is,

$$y = \text{C.F.}$$

i.e.,  $y = (c_1 + xc_2) e^{-x} + c_3 \cos 2x + c_4 \sin 2x$

**Illustration 12 :** Solve (i)  $(D^2 + 1)^2 (D - 1) y = 0$  (ii)  $(D^2 - 4) y = e^x + \sin 2x$

**Solution :** (i) Here the auxiliary equation is

$$(m^2 + 1)^2 (m - 1) = 0.$$

$\therefore m = \pm i, \pm i, 1$ .

$\therefore$  C.F.  $= c_1 e^x + (c_3 + c_4 x) \cos x + (c_5 + c_6 x) \sin x$ .

(ii) The auxiliary equation is  $m^2 - 4 = 0$  or  $m = 2, -2$ .

$\therefore$  C.F.  $= c_1 e^{2x} + c_2 e^{-2x}$ .

$$\text{And P.I.} = \frac{1}{D^2 - 4} e^x + \frac{1}{D^2 - 4} \sin 2x = \frac{e^x}{1 - 4} + \frac{\sin 2x}{-2^2 - 4} = -\frac{1}{2} e^x - \frac{1}{8} \sin 2x.$$

Hence the complete solution is

$$y = \text{C.F.} + \text{P.I.} = c_1 e^{2x} + c_2 e^{-2x} - \frac{1}{3} e^x - \frac{1}{8} \sin 2x.$$

**Illustration 13 :** Solve  $(D^4 + k^4) y = 0$ .

**Solution :** The auxiliary equation is

$$m^4 + k^4 = 0$$

$$\text{or} \quad (m^2 + k^2)^2 - 2k^2 m^2 = 0 \quad \text{or} \quad (m^2 + k^2)^2 - (\sqrt{2} km)^2 = 0$$

$$\text{or} \quad (m^2 + k^2 - \sqrt{2} km)(m^2 + k^2 + \sqrt{2} km) = 0$$

$$\text{or} \quad m^2 - \sqrt{2} km + k^2 = 0 \quad \text{and} \quad m^2 + \sqrt{2} km + k^2 = 0$$

$$\text{or} \quad m = \frac{\sqrt{2}k \pm \sqrt{(2k^2 - 4k^2)}}{2} \quad \text{and} \quad m = \frac{-\sqrt{2}k \pm \sqrt{(2k^2 - 4k^2)}}{2}$$

$$\text{or} \quad m = \frac{k}{\sqrt{2}} \pm i \frac{k}{\sqrt{2}} \quad \text{and} \quad -\frac{k}{\sqrt{2}} \pm i \frac{k}{\sqrt{2}}.$$

Hence the solution is

$$y = e^{kx/\sqrt{2}} \{c_1 \cos(kx/\sqrt{2}) + c_2 \sin(kx/\sqrt{2})\} \\ + e^{-kx/\sqrt{2}} \{c_3 \cos(kx/\sqrt{2}) + c_4 \sin(kx/\sqrt{2})\}.$$

**Illustration 14 :** Solve  $(D^4 + 8D^2 + 16) y = 0$ .

**Solution :** The auxiliary equation is

$$m^4 + 8m^2 + 16 = 0$$

$$\text{or} \quad (m^2 + 4)^2 = 0 \quad \text{or} \quad m^2 + 4 = 0 \quad \text{and} \quad m^2 + 4 = 0. \quad (\text{Note})$$

$$\therefore m = 0 \pm 2i, 0 \pm 2i.$$

Thus the imaginary roots are repeated twice. Hence the solution is

$$y = (c_1 + c_2 x) \cos 2x + (c_3 + c_4 x) \sin 2x, \text{ since } e^0 = 1.$$

## Comprehensive Exercise 1

Solve the following differential equations :

$$1. \quad \frac{d^2 y}{dx^2} + (a + b) \frac{dy}{dx} + aby = 0.$$

$$2. \quad \frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} - 4y = 0.$$

$$3. \quad \frac{d^2 y}{dx^2} + 5 \frac{dy}{dx} + 4y = 0.$$

4.  $\frac{d^3 y}{dx^3} + 6 \frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} - 10y = 0.$
5.  $\frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 4y = 0.$
6.  $(D^3 - 4D^2 + 5D - 2) y = 0.$
7.  $\frac{d^4 y}{dx^4} + 2 \frac{d^3 y}{dx^3} - 3 \frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 4y = 0.$
8.  $\frac{d^4 y}{dx^4} - \frac{d^3 y}{dx^3} - 9 \frac{d^2 y}{dx^2} - 11 \frac{dy}{dx} - 4y = 0.$
9.  $\frac{d^4 y}{dx^4} - k^4 y = 0.$

## Answers

1.  $y = c_1 e^{-ax} + c_2 e^{-bx}.$
2.  $y = c_1 e^{-x} + c_2 e^{4x}.$
3.  $y = c_1 e^{-x} + c_2 e^{-4x}.$
4.  $y = c_1 e^x + c_2 e^{-2x} + c_3 e^{-5x}.$
5.  $y = (c_1 + c_2 x) e^{2x}.$
6.  $y = (c_1 + c_2 x) e^x + c_3 e^{2x}.$
7.  $y = (c_1 + c_2 x) e^x + (c_3 + c_4 x) e^{-2x}.$
8.  $y = (c_1 + c_2 x + c_3 x^2) e^{-x} + c_4 e^{4x}.$
9.  $y = c_1 e^{kx} + c_2 e^{-kx} + c_3 \cos kx + c_4 \sin kx.$

### 10.4 The Particular Integral (P.I.)

As already shown in 10.1 the complete solution of

$$(D^n + a_1 D^{n-1} + a_2 D^{n-2} + \dots + a_{n-1} D + a_n) y = Q \quad \dots(1)$$

or  $F(D) y = Q$  is  $y = \text{C.F.} + \text{P.I.},$

where the C.F. consists of the general solution of the differential equation  $F(D) y = 0.$

The particular integral of the differential equation  $F(D) y = Q$  is  $\frac{1}{F(D)} Q.$  It is

obviously a function of  $x$  which when operated by  $F(D)$  gives  $Q.$

**Note.** If  $D$  stands for differentiation then  $1/D$  will stand for integration.

### 10.5 Particular Integral in some Special Cases

**Case I.** To find P.I. when  $Q$  is of the form  $e^{ax}$ , where  $a$  is any constant and  $F(a) \neq 0.$

$$\text{P.I.} = \frac{1}{F(D)} e^{ax} = \frac{1}{F(a)} e^{ax}, \text{ provided } F(a) \neq 0.$$

**Working Rule :** If  $\text{P.I.} = \{1 / F(D)\} e^{ax}$ , then put  $a$  for  $D$  in  $F(D)$  and we get P.I. provided  $F(a) \neq 0$ .

## Illustrative Examples

**Illustration 1 :** Solve  $\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = e^{5x}$ .

**Solution :** The given equation can be written as  
 $(D^2 - 3D + 2) y = e^{5x}$ , where  $d / dx \equiv D$ .

Here  $F(D) = D^2 - 3D + 2$  and  $Q = e^{5x}$ .

The auxiliary equation is

$$m^2 - 3m + 2 = 0 \quad \text{or} \quad (m - 1)(m - 2) = 0;$$

$$\therefore m = 1, 2.$$

$\therefore$  The C.F. =  $c_1 e^x + c_2 e^{2x}$ , the roots of the A.E. being distinct.

$$\text{And P.I.} = \frac{1}{F(D)} e^{ax} = \frac{1}{D^2 - 3D + 2} e^{5x} = \frac{e^{5x}}{5^2 - 3 \cdot 5 + 2} = \frac{e^{5x}}{12}.$$

[We have put 5 for  $D$  in  $F(D)$ , because here  $a = 5$ ]

Hence the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

$$\text{or} \quad y = c_1 e^x + c_2 e^{2x} + \frac{1}{12} e^{5x}.$$

**Illustration 2 :** Solve  $\frac{d^2 x}{dt^2} + \lambda \frac{dx}{dt} + \mu x = e^t$ .

[UPTU 2007]

**Solution :** The given equation can be written as,  
 $(D^2 + \lambda D + \mu) x = e^t$ .

The auxiliary equation is  $m^2 + \lambda m + \mu = 0$

$$\text{or} \quad m = \frac{-\lambda \pm \sqrt{\lambda^2 - 4\mu}}{2}$$

$$\therefore \text{C.F.} = c_1 e^{\left(\frac{-\lambda + \sqrt{\lambda^2 - 4\mu}}{2}\right)t} + c_2 e^{\left(\frac{-\lambda - \sqrt{\lambda^2 - 4\mu}}{2}\right)t}$$

$$\text{and} \quad \text{P.I.} = \frac{1}{D^2 + \lambda D + \mu} e^t = \frac{e^t}{1 + \lambda + \mu}.$$

Hence the general solution is  $y = (\text{C.F.}) + (\text{P.I.})$

or 
$$y = c_1 e^{\left(-\lambda + \frac{\sqrt{\lambda^2 - 4\mu}}{2}\right)t} + c_2 e^{\left(-\lambda - \frac{\sqrt{\lambda^2 - 4\mu}}{2}\right)t} + \frac{1}{1 + \lambda + \mu} e^t.$$

**Illustration 3 :** Solve  $\frac{d^2 y}{dx^2} + 31 \frac{dy}{dx} + 240 y = 272 e^{-x}$ .

**Solution :** The auxiliary equation is

$$m^2 + 31m + 240 = 0 \quad \text{or} \quad (m + 15)(m + 16) = 0.$$

$\therefore m = -15, -16.$

$\therefore \text{C.F.} = c_1 e^{-15x} + c_2 e^{-16x}.$

And 
$$\text{P.I.} = \frac{1}{D^2 + 31D + 240} (272 e^{-x})$$

$$\begin{aligned} &= 272 \cdot \frac{1}{D^2 + 31D + 240} e^{-x} = 272 \cdot \frac{1}{(-1)^2 + 31(-1) + 240} e^{-x} \\ &= \frac{272}{210} e^{-x} = \frac{136}{105} e^{-x}. \end{aligned}$$

Hence the general solution is  $y = (\text{C.F.}) + (\text{P.I.})$

i.e., 
$$y = c_1 e^{-15x} + c_2 e^{-16x} + \frac{136}{105} e^{-x}.$$

**Illustration 4 :** Solve  $\frac{d^2 y}{dx^2} - 5 \frac{dy}{dx} + 6y = e^{4x}$

[UPTU 2004, 06]

**Solution :** The given differential equation can be written as

$$(D^2 - 5D + 6) y = e^{4x}.$$

The auxiliary equation is

$$m^2 - 5m + 6 = 0 \quad \text{or} \quad (m - 2)(m - 3) = 0$$

or  $m = 2, 3.$

The  $\text{C.F.} = c_1 e^{2x} + c_2 e^{3x}.$

And 
$$\text{P.I.} = \frac{1}{D^2 - 5D + 6} e^{4x} = \frac{e^{4x}}{4^2 - 5 \cdot 4 + 6} = \frac{1}{2} e^{4x}.$$

Hence the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

or 
$$y = c_1 e^{2x} + c_2 e^{3x} + \frac{1}{2} e^{4x}.$$

**Illustration 5 :** Obtain the complete solution of the differential equation

$$\frac{d^2 y}{dx^2} - 7 \frac{dy}{dx} + 6y = e^{2x},$$

and determine the constant so that  $y = 0$  when  $x = 0$ .

**Solution :** The given equation is  $[D^2 - 7D + 6] y = e^{2x}.$

$\therefore$  auxiliary equation is  $m^2 - 7m + 6 = 0$

or  $(m-1)(m-6) = 0$  or  $m = 1, 6$ .

$\therefore$  C.F. =  $c_1 e^x + c_2 e^{6x}$ .

And P.I. =  $\frac{1}{D^2 - 7D + 6} e^{2x} = \frac{1}{2^2 - 7 \cdot 2 + 6} e^{2x} = -\frac{e^{2x}}{4}$ .

$\therefore$  the general solution is  $y = (\text{C.F.}) + (\text{P.I.})$

i.e.,  $y = c_1 e^x + c_2 e^{6x} - \frac{1}{4} e^{2x}$ . ...(1)

Now when  $y = 0, x = 0$ ;

$\therefore 0 = c_1 + c_2 - \frac{1}{4}$ , from (1)

so that  $c_2 = \left(\frac{1}{4} - c_1\right)$ .

Hence from (1), we have

$$y = c_1 e^x + \left(\frac{1}{4} - c_1\right) e^{6x} - \frac{1}{4} e^{2x} = c_1 (e^x - e^{6x}) + \frac{1}{4} (e^{6x} - e^{2x})$$

or  $y = c_1 (e^x - e^{6x}) + \frac{1}{4} e^{2x} (e^{4x} - 1)$  is the required solution.

## Comprehensive Exercise 2

Solve the following differential equations :

1.  $\frac{d^2 y}{dx^2} - 2k \frac{dy}{dx} + k^2 y = e^x$ .      2.  $\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + y = 2 e^{2x}$ .

3.  $\frac{d^2 y}{dx^2} + \frac{dy}{dx} + y = e^x$ .      4.  $[D^2 + D + 1] y = e^{-x}$ .

5.  $\frac{d^3 y}{dx^3} + 6 \frac{d^2 y}{dx^2} + 11 \frac{dy}{dx} + 6y = e^{2x}$ .

## Answers

1.  $y = (c_1 + c_2 x) e^{kx} + e^x / (1 - k)^2$ .

2.  $y = (c_1 + c_2 x) e^{-x} + \frac{2}{9} e^{2x}$ .

3.  $y = e^{-x/2} \left\{ c_1 \cos \left( x \frac{1}{2} \sqrt{3} \right) + c_2 \sin \left( x \frac{1}{2} \sqrt{3} \right) \right\} + \frac{1}{3} e^x$ .

4.  $y = c_1 e^{-x/2} \cos \{x(\sqrt{3}/2)\} + c_2 \{e^{-x}\}$ .

5.  $y = c_1 e^{-x} + c_2 e^{-2x} + c_3 e^{-3x} + \frac{1}{60} e^{2x}$ .

**Case II.** To find P.I. when  $Q$  is of the form  $\sin ax$  or  $\cos ax$  and  $F(-a^2) \neq 0$ .

$$\frac{1}{F(D^2)} \sin ax = \frac{1}{F(-a^2)} \sin ax, \text{ provided } F(-a^2) \neq 0.$$

$$\frac{1}{F(D^2)} \cos ax = \frac{1}{F(-a^2)} \cos ax, \text{ provided } F(-a^2) \neq 0.$$

**Working Rule.** If P.I. =  $\{1/F(D)\} \sin ax$  or  $\cos ax$ , put  $-a^2$  for  $D^2$ ,  $-a^2 D$  for  $D^3$ ,  $(-a^2)^2$  i.e.,  $a^4$  for  $D^4$ ,  $a^4 D$  for  $D^5$ ,  $-a^6$  for  $D^6$  etc. in  $F(D)$  and calculate the P.I.

**Note.** Linear factors in  $D$  of the form  $(pD \pm q)$  appearing in the denominator are removed by first multiplying the Nr. and Deno. by the conjugate factors  $(pD \mp q)$  and then putting  $-a^2$  for  $D^2$  in the denominator. The operation left in the numerator can be easily worked out because the operator  $D$  stands for differentiation with respect to  $x$ .

## Illustrative Examples

**Illustration 1:** Solve  $\frac{d^2 y}{dx^2} - \frac{dy}{dx} - 2y = \sin 2x$ .

[UPTU 2001]

**Solution :** The given equation is  $[D^2 - D - 2] y = \sin 2x$ .

$\therefore$  auxiliary equation is  $m^2 - m - 2 = 0$  or  $(m+1)(m-2) = 0$ .

$\therefore m = -1, 2$ .

$\therefore$  C.F. =  $c_1 e^{-x} + c_2 e^{2x}$ .

And P.I. =  $\frac{1}{D^2 - D - 2} \sin 2x = \frac{1}{-4 - D - 2} \sin 2x$ ,

putting  $-2^2$  i.e.,  $-4$  for  $D^2$

$$= -\frac{1}{D+6} \sin 2x = -\frac{(D-6)}{(D+6)(D-6)} \sin 2x = -\frac{(D-6)}{D^2-36} \sin 2x$$

$$= -\frac{(D-6)}{-4-36} \sin 2x, \text{ putting } -4 \text{ for } D^2$$

$$= \frac{1}{40} (D-6) \sin 2x = \frac{1}{40} \{D(\sin 2x) - 6 \sin 2x\}$$

$$= \frac{1}{40} (2 \cos 2x - 6 \sin 2x) = \frac{1}{20} \cos 2x - \frac{3}{20} \sin 2x.$$

$\therefore$  the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

or  $y = c_1 e^{-x} + c_2 e^{2x} + \frac{1}{20} \cos 2x - \frac{3}{20} \sin 2x$ .

**Illustration 2 :** Solve  $(D^2 + D + 1) y = \sin 2x$ .

[UPTU 2001 (Special)]



**Solution :** The auxiliary equation is  $m^2 + m + 1 = 0$ , giving

$$m = \frac{-1 \pm \sqrt{1-4 \cdot 1}}{2} = -\frac{1}{2} \pm \frac{i\sqrt{3}}{2}, \text{ (complex roots).}$$

$$\therefore \text{C.F.} = e^{(-1/2)x} \{c_1 \cos(x \frac{1}{2}\sqrt{3}) + c_2 \sin(x \frac{1}{2}\sqrt{3})\}.$$

$$\begin{aligned} \text{P.I.} &= \frac{1}{F(D)} \sin 2x = \frac{1}{D^2 + D + 1} \sin 2x = \frac{1}{-4 + D + 1} \sin 2x, \\ &\quad \text{putting } -2^2 \text{ i.e., } -4 \text{ for } D^2 \\ &= \frac{1}{D-3} \sin 2x = \frac{(D+3)}{(D-3)(D+3)} \sin 2x = \frac{D+3}{D^2-9} \sin 2x \\ &= \frac{D+3}{-4-9} \sin 2x, \quad \text{putting } -2^2 \text{ i.e., } -4 \text{ for } D^2 \\ &= \frac{D+3}{-13} (\sin 2x) = -\frac{1}{13} [D(\sin 2x) + 3 \sin 2x] \\ &= -\frac{1}{13} [2 \cos 2x + 3 \sin 2x]. \end{aligned}$$

Hence the complete solution is

$$y = (\text{C.F.}) + (\text{P.I.})$$

$$\begin{aligned} \text{or } y &= e^{(-1/2)x} \left\{ c_1 \cos\left(x \frac{1}{2}\sqrt{3}\right) + c_2 \sin\left(x \frac{1}{2}\sqrt{3}\right) \right\} \\ &\quad - \frac{1}{13} [2 \cos 2x + 3 \sin 2x]. \end{aligned}$$

**Illustration 3 :** Solve  $(D^2 - 5D + 6)y = \sin 3x$ .

**Solution :** Here the auxiliary equation is

$$m^2 - 5m + 6 = 0 \quad \text{or} \quad (m-2)(m-3) = 0.$$

$$\therefore m = 2, 3.$$

$$\therefore \text{C.F.} = c_1 e^{2x} + c_2 e^{3x}.$$

$$\begin{aligned} \text{And P.I.} &= \frac{1}{D^2 - 5D + 6} \sin 3x = \frac{1}{-3^2 - 5D + 6} \sin 3x \\ &= \frac{1}{-9 - 5D + 6} \sin 3x \\ &= \frac{-1}{5D + 3} \sin 3x = \frac{-(5D-3)}{(5D+3)(5D-3)} \sin 3x = \frac{-(5D-3)}{25D^2-9} \sin 3x \\ &= \frac{-(5D-3)}{25 \cdot (-3^2) - 9} \sin 3x = \frac{1}{234} \{5D(\sin 3x) - 3 \sin 3x\} \\ &= \frac{1}{234} (5 \cdot 3 \cos 3x - 3 \sin 3x) = \frac{1}{78} (5 \cos 3x - \sin 3x). \end{aligned}$$

$\therefore$  the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

or  $y = c_1 e^{2x} + c_2 e^{3x} + (1/78) (5 \cos 3x - \sin 3x) .$

**Illustration 4 :** Solve  $\frac{d^2 y}{dx^2} + 9y = \cos 2x + \sin 2x .$

**Solution :** The auxiliary equation is  $m^2 + 9 = 0 .$

$\therefore m = \pm 3i .$

$\therefore$  C.F.  $= c_1 \cos 3x + c_2 \sin 3x$  or  $= c_1 \cos (3x + c_2) .$

And P.I.  $= \frac{1}{(D^2 + 9)} (\cos 2x + \sin 2x)$

$$= \frac{1}{D^2 + 9} \cos 2x + \frac{1}{D^2 + 9} \sin 2x$$

$$= \frac{\cos 2x}{-2^2 + 9} + \frac{\sin 2x}{-2^2 + 9} = \frac{\cos 2x}{5} + \frac{\sin 2x}{5} = \frac{1}{5} (\cos 2x + \sin 2x) .$$

Hence the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

or  $y = c_1 \cos (3x + c_2) + \frac{1}{5} (\cos 2x + \sin 2x) .$

**Illustration 5 :** Solve  $\frac{d^2 y}{dx^2} - 4y = e^x + \sin 2x .$

**Solution :** The given differential equation can be written as

$$(D^2 - 4) y = e^x + \sin 2x .$$

The auxiliary equation is  $m^2 - 4 = 0 .$

$\therefore (m + 2) (m - 2) = 0$  or  $m = 2, -2 .$

$\therefore$  C.F.  $= c_1 e^{2x} + c_2 e^{-2x}$

and P.I.  $= \frac{1}{D^2 - 4} [e^x + \sin 2x]$

$$= \frac{1}{D^2 - 4} e^x + \frac{1}{D^2 - 4} \sin 2x$$

$$= \frac{e^x}{1^2 - 4} + \frac{\sin 2x}{-2^2 - 4} = -\frac{e^x}{3} - \frac{1}{8} \sin 2x .$$

## Comprehensive Exercise 3

Solve the following differential equations :

- $(d^2 y / dx^2) + 9y = \cos 4x .$
- $(d^2 y / dx^2) + y = \cos 2x .$
- $(D^2 - 3D + 2) y = \sin 3x .$
- $(D^2 - 2D + 5) y = \sin 3x .$
- $(D^2 - 3D + 2) y = \cos 3x .$
- $(D^3 + D^2 - D - 1) y = \cos 2x .$

$$7. \frac{d^2 y}{dx^2} - 8 \frac{dy}{dx} + 9y = 40 \sin 5x.$$

$$8. \frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + y = a \sin 2x.$$

$$9. \frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + 10y + 37 \sin 2x = 0.$$

## Answers

$$1. y = c_1 \cos 3x + c_2 \sin 3x - \frac{1}{7} \cos 4x.$$

$$2. y = c_1 \cos x + c_2 \sin x - \frac{1}{3} \cos 2x.$$

$$3. y = c_1 e^x + c_2 e^{2x} + \frac{1}{130} (9 \cos 3x - 7 \sin 3x).$$

$$4. y = e^{2x} (c_1 \cos 2x + c_2 \sin 2x) + (1/26) (3 \cos 3x - 2 \sin 3x).$$

$$5. y = c_1 e^x + c_2 e^{2x} - \frac{1}{130} (7 \cos 3x + 9 \sin 3x).$$

$$6. y = (c_1 + c_2 x) e^{-x} + c_3 e^x - (1/25) (2 \sin 2x + \cos 2x).$$

$$7. y = c_1 e^{4x} \cosh(x\sqrt{7} + c_2) + \frac{5}{29} (5 \cos 5x - 2 \sin 5x).$$

$$8. y = c_1 e^{2x} \cosh(\sqrt{3}x + c_2) + (1/73) a (8 \cos 2x - 3 \sin 2x).$$

$$9. y = e^{-x} (c_1 \cos 3x + c_2 \sin 3x) + \frac{37}{26} (2 \cos 2x - 3 \sin 2x).$$

**Case III.** To find P.I. when  $Q$  is of the form  $x^m$ , where  $m$  is a positive integer.

Consider first  $\{1/(D-a)\} x^m$ . We have

$$\begin{aligned} \frac{1}{(D-a)} x^m &= -\frac{1}{(a-D)} x^m = -\frac{1}{a\{1-(D/a)\}} x^m \\ &= -\frac{1}{a} \left(1 - \frac{D}{a}\right)^{-1} x^m \\ &= -\frac{1}{a} \left(1 + \frac{D}{a} + \frac{D^2}{a^2} + \dots\right) x^m, \text{ expanding by the binomial theorem} \\ &= -\frac{1}{a} \left\{x^m + \frac{1}{a} mx^{m-1} + \frac{1}{a^2} m(m-1)x^{m-2} + \dots\right\}. \end{aligned}$$

Here we observe that in the expansion by the binomial theorem, the terms of the expansion beyond the  $m$ th power of  $D$  need not be written since  $D^{m+1} x^m = 0$ ,  $D^{m+2} x^m = 0$ , etc.

**Working Rule.** In order to evaluate  $\{1/F(D)\} x^m$ , bring out common the lowest degree term in  $D$  from  $F(D)$  so that remaining factor in the denominator is of the form  $[1+f(D)]$  or  $[1-f(D)]$  which is taken in the numerator with a

negative index. Next we expand  $[1 \pm f(D)]^{-1}$  in powers of  $D$  by the binomial theorem and operate upon  $x^m$  with the expansion obtained.

This expansion should be done upto the term  $D^m$ , since  $D^{m+1}x^m = 0$  and all higher differential coefficients of  $x^m$  are zero. The whole process will be clear from the following examples.

The following binomial expansions should be remembered well.

$$(i) \quad (1 - D)^{-1} = 1 + D + D^2 + D^3 + D^4 + \dots$$

$$(ii) \quad (1 + D)^{-1} = 1 - D + D^2 - D^3 + D^4 - \dots$$

$$(iii) \quad (1 - D)^{-2} = 1 + 2D + 3D^2 + 4D^3 + \dots$$

$$(iv) \quad (1 + D)^{-2} = 1 - 2D + 3D^2 - 4D^3 + \dots$$

$$(v) \quad (1 - D)^{-3} = 1 + 3D + 6D^2 + 10D^3 + \dots$$

$$(vi) \quad (1 + D)^{-3} = 1 - 3D + 6D^2 - 10D^3 + \dots$$

$$(vii) \quad (1 + x)^{-n} = 1 - nx + \frac{n(n+1)}{2!}x^2 - \frac{n(n+1)(n+2)}{3!}x^3 + \dots$$

$$(viii) \quad (1 - x)^{-n} = 1 + nx + \frac{n(n+1)}{2!}x^2 + \frac{n(n+1)(n+2)}{3!}x^3 + \dots$$

## Illustrative Examples

**Illustration 1 :** Solve  $(d^2 y / dx^2) - 4y = x^2$ .

**Solution :** The auxiliary equation is  $m^2 - 4 = 0$  or  $m = \pm 2$ .

$$\therefore \quad \text{C.F.} = c_1 e^{2x} + c_2 e^{-2x}.$$

$$\text{And} \quad \text{P.I.} = \frac{1}{D^2 - 4} x^2 = \frac{1}{-4 \left[ 1 - \frac{1}{4} D^2 \right]} x^2 = -\frac{1}{4} \left[ 1 - \frac{1}{4} D^2 \right]^{-1} x^2$$

$$= -\frac{1}{4} \left[ 1 + \frac{1}{4} D^2 + \dots \right] x^2,$$

$$\begin{aligned} & \text{expanding by binomial theorem upto the terms containing } D^2 \\ &= -\frac{1}{4} \left[ x^2 + \frac{1}{4} D^2 (x^2) \right], \text{ because all the remaining terms vanish} \\ &= -\frac{1}{4} \left[ x^2 + \frac{1}{4} \cdot 2 \right] = -\frac{1}{4} \left[ x^2 + \frac{1}{2} \right]. \end{aligned}$$

Hence the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

$$\text{or} \quad y = c_1 e^{2x} + c_2 e^{-2x} - \frac{1}{4} \left( x^2 + \frac{1}{2} \right).$$

**Illustration 2 :** Solve  $\frac{d^3 y}{dx^3} - \frac{d^2 y}{dx^2} - 6 \frac{dy}{dx} = 1 + x^2$ .

**Solution :** The auxiliary equation is

$$m^3 - m^2 - 6m = 0 \quad \text{or} \quad m(m+2)(m-3) = 0.$$

$$\therefore m = 0, -2, 3.$$

$$\therefore \text{C.F.} = c_1 e^{0x} + c_2 e^{-2x} + c_3 e^{3x} = c_1 + c_2 e^{-2x} + c_3 e^{3x}.$$

And 
$$\text{P.I.} = \frac{1}{D^4 - D^2 - 6D} (1 + x^2) = \frac{1}{-6D \left[ 1 + \frac{1}{6}D - \frac{1}{6}D^2 \right]} (1 + x^2)$$

$$= -\frac{1}{6D} \left[ 1 - \frac{1}{6}(-D + D^2) \right]^{-1} (1 + x^2)$$

$$= -\frac{1}{6D} \left[ 1 + \frac{1}{6}(-D + D^2) + \frac{1}{36}(-D + D^2)^2 + \dots \right] (1 + x^2),$$

the terms in the expansion being needed only upto  $D^2$

$$= -\frac{1}{6D} \left[ 1 - \frac{1}{6}D + \frac{1}{6}D^2 + \frac{1}{36}D^2 + \dots \right] (1 + x^2)$$

$$= -\frac{1}{6D} \left[ 1 - \frac{1}{6}D + \frac{7}{36}D^2 + \dots \right] (1 + x^2)$$

$$= -\frac{1}{6D} \left[ (1 + x^2) - \frac{1}{6}D(1 + x^2) + \frac{7}{36}D^2(1 + x^2) \right],$$

because all other terms vanish

$$= -\frac{1}{6D} \left( 1 + x^2 - \frac{1}{3}x + \frac{7}{18} \right) = -\frac{1}{6D} \left( \frac{25}{18} - \frac{1}{3}x + x^2 \right)$$

$$= -\frac{1}{6} \int \left( \frac{25}{18} - \frac{1}{3}x + x^2 \right) dx \quad \left[ \because 1/D \equiv \int dx \right]$$

$$= -\frac{1}{6} \left[ \frac{25}{18}x - \frac{1}{6}x^2 + \frac{1}{3}x^3 \right].$$

Hence the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

or 
$$y = c_1 + c_2 e^{-2x} + c_3 e^{3x} - \frac{1}{6} \left[ \frac{25}{18}x - \frac{1}{6}x^2 + \frac{1}{3}x^3 \right].$$

**Illustration 3 :** Solve  $(D^3 + 8)y = x^4 + 2x + 1$ .

**Solution :** The auxiliary equation is

$$m^3 + 8 = 0 \quad \text{or} \quad (m+2)(m^2 - 2m + 4) = 0.$$

$$\therefore m = -2, 1 \pm i\sqrt{3}.$$

Hence 
$$\text{C.F.} = c_1 e^{-2x} + e^x \{c_2 \cos(x\sqrt{3}) + c_3 \sin(x\sqrt{3})\}.$$

And 
$$\text{P.I.} = \frac{1}{D^3 + 8} (x^4 + 2x + 1) = \frac{1}{8 \left[ 1 + \frac{1}{8}D^3 \right]} (x^4 + 2x + 1)$$

$$= \frac{1}{8} \left[ 1 + \frac{1}{8}D^3 \right]^{-1} (x^4 + 2x + 1)$$

$$= \frac{1}{8} \left[ 1 - \frac{1}{8} D^3 + \dots \right] (x^4 + 2x + 1),$$

the other terms in the expansion being of no need

$$\begin{aligned} &= \frac{1}{8} \left[ (x^4 + 2x + 1) - \frac{1}{8} D^3 (x^4 + 2x + 1) \right] \\ &= \frac{1}{8} [x^4 + 2x + 1 - 3x] = \frac{1}{8} (x^4 - x + 1). \end{aligned}$$

Hence the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

or 
$$y = c_1 e^{-2x} + e^x \{c_2 \cos(x\sqrt{3}) + c_3 \sin(x\sqrt{3})\} + \frac{1}{8} (x^4 - x + 1).$$

**Illustration 4 :** Solve  $(D^2 + D - 2)y = x + \sin x$ .

**Solution :** The auxiliary equation is

$$m^2 + m - 2 = 0 \quad \text{or} \quad (m - 1)(m + 2) = 0.$$

$$\therefore m = 1, -2.$$

$$\therefore \text{C.F.} = c_1 e^x + c_2 e^{-2x}.$$

And P.I. = 
$$\begin{aligned} &\frac{1}{D^2 + D - 2} (x + \sin x) = \frac{1}{(D^2 + D - 2)} x + \frac{1}{D^2 + D - 2} \sin x \\ &= \frac{1}{-2 \left[ 1 - \frac{1}{2} D - \frac{1}{2} D^2 \right]} x + \frac{1}{-1^2 + D - 2} \sin x \\ &= -\frac{1}{2} \left[ 1 - \left( \frac{1}{2} D + \frac{1}{2} D^2 \right) \right]^{-1} x + \frac{(D + 3)}{(D - 3)(D + 3)} \sin x \\ &= -\frac{1}{2} \left[ 1 + \frac{1}{2} D + \dots \right] x + \frac{(D + 3)}{D^2 - 9} \sin x = -\frac{1}{2} \left( x + \frac{1}{2} \right) + \frac{D + 3}{-1 - 9} \sin x \\ &= -\frac{1}{2} \left( x + \frac{1}{2} \right) - \frac{1}{10} \{D(\sin x) + 3 \sin x\} \\ &= -\frac{1}{2} x - \frac{1}{4} - \frac{1}{10} \{\cos x + 3 \sin x\}. \end{aligned}$$

Hence the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

or 
$$y = c_1 e^x + c_2 e^{-2x} - \frac{1}{2} x - \frac{1}{4} - \frac{1}{10} \cos x - \frac{3}{10} \sin x.$$

**Illustration 5 :** Solve  $(D^2 - 5D + 6)y = x + \sin 3x$ . [UPTU 2003, 10]

**Solution :** The auxiliary equation is  $m^2 - 5m + 6 = 0$ .

$$\therefore m = 2, 3.$$

Hence 
$$\text{C.F.} = c_1 e^{2x} + c_2 e^{3x}.$$

And 
$$\begin{aligned} \text{P.I.} &= \frac{1}{(D^2 - 5D + 6)} (x + \sin 3x) \\ &= \frac{1}{D^2 - 5D + 6} x + \frac{1}{(D^2 - 5D + 6)} \sin 3x. \end{aligned} \quad \dots (I)$$

Now

$$\begin{aligned}\frac{1}{D^2 - 5D + 6} x &= \frac{1}{6 \left[ 1 - \frac{5}{6} D + \frac{1}{6} D^2 \right]} x \\ &= (1/6) [1 - \{(5/6) D - (1/6) D^2\}]^{-1} x \\ &= (1/6) [1 + (5/6) D + \dots] x = (1/6) (x + 5/6).\end{aligned}$$

And

$$\begin{aligned}\frac{1}{[D^2 - 5D + 6]} \sin 3x &= \frac{1}{-3^2 - 5D + 6} \sin 3x, \text{ putting } -3^2 \text{ for } D^2 \\ &= \frac{1}{-(3+5D)} \sin 3x = -\frac{(3-5D)}{(3+5D)(3-5D)} \sin 3x \\ &= -\frac{(3-5D)}{9-25D^2} \sin 3x = -\frac{(3-5D)}{9-25(-9)} \sin 3x \\ &= -\frac{1}{234} [3 \sin 3x - 5D (\sin 3x)] \\ &= -\frac{1}{234} [3 \sin 3x - 15 \cos 3x].\end{aligned}$$

 $\therefore$  from (1),

$$\text{P.I.} = \frac{1}{6} \left( x + \frac{5}{6} \right) - \frac{1}{234} [3 \sin 3x - 15 \cos 3x].$$

Hence the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$ 

$$\text{or } y = c_1 e^{2x} + c_2 e^{3x} + \frac{1}{6} \left( x + \frac{5}{6} \right) - \frac{1}{234} (3 \sin 3x - 15 \cos 3x).$$

**Illustration 6 :** Solve  $\frac{d^2 y}{dx^2} - 4y = \cos^2 x$ .

[UPTU 2003]

**Solution :** The auxiliary equation is

$$m^2 - 4 = 0 \quad \text{or} \quad m = \pm 2.$$

$$\therefore \text{C.F.} = c_1 e^{2x} + c_2 e^{-2x}.$$

$$\begin{aligned}\text{And P.I.} &= \frac{1}{(D^2 - 4)} \cos^2 x = \frac{1}{(D^2 - 4)} \cdot \frac{1}{2} (2 \cos^2 x) = \frac{1}{(D^2 - 4)} \cdot \frac{1}{2} (1 + \cos 2x) \\ &= \frac{1}{2} \frac{1}{(D^2 - 4)} \cdot 1 + \frac{1}{2} \frac{1}{(D^2 - 4)} \cos 2x \quad \dots(1)\end{aligned}$$

$$\begin{aligned}\text{Now } \frac{1}{2} \frac{1}{(D^2 - 4)} 1 &= \frac{1}{2} \frac{1}{-4 \left( 1 - \frac{1}{4} D^2 \right)} 1 = -\frac{1}{8} \left[ 1 - \frac{1}{4} D^2 \right]^{-1} 1 \\ &= -\frac{1}{8} \left[ 1 + \frac{1}{4} D^2 + \dots \right] 1 = -\frac{1}{8}.\end{aligned}$$

$$\text{And } \frac{1}{2} \frac{1}{(D^2 - 4)} \cos 2x = \frac{1}{2} \frac{1}{(-2^2 - 4)} \cos 2x = -\frac{1}{16} \cos 2x.$$

Substituting these values in (1), the

$$\text{P.I.} = -\frac{1}{8} - \frac{1}{16} \cos 2x.$$

Hence the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

$$\text{or } y = c_1 e^{2x} + c_2 e^{-2x} - \frac{1}{8} - \frac{1}{16} \cos 2x.$$

**Illustration 7 :** Solve  $\frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 4y = x^2 + e^x + \cos 2x$ .

**Solution :** The auxiliary equation is  $m^2 - 4m + 4 = 0$ .

$$\therefore m = 2, 2.$$

$$\text{Hence C.F.} = (c_1 + c_2 x) e^{2x}.$$

$$\text{And P.I.} = \frac{1}{(D^2 - 4D + 4)} (x^2 + e^x + \cos 2x) = P_1 + P_2 + P_3, (\text{say}),$$

$$\begin{aligned} \text{where } P_1 &= \frac{1}{D^2 - 4D + 4} x^2 = \frac{1}{(D-2)^2} x^2 = \frac{1}{4} \left[ 1 - \frac{1}{2} D \right]^{-2} x^2 \\ &= \frac{1}{4} \left[ 1 + D + \frac{3}{4} D^2 + \dots \right] x^2 = \frac{1}{4} \left[ x^2 + 2x + \frac{3}{2} \right], \end{aligned}$$

$$P_2 = \frac{1}{(D^2 - 4D + 4)} e^x = \frac{e^x}{1^2 - 4 \cdot 1 + 4} = \frac{e^x}{1} = e^x,$$

$$\begin{aligned} \text{and } P_3 &= \frac{1}{(D^2 - 4D + 4)} \cos 2x = \frac{1}{-2^2 - 4D + 4} \cos 2x = -\frac{1}{4D} \cos 2x \\ &= -\frac{1}{4} \int \cos 2x \, dx = -\frac{1}{4} \cdot \frac{1}{2} \sin 2x = -\frac{1}{8} \sin 2x. \end{aligned}$$

Hence the complete solution is  $y = \text{C.F.} + \text{P.I.}$

$$\text{or } y = \text{C.F.} + P_1 + P_2 + P_3$$

$$\text{or } y = (c_1 + c_2 x) e^{2x} + \frac{1}{4} \left( x^2 + 2x + \frac{3}{2} \right) + e^x - \frac{1}{8} \sin 2x.$$

**Illustration 8 :** Solve  $(D^2 - 3D + 2) y = 6e^{-3x} + \sin 2x$  (UPTU 2010)

**Solution :** The auxiliary equation is

$$m^2 - 3m + 2 = 0$$

$$\therefore m = 1, 2$$

$$\text{C.F.} = C_1 e^x + C_2 e^{2x}$$

$$\begin{aligned} \text{And P.I.} &= \frac{1}{D^2 - 3D + 2} (6e^{-3x} + \sin 2x) \\ &= \frac{1}{D^2 - 3D + 2} 6e^{-3x} + \frac{1}{D^2 - 3D + 2} \sin 2x \quad \dots(1) \end{aligned}$$

$$\text{Now } \frac{1}{D^2 - 3D + 2} 6e^{-3x} = \frac{6}{(-3)^2 - 3(-3) + 2} e^{-3x},$$

putting  $(-3)$  for  $D$



$$\begin{aligned}
 &= \frac{3}{10} e^{-3x} \\
 \text{and } \frac{1}{D^2 - 3D + 2} \sin 2x &= \frac{1}{-2^2 - 3D + 2} \sin 2x, \text{ putting } -2^2 \text{ for } D^2 \\
 &= \frac{1}{-(2 + 3D)} \sin 2x = \frac{-(2 - 3D)}{(2 + 3D)(2 - 3D)} \sin 2x \\
 &= \frac{-(2 - 3D)}{4 - 9D^2} \sin 2x = \frac{-(2 - 3D)}{4 - 9(-4)} \sin 2x \\
 &= -\frac{1}{40} (2 - 3D) \sin 2x \\
 &= -\frac{1}{40} [2 \sin 2x - 3D (\sin 2x)] \\
 &= -\frac{1}{40} [2 \sin 2x - 6 \cos 2x]
 \end{aligned}$$

∴ from (1), we have

$$\text{P.I.} = \frac{3}{10} e^{-3x} - \frac{1}{40} (2 \sin 2x - 6 \cos 2x)$$

Hence the complete solution is

$$y = \text{C.F.} + \text{P.I.}$$

$$\text{or } y = C_1 e^x + C_2 e^{2x} + \frac{3}{10} e^{-3x} - \frac{1}{40} (2 \sin 2x - 6 \cos 2x).$$

## Comprehensive Exercise 4

**Solve the following differential equations**

1.  $(D^2 + D - 6)y = x$ .
2.  $(d^2 y / dx^2) + (dy / dx) - 2y = 2x$ .
3.  $(d^3 y / dx^3) + 3(d^2 y / dx^2) + 2(dy / dx) = x^2$ .
4.  $\frac{d^3 y}{dx^3} - 4 \frac{d^2 y}{dx^2} + 5 \frac{dy}{dx} - 2 = 0$ .
5.  $(D^2 + 2D + 1)y = 2x + x^2$ .
6.  $(D^2 + 3D + 2)y = x^2$ .
7.  $(D^4 + D^2 + 16)y = 16x^2 + 256$ .
8.  $\frac{d^2 y}{dx^2} + 4y = \cos 3x + \sin 3x$ .
9.  $(D^2 - 4D + 3)y = e^{-x} + 5$ .
10.  $(d^2 y / dx^2) - 4y = \sin^2 x$ .
11.  $(D^2 - 2D + 3)y = \cos x + x^2$ .
12.  $\frac{d^3 y}{dx^3} + 2 \frac{d^2 y}{dx^2} + \frac{dy}{dx} = e^{2x} + x^2 + x$ .
13. Solve the equation  $y'' + y' - 2y = x$ , provided that  $y = \frac{dy}{dx} = 0$ , when  $x = 0$ .

14. Solve the equation  $(d^2 y / dx^2) = a + bx + cx^2$ , given that  $(dy / dx) = 0$ , when  $x = 0$  and  $y = d$ , when  $x = 0$ .

## Answers

1.  $y = c_1 e^{2x} + c_2 e^{-3x} - \frac{1}{36} (6x + 1) .$
2.  $y = c_1 e^x + c_2 e^{-2x} - x - \frac{1}{2} .$
3.  $y = c_1 + c_2 e^{-x} + c_3 e^{-2x} + \frac{1}{12} x (2x^2 - 9x + 21) .$
4.  $y = c_1 + c_2 e^{2x} \cos (x + c_3) + \frac{2}{5} x .$
5.  $y = (c_1 + c_2 x) e^{-x} + (x^2 - 2x + 2) .$
6.  $y = c_1 e^{-x} + c_2 e^{-2x} + \frac{1}{2} x^2 - \frac{3}{2} x + \frac{7}{4} .$
7.  $y = c_1 e^{-(1/2)\sqrt{7} \cdot x} \sin \left( \frac{3}{2} x + c_2 \right) + c_3 e^{(1/2)\sqrt{7} x} \sin \left( \frac{3}{2} x + c_4 \right) + x^2 + \frac{127}{8} .$
8.  $c_1 \cos 2x + c_2 \sin 2x - \frac{1}{5} (\cos 3x + \sin 3x) .$
9.  $y = c_1 e^x + c_2 e^{3x} + \frac{1}{8} e^{-x} + \frac{5}{3} .$
10.  $y = c_1 e^{2x} + c_2 e^{-2x} - \frac{1}{8} + \frac{1}{16} \cos 2x .$
11.  $y = e^x [c_1 \cos \sqrt{2}x + c_2 \sin \sqrt{2}x] + \frac{1}{3} x^2 + \frac{4}{9} x + \frac{2}{27} + \frac{1}{4} (\cos x - \sin x) .$
12.  $y = c_1 + (c_2 + c_3 x) e^{-x} + \frac{1}{18} e^{2x} + \frac{1}{3} x^3 - \frac{3}{2} x^2 + 4x .$
13.  $y = \frac{1}{3} e^x - \frac{1}{12} e^{-2x} - \frac{1}{2} x - \frac{1}{4} .$
14.  $y = d + \frac{1}{2} ax^2 + \frac{1}{6} bx^3 + \frac{1}{12} cx^4 .$

### 10.6 To Find P.I. when $Q = e^{ax} V$ , where $V$ is any Function of $x$ .

$$\frac{1}{F(D)} (e^{ax} V) = e^{ax} \frac{1}{F(D+a)} V .$$

**Working Rule :** Replace  $D$  by  $(D + a)$  and take out  $e^{ax}$  before the operator  $1 / F(D)$ . Then determine  $\{1 / F(D + a)\} V$  by the methods discussed in 10.4.

## Illustrative Examples

**Illustration 1:** Solve  $(D^2 - 2D + 1)y = x^2 e^{3x}$ .

**Solution :** The auxiliary equation is

$$m^2 - 2m + 1 = 0 \quad \text{or} \quad (m - 1)^2 = 0.$$

$$\therefore m = 1, 1.$$

$$\therefore \text{C.F.} = (c_1 + c_2 x) e^x.$$

And 
$$\text{P.I.} = \frac{1}{D^2 - 2D + 1} x^2 e^{3x} = \frac{1}{(D - 1)^2} e^{3x} x^2$$

$$= e^{3x} \frac{1}{\{(D + 3) - 1\}^2} x^2,$$

putting  $D + 3$  for  $D$  and bringing  $e^{3x}$  before the operator

$$= e^{3x} \frac{1}{(D + 2)^2} x^2 = e^{3x} \frac{1}{4 \left(1 + \frac{1}{2} D\right)^2} x^2$$

$$= \frac{1}{4} e^{3x} \left(1 + \frac{1}{2} D\right)^{-2} x^2$$

$$= \frac{1}{4} e^{3x} \left[1 - 2 \cdot \frac{1}{2} D + 3 \cdot \frac{1}{4} D^2 + \dots\right] x^2,$$

expanding by binomial theorem

$$= \frac{1}{4} e^{3x} \left[1 - D + \frac{3}{4} D^2 + \dots\right] x^2$$

$$= \frac{1}{4} e^{3x} \{x^2 - 2x + \frac{3}{4} \cdot 2\} = \frac{1}{4} e^{3x} \{x^2 - 2x + \frac{3}{2}\}.$$

Hence the required solution is  $y = (\text{C.F.}) + (\text{P.I.})$

or 
$$y = (c_1 + c_2 x) e^x + \frac{1}{4} e^{3x} \left\{x^2 - 2x + \frac{3}{2}\right\}.$$

**Illustration 2:** Solve  $(D^2 - 4D + 4)y = x^3 e^{2x}$ .

[UPTU 2010]

**Solution :** The auxiliary equation is

$$m^2 - 4m + 4 = 0 \quad \text{or} \quad (m - 2)^2 = 0.$$

$$\therefore m = 2, 2.$$

$$\therefore \text{C.F.} = (c_1 + c_2 x) e^{2x}.$$

And 
$$\text{P.I.} = \frac{1}{D^2 - 4D + 4} x^3 e^{2x}$$

$$= \frac{1}{(D - 2)^2} x^3 e^{2x} = e^{2x} \frac{1}{(D + 2 - 2)^2} x^3, \text{ by 10.5}$$

$$= e^{2x} \frac{1}{D^2} x^3 = e^{2x} \frac{1}{D} \left( \frac{x^4}{4} \right)$$

$$= e^{2x} \frac{x^5}{20}$$

Hence the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

or  $y = (c_1 + c_2 x) e^{2x} + e^{2x} \frac{x^5}{20}.$

**Illustration 3 :** Solve  $\frac{d^2 y}{dx^2} + 4 \frac{dy}{dx} - 12 y = (x - 1) e^{2x}.$

**Solution :** The auxiliary equation is  $m^2 + 4m - 12 = 0$

or  $(m - 2)(m + 6) = 0.$

$\therefore m = 2, -6.$

$\therefore \text{C.F.} = c_1 e^{2x} + c_2 e^{-6x}.$

And  $\text{P.I.} = \frac{1}{D^2 + 4D - 12} e^{2x} (x - 1)$

$$= e^{2x} \frac{1}{(D + 2)^2 + 4(D + 2) - 12} (x - 1)$$

$$= e^{2x} \frac{1}{D^2 + 8D} (x - 1) = e^{2x} \frac{1}{8D \left(1 + \frac{1}{8} D\right)} (x - 1)$$

$$= \frac{1}{8} e^{2x} \frac{1}{D} \left(1 + \frac{1}{8} D\right)^{-1} (x - 1)$$

$$= \frac{1}{8} e^{2x} \frac{1}{D} \left(1 - \frac{1}{8} D + \dots\right) (x - 1)$$

$$= \frac{1}{8} e^{2x} \frac{1}{D} \left(x - 1 - \frac{1}{8}\right), \text{ because all other terms vanish}$$

$$= \frac{1}{8} e^{2x} \frac{1}{D} \left(x - \frac{9}{8}\right) = \frac{1}{8} e^{2x} \int \left(x - \frac{9}{8}\right) dx$$

$$= \frac{1}{8} e^{2x} \left(\frac{x^2}{2} - \frac{9}{8} x\right) = \frac{1}{16} x^2 e^{2x} - \frac{9}{64} x e^{2x}.$$

Hence the general solution is

$$y = c_1 e^{2x} + c_2 e^{-6x} + (1/16) x^2 e^{2x} - (9/64) x e^{2x}.$$

**Illustration 4 :** Solve  $(D^2 + 2D + 1) y = e^{-x} / (x + 2).$

**Solution :** The auxiliary equation is  $m^2 + 2m + 1 = 0.$

$\therefore m = -1, -1.$

$\therefore \text{C.F.} = (c_1 + c_2 x) e^{-x}.$

And 
$$\begin{aligned} \text{P.I.} &= \frac{1}{D^2 + 2D + 1} \left\{ \frac{e^{-x}}{x+2} \right\} = \frac{1}{(D+1)^2} \frac{e^{-x}}{x+2} \\ &= e^{-x} \frac{1}{(D-1+1)^2} \frac{1}{(x+2)} = e^{-x} \frac{1}{D^2} \frac{1}{(x+2)} \\ &= e^{-x} \frac{1}{D} \int \frac{1}{x+2} dx = e^{-x} \frac{1}{D} \log(x+2) \\ &= e^{-x} \int \log(x+2) \cdot 1 dx = e^{-x} [x \log(x+2) - \int \{x/(x+2)\} dx], \\ &\quad \text{(integrating by parts taking 1 as second function)} \\ &= e^{-x} \{x \log(x+2) - x + 2 \log(x+2)\} \\ &= e^{-x} \{x \log(x+2) + 2 \log(x+2)\}, \text{ omitting the term } -xe^{-x} \end{aligned}$$

because such a term has already been included in the C.F. in  $c_2 x e^{-x}$ .

Hence complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

or 
$$y = (c_1 + c_2 x) e^{-x} + e^{-x} \{x \log(x+2) + 2 \log(x+2)\}.$$

**Illustration 5 :** If  $\frac{d^2 y}{dx^2} + y = x^2 e^x$ , find  $y$ .

**Solution :** The auxiliary equation is  $(m^2 + 1) = 0$ .

$\therefore m = \pm i$ .

$\therefore \text{C.F.} = c_1 \cos x + c_2 \sin x$ .

And 
$$\begin{aligned} \text{P.I.} &= \frac{1}{D^2 + 1} x^2 e^x = e^x \frac{1}{(D+1)^2 + 1} \cdot x^2 = e^x \frac{1}{(D^2 + 2D + 1 + 1)} \cdot x^2 \\ &= e^x \frac{1}{D^2 + 2D + 2} \cdot x^2 = e^x \frac{1}{2 \left[ 1 + \left( D + \frac{D^2}{2} \right) \right]} \cdot x^2 = \frac{e^x}{2} \left[ 1 + \left( D + \frac{D^2}{2} \right) \right]^{-1} x^2 \\ &= \frac{e^x}{2} \left[ 1 - \left( D + \frac{D^2}{2} \right) + \left( D + \frac{D^2}{2} \right)^2 - \dots \right] x^2 = \frac{e^x}{2} \left[ 1 - D - \frac{D^2}{2} + D^2 \right] x^2 \\ &= \frac{e^x}{2} \left[ x^2 - D(x^2) + \frac{1}{2} D^2(x^2) \right] \\ &= \frac{e^x}{2} \left[ x^2 - 2x + \frac{1}{2} (2) \right] = \frac{e^x}{2} [x^2 - 2x + 1]. \end{aligned}$$

Hence the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

or 
$$y = c_1 \cos x + c_2 \sin x + \frac{e^x}{2} (x^2 - 2x + 1).$$

**Illustration 6 :** Solve  $(D^2 - 2D + 5) y = e^{2x} \sin x$ .

[UPTU 2008]

**Solution :** The auxiliary equation is  $m^2 - 2m + 5 = 0$ .

$\therefore m = \{2 \pm \sqrt{4 - 20}\} / 2 = 1 \pm 2i$ .

Hence 
$$\text{C.F.} = e^x (c_1 \cos 2x + c_2 \sin 2x).$$

And 
$$\begin{aligned} \text{P.I.} &= \frac{1}{D^2 - 2D + 5} e^{2x} \sin x \\ &= e^{2x} \frac{1}{(D+2)^2 - 2(D+2) + 5} \sin x = e^{2x} \frac{1}{D^2 + 2D + 5} \sin x \\ &= e^{2x} \frac{1}{(-1^2 + 2D + 5)} \sin x, \text{ putting } -1^2 \text{ for } D^2 \\ &= \frac{1}{2} e^{2x} \frac{1}{D+2} \sin x = \frac{1}{2} e^{2x} \frac{D-2}{(D-2)(D+2)} \sin x \\ &= \frac{1}{2} e^{2x} \frac{D-2}{(D^2-4)} \sin x = \frac{1}{2} e^{2x} \frac{D-2}{-1-4} \sin x \\ &= -\frac{1}{10} e^{2x} (D-2) \sin x = -\frac{1}{10} e^{2x} \{D(\sin x) - 2 \sin x\} \\ &= -\frac{1}{10} e^{2x} (\cos x - 2 \sin x) . \end{aligned}$$

Hence the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

or 
$$y = e^x (c_1 \cos 2x + c_2 \sin 2x) - \frac{1}{10} e^{2x} (\cos x - 2 \sin x) .$$

## Comprehensive Exercise 5

Solve the following differential equations :

1.  $(D+1)^3 y = x^2 e^{-x} .$
2.  $(D^2 - 2D + 1) y = x^2 e^x .$
3.  $(D^2 - 2D + 1) y = x^2 e^{2x} .$
4.  $(D^2 - 3D + 2) y = x e^x .$
5.  $(D^2 + 2D + 2) y = x e^x .$
6.  $(D^2 - 4D + 1) y = e^{2x} \sin x .$
7.  $(D^2 - 4D + 4) y = e^{2x} \sin 3x .$
8.  $(D^4 - 1) y = e^x \cos x .$
9.  $(D^2 - 2D + 4) y = e^x \cos x .$
10.  $(D^2 - 5D + 6) y = e^{2x} \sin 2x .$

## Answers

1.  $y = (c_1 + c_2 x + c_3 x^2) e^{-x} + \frac{1}{60} x^5 e^{-x} .$
2.  $y = (c_1 + c_2 x) e^x + \frac{1}{12} x^4 e^x .$
3.  $y = (c_1 + c_2 x) e^x + e^{2x} (x^2 - 4x + 6) .$
4.  $y = c_1 e^x + c_2 e^{2x} - e^x \left( \frac{1}{2} x^2 + x \right) .$
5.  $y = e^{-x} (c_1 \cos x + c_2 \sin x) + \frac{1}{5} e^x \left( x - \frac{4}{5} \right) .$

6.  $y = e^{2x} (c_1 \cosh \sqrt{3x} + c_2 \sinh \sqrt{3x}) - \frac{1}{4} e^{2x} \sin x.$
7.  $y = (c_1 + c_2 x) e^{2x} - \frac{1}{9} e^{2x} \sin 3x.$
8.  $y = c_1 e^x + c_2 e^{-x} + c_3 \cos x + c_4 \sin x - \frac{1}{5} e^x \cos x.$
9.  $y = e^x (c_1 \cos \sqrt{3x} + c_2 \sin \sqrt{3x}) + \frac{1}{2} e^x \cos x.$
10.  $y = c_1 e^{2x} + c_2 e^{3x} + \frac{1}{10} e^{2x} (\cos 2x - 2 \sin 2x).$

## 10.7 To Find P.I. when $Q = e^{ax}$ and $F(a) = 0$ .

We have  $\text{P.I.} = \frac{1}{F(D)} e^{ax} = \frac{1}{F(D)} e^{ax} \cdot 1$ , if  $F(a) = 0$ .

Now this is of the form  $e^{ax} V$ . Hence, by putting  $(D + a)$  for  $D$  in the operator and by making  $e^{ax}$  free from the operator, we have

$$\text{P.I.} = e^{ax} \frac{1}{F(D+a)} 1.$$

Now this can be evaluated by using the method for finding P.I. in the case of  $x^m$ .

## Illustrative Examples

**Illustration 1 :** Solve  $\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = e^x.$

**Solution :** The auxiliary equation is

$$m^2 - 3m + 2 = 0 \quad \text{or} \quad (m-1)(m-2) = 0.$$

$$\therefore m = 1, 2.$$

$$\text{Hence C.F.} = c_1 e^x + c_2 e^{2x}.$$

$$\text{And } \text{P.I.} = \frac{1}{D^2 - 3D + 2} e^x = \frac{1}{(D-2)(D-1)} e^x$$

$$= \frac{1}{(1-2)(D-1)} e^x \quad [\text{putting 1 for } D \text{ in the factor } D-2]$$

because it does not vanish by doing so]

$$= -\frac{1}{D-1} e^x \cdot 1 \quad [\text{Note that } D-1 \text{ becomes zero}]$$

by putting 1 for  $D$ ; so here we shall apply the method for  $e^{ax} V$  by taking 1 for  $V$ .]

$$= -e^x \frac{1}{(D+1)-1} 1 = -e^x \frac{1}{D} 1 = -e^x x.$$

Hence the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

or 
$$y = c_1 e^x + c_2 e^{2x} - x e^x.$$

**Note.** While finding P.I. in the case of  $e^{ax}$  if  $F(D)$  becomes zero by putting  $a$  for  $D$ , we factorise  $F(D)$ . Then we put  $D = a$  in the factors which do not vanish by doing so. The remaining operator is then dealt with by using the method for  $e^{ax} V$  on taking 1 for  $V$ .

**Illustration 2 :** Solve  $(D^3 - 7D + 6)y = e^{2x}$ .

[UPTU 2002]

**Solution :** The auxiliary equation is

$$m^3 - 7m + 6 = 0 \quad \text{or} \quad (m - 1)(m - 2)(m + 3) = 0.$$

$\therefore m = 1, 2, -3.$

$\therefore \text{C.F.} = c_1 e^x + c_2 e^{2x} + c_3 e^{-3x}.$

And 
$$\begin{aligned} \text{P.I.} &= \frac{1}{(D^3 - 7D + 6)} e^{2x} = \frac{1}{(D - 1)(D - 2)(D + 3)} e^{2x} \\ &= \frac{1}{(2 - 1)(D - 2)(2 + 3)} e^{2x}, \end{aligned}$$

putting 2 for  $D$  in all the factors except  $D - 2$

$$\begin{aligned} &= \frac{1}{5} \frac{1}{D - 2} e^{2x} \cdot 1 = \frac{1}{5} e^{2x} \frac{1}{(D + 2) - 2} 1 = \frac{1}{5} e^{2x} \frac{1}{D} 1 \\ &= \frac{1}{5} x e^{2x}. \end{aligned}$$

Hence the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

or 
$$y = c_1 e^x + c_2 e^{2x} + c_3 e^{-3x} + \frac{1}{5} x e^{2x}.$$

**Illustration 3 :** Solve  $(D^2 - 5D + 6)y = e^{3x}$ .

[UPTU 2002]

**Solution :** The auxiliary equation is  $m^2 - 5m + 6 = 0$ .

$\therefore m = 2, 3.$

Thus,  $\text{C.F.} = c_1 e^{2x} + c_2 e^{3x}$

and 
$$\begin{aligned} \text{P.I.} &= \frac{1}{(D^2 - 5D + 6)} e^{3x} \\ &= \frac{1}{(D - 2)(D - 3)} e^{3x} = \frac{1}{(3 - 2)(D - 3)} e^{3x}, \end{aligned}$$

putting 3 for  $D$  except  $(D - 3)$

$$\begin{aligned} &= \frac{1}{(D - 3)} e^{3x} \cdot 1 = e^{3x} \frac{1}{(D + 3) - 3} 1 = e^{3x} \frac{1}{D} 1 \\ &= x e^{3x}. \end{aligned}$$

Hence the complete solution is

$$y = (\text{C.F.}) + (\text{P.I.}) \quad \text{or} \quad y = c_1 e^{2x} + c_2 e^{3x} + x e^{3x}.$$



**Illustration 4 :** Solve  $(D^2 + 4D + 4) y = e^{2x} - e^{-2x}$ .

**Solution :** The auxiliary equation is  $(m + 2)^2 = 0$ .

$$\therefore m = -2, -2.$$

Hence C.F. =  $(c_1 + c_2 x) e^{-2x}$ .

And P.I. =  $\frac{1}{(D+2)^2} (e^{2x} - e^{-2x}) = \frac{1}{(D+2)^2} e^{2x} - \frac{1}{(D+2)^2} e^{-2x}$ .

Now  $\frac{1}{(D+2)^2} e^{2x} = \frac{e^{2x}}{(2+2)^2} = \frac{1}{16} e^{2x}$ . [  $\because$  here,  $F(a) \neq 0$  ]

Also  $\frac{1}{(D+2)^2} e^{-2x} = \frac{1}{(D+2)^2} e^{-2x} \cdot 1$ , [  $\because$  here  $F(a) = 0$  ]  
 $= e^{-2x} \frac{1}{\{(D-2)+2\}^2} 1 = e^{-2x} \frac{1}{D^2} 1 = e^{-2x} \frac{1}{D} x = \frac{1}{2} x^2 e^{-2x}$ .

Hence the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

or  $y = (c_1 + c_2 x) e^{-2x} + \frac{1}{16} e^{2x} - \frac{1}{2} x^2 e^{-2x}$ .

**Illustration 5 :** Solve  $D^2 y - 3Dy + 2y = \cosh x$ .

**Solution :** The auxiliary equation is

$$m^2 - 3m + 2 = 0 \quad \text{or} \quad (m-1)(m-2) = 0.$$

$$\therefore m = 1, 2.$$

$$\therefore \text{C.F.} = c_1 e^x + c_2 e^{2x}.$$

And P.I. =  $\frac{1}{D^2 - 3D + 2} \cosh x = \frac{1}{(D^2 - 3D + 2)} \left( \frac{e^x + e^{-x}}{2} \right)$   
 $= \frac{1}{2} \frac{1}{D^2 - 3D + 2} e^x + \frac{1}{2} \frac{1}{D^2 - 3D + 2} e^{-x}$ .

Now  $\frac{1}{2} \frac{1}{D^2 - 3D + 2} e^x = \frac{1}{2} \frac{1}{(D-1)(D-2)} e^x$   
 $= \frac{1}{2} \frac{1}{(D-1)(1-2)} e^x = -\frac{1}{2} \frac{1}{D-1} e^x \cdot 1$   
 $= -\frac{1}{2} e^x \frac{1}{D+1-1} 1 = -\frac{1}{2} e^x \frac{1}{D} 1 = -\frac{1}{2} x e^x$ .

Also  $\frac{1}{2} \frac{1}{D^2 - 3D + 2} e^{-x} = \frac{1}{2} \frac{1}{(-1)^2 - 3(-1) + 2} e^{-x}$ , putting  $-1$  for  $D$   
 $= \frac{1}{12} e^{-x}$ .

$$\therefore \text{P.I.} = -\frac{1}{2} x e^x + \frac{1}{12} e^{-x}.$$

Hence the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

or 
$$y = c_1 e^x + c_2 e^{2x} - \frac{1}{2} x e^x + \frac{1}{12} e^{-x}.$$

**Illustration 6 :** Solve  $(D^2 - 4D + 4) y = e^{2x} + \sin 2x$ .

**Solution :** The auxiliary equation is

$$m^2 - 4m + 4 = 0 \quad \text{or} \quad (m - 2)^2 = 0.$$

$\therefore m = 2, 2.$

Hence C.F. =  $(c_1 + c_2 x) e^{2x}.$

And P.I. =  $\frac{1}{(D - 2)^2} e^{2x} + \frac{1}{D^2 - 4D + 4} \sin 2x.$

Now 
$$\begin{aligned} \frac{1}{(D - 2)^2} e^{2x} &= \frac{1}{(D - 2)^2} e^{2x} \cdot 1 = e^{2x} \frac{1}{(D + 2 - 2)^2} 1 \\ &= e^{2x} \frac{1}{D^2} (1) = e^{2x} \frac{1}{D} (x) = e^{2x} \cdot \frac{x^2}{2} = \frac{1}{2} x^2 e^{2x}, \end{aligned}$$

and 
$$\frac{1}{D^2 - 4D + 4} \sin 2x = \frac{1}{-2^2 - 4D + 4} \sin 2x,$$

putting  $-2^2$  for  $D^2$

$$= \frac{1}{-4D} \sin 2x = -\frac{1}{4} \int \sin 2x \, dx = -\frac{1}{4} \cdot \frac{1}{2} (-\cos 2x) = \frac{1}{8} \cos 2x.$$

Hence the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

or 
$$y = (c_1 + c_2 x) e^{2x} + \frac{1}{2} x^2 e^{2x} + \frac{1}{8} \cos 2x.$$

**Illustration 7 :** Solve  $(D^2 - 4D + 4) y = 8(x^2 + e^{2x} + \sin 2x)$ . [UPTU 2003]

**Solution :** Proceeding as in illustration 6 above, here

$$\text{C.F.} = (c_1 + c_2 x) e^{2x}.$$

And P.I. = 
$$\frac{1}{D^2 - 4D + 4} \{8(x^2 + e^{2x} + \sin 2x)\}$$

$$= 8 \frac{1}{D^2 - 4D + 4} x^2 + 8 \frac{1}{D^2 - 4D + 4} e^{2x} + 8 \frac{1}{D^2 - 4D + 4} \sin 2x.$$

Now 
$$\begin{aligned} 8 \frac{1}{D^2 - 4D + 4} x^2 &= 8 \frac{1}{(D - 2)^2} x^2 = 8 \cdot \frac{1}{4(1 - \frac{1}{2}D)^2} x^2 \\ &= 2(1 - \frac{1}{2}D)^{-2} x^2 = 2 \{1 + 2 \cdot \frac{1}{2}D + 3 \cdot \frac{1}{4}D^2 + \dots\} x^2 \\ &= 2(1 + D + \frac{3}{4}D^2 + \dots) x^2 = 2(x^2 + 2x + \frac{3}{4} \cdot 2) = 2(x^2 + 2x + \frac{3}{2}), \\ 8 \frac{1}{D^2 - 4D + 4} e^{2x} &= 8 \frac{1}{(D - 2)^2} e^{2x} \cdot 1 = 4 x^2 e^{2x}, \end{aligned}$$

and 
$$8 \frac{1}{D^2 - 4D + 4} \sin 2x = 8 \cdot \frac{1}{8} \cos 2x. \quad [\text{See Illustration 6}]$$

$$\therefore \text{P.I.} = 2 \left( x^2 + 2x + \frac{3}{2} \right) + 4x^2 e^{2x} + \cos 2x.$$

Hence the complete solution is

$$y = (\text{C.F.}) + (\text{P.I.})$$

$$\text{or } y = (c_1 + c_2 x) e^{2x} + 2 \left( x^2 + 2x + \frac{3}{2} \right) + 4x^2 e^{2x} + \cos 2x.$$

**Illustration 8 :** Solve  $(D^3 - 5D^2 + 7D - 3)y = e^{2x} \cosh x$ .

**Solution :** The auxiliary equation is  $m^3 - 5m^2 + 7m - 3 = 0$

$$\text{or } (m-1)(m^2 - 4m + 3) = 0 \text{ or } (m-1)\{(m-1)(m-3)\} = 0.$$

$$\therefore m = 1, 1, 3.$$

$$\text{Hence C.F.} = (c_1 + c_2 x) e^x + c_3 e^{3x}.$$

$$\begin{aligned} \text{And P.I.} &= \frac{1}{D^3 - 5D^2 + 7D - 3} e^{2x} \cosh x \\ &= \frac{1}{(D-1)^2 (D-3)} e^{2x} \left( \frac{e^x + e^{-x}}{2} \right), \left\{ \because \cosh x = \frac{e^x + e^{-x}}{2} \right\} \\ &= \frac{1}{(D-1)^2 (D-3)} \left( \frac{1}{2} e^{3x} + \frac{1}{2} e^x \right) \\ &= \frac{1}{2} \frac{1}{(D-1)^2 (D-3)} e^{3x} + \frac{1}{2} \frac{1}{(D-1)^2 (D-3)} e^x. \quad \dots(1) \end{aligned}$$

$$\begin{aligned} \text{Now } \frac{1}{2} \frac{1}{(D-1)^2 (D-3)} e^{3x} &= \frac{1}{2} \frac{1}{(3-1)^2 (D-3)} e^{3x}, \\ &\quad \text{putting 3 for } D \text{ in the factor } (D-1)^2 \\ &= \frac{1}{8} \frac{1}{D-3} e^{3x} \cdot 1 = \frac{1}{8} e^{3x} \frac{1}{D+3-3} \cdot 1 = \frac{1}{8} e^{3x} \frac{1}{D} \cdot 1 = \frac{1}{8} x e^{3x}. \end{aligned}$$

$$\begin{aligned} \text{Again } \frac{1}{2} \frac{1}{(D-1)^2 (D-3)} e^x &= \frac{1}{2} \frac{1}{(D-1)^2 (1-3)} e^x, \\ &\quad \text{putting 1 for } D \text{ in the factor } D-3 \\ &= -\frac{1}{4} \frac{1}{(D-1)^2} e^x \cdot 1 = -\frac{1}{4} e^x \frac{1}{\{(D+1)-1\}^2} \cdot 1 \\ &= -\frac{1}{4} e^x \frac{1}{D^2} \cdot 1 = -\frac{1}{4} e^x \frac{1}{D} x \\ &= -\frac{1}{4} e^x \cdot \frac{1}{2} x^2 = -\frac{1}{8} x^2 e^x. \end{aligned}$$

$$\therefore \text{from (1), P.I.} = \frac{1}{8} x e^{3x} - \frac{1}{8} x^2 e^x.$$

Hence the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

$$\text{or } y = (c_1 + c_2 x) e^x + c_3 e^{3x} + \frac{1}{8} x e^{3x} - \frac{1}{8} x^2 e^x.$$

## Comprehensive Exercise 6

Solve the following differential equations :

1.  $(D^2 + D - 2) y = e^x$ .
2.  $(D^2 - 2D + 1) y = e^x$ .
3.  $(D^2 + D - 6) y = e^{2x}$ .
4.  $(D^2 - 4D + 3) y = 2e^{3x}$ .
5.  $(D^3 + 3D^2 + 3D + 1) y = e^{-x}$ .
6.  $(d^2 y / dx^2) - 5 (dy / dx) + 6y = e^x + e^{2x}$ .
7.  $(D^2 - a^2) = \cosh ax$ .
8.  $(2D + 1)^2 (D + 2) (D^2 + 4) y = 51 \cdot e^{-x/2}$ .

## Answers

1.  $y = c_1 e^x + c_2 e^{-2x} + \frac{1}{3} x e^x$ .
2.  $y = (c_1 + c_2 x) e^x + \frac{1}{2} x^2 e^x$ .
3.  $y = c_1 e^{2x} + c_2 e^{-3x} + \frac{1}{5} x e^{2x}$ .
4.  $y = c_1 e^x + c_2 e^{3x} + x e^{3x}$ .
5.  $y = (c_1 + c_2 x + c_3 x^2) e^{-x} + \frac{1}{6} x^3 e^{-x}$ .
6.  $y = c_1 e^{2x} + c_2 e^{3x} + \frac{1}{2} e^x - x e^{2x}$ .
7.  $y = c_1 e^{ax} + c_2 e^{-ax} + (x / 2a) \sinh ax$ .
8.  $y = c_1 e^{-2x} + (c_2 + c_3 x) e^{-x/2} + c_4 \cos 2x + c_5 \sin 2x + x^2 e^{-x/2}$ .

### 10.8 To Find P.I. when $Q = \sin ax$ or $\cos ax$ and $F(-a^2) = 0$ .

To find out these types of particular integrals, it is convenient to replace  $\sin ax$  or  $\cos ax$  by the exponential value and then apply 10.6. Thus the particular integral of  $\cos ax$

i.e.,  $\frac{1}{F(D^2)} \cos ax$ , when  $F(-a^2) = 0$ , can be written as

$$= \frac{1}{F(D^2)} (\text{Real part of } e^{iax}), \quad [\because e^{iax} = \cos ax + i \sin ax]$$

$$= \text{Real Part of } \frac{1}{F(D^2)} e^{iax} \cdot 1, \text{ which is now of the form of 10.5.}$$

Similarly, the particular integral of  $\sin ax$

i.e.,

$$\begin{aligned} & \frac{1}{F(D^2)} \sin ax, \text{ when } F(-a^2) = 0 \\ &= \frac{1}{F(D^2)} (\text{Imaginary part of } e^{iax}), \quad [\because e^{iax} = \cos ax + i \sin ax] \\ &= \text{Imaginary part of } \frac{1}{F(D^2)} e^{iax} \cdot 1, \end{aligned}$$

which is now of the form  $\frac{1}{F(D)} e^{ax}$  V of 10.5.

Now the method of 10.5 is to be followed.

## Illustrative Examples

**Illustration 1 (a) :** Solve  $\frac{d^2 y}{dx^2} + 9y = \cos 3x$ .

[UPTU 2002]

**Solution :** The auxiliary equation is

$$m^2 + 9 = 0 \quad \text{i.e.,} \quad m^2 + 3^2 = 0 \quad \text{i.e.,} \quad m = 0 \pm 3i.$$

$$\therefore \text{C.F.} = e^{0x} (c_1 \cos 3x + c_2 \sin 3x) = c_1 \cos 3x + c_2 \sin 3x.$$

And 
$$\text{P.I.} = \frac{1}{D^2 + 3^2} \cos 3x$$

$$= \text{the real part in } \frac{1}{D^2 + 3^2} (\cos 3x + i \sin 3x) \quad [\text{Note}]$$

$$= \text{the real part in } \frac{1}{D^2 + 3^2} e^{3ix},$$

by Euler's theorem of trigonometry.

Now 
$$\frac{1}{D^2 + 3^2} e^{3ix} = \frac{1}{(D + 3i)(D - 3i)} e^{3ix}$$

$$= \frac{1}{(3i + 3i)(D - 3i)} e^{3ix}, \text{ putting } 3i \text{ for } D \text{ in the factor } D + 3i$$

$$= \frac{1}{6i(D - 3i)} e^{3ix} \cdot 1 \quad [\text{Note}]$$

$$= \frac{1}{2i} e^{3ix} \frac{1}{(D + 3i) - 3i} \cdot 1 = \frac{1}{6i} e^{3ix} \frac{1}{D} \cdot 1$$

$$= \frac{1}{6i} e^{3ix} \cdot x = \frac{x}{6i} (\cos 3x + i \sin 3x), \quad [\because e^{iax} = \cos ax + i \sin ax]$$

$$= -\frac{ix}{6} (\cos 3x + i \sin 3x), \quad \left[ \because \frac{1}{i} = \frac{i}{i^2} = \frac{i}{-1} = -i \right]$$

$$= -i \frac{x}{6} \cos 3x + \frac{x}{6} \sin 3x, \quad [\because i^2 = -1]$$

$$\therefore \frac{1}{D^2 + 3^2} \cos 3x = \text{the real part in } \left( \frac{x}{6} \sin 3x - i \frac{x}{6} \cos 3x \right)$$

$$= \frac{x}{6} \sin 3x.$$

Hence the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

or  $y = c_1 \cos 3x + c_2 \sin 3x + \frac{1}{6} x \sin 3x.$

**Remember :** While solving the differential equation  $(D^2 + a^2) y = \cos ax$ , the P.I.

$$\frac{1}{D^2 + a^2} \cos ax = \frac{x}{2a} \sin ax = \frac{x}{2} \int \cos ax \, dx.$$

**Illustration 1 (b) :** Solve  $\frac{d^2 y}{dx^2} + a^2 y = \cos ax.$

**Solution :** Proceed exactly as in Illustration 1 (a).

**Ans.**  $y = c_1 \cos ax + c_2 \sin ax + (x / 2a) \sin ax.$

**Illustration 2 :** Solve  $(D^2 + a^2) y = \sin ax.$

**Solution :** Here as in Illustration 1, C.F. =  $c_1 \cos ax + c_2 \sin ax.$

Also P.I. =  $\frac{1}{D^2 + a^2} \sin ax$

$$= \text{the coefficient of } i \text{ in } \frac{1}{D^2 + a^2} (\cos ax + i \sin ax) \quad [\text{Note}]$$

$$= \text{the coefficient of } i \text{ in } \frac{1}{D^2 + a^2} e^{iax}.$$

Now  $\frac{1}{D^2 + a^2} e^{iax} = -i \frac{x}{2a} \cos ax + \frac{x}{2a} \sin ax.$

[Proceed as in Illustration 1]

$$\therefore \frac{1}{D^2 + a^2} \sin ax$$

$$= \text{the coefficient of } i \text{ in } \left( -i \frac{x}{2a} \cos ax + \frac{x}{2a} \sin ax \right)$$

$$= -\frac{x}{2a} \cos ax.$$

Hence the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

or  $y = c_1 \cos ax + c_2 \sin ax - (x / 2a) \cos ax.$

**Remember :**  $\frac{1}{D^2 + a^2} \sin ax = -\frac{x}{2a} \cos ax = \frac{x}{2} \int \sin ax \, dx.$

**Illustration 3 :** Solve  $(D^2 + 4) y = \sin^2 x.$

**Solution :** The auxiliary equation is  $m^2 + 2 = 0.$

$$\therefore m = 0 \pm 2i.$$

$$\text{Hence C.F.} = c_1 \cos 2x + c_2 \sin 2x. \quad [\because e^{0x} = 1]$$

$$\begin{aligned} \text{And P.I.} &= \frac{1}{D^2 + 4} \sin^2 x = \frac{1}{D^2 + 4} \cdot \frac{1}{2} (2 \sin^2 x) \\ &= \frac{1}{2} \frac{1}{(D^2 + 4)} (1 - \cos 2x) \\ &= \frac{1}{2} \frac{1}{D^2 + 4} 1 - \frac{1}{2} \frac{1}{D^2 + 4} \cos 2x. \end{aligned}$$

$$\text{Now } \frac{1}{2} \frac{1}{D^2 + 4} 1 = \frac{1}{8} \left( 1 + \frac{1}{4} D^2 \right)^{-1} 1 = \frac{1}{8} \left( 1 - \frac{1}{4} D^2 + \dots \right) 1 = \frac{1}{8}.$$

$$\text{Again } \frac{1}{2} \frac{1}{D^2 + 4} \cos 2x = \frac{1}{2} \cdot \frac{1}{4} x \sin 2x. \quad [\text{Refer Illustration 1.}]$$

Hence the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

$$\text{or } y = c_1 \cos 2x + c_2 \sin 2x + \frac{1}{8} - \frac{1}{8} x \sin 2x.$$

**Illustration 4 :** Solve  $(D^2 + 1)y = \sin x \sin 2x$ .

**Solution :** The auxiliary equation is  $m^2 + 1 = 0$  or  $m = \pm i$ .

$$\therefore \text{C.F.} = c_1 \cos x + c_2 \sin x.$$

$$\begin{aligned} \text{And P.I.} &= \frac{1}{D^2 + 1} (\sin x \sin 2x) = \frac{1}{D^2 + 1} \cdot \frac{1}{2} (2 \sin x \sin 2x) \\ &= \frac{1}{2} \frac{1}{D^2 + 1} (\cos x - \cos 3x) = \frac{1}{2} \frac{1}{D^2 + 1} \cos x - \frac{1}{2} \frac{1}{D^2 + 1} \cos 3x. \end{aligned}$$

$$\begin{aligned} \text{Now } \frac{1}{D^2 + 1} \cos x &= \text{the real part in } \frac{1}{D^2 + 1} e^{ix} \\ &= \text{the real part in } \frac{1}{(D + i)(D - i)} e^{ix} \\ &= \text{the real part in } \frac{1}{(i + i)(D - i)} e^{ix} \\ &= \text{the real part in } \frac{1}{2i(D - i)} e^{ix} \cdot 1 \\ &= \text{the real part in } \frac{1}{2i} e^{ix} \frac{1}{D + i - i} 1 \\ &= \text{the real part in } \frac{1}{2i} e^{ix} \frac{1}{D} 1 = \text{the real part in } \frac{1}{2i} e^{ix} x \\ &= \text{the real part in } \frac{x}{2i} (\cos x + i \sin x) \\ &= \text{the real part in } -\frac{1}{2} xi (\cos x + i \sin x) \end{aligned}$$

$$= \text{the real part in } -i \frac{1}{2} x \cos x + \frac{1}{2} x \sin x = \frac{1}{2} x \sin x.$$

$$\text{Again } \frac{1}{D^2 + 1} \cos 3x = \frac{1}{-3^2 + 1} \cos 3x = -\frac{1}{8} \cos 3x.$$

$$\therefore \text{the P.I.} = \frac{1}{2} \cdot \frac{1}{2} x \sin x - \frac{1}{2} \cdot \left(-\frac{1}{8} \cos 3x\right) = \frac{1}{4} x \sin x + \frac{1}{16} \cos 3x.$$

Hence the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

$$\text{or } y = c_1 \cos x + c_2 \sin x + \frac{1}{4} x \sin x + \frac{1}{16} \cos 3x.$$

**Illustration 5 :** Solve the differential equation

$$(D - 1)^2 (D^2 + 1)^2 y = \sin^2 \frac{1}{2} x + e^x.$$

[UPTU 2002]

**Solution :** Here the auxiliary equation is

$$(m - 1)^2 (m^2 + 1)^2 = 0.$$

$$\therefore m = 1, 1, \pm i, \pm i.$$

$$\therefore \text{C.F.} = (c_1 + c_2 x) e^x + (c_3 + c_4 x) \cos x + (c_5 + c_6 x) \sin x.$$

$$\begin{aligned} \text{Also P.I.} &= \frac{1}{(D - 1)^2 (D^2 + 1)^2} \left[ e^x + \sin^2 \frac{1}{2} x \right] \\ &= \frac{1}{(D - 1)^2 (D^2 + 1)^2} \left[ e^x + \frac{1}{2} (1 - \cos x) \right] \\ &= \frac{1}{(D - 1)^2 (D^2 + 1)^2} \frac{1}{2} + \frac{1}{(D - 1)^2 (D^2 + 1)^2} e^x \\ &\quad - \frac{1}{(D - 1)^2 (D^2 + 1)^2} \frac{1}{2} \cos x. \end{aligned}$$

$$\begin{aligned} \text{Now } \frac{1}{(D - 1)^2 (D^2 + 1)^2} \frac{1}{2} &= \frac{1}{2} \frac{1}{(D - 1)^2 (D^2 + 1)^2} 1 \\ &= \frac{1}{2} \frac{1}{(D - 1)^2 (D^2 + 1)^2} e^{0x} = \frac{1}{2} \frac{1}{(0 - 1)^2 (0 + 1)^2} e^{0x} = \frac{1}{2}, \\ \frac{1}{(D - 1)^2 (D^2 + 1)^2} e^x &= \frac{1}{(1 + 1)^2 (D - 1)^2} e^x \cdot 1 \\ &= \frac{1}{4} e^x \frac{1}{(D + 1 - 1)^2} 1 = \frac{1}{4} e^x \frac{1}{D^2} 1 = \frac{1}{4} e^x \cdot \frac{1}{2} x^2 = \frac{1}{8} x^2 e^x, \end{aligned}$$

$$\begin{aligned} \text{and } -\frac{1}{(D - 1)^2 (D^2 + 1)^2} \frac{1}{2} \cos x &= -\frac{1}{2} \frac{1}{(D^2 - 2D + 1) (D^2 + 1)^2} \cos x \\ &= -\frac{1}{2} \frac{1}{(-1 - 2D + 1) (D^2 + 1)^2} \cos x = \frac{1}{4} \frac{1}{(D^2 + 1)^2 D} \cos x \\ &= \frac{1}{4} \frac{1}{(D^2 + 1)^2} \sin x \quad \left[ \because \frac{1}{D} \cos x = \sin x \right] \end{aligned}$$



$$\begin{aligned}
 &= \text{the coefficient of } i \text{ in } \frac{1}{4(D^2 + 1)^2} e^{ix} \\
 &= \text{the coeff. of } i \text{ in } \frac{1}{4(D + i)^2 (D - i)^2} e^{ix} \\
 &= \text{the coeff. of } i \text{ in } \frac{1}{4(i + i)^2 (D - i)^2} e^{ix} \cdot 1 \\
 &= \text{the coeff. of } i \text{ in } -\frac{1}{16} e^{ix} \frac{1}{(D + i - i)^2} 1 \\
 &= \text{the coeff. of } i \text{ in } -\frac{1}{16} e^{ix} \frac{1}{D^2} 1 \\
 &= \text{the coeff. of } i \text{ in } -\frac{x^2}{32} (\cos x + i \sin x) = -\frac{x^2}{32} \sin x.
 \end{aligned}$$

$$\therefore \text{P.I.} = \frac{1}{2} + \frac{1}{8} x^2 e^x - \frac{1}{32} x^2 \sin x.$$

Hence the general solution is  $y = (\text{C.F.}) + (\text{P.I.})$

$$\begin{aligned}
 \text{or } y &= (c_1 + c_2 x) e^x + (c_3 + c_4 x) \cos x + (c_5 + c_6 x) \sin x \\
 &\quad + \frac{1}{2} + \frac{1}{8} x^2 e^x - \frac{1}{32} x^2 \sin x.
 \end{aligned}$$

**Illustration 6 :** Solve  $(D^3 + a^2 D) y = \sin ax$ .

**Solution :** The auxiliary equation is  $m^3 + a^2 m = 0$

$$\text{or } m(m^2 + a^2) = 0, \text{ i.e., } m = 0, \pm ai.$$

$$\therefore \text{C.F.} = c_1 + c_2 \cos ax + c_3 \sin ax. \quad [\because e^{0x} = 1]$$

$$\begin{aligned}
 \text{And P.I.} &= \frac{1}{D^3 + a^2 D} \sin ax = \frac{1}{(D^2 + a^2) D} \sin ax \\
 &= \frac{1}{(D^2 + a^2)} \left\{ \frac{1}{D} \sin ax \right\} = \frac{1}{D^2 + a^2} \left( -\frac{\cos ax}{a} \right), \\
 &\quad [\because (1/D) \sin ax = \int \sin ax \, dx] \\
 &= -\frac{1}{a} \frac{1}{D^2 + a^2} \cos ax.
 \end{aligned}$$

$$\begin{aligned}
 \text{Now } \frac{1}{D^2 + a^2} \cos ax &= \text{the real part in } \frac{1}{D^2 + a^2} (\cos ax + i \sin ax) \\
 &= \frac{x}{2a} \sin ax. \quad [\text{Proceed as in Illustration 1 (a)}]
 \end{aligned}$$

$$\therefore \text{the P.I.} = -\frac{1}{a} \cdot \frac{x}{2a} \sin ax = -\frac{x}{2a^2} \sin ax.$$

Hence the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

$$\text{or } y = c_1 + c_2 \cos ax + c_3 \sin ax - (1/2a) x \sin ax.$$

## Comprehensive Exercise 7

Solve the following differential equations :

1.  $(D^2 + 4) y = \cos 2x$  .
2.  $(D^2 + 4) y = \sin 2x$  .
3.  $(D^2 + 9) (D^2 + 1) y = \cos 3x$  .
4.  $\frac{d^2 y}{dx^2} + \frac{dy}{dx} = \sin x \cdot \sin 2x$  .

## Answers

1.  $y = c_1 \cos 2x + c_2 \sin 2x + \frac{1}{4} x \sin 2x$  .
2.  $y = c_1 \cos 2x + c_2 \sin 2x - \frac{1}{4} x \cos 2x$  .
3.  $y = c_1 \cos 3x + c_2 \sin 3x + c_3 \cos x + c_4 \sin x - \frac{1}{48} x \sin 3x$  .
4.  $y = c_1 + c_2 e^{-x} + \frac{1}{4} \sin x - \frac{1}{4} \cos x - \frac{1}{60} \sin 3x + \frac{1}{20} \cos 3x$  .

### 10.9 To Find P.I. when $Q = xV$ , where $V$ is any Function of $x$

$$\frac{1}{F(D)} (xV) = x \frac{1}{F(D)} V - \frac{F'(D)}{\{F(D)\}^2} V \quad [\text{Remember}]$$

$$\text{or } \frac{1}{F(D)} (xV) = x \frac{1}{F(D)} V + \left\{ \frac{d}{dD} \frac{1}{F(D)} \right\} V .$$

Proceeding by *repeated application* of the above formula  $\{1/F(D)\} x^m V$  can be determined.

**Important Note.** The method of 10.8 cannot be used when  $V$  is of the form  $\cos ax$  or  $\sin ax$  and  $F(-a^2) = 0$  i.e.,  $F(D^2)$  vanishes by putting  $-a^2$  for  $D^2$ . In that case P.I. is found by the method discussed in 10.7.

## Illustrative Examples

**Illustration 1 :** Solve  $\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + y = x \sin x$  .

**Solution :** The auxiliary equation is  $m^2 - 2m + 1 = 0$  .

$\therefore m = 1, 1$  .

Hence

$$\text{C.F.} = (c_1 + c_2 x) e^x.$$

And

$$\text{P.I.} = \frac{1}{(D-1)^2} x \sin x = x \frac{1}{(D-1)^2} \sin x - \frac{2}{(D-1)^3} \sin x, \text{ by 10.8}$$

$$= x \frac{1}{D^2 - 2D + 1} \sin x - \frac{2}{D^3 - 3D^2 + 3D - 1} \sin x$$

$$= x \cdot \frac{1}{-2D} \sin x - \frac{2}{-D + 3 + 3D - 1} \sin x,$$

putting  $-1^2$  i.e.,  $-1$  for  $D^2$

$$= -\frac{x}{2} \cdot \frac{1}{D} \sin x - \frac{2}{2(D+1)} \sin x = \frac{x}{2} \cos x - \frac{1}{D+1} \sin x$$

$$= \frac{x}{2} \cos x - \frac{(D-1)}{(D+1)(D-1)} \sin x = \frac{x \cos x}{2} - \frac{D-1}{D^2-1} \sin x$$

$$= \frac{1}{2} x \cos x + \frac{1}{2} (D-1) \sin x = \frac{1}{2} x \cos x + \frac{1}{2} (\cos x - \sin x).$$

Hence the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

$$\text{or } y = (c_1 + c_2 x) e^x + \frac{1}{2} x \cos x + \frac{1}{2} (\cos x - \sin x).$$

**Illustration. 2 :** Solve  $\frac{d^2 y}{dx^2} + 4y = x \sin x$ .

**Solution :** The auxiliary equation is  $m^2 + 4 = 0$ .

Therefore  $m = 0 \pm 2i$ .

$$\therefore \text{C.F.} = c_1 \cos 2x + c_2 \sin 2x. \quad [\because e^{0x} = 1]$$

$$\text{And } \text{P.I.} = \frac{1}{D^2 + 4} x \sin x = x \frac{1}{D^2 + 4} \sin x - \frac{2D}{(D^2 + 4)^2} \sin x, \text{ by 10.8}$$

$$= \frac{x \sin x}{-1^2 + 4} - \frac{2D}{(-1^2 + 4)^2} \sin x, \text{ putting } -1^2 \text{ for } D^2$$

$$= \frac{1}{3} x \sin x - \frac{2}{9} D(\sin x) = \frac{1}{3} x \sin x - \frac{2}{9} \cos x.$$

Hence the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

$$\text{or } y = c_1 \cos 2x + c_2 \sin 2x + \frac{1}{3} x \sin x - \frac{2}{9} \cos x.$$

**Illustration 3 :** Solve  $(D^4 - 1)y = x \sin x$ .

**Solution :** The auxiliary equation is

$$m^4 - 1 = 0 \quad \text{or} \quad (m^2 - 1)(m^2 + 1) = 0.$$

$$\therefore m = 1, -1, 0 \pm i.$$

$$\therefore \text{C.F.} = c_1 e^x + c_2 e^{-x} + e^{0x} (c_3 \cos x + c_4 \sin x).$$

$$\text{And } \text{P.I.} = \frac{1}{D^4 - 1} x \sin x.$$

Here we cannot use the method given in 10.8 because  $D^4 - 1$  vanishes by putting  $-1^2$  i.e.,  $-1$  for  $D^2$ . So here we shall proceed by the method given in 10.7.

$$\begin{aligned}
 \therefore \quad \text{P.I.} &= \text{Imaginary part of } \frac{1}{D^4 - 1} x e^{ix}, \quad [\because e^{ix} = \cos x + i \sin x] \\
 &= \text{I.P. of } e^{ix} \frac{1}{(D + i)^4 - 1} x, \text{ by using the method for } e^{ax} V \\
 &= \text{I.P. of } e^{ix} \frac{1}{(D^4 + 4iD^3 + 6i^2D^2 + 4i^3D + i^4 - 1)} x \\
 &= \text{I.P. of } e^{ix} \frac{1}{D^4 + 4iD^3 - 6D^2 - 4iD} x \quad [\because i^2 = -1] \\
 &= \text{I.P. of } e^{ix} \frac{1}{-4iD [1 + (3D/2i) - D^2 - (D^3/4i)]} x \\
 &= \text{I.P. of } \frac{e^{ix}}{-4i} \cdot \frac{1}{D} \left\{ 1 + \left( \frac{3D}{2i} - \dots \right) \right\}^{-1} x \\
 &= \text{I.P. of } \frac{e^{ix}}{-4i} \cdot \frac{1}{D} \left\{ 1 - \frac{3D}{2i} + \dots \right\} x \\
 &= \text{I.P. of } \frac{e^{ix}}{-4i} \frac{1}{D} \left( x - \frac{3}{2i} \right) \quad \left[ \because D \equiv \frac{d}{dx} \right] \\
 &= \text{I.P. of } \frac{e^{ix}}{-4i} \left( \frac{x^2}{2} - \frac{3x}{2i} \right) \\
 &\quad [\because (1/D) \text{ stands for integration w.r.t. } x] \\
 &= \text{I.P. of } \frac{1}{4} i e^{ix} \left( \frac{1}{2} x^2 + \frac{3}{2} i x \right) \quad [\because i^2 = -1 \Rightarrow 1/i = -i] \\
 &= \text{I.P. of } \frac{1}{4} i (\cos x + i \sin x) \left( \frac{1}{2} x^2 + \frac{3}{2} i x \right) \\
 &= \frac{1}{4} \left( \frac{1}{2} x^2 \cos x - \frac{3}{2} x \sin x \right) = \frac{1}{8} (x^2 \cos x - 3x \sin x).
 \end{aligned}$$

Hence the complete solution is

$$y = (\text{C.F.}) + (\text{P.I.})$$

$$\text{or} \quad y = c_1 e^x + c_2 e^{-x} + c_3 \cos x + c_4 \sin x + \frac{1}{8} (x^2 \cos x - 3x \sin x).$$

**Illustration 4 :** Solve  $(D^4 + 2D^2 + 1)y = x^2 \cos x$ . [UPTU 2007]

**Solution :** The auxiliary equation is  $m^4 + 2m^2 + 1 = 0$

$$\text{or} \quad (m^2 + 1)^2 = 0 \text{ giving } m = \pm i, \pm i.$$

$$\therefore \quad \text{C.F.} = (c_1 + c_2 x) \cos x + (c_3 + c_4 x) \sin x. \quad [\because e^{0x} = 1]$$

$$\text{And} \quad \text{P.I.} = \frac{1}{D^4 + 2D^2 + 1} x^2 \cos x$$

$$\begin{aligned}
&= \text{Real part of } \frac{1}{(D^2 + 1)^2} x^2 e^{ix} \quad [\because e^{ix} = \cos x + i \sin x] \\
&= \text{R.P. of } e^{ix} \frac{1}{\{(D+i)^2 + 1\}^2} x^2 \quad [\text{By 10.5}] \\
&= \text{R.P. of } e^{ix} \frac{1}{(D^2 + 2iD)^2} x^2 \quad [\because i^2 = -1] \\
&= \text{R.P. of } e^{ix} \frac{1}{4i^2 D^2 [1 + (D/2i)]^2} x^2 \\
&= \text{R.P. of } -\frac{1}{4} e^{ix} \frac{1}{D^2} \left[1 + \frac{D}{2i}\right]^{-2} x^2 \quad [\because i^2 = -1] \\
&= \text{R.P. of } -\frac{1}{4} e^{ix} \frac{1}{D^2} \left[1 - \frac{1}{2} iD\right]^{-2} x^2 \quad \left[\because \frac{1}{i} = -i\right] \\
&= \text{R.P. of } -\frac{1}{4} e^{ix} \frac{1}{D^2} \left[1 + 2 \cdot \frac{1}{2} iD + 3 \cdot \frac{1}{4} i^2 D^2 + \dots\right] x^2, \\
&\quad \text{expanding by binomial theorem} \\
&= \text{R.P. of } -\frac{1}{4} e^{ix} \frac{1}{D^2} \left[1 + iD - \frac{3}{4} D^2 + \dots\right] x^2 \\
&= \text{R.P. of } -\frac{1}{4} e^{ix} \left[\frac{1}{D^2} + \frac{i}{D} - \frac{3}{4} + \text{terms in } D, D^2, \text{ and so on}\right] x^2 \\
&= \text{R.P. of } -\frac{1}{4} e^{ix} \left[\frac{1}{3} \frac{x^4}{4} + i \frac{1}{3} x^3 - \frac{3}{4} x^2 + \text{terms in } x^1, x^0\right], \\
&\quad (\because 1/D \text{ stands for integration w.r.t. } x) \\
&= \text{R.P. of } -\frac{1}{4} (\cos x + i \sin x) \left(\frac{1}{12} x^4 + \frac{1}{3} ix^3 - \frac{3}{4} x^2 + \text{terms in } x^1, x^0\right) \\
&= -\frac{1}{4} \left(\frac{1}{12} x^4 - \frac{3}{4} x^2\right) \cos x + \frac{1}{4} \left(\frac{1}{3} x^3\right) \sin x \\
&\quad + \text{terms already included in the C.F.} \\
&= -\frac{1}{48} (x^4 - 9x^2) \cos x + \frac{1}{12} x^3 \sin x,
\end{aligned}$$

neglecting the terms already included in the C.F.

Hence the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

or 
$$y = (c_1 + c_2 x) \cos x + (c_3 + c_4 x) \sin x - \frac{1}{48} (x^4 - 9x^2) \cos x + \frac{1}{12} x^3 \sin x.$$

**Note.** 
$$\frac{1}{F(D)} (e^{ax} \cdot x \cdot \sin bx) = \frac{1}{F(D)} e^{ax} \cdot (x \sin bx) = e^{ax} \frac{1}{F(D+a)} x \sin bx$$

$$= e^{ax} \left[ x \cdot \frac{1}{F(D+a)} \sin bx + \left\{ \frac{d}{dD} \cdot \frac{1}{F(D+a)} \right\} \sin bx \right].$$

**Illustration 5 :** Solve  $(D^2 - 2D + 1) y = xe^x \sin x$ .

**Solution :** The auxiliary equation is  $m^2 - 2m + 1 = 0$  or  $(m - 1)^2 = 0$ .

$$\therefore m = 1, 1.$$

$$\therefore \text{C.F.} = (c_1 + c_2 x) e^x.$$

$$\text{And P.I.} = \frac{1}{D^2 - 2D + 1} xe^x \sin x = \frac{1}{(D - 1)^2} e^x x \sin x$$

$$= e^x \frac{1}{\{(D + 1) - 1\}^2} x \sin x, \text{ by 10.5}$$

$$= e^x \frac{1}{D^2} (x \sin x) = e^x \cdot \frac{1}{D} \int x \sin x dx,$$

[ $\because (1/D)$  stands for integration w.r.t.  $x$ ]

$$= e^x \frac{1}{D} (-x \cos x + \sin x), \text{ integrating by parts}$$

$$= e^x \int (-x \cos x + \sin x) dx$$

$$= e^x \left[ \int -x \cos x dx + \int \sin x dx \right]$$

$$= e^x [-x \sin x + \int \sin x dx + \int \sin x dx]$$

$$= e^x (-x \sin x - 2 \cos x).$$

Hence the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

$$\text{or } y = (c_1 + c_2 x) e^x - e^x (x \sin x + 2 \cos x).$$

**Illustration 6 :** Solve  $\frac{d^2 y}{dx^2} - y = x \sin x + (1 + x^2) e^x$ . [UPTU 2001 (Special)]

**Solution :** The auxiliary equation is  $m^2 - 1 = 0$ .

$$\therefore m = \pm 1.$$

$$\text{Hence C.F.} = c_1 e^x + c_2 e^{-x}.$$

$$\begin{aligned} \text{And P.I.} &= \frac{1}{(D^2 - 1)} \{x \sin x + (1 + x^2) e^x\} \\ &= \frac{1}{(D^2 - 1)} x \sin x + \frac{1}{(D^2 - 1)} (1 + x^2) e^x = P_1 + P_2 \text{ (say), ... (1)} \end{aligned}$$

$$\begin{aligned} \text{where } P_1 &= \frac{1}{(D^2 - 1)} x \sin x = \text{Imaginary part of } \frac{1}{D^2 - 1} x e^{ix} \\ &= \text{I.P. of } e^{ix} \frac{1}{(D + i)^2 - 1} x = \text{I.P. of } e^{ix} \frac{1}{D^2 + 2iD - 2} x \\ &= \text{I.P. of } e^{ix} \cdot (1/-2) [1 - (iD + \frac{1}{2} D^2)]^{-1} x \end{aligned}$$

$$\begin{aligned}
 &= \text{I.P. of } -\frac{1}{2} e^{ix} [1 + iD + \dots] x = \text{I.P. of } -\frac{1}{2} e^{ix} [x + i] \\
 &= \text{I.P. of } -\frac{1}{2} (\cos x + i \sin x) (x + i) = -\frac{1}{2} (\cos x + x \sin x),
 \end{aligned}$$

and 
$$\begin{aligned}
 P_2 &= \frac{1}{D^2 - 1} (1 + x^2) e^x = e^x \frac{1}{\{(D + 1)^2 + 1\}} (1 + x^2) \\
 &= e^x \frac{1}{D^2 + 2D} (1 + x^2) = e^x \frac{1}{2D (1 + \frac{1}{2} D)} (1 + x^2) \\
 &= \frac{e^x}{2} \cdot \frac{1}{D} \left(1 + \frac{D}{2}\right)^{-1} (1 + x^2) = \frac{e^x}{2} \cdot \frac{1}{D} \left(1 - \frac{D}{2} + \frac{D^2}{4} - \dots\right) (1 + x^2) \\
 &= \frac{e^x}{2} \cdot \frac{1}{D} \left[(1 + x^2) - \frac{1}{2} (2x) + \frac{1}{4} \cdot 2\right] = \frac{e^x}{2} \cdot \frac{1}{D} \left[x^2 - x + \frac{3}{2}\right] \\
 &= \frac{1}{2} e^x \left[\frac{1}{3} x^3 - \frac{1}{2} x^2 + \frac{3}{2} x\right] = \frac{1}{12} e^x (2x^3 - 3x^2 + 9x).
 \end{aligned}$$

$\therefore$  from (1),

$$\text{P.I.} = -\frac{1}{2} (\cos x + x \sin x) + (1/12) e^x (2x^3 - 3x^2 + 9x).$$

Hence the complete solution is

$$y = (\text{C.F.}) + (\text{P.I.})$$

or 
$$y = c_1 e^x + c_2 e^{-x} - \frac{1}{2} (\cos x + x \sin x) + (1/12) e^x (2x^3 - 3x^2 + 9x).$$

**Illustration 7 :** Solve  $(D^2 - 4D + 4) y = 8 x^2 e^{2x} \sin 2x$ . (UPTU 2005)

**Solution :** The auxiliary equation is

$$m^2 - 4m + 4 = 0 \quad \text{or} \quad (m - 2)^2 = 0. \quad \therefore m = 2, 2.$$

$\therefore$  C.F. =  $(c_1 + c_2 x) e^{2x}$ .

And 
$$\begin{aligned}
 \text{P.I.} &= \frac{1}{D^2 - 4D + 4} 8 x^2 e^{2x} \sin 2x = 8 \frac{1}{(D - 2)^2} e^{2x} x^2 \sin 2x \\
 &= 8 e^{2x} \frac{1}{\{(D + 2) - 2\}^2} x^2 \sin 2x, \text{ by 10.5} \\
 &= 8 e^{2x} \frac{1}{D^2} x^2 \sin 2x \\
 &= \text{I.P. of } 8 e^{2x} \frac{1}{D^2} x^2 e^{i2x}, \quad [\because e^{i2x} = \cos 2x + i \sin 2x] \\
 &= \text{I.P. of } 8 e^{2x} e^{i2x} \frac{1}{(D + 2i)^2} x^2, \text{ by 10.5}
 \end{aligned}$$

$$\begin{aligned}
 &= \text{I.P. of } 8e^{2x} e^{i2x} \frac{1}{4i^2 [1 + (D/2i)]^2} x^2 \\
 &= \text{I.P. of } -2e^{2x} e^{i2x} [1 - (i/2) D]^{-2} x^2, \\
 &\quad [\because i^2 = -1, (1/i) = -i] \\
 &= \text{I.P. of } -2e^{2x} e^{i2x} [1 + 2(i/2) D + 3(i^2/4) D^2 + \dots] x^2 \\
 &= \text{I.P. of } -2e^{2x} e^{i2x} [1 + iD - (3/4) D^2 + \dots] x^2 \\
 &= \text{I.P. of } -2e^{2x} (\cos 2x + i \sin 2x) \{x^2 + i2x - (3/2)\} \\
 &= -2e^{2x} [2x \cos 2x + \{x^2 - (3/2)\} \sin 2x] \\
 &= e^{2x} [-2x^2 \sin 2x + 3 \sin 2x - 4x \cos 2x].
 \end{aligned}$$

Hence the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

or  $y = (c_1 + c_2 x) e^{2x} + e^{2x} (-2x^2 \sin 2x + 3 \sin 2x - 4x \cos 2x).$

## Comprehensive Exercise 8

Solve the following differential equations :

1.  $d^2 y / dx^2 + 9y = x \sin x.$
2.  $(D^2 + 2D + 1) y = x \cos x.$
3.  $(D^2 + m^2) y = x \cos mx.$
4.  $(D^2 + 4) y = x \sin 2x.$
5.  $(D^2 + D) y = x \cos x.$
6.  $(D^2 - 1) y = x^2 \cos x.$
7.  $(D^2 + 1) y = x^2 \sin 2x.$
8.  $(D^4 - 1) y = x \sin 2x.$
9.  $(D^2 + 4) x = t^2 + \sin 2t.$

## Answers

1.  $y = c_1 \cos 3x + c_2 \sin 3x + \frac{1}{8} x \sin x - \frac{1}{32} \cos x.$
2.  $y = (c_1 + c_2 x) e^{-x} + \frac{1}{2} x \sin x - \frac{1}{2} \sin x + \frac{1}{2} \cos x.$
3.  $y = c_1 \cos mx + c_2 \sin mx + (x/4m^2) \cos mx + (x^2/4m) \sin mx.$



4.  $y = c_1 \cos 2x + c_2 \sin 2x - \frac{1}{8} x^2 \cos 2x + \frac{1}{16} x \sin 2x.$
5.  $y = c_1 + c_2 e^{-x} - \frac{1}{2} x (\cos x - \sin x) + \cos x + \frac{1}{2} \sin x.$
6.  $y = c_1 e^x + c_2 e^{-x} - \frac{1}{2} (x^2 - 1) \cos x + x \sin x.$
7.  $y = c_1 \cos x + c_2 \sin x - \frac{1}{27} [24x \cos 2x - (9x^2 - 26) \sin 2x].$
8.  $y = c_1 e^x + c_2 e^{-x} + c_3 \cos x + c_4 \sin x + \frac{1}{15} \left( \frac{32}{15} \cos 2x + x \sin 2x \right).$
9.  $y = c_1 \cos 2t + c_2 \sin 2t + \frac{1}{4} t^2 - \frac{1}{8} - \frac{1}{4} t \cos 2t.$

## 10.10 The Operator $\frac{1}{D-\alpha}$ and $\frac{1}{D+\alpha}$ , $\alpha$ being a Constant

If  $Q$  is any function of  $x$ , then

$$\frac{1}{D-\alpha} Q = e^{\alpha x} \int e^{-\alpha x} Q dx \text{ and } \frac{1}{D+\alpha} Q = e^{-\alpha x} \int e^{\alpha x} Q dx.$$

## Illustrative Examples

**Illustration 1 :** Solve  $\frac{d^2 y}{dx^2} + a^2 y = \sec ax.$

**Solution :** The auxiliary equation is  $m^2 + a^2 = 0$ , or  $m = \pm ia.$

$\therefore$  C.F. =  $c_1 \cos ax + c_2 \sin ax.$

And P.I. =  $\frac{1}{D^2 + a^2} \sec ax = \frac{1}{(D+ia)(D-ia)} \sec ax$

$$= \frac{1}{2ia} \left[ \frac{1}{D-ia} - \frac{1}{D+ia} \right] \sec ax$$

[By resolving into partial fractions]

$$= \frac{1}{2ia} \left[ \frac{1}{D-ia} \sec ax - \frac{1}{D+ia} \sec ax \right]$$

$$= \frac{1}{2ia} [e^{iax} \int e^{-iax} \sec ax dx - e^{-iax} \int e^{iax} \sec ax dx], \text{ by 10.9}$$

$$= \frac{1}{2ia} \left[ e^{iax} \int \frac{\cos ax - i \sin ax}{\cos ax} dx - e^{-iax} \int \frac{\cos ax + i \sin ax}{\cos ax} dx \right],$$

$$[\because e^{-iax} = \cos ax - i \sin ax \text{ and } e^{iax} = \cos ax + i \sin ax]$$

$$\begin{aligned}
 &= \frac{1}{2ia} \left[ e^{iax} \left\{ x + \frac{i}{a} \log \cos ax \right\} - e^{-iax} \left\{ x - \frac{i}{a} \log \cos ax \right\} \right] \\
 &= \frac{x}{a} \left\{ \frac{e^{iax} - e^{-iax}}{2i} \right\} + \frac{1}{a^2} (\log \cos ax) \left\{ \frac{e^{iax} + e^{-iax}}{2} \right\} \\
 &= \frac{x}{a} \sin ax + \frac{1}{a^2} (\log \cos ax) \cos ax.
 \end{aligned}$$

Hence the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

or 
$$y = c_1 \cos ax + c_2 \sin ax + \frac{x}{a} \sin ax + \frac{1}{a^2} \cos ax \cdot \log (\cos ax).$$

**Illustration 2 :** Solve  $\frac{d^2 y}{dx^2} + 9y = \sec 3x$ .

[UPTU 2002]

**Solution :** Proceeding as in illustration 1, we have

$$\text{C.F.} = c_1 \cos 3x + c_2 \sin 3x.$$

And 
$$\begin{aligned} \text{P.I.} &= \frac{1}{D^2 + 9} \sec 3x = \frac{1}{(D + 3i)(D - 3i)} \sec 3x \\ &= \frac{1}{6i} \left[ \frac{1}{D - 3i} - \frac{1}{D + 3i} \right] \sec 3x, \end{aligned}$$

by resolving into partial fractions

$$\begin{aligned}
 &= \frac{1}{6i} \left[ \frac{1}{D - 3i} \sec 3x - \frac{1}{D + 3i} \sec 3x \right] \\
 &= \frac{1}{6i} [e^{3ix} \int e^{-3ix} \sec 3x \, dx - e^{-3ix} \int e^{3ix} \sec 3x \, dx], \text{ by 10.9} \\
 &= \frac{1}{6i} \left[ e^{3ix} \int \frac{\cos 3x - i \sin 3x}{\cos 3x} \, dx - e^{-3ix} \int \frac{\cos 3x + i \sin 3x}{\cos 3x} \, dx \right] \\
 &= \frac{1}{6i} \left[ e^{3ix} \left\{ x + \frac{i}{3} \log \cos 3x \right\} - e^{-3ix} \left\{ x - \frac{i}{3} \log \cos 3x \right\} \right] \\
 &= \frac{x}{3} \cdot \frac{e^{3ix} - e^{-3ix}}{2i} + \frac{(\log \cos 3x)}{9} \cdot \frac{e^{3ix} + e^{-3ix}}{2}. \quad (\text{Note})
 \end{aligned}$$

Now putting  $\frac{e^{3ix} - e^{-3ix}}{2i} = \sin 3x$  and  $\frac{e^{3ix} + e^{-3ix}}{2} = \cos 3x$ , we get

$$\text{P.I.} = \frac{1}{3} x \sin 3x + \frac{1}{9} \cos 3x \log (\cos 3x).$$

Hence the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

or 
$$y = c_1 \cos 3x + c_2 \sin 3x + \frac{1}{3} x \sin 3x + \frac{1}{9} \cos 3x \log (\cos 3x).$$

**Illustration 3 :** Solve  $(D^2 + a^2) y = \operatorname{cosec} ax$ .

**Solution :** The auxiliary equation is  $m^2 + a^2 = 0$  or  $m = \pm ai$ .

$\therefore$  C.F. =  $c_1 \cos ax + c_2 \sin ax$ .

And P.I. =  $\frac{1}{D^2 + a^2} \operatorname{cosec} ax = \frac{1}{(D + ia)(D - ia)} \operatorname{cosec} ax$

$$= \frac{1}{2ai} \left[ \frac{1}{D - ia} - \frac{1}{D + ia} \right] \operatorname{cosec} ax$$

$$= \frac{1}{2ai} \left[ \frac{1}{D - ia} \operatorname{cosec} ax - \frac{1}{D + ia} \operatorname{cosec} ax \right]$$

$$= \frac{1}{2ai} \left[ e^{iax} \int e^{-iax} \operatorname{cosec} ax \, dx - e^{-iax} \int e^{iax} \operatorname{cosec} ax \, dx \right], \quad [\text{by 10.9}]$$

$$= \frac{1}{2ai} \left[ e^{iax} \int \frac{\cos ax - i \sin ax}{\sin ax} \, dx - e^{-iax} \int \frac{\cos ax + i \sin ax}{\sin ax} \, dx \right],$$

$$[\because e^{-iax} = \cos ax - i \sin ax \text{ and } e^{iax} = \cos ax + i \sin ax]$$

$$= \frac{1}{2ai} \left[ e^{iax} \left( \frac{1}{a} \log \sin ax - ix \right) - e^{-iax} \left( \frac{1}{a} \log \sin ax + ix \right) \right]$$

$$= \frac{1}{a^2} (\log \sin ax) \left( \frac{e^{iax} - e^{-iax}}{2i} \right) - \frac{x}{a} \left( \frac{e^{iax} + e^{-iax}}{2} \right)$$

$$= \frac{1}{a^2} \sin ax (\log \sin ax) - \frac{x}{a} \cos ax.$$

Hence the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

or  $y = c_1 \cos ax + c_2 \sin ax + (1/a^2) \sin ax \log \sin ax - (x/a) \cos ax$ .

**Illustration 4 :** Solve  $(D^2 + a^2) y = \tan ax$ .

**Solution :** Here the C.F. =  $c_1 \cos ax + c_2 \sin ax$ , as in Illustration 3.

And P.I. =  $\frac{1}{D^2 + a^2} \tan ax = \frac{1}{(D + ia)(D - ia)} \tan ax$

$$= \frac{1}{2ai} \left[ \frac{1}{D - ia} - \frac{1}{D + ia} \right] \tan ax$$

$$= \frac{1}{2ai} \left[ \frac{1}{D - ia} \tan ax - \frac{1}{D + ia} \tan ax \right].$$

Now  $\frac{1}{D - ia} \tan ax = e^{iax} \int e^{-iax} \tan ax \, dx, \quad [\text{by 10.9}]$

$$= e^{iax} \int (\cos ax - i \sin ax) \cdot \frac{\sin ax}{\cos ax} \, dx$$


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$$\begin{aligned}
 &= e^{iax} \int \left( \sin ax - i \frac{\sin^2 ax}{\cos ax} \right) dx \\
 &= e^{iax} \int \left( \sin ax - i \frac{1 - \cos^2 ax}{\cos ax} \right) dx \\
 &= e^{iax} \int \{ \sin ax - i \sec ax + i \cos ax \} dx \\
 &= e^{iax} \left( -\frac{\cos ax}{a} \right) - ie^{iax} \int \sec ax \, dx + ie^{iax} \left( \frac{\sin ax}{a} \right). \quad \dots(1)
 \end{aligned}$$

Again  $\frac{1}{D + ia} \tan ax = e^{-iax} \int e^{iax} \tan ax \, dx,$  [by 10.9]

$$\begin{aligned}
 &= e^{-iax} \int (\cos ax + i \sin ax) \cdot \frac{\sin ax}{\cos ax} dx \\
 &= e^{-iax} \int \left( \sin ax + i \frac{\sin^2 ax}{\cos ax} \right) dx \\
 &= e^{-iax} \int \left( \sin ax + i \frac{1 - \cos^2 ax}{\cos ax} \right) dx \\
 &= e^{-iax} \int \{ \sin ax + i \sec ax - i \cos ax \} dx \\
 &= e^{-iax} \left( -\frac{\cos ax}{a} \right) + ie^{-iax} \int \sec ax \, dx - ie^{-iax} \left( \frac{\sin ax}{a} \right) \quad \dots(2)
 \end{aligned}$$

Subtracting (2) from (1) and dividing by  $2ai$ , we find that the P.I.

$$\begin{aligned}
 &= -\frac{e^{iax} - e^{-iax}}{2i} \cdot \frac{\cos ax}{a^2} - \frac{1}{a} \frac{e^{iax} + e^{-iax}}{2} \int \sec ax \, dx \\
 &\quad + \frac{e^{iax} + e^{-iax}}{2} \cdot \frac{\sin ax}{a^2} \\
 &= -\frac{\sin ax \cos ax}{a^2} - \frac{1}{a} (\cos ax) \cdot \frac{1}{a} \log \tan \left( \frac{1}{4} \pi + \frac{1}{2} ax \right) \\
 &\quad + \frac{\cos ax \sin ax}{a^2} \\
 &= (-1/a^2) \cos ax \log \tan \left( \frac{1}{4} \pi + \frac{1}{2} ax \right).
 \end{aligned}$$

Hence the complete solution is  $y = (\text{C.F.}) + (\text{P.I.})$

or  $y = c_1 \cos ax + c_2 \sin ax - (1/a^2) \cos ax \cdot \log \tan \left( \frac{1}{4} \pi + \frac{1}{2} ax \right).$

## Comprehensive Exercise 9

### Fill in the Blanks

Fill in the blanks "....." so that the following statements are complete and correct.

- The general solution of the equation  $f(D)y = 0$  is called the ..... of the equation  $f(D)y = Q$ .
- The solution of the differential equation  $\frac{d^2 y}{dx^2} + a^2 y = 0$  is ..... .
- The complementary function (C.F.) of the differential equation  $\frac{d^2 y}{dx^2} - 5 \frac{dy}{dx} + 4y = 0$  is ..... .
- The particular integral (P.I.) of the differential equation  $\frac{d^3 y}{dx^3} - 4 \frac{d^2 y}{dx^2} + 5 \frac{dy}{dx} - 2 = 0$  is ..... .
- The particular integral of the differential equation  $f(D)y = Q$  will be ..... .
- The number of arbitrary constants in general solution of second order differential equation is ..... [UPTU 2011]

### Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

- The solution of differential equation  $\frac{d^2 y}{dx^2} + y = 0$  is
  - $c_1 e^{-x} + c_2 e^x$
  - $c_1 \cos x + c_2 \sin x$
  - $(c_1 + c_2 x) \cos x + (c_3 + c_4 x) \sin x$
  - None of these.
- P.I. of the differential equation  $(D^2 + D + 1)y = e^x$  is
  - $\frac{1}{3} e^x$
  - $3 e^x$
  - $e^x$
  - None of these.

9. For the differential equation  $F(D) y = e^{ax}$ , if  $F(a) = 0$ , then

P.I. =  $\frac{1}{F(D)} e^{ax}$  is given by

- (a)  $\frac{1}{F(a)} e^{ax}$  (b)  $\frac{1}{F(-a)} e^{ax}$   
 (c)  $e^{ax} \frac{1}{F(D+a)} 1$  (d) None of these.

10. For the particular integral  $\frac{1}{f(D^2)} \sin ax$  when  $f(-a^2) = 0$ , which one of the

following is correct

- (a)  $\frac{1}{f(D^2)} \sin ax = \frac{1}{f(-a^2)} \sin ax$   
 (b)  $\frac{1}{D^2 + a^2} \sin ax = -\frac{x}{2a} \cos ax$   
 (c)  $\frac{1}{D^2 + a^2} \sin ax = \frac{x}{2a} \cos ax$   
 (d) None of these.

11. The solution of the differential equation  $\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = e^x$  is

- (a)  $y = c_1 e^x + c_2 e^{3x} + x$   
 (b)  $y = (c_1 + c_2) e^x - xe^x$   
 (c)  $y = c_1 e^x + c_2 e^{2x} - xe^x$   
 (d)  $y = c_1 x + c_2 e^x - e^{2x}$ .

12. The solution of the differential equation  $\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + y = 0$  is

- (a)  $y = (c_1 + c_2 x) e^{-x}$  (b)  $y = (c_1 - c_2 x) e^{-x}$   
 (c)  $(c_1 + c_2 x) e^x$  (d) None of these [UPTU 2011]

### True or False

Write “T” for true and “F” for false statement.

13. A linear differential equation is an equation in which the dependent variable and its derivatives appear only in the first degree.

[UPTU 2011]

14.  $\frac{d^4 y}{dx^4} - 5 \frac{d^3 y}{dx^3} - 2 \frac{dy}{dx} + 4xy = 0$  is a linear differential equation with constant coefficients.

15.  $e^{-x} (c_1 \cos \sqrt{3} x + c_2 \sin \sqrt{3} x) + c_3 e^{2x}$  is the solution of the differential equation  $\frac{d^3 y}{dx^3} - 8y = 0$ .

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## Answers

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- |                    |                                     |                            |
|--------------------|-------------------------------------|----------------------------|
| 1. C.F.            | 2. $y = c_1 \cos ax + c_2 \sin ax.$ | 3. $c_1 e^x + c_2 e^{4x}.$ |
| 4. $\frac{2}{5}x.$ | 5. $\frac{1}{F(D)}Q.$               | 6. Two                     |
| 8. (a).            | 9. (c).                             | 10. (b).                   |
| 12. (d)            | 13. $T.$                            | 15. $T.$                   |
7. (b).  
11. (c).



LOGARITHMS

|    | 0    | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 9    | 1 2 3            | 4 5 6                | 7 8 9                |
|----|------|------|------|------|------|------|------|------|------|------|------------------|----------------------|----------------------|
| 10 | 0000 | 0043 | 0086 | 0128 | 0170 | 0212 | 0253 | 0294 | 0334 | 0374 | 5 9 13<br>4 8 12 | 17 21 26<br>16 20 24 | 30 34 38<br>28 32 36 |
| 11 | 0414 | 0453 | 0492 | 0531 | 0569 | 0607 | 0645 | 0682 | 0719 | 0775 | 4 8 12<br>4 7 11 | 16 20 23<br>15 18 22 | 27 31 35<br>26 29 33 |
| 12 | 0792 | 0828 | 0864 | 0899 | 0934 | 0969 | 1004 | 1038 | 1072 | 1106 | 3 7 11<br>3 7 10 | 14 18 21<br>14 17 20 | 25 28 32<br>24 27 31 |
| 13 | 1139 | 1173 | 1206 | 1239 | 1271 | 1303 | 1335 | 1367 | 1399 | 1430 | 3 6 10<br>3 7 10 | 13 16 19<br>13 16 19 | 23 26 29<br>22 25 29 |
| 14 | 1461 | 1492 | 1523 | 1553 | 1584 | 1614 | 1644 | 1673 | 1703 | 1732 | 3 6 9<br>3 6 9   | 12 15 19<br>12 14 17 | 22 25 28<br>20 23 26 |
| 15 | 1761 | 1790 | 1818 | 1847 | 1875 | 1903 | 1932 | 1959 | 1987 | 2014 | 3 6 9<br>3 6 8   | 11 14 17<br>10 14 17 | 20 23 26<br>19 22 25 |
| 16 | 2041 | 2068 | 2095 | 2122 | 2148 | 2175 | 2201 | 2227 | 2253 | 2279 | 3 6 8<br>3 5 8   | 11 14 16<br>10 13 16 | 19 22 24<br>18 21 23 |
| 17 | 2304 | 2330 | 2355 | 2380 | 2405 | 2430 | 2455 | 2480 | 2504 | 2529 | 3 5 8<br>3 5 8   | 10 13 15<br>10 12 15 | 18 20 23<br>17 20 22 |
| 18 | 2553 | 2577 | 2601 | 2625 | 2648 | 2672 | 2695 | 2718 | 2742 | 2765 | 2 5 7<br>2 4 7   | 9 12 14<br>9 11 14   | 17 19 21<br>16 18 21 |
| 19 | 2788 | 2810 | 2833 | 2856 | 2878 | 2900 | 2923 | 2945 | 2967 | 2989 | 2 4 7<br>2 4 6   | 9 11 13<br>8 11 13   | 16 18 20<br>15 17 19 |
| 20 | 3010 | 3032 | 3054 | 3075 | 3096 | 3118 | 3139 | 3160 | 3181 | 3201 | 2 4 6            | 8 11 13              | 15 17 19             |
| 21 | 3222 | 3243 | 3263 | 3284 | 3304 | 3324 | 3345 | 3365 | 3385 | 3404 | 2 4 6            | 8 10 12              | 14 16 18             |
| 22 | 3424 | 3444 | 3464 | 3483 | 3502 | 3522 | 3541 | 3560 | 3579 | 3598 | 2 4 6            | 8 10 12              | 14 15 17             |
| 23 | 3617 | 3636 | 3655 | 3674 | 3692 | 3711 | 3729 | 3747 | 3766 | 3784 | 2 4 6            | 7 9 11               | 13 15 17             |
| 24 | 3802 | 3820 | 3838 | 3856 | 3874 | 3892 | 3909 | 3927 | 3945 | 3962 | 2 4 5            | 7 9 11               | 12 14 16             |
| 25 | 3979 | 3997 | 4014 | 4031 | 4048 | 4065 | 4082 | 4099 | 4116 | 4133 | 2 3 5            | 7 9 10               | 12 14 15             |
| 26 | 4150 | 4166 | 4183 | 4200 | 4216 | 4232 | 4249 | 4265 | 4281 | 4298 | 2 3 5            | 7 8 10               | 11 13 15             |
| 27 | 4314 | 4330 | 4346 | 4362 | 4378 | 4293 | 4409 | 4425 | 4440 | 4456 | 2 3 5            | 6 8 9                | 11 13 14             |
| 28 | 4472 | 4487 | 4502 | 4518 | 4533 | 4548 | 4564 | 4579 | 4594 | 4609 | 2 3 5            | 6 8 9                | 11 12 14             |
| 29 | 4624 | 4639 | 4654 | 4669 | 4683 | 4698 | 4713 | 4728 | 4742 | 4757 | 1 3 4            | 6 7 9                | 10 12 13             |
| 30 | 4771 | 4786 | 4800 | 4814 | 4829 | 4843 | 4857 | 4871 | 4886 | 4900 | 1 3 4            | 6 7 9                | 10 11 13             |
| 31 | 4914 | 4928 | 4942 | 4955 | 4969 | 4983 | 4997 | 5011 | 5024 | 5038 | 1 3 4            | 6 7 8                | 10 11 12             |
| 32 | 5051 | 5065 | 5079 | 5092 | 5105 | 5119 | 5132 | 5145 | 5159 | 5172 | 1 3 4            | 5 7 8                | 9 11 12              |
| 33 | 5185 | 5198 | 5211 | 5224 | 5237 | 5250 | 5263 | 5276 | 5289 | 5302 | 1 3 4            | 5 6 8                | 9 10 12              |
| 34 | 5315 | 5328 | 5340 | 5353 | 5366 | 5378 | 5391 | 5403 | 5416 | 5428 | 1 3 4            | 5 6 8                | 9 10 11              |
| 35 | 5441 | 5453 | 5456 | 5478 | 5490 | 5502 | 5514 | 5527 | 5539 | 5551 | 1 2 4            | 5 6 7                | 9 10 11              |
| 36 | 5563 | 5575 | 5587 | 5599 | 5611 | 5623 | 5635 | 5647 | 5658 | 5670 | 1 2 4            | 5 6 7                | 8 10 11              |
| 37 | 5682 | 5694 | 5705 | 5717 | 5729 | 5740 | 5752 | 5763 | 5775 | 5786 | 1 2 3            | 5 6 7                | 8 9 10               |
| 38 | 5798 | 5809 | 5821 | 5832 | 5843 | 5855 | 5866 | 5877 | 5888 | 5899 | 1 2 3            | 5 6 7                | 8 9 10               |
| 39 | 5911 | 5922 | 5933 | 5944 | 5955 | 5966 | 5977 | 5988 | 5999 | 6010 | 1 2 3            | 4 5 7                | 8 9 10               |
| 40 | 6021 | 6031 | 6042 | 6053 | 6064 | 6075 | 6085 | 6096 | 6107 | 6117 | 1 2 3            | 4 5 6                | 8 9 10               |
| 41 | 6128 | 6138 | 6149 | 6160 | 6170 | 6180 | 6191 | 6201 | 6212 | 6222 | 1 2 3            | 4 5 6                | 7 8 9                |
| 42 | 6232 | 6243 | 6253 | 6263 | 6274 | 6284 | 6294 | 6304 | 6314 | 6325 | 1 2 3            | 4 5 6                | 7 8 9                |
| 43 | 6335 | 6345 | 6355 | 6365 | 6375 | 6385 | 6395 | 6405 | 6415 | 6425 | 1 2 3            | 4 5 6                | 7 8 9                |
| 44 | 6435 | 6444 | 6454 | 6464 | 6474 | 6484 | 6493 | 6503 | 6513 | 6522 | 1 2 3            | 4 5 6                | 7 8 9                |
| 45 | 6532 | 6542 | 6551 | 6561 | 6571 | 6580 | 6590 | 6599 | 6609 | 6618 | 1 2 3            | 4 5 6                | 7 8 9                |
| 46 | 6628 | 6637 | 6646 | 6656 | 6665 | 6675 | 6684 | 6693 | 6702 | 6712 | 1 2 3            | 4 5 6                | 7 7 8                |
| 47 | 6721 | 6730 | 6739 | 6749 | 6758 | 6767 | 6776 | 6785 | 6794 | 6803 | 1 2 3            | 4 5 6                | 6 7 8                |
| 48 | 6812 | 6821 | 6830 | 6839 | 6848 | 6857 | 6866 | 6875 | 6884 | 6893 | 1 2 3            | 4 5 5                | 6 7 8                |
| 49 | 6902 | 6911 | 6920 | 6928 | 6937 | 6946 | 6955 | 6964 | 6972 | 6981 | 1 2 3            | 4 4 5                | 6 7 8                |



## LOGARITHMS

|    | 0    | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 9    | 1 2 3 | 4 5 6 | 7 8 9 |
|----|------|------|------|------|------|------|------|------|------|------|-------|-------|-------|
| 50 | 6990 | 6998 | 7007 | 7016 | 7024 | 7033 | 7042 | 7050 | 7059 | 7067 | 1 2 3 | 3 4 5 | 6 7 8 |
| 51 | 7076 | 7084 | 7093 | 7101 | 7110 | 7118 | 7126 | 7135 | 7143 | 7152 | 1 2 3 | 3 4 5 | 6 7 8 |
| 52 | 7160 | 7168 | 7177 | 7185 | 7193 | 7202 | 7210 | 7218 | 7226 | 7235 | 1 2 2 | 3 4 5 | 6 7 7 |
| 53 | 7243 | 7251 | 7259 | 7267 | 7275 | 7284 | 7292 | 7300 | 7308 | 7316 | 1 2 2 | 3 4 5 | 6 6 7 |
| 54 | 7324 | 7332 | 7340 | 7348 | 7356 | 7364 | 7372 | 7380 | 7388 | 7396 | 1 2 2 | 3 4 5 | 6 6 7 |
| 55 | 7404 | 7412 | 7419 | 7427 | 7435 | 7443 | 7451 | 7459 | 7466 | 7474 | 1 2 2 | 3 4 5 | 5 6 7 |
| 56 | 7482 | 7490 | 7497 | 7505 | 7513 | 7520 | 7528 | 7536 | 7543 | 7551 | 1 2 2 | 3 4 5 | 5 6 7 |
| 57 | 7559 | 7566 | 7574 | 7582 | 7589 | 7597 | 7604 | 7612 | 7619 | 7627 | 1 2 2 | 3 4 5 | 5 6 7 |
| 58 | 7634 | 7642 | 7649 | 7657 | 7664 | 7672 | 7679 | 7686 | 7694 | 7701 | 1 1 2 | 3 4 4 | 5 6 7 |
| 59 | 7709 | 7716 | 7723 | 7731 | 7738 | 7745 | 7752 | 7760 | 7767 | 7774 | 1 1 2 | 3 4 4 | 5 6 7 |
| 60 | 7782 | 7789 | 7796 | 7803 | 7810 | 7818 | 7825 | 7832 | 7839 | 7846 | 1 1 2 | 3 4 4 | 5 6 6 |
| 61 | 7853 | 7860 | 7868 | 7875 | 7882 | 7889 | 7896 | 7903 | 7910 | 7917 | 1 1 2 | 3 4 4 | 5 6 6 |
| 62 | 7924 | 7931 | 7938 | 7945 | 7952 | 7959 | 7966 | 7973 | 7980 | 7987 | 1 1 2 | 3 3 4 | 5 6 6 |
| 63 | 7993 | 8000 | 8007 | 8014 | 8021 | 8028 | 8035 | 8041 | 8048 | 8055 | 1 1 2 | 3 3 4 | 5 5 6 |
| 64 | 8062 | 8069 | 8075 | 8082 | 8089 | 8096 | 8102 | 8109 | 8116 | 8122 | 1 1 2 | 3 3 4 | 5 5 6 |
| 65 | 8129 | 8136 | 8142 | 8149 | 8156 | 8162 | 8269 | 8176 | 8182 | 8189 | 1 1 2 | 3 3 4 | 5 5 6 |
| 66 | 8195 | 8202 | 8209 | 8215 | 8222 | 8228 | 8235 | 8241 | 8248 | 8254 | 1 1 2 | 3 3 4 | 5 5 6 |
| 67 | 8261 | 8267 | 8274 | 8280 | 8287 | 8293 | 8299 | 8306 | 8312 | 8319 | 1 1 2 | 3 3 4 | 5 5 6 |
| 68 | 8325 | 8331 | 8338 | 8344 | 8351 | 8357 | 8363 | 8370 | 8376 | 8382 | 1 1 2 | 3 3 4 | 4 5 6 |
| 69 | 8388 | 8395 | 8401 | 8407 | 8414 | 8420 | 8426 | 8432 | 8439 | 8445 | 1 1 2 | 2 3 4 | 4 5 6 |
| 70 | 8451 | 8457 | 8463 | 8470 | 8476 | 8482 | 8488 | 8494 | 8500 | 8506 | 1 1 2 | 2 3 4 | 4 5 6 |
| 71 | 8513 | 8519 | 8525 | 8531 | 8537 | 8543 | 8549 | 8555 | 8561 | 8567 | 1 1 2 | 2 3 4 | 4 5 6 |
| 72 | 8573 | 8579 | 8585 | 8591 | 8597 | 8603 | 8609 | 8615 | 8621 | 8627 | 1 1 2 | 2 3 4 | 4 5 5 |
| 73 | 8633 | 8639 | 8645 | 8651 | 8657 | 8663 | 8669 | 8675 | 8681 | 8686 | 1 1 2 | 2 3 4 | 4 5 5 |
| 74 | 8692 | 8698 | 8704 | 8710 | 8716 | 8722 | 8727 | 8733 | 8739 | 8745 | 1 1 2 | 2 3 4 | 4 5 5 |
| 75 | 8751 | 8756 | 8762 | 8768 | 8774 | 8779 | 8785 | 8791 | 8797 | 8802 | 1 1 2 | 2 3 3 | 4 5 5 |
| 76 | 8808 | 8814 | 8820 | 8825 | 8831 | 8837 | 8842 | 8848 | 8854 | 8859 | 1 1 2 | 2 3 3 | 4 5 5 |
| 77 | 8865 | 8871 | 8876 | 8882 | 8887 | 8893 | 8899 | 8904 | 8910 | 8915 | 1 1 2 | 2 3 3 | 4 4 5 |
| 78 | 8921 | 8927 | 8932 | 8938 | 8943 | 8949 | 8954 | 8960 | 8965 | 8971 | 1 1 2 | 2 3 3 | 4 4 5 |
| 79 | 8976 | 8982 | 8987 | 8993 | 8998 | 9004 | 9009 | 9014 | 9020 | 9025 | 1 1 2 | 2 3 3 | 4 4 5 |
| 80 | 9031 | 9036 | 9042 | 9047 | 9053 | 9058 | 9063 | 9069 | 9074 | 9079 | 1 1 2 | 2 3 3 | 4 4 5 |
| 81 | 9085 | 9090 | 9096 | 9101 | 9106 | 9112 | 9117 | 9122 | 9128 | 9133 | 1 1 2 | 2 3 3 | 4 4 5 |
| 82 | 9138 | 9143 | 9149 | 9154 | 9159 | 9165 | 9170 | 9175 | 9180 | 9186 | 1 1 2 | 2 3 3 | 4 4 5 |
| 83 | 9191 | 9196 | 9201 | 9206 | 9212 | 9217 | 9222 | 9227 | 9232 | 9238 | 1 1 2 | 2 3 3 | 4 4 5 |
| 84 | 9243 | 9248 | 9253 | 9258 | 9263 | 9269 | 9274 | 9279 | 9284 | 9289 | 1 1 2 | 2 3 3 | 4 4 5 |
| 85 | 9294 | 9299 | 9304 | 9309 | 9315 | 9320 | 9325 | 9330 | 9335 | 9340 | 1 1 2 | 2 3 3 | 4 4 5 |
| 86 | 9345 | 9350 | 9355 | 9360 | 9365 | 9370 | 9375 | 9380 | 9385 | 9390 | 1 1 2 | 2 3 3 | 4 4 5 |
| 87 | 9395 | 9400 | 9405 | 9410 | 9415 | 9420 | 9425 | 9430 | 9435 | 9440 | 0 1 1 | 2 2 3 | 3 4 4 |
| 88 | 9445 | 9450 | 9455 | 9460 | 9465 | 9469 | 9474 | 9479 | 9484 | 9489 | 0 1 1 | 2 2 3 | 3 4 4 |
| 89 | 9494 | 9499 | 9504 | 9509 | 9513 | 9518 | 9523 | 9528 | 9533 | 9538 | 0 1 1 | 2 2 3 | 3 4 4 |
| 90 | 9542 | 9547 | 9552 | 9557 | 9562 | 9566 | 9571 | 9576 | 9581 | 9586 | 0 1 1 | 2 2 3 | 3 4 4 |
| 91 | 9590 | 9595 | 9600 | 9605 | 9609 | 9614 | 9619 | 9624 | 9628 | 9633 | 0 1 1 | 2 2 3 | 3 4 4 |
| 92 | 9638 | 9643 | 9647 | 9652 | 9657 | 9661 | 9666 | 9671 | 9675 | 9680 | 0 1 1 | 2 2 3 | 3 4 4 |
| 93 | 9685 | 9689 | 9694 | 9699 | 9703 | 9708 | 9713 | 9717 | 9722 | 9727 | 0 1 1 | 2 2 3 | 3 4 4 |
| 94 | 9731 | 9736 | 9741 | 9745 | 9750 | 9754 | 9759 | 9763 | 9768 | 9773 | 0 1 1 | 2 2 3 | 3 4 4 |
| 95 | 9777 | 9782 | 9786 | 9791 | 9795 | 9800 | 9805 | 9809 | 9814 | 9818 | 0 1 1 | 2 2 3 | 3 4 4 |
| 96 | 9823 | 9827 | 9832 | 9836 | 9841 | 9845 | 9850 | 9854 | 9859 | 9863 | 0 1 1 | 2 2 3 | 3 4 4 |
| 97 | 9868 | 9872 | 9877 | 9881 | 9886 | 9890 | 9894 | 9899 | 9903 | 9908 | 0 1 1 | 2 2 3 | 3 4 4 |
| 98 | 9912 | 9917 | 9921 | 9926 | 9930 | 9934 | 9939 | 9943 | 9948 | 9952 | 0 1 1 | 2 2 3 | 3 4 4 |
| 99 | 9956 | 9961 | 9965 | 9969 | 9974 | 9978 | 9983 | 9987 | 9991 | 9996 | 0 1 1 | 2 2 3 | 3 3 4 |

## ANTILOGARITHMS

|     | 0    | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 9    | 1 2 3 | 4 5 6 | 7 8 9 |
|-----|------|------|------|------|------|------|------|------|------|------|-------|-------|-------|
| .00 | 1000 | 1002 | 1005 | 1007 | 1009 | 1012 | 1014 | 1016 | 1019 | 1021 | 0 0 1 | 1 1 1 | 2 2 2 |
| .01 | 1023 | 1026 | 1028 | 1030 | 1033 | 1035 | 1038 | 1040 | 1042 | 1045 | 0 0 1 | 1 1 1 | 2 2 2 |
| .02 | 1047 | 1050 | 1052 | 1054 | 1057 | 1059 | 1062 | 1064 | 1067 | 1069 | 0 0 1 | 1 1 1 | 2 2 2 |
| .03 | 1072 | 1074 | 1076 | 1079 | 1081 | 1084 | 1086 | 1089 | 1091 | 1094 | 0 0 1 | 1 1 1 | 2 2 2 |
| .04 | 1096 | 1099 | 1102 | 1104 | 1107 | 1109 | 1112 | 1114 | 1117 | 1119 | 0 1 1 | 1 1 2 | 2 2 2 |
| .05 | 1122 | 1125 | 1127 | 1130 | 1132 | 1135 | 1138 | 1140 | 1143 | 1146 | 0 1 1 | 1 1 2 | 2 2 2 |
| .06 | 1148 | 1151 | 1153 | 1156 | 1159 | 1161 | 1164 | 1167 | 1169 | 1172 | 0 1 1 | 1 1 2 | 2 2 2 |
| .07 | 1175 | 1178 | 1180 | 1183 | 1186 | 1189 | 1191 | 1194 | 1197 | 1199 | 0 1 1 | 1 1 2 | 2 2 2 |
| .08 | 1202 | 1205 | 1208 | 1211 | 1213 | 1216 | 1219 | 1222 | 1225 | 1227 | 0 1 1 | 1 2 2 | 2 2 3 |
| .09 | 1230 | 1233 | 1236 | 1139 | 1242 | 1245 | 1247 | 1250 | 1253 | 1256 | 0 1 1 | 1 2 2 | 2 2 3 |
| .10 | 1259 | 1262 | 1265 | 1268 | 1271 | 1274 | 1276 | 1279 | 1282 | 1285 | 0 1 1 | 1 2 2 | 2 2 3 |
| .11 | 1288 | 1291 | 1294 | 1297 | 1300 | 1303 | 1306 | 1309 | 1312 | 1315 | 0 1 1 | 1 2 2 | 2 2 3 |
| .12 | 1318 | 1321 | 1324 | 1327 | 1330 | 1334 | 1337 | 1340 | 1343 | 1346 | 0 1 1 | 1 2 2 | 2 2 3 |
| .13 | 1349 | 1352 | 1355 | 1358 | 1361 | 1365 | 1368 | 1371 | 1374 | 1377 | 0 1 1 | 1 2 2 | 2 3 3 |
| .14 | 1380 | 1384 | 1387 | 1390 | 1393 | 1396 | 1400 | 1403 | 1406 | 1409 | 0 1 1 | 1 2 2 | 2 3 3 |
| .15 | 1413 | 1416 | 1419 | 1422 | 1426 | 1429 | 1432 | 1435 | 1439 | 1442 | 0 1 1 | 1 2 2 | 2 3 3 |
| .16 | 1445 | 1449 | 1452 | 1455 | 1459 | 1462 | 1466 | 1469 | 1472 | 1476 | 0 1 1 | 1 2 2 | 2 3 3 |
| .17 | 1479 | 1483 | 1486 | 1489 | 1493 | 1496 | 1500 | 1503 | 1507 | 1510 | 0 1 1 | 1 2 2 | 2 3 3 |
| .18 | 1514 | 1517 | 1521 | 1524 | 1528 | 1531 | 1535 | 1538 | 1542 | 1545 | 0 1 1 | 1 2 2 | 2 3 3 |
| .19 | 1549 | 1552 | 1556 | 1560 | 1563 | 1567 | 1570 | 1574 | 1578 | 1581 | 0 1 1 | 1 2 2 | 3 3 3 |
| .20 | 1585 | 1589 | 1592 | 1596 | 1600 | 1603 | 1607 | 1611 | 1614 | 1618 | 0 1 1 | 1 2 2 | 3 3 3 |
| .21 | 1622 | 1626 | 1629 | 1633 | 1637 | 1641 | 1644 | 1648 | 1652 | 1656 | 0 1 1 | 2 2 2 | 3 3 3 |
| .22 | 1660 | 1663 | 1667 | 1671 | 1675 | 1679 | 1683 | 1687 | 1690 | 1694 | 0 1 1 | 2 2 2 | 3 3 3 |
| .23 | 1698 | 1702 | 1706 | 1710 | 1714 | 1718 | 1722 | 1726 | 1730 | 1734 | 0 1 1 | 2 2 2 | 3 3 4 |
| .24 | 1738 | 1742 | 1746 | 1750 | 1754 | 1758 | 1762 | 1766 | 1770 | 1774 | 0 1 1 | 2 2 2 | 3 3 4 |
| .25 | 1778 | 1782 | 1786 | 1791 | 1795 | 1799 | 1803 | 1807 | 1811 | 1816 | 0 1 1 | 2 2 2 | 3 3 4 |
| .26 | 1820 | 1824 | 1828 | 1832 | 1837 | 1841 | 1845 | 1849 | 1854 | 1858 | 0 1 1 | 2 2 3 | 3 3 4 |
| .27 | 1862 | 1866 | 1871 | 1875 | 1879 | 1884 | 1888 | 1892 | 1897 | 1901 | 0 1 1 | 2 2 3 | 3 3 4 |
| .28 | 1905 | 1910 | 1914 | 1919 | 1923 | 1928 | 1932 | 1936 | 1941 | 1945 | 0 1 1 | 2 2 3 | 3 4 4 |
| .29 | 1950 | 1954 | 1959 | 1963 | 1968 | 1972 | 1977 | 1982 | 1986 | 1991 | 0 1 1 | 2 2 3 | 3 4 4 |
| .30 | 1995 | 2000 | 2004 | 2009 | 2014 | 2018 | 2023 | 2028 | 2032 | 2037 | 0 1 1 | 2 2 3 | 3 4 4 |
| .31 | 2042 | 2046 | 2051 | 2056 | 2061 | 2065 | 2070 | 2075 | 2080 | 2084 | 0 1 1 | 2 2 3 | 3 4 4 |
| .32 | 2089 | 2094 | 2099 | 2104 | 2109 | 2113 | 2118 | 2123 | 2128 | 2133 | 0 1 1 | 2 2 3 | 3 4 4 |
| .33 | 2138 | 2143 | 2148 | 2153 | 2158 | 2163 | 2168 | 2173 | 2178 | 2183 | 0 1 1 | 2 2 3 | 3 4 4 |
| .34 | 2188 | 2193 | 2198 | 2203 | 2208 | 2213 | 2218 | 2223 | 2228 | 2234 | 1 1 2 | 2 3 3 | 4 4 5 |
| .35 | 2239 | 2244 | 2249 | 2254 | 2259 | 2265 | 2270 | 2275 | 2280 | 2286 | 1 1 2 | 2 3 3 | 4 4 5 |
| .36 | 2291 | 2296 | 2301 | 2307 | 2312 | 2317 | 2323 | 2328 | 2333 | 2339 | 1 1 2 | 2 3 3 | 4 4 5 |
| .37 | 2344 | 2350 | 2355 | 2360 | 2366 | 2371 | 2377 | 2382 | 2388 | 2393 | 1 1 2 | 2 3 3 | 4 4 5 |
| .38 | 2399 | 2404 | 2410 | 2415 | 2421 | 2427 | 2432 | 2438 | 2443 | 2449 | 1 1 2 | 2 3 3 | 4 4 5 |
| .39 | 2455 | 2460 | 2466 | 2472 | 2477 | 2483 | 2489 | 2495 | 2500 | 2506 | 1 1 2 | 2 3 3 | 4 5 5 |
| .40 | 2512 | 2518 | 2523 | 2529 | 2535 | 2541 | 2547 | 2553 | 2559 | 2564 | 1 1 2 | 2 3 4 | 4 5 5 |
| .41 | 2570 | 2576 | 2582 | 2588 | 2594 | 2600 | 2606 | 2612 | 2618 | 2624 | 1 1 2 | 2 3 4 | 4 5 5 |
| .42 | 2630 | 2636 | 2642 | 2649 | 2655 | 2661 | 2667 | 2673 | 2679 | 2685 | 1 1 2 | 2 3 4 | 4 5 6 |
| .43 | 2692 | 2698 | 2704 | 2710 | 2716 | 2723 | 2729 | 2735 | 2742 | 2748 | 1 1 2 | 3 3 4 | 4 5 6 |
| .44 | 2754 | 2761 | 2767 | 2773 | 2780 | 2786 | 2793 | 2799 | 2805 | 2812 | 1 1 2 | 3 3 4 | 4 5 6 |
| .45 | 2818 | 2825 | 2831 | 2838 | 2844 | 2851 | 2858 | 2864 | 2871 | 2877 | 1 1 2 | 3 3 4 | 5 5 6 |
| .46 | 2884 | 2891 | 2897 | 2904 | 2911 | 2917 | 2924 | 2931 | 2938 | 2944 | 1 1 2 | 3 3 4 | 5 5 6 |
| .47 | 2951 | 2958 | 2965 | 2972 | 2979 | 2985 | 2992 | 2999 | 3006 | 3013 | 1 1 2 | 3 3 4 | 5 5 6 |
| .48 | 3020 | 3027 | 3034 | 3041 | 3048 | 3055 | 3062 | 3069 | 3076 | 3083 | 1 1 2 | 3 4 4 | 5 6 6 |
| .49 | 3090 | 3097 | 3105 | 3112 | 3119 | 3126 | 3133 | 3141 | 3148 | 3155 | 1 1 2 | 3 4 4 | 5 6 6 |

## ANTILOGARITHMS

|     | 0    | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 9    | 1 | 2 | 3 | 4 | 5  | 6  | 7  | 8  | 9  |
|-----|------|------|------|------|------|------|------|------|------|------|---|---|---|---|----|----|----|----|----|
| .50 | 3162 | 3170 | 3177 | 3184 | 3192 | 3199 | 3206 | 3214 | 3221 | 3228 | 1 | 1 | 2 | 3 | 4  | 5  | 5  | 6  | 7  |
| .51 | 3236 | 3243 | 3251 | 3258 | 3266 | 3273 | 3281 | 3289 | 3296 | 3304 | 1 | 2 | 2 | 3 | 4  | 5  | 5  | 6  | 7  |
| .52 | 3311 | 3319 | 3327 | 3334 | 3342 | 3350 | 3357 | 3365 | 3373 | 3381 | 1 | 2 | 2 | 3 | 4  | 5  | 5  | 6  | 7  |
| .53 | 3388 | 3396 | 3404 | 3412 | 3420 | 3428 | 3436 | 3443 | 3451 | 3459 | 1 | 2 | 2 | 3 | 4  | 5  | 6  | 6  | 7  |
| .54 | 3467 | 3475 | 3483 | 3491 | 3499 | 3508 | 3516 | 3524 | 3532 | 3540 | 1 | 2 | 2 | 3 | 4  | 5  | 6  | 6  | 7  |
| .55 | 3548 | 3556 | 3565 | 3573 | 3581 | 3589 | 3597 | 3606 | 3614 | 3622 | 1 | 2 | 2 | 3 | 4  | 5  | 6  | 7  | 7  |
| .56 | 3631 | 3639 | 3648 | 3656 | 3664 | 3673 | 3681 | 3690 | 3698 | 3707 | 1 | 2 | 3 | 3 | 4  | 5  | 6  | 7  | 8  |
| .57 | 3715 | 3724 | 3733 | 3741 | 3750 | 3758 | 3767 | 3776 | 3784 | 3793 | 1 | 2 | 3 | 3 | 4  | 5  | 6  | 7  | 8  |
| .58 | 3802 | 3811 | 3819 | 3828 | 3837 | 3846 | 3855 | 3864 | 3873 | 3882 | 1 | 2 | 3 | 4 | 4  | 5  | 6  | 7  | 8  |
| .59 | 3890 | 3899 | 3908 | 3917 | 3926 | 3936 | 3945 | 3954 | 3963 | 3972 | 1 | 2 | 3 | 4 | 5  | 5  | 6  | 7  | 8  |
| .60 | 3981 | 3990 | 3999 | 4009 | 4018 | 4027 | 4036 | 4046 | 4055 | 4064 | 1 | 2 | 3 | 4 | 5  | 6  | 6  | 7  | 8  |
| .61 | 4074 | 4083 | 4093 | 4102 | 4111 | 4121 | 4130 | 4140 | 4150 | 4159 | 1 | 2 | 3 | 4 | 5  | 6  | 7  | 8  | 9  |
| .62 | 4169 | 4178 | 4188 | 4198 | 4207 | 4217 | 4227 | 4236 | 4246 | 4256 | 1 | 2 | 3 | 4 | 5  | 6  | 7  | 8  | 9  |
| .63 | 4266 | 4276 | 4285 | 4295 | 4305 | 4315 | 4325 | 4335 | 4345 | 4355 | 1 | 2 | 3 | 4 | 5  | 6  | 7  | 8  | 9  |
| .64 | 4365 | 4375 | 4385 | 4395 | 4406 | 4416 | 4426 | 4436 | 4446 | 4457 | 1 | 2 | 3 | 4 | 5  | 6  | 7  | 8  | 9  |
| .65 | 4467 | 4477 | 4487 | 4498 | 4508 | 4519 | 4529 | 4539 | 4550 | 4560 | 1 | 2 | 3 | 4 | 5  | 6  | 7  | 8  | 9  |
| .66 | 4571 | 4581 | 4592 | 4603 | 4613 | 4624 | 4634 | 4645 | 4556 | 4667 | 1 | 2 | 3 | 4 | 5  | 6  | 7  | 9  | 10 |
| .67 | 4677 | 4688 | 4699 | 4710 | 4721 | 4732 | 4742 | 4753 | 4764 | 4675 | 1 | 2 | 3 | 4 | 5  | 7  | 8  | 9  | 10 |
| .68 | 4786 | 4797 | 4808 | 4819 | 4831 | 4842 | 4853 | 4864 | 4875 | 4887 | 1 | 2 | 3 | 4 | 6  | 7  | 8  | 9  | 10 |
| .69 | 4898 | 4909 | 4920 | 4932 | 4943 | 4955 | 4966 | 4977 | 4989 | 5000 | 1 | 2 | 3 | 5 | 6  | 7  | 8  | 9  | 10 |
| .70 | 5012 | 5023 | 5035 | 5047 | 5058 | 5070 | 5082 | 5093 | 5105 | 5117 | 1 | 2 | 4 | 5 | 6  | 7  | 8  | 9  | 11 |
| .71 | 5129 | 5140 | 5152 | 5164 | 5176 | 5188 | 5200 | 5212 | 5224 | 5236 | 1 | 2 | 4 | 5 | 6  | 7  | 8  | 10 | 11 |
| .72 | 5248 | 5260 | 5272 | 5284 | 5297 | 5309 | 5321 | 5333 | 5346 | 5358 | 1 | 2 | 4 | 5 | 6  | 7  | 9  | 10 | 11 |
| .73 | 5370 | 5383 | 5395 | 5408 | 5420 | 5433 | 5445 | 5458 | 5470 | 5483 | 1 | 3 | 4 | 5 | 6  | 8  | 9  | 10 | 11 |
| .74 | 5495 | 5508 | 5521 | 5534 | 5546 | 5559 | 5572 | 5585 | 5598 | 5610 | 1 | 3 | 4 | 5 | 6  | 8  | 9  | 10 | 12 |
| .75 | 5623 | 5636 | 5649 | 5662 | 5675 | 5689 | 5702 | 5715 | 5728 | 5741 | 1 | 3 | 4 | 5 | 7  | 8  | 9  | 10 | 12 |
| .76 | 5754 | 5768 | 5781 | 5794 | 5808 | 5821 | 5834 | 5848 | 5861 | 5875 | 1 | 3 | 4 | 5 | 7  | 8  | 9  | 11 | 12 |
| .77 | 5888 | 5902 | 5916 | 5929 | 5943 | 5957 | 5970 | 5984 | 5998 | 6012 | 1 | 3 | 4 | 5 | 7  | 8  | 10 | 11 | 12 |
| .78 | 6026 | 6039 | 6053 | 6067 | 6081 | 6095 | 6109 | 6124 | 6138 | 6152 | 1 | 3 | 4 | 6 | 7  | 8  | 10 | 11 | 13 |
| .79 | 6166 | 6180 | 6194 | 6209 | 6223 | 6237 | 6252 | 6266 | 6281 | 6295 | 1 | 3 | 4 | 6 | 7  | 9  | 10 | 11 | 13 |
| .80 | 6310 | 6324 | 6339 | 6353 | 6368 | 6383 | 6397 | 6412 | 6427 | 6442 | 1 | 3 | 4 | 6 | 7  | 9  | 10 | 12 | 13 |
| .81 | 6457 | 6471 | 6486 | 6501 | 6516 | 6531 | 6546 | 6561 | 6577 | 6592 | 2 | 3 | 5 | 6 | 8  | 9  | 11 | 12 | 14 |
| .82 | 6607 | 6622 | 6637 | 6653 | 6668 | 6683 | 6699 | 6714 | 6730 | 6745 | 2 | 3 | 5 | 6 | 8  | 9  | 11 | 12 | 14 |
| .83 | 6761 | 6776 | 6792 | 6803 | 6823 | 6839 | 6855 | 6871 | 6887 | 6902 | 2 | 3 | 5 | 6 | 8  | 9  | 11 | 13 | 14 |
| .84 | 6918 | 6934 | 6950 | 6966 | 6982 | 6998 | 7015 | 7031 | 7047 | 7063 | 2 | 3 | 5 | 6 | 8  | 10 | 11 | 13 | 15 |
| .85 | 7079 | 7096 | 7112 | 7129 | 7145 | 7161 | 7178 | 7194 | 7211 | 7228 | 2 | 3 | 5 | 7 | 8  | 10 | 12 | 13 | 15 |
| .86 | 7244 | 7261 | 7278 | 7295 | 7311 | 7328 | 7345 | 7362 | 7379 | 7396 | 2 | 3 | 5 | 7 | 8  | 10 | 12 | 13 | 15 |
| .87 | 7413 | 7430 | 7447 | 7464 | 7482 | 7499 | 7516 | 7534 | 7551 | 7568 | 2 | 3 | 5 | 7 | 9  | 10 | 12 | 14 | 16 |
| .88 | 7686 | 7603 | 7621 | 7638 | 7656 | 7674 | 7691 | 7709 | 7727 | 7745 | 2 | 4 | 5 | 7 | 9  | 11 | 12 | 14 | 16 |
| .89 | 7762 | 7780 | 7798 | 7816 | 7834 | 7852 | 7870 | 7889 | 7907 | 7925 | 2 | 4 | 5 | 7 | 9  | 11 | 13 | 14 | 16 |
| .90 | 7943 | 7962 | 7980 | 7998 | 8017 | 8035 | 8054 | 8072 | 8091 | 8110 | 2 | 4 | 6 | 7 | 9  | 11 | 13 | 15 | 17 |
| .91 | 8128 | 8147 | 8166 | 8185 | 8204 | 8222 | 8241 | 8260 | 8279 | 8299 | 2 | 4 | 6 | 8 | 9  | 11 | 13 | 15 | 17 |
| .92 | 8318 | 8337 | 8356 | 8375 | 8395 | 8414 | 8433 | 8453 | 8472 | 8492 | 2 | 4 | 6 | 8 | 10 | 12 | 14 | 15 | 17 |
| .93 | 8511 | 8531 | 8551 | 8570 | 8590 | 8610 | 8630 | 8650 | 8670 | 8690 | 2 | 4 | 6 | 8 | 10 | 12 | 14 | 16 | 18 |
| .94 | 8710 | 8730 | 8750 | 8770 | 8790 | 8810 | 8831 | 8851 | 8872 | 8892 | 2 | 4 | 6 | 8 | 10 | 12 | 14 | 16 | 18 |
| .95 | 8913 | 8933 | 8954 | 8974 | 8995 | 9016 | 9036 | 9057 | 9078 | 9099 | 2 | 4 | 6 | 8 | 10 | 12 | 15 | 17 | 19 |
| .96 | 9120 | 9141 | 9162 | 9183 | 9204 | 9226 | 9247 | 9268 | 9290 | 9311 | 2 | 4 | 6 | 8 | 11 | 13 | 15 | 17 | 19 |
| .97 | 9333 | 9354 | 9376 | 9397 | 9419 | 9441 | 9462 | 9484 | 9506 | 9528 | 2 | 4 | 7 | 9 | 11 | 13 | 15 | 17 | 20 |
| .98 | 9550 | 9572 | 9594 | 9616 | 9638 | 9661 | 9683 | 9705 | 9727 | 9750 | 2 | 4 | 7 | 9 | 11 | 13 | 16 | 18 | 20 |
| .99 | 9772 | 9795 | 9817 | 9840 | 9863 | 9886 | 9908 | 9931 | 9954 | 9977 | 2 | 5 | 7 | 9 | 11 | 13 | 16 | 18 | 20 |

## INTEGRATION

## § 1. Integration

The process inverse to differentiation is defined as **integration**.

Thus if  $\frac{d}{dx} F(x) = f(x)$ , we say that  $F(x)$  is an *integral* or a *primitive* of  $f(x)$ , and, in symbols, we write  $\int f(x) dx = F(x)$ . The letter  $x$  in  $dx$  denotes that the integration is to be performed with respect to the variable  $x$ .

The letter  $x$  in  $dx$  denotes that the integration is to be performed with respect to the variable  $x$ .

The process of determining an integral of a function is called **integration** and the function to be integrated is called **integrand**.

## § 2. Constant of Integration

As the differential coefficient of a constant is zero, if  $\frac{d}{dx} F(x) = f(x)$ ; we have

$$\frac{d}{dx} [F(x) + c] = f(x),$$

therefore  $\int f(x) dx = F(x) + c$ .

This constant  $c$  is called a *constant of integration* and can take any constant value. Also  $\int f(x) dx$  is called the **indefinite integral** of  $f(x)$  w.r.t. 'x'; for by giving different values to the constant of integration, the indefinite nature is preserved.

## § 3. Some Properties of Integral

(i)  $\frac{d}{dx} \left\{ \int f(x) dx \right\} = f(x).$

(ii) The integral of the product of a constant and a function is equal to the product of the constant and the integral of the function.

Thus if  $\lambda$  is a constant, then

$$\int \lambda f(x) dx = \lambda \int f(x) dx.$$

- (iii) The integral of a sum or difference of a finite number of functions is equal to the sum or difference of the integrals of the functions.

Symbolically,

$$\int [f_1(x) \pm f_2(x) \pm \dots \pm f_n(x)] dx = \int f_1(x) dx \pm \int f_2(x) dx \pm \dots \pm \int f_n(x) dx.$$

#### § 4. Standard Results.

- (i)  $\frac{d}{dx} \left( \frac{x^{n+1}}{n+1} \right) = x^n, n \neq -1 \quad \rightarrow \quad \int x^n dx = \frac{x^{n+1}}{n+1} + c,$
- (ii)  $\frac{d}{dx} (\log |x|) = \frac{1}{x}, x \neq 0 \quad \rightarrow \quad \int \frac{1}{x} dx = \log |x| + c,$
- (iii)  $\frac{d}{dx} (e^x) = e^x \quad \rightarrow \quad \int e^x dx = e^x + c,$
- (iv)  $\frac{d}{dx} \left( \frac{a^x}{\log a} \right) = a^x, a > 0, a \neq 1 \quad \rightarrow \quad \int a^x dx = \frac{a^x}{\log a} + c,$
- (v)  $\frac{d}{dx} (\sin x) = \cos x \quad \rightarrow \quad \int \cos x dx = \sin x + c,$
- (vi)  $\frac{d}{dx} (-\cos x) = \sin x \quad \rightarrow \quad \int \sin x dx = -\cos x + c,$
- (vii)  $\frac{d}{dx} (\tan x) = \sec^2 x \quad \rightarrow \quad \int \sec^2 x dx = \tan x + c,$
- (viii)  $\frac{d}{dx} (-\cot x) = \operatorname{cosec}^2 x \quad \rightarrow \quad \int \operatorname{cosec}^2 x dx = -\cot x + c,$
- (ix)  $\frac{d}{dx} (\sec x) = \sec x \tan x \quad \rightarrow \quad \int \sec x \tan x dx = \sec x + c,$
- (x)  $\frac{d}{dx} (-\operatorname{cosec} x) = \operatorname{cosec} x \cot x \quad \rightarrow \quad \int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + c,$
- (xi)  $\frac{d}{dx} (\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}} \quad \rightarrow \quad \int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} x + c,$

$$(xii) \frac{d}{dx} (\tan^{-1} x) = \frac{1}{1+x^2} \quad \rightarrow \quad \int \frac{1}{1+x^2} dx = \tan^{-1} x + c,$$

$$(xiii) \frac{d}{dx} (\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}} \quad \rightarrow \quad \int \frac{1}{x\sqrt{x^2-1}} dx = \sec^{-1} x + c.$$

## Illustrative Examples

**Illustration 1:** Evaluate :

$$(i) \int \left( 5x^3 + 2x^{-5} - 7x + \frac{1}{\sqrt{x}} + \frac{5}{x} \right) dx \quad (ii) \int \frac{x^3 + 5x^2 + 4x + 1}{x^2} dx$$

$$(iii) \int (3 \sin x - 2 \cos x + 4 \sec^2 x - 5 \operatorname{cosec}^2 x) dx$$

$$(iv) \int (1-x)(2+3x)(5-4x) dx.$$

**Solution :**

$$(i) \text{ Let } I = \int \left( 5x^3 + 2x^{-5} - 7x + \frac{1}{\sqrt{x}} + \frac{5}{x} \right) dx.$$

$$\text{Then } I = 5 \int x^3 dx + 2 \int x^{-5} dx - 7 \int x dx + \int x^{-1/2} dx + 5 \int \frac{1}{x} dx$$

$$= 5 \cdot \frac{x^4}{4} + 2 \cdot \frac{x^{-4}}{(-4)} - 7 \cdot \frac{x^2}{2} + \frac{x^{1/2}}{(1/2)} + 5 \log |x| + c$$

$$= \frac{5x^4}{4} - \frac{1}{2x^4} - \frac{7x^2}{2} + 2\sqrt{x} + 5 \log |x| + c.$$

$$(ii) \text{ Let } I = \int \frac{x^3 + 5x^2 + 4x + 1}{x^2} dx.$$

$$\text{Then } I = \int \left( x + 5 + \frac{4}{x} + \frac{1}{x^2} \right) dx, \text{ dividing each term by } x^2$$

$$= \int x dx + 5 \int dx + 4 \int \frac{1}{x} dx + \int x^{-2} dx$$

$$= \frac{x^2}{2} + 5x + 4 \log |x| - \frac{1}{x} + c.$$

$$(iii) \text{ Here } I = \int (3 \sin x - 2 \cos x + 4 \sec^2 x - 5 \operatorname{cosec}^2 x) dx$$

$$= 3 \int \sin x dx - 2 \int \cos x dx + 4 \int \sec^2 x dx - 5 \int \operatorname{cosec}^2 x dx$$

$$= -3 \cos x - 2 \sin x + 4 \tan x + 5 \cot x + c.$$

$$\begin{aligned}
 \text{(iv) Here } I &= \int (1-x)(2+3x)(5-4x) dx = \int (10-3x-19x^2+12x^3) dx \\
 &= 10 \int dx - 3 \int x dx - 19 \int x^2 dx + 12 \int x^3 dx \\
 &= 10x - 3 \cdot \frac{x^2}{2} - 19 \cdot \frac{x^3}{3} + 12 \cdot \frac{x^4}{4} + c = 10x - \frac{3x^2}{2} - \frac{19x^3}{3} \\
 &\quad + 3x^4 + c.
 \end{aligned}$$

**Illustration 2 :** Evaluate :

$$\begin{aligned}
 \text{(i) } \int \tan^2 x dx & \qquad \qquad \qquad \text{(ii) } \int \cot^2 x dx \\
 \text{(iii) } \int \frac{\cos x}{\sin^2 x} dx & \text{ [UPTU 2001(Special)]} \qquad \text{(iv) } \int (\tan x + \cot x)^2 dx \\
 \text{(v) } \int \frac{1}{\sin^2 x \cos^2 x} dx & \qquad \qquad \qquad \text{(vi) } \int \frac{\cos 2x}{\sin^2 x \cos^2 x} dx \\
 \text{(vii) } \int \frac{\sec x}{(\sec x + \tan x)} dx.
 \end{aligned}$$

**Solution :**

$$\begin{aligned}
 \text{(i) Here } I &= \int \tan^2 x dx = \int (\sec^2 x - 1) dx \\
 &= \int \sec^2 x dx - \int dx = \tan x - x + c. \\
 \text{(ii) Here } I &= \int \cot^2 x dx = \int (\operatorname{cosec}^2 x - 1) dx \\
 &= \int \operatorname{cosec}^2 x dx - \int dx = -\cot x - x + c. \\
 \text{(iii) Here } I &= \int \frac{\cos x}{\sin^2 x} dx = \int \left( \frac{\cos x}{\sin x} \cdot \frac{1}{\sin x} \right) dx = \int \cot x \operatorname{cosec} x dx \\
 &= -\operatorname{cosec} x + c. \\
 \text{(iv) Here } I &= \int (\tan x + \cot x)^2 dx \\
 &= \int (\tan^2 x + \cot^2 x + 2 \tan x \cdot \cot x) dx = \int (\tan^2 x + \cot^2 x + 2) dx \\
 &= \int \{(1 + \tan^2 x) + (1 + \cot^2 x)\} dx \\
 &= \int (\sec^2 x + \operatorname{cosec}^2 x) dx = \int \sec^2 x dx + \int \operatorname{cosec}^2 x dx \\
 &= \tan x - \cot x + c. \\
 \text{(v) Here } I &= \int \frac{1}{\sin^2 x \cos^2 x} dx = \int \left( \frac{\sin^2 x + \cos^2 x}{\sin^2 x \cos^2 x} \right) dx = \int \left( \frac{1}{\cos^2 x} + \frac{1}{\sin^2 x} \right) dx \\
 &= \int \sec^2 x dx + \int \operatorname{cosec}^2 x dx = \tan x - \cot x + c.
 \end{aligned}$$

$$\begin{aligned} \text{(vi) Here } I &= \int \frac{\cos 2x}{\sin^2 x \cos^2 x} dx = \int \frac{\cos^2 x - \sin^2 x}{\sin^2 x \cos^2 x} dx = \int \left( \frac{1}{\sin^2 x} - \frac{1}{\cos^2 x} \right) dx \\ &= \int \operatorname{cosec}^2 x dx - \int \sec^2 x dx = -\cot x - \tan x + c. \end{aligned}$$

$$\begin{aligned} \text{(vii) Here } I &= \int \frac{\sec x}{\sec x + \tan x} dx = \int \frac{\sec x (\sec x - \tan x)}{(\sec x + \tan x)(\sec x - \tan x)} dx \\ &= \int \frac{(\sec^2 x - \sec x \tan x)}{\sec^2 x - \tan^2 x} dx = \int (\sec^2 x - \sec x \tan x) dx \\ &= \int \sec^2 x dx - \int \sec x \tan x dx = \tan x - \sec x + c. \end{aligned}$$

**Illustration 3:** Evaluate :

$$(i) \int \sqrt{1 + \sin 2x} dx$$

$$(ii) \int \frac{dx}{1 + \sin x}.$$

**Solution :**

$$(i) \text{ Let } I = \int \sqrt{1 + \sin 2x} dx.$$

$$\begin{aligned} \text{Then } I &= \int \sqrt{\cos^2 x + \sin^2 x + 2 \sin x \cos x} dx = \int \sqrt{(\cos x + \sin x)^2} dx \\ &= \int (\cos x + \sin x) dx = \int \cos x dx + \int \sin x dx = \sin x - \cos x + c \end{aligned}$$

$$\begin{aligned} \text{(ii) Here } I &= \int \frac{dx}{1 + \sin x} = \int \frac{1}{(1 + \sin x)} \cdot \frac{(1 - \sin x)}{(1 - \sin x)} dx \\ &= \int \frac{1 - \sin x}{1 - \sin^2 x} dx = \int \frac{1 - \sin x}{\cos^2 x} dx = \int \left( \frac{1}{\cos^2 x} - \frac{\sin x}{\cos^2 x} \right) dx \\ &= \int (\sec^2 x - \sec x \tan x) dx \\ &= \int \sec^2 x dx - \int \sec x \tan x dx = \tan x - \sec x + c. \end{aligned}$$

## Comprehensive Exercise 1

**Integrate the following functions with respect to x :**

$$1. (i) x^4$$

$$(ii) x^{-7}$$

$$(iii) x^{5/3}$$

$$(iv) 2^x$$

$$(v) 3\sqrt{x^2}$$

$$(vi) \frac{1}{4\sqrt{x^3}}$$

$$(vii) 3^2 \log_3 x$$

$$(viii) \log_x x$$



$$(ix) \ x^3 + 5x^2 - 4 + \frac{7}{x} + \frac{2}{\sqrt{x}}$$

$$(x) \ 2^x + \frac{5}{x} - \frac{1}{x^{1/3}}.$$

$$2. (i) \ (1-x)\sqrt{x}$$

$$(ii) \ \sqrt{x}(ax^2 + bx + c)$$

$$(iii) \ \frac{x}{m} + \frac{m}{x} + x^m + m^x$$

$$(iv) \ \frac{a}{x^2} + \frac{b}{x} + c$$

$$(v) \ (2-5x)(3+2x)(1-x)$$

$$(vi) \ \frac{(1+x)^2}{\sqrt{x}}$$

$$(vii) \ \frac{x^2-1}{x^2+1}$$

$$(viii) \ \frac{x^6-1}{x^2+1}$$

$$3. (i) \ 3 \sin x - 4 \cos x + \frac{5}{\cos^2 x} - \frac{6}{\sin^2 x} + \tan^2 x - \cot^2 x$$

$$(ii) \ 1 + \frac{1}{1+x^2} - \frac{2}{\sqrt{1-x^2}} + \frac{5}{x\sqrt{x^2-1}} + 3^x$$

$$(iii) \ \sec x (\sec x + \tan x)$$

$$(iv) \ \frac{\cot x}{\sin x} - \tan^2 x - \frac{\tan x}{\cos x} + \frac{2}{\cos^2 x}$$

$$(v) \ \operatorname{cosec} x (\operatorname{cosec} x - \cot x)$$

$$(vi) \ \frac{4-5 \cos x}{\sin^2 x}$$

$$(vii) \ \frac{1+2 \sin x}{\cos^2 x}.$$

**Evaluate the following integrals :**

$$4. (i) \ \int \sqrt{1+\cos 2x} \, dx$$

$$(ii) \ \int \frac{1}{(1-\cos x)} \, dx$$

$$(iii) \ \int \frac{1}{(1+\cos 2x)} \, dx$$

$$(iv) \ \int \frac{\cos x}{1+\cos x} \, dx$$

$$(v) \ \int \frac{\sin x}{1-\sin x} \, dx$$

$$(vi) \ \int \frac{\tan x}{\sec x + \tan x} \, dx$$

$$(vii) \ \int \frac{\operatorname{cosec} x}{\operatorname{cosec} x - \cot x} \, dx$$

$$(viii) \ \int \frac{\cot x}{\operatorname{cosec} x - \cot x} \, dx.$$

## Answers

$$1. (i) \ \frac{x^5}{5} + c. \quad (ii) \ -\frac{1}{6x^6} + c. \quad (iii) \ \frac{3}{8}x^{8/3} + c. \quad (iv) \ \frac{2^x}{\log 2} + c.$$

$$(v) \ \frac{3}{5}x^{5/3} + c. \quad (vi) \ 4x^{1/4} + c. \quad (vii) \ \frac{x^3}{3} + c. \quad (viii) \ x + c.$$

$$(ix) \ \frac{x^4}{4} + \frac{5}{3}x^3 - 4x + 7 \log|x| + 4\sqrt{x} + c.$$

$$(x) \frac{2^x}{\log 2} + 5 \log |x| - \frac{3}{2} x^{2/3} + c.$$

$$2. \quad (i) \frac{2}{3} x^{3/2} - \frac{2}{5} x^{5/2} + c. \quad (ii) \frac{2ax^{7/2}}{7} + \frac{2bx^{5/2}}{5} + \frac{2cx^{3/2}}{3} + k.$$

$$(iii) -\frac{a}{x} + b \log |x| + cx + k. \quad (iv) \frac{x^2}{2m} + m \log |x| + \frac{x^{m+1}}{m+1} + \frac{m^x}{\log m} + c.$$

$$(v) 6x - \frac{17x^2}{2} + \frac{x^3}{3} + \frac{5x^4}{2} + c. \quad (vi) 2\sqrt{x} + \frac{4}{3} x^{3/2} + \frac{2}{5} x^{5/2} + c.$$

$$(vii) x - 2 \tan^{-1} x + c. \quad (viii) \frac{x^5}{5} - \frac{x^3}{3} + x - 2 \tan^{-1} x + c.$$

$$3. \quad (i) -3 \cos x - 4 \sin x + 6 \tan x + 7 \cot x + c.$$

$$(ii) x + \tan^{-1} x - 2 \sin^{-1} x + 5 \sec^{-1} x + \frac{3^x}{\log 3} + c.$$

$$(iii) \tan x + \sec x + c.$$

$$(iv) -\operatorname{cosec} x + \tan x + x - \sec x + c.$$

$$(v) -\cot x + \operatorname{cosec} x + c.$$

$$(vi) -4 \cot x + 5 \operatorname{cosec} x + c.$$

$$(vii) \tan x + 2 \sec x + c.$$

$$4. \quad (i) \sqrt{2} \sin x + c.$$

$$(ii) -\cot x - \operatorname{cosec} x + c.$$

$$(iii) \frac{1}{2} \tan x + c.$$

$$(iv) x + \cot x - \operatorname{cosec} x + c.$$

$$(v) \sec x + \tan x - x + c.$$

$$(vi) \sec x - \tan x + x + c.$$

$$(vii) -\cot x - \operatorname{cosec} x + c.$$

$$(viii) -\operatorname{cosec} x - \cot x - x + c.$$

### § 5. Extended Forms of Fundamental Formulae.

$$(i) \int (ax + b)^n dx = \frac{1}{a} \cdot \frac{(ax + b)^{n+1}}{n+1} + c, \quad n \neq -1.$$

$$(ii) \int \frac{1}{ax + b} dx = \frac{1}{a} \cdot \log |ax + b| + c.$$

$$(iii) \int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + c.$$

$$(iv) \int a^{bx+c} dx = \frac{1}{b} \cdot \frac{a^{bx+c}}{\log a} + k, \quad a > 0 \text{ and } a \neq 1.$$

$$(v) \int \sin(ax + b) dx = -\frac{1}{a} \cos(ax + b) + c.$$

$$(vi) \int \cos(ax + b) dx = \frac{1}{a} \sin(ax + b) + c.$$

$$(vii) \int \sec^2(ax + b) dx = \frac{1}{a} \tan(ax + b) + c.$$

$$(viii) \int \operatorname{cosec}^2(ax+b) dx = -\frac{1}{a} \cot(ax+b) + c.$$

$$(ix) \int \sec(ax+b) \tan(ax+b) dx = \frac{1}{a} \sec(ax+b) + c.$$

$$(x) \int \operatorname{cosec}(ax+b) \cot(ax+b) dx = -\frac{1}{a} \operatorname{cosec}(ax+b) + c.$$

## § 6. Methods of Integration.

- (i) Integration by substitution,
- (ii) Integration by parts,
- (iii) Integration by decomposition into sum,
- (iv) Integration by successive reduction.

## § 7. Integration By Substitution.

A change in the variable of integration often reduces an integral to one of the fundamental integrals. The method in which we change the variable to some other variable is called the **Method of substitution**.

**Rule to remember :**

To evaluate  $\int f\{\phi(x)\} \cdot \phi'(x) dx$ , put  $\phi(x) = t$  and  $\phi'(x) dx = dt$ ,

where  $\phi'(x)$  is the differential coefficient of  $\phi(x)$  w.r.t.  $x$ .

**Important.** The success of the method of substitution depends on choosing the substitution  $x = \phi(t)$  so that the new integrand  $f\{\phi(t)\} \cdot \phi'(t)$  is of a form whose integral is known. This is done by guess rather than in according with some rule. However, try to put that expression of  $x$  equal to  $t$  whose differential coefficient is multiplied with  $dx$ .

**Three important forms of integrals.**

$$1. \int \frac{f'(x)}{f(x)} dx = \log |f(x)| + c.$$

**Remember :** The integral of a fraction whose numerator is the exact derivative of its denominator is equal to the logarithm of modulus of its denominator.

For example  $\int \frac{4x^3}{1+x^4} dx = \log |1+x^4| + c$ , as in this case numerator is the exact derivative of the denominator.

**Integrals of  $\tan x$ ,  $\cot x$ ,  $\sec x$  and  $\operatorname{cosec} x$ .**

$$(i) \int \tan x dx = \int \frac{\sin x}{\cos x} dx = - \int \frac{(-\sin x)}{\cos x} dx, \text{ adjusting the numerator as}$$

the exact diff. coeffi. of the denominator

$$= -\log |\cos x| + c = \log |\sec x| + c.$$

[UPTU 2002]

$$(ii) \text{ Similarly, } \int \cot x dx = \int \frac{\cos x}{\sin x} dx = \log |\sin x| + c.$$

$$\begin{aligned}
 \text{(iii)} \quad \int \operatorname{cosec} x \, dx &= \int \frac{dx}{\sin x} = \int \frac{dx}{2 \sin \frac{1}{2} x \cos \frac{1}{2} x} = \int \frac{\sec^2 \frac{1}{2} x \, dx}{2 \tan \frac{1}{2} x}, \\
 &\quad \left[ \text{dividing Nr. and Dr. by } \cos^2 \frac{1}{2} x \right] \\
 &= \log \left| \tan \frac{1}{2} x \right| + c. \\
 &\quad \left[ \dots \frac{1}{2} \sec^2 \frac{1}{2} x \text{ is the diff. coeffi. of } \tan \frac{1}{2} x \right]
 \end{aligned}$$

$$2. \quad \int [f(x)]^n f'(x) \, dx = \frac{[f(x)]^{n+1}}{n+1} + c, \quad \text{when } n \neq -1.$$

**Remember :** If the integrand consists of the product of a constant power of a function  $f(x)$  and the derivative  $f'(x)$  of  $f(x)$ , to obtain the integral we increase the index. This is known as **Power formula**.

$$3. \quad \int f'(ax+b) \, dx = \frac{f(ax+b)}{a} + c.$$

**Note.** To evaluate integrals of the type

$$\int \frac{dx}{a+b \cos x}, \int \frac{dx}{a+b \sin x}, \int \frac{dx}{a \sin x + b \cos x}, \int \frac{dx}{a \sin x + b \cos x + c};$$

we write  $\sin x = \frac{2 \tan(x/2)}{1 + \tan^2(x/2)}$  and  $\cos x = \frac{1 - \tan^2(x/2)}{1 + \tan^2(x/2)}$

In numerator replace  $1 + \tan^2 \frac{x}{2}$  by  $\sec^2 \frac{x}{2}$ .

Now put  $\tan \frac{x}{2} = t$ , so that  $\sec^2 \frac{x}{2} dx = 2 dt$ .

## Illustrative Examples

**Illustration 1:** Evaluate :

$$\begin{aligned}
 \text{(i)} \quad & \int \frac{\tan^{-1} x}{1+x^2} dx & \text{(ii)} \quad & \int \frac{e^{\tan^{-1} x}}{1+x^2} dx & \text{(iii)} \quad & \int \frac{\sin \sqrt{x}}{\sqrt{x}} dx \\
 \text{(iv)} \quad & \int \frac{\log x}{x} dx & \text{(v)} \quad & \int \frac{\sec^2(\log x)}{x} dx & \text{(vi)} \quad & \int \frac{e^x - \sin x}{e^x + \cos x} dx \quad [\text{UPTU 2001}]
 \end{aligned}$$

**Solution:**

(i) Put  $\tan^{-1} x = t$ , so that  $\frac{1}{1+x^2} dx = dt$ .

$$\therefore \int \frac{\tan^{-1} x}{1+x^2} dx = \int t \, dt = \frac{t^2}{2} + c = \frac{(\tan^{-1} x)^2}{2} + c.$$

(ii) Put  $\tan^{-1} x = t$ , so that  $\frac{1}{1+x^2} dx = dt$ .

$$\therefore \int \frac{e^{\tan^{-1} x}}{1+x^2} dx = \int e^t dt = e^t + c = e^{\tan^{-1} x} + c.$$

(iii) Put  $\sqrt{x} = t$ , so that  $\frac{1}{2} x^{-1/2} dx = dt$  or  $\frac{1}{\sqrt{x}} dx = 2 dt$ .

$$\therefore \int \frac{\sin \sqrt{x}}{\sqrt{x}} dx = \int 2 \sin t dt = 2(-\cos t) + c = -2 \cos \sqrt{x} + c.$$

(iv) Put  $\log x = t$ , so that  $\frac{1}{x} dx = dt$ .

$$\therefore \int \frac{\log x}{x} dx = \int t dt = \frac{1}{2} t^2 + c = \frac{1}{2} (\log x)^2 + c.$$

(v) Put  $\log x = t$ , so that  $\frac{1}{x} dx = dt$ .

$$\therefore \int \frac{\sec^2 (\log x)}{x} dx = \int \sec^2 t dt = \tan t + c = \tan (\log x) + c.$$

(vi) Put  $e^x + \cos x = t$ , so that  $(e^x - \sin x) dx = dt$ .

$$\therefore \int \frac{e^x - \sin x}{e^x + \cos x} dx = \int \frac{1}{t} dt = \log t + c = \log (e^x + \cos x) + c.$$

**Illustration 2:** Evaluate :

(i)  $\int \frac{x^2 dx}{1+x^6}$  [UPTU 2001]

(ii)  $\int \frac{x^8}{(1-x^3)^{1/3}} dx$

(iii)  $\int \frac{dx}{x \sqrt{x^6 - 1}}$

(iv)  $\int \frac{(1 + \log x)^2}{x} dx$  [UPTU 2002]

(v)  $\int \frac{\tan x}{\log \sec x} dx.$  [UPTU 2003]

**Solution:**

(i) Put  $x^3 = t$ , so that  $3x^2 dx = dt$ .

$$\therefore \frac{1}{3} \int \frac{3x^2 dx}{1+x^6} = \frac{1}{3} \int \frac{dt}{1+t^2} = \frac{1}{3} \tan^{-1} t + c = \frac{1}{3} \tan^{-1} x^3 + c.$$

(ii) Put  $(1 - x^3) = t$ , so that  $x^3 = 1 - t$  and  $3x^2 dx = -dt$ .

$$\begin{aligned} \therefore \int \frac{x^8}{(1-x^3)^{1/3}} dx &= \int \frac{x^6 \cdot x^2}{(1-x^3)^{1/3}} dx = -\frac{1}{3} \int \frac{(1-t)^2 dt}{t^{1/3}} \\ &= -\frac{1}{3} \int \frac{(1-2t+t^2)}{t^{1/3}} dt = -\frac{1}{3} \int t^{-1/3} dt + \frac{2}{3} \int t^{2/3} dt - \frac{1}{3} \int t^{5/3} dt \\ &= -\frac{1}{2} t^{2/3} + \frac{2}{5} t^{5/3} - \frac{1}{8} t^{8/3} + c \end{aligned}$$

$$= -\frac{1}{2}(1-x^3)^{2/3} + \frac{2}{5}(1-x^3)^{5/3} - \frac{1}{8}(1-x^3)^{8/3} + c.$$

(iii) Put  $x^3 = t$ , so that  $3x^2 dx = dt$  or  $x^2 dx = \frac{1}{3} dt$ .

$$\begin{aligned} \therefore \int \frac{dx}{x\sqrt{x^6-1}} &= \int \frac{x^2 dx}{x^3\sqrt{x^6-1}} \quad [\text{Multiplying Nr. and Dr. by } x^2] \\ &= \frac{1}{3} \int \frac{dt}{t\sqrt{t^2-1}} = \frac{1}{3} \sec^{-1} t + c = \frac{1}{3} \sec^{-1} x^3 + c. \end{aligned}$$

(iv) Put  $1 + \log x = t$ , so that  $\frac{1}{x} dx = dt$

$$\therefore \int \frac{(1 + \log x)^2 dx}{x} = \int t^2 dt = \frac{t^3}{3} + c = \frac{1}{3}(1 + \log x)^3 + c.$$

(v) Put  $\log \sec x = t$ , so that  $\frac{1}{\sec x} \sec x \tan x dx = dt$  or  $\tan x dx = dt$

$$\therefore \int \frac{\tan x}{\log \sec x} dx = \int \frac{1}{t} dt = \log t + c = \log (\log \sec x) + c.$$

**Illustration 3:** Evaluate :

$$(i) \int \cos^5 x dx$$

$$(ii) \int \sin^7 x dx$$

$$(iii) \int \sqrt{\sin x} \cos x dx$$

$$(iv) \int \sec^4 x \tan x dx.$$

$$(v) \int \frac{dx}{(5 + 4 \cos x)} \quad [\text{UPTU 2001}]$$

$$(vi) \int \frac{dx}{a^2 \cos^2 x + b^2 \sin^2 x} \quad [\text{UPTU 2008}]$$

**Solution:**

$$(i) \text{ Let } I = \int \cos^5 x dx = \int \cos^4 x \cdot \cos x dx = \int (1 - \sin^2 x)^2 \cos x dx.$$

Put  $\sin x = t$ , so that  $\cos x dx = dt$ .

$$\begin{aligned} \therefore I &= \int (1 - t^2)^2 dt = \int (1 - 2t^2 + t^4) dt = t - 2\frac{t^3}{3} + \frac{t^5}{5} + c \\ &= \sin x - \frac{2}{3} \sin^3 x + \frac{1}{5} \sin^5 x + c. \end{aligned}$$

$$(ii) \text{ Let } I = \int \sin^7 x dx = \int \sin^6 x \cdot \sin x dx = \int (1 - \cos^2 x)^3 \sin x dx.$$

Put  $\cos x = t$ , so that  $(-\sin x) dx = dt$  or  $\sin x dx = -dt$ .

$$\begin{aligned}\therefore I &= -\int (1-t^2)^3 dt = \int (t^6 - 3t^4 + 3t^2 - 1) dt \\ &= \frac{t^7}{7} - 3\frac{t^5}{5} + t^3 - t + c = \frac{\cos^7 x}{7} - \frac{3}{5}\cos^5 x + \cos^3 x - \cos x + c\end{aligned}$$

(iii) Let  $I = \int \sqrt{\sin x} \cos x dx$ .

Put  $\sin x = t$ , so that  $\cos x dx = dt$ .

$$\therefore I = \int \sqrt{t} dt = \frac{2}{3} t^{3/2} + c = \frac{2}{3} (\sin x)^{3/2} + c.$$

(iv) Let  $I = \int \sec^4 x \tan x dx = \int \sec^2 x \cdot \sec^2 x \cdot \tan x dx$   
 $= \int (1 + \tan^2 x) \sec^2 x \tan x dx.$

Put  $\tan x = t$ , so that  $\sec^2 x dx = dt$ .

$$\therefore I = \int (1+t^2) t dt = \int (t + t^3) dt = \frac{t^2}{2} + \frac{t^4}{4} + c = \frac{1}{2} \tan^2 x + \frac{1}{4} \tan^4 x + c.$$

(v) Let  $I = \int \frac{dx}{5 + 4 \cos x}.$

Writing  $\cos x = \frac{1 - \tan^2 (x/2)}{1 + \tan^2 (x/2)}$ , we have

$$\begin{aligned}I &= \int \frac{dx}{5 + 4 \left\{ \frac{1 - \tan^2 (x/2)}{1 + \tan^2 (x/2)} \right\}} = \int \frac{1 + \tan^2 (x/2)}{9 + \tan^2 (x/2)} dx = \int \frac{\sec^2 (x/2)}{9 + \tan^2 (x/2)} dx \\ &= \int \frac{2 dt}{9 + t^2}, \quad \text{putting } \tan \frac{1}{2} x = t, \text{ so that } \sec^2 \frac{x}{2} dx = 2 dt\end{aligned}$$

(vi) Here  $I = \int \frac{dx}{a^2 \cos^2 x + b^2 \sin^2 x} = \int \frac{\sec^2 x dx}{a^2 + b^2 \tan^2 x}$

dividing the numerator and the denominator by  $\cos^2 x$ . Put  $\tan x = t$ , so that  $\sec^2 x dx = dt$ .

$$\begin{aligned}\therefore I &= \int \frac{dt}{a^2 + b^2 t^2} = \frac{1}{b^2} \int \frac{dt}{(a^2/b^2) + t^2} = \frac{1}{b^2} \int \frac{dt}{(a/b)^2 + t^2} \\ &= \frac{1}{b^2} \cdot \frac{1}{a/b} \tan^{-1} \frac{t}{a/b} + c = \frac{1}{ab} \tan^{-1} \left( \frac{tb}{a} \right) + c \\ &= \frac{1}{ab} \tan^{-1} \left( \frac{b \tan x}{a} \right) + c.\end{aligned}$$


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**Illustration 4:** Evaluate :

(i)  $\int \frac{\sin(x-a)}{\sin x} dx$

(ii)  $\int \frac{\sin x}{\sin(x-a)} dx.$

**[UPTU 2007]****Solution:**

$$\begin{aligned}
 \text{(i) Here } I &= \int \frac{\sin(x-a)}{\sin x} dx = \int \frac{\sin x \cos a - \cos x \sin a}{\sin x} dx \\
 &= \int \cos a \, dx - \int \sin a \cot x \, dx = \cos a \int dx - \sin a \int \cot x \, dx \\
 &= x \cos a - \sin a \cdot \log |\sin x| + c.
 \end{aligned}$$

(ii) Put  $(x-a) = t$ , so that  $x = t+a$  and  $dx = dt$ .

$$\begin{aligned}
 \therefore \int \frac{\sin x}{\sin(x-a)} dx &= \int \frac{\sin(t+a)}{\sin t} dt = \int \frac{\sin t \cos a + \cos t \sin a}{\sin t} dt \\
 &= \cos a \int dt + \sin a \int \cot t \, dt = \cos a \cdot t + \sin a \cdot \log |\sin t| + c \\
 &= (x-a) \cos a + \sin a \cdot \log |\sin(x-a)| + c.
 \end{aligned}$$

## Comprehensive Exercise 2

**Evaluate the following integrals :**

1. (i)  $\int x^3 \cos x^4 \, dx$

(ii)  $\int \sin(ax+b) \cos(ax+b) \, dx$

(iii)  $\int \frac{\sin x}{(3+4 \cos x)^2} dx$

(iv)  $\int \frac{\sin(2 \tan^{-1} x)}{1+x^2} dx$

(v)  $\int \frac{\sin^{-1} x}{\sqrt{1-x^2}} dx$

(vi)  $\int \frac{x \tan^{-1} x^2}{1+x^4} dx.$

2. (i)  $\int e^{\tan x} \sec^2 x \, dx$

(ii)  $\int \frac{e^x}{(1+e^{2x})} dx$

(iii)  $\int \frac{\cos \sqrt{x}}{\sqrt{x}} dx$

(iv)  $\int \frac{\cos(\log x)}{x} dx$

(v)  $\int \frac{e^{m \tan^{-1} x}}{1+x^2} dx$

(vi)  $\int \frac{1}{x^2} e^{-1/x} dx.$



3. (i)  $\int \frac{\sin x}{1 + \cos x} dx$

(ii)  $\int \frac{4x - 5}{2x^2 - 5x + 1} dx$

(iii)  $\int \frac{\cot x}{\log \sin x} dx$

(iv)  $\int (4x + 2) \sqrt{x^2 + x + 1} dx$

(v)  $\int \frac{3x^2}{(1 + x^6)} dx$

(vi)  $\int \sin^3 x \cos x dx.$

## Answers

1. (i)  $\frac{1}{4} \sin x^4 + c$

(ii)  $\frac{\sin^2 (ax + b)}{2a} + c$

(iii)  $\frac{1}{4(3 + 4 \cos x)} + c$

(iv)  $-\frac{1}{2} \cos (2 \tan^{-1} x) + c$

(v)  $\frac{1}{2} (\sin^{-1} x)^2 + c$

(vi)  $\frac{1}{4} (\tan^{-1} x^2)^2 + c.$

2. (i)  $e^{\tan x} + c$

(ii)  $\tan^{-1} (e^x) + c$

(iii)  $2 \sin \sqrt{x} + c$

(iv)  $\sin (\log x) + c$

(v)  $\frac{1}{m} e^{m \tan^{-1} x} + c$

(vi)  $e^{-1/x} + c.$

3. (i)  $-\log |1 + \cos x| + c$

(ii)  $\log |2x^2 - 5x + 1| + c$

(iii)  $\log |\log \sin x| + c$

(iv)  $\frac{4}{3} (x^2 + x + 1)^{3/2} + c$

(v)  $\tan^{-1} x^3 + c$

(vi)  $\frac{\sin^4 x}{4} + c.$

### § 8. Integral of the Product of two Functions.

**Integration by Parts.** Let  $f_1(x)$  and  $f_2(x)$  be two functions of  $x$ . Then

$$\int f_1(x) f_2(x) dx = f_1(x) \cdot \int f_2(x) dx - \int \left[ \left\{ \frac{d}{dx} f_1(x) \right\} \cdot \int f_2(x) dx \right] dx$$

*i.e.*, the integral of the product of two functions

= first function  $\times$  integral of second function

– integral of {diff. coeffi. of first function

$\times$  integral of second function}.

**Note 1.** Care must be taken in choosing the first function and the second function. Obviously we must take that function as the second function whose integral is well known to us. Thus, to evaluate  $\int x \log x \, dx$  we shall take  $x$  as the second function because we so far do not know the integral of  $\log x$ . But to evaluate  $\int x \sin x \, dx$  we must take  $\sin x$  as the second function and  $x$  as the first function. Here, if we take  $x$  as the second function, then the new integral will become more complicated. Thus, to evaluate integrals of the type  $\int x^2 e^x \, dx$ ,  $\int x^3 \cos x \, dx$  etc., the function of the type  $x^n$  must be taken as the first function. In certain cases we can take unity (*i.e.*, 1) as the second function. Thus, to evaluate  $\int \log x \, dx$  we shall take 1 as the second function. To evaluate  $\int e^x \sin x \, dx$  we can take either  $e^x$  or  $\sin x$  as the second function.

**Note 2.** The formula of integration by parts can be applied more than once if necessary.

## Illustrative Examples

**Illustration 1:** Evaluate :

$$(i) \int x \sin 2x \, dx$$

$$(ii) \int x^2 e^x \, dx$$

$$(iii) \int \log x \, dx \quad [\text{UPTU 2009}]$$

$$(iv) \int \sin^{-1} x \, dx.$$

**Solution:**

(i) Integrating by parts, taking  $x$  as the first function, we have

$$\begin{aligned} \int x \sin 2x \, dx &= x \int \sin 2x \, dx - \int \left\{ \frac{d}{dx}(x) \cdot \int \sin 2x \, dx \right\} dx \\ &= x \left( \frac{-\cos 2x}{2} \right) - \int 1 \cdot \left( \frac{-\cos 2x}{2} \right) dx = -\frac{x \cos 2x}{2} + \frac{1}{2} \int \cos 2x \, dx \\ &= -\frac{x \cos 2x}{2} + \frac{1}{2} \cdot \frac{\sin 2x}{2} + c = -\frac{x \cos 2x}{2} + \frac{1}{4} \sin 2x + c. \end{aligned}$$

(ii) Integrating by parts taking  $e^x$  as the second function, we have

$$\begin{aligned} \int x^2 e^x \, dx &= x^2 \int e^x \, dx - \left[ \frac{d}{dx}(x^2) \cdot \int e^x \, dx \right] dx \\ &= x^2 e^x - \int 2x \cdot e^x \, dx = x^2 e^x - 2 \int x e^x \, dx \\ &= x^2 e^x - 2 \left\{ x \int e^x \, dx - \int \left[ \frac{d}{dx}(x) \cdot \int e^x \, dx \right] dx \right\} \end{aligned}$$

[Again integrating by parts]

$$= x^2 e^x - 2 \{x e^x - \int 1 \cdot e^x dx\} = x^2 e^x - 2 \{x e^x - e^x\} + c.$$

(iii) As there is only one function here, unity should be taken as the second function.

$$\begin{aligned} \text{We have } \int \log x \, dx &= \int (\log x) \cdot 1 \, dx \\ &= (\log x) \cdot x - \int \left\{ \frac{1}{x} \cdot x \right\} dx = x \log x - \int dx \\ &= x \log x - x + c = x (\log x - 1) + c. \end{aligned}$$

(iv) As there is only one function here, unity should be taken as the 2nd function.

$$\text{We have } \int \sin^{-1} x \, dx = \int (\sin^{-1} x) \cdot 1 \, dx.$$

Integrating by parts regarding 1 as the second function, we have

$$\begin{aligned} \int (\sin^{-1} x) \cdot 1 \, dx &= (\sin^{-1} x) \cdot \int 1 \, dx - \int \left\{ \frac{d}{dx} (\sin^{-1} x) \cdot \int 1 \, dx \right\} dx \\ &= (\sin^{-1} x) \cdot x - \int \frac{1}{\sqrt{1-x^2}} \cdot x \, dx \\ &= x \sin^{-1} x + \frac{1}{2} \int (1-x^2)^{-1/2} (-2x) \, dx \quad [\text{Note}] \\ &= x \sin^{-1} x + \frac{1}{2} \frac{(1-x^2)^{1/2}}{1/2} + c \quad [\text{By power formula}] \\ &= x \sin^{-1} x + \sqrt{1-x^2} + c. \end{aligned}$$

**Illustration 2:** Evaluate :

$$(i) \int x^2 \sin^{-1} x \, dx \quad (ii) \int x^3 \sin x^2 \, dx \quad [\text{UPTU 2004}]$$

$$(iii) \int \frac{x^2 \tan^{-1} x}{1+x^2} \, dx. \quad [\text{UPTU 2004}] \quad (iv) \int -e^x \sin x \, dx \quad [\text{UPTU 2008}]$$

**Solution:**

(i) Integrating by parts taking  $x^2$  as the second function, we have

$$\begin{aligned} \int x^2 \sin^{-1} x \, dx &= (\sin^{-1} x) \cdot \frac{x^3}{3} - \int \frac{1}{\sqrt{1-x^2}} \cdot \frac{x^3}{3} \, dx \\ &= \frac{x^3 \sin^{-1} x}{3} - \frac{1}{3} \int \frac{x^3}{\sqrt{1-x^2}} \, dx = \frac{x^3 \sin^{-1} x}{3} - \frac{1}{3} \int \frac{x \cdot x^2}{\sqrt{1-x^2}} \, dx \\ &= \frac{x^3 \sin^{-1} x}{3} + \frac{1}{3} \int \frac{t(1-t^2)}{t} \, dt, \text{ where } (1-x^2) = t^2 \cdot 3 \\ &= \frac{x^3 \sin^{-1} x}{3} + \frac{1}{3} \left[ \int dt - \int t^2 \, dt \right] = \frac{x^3 \sin^{-1} x}{3} + \frac{1}{3} \left[ t - \frac{t^3}{3} \right] + c \end{aligned}$$

$$= \frac{x^3 \sin^{-1} x}{3} + \frac{1}{3} \sqrt{1-x^2} - \frac{1}{9} (1-x^2)^{3/2} + c.$$

(ii) We have  $\int x^3 \sin x^2 dx = \int x^2 \cdot x \sin x^2 dx = \int x^2 \sin x^2 x dx$ .

Put  $x^2 = t$  so that  $2x dx = dt$  or  $x dx = \frac{1}{2} dt$ .

$$\begin{aligned} \therefore \int x^2 \sin x^2 x dx &= \frac{1}{2} \int t \sin t dt = \frac{1}{2} [-t \cos t + \int \cos t dt] \\ &= \frac{1}{2} [-t \cos t + \sin t] + c = \frac{1}{2} [-x^2 \cos x^2 + \sin x^2] + c. \end{aligned}$$

(iii) Put  $\tan^{-1} x = t$  so that  $x = \tan t$ . Also  $dx = \sec^2 t dt$ .

$$\begin{aligned} \therefore \int \frac{x^2 \tan^{-1} x}{1+x^2} dx &= \int \frac{\tan^2 t \cdot t \cdot \sec^2 t dt}{1+\tan^2 t} \\ &= \int t (\sec^2 t - 1) dt = \int t \sec^2 t dt - \int t dt \\ &= t \int \sec^2 t dt - \int \left[ \frac{d}{dt} t \int \sec^2 t dt \right] dt - \frac{t^2}{2} + c \\ &= t \tan t - \int \tan t dt - \frac{t^2}{2} + c = t \tan t - \log |\sec t| - \frac{t^2}{2} + c \\ &= t \tan t - \log |\sqrt{1+\tan^2 t}| - \frac{t^2}{2} + c \\ &= x \tan^{-1} x - \log |\sqrt{1+x^2}| - \frac{(\tan^{-1} x)^2}{2} + c. \end{aligned}$$

(iv) We have  $\int -e^x \sin x dx = -[e^x (-\cos x) - \int e^x (-\cos x) dx]$ ,

integrating by parts taking  $\sin x$  as the second function

$$\begin{aligned} &= +e^x \cos x - \int e^x \cos x dx \\ &= e^x \cos x - [e^x \sin x - \int e^x \sin x dx], \end{aligned}$$

again integrating by parts taking  
cos x as the second function

$$= e^x \cos x - e^x \sin x + \int e^x \sin x dx$$

Transposing the last term on the right hand side to the left and dividing by 2, we get

$$\int -e^x \sin x dx = \frac{1}{2} e^x (\cos x - \sin x) + c.$$

## Comprehensive Exercise 3

Evaluate the following integrals :

1.  $\int x \cos 2x dx$ .

2.  $\int x e^x dx$ .

3.  $\int x \sin 3x \, dx.$

5.  $\int x \sin^2 x \, dx.$

7.  $\int x^2 \cos x \, dx.$

9.  $\int \operatorname{cosec}^3 x \, dx.$

11.  $\int \sec^{-1} x \, dx.$

13.  $\int \frac{x \sin^{-1} x}{\sqrt{1-x^2}} \, dx. \quad [\text{UPTU 2006}]$

4.  $\int \frac{x}{\sin^2 x} \, dx.$

6.  $\int x \cos^3 x \, dx.$

8.  $\int x^2 e^{3x} \, dx.$

10.  $\int \log (1+x^2) \, dx.$

12.  $\int \tan^{-1} x \, dx$

14.  $\int \frac{x \tan^{-1} x}{(1+x^2)^{3/2}} \, dx.$

## Answers

1.  $\frac{1}{2} x \sin 2x + \frac{1}{4} \cos 2x + c.$

2.  $(x-1)e^x + c.$

3.  $\frac{-x \cos 3x}{3} + \frac{\sin 3x}{9} + c.$

4.  $-x \cot x + \log \infty \sin x \infty + c.$

5.  $\frac{x^2}{4} - \frac{x \sin 2x}{4} - \frac{\cos 2x}{8} + c.$

6.  $\frac{1}{12} x \sin 3x + \frac{1}{36} \cos 3x + \frac{3}{4} x \sin x + \frac{3}{4} \cos x + c.$

7.  $x^2 \sin x + 2x \cos x - 2 \sin x + c.$

8.  $\left( \frac{x^2}{3} - \frac{2x}{9} + \frac{2}{27} \right) e^{3x} + c.$

9.  $-\frac{1}{2} \operatorname{cosec} x \cot x + \frac{1}{2} \log \left| \tan \frac{x}{2} \right| + c.$

10.  $x \log (1+x^2) - 2x + 2 \tan^{-1} x + c.$

11.  $x \sec^{-1} x - \log |x + \sqrt{x^2 - 1}| + c.$

12.  $x \tan^{-1} x - \frac{1}{2} \log |1+x^2| + c.$

13.  $-(\sin^{-1} x) \sqrt{1-x^2} + x + c.$

14.  $\frac{x - \tan^{-1} x}{\sqrt{1+x^2}} + c.$

§ 9. Integration by Parts as Applied to the Functions of the type  $e^x [f(x) + f'(x)]$ .

$$\text{Let } I = \int e^x [f(x) + f'(x)] \, dx = \int e^x f(x) \, dx + \int e^x f'(x) \, dx.$$

Integrating the first integral by parts regarding  $e^x$  as the 2nd function, we have

$$\begin{aligned} I &= f(x) \cdot e^x - \int f'(x) \cdot e^x \, dx + \int e^x f'(x) \, dx \\ &= e^x f(x) + c. \end{aligned}$$

[Note that we have left the other integral unchanged because the last two integrals cancel each other.]

§ 10. Integrals of  $e^{ax} \sin bx$  and  $e^{ax} \cos bx$ .

$$\int e^{ax} \sin bx \, dx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx) + c$$

and 
$$\int e^{ax} \cos bx \, dx = \frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx) + c.$$

Alternative forms of  $\int e^{ax} \sin bx \, dx$  and  $\int e^{ax} \cos bx \, dx$ .

$$\int e^{ax} \sin bx \, dx = \frac{e^{ax}}{\sqrt{a^2 + b^2}} \sin \left( bx - \tan^{-1} \frac{b}{a} \right) + c$$

and 
$$\int e^{ax} \cos bx \, dx = \frac{e^{ax}}{\sqrt{a^2 + b^2}} \cos \left( bx - \tan^{-1} \frac{b}{a} \right) + c.$$

## Illustrative Examples

**Illustration 1:** Evaluate :

(i)  $\int e^x \left( \frac{1}{x} - \frac{1}{x^2} \right) dx$

(ii)  $\int e^x \frac{x}{(1+x)^2} dx$  [UPTU 2007]

(iii)  $\int e^x (\cot x + \log \sin x) dx$

(iv)  $\int [\sin (\log x) + \cos (\log x)] dx.$

**Solution:**

(i) We have  $\int e^x \left( \frac{1}{x} - \frac{1}{x^2} \right) dx$

$$= \int e^x \{ f(x) + f'(x) \} dx, \text{ where } f(x) = \frac{1}{x}$$

$$= \int e^x f(x) dx + \int e^x f'(x) dx = e^x f(x) - \int e^x f'(x) dx + \int e^x f'(x) dx,$$

applying integration by parts to the

first integral taking  $e^x$  as the second function

$$= e^x f(x) + c = \frac{1}{x} e^x + c.$$

(ii) We have  $\int e^x \frac{x}{(x+1)^2} dx = \int e^x \frac{(x+1) - 1}{(x+1)^2} dx = \int e^x \left[ \frac{1}{x+1} - \frac{1}{(x+1)^2} \right] dx$

$$= \int e^x [f(x) + f'(x)] dx, \text{ where } f(x) = \frac{1}{x+1}$$

$$= \int e^x f(x) dx + \int e^x f'(x) dx$$

$$= \int e^x f(x) dx - \int e^x f'(x) dx + \int e^x f'(x) dx,$$

applying integration by parts to the

first integral taking  $e^x$  as the second function

$$= e^x f(x) + c = e^x \frac{1}{x+1} + c.$$

(iii) We have  $\int e^x (\cot x + \log \sin x) dx$

$$= \int e^x (\log \sin x + \cot x) dx = e^x [f(x) + f'(x)] dx,$$

where  $f(x) = \log \sin x$

so that  $f'(x) = (1/\sin x) \cos x = \cot x$

$$= e^x f(x) + c = e^x \log \sin x + c.$$

(iv) Let  $I = \int [\sin(\log x) + \cos(\log x)] dx$ .

Put  $\log x = t$  or  $x = e^t$  so that  $dx = e^t dt$ .

$$\therefore I = \int e^t [\sin t + \cos t] dt = \int e^t [f(t) + f'(t)] dt,$$

where  $f(t) = \sin t$  so that  $f'(t) = \cos t$

$$= e^t f(t) + c = e^{\log x} \sin(\log x) + c = x \sin(\log x) + c.$$

**Illustration 2:** Evaluate :

(i)  $\int e^x \cos^2 x dx$

(ii)  $\int \sin(\log x) dx$

(iii)  $\int e^{2x} (-\sin x + 2 \cos x) dx$ . [UPTU 2004]

**Solution:**

(i) We have  $\int e^x \cos^2 x dx = \int e^x \left( \frac{1 + \cos 2x}{2} \right) dx$

$$= \frac{1}{2} \int e^x dx + \frac{1}{2} \int e^x \cos 2x dx = \frac{1}{2} e^x + \frac{1}{2} I,$$

$$\text{where } I = \int e^x \cos 2x dx.$$

Integrating by parts taking  $\cos 2x$  as the second function, we get

$$\begin{aligned} I &= e^x \left( \frac{\sin 2x}{2} \right) - \int e^x \cdot \frac{\sin 2x}{2} dx \\ &= \frac{1}{2} e^x \sin 2x - \frac{1}{2} \int e^x \sin 2x dx \\ &= \frac{1}{2} e^x \sin 2x - \frac{1}{2} \left[ e^x \left( -\frac{\cos 2x}{2} \right) - \int e^x \left( -\frac{\cos 2x}{2} \right) dx \right], \end{aligned}$$

[Again integrating by parts taking  $\sin 2x$  as the 2nd function]

$$\begin{aligned} &= \frac{1}{2} e^x \sin 2x + \frac{1}{4} e^x \cos 2x - \frac{1}{4} \int e^x \cos 2x dx \\ &= \frac{1}{2} e^x \sin 2x + \frac{1}{4} e^x \cos 2x - \frac{1}{4} I \end{aligned}$$

$$\text{or } I + \frac{1}{4} I = \frac{1}{2} e^x \sin 2x + \frac{1}{4} e^x \cos 2x \quad \text{or } I = \frac{1}{5} e^x (2 \sin 2x + \cos 2x) + c.$$

$$\text{Thus, } \int e^x \cos^2 x dx = \frac{1}{2} e^x + \frac{1}{10} e^x (2 \sin 2x + \cos 2x) + c.$$

(ii) Let  $I = \int \sin (\log x) dx$ .

Put  $\log x = t$  or  $x = e^t$  so that  $dx = e^t dt$ .

Then  $I = \int e^t \sin t dt = e^t (-\cos t) - \int e^t (-\cos t) dt$ ,

[Integrating by parts taking  $\sin t$  as the second function]

$$= -e^t \cos t + \int e^t \cos t dt$$

$$= -e^t \cos t + e^t \sin t - \int e^t \sin t dt,$$

[Again integrating by parts taking  $\cos t$  as the second function]

$$= -e^t \cos t + e^t \sin t - I$$

or  $2I = -e^t \cos t + e^t \sin t$  or  $I = \frac{1}{2} e^t (\sin t - \cos t) + c$ .

Thus,  $\int \sin (\log x) dx = \frac{x}{2} \{\sin (\log x) - \cos (\log x)\} + c$ .

(iii) We have  $\int e^{2x} (-\sin x + 2 \cos x) dx = \int (-e^{2x} \sin x + 2e^{2x} \cos x) dx$   
 $= \frac{-e^{2x}}{2^2 + 1^2} (2 \sin x - \cos x) + \frac{2e^{2x}}{2^2 + 1^2} (2 \cos x + \sin x) + c$

$$\left[ \begin{aligned} \because \int e^{ax} \sin bx dx &= \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx) + c \\ \text{and } e^{ax} \cos bx dx &= \frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx) + c \end{aligned} \right]$$

$$= \frac{e^{2x}}{5} [-2 \sin x + \cos x + 4 \cos x + 2 \sin x] + c = e^{2x} \cos x + c.$$

## Comprehensive Exercise 4

Evaluate the following integrals :

1.  $\int e^x (\sin x + \cos x) dx$ .

2.  $\int e^x (\tan x + \log \sec x) dx$ .

3.  $\int e^x (\cot x - \operatorname{cosec}^2 x) dx$ .

4.  $\int e^x \left( \frac{1}{x^2} - \frac{2}{x^3} \right) dx$ .

5.  $\int e^x \left\{ \frac{2-x}{(1-x)^2} \right\} dx$ .

6.  $\int e^x \left( \log x + \frac{1}{x^2} \right) dx$ .

7.  $\int e^{3x} \sin 4x dx$ .

8.  $\int e^{-x} \cos x dx$ .

9.  $\int e^{ax} \sin (bx + c) dx$ .

10.  $\int e^{2x} \cos (3x + 4) dx$ .

## Answers

1.  $e^x \sin x + c$ .

2.  $e^x \log (\sec x) + c$ .



3.  $e^x \cot x + c.$

4.  $\frac{e^x}{x^2} + c.$

5.  $\frac{e^x}{1-x} + c.$

6.  $e^x \left( \log x - \frac{1}{x} \right) + c.$

7.  $\frac{e^{3x}}{5} \sin \left[ 4x - \tan^{-1} \frac{4}{3} \right] + c.$

8.  $\frac{e^{-x}}{2} (-\cos x + \sin x) + c.$

9.  $\frac{e^{ax}}{\sqrt{a^2 + b^2}} \sin \left\{ bx + c - \tan^{-1} \frac{b}{a} \right\} + k.$

10.  $\frac{e^{2x}}{13} \{2 \cos (3x + 4) + 3 \sin (3x + 4)\} + c.$

**§ 11. Three Special Integrals.**

$$\int \frac{dx}{a^2 + x^2} ; \int \frac{dx}{x^2 - a^2}, (x > a) ; \int \frac{dx}{a^2 - x^2}, (x < a).$$

(i) To evaluate  $\int \frac{dx}{a^2 + x^2}$ . **[UPTU 2003]**

Put  $x = a \tan \theta$ ; so that  $dx = a \sec^2 \theta d\theta$ .

Also  $a^2 + x^2 = a^2 + a^2 \tan^2 \theta = a^2 (1 + \tan^2 \theta) = a^2 \sec^2 \theta$ .

$$\therefore \int \frac{dx}{a^2 + x^2} = \int \frac{a \sec^2 \theta d\theta}{a^2 \sec^2 \theta} = \frac{1}{a} \int d\theta = \frac{1}{a} \theta + c = \frac{1}{a} \tan^{-1} \frac{x}{a} + c.$$

Thus,  $\int \frac{dx}{a^2 + x^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} + c.$

(ii) To evaluate  $\int \frac{dx}{x^2 - a^2}, (x > a).$

We know that  $\frac{1}{x^2 - a^2} = \frac{1}{(x-a)(x+a)}$

$$= \frac{1}{2a} \left[ \frac{(x+a) - (x-a)}{(x-a)(x+a)} \right] = \frac{1}{2a} \left[ \frac{1}{(x-a)} - \frac{1}{(x+a)} \right].$$

$$\begin{aligned} \therefore \int \frac{dx}{x^2 - a^2} &= \int \frac{1}{2a} \left[ \frac{1}{(x-a)} - \frac{1}{(x+a)} \right] dx \\ &= \frac{1}{2a} \left[ \int \frac{1}{(x-a)} dx - \int \frac{1}{(x+a)} dx \right] \\ &= \frac{1}{2a} [\log|x-a| - \log|x+a|] + c = \frac{1}{2a} \log \left| \frac{x-a}{x+a} \right| + c \end{aligned}$$

Thus,  $\int \frac{1}{x^2 - a^2} dx = \frac{1}{2a} \log \left| \frac{x - a}{x + a} \right| + c, x > a.$

(iii) To evaluate  $\int \frac{dx}{a^2 - x^2}, (x < a).$

We know that

$$\frac{1}{a^2 - x^2} = \frac{1}{(a - x)(a + x)} = \frac{1}{2a} \left[ \frac{(a + x) + (a - x)}{(a - x)(a + x)} \right] = \frac{1}{2a} \left[ \frac{1}{(a - x)} + \frac{1}{(a + x)} \right].$$

$$\begin{aligned} \therefore \int \frac{dx}{a^2 - x^2} &= \int \frac{1}{2a} \left[ \frac{1}{(a - x)} + \frac{1}{(a + x)} \right] dx \\ &= \frac{1}{2a} \left[ \int \frac{1}{(a - x)} dx + \int \frac{1}{(a + x)} dx \right] \\ &= \frac{1}{2a} [-\log |a - x| + \log |a + x|] + c = \frac{1}{2a} \log \left| \frac{a + x}{a - x} \right| + c \end{aligned}$$

Thus,  $\int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \log \left| \frac{a + x}{a - x} \right| + c, x < a.$

## § 12. Integration by Partial Fractions.

The following table gives an idea what kind of partial fractions are to be taken for what kind of factors in the denominator :

| S.No. | Factor in the denominator | Form of the partial fraction                                      |
|-------|---------------------------|-------------------------------------------------------------------|
| (i)   | $(x - a)(x - b)$          | $\frac{A}{(x - a)} + \frac{B}{(x - b)}$                           |
| (ii)  | $(x - a)^2(x - b)$        | $\frac{A}{(x - a)} + \frac{B}{(x - a)^2} + \frac{C}{(x - b)}$     |
| (iii) | $(x - a)^3$               | $\frac{A}{(x - a)} + \frac{B}{(x - a)^2} + \frac{C}{(x - a)^3}$   |
| (iv)  | $(ax^2 + bx + c)$         | $\frac{Ax + B}{ax^2 + bx + c}$                                    |
| (v)   | $(ax^2 + bx + c)^2$       | $\frac{Ax + B}{ax^2 + bx + c} + \frac{Cx + D}{(ax^2 + bx + c)^2}$ |

**Note.** There are as many constants to be determined as the degree of the denominator.

## Illustrative Examples

**Illustration 1:** Resolve  $\frac{(x - 1)}{(x - 3)(x - 2)}$  into partial fractions.

**Solution:** Let  $\frac{x-1}{(x-3)(x-2)} = \frac{A}{(x-3)} + \frac{B}{(x-2)} = \frac{A(x-2) + B(x-3)}{(x-3)(x-2)}$ .

Clearly  $x-1 \equiv A(x-2) + B(x-3)$ . ... (1)

Comparing the coefficients of  $x$  and the constant terms on both sides of (1), we get

$$1 = A + B \quad \dots (2)$$

and  $-1 = -2A - 3B \quad \dots (3)$

Solving (2) and (3), we get  $A = 2, B = -1$ .

$$\therefore \frac{(x-1)}{(x-3)(x-2)} = \frac{2}{(x-3)} - \frac{1}{(x-2)}.$$

**Important Note.** An easy way to find the constants  $A$  and  $B$  etc. corresponding to linear non-repeated factors is like this : The factor below  $A$  is  $(x-3)$ . The equation  $x-3=0$  gives  $x=3$ . Now suppress  $(x-3)$  in the given fraction  $\frac{(x-1)}{(x-3)(x-2)}$  and put  $x=3$  in the remaining fraction  $\frac{(x-1)}{(x-2)}$  to get  $A$ . Thus,  $A = \frac{3-1}{3-2} = 2$ .

Similarly  $B = \frac{2-1}{2-3} = -1$ .

**Illustration 2:** Evaluate :

(i)  $\int \frac{dx}{x+x^2}$  [UPTU 2005]

(ii)  $\int \frac{dx}{x-x^3}$  [UPTU 2004]

**Solution:**

(i) Here  $\frac{1}{x+x^2} = \frac{1}{x(1+x)} = \frac{1}{x} - \frac{1}{1+x}$ , resolving into partial fractions

$$\therefore \int \frac{dx}{x+x^2} = \int \left( \frac{1}{x} - \frac{1}{1+x} \right) dx = \int \frac{1}{x} dx - \int \frac{1}{1+x} dx$$

$$= \log |x| - \log |1+x| + c = \log \left| \frac{x}{1+x} \right| + c.$$

(ii) Here  $\frac{1}{x-x^3} = \frac{1}{x(1-x^2)} = \frac{1}{x(1-x)(1+x)} \equiv \frac{A}{x} + \frac{B}{(1-x)} + \frac{C}{(1+x)}$ , (say).

To find  $A$  suppress  $x$  in the given fraction and put  $x=0$  in the remaining fraction.

$$\therefore A = \frac{1}{(1-0)(1+0)} = 1.$$

To find  $B$  suppress  $(1-x)$  in the given fraction and put  $x=1$  in the remaining fraction.

$$\therefore B = \frac{1}{1(1+1)} = \frac{1}{2}. \quad \text{Similarly, } C = \frac{1}{2}.$$

$$\text{Thus, } \frac{1}{x-x^3} = \frac{1}{x} + \frac{1}{2(1-x)} + \frac{1}{2(1+x)}.$$

$$\begin{aligned} \therefore \int \frac{dx}{(x-x^3)} &= \int \frac{1}{x} dx + \frac{1}{2} \int \frac{1}{(1-x)} dx + \frac{1}{2} \int \frac{1}{(1+x)} dx \\ &= \log x - \frac{1}{2} \log(1-x) + \frac{1}{2} \log(1+x) + c \\ &= \log x + \frac{1}{2} \log \frac{(1+x)}{(1-x)} + c = \frac{1}{2} \log \frac{x^2(1+x)}{(1-x)} + c. \end{aligned}$$

**Illustration 3:** Evaluate  $\int \frac{x^3}{(x-1)(x-2)(x-3)} dx$ . [UPTU 2006]

**Solution:** Here since the numerator is not of a lower degree than the denominator, we divide the numerator by the denominator till the remainder is of lesser degree than the denominator. We orally see that the quotient is 1.

We need not find out the actual value of the remainder because ultimately we have to break the fraction into partial fractions. Note that the denominators of the partial fractions depend only upon the denominator of the given fraction. So let

$$\frac{x^3}{(x-1)(x-2)(x-3)} \equiv 1 + \frac{A}{(x-1)} + \frac{B}{(x-2)} + \frac{C}{(x-3)}.$$

$$\text{We have } A = \frac{1^3}{(1-2)(1-3)} = \frac{1}{2}, \quad B = \frac{2^3}{(2-1)(2-3)} = -8,$$

$$\text{and } C = \frac{3^3}{(3-1)(3-2)} = \frac{27}{2}.$$

$$\therefore \frac{x^3}{(x-1)(x-2)(x-3)} = 1 + \frac{1}{2(x-1)} - \frac{8}{(x-2)} + \frac{27}{2(x-3)}.$$

$$\begin{aligned} \text{Hence, } \int \frac{x^3 dx}{(x-1)(x-2)(x-3)} &= \int 1 \cdot dx + \int \frac{dx}{2(x-1)} - \int \frac{8 dx}{(x-2)} + \int \frac{27 dx}{2(x-3)} \\ &= x + \frac{1}{2} \log|x-1| - 8 \log|x-2| + \frac{27}{2} \log|x-3| + c. \end{aligned}$$

**Illustration 4:** Evaluate  $\int \frac{x^2}{(x^2+2)(x^2+3)} dx$ .

**Solution:** Let  $y = x^2$ .

Then  $\frac{x^2}{(x^2 + 2)(x^2 + 3)} = \frac{y}{(y + 2)(y + 3)} \equiv \frac{A}{(y + 2)} + \frac{B}{(y + 3)}$ , (say).

We have  $A = \text{the value of } \frac{y}{y + 3}, \text{ when } y \text{ is } -2,$

$$= -2$$

and  $B = \text{the value of } \frac{y}{y + 2}, \text{ when } y \text{ is } -3, = 3.$

Thus,  $\frac{x^2}{(x^2 + 2)(x^2 + 3)} = \frac{-2}{y + 2} + \frac{3}{y + 3} = \frac{-2}{x^2 + 2} + \frac{3}{x^2 + 3}.$

$$\begin{aligned} \therefore \int \frac{x^2}{(x^2 + 2)(x^2 + 3)} dx &= -2 \int \frac{dx}{x^2 + 2} + 3 \int \frac{dx}{x^2 + 3} \\ &= -2 \cdot \frac{1}{\sqrt{2}} \tan^{-1} \frac{x}{\sqrt{2}} + 3 \cdot \frac{1}{\sqrt{3}} \tan^{-1} \frac{x}{\sqrt{3}} + c \\ &= -\sqrt{2} \tan^{-1} \frac{x}{\sqrt{2}} + \sqrt{3} \tan^{-1} \frac{x}{\sqrt{3}} + c. \end{aligned}$$

**Note.** In the above example, the substitution was made only for the partial fraction part and not for the integration part.

**Illustration 5:** Evaluate  $\int \frac{(3x + 1)}{(x - 2)^2 (x + 2)} dx.$

**Solution:** Let  $\frac{(3x + 1)}{(x - 2)^2 (x + 2)} \equiv \frac{A}{(x - 2)} + \frac{B}{(x - 2)^2} + \frac{C}{x + 2}$

or  $(3x + 1) \equiv A(x - 2)(x + 2) + B(x + 2) + C(x - 2)^2 \quad \dots(1)$

Then  $B = \text{the value of } \frac{3x + 1}{x + 2}, \text{ when } x \text{ is } -2 = \frac{7}{4},$

$C = \text{the value of } \frac{3x + 1}{(x - 2)^2}, \text{ when } x \text{ is } -2 = -\frac{5}{16}.$

Comparing the coefficients of  $x^2$  on both sides of (1), we get

$$A + C = 0 \rightarrow A = -C = \frac{5}{16}.$$

Thus,  $\frac{(3x + 1)}{(x - 2)^2 (x + 2)} = \frac{5}{16(x - 2)} + \frac{7}{4(x - 2)^2} - \frac{5}{16(x + 2)}.$

$$\begin{aligned} \therefore \int \frac{(3x + 1)}{(x - 2)^2 (x + 2)} dx &= \frac{5}{16} \int \frac{dx}{x - 2} + \frac{7}{4} \int \frac{dx}{(x - 2)^2} - \frac{5}{16} \int \frac{dx}{x + 2} \\ &= \frac{5}{16} \log \infty x - 2\infty - \frac{7}{4(x - 2)} - \frac{5}{16} \log \infty x + 2\infty + c. \end{aligned}$$

**Illustration 6:** Evaluate  $\int \frac{\cos^3 x + \cos^5 x}{\sin^2 x + \sin^4 x} dx.$

**Solution:**

$$\begin{aligned}
 \int \frac{(\cos^3 x + \cos^5 x)}{(\sin^2 x + \sin^4 x)} dx &= \int \frac{\cos^2 x (1 + \cos^2 x)}{\sin^2 x (1 + \sin^2 x)} \cos x dx \\
 &= \int \frac{(1 - \sin^2 x)(2 - \sin^2 x)}{\sin^2 x (1 + \sin^2 x)} \cos x dx \\
 &= \int \frac{(1 - t^2)(2 - t^2)}{t^2 (1 + t^2)} dt \quad [\text{Putting } \sin x = t \text{ so that } \cos x dx = dt] \\
 &= \int \left[ 1 + \frac{2}{t^2} - \frac{6}{1 + t^2} \right] dt \quad [\text{Resolving into partial fractions}] \\
 &= \int dt + 2 \int \frac{dt}{t^2} - 6 \int \frac{1}{1 + t^2} dt \\
 &= t - \frac{2}{t} - 6 \tan^{-1} t + c \\
 &= \sin x - \frac{2}{\sin x} - 6 \tan^{-1} (\sin x) + c \\
 &= \sin x - 2 \operatorname{cosec} x - 6 \tan^{-1} (\sin x) + c.
 \end{aligned}$$

**A special Type of Integral.** Here we shall give a very interesting method for evaluating the integral of a fraction in which the denominator is of degree 4 and the numerator is either of degree 2 or is constant. Moreover the odd powers of  $x$  occur neither in the numerator nor in the denominator.

**Method.** Divide the numerator and denominator by  $x^2$  and put  $x + \frac{1}{x} = t$  or  $x - \frac{1}{x} = t$ , whichever on differentiation gives the numerator of the resulting integrand.

**Illustration 7:** Evaluate :

$$(i) \int \frac{x^2 + 1}{x^4 + 1} dx \quad [\text{UPTU 2005}] \quad (ii) \int \frac{x^2 - 1}{x^4 + x^2 + 1} dx.$$

**Solution:**

$$(i) \text{ Let } I = \int \frac{x^2 + 1}{x^4 + 1} dx.$$

Here both the numerator and the denominator do not contain odd powers of  $x$ . Also the numerator is of degree 2 and the denominator is of degree 4. So dividing the numerator and the denominator by  $x^2$ , we get

$$I = \int \frac{1 + (1/x^2)}{x^2 + (1/x^2)} dx = \int \frac{1 + (1/x^2)}{[x - (1/x)]^2 + 2}$$

$$\left[ \text{Note that } \frac{d}{dx} \left( x - \frac{1}{x} \right) = 1 + \frac{1}{x^2} \right]$$

Now put  $x + (1/x) = t$ , so that  $\{1 + (1/x^2)\} dx = dt$

$$\begin{aligned}\therefore I &= \int \frac{dt}{t^2 + t} = \frac{1}{\sqrt{2}} \tan^{-1} \left( \frac{t}{\sqrt{2}} \right) + c = \frac{1}{\sqrt{2}} \tan^{-1} \left\{ \frac{x - (1/x)}{\sqrt{2}} \right\} + c \\ &= \frac{1}{\sqrt{2}} \tan^{-1} \left( \frac{x^2 - 1}{x \sqrt{2}} \right) + c.\end{aligned}$$

(ii) We have  $I = \int \frac{x^2 - 1}{x^4 + x^2 + 1} dx$ , [Note the form of the integrand]

$$= \int \frac{1 - (1/x^2)}{x^2 + 1 + (1/x^2)} dx,$$

dividing the numerator and the denominator by  $x^2$

$$= \int \frac{1 - (1/x^2)}{x + (1/x)^2 - 1} dx, \quad \left[ \text{Note that } \frac{d}{dx} \{x + (1/x)\} = 1 - (1/x^2) \right]$$

Now put  $x + (1/x) = t$ , so that  $\{1 - (1/x^2)\} dx = dt$

$$\begin{aligned}\therefore I &= \int \frac{dt}{t^2 - 1} = \frac{1}{2} \log \left| \frac{t-1}{t+1} \right| + c = \frac{1}{2} \log \left| \frac{x + (1/x) - 1}{x + (1/x) + 1} \right| \\ &= \frac{1}{2} \log \left| \frac{x^2 - x + 1}{x^2 + x + 1} \right| + c\end{aligned}$$

**Illustration 8:** Evaluate  $\int (\sqrt{\tan \theta} + \sqrt{\cot \theta}) d\theta$ .

**Solution:** We have  $I = \int \left\{ \sqrt{\tan \theta} + \frac{1}{\sqrt{\tan \theta}} \right\} d\theta = \int \frac{\tan \theta + 1}{\sqrt{\tan \theta}} d\theta$ .

Put  $\tan \theta = x^2$ , so that  $\sec^2 \theta d\theta = 2x dx$  or  $d\theta = \frac{2x dx}{1 + x^4}$ .

$$\begin{aligned}\therefore I &= \int \frac{x^2 + 1}{x} \cdot \frac{2x dx}{1 + x^4} = 2 \int \frac{x^2 + 1}{x^4 + 1} dx \\ &= \sqrt{2} \tan^{-1} \left\{ \frac{\tan \theta - 1}{\sqrt{2 \tan \theta}} \right\} + c. \quad [\text{Proceeding as in illustration 7 (i)}]\end{aligned}$$

**Illustration 9:** Evaluate  $\int \sqrt{\cot x} dx$ .

**Solution:** Let  $I = \int \sqrt{\cot x} dx$ .

Put  $\cot x = t^2$ , so that  $-\operatorname{cosec}^2 x dx = 2t dt$

or 
$$dx = \frac{-2t dt}{\operatorname{cosec}^2 x} = \frac{-2t dt}{1 + \cot^2 x} = \frac{-2t dt}{1 + t^4}.$$

$$\therefore I = \int \frac{t(-2t)dt}{1+t^4} = - \int \frac{2t^2 dt}{1+t^4} = - \int \frac{2dt}{t^2 + (1/t^2)}$$

dividing the numerator and the denominator by  $t^2$

$$I = \int \frac{\{1 + (1/t^2)\} + \{1 - (1/t^2)\}}{t^2 + 1(1/t^2)} dt$$

$$I = \int \frac{\{1 + (1/t^2)\}}{\{t - (1/t)\}^2 + 2} dt - \int \frac{\{1 - (1/t^2)\}}{\{t + (1/t)\}^2 - 2} dt$$

In the first integral put  $t - (1/t) = z$ , so that  $\{1 - (1/t^2)\} dt = dz$ .

In the second integral put  $t + (1/t) = u$ , so that  $\{1 + (1/t^2)\} dt = du$ .

$$\begin{aligned} \text{Hence, } I &= - \int \frac{dz}{z^2 + 2} - \int \frac{du}{u^2 - 2} = - \frac{1}{\sqrt{2}} \tan^{-1} \frac{z}{\sqrt{2}} - \frac{1}{2\sqrt{2}} \log \left| \frac{u - \sqrt{2}}{u + \sqrt{2}} \right| + c \\ &= - \frac{1}{\sqrt{2}} \tan^{-1} \left\{ \frac{t - (1/t)}{\sqrt{2}} \right\} - \frac{1}{2\sqrt{2}} \log \left| \frac{t + (1/t) - \sqrt{2}}{t + (1/t) + \sqrt{2}} \right| + c \\ &= - \frac{1}{\sqrt{2}} \tan^{-1} \left\{ \frac{t^2 - 1}{t\sqrt{2}} \right\} - \frac{1}{2\sqrt{2}} \log \left| \frac{t^2 - \sqrt{2}t + 1}{t^2 + \sqrt{2}t + 1} \right| + c \\ &= - \frac{1}{\sqrt{2}} \tan^{-1} \left\{ \frac{\cot x - 1}{\sqrt{2} \cot x} \right\} - \frac{1}{2\sqrt{2}} \log \left| \frac{\cot x - \sqrt{2} \cot x + 1}{\cot x + \sqrt{2} \cot x + 1} \right| + c. \end{aligned}$$

## Comprehensive Exercise 5

Evaluate the following integrals :

1.  $\int \frac{(1-x^2)}{x(1-2x)} dx.$

2.  $\int \frac{(2x+1)}{(x+1)(x-2)} dx.$

3.  $\int \frac{dx}{x(x+2)}.$

4.  $\int \frac{x}{(x+2)(x+3)} dx.$

5.  $\int \frac{x}{(x+2)(3-2x)} dx.$

6.  $\int \frac{2x+5}{x^2-x-2} dx.$



$$7. \int \frac{x^3 - x - 2}{1 - x^2} dx.$$

$$8. \int \frac{2x - 3}{(x^2 - 1)(2x + 3)} dx.$$

$$9. \int \frac{dx}{x(x^n + 1)}.$$

$$10. \int \frac{dx}{(1 - e^x)}.$$

$$11. \int \frac{\sin x}{\sin 4x} dx.$$

$$12. \int \sqrt{\tan x} dx.$$

## Answers

$$1. \frac{1}{2}x + \log |x| - \frac{3}{4} \log |1 - 2x| + c.$$

$$2. \frac{1}{3} \log |x + 1| + \frac{5}{3} \log |x - 2| + c.$$

$$3. \frac{1}{2} \log \left| \frac{x}{x + 2} \right| + c.$$

$$4. 3 \log |x + 3| - 2 \log |x + 2| + c.$$

$$5. -\frac{2}{7} \log |x + 2| - \frac{3}{14} \log |3 - 2x| + c.$$

$$6. 3 \log |x - 2| - \log |x + 1| + c.$$

$$7. -\frac{x^2}{2} + \log \left| \frac{1 - x}{1 + x} \right| + c.$$

$$8. \frac{5}{2} \log |x + 1| - \frac{1}{10} \log |x - 1| - \frac{12}{5} \log |2x + 3| + c.$$

$$9. \frac{1}{n} \log |x^n - 1| - \frac{1}{n} \log |x^n + 1| + c.$$

$$10. \log |e^x| - \log |1 - e^x| + c.$$

$$11. -\frac{1}{8} \log \left| \frac{1 + \sin x}{1 - \sin x} \right| + \frac{1}{4\sqrt{2}} \log \left| \frac{1 + \sqrt{2} \sin x}{1 - \sqrt{2} \sin x} \right| + c.$$

$$12. \frac{1}{\sqrt{2}} \tan^{-1} \left\{ \frac{\tan x - 1}{\sqrt{2} \tan x} \right\} + \frac{1}{2\sqrt{2}} \log \left| \frac{\tan x - \sqrt{2 \tan x} + 1}{\tan x + \sqrt{2 \tan x} + 1} \right| + c.$$

## Comprehensive Exercise 6

### Fill in the Blank(s).

Fill in the blanks ".....", so that the following statements are complete and correct.

$$1. \text{ The value of } \int \operatorname{cosec} x \cot x dx = \dots\dots\dots + c.$$

$$2. \text{ The value of } \int a^x dx = \dots\dots\dots + c.$$

3. The value of  $\int \tan^2 x \, dx = \dots\dots\dots + c$ .
4. The value of  $\int e^{(1-3x)} \, dx = \dots\dots\dots + c$ .
5. The value of  $\int [f(x)]^n f'(x) \, dx = \frac{[f(x)]^{n+1}}{n+1} + c$ , when  $n \neq -1$  is known as ..... formula.
6. The value of  $\int \sin 2x \, dx = \dots\dots\dots + c$ . **[UPTU 2009]**
7. The value of  $\int \log x \, dx = \dots\dots\dots + c$ . **[UPTU 2009]**
8. The value of  $\int \frac{\sec^2 (\log x)}{x} \, dx = \dots\dots\dots + c$ .

### Multiple Choice Questions.

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

9. The value of  $\int \sec (ax + b) \tan (ax + b) \, dx$  is
 

|                                     |                                     |
|-------------------------------------|-------------------------------------|
| (a) $\frac{1}{a} \sec (ax + b) + c$ | (b) $\frac{1}{b} \sec (ax + b) + c$ |
| (c) $\frac{1}{a} \tan (ax + b) + c$ | (d) $\tan (ax + b) + c$             |
10.  $\int \frac{1}{\sqrt{(a^2 - x^2)}} \, dx$  is equal to
 

|                             |                             |
|-----------------------------|-----------------------------|
| (a) $\sin^{-1} (a - x) + c$ | (b) $\sin^{-1} (x - a) + c$ |
| (c) $\sin^{-1} (x / a) + c$ | (d) $\sin^{-1} (a / x) + c$ |
11. The value of  $\int \frac{\sqrt{(\tan x)}}{\sin x \cos x} \, dx$  is
 

|                               |                             |
|-------------------------------|-----------------------------|
| (a) $2 \sqrt{(\tan x)} + c$   | (b) $2 \sqrt{(\sec x)} + c$ |
| (c) $2 / \sqrt{(\tan x)} + c$ | (d) none of these           |
12. The value of  $\int \sqrt{(1 + \sin x)} \, dx$  is
 

|                                       |                                       |
|---------------------------------------|---------------------------------------|
| (a) $2 (\sin x / 2 - \cos x / 2) + c$ | (b) $2 (\sin x / 2 + \cos x / 2) + c$ |
| (c) $\sin x / 2 - \cos x / 2 + c$     | (d) none of these.                    |
13. The value of  $\int \frac{\cos \sqrt{x}}{\sqrt{x}} \, dx$  is
 

|                           |                                   |
|---------------------------|-----------------------------------|
| (a) $2 \cos \sqrt{x} + c$ | (b) $\sqrt{\{(\cos x) / x\}} + c$ |
| (c) $\sin \sqrt{x} + c$   | (d) $2 \sin \sqrt{x} + c$         |
14. The value of  $\int e^{ax} \sin bx \, dx$  is
 

|                                                            |                                                            |
|------------------------------------------------------------|------------------------------------------------------------|
| (a) $\frac{e^{bx}}{a^2 + b^2} (a \sin bx - b \cos bx) + c$ | (b) $\frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx) + c$ |
|------------------------------------------------------------|------------------------------------------------------------|

$$(c) \frac{e^{ax}}{a^2 + b^2} (a \sin bx + b \cos bx) + c \quad (d) \frac{e^{ax}}{a^2 + b^2} (b \cos bx - a \sin bx) + c$$

### True or False.

Write 'T' for true and 'F' for false statement.

15.  $\int 2^{(x+3)} dx = \frac{2^{(x+3)}}{\log 2} + c.$
16.  $\int f_1(x) \cdot f_2(x) dx = f_1(x) \cdot \int f_2(x) dx + \int \left[ \left\{ \frac{d}{dx} f_1(x) \right\} \cdot \int f_2(x) dx \right] dx.$
17.  $\int \frac{dx}{a^2 + x^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} + c.$
18.  $\int \frac{(a^x + b^x)^2}{a^x b^x} dx = \frac{(a/b)^x}{\log(a/b)} + \frac{(b/a)^x}{\log(b/a)} + 2x + c.$
19.  $\int \sqrt{1 - \sin 2x} dx = \sin x - \cos x + c.$
20.  $\int e^{2x-3} dx = \frac{1}{2} e^{2x-3} + c.$

## Answers

- |                               |                                       |
|-------------------------------|---------------------------------------|
| 1. $-\operatorname{cosec} x.$ | 2. $\frac{a^x}{\log a}$               |
| 3. $\tan x - x.$              | 4. $\frac{-1}{3} e^{(1-3x)}.$         |
| 5. power                      | 6. $-\left(\frac{\cos 2x}{2}\right).$ |
| 7. $x(\log x - 1).$           | 8. $\tan(\log x).$                    |
| 9. (a)                        | 10. (c)                               |
| 11. (a)                       | 12. (a)                               |
| 13. (d)                       | 14. (b)                               |
| 15. T                         | 16. F                                 |
| 17. T                         | 18. T                                 |
| 19. F                         | 20. T                                 |